

1 Multi-Feature Configuration Examples

- 1.1 Deploying VRRP on a Data Center Network with 3-Layer Architecture
- 1.2 Deploying VRRP on a Data Center Network with 2-Layer Architecture
- 1.3 Configuring a Data Center Network Based on VS and VRRP
- 1.4 Example for Configuring IPv4 VXLAN in Distributed Gateway Deployment Mode
- 1.5 Example for Configuring IPv6 VXLAN in Distributed Gateway Deployment Mode
- 1.6 Example for Configuring VXLAN in Centralized Gateway Mode
- 1.7 Example for Configuring a VXLAN Network Using a Two-Layer Architecture in Distributed Gateway Deployment Mode in a DC
- 1.8 Example for Configuring NFVI Distributed Gateways (Asymmetric Mode)
- 1.9 Example for Configuring NFVI Distributed Gateways (Symmetric Mode)
- 1.10 Example for Configuring M-LAG Lite
- 1.11 Example for Configuring IPv6 M-LAG Lite
- 1.12 Example for Configuring VRRP over VXLAN with Inter-DC Master and Backup Gateways
- 1.13 Configuring M-LAG Dual-Active Distributed Gateways as DHCPv4/DHCPv6 Relay Agents
- 1.14 Configuring M-LAG and Transparent Firewalls
- 1.15 Configuring M-LAG, VSs, and Firewalls in Bypass Mode
- 1.16 Configuring M-LAG and VRRP Across Data Centers
- 1.17 Configuring M-LAG and VRRP/VRRP6 Across Data Centers
- 1.18 Example for Configuring Dynamic Routing Protocols (OSPF/OSPFv3/BGP/BGP4+) for Communication with an M-LAG
- 1.19 Example for Configuring IPv4 Layer 3 Multicast Over VXLAN When the Multicast Source is Located on an External Network

[1.20 Configuring an Intelligent Lossless Network](#)

[1.21 Example for Configuring the INC Function for an Intelligent Lossless Network](#)

1.1 Deploying VRRP on a Data Center Network with 3-Layer Architecture

Applicable Products and Versions

- CloudEngine series switches running V100R001C00 or later versions
- USG5500 series products running V300R001
- For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

The data center uses the access, aggregation, and core layers. The requirements are as follows:

- To ensure service reliability, deploy link redundancy between access and aggregation layers. When one uplink is disconnected, traffic can be switched to another uplink.
- Loops caused by link redundancy between access and aggregation layers are eliminated.
- Connect the core device to the firewall to filter service traffic.
- Deploy OSPF at aggregation and core layers to implement Layer 3 connectivity.

Figure 1-1 Networking for deploying VRRP on a data center network with 3-layer architecture

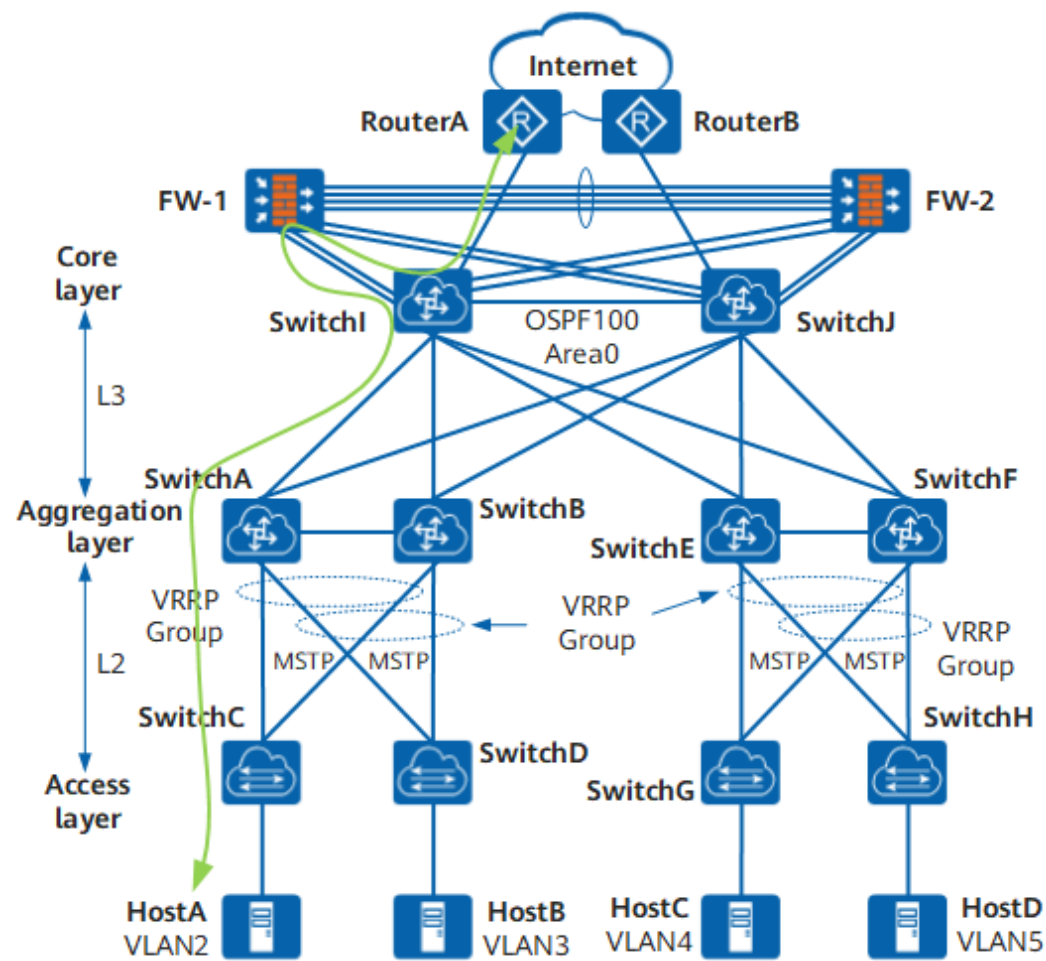


Table 1-1 Data preparation (SwitchA, SwitchB, SwitchC, and SwitchD are used as examples)

Device	VLAN and IP Address	Interface Number	Description
SwitchA	VLAN: 2 IP Address: 10.1.2.102/24 Virtual IP Address: 10.1.2.100	10GE1/0/1	TO-CE6800-SWITCHC
		10GE1/0/3	TO-CE12800-SWITCHB
	VLAN: 3 IP Address: 10.1.3.102/24 Virtual IP Address: 10.1.3.100	10GE1/0/2	TO-CE6800-SWITCHD
		10GE1/0/3	TO-CE12800-SWITCHB

Device	VLAN and IP Address	Interface Number	Description
	VLAN: 6 IP Address: 10.1.6.102/24	10GE1/0/4	TO-CE12800-SWITCHI
	VLAN: 7 IP Address: 10.1.7.102/24	10GE1/0/5	TO-CE12800-SWITCHJ
SwitchB	VLAN: 2 IP Address: 10.1.2.103/24 Virtual IP Address: 10.1.2.100	10GE1/0/2	TO-CE6800-SWITCHC
		10GE1/0/3	TO-CE12800-SWITCHA
	VLAN: 3 IP Address: 10.1.3.103/24 Virtual IP Address: 10.1.3.100	10GE1/0/1	TO-CE6800-SWITCHD
		10GE1/0/3	TO-CE12800-SWITCHA
	VLAN: 6 IP Address: 10.1.6.103/24	10GE1/0/4	TO-CE12800-SWITCHI
	VLAN: 7 IP Address: 10.1.7.103/24	10GE1/0/5	TO-CE12800-SWITCHJ
SwitchC	VLAN: 2	10GE1/0/1	TO-CE12800-SWITCHA
		10GE1/0/2	TO-CE12800-SWITCHB
		10GE1/0/3	TO-HOSTA
SwitchD	VLAN: 3	10GE1/0/1	TO-CE12800-SWITCHB
		10GE1/0/2	TO-CE12800-SWITCHA
		10GE1/0/3	TO-HOSTB
SwitchI	VLAN: 6 IP Address: 10.1.6.104/24	10GE1/0/1	TO-CE12800-SWITCHA
		10GE1/0/2	TO-CE12800-SWITCHB

Device	VLAN and IP Address	Interface Number	Description
		10GE1/0/3	TO-CE12800-SWITCHE
		10GE1/0/4	TO-CE12800-SWITCHF
	VLAN: 8 IP Address: 10.1.8.104/24	10GE1/0/5	TO-ROUTERA
	VLAN: 9 IP Address: 172.16.1.2/24	10GE1/0/6	TO-FW-1
	VLAN: 10 IP Address: 172.16.2.2/24	10GE1/0/7	TO-FW-1
	VLAN: 11 IP Address: 172.16.3.2/24	10GE1/0/8	TO-FW-2
	VLAN: 12 IP Address: 172.16.4.2/24	10GE1/0/9	TO-FW-2
	VLAN: 13 IP Address: 10.1.13.102/24	10GE1/0/14	TO-CE12800-SWITCHJ
SwitchJ	VLAN: 7 IP Address: 10.1.7.104/24	10GE1/0/1	TO-CE12800-SWITCHA
		10GE1/0/2	TO-CE12800-SWITCHB
		10GE1/0/3	TO-CE12800-SWITCHE
		10GE1/0/4	TO-CE12800-SWITCHF
	VLAN: 8 IP Address: 10.1.8.105/24	10GE1/0/5	TO-ROUTERB
	VLAN: 9 IP Address: 172.16.6.2/24	10GE1/0/6	TO-FW-1

Device	VLAN and IP Address	Interface Number	Description
	VLAN: 10 IP Address: 172.16.7.2/24	10GE1/0/7	TO-FW-1
	VLAN: 11 IP Address: 172.16.8.2/24	10GE1/0/8	TO-FW-2
	VLAN: 12 IP Address: 172.16.9.2/24	10GE1/0/9	TO-FW-2
	VLAN: 13 IP Address: 10.1.13.103/24	10GE1/0/14	TO-CE12800-SWITCHI
FW-1	172.16.1.1/24	GE1/0/1	TO-SWITCHI-Upstream
	172.16.2.1/24	GE1/0/2	TO-SWITCHI-Downstream
	172.16.6.1/24	GE1/0/3	TO-SWITCHJ-Upstream
	172.16.7.1/24	GE1/0/4	TO-SWITCHJ-Downstream
	172.16.5.1/24	Eth-Trunk1: GE2/0/0	TO-FW-2-HRP
		Eth-Trunk1: GE2/0/1	
		Eth-Trunk1: GE2/0/2	
		Eth-Trunk1: GE2/0/3	
	172.16.100.1/24	Loopback1	NA
	172.16.100.2/24	Loopback2	NA
	172.16.100.3/24	Loopback3	NA
	172.16.100.4/24	Loopback4	NA
FW-2	172.16.8.1/24	GE1/0/1	TO-SWITCHJ-Upstream
	172.16.9.1/24	GE1/0/2	TO-SWITCHJ-Downstream

Device	VLAN and IP Address	Interface Number	Description
	172.16.3.1/24	GE1/0/3	TO-SWITCHI- Upstream
	172.16.4.1/24	GE1/0/4	TO-SWITCHI- Downstream
	172.16.10.1/24	Eth-Trunk1: GE2/0/0	TO-FW-1-HRP
		Eth-Trunk1: GE2/0/1	
		Eth-Trunk1: GE2/0/2	
		Eth-Trunk1: GE2/0/3	
	172.16.100.1/24	Loopback1	NA
	172.16.100.2/24	Loopback2	NA
	172.16.100.3/24	Loopback3	NA
	172.16.100.4/24	Loopback4	NA

Requirement Analysis

- Deploy VRRP between SwitchA and SwitchB to implement link redundancy at the aggregation layer.
- Deploy MSTP between SwitchA, SwitchB, and SwitchC to eliminate loops at the access and aggregation layers.
- Configure HSB for FW-1 and FW-2 so that traffic forwarded by SwitchI or SwitchJ is filtered by the firewall and then reaches the data center or Internet.
- Deploy OSPF between SwitchA, SwitchB, SwitchI, and SwitchJ to implement Layer 3 connectivity. SwitchA and SwitchB are aggregation devices. SwitchI and SwitchJ are core devices.

Procedure

Step 1 Configure basic MSTP functions.

NOTE

Two devices belong to the same MST region when the following parameters on the two devices are the same:

- Name of the MST region
- Mapping between VLANs and MSTIs
- Revision level of the MST region

1. Configure SwitchA, SwitchB, and SwitchC in the MST region **RG1** and create MSTI 1 and MSTI 2.

Configure an MST region on SwitchA.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchA
[*HUAWEI] commit
[~SwitchA] stp region-configuration
[~SwitchA-mst-region] region-name RG1
[*SwitchA-mst-region] instance 1 vlan 2
[*SwitchA-mst-region] instance 2 vlan 3
[*SwitchA-mst-region] commit
[~SwitchA-mst-region] quit
```

Configure an MST region on SwitchB.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchB
[*HUAWEI] commit
[~SwitchB] stp region-configuration
[~SwitchB-mst-region] region-name RG1
[*SwitchB-mst-region] instance 1 vlan 2
[*SwitchB-mst-region] instance 2 vlan 3
[*SwitchB-mst-region] commit
[~SwitchB-mst-region] quit
```

Configure an MST region on SwitchC.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchC
[*HUAWEI] commit
[~SwitchC] stp region-configuration
[~SwitchC-mst-region] region-name RG1
[*SwitchC-mst-region] instance 1 vlan 2
[*SwitchC-mst-region] instance 2 vlan 3
[*SwitchC-mst-region] commit
[~SwitchC-mst-region] quit
```

Configure an MST region on SwitchD.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchD
[*HUAWEI] commit
[~SwitchD] stp region-configuration
[~SwitchD-mst-region] region-name RG1
[*SwitchD-mst-region] instance 1 vlan 2
[*SwitchD-mst-region] instance 2 vlan 3
[*SwitchD-mst-region] commit
[~SwitchD-mst-region] quit
```

2. Configure root bridges and secondary root bridges of MSTI 1 and MSTI 2 in the MST region **RG1**.

- Configure the root bridge and secondary root bridge of MSTI 1.

Configure SwitchA as the root bridge in MSTI 1.

```
[~SwitchA] stp instance 1 root primary
[*SwitchA] commit
```

Configure SwitchB as the secondary root bridge in MSTI 1.

```
[~SwitchB] stp instance 1 root secondary
[*SwitchB] commit
```

- Configure the root bridge and secondary root bridge of MSTI 2.

Configure SwitchB as the root bridge in MSTI 2.

```
[~SwitchB] stp instance 2 root primary
[*SwitchB] commit
```

Configure SwitchA as the secondary root bridge in MSTI 2.

```
[~SwitchA] stp instance 2 root secondary
[*SwitchA] commit
```

3. Set the path costs of the interfaces to be blocked in MSTI 1 and MSTI 2 to be greater than the default value.

 NOTE

- The path cost range depends on the algorithm. Huawei proprietary algorithm is used as an example. Set the path costs of the interfaces to be blocked in MSTI 1 and MSTI 2 to 20000.
- Switching devices on the same network must use the same algorithm to calculate the path cost of interfaces.

Configure SwitchA to use Huawei proprietary algorithm to calculate the path cost.

```
[~SwitchA] stp pathcost-standard legacy
[*SwitchA] commit
```

Configure SwitchB to use Huawei proprietary algorithm to calculate the path cost.

```
[~SwitchB] stp pathcost-standard legacy
[*SwitchB] commit
```

Configure SwitchC to use Huawei proprietary algorithm to calculate the path cost and set the path cost of 10GE1/0/2 to 20000 in MSTI 1.

```
[~SwitchC] stp pathcost-standard legacy
[*SwitchC] interface 10ge 1/0/2
[*SwitchC-10GE1/0/2] description TO-CE12800-SWITCHB
[*SwitchC-10GE1/0/2] stp instance 1 cost 20000
[*SwitchC-10GE1/0/2] commit
[~SwitchC-10GE1/0/2] quit
```

Configure SwitchD to use Huawei proprietary algorithm to calculate the path cost and set the path cost of 10GE1/0/2 to 20000 in MSTI 2.

```
[~SwitchD] stp pathcost-standard legacy
[*SwitchD] interface 10ge 1/0/2
[*SwitchD-10GE1/0/2] description TO-CE12800-SWITCHA
[*SwitchD-10GE1/0/2] stp instance 2 cost 20000
[*SwitchD-10GE1/0/2] commit
[~SwitchD-10GE1/0/2] quit
```

4. Enable MSTP to eliminate loops.

 NOTE

MSTP is enabled by default.

- Enable MSTP globally.

Enable MSTP on SwitchA.

```
[~SwitchA] stp enable
[*SwitchA] commit
```

Enable MSTP on SwitchB.

```
[~SwitchB] stp enable
[*SwitchB] commit
```

Enable MSTP on SwitchC.

```
[~SwitchC] stp enable
[*SwitchC] commit
```

Enable MSTP on SwitchD.

```
[~SwitchD] stp enable
[*SwitchD] commit
```

- Configure ports connected to hosts as edge ports.

Configure 10GE1/0/3 of SwitchC as an edge port.

```
[~SwitchC] interface 10ge 1/0/3
[*SwitchC-10GE1/0/3] description TO-HOSTA
[*SwitchC-10GE1/0/3] stp edged-port enable
```

```
[*SwitchC-10GE1/0/3] commit
[~SwitchC-10GE1/0/3] quit

# Configure 10GE1/0/3 of SwitchD as an edge port.

[~SwitchD] interface 10ge 1/0/3
[*SwitchD-10GE1/0/3] description TO-HOSTB
[*SwitchD-10GE1/0/3] stp edged-port enable
[*SwitchD-10GE1/0/3] commit
[~SwitchD-10GE1/0/3] quit
```

Step 2 Enable protection functions on the designated port of each root bridge in each MSTI. Here, root protection is used.

Enable root protection on 10GE1/0/1 of SwitchA.

```
[~SwitchA] interface 10ge 1/0/1
[~SwitchA-10GE1/0/1] description TO-CE6800-SWITCHC
[*SwitchA-10GE1/0/1] stp root-protection
[*SwitchA-10GE1/0/1] commit
[~SwitchA-10GE1/0/1] quit
```

Enable root protection on 10GE1/0/1 of SwitchB.

```
[~SwitchB] interface 10ge 1/0/1
[~SwitchB-10GE1/0/1] description TO-CE6800-SWITCHD
[*SwitchB-10GE1/0/1] stp root-protection
[*SwitchB-10GE1/0/1] commit
[~SwitchB-10GE1/0/1] quit
```

Step 3 Configure Layer 2 forwarding on switches on the ring network.

- Create VLAN 2 and VLAN 3 on SwitchA, SwitchB, and SwitchC.

Create VLAN 2 and VLAN 3 on SwitchA.

```
[~SwitchA] vlan batch 2 to 3
```

Create VLAN 2 and VLAN 3 on SwitchB.

```
[~SwitchB] vlan batch 2 to 3
```

Create VLAN 2 on SwitchC.

```
[~SwitchC] vlan batch 2
```

Create VLAN 3 on SwitchD.

```
[~SwitchD] vlan batch 3
```

- Add interfaces connecting to the ring to VLANs.

Add 10GE1/0/1 on SwitchA to VLAN 2.

```
[~SwitchA] interface 10ge 1/0/1
[~SwitchA-10GE1/0/1] port link-type trunk
[*SwitchA-10GE1/0/1] undo port trunk allow-pass vlan 1
[*SwitchA-10GE1/0/1] port trunk allow-pass vlan 2
[*SwitchA-10GE1/0/1] commit
[~SwitchA-10GE1/0/1] quit
```

Add 10GE1/0/2 on SwitchA to VLAN 3.

```
[~SwitchA] interface 10ge 1/0/2
[~SwitchA-10GE1/0/2] description TO-CE6800-SWITCHD
[*SwitchA-10GE1/0/2] port link-type trunk
[*SwitchA-10GE1/0/2] port trunk allow-pass vlan 3
[*SwitchA-10GE1/0/2] commit
[~SwitchA-10GE1/0/2] quit
```

Add 10GE1/0/3 on SwitchA to VLAN 2 and VLAN 3.

```
[~SwitchA] interface 10ge 1/0/3
[~SwitchA-10GE1/0/3] description TO-CE12800-SWITCHB
[*SwitchA-10GE1/0/3] port link-type trunk
[*SwitchA-10GE1/0/3] undo port trunk allow-pass vlan 1
[*SwitchA-10GE1/0/3] port trunk allow-pass vlan 2 to 3
```

```
[*SwitchA-10GE1/0/3] commit
[~SwitchA-10GE1/0/3] quit

# Add 10GE1/0/1 on SwitchB to VLAN 3.

[~SwitchB] interface 10ge 1/0/1
[~SwitchB-10GE1/0/1] port link-type trunk
[*SwitchB-10GE1/0/1] undo port trunk allow-pass vlan 1
[*SwitchB-10GE1/0/1] port trunk allow-pass vlan 3
[*SwitchB-10GE1/0/1] commit
[~SwitchB-10GE1/0/1] quit

# Add 10GE1/0/2 on SwitchB to VLAN 2.

[~SwitchB] interface 10ge 1/0/2
[~SwitchB-10GE1/0/2] description TO-CE6800-SWITCHC
[*SwitchB-10GE1/0/2] port link-type trunk
[*SwitchB-10GE1/0/2] undo port trunk allow-pass vlan 1
[*SwitchB-10GE1/0/2] port trunk allow-pass vlan 2
[*SwitchB-10GE1/0/2] commit
[~SwitchB-10GE1/0/2] quit

# Add 10GE1/0/3 on SwitchB to VLAN 2 and VLAN 3.

[~SwitchB] interface 10ge 1/0/3
[~SwitchB-10GE1/0/3] description TO-CE12800-SWITCHA
[*SwitchB-10GE1/0/3] port link-type trunk
[*SwitchB-10GE1/0/3] undo port trunk allow-pass vlan 1
[*SwitchB-10GE1/0/3] port trunk allow-pass vlan 2 to 3
[*SwitchB-10GE1/0/3] commit
[~SwitchB-10GE1/0/3] quit

# Add 10GE1/0/1 on SwitchC to VLAN 2.

[~SwitchC] interface 10ge 1/0/1
[~SwitchC-10GE1/0/1] description TO-CE12800-SWITCHA
[*SwitchC-10GE1/0/1] port link-type trunk
[*SwitchC-10GE1/0/1] undo port trunk allow-pass vlan 1
[*SwitchC-10GE1/0/1] port trunk allow-pass vlan 2
[*SwitchC-10GE1/0/1] commit
[~SwitchC-10GE1/0/1] quit

# Add 10GE1/0/2 on SwitchC to VLAN 2.

[~SwitchC] interface 10ge 1/0/2
[~SwitchC-10GE1/0/2] port link-type trunk
[*SwitchC-10GE1/0/2] undo port trunk allow-pass vlan 1
[*SwitchC-10GE1/0/2] port trunk allow-pass vlan 2
[*SwitchC-10GE1/0/2] commit
[~SwitchC-10GE1/0/2] quit

# Add 10GE1/0/3 on SwitchC to VLAN 2.

[~SwitchC] interface 10ge 1/0/3
[~SwitchC-10GE1/0/3] port link-type access
[*SwitchC-10GE1/0/3] port default vlan 2
[*SwitchC-10GE1/0/3] commit
[~SwitchC-10GE1/0/3] quit

# Add 10GE1/0/1 on SwitchD to VLAN 3.

[~SwitchD] interface 10ge 1/0/1
[~SwitchD-10GE1/0/1] description TO-CE12800-SWITCHB
[*SwitchD-10GE1/0/1] port link-type trunk
[*SwitchD-10GE1/0/1] undo port trunk allow-pass vlan 1
[*SwitchD-10GE1/0/1] port trunk allow-pass vlan 3
[*SwitchD-10GE1/0/1] commit
[~SwitchD-10GE1/0/1] quit

# Add 10GE1/0/2 on SwitchD to VLAN 3.

[~SwitchD] interface 10ge 1/0/2
[~SwitchD-10GE1/0/2] port link-type trunk
[*SwitchD-10GE1/0/2] undo port trunk allow-pass vlan 1
[*SwitchD-10GE1/0/2] port trunk allow-pass vlan 3
[*SwitchD-10GE1/0/2] commit
[~SwitchD-10GE1/0/2] quit
```

Add 10GE1/0/3 on SwitchD to VLAN 3.

```
[~SwitchD] interface 10ge 1/0/3
[~SwitchD-10GE1/0/3] port link-type access
[*SwitchD-10GE1/0/3] port default vlan 3
[*SwitchD-10GE1/0/3] commit
[~SwitchD-10GE1/0/3] quit
```

Step 4 Configure VRRP groups.

Configure VRRP group 1 on SwitchA and SwitchB, set the priority of SwitchA to 120 and the preemption delay to 20s, and set the default priority for SwitchB.

```
[~SwitchA] interface vlanif 2
[*SwitchA-Vlanif2] vrrp vrid 1 virtual-ip 10.1.2.100
[*SwitchA-Vlanif2] vrrp vrid 1 priority 120
[*SwitchA-Vlanif2] vrrp vrid 1 preempt-mode timer delay 20
[*SwitchA-Vlanif2] commit
[~SwitchA-Vlanif2] quit
[~SwitchB] interface vlanif 2
[*SwitchB-Vlanif2] vrrp vrid 1 virtual-ip 10.1.2.100
[*SwitchB-Vlanif2] commit
[~SwitchB-Vlanif2] quit
```

Configure VRRP group 2 on SwitchA and SwitchB, set the priority of SwitchB to 120 and the preemption delay to 20s, and set the default priority for SwitchA.

```
[~SwitchB] interface vlanif 3
[*SwitchB-Vlanif3] vrrp vrid 2 virtual-ip 10.1.3.100
[*SwitchB-Vlanif3] vrrp vrid 2 priority 120
[*SwitchB-Vlanif3] vrrp vrid 2 preempt-mode timer delay 20
[*SwitchB-Vlanif3] commit
[~SwitchB-Vlanif3] quit
[~SwitchA] interface vlanif 3
[*SwitchA-Vlanif3] vrrp vrid 2 virtual-ip 10.1.3.100
[*SwitchA-Vlanif3] commit
[~SwitchA-Vlanif3] quit
```

Set virtual IP address 10.1.2.100 of VRRP group 1 as the default gateway of HostA, and virtual IP address 10.1.3.100 of VRRP group 2 as the default gateway of HostB.

Step 5 Configure devices to ensure network connectivity.

Assign an IP address to each interface. SwitchA is used as an example. The configurations of SwitchB, SwitchI, and SwitchJ are similar to the configuration of SwitchA.

```
[~SwitchA] vlan batch 6 7
[*SwitchA] interface 10ge 1/0/4
[*SwitchA-10GE1/0/4] description TO-CE12800-SWITCHI
[*SwitchA-10GE1/0/4] port link-type trunk
[*SwitchA-10GE1/0/4] undo port trunk allow-pass vlan 1
[*SwitchA-10GE1/0/4] port trunk allow-pass vlan 6
[*SwitchA-10GE1/0/4] quit
[*SwitchA] interface 10ge 1/0/5
[*SwitchA-10GE1/0/5] description TO-CE12800-SWITCHJ
[*SwitchA-10GE1/0/5] port link-type trunk
[*SwitchA-10GE1/0/5] undo port trunk allow-pass vlan 1
[*SwitchA-10GE1/0/5] port trunk allow-pass vlan 7
[*SwitchA-10GE1/0/5] quit
[*SwitchA] interface vlanif 2
[*SwitchA-Vlanif2] ip address 10.1.2.102 24
[*SwitchA-Vlanif2] quit
[*SwitchA] interface vlanif 3
[*SwitchA-Vlanif3] ip address 10.1.3.102 24
[*SwitchA-Vlanif3] quit
[*SwitchA] interface vlanif 6
```

```
[*SwitchA-Vlanif6] ip address 10.1.6.102 24
[*SwitchA-Vlanif6] quit
[*SwitchA] interface vlanif 7
[*SwitchA-Vlanif7] ip address 10.1.7.102 24
[*SwitchA-Vlanif7] quit
[*SwitchA] commit
```

Configure OSPF between SwitchA, SwitchB, SwitchI, SwitchJ, and router.

```
[~SwitchA] ospf 1
[*SwitchA-ospf-1] area 0
[*SwitchA-ospf-1-area-0.0.0.0] network 10.1.2.0 0.0.0.255
[*SwitchA-ospf-1-area-0.0.0.0] network 10.1.3.0 0.0.0.255
[*SwitchA-ospf-1-area-0.0.0.0] network 10.1.6.0 0.0.0.255
[*SwitchA-ospf-1-area-0.0.0.0] network 10.1.7.0 0.0.0.255
[*SwitchA-ospf-1-area-0.0.0.0] quit
[*SwitchA-ospf-1] quit
[*SwitchA] commit
```

Step 6 Configure firewalls.

Configure hot standby on FW-1 and FW-2. Packets from SwitchI and SwitchJ are processed by FW-1 and FW-2, and then are forwarded to the data center or Internet.

FW-1 and FW-2 work in load balancing mode. If a firewall fails, service packets are switched to another firewall.

In this example, FW-1 and FW-2 are Huawei USG security gateways.

1. Perform basic configurations including the device name, interface, and security zone on FW-1.

```
<USG> system-view
[USG] sysname FW-1
[FW-1] interface GigabitEthernet 1/0/1
[FW-1-GigabitEthernet1/0/1] ip address 172.16.1.1 24
[FW-1-GigabitEthernet1/0/1] quit
[FW-1] interface GigabitEthernet 1/0/2
[FW-1-GigabitEthernet1/0/2] ip address 172.16.2.1 24
[FW-1-GigabitEthernet1/0/2] quit
[FW-1] interface GigabitEthernet 1/0/3
[FW-1-GigabitEthernet1/0/3] ip address 172.16.6.1 24
[FW-1-GigabitEthernet1/0/3] quit
[FW-1] interface GigabitEthernet 1/0/4
[FW-1-GigabitEthernet1/0/4] ip address 172.16.7.1 24
[FW-1-GigabitEthernet1/0/4] quit

[FW-1] interface Eth-Trunk 1
[FW-1-Eth-Trunk1] trunkport GigabitEthernet 2/0/0 2/0/1 2/0/2 2/0/3
[FW-1-Eth-Trunk1] ip address 172.16.5.1 24
[FW-1-Eth-Trunk1] quit

[FW-1] firewall zone trust
[FW-1-zone-trust] add interface GigabitEthernet 1/0/1
[FW-1-zone-trust] add interface GigabitEthernet 1/0/3
[FW-1-zone-trust] quit
[FW-1] firewall zone untrust
[FW-1-zone-untrust] add interface GigabitEthernet 1/0/2
[FW-1-zone-untrust] add interface GigabitEthernet 1/0/4
[FW-1-zone-untrust] quit
[FW-1] firewall zone dmz
[FW-1-zone-dmz] add interface Eth-Trunk 1
[FW-1-zone-dmz] quit

[FW-1] interface LoopBack 1
[FW-1-LoopBack1] ip address 172.16.100.1 32
[FW-1-LoopBack1] quit
[FW-1] interface LoopBack 2
```

```
[FW-1-LoopBack2] ip address 172.16.100.2 32
[FW-1-LoopBack2] quit
[FW-1] interface LoopBack 3
[FW-1-LoopBack3] ip address 172.16.100.3 32
[FW-1-LoopBack3] quit
[FW-1] interface LoopBack 4
[FW-1-LoopBack4] ip address 172.16.100.4 32
[FW-1-LoopBack4] quit
```

2. Perform basic configurations including the device name, interface, and security zone on FW-2.

```
<USG> system-view
[USG] sysname FW-2
[FW-2] interface GigabitEthernet 1/0/1
[FW-2-GigabitEthernet1/0/1] ip address 172.16.8.1 24
[FW-2-GigabitEthernet1/0/1] quit
[FW-2] interface GigabitEthernet 1/0/2
[FW-2-GigabitEthernet1/0/2] ip address 172.16.9.1 24
[FW-2-GigabitEthernet1/0/2] quit
[FW-2] interface GigabitEthernet 1/0/3
[FW-2-GigabitEthernet1/0/3] ip address 172.16.3.1 24
[FW-2-GigabitEthernet1/0/3] quit
[FW-2] interface GigabitEthernet 1/0/4
[FW-2-GigabitEthernet1/0/4] ip address 172.16.4.1 24
[FW-2-GigabitEthernet1/0/4] quit

[FW-2] interface Eth-Trunk 1
[FW-2-Eth-Trunk1] trunkport GigabitEthernet 2/0/0 2/0/1 2/0/2 2/0/3
[FW-2-Eth-Trunk1] ip address 172.16.10.1 24
[FW-2-Eth-Trunk1] quit

[FW-2] firewall zone trust
[FW-2-zone-trust] add interface GigabitEthernet 1/0/1
[FW-2-zone-trust] add interface GigabitEthernet 1/0/3
[FW-2-zone-trust] quit
[FW-2] firewall zone untrust
[FW-2-zone-untrust] add interface GigabitEthernet 1/0/2
[FW-2-zone-untrust] add interface GigabitEthernet 1/0/4
[FW-2-zone-untrust] quit
[FW-2] firewall zone dmz
[FW-2-zone-dmz] add interface Eth-Trunk 1
[FW-2-zone-dmz] quit

[FW-2] interface LoopBack 1
[FW-2-LoopBack1] ip address 172.16.100.1 32
[FW-2-LoopBack1] quit
[FW-2] interface LoopBack 2
[FW-2-LoopBack2] ip address 172.16.100.2 32
[FW-2-LoopBack2] quit
[FW-2] interface LoopBack 3
[FW-2-LoopBack3] ip address 172.16.100.3 32
[FW-2-LoopBack3] quit
[FW-2] interface LoopBack 4
[FW-2-LoopBack4] ip address 172.16.100.4 32
[FW-2-LoopBack4] quit
```

3. Configure OSPF on FW-1 and FW-2. Specify a unique router ID for each process. In addition, the router IDs of OSPF processes on active and standby firewalls must be different to prevent OSPF flapping.

```
[FW-1] ospf 1 router-id 172.16.100.1
[FW-1-ospf-1] area 0
[FW-1-ospf-1-area-0.0.0.0] network 172.16.1.0 0.0.0.255
[FW-1-ospf-1-area-0.0.0.0] network 172.16.100.1 0.0.0.0
[FW-1-ospf-1-area-0.0.0.0] quit
[FW-1-ospf-1] quit
[FW-1] ospf 2 router-id 172.16.100.2
[FW-1-ospf-2] area 0
[FW-1-ospf-2-area-0.0.0.0] network 172.16.2.0 0.0.0.255
[FW-1-ospf-2-area-0.0.0.0] network 172.16.100.2 0.0.0.0
```

```
[FW-1-ospf-2-area-0.0.0.0] quit
[FW-1-ospf-2] quit
[FW-1] ospf 3 router-id 172.16.100.3
[FW-1-ospf-3] area 0
[FW-1-ospf-3-area-0.0.0.0] network 172.16.6.0 0.0.0.255
[FW-1-ospf-3-area-0.0.0.0] network 172.16.100.3 0.0.0.0
[FW-1-ospf-3-area-0.0.0.0] quit
[FW-1-ospf-3] quit
[FW-1] ospf 4 router-id 172.16.100.4
[FW-1-ospf-4] area 0
[FW-1-ospf-4-area-0.0.0.0] network 172.16.7.0 0.0.0.255
[FW-1-ospf-4-area-0.0.0.0] network 172.16.100.4 0.0.0.0
[FW-1-ospf-4-area-0.0.0.0] quit
[FW-1-ospf-4] quit

[FW-2] ospf 1 router-id 172.16.100.6
[FW-2-ospf-1] area 0
[FW-2-ospf-1-area-0.0.0.0] network 172.16.8.0 0.0.0.255
[FW-2-ospf-1-area-0.0.0.0] network 172.16.100.1 0.0.0.0
[FW-2-ospf-1-area-0.0.0.0] quit
[FW-2-ospf-1] quit
[FW-2] ospf 2 router-id 172.16.100.7
[FW-2-ospf-2] area 0
[FW-2-ospf-2-area-0.0.0.0] network 172.16.9.0 0.0.0.255
[FW-2-ospf-2-area-0.0.0.0] network 172.16.100.2 0.0.0.0
[FW-2-ospf-2-area-0.0.0.0] quit
[FW-2-ospf-2] quit
[FW-2] ospf 3 router-id 172.16.100.8
[FW-2-ospf-3] area 0
[FW-2-ospf-3-area-0.0.0.0] network 172.16.3.0 0.0.0.255
[FW-2-ospf-3-area-0.0.0.0] network 172.16.100.3 0.0.0.0
[FW-2-ospf-3-area-0.0.0.0] quit
[FW-2-ospf-3] quit
[FW-2] ospf 4 router-id 172.16.100.9
[FW-2-ospf-4] area 0
[FW-2-ospf-4-area-0.0.0.0] network 172.16.4.0 0.0.0.255
[FW-2-ospf-4-area-0.0.0.0] network 172.16.100.4 0.0.0.0
[FW-2-ospf-4-area-0.0.0.0] quit
[FW-2-ospf-4] quit
```

4. Configure HSB on FW-1 and FW-2.

- Configure HSB on FW-1.

```
[FW-1] hrp track interface GigabitEthernet 1/0/1
[FW-1] hrp track interface GigabitEthernet 1/0/2
[FW-1] hrp track interface GigabitEthernet 1/0/3
[FW-1] hrp track interface GigabitEthernet 1/0/4
[FW-1] hrp adjust ospf-cost enable
[FW-1] hrp interface Eth-Trunk 1 remote 172.16.10.1
[FW-1] hrp enable
[FW-1] hrp mirror session enable
```

- Configure HSB on FW-2.

```
[FW-2] hrp track interface GigabitEthernet 1/0/1
[FW-2] hrp track interface GigabitEthernet 1/0/2
[FW-2] hrp track interface GigabitEthernet 1/0/3
[FW-2] hrp track interface GigabitEthernet 1/0/4
[FW-2] hrp adjust ospf-cost enable
[FW-2] hrp interface Eth-Trunk 1 remote 172.16.5.1
[FW-2] hrp enable
[FW-2] hrp mirror session enable
```

5. Configure security policies and intrusion prevention.

```
HRP_M[FW-1] policy interzone trust untrust outbound
HRP_M[FW-1-policy-interzone-trust-untrust-outbound] policy 1
HRP_M[FW-1-policy-interzone-trust-untrust-outbound-1] policy source 10.1.2.0 mask 24
HRP_M[FW-1-policy-interzone-trust-untrust-outbound-1] policy source 10.1.3.0 mask 24
HRP_M[FW-1-policy-interzone-trust-untrust-outbound-1] policy source 10.1.4.0 mask 24
HRP_M[FW-1-policy-interzone-trust-untrust-outbound-1] policy source 10.1.5.0 mask 24
HRP_M[FW-1-policy-interzone-trust-untrust-outbound-1] action permit
```

```
HRP_M[FW-1-policy-interzone-trust-untrust-outbound-1] profile ips default
HRP_M[FW-1-policy-interzone-trust-untrust-outbound-1] quit
HRP_M[FW-1-policy-interzone-trust-untrust-outbound] quit
HRP_M[FW-1] policy interzone trust untrust inbound
HRP_M[FW-1-policy-interzone-trust-untrust-inbound] policy 1
HRP_M[FW-1-policy-interzone-trust-untrust-inbound-1] policy destination 10.1.2.0 mask 24
HRP_M[FW-1-policy-interzone-trust-untrust-inbound-1] policy destination 10.1.3.0 mask 24
HRP_M[FW-1-policy-interzone-trust-untrust-inbound-1] policy destination 10.1.4.0 mask 24
HRP_M[FW-1-policy-interzone-trust-untrust-inbound-1] policy destination 10.1.5.0 mask 24
HRP_M[FW-1-policy-interzone-trust-untrust-inbound-1] policy service service-set ftp http
HRP_M[FW-1-policy-interzone-trust-untrust-inbound-1] action permit
HRP_M[FW-1-policy-interzone-trust-untrust-inbound-1] profile ips default
HRP_M[FW-1-policy-interzone-trust-untrust-inbound-1] quit
HRP_M[FW-1-policy-interzone-trust-untrust-inbound] quit
HRP_M[FW-1] ips enable
```

6. Configure attack defense.

NOTE

The attack defense threshold is used for reference. Set this value according to actual network traffic.

```
HRP_M[FW-1] firewall defend syn-flood enable
HRP_M[FW-1] firewall defend syn-flood enable
HRP_M[FW-1] firewall defend syn-flood zone untrust max-rate 20000
HRP_M[FW-1] firewall defend udp-flood enable
HRP_M[FW-1] firewall defend udp-flood zone untrust max-rate 1500
HRP_M[FW-1] firewall defend icmp-flood enable
HRP_M[FW-1] firewall defend icmp-flood zone untrust max-rate 20000
HRP_M[FW-1] firewall blacklist enable
HRP_M[FW-1] firewall defend ip-sweep enable
HRP_M[FW-1] firewall defend ip-sweep max-rate 4000
HRP_M[FW-1] firewall defend port-scan enable
HRP_M[FW-1] firewall defend port-scan max-rate 4000
HRP_M[FW-1] firewall defend ip-fragment enable
HRP_M[FW-1] firewall defend ip-spoofing enable
```

Step 7 Configure Policy-Based Routing (PBR) so that traffic passing through SwitchI and SwitchJ are redirected to the firewall for filtering.

SwitchI is used as an example. The configuration of SwitchJ is similar to the configuration of SwitchI.

```
[~SwitchI] acl 3001
[*SwitchI-acl4-advance-3001] rule 5 permit ip source 10.1.2.0 24
[*SwitchI-acl4-advance-3001] rule 10 permit ip source 10.1.3.0 24
[*SwitchI-acl4-advance-3001] rule 15 permit ip source 10.1.4.0 24
[*SwitchI-acl4-advance-3001] rule 20 permit ip source 10.1.5.0 24
[*SwitchI-acl4-advance-3001] commit
[~SwitchI-acl4-advance-3001] quit
[~SwitchI] traffic classifier c1
[*SwitchI-classifier-c1] if-match acl 3001
[*SwitchI-classifier-c1] quit
[*SwitchI] commit
[~SwitchI] traffic behavior b1
[*SwitchI-behavior-b1] redirect load-balance nexthop 172.16.100.1 172.16.100.3
[*SwitchI-behavior-b1] quit
[*SwitchI] commit
[~SwitchI] traffic policy p1
[*SwitchI-trafficpolicy-p1] classifier c1 behavior b1
[*SwitchI-trafficpolicy-p1] quit
[*SwitchI] commit
[~SwitchI] interface 10ge 1/0/1
[~SwitchI-10GE1/0/1] traffic-policy p1 inbound
[*SwitchI-10GE1/0/1] quit
[*SwitchI] commit
[~SwitchI] interface 10ge 1/0/2
[~SwitchI-10GE1/0/2] traffic-policy p1 inbound
[*SwitchI-10GE1/0/2] quit
```



```
[*SwitchI] commit
[~SwitchI] interface 10ge 1/0/3
[~SwitchI-10GE1/0/3] traffic-policy p1 inbound
[*SwitchI-10GE1/0/3] quit
[*SwitchI] commit
[~SwitchI] interface 10ge 1/0/4
[~SwitchI-10GE1/0/4] traffic-policy p1 inbound
[*SwitchI-10GE1/0/4] quit
[*SwitchI] commit
[~SwitchI] interface 10ge 1/0/14
[~SwitchI-10GE1/0/14] traffic-policy p1 inbound
[*SwitchI-10GE1/0/14] quit
[*SwitchI] commit
[~SwitchI] acl 3003
[*SwitchI-acl4-advance-3003] rule 5 permit ip destination 10.1.2.0 24
[*SwitchI-acl4-advance-3003] rule 10 permit ip destination 10.1.3.0 24
[*SwitchI-acl4-advance-3003] rule 15 permit ip destination 10.1.4.0 24
[*SwitchI-acl4-advance-3003] rule 10 permit ip destination 10.1.5.0 24
[*SwitchI-acl4-advance-3003] commit
[~SwitchI-acl4-advance-3003] quit
[~SwitchI] traffic classifier c3
[*SwitchI-classifier-c3] if-match acl 3003
[*SwitchI-classifier-c3] quit
[*SwitchI] commit
[~SwitchI] traffic behavior b3
[*SwitchI-behavior-b3] redirect load-balance nexthop 172.16.100.2 172.16.100.4
[*SwitchI-behavior-b3] quit
[*SwitchI] commit
[~SwitchI] traffic policy p2
[*SwitchI-trafficpolicy-p2] classifier c3 behavior b3
[*SwitchI-trafficpolicy-p2] quit
[*SwitchI] commit
[~SwitchI] interface 10ge 1/0/5
[~SwitchI-10GE1/0/5] traffic-policy p2 inbound
[*SwitchI-10GE1/0/5] quit
[*SwitchI] commit
```

----End

Verifying the Configuration

1. # After the configuration is complete, run the **display vrrp** command on SwitchA. You can see that SwitchA is the master in VRRP group 1 and the backup in VRRP group 2.

```
<SwitchA> display vrrp verbose
Vlanif2 | Virtual Router 1
  State : Master
  Virtual IP : 10.1.2.100
  Master IP : 10.1.2.102
  PriorityRun : 120
  PriorityConfig : 120
  MasterPriority : 120
  Preempt : YES   Delay Time : 20 s   Remain : --
  TimerRun : 1 s
  TimerConfig : 1 s
  Auth Type : NONE
  Virtual MAC : 0000-5e00-0101
  Check TTL : YES
  Config Type : normal-vrrp
  Create Time : 2013-05-11 11:39:18
  Last Change Time : 2013-05-26 11:38:58

Vlanif3 | Virtual Router 2
  State : Backup
  Virtual IP : 10.1.3.100
  Master IP : 10.1.3.103
  PriorityRun : 100
  PriorityConfig : 100
```

```
MasterPriority : 120
Preempt : YES   Delay Time : 0 s   Remain : --
TimerRun : 1 s
TimerConfig : 1 s
Auth type : NONE
Virtual MAC : 0000-5e00-0102
Check TTL : YES
Config Type : normal-vrrp
Create Time : 2013-05-11 11:40:18
Last Change Time : 2013-05-26 11:48:58
```

2. Run the **display vrrp** command on SwitchB. You can see that SwitchB is the backup in VRRP group 1 and the master in VRRP group 2.

```
<SwitchB> display vrrp verbose
Vlanif2 | Virtual Router 1
  State : Backup
  Virtual IP : 10.1.2.100
  Master IP : 10.1.2.102
  PriorityRun : 100
  PriorityConfig : 100
  MasterPriority : 120
  Preempt : YES   Delay Time : 0 s   Remain : --
  TimerRun : 1 s
  TimerConfig : 1 s
  Auth Type : NONE
  Virtual MAC : 0000-5e00-0101
  Check TTL : YES
  Config Type : normal-vrrp
  Create Time : 2012-05-11 11:39:18
  Last Change Time : 2012-05-26 11:38:58

Vlanif3 | Virtual Router 2
  State : Master
  Virtual IP : 10.1.3.100
  Master IP : 10.1.3.103
  PriorityRun : 120
  PriorityConfig : 120
  MasterPriority : 120
  Preempt : YES   Delay Time : 20 s   Remain : --
  TimerRun : 1 s
  TimerConfig : 1 s
  Auth type : NONE
  Virtual MAC : 0000-5e00-0102
  Check TTL : YES
  Config Type : normal-vrrp
  Create Time : 2012-05-11 11:40:18
  Last Change Time : 2012-05-26 11:48:58
```

Configuration Files

- Configuration file of SwitchA

```
#
sysname SwitchA
#
vlan batch 2 to 3 6 to 7
#
stp instance 1 root primary
stp instance 2 root secondary
stp pathcost-standard legacy
#
stp region-configuration
region-name RG1
instance 1 vlan 2
instance 2 vlan 3
#
interface Vlanif2
ip address 10.1.2.102 255.255.255.0
vrrp vrid 1 virtual-ip 10.1.2.100
```

```
vrrp vrid 1 priority 120
vrrp vrid 1 preempt timer delay 20
#
interface Vlanif3
ip address 10.1.3.102 255.255.255.0
vrrp vrid 2 virtual-ip 10.1.3.100
#
interface Vlanif6
ip address 10.1.6.102 255.255.255.0
#
interface Vlanif7
ip address 10.1.7.102 255.255.255.0
#
interface 10GE1/0/1
description TO-CE6800-SWITCHC
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2
stp root-protection
#
interface 10GE1/0/2
description TO-CE6800-SWITCHD
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 3
#
interface 10GE1/0/3
description TO-CE12800-SWITCHB
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2 to 3
#
interface 10GE1/0/4
description TO-CE12800-SWITCHI
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 6
#
interface 10GE1/0/5
description TO-CE12800-SWITCHJ
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 7
#
ospf 1
area 0.0.0.0
network 10.1.2.0 0.0.0.255
network 10.1.3.0 0.0.0.255
network 10.1.6.0 0.0.0.255
network 10.1.7.0 0.0.0.255
#
return
```

- Configuration file of SwitchB

```
#
sysname SwitchB
#
vlan batch 2 to 3 6 to 7
#
stp instance 1 root secondary
stp instance 2 root primary
stp pathcost-standard legacy
#
stp region-configuration
region-name RG1
instance 1 vlan 2
instance 2 vlan 3
#
interface Vlanif2
ip address 10.1.2.103 255.255.255.0
```

```
vrrp vrid 1 virtual-ip 10.1.2.100
#
interface Vlanif3
ip address 10.1.3.103 255.255.255.0
vrrp vrid 2 virtual-ip 10.1.3.100
vrrp vrid 2 priority 120
vrrp vrid 2 preempt timer delay 20
#
interface Vlanif6
ip address 10.1.6.103 255.255.255.0
#
interface Vlanif7
ip address 10.1.7.103 255.255.255.0
#
interface 10GE1/0/1
description TO-CE6800-SWITCHD
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 3
stp root-protection
#
interface 10GE1/0/2
description TO-CE6800-SWITCHC
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2
#
interface 10GE1/0/3
description TO-CE12800-SWITCHA
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2 to 3
#
interface 10GE1/0/4
description TO-CE12800-SWITCHI
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 6
#
interface 10GE1/0/5
description TO-CE12800-SWITCHJ
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 7
#
ospf 1
area 0.0.0.0
network 10.1.2.0 0.0.0.255
network 10.1.3.0 0.0.0.255
network 10.1.6.0 0.0.0.255
network 10.1.7.0 0.0.0.255
#
return
```

- Configuration file of SwitchC

```
#
sysname SwitchC
#
vlan batch 2
#
stp pathcost-standard legacy
#
stp region-configuration
region-name RG1
instance 1 vlan 2
instance 2 vlan 3
#
interface 10GE1/0/1
description TO-CE12800-SWITCHA
port link-type trunk
```

```
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2
#
interface 10GE1/0/2
description TO-CE12800-SWITCHB
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2
stp instance 1 cost 20000
#
interface 10GE1/0/3
description TO-HOSTA
port default vlan 2
stp disable
#
return
```

- Configuration file of SwitchD

```
#
sysname SwitchD
#
vlan batch 3
#
stp pathcost-standard legacy
#
stp region-configuration
region-name RG1
instance 1 vlan 2
instance 2 vlan 3
#
interface 10GE1/0/1
description TO-CE12800-SWITCHB
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 3
#
interface 10GE1/0/2
description TO-CE12800-SWITCHA
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 3
stp instance 2 cost 20000
#
interface 10GE1/0/3
description TO-HOSTB
port default vlan 3
stp disable
#
return
```

- Configuration file of SwitchI

```
#
sysname SwitchI
#
vlan batch 6 8 to 13
#
acl number 3001
rule 5 permit ip source 10.1.2.0 0.0.0.255
rule 10 permit ip source 10.1.3.0 0.0.0.255
rule 15 permit ip source 10.1.4.0 0.0.0.255
rule 20 permit ip source 10.1.5.0 0.0.0.255
#
acl number 3003
rule 5 permit ip destination 10.1.2.0 0.0.0.255
rule 10 permit ip destination 10.1.3.0 0.0.0.255
rule 15 permit ip destination 10.1.4.0 0.0.0.255
rule 20 permit ip destination 10.1.5.0 0.0.0.255
#
traffic classifier c1 type or
if-match acl 3001
```

```
#
traffic classifier c3 type or
if-match acl 3003
#
traffic behavior b1
redirect load-balance nexthop 172.16.100.1 172.16.100.3
#
traffic behavior b3
redirect load-balance nexthop 172.16.100.2 172.16.100.4
#
traffic policy p1
classifier c1 behavior b1 precedence 5
#
traffic policy p2
classifier c3 behavior b3 precedence 5
#
interface Vlanif6
ip address 10.1.6.104 255.255.255.0
#
interface Vlanif8
ip address 10.1.8.104 255.255.255.0
#
interface Vlanif9
ip address 172.16.1.2 255.255.255.0
#
interface Vlanif10
ip address 172.16.2.2 255.255.255.0
#
interface Vlanif11
ip address 172.16.3.2 255.255.255.0
#
interface Vlanif12
ip address 172.16.4.2 255.255.255.0
#
interface Vlanif13
ip address 10.1.13.102 255.255.255.0
#
interface 10GE1/0/1
description TO-CE12800-SWITCHA
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 6
traffic-policy p1 inbound
#
interface 10GE1/0/2
description TO-CE12800-SWITCHB
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 6
traffic-policy p1 inbound
#
interface 10GE1/0/3
description TO-CE12800-SWITCHE
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 6
traffic-policy p1 inbound
#
interface 10GE1/0/4
description TO-CE12800-SWITCHF
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 6
traffic-policy p1 inbound
#
interface 10GE1/0/5
description TO-ROUTERA
port link-type trunk
undo port trunk allow-pass vlan 1
```

```
port trunk allow-pass vlan 8
traffic-policy p2 inbound
#
interface 10GE1/0/6
description TO-FW-1
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 9
#
interface 10GE1/0/7
description TO-FW-1
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 10
#
interface 10GE1/0/8
description TO-FW-2
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
#
interface 10GE1/0/9
description TO-FW-2
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 12
#
interface 10GE1/0/14
description TO-CE12800-SWITCHJ
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 13
traffic-policy p1 inbound
#
ospf 1
area 0.0.0.0
network 10.1.6.0 0.0.0.255
network 10.1.8.0 0.0.0.255
network 10.1.13.0 0.0.0.255
network 172.16.1.0 0.0.0.255
network 172.16.2.0 0.0.0.255
network 172.16.3.0 0.0.0.255
network 172.16.4.0 0.0.0.255
#
return
```

- Configuration file of SwitchJ

```
#
sysname SwitchJ
#
vlan batch 7 to 13
#
acl number 3001
rule 5 permit ip source 10.1.2.0 0.0.0.255
rule 10 permit ip source 10.1.3.0 0.0.0.255
rule 15 permit ip source 10.1.4.0 0.0.0.255
rule 20 permit ip source 10.1.5.0 0.0.0.255
#
acl number 3003
rule 5 permit ip destination 10.1.2.0 0.0.0.255
rule 10 permit ip destination 10.1.3.0 0.0.0.255
rule 15 permit ip destination 10.1.4.0 0.0.0.255
rule 20 permit ip destination 10.1.5.0 0.0.0.255
#
traffic classifier c1 type or
if-match acl 3001
#
traffic classifier c3 type or
if-match acl 3003
#
```

```
traffic behavior b1
  redirect load-balance nexthop 172.16.100.1 172.16.100.3
#
traffic behavior b3
  redirect load-balance nexthop 172.16.100.2 172.16.100.4
#
traffic policy p1
  classifier c1 behavior b1 precedence 5
#
traffic policy p2
  classifier c3 behavior b3 precedence 5
#
interface Vlanif7
  ip address 10.1.7.105 255.255.255.0
#
interface Vlanif8
  ip address 10.1.8.105 255.255.255.0
#
interface Vlanif9
  ip address 172.16.6.2 255.255.255.0
#
interface Vlanif10
  ip address 172.16.7.2 255.255.255.0
#
interface Vlanif11
  ip address 172.16.8.2 255.255.255.0
#
interface Vlanif12
  ip address 172.16.9.2 255.255.255.0
#
interface Vlanif13
  ip address 10.1.13.103 255.255.255.0
#
interface 10GE1/0/1
  description TO-CE12800-SWITCHA
  port link-type trunk
  undo port trunk allow-pass vlan 1
  port trunk allow-pass vlan 7
  traffic-policy p1 inbound
#
interface 10GE1/0/2
  description TO-CE12800-SWITCHB
  port link-type trunk
  undo port trunk allow-pass vlan 1
  port trunk allow-pass vlan 7
  traffic-policy p1 inbound
#
interface 10GE1/0/3
  description TO-CE12800-SWITCHC
  port link-type trunk
  undo port trunk allow-pass vlan 1
  port trunk allow-pass vlan 7
  traffic-policy p1 inbound
#
interface 10GE1/0/4
  description TO-CE12800-SWITCHD
  port link-type trunk
  undo port trunk allow-pass vlan 1
  port trunk allow-pass vlan 7
  traffic-policy p1 inbound
#
interface 10GE1/0/5
  description TO-ROUTERB
  port link-type trunk
  undo port trunk allow-pass vlan 1
  port trunk allow-pass vlan 8
  traffic-policy p2 inbound
#
interface 10GE1/0/6
```



```
description TO-FW-1
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 9
#
interface 10GE1/0/7
description TO-FW-1
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 10
#
interface 10GE1/0/8
description TO-FW-2
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
#
interface 10GE1/0/9
description TO-FW-2
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 12
#
interface 10GE1/0/14
description TO-CE12800-SWITCH1
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 13
traffic-policy p1 inbound
#
ospf 1
area 0.0.0.0
network 10.1.7.0 0.0.0.255
network 10.1.8.0 0.0.0.255
network 10.1.11.0 0.0.0.255
network 172.16.6.0 0.0.0.255
network 172.16.7.0 0.0.0.255
network 172.16.8.0 0.0.0.255
network 172.16.9.0 0.0.0.255
#
return
```

- Configuration file of FW-1

```
#
sysname FW-1
#
firewall packet-filter default permit interzone local dmz direction inbound
firewall packet-filter default permit interzone local dmz direction outbound
#
firewall defend port-scan enable
firewall defend ip-sweep enable
firewall defend ip-fragment enable
firewall defend icmp-flood enable
firewall defend udp-flood enable
firewall defend syn-flood enable
firewall defend ip-spoofing enable
firewall defend action discard
firewall defend icmp-flood zone untrust max-rate 20000
firewall defend udp-flood zone untrust max-rate 1500
firewall defend syn-flood zone untrust max-rate 20000
#
hrp enable
hrp adjust ospf-cost enable
hrp interface Eth-Trunk1 remote 172.16.10.1
hrp mirror session enable
hrp track interface GigabitEthernet 1/0/1
hrp track interface GigabitEthernet 1/0/2
hrp track interface GigabitEthernet 1/0/3
hrp track interface GigabitEthernet 1/0/4
#
```

```
ips enable
#
interface Eth-Trunk1
 ip address 172.16.5.1 255.255.255.0
#
interface GigabitEthernet1/0/1
 description TO-CE12800-SwitchI-Upstream
 ip address 172.16.1.1 255.255.255.0
 undo shutdown
#
interface GigabitEthernet1/0/2
 description TO-CE12800-SwitchI-Downstream
 ip address 172.16.2.1 255.255.255.0
 undo shutdown
#
interface GigabitEthernet1/0/3
 description TO-CE12800-SwitchJ-Upstream
 ip address 172.16.6.1 255.255.255.0
 undo shutdown
#
interface GigabitEthernet1/0/4
 description TO-CE12800-SwitchJ-Downstream
 ip address 172.16.7.1 255.255.255.0
 undo shutdown
#
interface GigabitEthernet2/0/0
 description TO-FW-2-HRP
 undo shutdown
 eth-trunk 1
#
interface GigabitEthernet2/0/1
 description TO-FW-2-HRP
 undo shutdown
 eth-trunk 1
#
interface GigabitEthernet2/0/2
 description TO-FW-2-HRP
 undo shutdown
 eth-trunk 1
#
interface GigabitEthernet2/0/3
 description TO-FW-2-HRP
 undo shutdown
 eth-trunk 1
#
interface LoopBack 1
 ip address 172.16.100.1 32
#
interface LoopBack 2
 ip address 172.16.100.2 32
#
interface LoopBack 3
 ip address 172.16.100.3 32
#
interface LoopBack 4
 ip address 172.16.100.4 32
#
profile type ips name default
 signature-set name default
  os both
  target both
  severity low medium high
  protocol all
  category all
#
firewall zone trust
 set priority 85
 add interface GigabitEthernet 1/0/1
 add interface GigabitEthernet 1/0/3
```

```
#
firewall zone untrust
set priority 5
add interface GigabitEthernet 1/0/2
add interface GigabitEthernet 1/0/4
#
firewall zone dmz
set priority 50
add interface Eth-Trunk1
#
firewall interzone trust untrust
detect ftp
#
policy interzone trust untrust inbound
policy 1
action permit
profile ips default
policy service service-set ftp
policy service service-set http
policy destination 10.1.2.0 mask 24
policy destination 10.1.3.0 mask 24
policy destination 10.1.4.0 mask 24
policy destination 10.1.5.0 mask 24
#
policy interzone trust untrust outbound
policy 1
action permit
profile ips default
policy source 10.1.2.0 mask 24
policy source 10.1.3.0 mask 24
policy source 10.1.4.0 mask 24
policy source 10.1.5.0 mask 24
#
ospf 1 router-id 172.16.100.1
area 0.0.0.0
network 172.16.1.0 0.0.0.255
network 172.16.100.1 0.0.0.0
#
ospf 2 router-id 172.16.100.2
area 0.0.0.0
network 172.16.2.0 0.0.0.255
network 172.16.100.2 0.0.0.0
#
ospf 3 router-id 172.16.100.3
area 0.0.0.0
network 172.16.6.0 0.0.0.255
network 172.16.100.3 0.0.0.0
#
ospf 4 router-id 172.16.100.4
area 0.0.0.0
network 172.16.7.0 0.0.0.255
network 172.16.100.4 0.0.0.0
#
return
```

- Configuration file of FW-2

```
#
sysname FW-2
#
firewall packet-filter default permit interzone local dmz direction inbound
firewall packet-filter default permit interzone local dmz direction outbound
#
firewall defend port-scan enable
firewall defend ip-sweep enable
firewall defend ip-fragment enable
firewall defend icmp-flood enable
firewall defend udp-flood enable
firewall defend syn-flood enable
firewall defend ip-spoofing enable
firewall defend action discard
```

```

firewall defend icmp-flood zone untrust max-rate 20000
firewall defend udp-flood zone untrust max-rate 1500
firewall defend syn-flood zone untrust max-rate 20000
#
hrp enable
hrp adjust ospf-cost enable
hrp interface Eth-Trunk1 remote 172.16.5.1
hrp mirror session enable
hrp track interface GigabitEthernet 1/0/1
hrp track interface GigabitEthernet 1/0/2
hrp track interface GigabitEthernet 1/0/3
hrp track interface GigabitEthernet 1/0/4
#
ips enable
#
interface Eth-Trunk1
ip address 172.16.10.1 255.255.255.0
#
interface GigabitEthernet1/0/1
description TO-CE12800-SwitchI-Upstream
ip address 172.16.8.1 255.255.255.0
undo shutdown
#
interface GigabitEthernet1/0/2
description TO-CE12800-SwitchI-Downstream
ip address 172.16.9.1 255.255.255.0
undo shutdown
#
interface GigabitEthernet1/0/3
description TO-CE12800-SwitchJ-Upstream
ip address 172.16.3.1 255.255.255.0
undo shutdown
#
interface GigabitEthernet1/0/4
description TO-CE12800-SwitchJ-Downstream
ip address 172.16.4.1 255.255.255.0
undo shutdown
#
interface GigabitEthernet2/0/0
description TO-FW-1-HRP
undo shutdown
eth-trunk 1
#
interface GigabitEthernet2/0/1
description TO-FW-1-HRP
undo shutdown
eth-trunk 1
#
interface GigabitEthernet2/0/2
description TO-FW-1-HRP
undo shutdown
eth-trunk 1
#
interface GigabitEthernet2/0/3
description TO-FW-1-HRP
undo shutdown
eth-trunk 1
#
interface LoopBack 1
ip address 172.16.100.1 32
#
interface LoopBack 2
ip address 172.16.100.2 32
#
interface LoopBack 3
ip address 172.16.100.3 32
#
interface LoopBack 4
ip address 172.16.100.4 32

```

```
#
profile type ips name default
signature-set name default
os both
target both
severity low medium high
protocol all
category all
#
firewall zone trust
set priority 85
add interface GigabitEthernet 1/0/1
add interface GigabitEthernet 1/0/3
#
firewall zone untrust
set priority 5
add interface GigabitEthernet 1/0/2
add interface GigabitEthernet 1/0/4
#
firewall zone dmz
set priority 50
add interface Eth-Trunk1
#
firewall interzone trust untrust
detect ftp
#
policy interzone trust untrust inbound
policy 1
action permit
profile ips default
policy service service-set ftp
policy service service-set http
policy destination 10.1.2.0 mask 24
policy destination 10.1.3.0 mask 24
policy destination 10.1.4.0 mask 24
policy destination 10.1.5.0 mask 24
#
policy interzone trust untrust outbound
policy 1
action permit
profile ips default
policy source 10.1.2.0 mask 24
policy source 10.1.3.0 mask 24
policy source 10.1.4.0 mask 24
policy source 10.1.5.0 mask 24
#
ospf 1 router-id 172.16.100.6
area 0.0.0.0
network 172.16.8.0 0.0.0.255
network 172.16.100.1 0.0.0.0
#
ospf 2 router-id 172.16.100.7
area 0.0.0.0
network 172.16.9.0 0.0.0.255
network 172.16.100.2 0.0.0.0
#
ospf 3 router-id 172.16.100.8
area 0.0.0.0
network 172.16.3.0 0.0.0.255
network 172.16.100.3 0.0.0.0
#
ospf 4 router-id 172.16.100.9
area 0.0.0.0
network 172.16.4.0 0.0.0.255
network 172.16.100.4 0.0.0.0
#
return
```

1.2 Deploying VRRP on a Data Center Network with 2-Layer Architecture

Applicable Products and Versions

- CloudEngine series switches running V100R001C00 or later versions
- For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

Access switches are dual-homed to core devices on a data center network. The requirements are as follows:

- To ensure service reliability, deploy link redundancy. When one uplink is disconnected, traffic can be switched to another uplink.
- Loops caused by link redundancy on the ring network are eliminated.

Figure 1-2 Networking for deploying VRRP on a data center network with 2-layer architecture

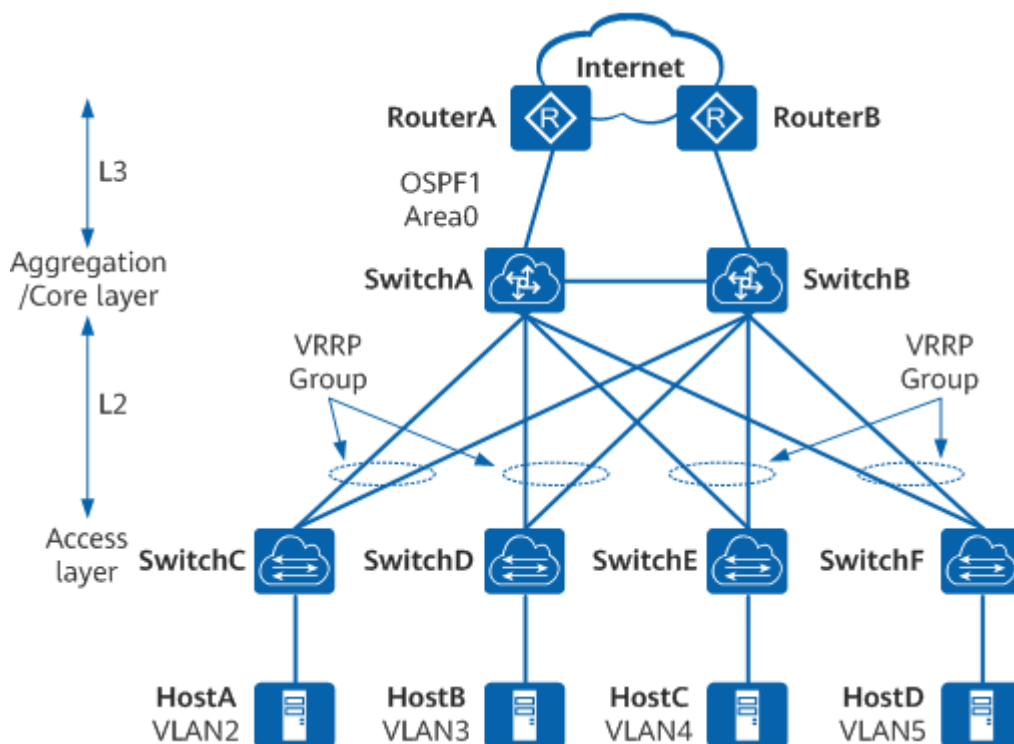


Table 1-2 Data preparation

Device	VLAN and IP Address	Interface	Description
SwitchA	VLAN: 2 IP Address: 10.1.2.102/24 Virtual IP Address: 10.1.2.100	10GE1/0/1	TO-CE6800-SWITCHC
		10GE1/0/5	TO-CE12800-SWITCHB
	VLAN: 3 IP Address: 10.1.3.102/24 Virtual IP Address: 10.1.3.100	10GE1/0/2	TO-CE6800-SWITCHD
		10GE1/0/5	TO-CE12800-SWITCHB
	VLAN: 4 IP Address: 10.1.4.102/24 Virtual IP Address: 10.1.4.100	10GE1/0/3	TO-CE6800-SWITCHE
		10GE1/0/5	TO-CE12800-SWITCHB
	VLAN: 5 IP Address: 10.1.5.102/24 Virtual IP Address: 10.1.5.100	10GE1/0/4	TO-CE6800-SWITCHF
		10GE1/0/5	TO-CE12800-SWITCHB
	VLAN: 6 IP Address: 10.1.6.103/24	10GE1/0/6	TO-ROUTERA
SwitchB	VLAN: 2 IP Address: 10.1.2.103/24 Virtual IP Address: 10.1.2.100	10GE1/0/4	TO-CE6800-SWITCHC
		10GE1/0/5	TO-CE12800-SWITCHA
	VLAN: 3 IP Address: 10.1.3.103/24 Virtual IP Address: 10.1.3.100	10GE1/0/3	TO-CE6800-SWITCHD
		10GE1/0/5	TO-CE12800-SWITCHA
	VLAN: 4 IP Address: 10.1.4.103/24 Virtual IP Address: 10.1.4.100	10GE1/0/2	TO-CE6800-SWITCHE
		10GE1/0/5	TO-CE12800-SWITCHA

Device	VLAN and IP Address	Interface	Description
	VLAN: 5 IP Address: 10.1.5.103/24 Virtual IP Address: 10.1.5.100	10GE1/0/1	TO-CE6800-SWITCHF
		10GE1/0/5	TO-CE12800-SWITCHA
	VLAN: 7 IP Address: 10.1.7.103/24	10GE1/0/6	TO-ROUTERB
SwitchC	VLAN: 2	10GE1/0/1	TO-CE12800-SWITCHA
		10GE1/0/2	TO-CE12800-SWITCHB
		10GE1/0/3	TO-HOSTA
SwitchD	VLAN: 3	10GE1/0/1	TO-CE12800-SWITCHB
		10GE1/0/2	TO-CE12800-SWITCHA
		10GE1/0/3	TO-HOSTB
SwitchE	VLAN: 4	10GE1/0/1	TO-CE12800-SWITCHA
		10GE1/0/2	TO-CE12800-SWITCHB
		10GE1/0/3	TO-HOSTC
SwitchF	VLAN: 5	10GE1/0/1	TO-CE12800-SWITCHB
		10GE1/0/2	TO-CE12800-SWITCHA
		10GE1/0/3	TO-HOSTD

Requirement Analysis

- Deploy VRRP between SwitchA and SwitchB to implement link redundancy at the aggregation layer.
- Deploy MSTP between access and core layers to eliminate loops.

Procedure

Step 1 Configure basic MSTP functions (SwitchA, SwitchB, SwitchC, and SwitchD are used as examples).

NOTE

Two devices belong to the same MST region when the following parameters on the two devices are the same:

- Name of the MST region
- Mapping between VLANs and MSTIs
- Revision level of the MST region

1. Configure SwitchA, SwitchB, SwitchC, and SwitchD in the MST region **RG1** and create MSTI 1 and MSTI 2.

Configure an MST region on SwitchA.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchA
[*HUAWEI] commit
[~SwitchA] stp region-configuration
[~SwitchA-mst-region] region-name RG1
[~SwitchA-mst-region] instance 1 vlan 2
[~SwitchA-mst-region] instance 2 vlan 3
[*SwitchA-mst-region] commit
[~SwitchA-mst-region] quit
```

Configure an MST region on SwitchB.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchB
[*HUAWEI] commit
[~SwitchB] stp region-configuration
[~SwitchB-mst-region] region-name RG1
[~SwitchB-mst-region] instance 1 vlan 2
[~SwitchB-mst-region] instance 2 vlan 3
[*SwitchB-mst-region] commit
[~SwitchB-mst-region] quit
```

Configure an MST region on SwitchC.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchC
[*HUAWEI] commit
[~SwitchC] stp region-configuration
[~SwitchC-mst-region] region-name RG1
[~SwitchC-mst-region] instance 1 vlan 2
[~SwitchC-mst-region] instance 2 vlan 3
[*SwitchC-mst-region] commit
[~SwitchC-mst-region] quit
```

Configure an MST region on SwitchD.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchD
[*HUAWEI] commit
[~SwitchD] stp region-configuration
[~SwitchD-mst-region] region-name RG1
[~SwitchD-mst-region] instance 1 vlan 2
[~SwitchD-mst-region] instance 2 vlan 3
[*SwitchD-mst-region] commit
[~SwitchD-mst-region] quit
```

2. Configure root bridges and secondary root bridges of MSTI 1 and MSTI 2 in the MST region **RG1**.

- Configure the root bridge and secondary root bridge of MSTI 1.

Configure SwitchA as the root bridge in MSTI 1.

```
[~SwitchA] stp instance 1 root primary
[*SwitchA] commit
```

Configure SwitchB as the secondary root bridge in MSTI 1.

```
[~SwitchB] stp instance 1 root secondary
[*SwitchB] commit
```

- Configure the root bridge and secondary root bridge of MSTI 2.

Configure SwitchB as the root bridge in MSTI 2.

```
[~SwitchB] stp instance 2 root primary  
[*SwitchB] commit
```

Configure SwitchA as the secondary root bridge in MSTI 2.

```
[~SwitchA] stp instance 2 root secondary  
[*SwitchA] commit
```

3. Set the path costs of the interfaces to be blocked in MSTI 1 and MSTI 2 to be greater than the default value.

NOTE

- The path cost range depends on the algorithm. Huawei proprietary algorithm is used as an example. Set the path costs of the interfaces to be blocked in MSTI 1 and MSTI 2 to 20000.
- Switching devices on the same network must use the same algorithm to calculate the path cost of interfaces.

Configure SwitchA to use Huawei proprietary algorithm to calculate the path cost.

```
[~SwitchA] stp pathcost-standard legacy  
[*SwitchA] commit
```

Configure SwitchB to use Huawei proprietary algorithm to calculate the path cost.

```
[~SwitchB] stp pathcost-standard legacy  
[*SwitchB] commit
```

Configure SwitchC to use Huawei proprietary algorithm to calculate the path cost and set the path cost of 10GE1/0/2 to 20000 in MSTI 1.

```
[~SwitchC] stp pathcost-standard legacy  
[*SwitchC] interface 10ge 1/0/2  
[*SwitchC-10GE1/0/2] description TO-CE12800-SWITCHB  
[*SwitchC-10GE1/0/2] stp instance 1 cost 20000  
[*SwitchC-10GE1/0/2] commit  
[~SwitchC-10GE1/0/2] quit
```

Configure SwitchD to use Huawei proprietary algorithm to calculate the path cost and set the path cost of 10GE1/0/2 to 20000 in MSTI 2.

```
[~SwitchD] stp pathcost-standard legacy  
[*SwitchD] interface 10ge 1/0/2  
[*SwitchD-10GE1/0/2] description TO-CE12800-SWITCHA  
[*SwitchD-10GE1/0/2] stp instance 2 cost 20000  
[*SwitchD-10GE1/0/2] commit  
[~SwitchD-10GE1/0/2] quit
```

4. Enable MSTP to eliminate loops.

NOTE

MSTP is enabled by default.

- Enable MSTP globally.
Enable MSTP on SwitchA.

```
[~SwitchA] stp enable  
[*SwitchA] commit
```

Enable MSTP on SwitchB.

```
[~SwitchB] stp enable  
[*SwitchB] commit
```

Enable MSTP on SwitchC.

```
[~SwitchC] stp enable
[*SwitchC] commit
```

Enable MSTP on SwitchD.

```
[~SwitchD] stp enable
[*SwitchD] commit
```

- Configure ports connected to hosts as edge ports.

Configure 10GE1/0/3 of SwitchC as an edge port.

```
[~SwitchC] interface 10ge 1/0/3
[~SwitchC-10GE1/0/3] description TO-HOSTA
[*SwitchC-10GE1/0/3] stp edged-port enable
[*SwitchC-10GE1/0/3] commit
[~SwitchC-10GE1/0/3] quit
```

Configure 10GE1/0/3 of SwitchD as an edge port.

```
[~SwitchD] interface 10ge 1/0/3
[~SwitchD-10GE1/0/3] description TO-HOSTB
[*SwitchD-10GE1/0/3] stp edged-port enable
[*SwitchD-10GE1/0/3] commit
[~SwitchD-10GE1/0/3] quit
```

Step 2 Enable protection functions on the designated port of each root bridge in each MSTI. Here, root protection is used (SwitchA, SwitchB, SwitchC, and SwitchD are used as examples).

Enable root protection on 10GE1/0/1 of SwitchA.

```
[~SwitchA] interface 10ge 1/0/1
[~SwitchA-10GE1/0/1] TO-CE6800-SWITCHC
[*SwitchA-10GE1/0/1] stp root-protection
[*SwitchA-10GE1/0/1] commit
[~SwitchA-10GE1/0/1] quit
```

Enable root protection on 10GE1/0/3 of SwitchB.

```
[~SwitchB] interface 10ge 1/0/3
[~SwitchB-10GE1/0/3] TO-CE6800-SWITCHD
[*SwitchB-10GE1/0/3] stp root-protection
[*SwitchB-10GE1/0/3] commit
[~SwitchB-10GE1/0/3] quit
```

Step 3 Configure Layer 2 forwarding on switches on the ring network (SwitchA, SwitchB, SwitchC, and SwitchD are used as examples).

- Create VLAN 2 and VLAN 3 on SwitchA, SwitchB, and SwitchC.

Create VLAN 2 and VLAN 3 on SwitchA.

```
[~SwitchA] vlan batch 2 to 3
```

Create VLAN 2 and VLAN 3 on SwitchB.

```
[~SwitchB] vlan batch 2 to 3
```

Create VLAN 2 on SwitchC.

```
[~SwitchC] vlan batch 2
```

Create VLAN 3 on SwitchD.

```
[~SwitchD] vlan batch 3
```

- Add interfaces connecting to the ring to VLANs.

Add 10GE1/0/1 on SwitchA to VLAN 2.

```
[~SwitchA] interface 10ge 1/0/1
[~SwitchA-10GE1/0/1] port link-type trunk
[*SwitchA-10GE1/0/1] port trunk allow-pass vlan 2
[*SwitchA-10GE1/0/1] undo port trunk allow-pass vlan 1
[*SwitchA-10GE1/0/1] commit
[~SwitchA-10GE1/0/1] quit
```

Add 10GE1/0/2 on SwitchA to VLAN 3.

```
[~SwitchA] interface 10ge 1/0/2
[~SwitchA-10GE1/0/2] description TO-CE6800-SWITCHD
[*SwitchA-10GE1/0/2] port link-type trunk
[*SwitchA-10GE1/0/2] port trunk allow-pass vlan 3
[*SwitchA-10GE1/0/2] undo port trunk allow-pass vlan 1
[*SwitchA-10GE1/0/2] commit
[~SwitchA-10GE1/0/2] quit
```

Add 10GE1/0/5 on SwitchA to VLAN 2 and VLAN 3.

```
[~SwitchA] interface 10ge 1/0/5
[~SwitchA-10GE1/0/5] description TO-CE12800-SWITCHB
[*SwitchA-10GE1/0/5] port link-type trunk
[*SwitchA-10GE1/0/5] port trunk allow-pass vlan 2 to 3
[*SwitchA-10GE1/0/5] undo port trunk allow-pass vlan 1
[*SwitchA-10GE1/0/5] commit
[~SwitchA-10GE1/0/5] quit
```

Add 10GE1/0/3 on SwitchB to VLAN 3.

```
[~SwitchB] interface 10ge 1/0/3
[~SwitchB-10GE1/0/3] port link-type trunk
[SwitchB-10GE1/0/3] port trunk allow-pass vlan 3
[*SwitchB-10GE1/0/3] undo port trunk allow-pass vlan 1
[*SwitchB-10GE1/0/3] commit
[~SwitchB-10GE1/0/3] quit
```

Add 10GE1/0/4 on SwitchB to VLAN 2.

```
[~SwitchB] interface 10ge 1/0/4
[~SwitchB-10GE1/0/4] description TO-CE6800-SWITCHC
[*SwitchB-10GE1/0/4] port link-type trunk
[*SwitchB-10GE1/0/4] port trunk allow-pass vlan 2
[*SwitchB-10GE1/0/4] undo port trunk allow-pass vlan 1
[*SwitchB-10GE1/0/4] commit
[~SwitchB-10GE1/0/4] quit
```

Add 10GE1/0/5 on SwitchB to VLAN 2 and VLAN 3.

```
[~SwitchB] interface 10ge 1/0/5
[~SwitchB-10GE1/0/5] description TO-CE12800-SWITCHB
[*SwitchB-10GE1/0/5] port link-type trunk
[*SwitchB-10GE1/0/5] port trunk allow-pass vlan 2 to 3
[*SwitchB-10GE1/0/5] undo port trunk allow-pass vlan 1
[*SwitchB-10GE1/0/5] commit
[~SwitchB-10GE1/0/5] quit
```

Add 10GE1/0/1 on SwitchC to VLAN 2.

```
[~SwitchC] interface 10ge 1/0/1
[~SwitchC-10GE1/0/1] description TO-CE12800-SWITCHA
[*SwitchC-10GE1/0/1] port trunk allow-pass vlan 2
[*SwitchC-10GE1/0/1] undo port trunk allow-pass vlan 1
[*SwitchC-10GE1/0/1] commit
[~SwitchC-10GE1/0/1] quit
```

Add 10GE1/0/2 on SwitchC to VLAN 2.

```
[~SwitchC] interface 10ge 1/0/2
[~SwitchC-10GE1/0/2] port link-type trunk
[*SwitchC-10GE1/0/2] port trunk allow-pass vlan 2
[*SwitchC-10GE1/0/2] undo port trunk allow-pass vlan 1
[*SwitchC-10GE1/0/2] commit
[~SwitchC-10GE1/0/2] quit
```

Add 10GE1/0/3 on SwitchC to VLAN 2.

```
[~SwitchC] interface 10ge 1/0/3
[~SwitchC-10GE1/0/3] port link-type access
[*SwitchC-10GE1/0/3] port default vlan 2
[*SwitchC-10GE1/0/3] commit
[~SwitchC-10GE1/0/3] quit
```

Add 10GE1/0/1 on SwitchD to VLAN 3.

```
[~SwitchD] interface 10ge 1/0/1
[~SwitchD-10GE1/0/1] description TO-CE12800-SWITCHB
[*SwitchD-10GE1/0/1] port link-type trunk
[*SwitchD-10GE1/0/1] port trunk allow-pass vlan 3
[*SwitchD-10GE1/0/1] undo port trunk allow-pass vlan 1
[*SwitchD-10GE1/0/1] commit
[~SwitchD-10GE1/0/1] quit
```

Add 10GE1/0/2 on SwitchD to VLAN 3.

```
[~SwitchD] interface 10ge 1/0/2
[~SwitchD-10GE1/0/2] port link-type trunk
[*SwitchD-10GE1/0/2] port trunk allow-pass vlan 3
[*SwitchD-10GE1/0/2] commit
[~SwitchD-10GE1/0/2] quit
```

Add 10GE1/0/3 on SwitchD to VLAN 3.

```
[~SwitchD] interface 10ge 1/0/3
[~SwitchD-10GE1/0/3] port link-type access
[*SwitchD-10GE1/0/3] port default vlan 3
[*SwitchD-10GE1/0/3] commit
[~SwitchD-10GE1/0/3] quit
```

Step 4 Configure devices to ensure network connectivity.

Assign an IP address to each interface. SwitchA is used as an example. The configurations of other devices are similar to the configuration of SwitchA.

```
[~SwitchA] vlan batch 6
[*SwitchA] interface 10ge 1/0/6
[*SwitchA-10GE1/0/6] description TO-ROUTERA
[*SwitchA-10GE1/0/6] port link-type trunk
[*SwitchA-10GE1/0/6] port trunk allow-pass vlan 6
[*SwitchA-10GE1/0/6] undo port trunk allow-pass vlan 1
[*SwitchA-10GE1/0/6] quit
[*SwitchA] interface vlanif 2
[*SwitchA-Vlanif2] ip address 10.1.2.102 24
[*SwitchA-Vlanif2] quit
[*SwitchA] interface vlanif 3
[*SwitchA-Vlanif3] ip address 10.1.3.102 24
[*SwitchA-Vlanif3] quit
[*SwitchA] interface vlanif 4
[*SwitchA-Vlanif4] ip address 10.1.4.102 24
[*SwitchA-Vlanif4] quit
[*SwitchA] interface vlanif 5
[*SwitchA-Vlanif5] ip address 10.1.5.102 24
[*SwitchA-Vlanif5] quit
[*SwitchA] interface vlanif 6
[*SwitchA-Vlanif6] ip address 10.1.6.102 24
[*SwitchA-Vlanif6] quit
[*SwitchA] commit
```

Configure OSPF between core devices and egress router. SwitchA is used as an example. The configurations of other devices are similar to the configuration of SwitchA.

```
[~SwitchA] ospf 1
[*SwitchA-ospf-1] area 0
[*SwitchA-ospf-1-area-0.0.0.0] network 10.1.2.0 0.0.0.255
[*SwitchA-ospf-1-area-0.0.0.0] network 10.1.3.0 0.0.0.255
[*SwitchA-ospf-1-area-0.0.0.0] network 10.1.4.0 0.0.0.255
[*SwitchA-ospf-1-area-0.0.0.0] network 10.1.5.0 0.0.0.255
[*SwitchA-ospf-1-area-0.0.0.0] network 10.1.6.0 0.0.0.255
[*SwitchA-ospf-1-area-0.0.0.0] quit
[*SwitchA-ospf-1] quit
[*SwitchA] commit
```

Step 5 Configure VRRP groups.

Configure VRRP group 1 on SwitchA and SwitchB, set the priority of SwitchA to 120 and the preemption delay to 20s, and set the default priority for SwitchB.

```
[~SwitchA] interface vlanif 2
[~SwitchA-Vlanif2] vrrp vrid 1 virtual-ip 10.1.2.100
[*SwitchA-Vlanif2] vrrp vrid 1 priority 120
[*SwitchA-Vlanif2] vrrp vrid 1 preempt-mode timer delay 20
[*SwitchA-Vlanif2] commit
[~SwitchA-Vlanif2] quit
[~SwitchB] interface vlanif 2
[~SwitchB-Vlanif2] vrrp vrid 1 virtual-ip 10.1.2.100
[*SwitchB-Vlanif2] commit
[~SwitchB-Vlanif2] quit
```

Configure VRRP group 2 on SwitchA and SwitchB, set the priority of SwitchB to 120 and the preemption delay to 20s, and set the default priority for SwitchA.

```
[~SwitchB] interface vlanif 3
[~SwitchB-Vlanif3] vrrp vrid 2 virtual-ip 10.1.3.100
[*SwitchB-Vlanif3] vrrp vrid 2 priority 120
[*SwitchB-Vlanif3] vrrp vrid 2 preempt-mode timer delay 20
[*SwitchB-Vlanif3] commit
[~SwitchB-Vlanif3] quit
[~SwitchA] interface vlanif 3
[~SwitchA-Vlanif3] vrrp vrid 2 virtual-ip 10.1.3.100
[*SwitchA-Vlanif3] commit
[~SwitchA-Vlanif3] quit
```

Configure VRRP group 3 on SwitchA and SwitchB, set the priority of SwitchA to 120 and the preemption delay to 20s, and set the default priority for SwitchB.

```
[~SwitchA] interface vlanif 4
[~SwitchA-Vlanif4] vrrp vrid 3 virtual-ip 10.1.4.100
[*SwitchA-Vlanif4] vrrp vrid 3 priority 120
[*SwitchA-Vlanif4] vrrp vrid 3 preempt-mode timer delay 20
[*SwitchA-Vlanif4] commit
[~SwitchA-Vlanif4] quit
[~SwitchB] interface vlanif 4
[~SwitchB-Vlanif4] vrrp vrid 3 virtual-ip 10.1.4.100
[*SwitchB-Vlanif4] commit
[~SwitchB-Vlanif4] quit
```

Configure VRRP group 4 on SwitchA and SwitchB, set the priority of SwitchB to 120 and the preemption delay to 20s, and set the default priority for SwitchA.

```
[~SwitchB] interface vlanif 5
[~SwitchB-Vlanif5] vrrp vrid 4 virtual-ip 10.1.5.100
[*SwitchB-Vlanif5] vrrp vrid 4 priority 120
[*SwitchB-Vlanif5] vrrp vrid 4 preempt-mode timer delay 20
[*SwitchB-Vlanif5] commit
[~SwitchB-Vlanif5] quit
[~SwitchA] interface vlanif 5
[~SwitchA-Vlanif5] vrrp vrid 4 virtual-ip 10.1.5.100
[*SwitchA-Vlanif5] commit
[~SwitchA-Vlanif5] quit
```

Set virtual IP address 10.1.2.100 of VRRP group 1 as the default gateway of HostA, virtual IP address 10.1.3.100 of VRRP group 2 as the default gateway of HostB, virtual IP address 10.1.4.100 of VRRP group 3 as the default gateway of HostC, and virtual IP address 10.1.5.100 of VRRP group 4 as the default gateway of HostD.

----End

Verifying the Configuration

1. # After the configuration is complete, run the **display vrrp** command on SwitchA. You can see that SwitchA is the master in VRRP group 1 and the backup in VRRP group 2.

```
<SwitchA> display vrrp verbose
Vlanif2 | Virtual Router 1
  State : Master
  Virtual IP : 10.1.2.100
  Master IP : 10.1.2.102
  PriorityRun : 120
  PriorityConfig : 120
  MasterPriority : 120
  Preempt : YES Delay Time : 20 s Remain : --
  TimerRun : 1 s
  TimerConfig : 1 s
  Auth Type : NONE
  Virtual MAC : 0000-5e00-0101
  Check TTL : YES
  Config Type : normal-vrrp
  Create Time : 2013-05-11 11:39:18
  Last Change Time : 2013-05-26 11:38:58

Vlanif3 | Virtual Router 2
  State : Backup
  Virtual IP : 10.1.3.100
  Master IP : 10.1.3.103
  PriorityRun : 100
  PriorityConfig : 100
  MasterPriority : 120
  Preempt : YES Delay Time : 0 s Remain : --
  TimerRun : 1 s
  TimerConfig : 1 s
  Auth type : NONE
  Virtual MAC : 0000-5e00-0102
  Check TTL : YES
  Config Type : normal-vrrp
  Create Time : 2013-05-11 11:40:18
  Last Change Time : 2013-05-26 11:48:58
```

2. Run the **display vrrp** command on SwitchB. You can see that SwitchB is the backup in VRRP group 1 and the master in VRRP group 2.

```
<SwitchB> display vrrp verbose
Vlanif2 | Virtual Router 1
  State : Backup
  Virtual IP : 10.1.2.100
  Master IP : 10.1.2.102
  PriorityRun : 100
  PriorityConfig : 100
  MasterPriority : 120
  Preempt : YES Delay Time : 0 s Remain : --
  TimerRun : 1 s
  TimerConfig : 1 s
  Auth Type : NONE
  Virtual MAC : 0000-5e00-0101
  Check TTL : YES
  Config Type : normal-vrrp
  Create Time : 2012-05-11 11:39:18
  Last Change Time : 2012-05-26 11:38:58

Vlanif3 | Virtual Router 2
  State : Master
  Virtual IP : 10.1.3.100
  Master IP : 10.1.3.103
  PriorityRun : 120
  PriorityConfig : 120
  MasterPriority : 120
  Preempt : YES Delay Time : 20 s Remain : --
```

```
TimerRun : 1 s
TimerConfig : 1 s
Auth type : NONE
Virtual MAC : 0000-5e00-0102
Check TTL : YES
Config Type : normal-vrrp
Create Time : 2012-05-11 11:40:18
Last Change Time : 2012-05-26 11:48:58
```

Configuration Files

- Configuration file of SwitchA

```
#
sysname SwitchA
#
vlan batch 2 to 6
#
stp instance 1 root primary
stp instance 2 root secondary
stp instance 3 root primary
stp instance 4 root secondary
stp pathcost-standard legacy
#
stp region-configuration
region-name RG1
instance 1 vlan 2
instance 2 vlan 3
instance 3 vlan 4
instance 4 vlan 5
#
interface Vlanif2
ip address 10.1.2.102 255.255.255.0
vrrp vrid 1 virtual-ip 10.1.2.100
vrrp vrid 1 priority 120
vrrp vrid 1 preempt-mode timer delay 20
#
interface Vlanif3
ip address 10.1.3.102 255.255.255.0
vrrp vrid 2 virtual-ip 10.1.3.100
#
interface Vlanif4
ip address 10.1.4.102 255.255.255.0
vrrp vrid 3 virtual-ip 10.1.4.100
vrrp vrid 3 priority 120
vrrp vrid 3 preempt-mode timer delay 20
#
interface Vlanif5
ip address 10.1.5.102 255.255.255.0
vrrp vrid 4 virtual-ip 10.1.5.100
#
interface Vlanif6
ip address 10.1.6.102 255.255.255.0
#
interface 10GE1/0/1
description TO-CE6800-SWITCHC
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2
stp root-protection
#
interface 10GE1/0/2
description TO-CE6800-SWITCHD
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 3
#
interface 10GE1/0/3
description TO-CE6800-SWITCHE
port link-type trunk
```



```
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 4
stp root-protection
#
interface 10GE1/0/4
description TO-CE6800-SWITCHF
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 5
#
interface 10GE1/0/5
description TO-CE12800-SWITCHB
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2 to 5
#
interface 10GE1/0/6
description TO-ROUTERA
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 6
#
ospf 1
area 0.0.0.0
network 10.1.2.0 0.0.0.255
network 10.1.3.0 0.0.0.255
network 10.1.4.0 0.0.0.255
network 10.1.5.0 0.0.0.255
network 10.1.6.0 0.0.0.255
#
return
```

- Configuration file of SwitchB

```
#
sysname SwitchB
#
vlan batch 2 to 5 7
#
stp instance 1 root secondary
stp instance 2 root primary
stp instance 3 root secondary
stp instance 4 root primary
stp pathcost-standard legacy
#
stp region-configuration
region-name RG1
instance 1 vlan 2
instance 2 vlan 3
instance 3 vlan 4
instance 4 vlan 5
#
interface Vlanif2
ip address 10.1.2.103 255.255.255.0
vrp vrid 1 virtual-ip 10.1.2.100
#
interface Vlanif3
ip address 10.1.3.103 255.255.255.0
vrp vrid 2 virtual-ip 10.1.3.100
vrp vrid 2 priority 120
vrp vrid 2 preempt-mode timer delay 20
#
interface Vlanif4
ip address 10.1.4.103 255.255.255.0
vrp vrid 3 virtual-ip 10.1.4.100
#
interface Vlanif5
ip address 10.1.5.103 255.255.255.0
vrp vrid 4 virtual-ip 10.1.5.100
vrp vrid 4 priority 120
vrp vrid 4 preempt-mode timer delay 20
```

```
#
interface Vlanif7
ip address 10.1.7.103 255.255.255.0
#
interface 10GE1/0/1
description TO-CE6800-SWITCHF
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 5
stp root-protection
#
interface 10GE1/0/2
description TO-CE6800-SWITCHE
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 4
#
interface 10GE1/0/3
description TO-CE6800-SWITCHD
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 3
stp root-protection
#
interface 10GE1/0/4
description TO-CE6800-SWITCHC
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2
#
interface 10GE1/0/5
description TO-CE12800-SWITCHA
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2 to 5
#
interface 10GE1/0/6
description TO-CE12800-SWITCHA
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 7
#
ospf 1
area 0.0.0.0
network 10.1.2.0 0.0.0.255
network 10.1.3.0 0.0.0.255
network 10.1.4.0 0.0.0.255
network 10.1.5.0 0.0.0.255
network 10.1.7.0 0.0.0.255
#
return
```

- Configuration file of SwitchC

```
#
sysname SwitchC
#
vlan batch 2
#
stp pathcost-standard legacy
#
stp region-configuration
region-name RG1
instance 1 vlan 2
instance 2 vlan 3
instance 3 vlan 4
instance 4 vlan 5
#
interface 10GE1/0/1
description TO-CE12800-SWITCHA
port link-type trunk
```

```
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2
#
interface 10GE1/0/2
description TO-CE12800-SWITCHB
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 2
stp instance 1 cost 20000
#
interface 10GE1/0/3
description TO-HOSTA
port default vlan 2
stp edged-port enable
#
return
```

- Configuration file of SwitchD

```
#
sysname SwitchD
#
vlan batch 3
#
stp pathcost-standard legacy
#
stp region-configuration
region-name RG1
instance 1 vlan 2
instance 2 vlan 3
instance 3 vlan 4
instance 4 vlan 5
#
interface 10GE1/0/1
description TO-CE12800-SWITCHB
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 3
#
interface 10GE1/0/2
description TO-CE12800-SWITCHA
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 3
stp instance 2 cost 20000
#
interface 10GE1/0/3
description TO-HOSTB
port default vlan 3
stp edged-port enable
#
return
```

- Configuration file of SwitchE

```
#
sysname SwitchE
#
vlan batch 4
#
stp pathcost-standard legacy
#
stp region-configuration
region-name RG1
instance 1 vlan 2
instance 2 vlan 3
instance 3 vlan 4
instance 4 vlan 5
#
interface 10GE1/0/1
description TO-CE12800-SWITCHA
port link-type trunk
```

```
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 4
#
interface 10GE1/0/2
description TO-CE12800-SWITCHB
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 4
stp instance 1 cost 20000
#
interface 10GE1/0/3
description TO-HOSTC
port default vlan 4
stp edged-port enable
#
return
```

- Configuration file of SwitchF

```
#
sysname SwitchF
#
vlan batch 5
#
stp pathcost-standard legacy
#
stp region-configuration
region-name RG1
instance 1 vlan 2
instance 2 vlan 3
instance 3 vlan 4
instance 4 vlan 5
#
interface 10GE1/0/1
description TO-CE12800-SWITCHB
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 5
#
interface 10GE1/0/2
description TO-CE12800-SWITCHA
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 5
stp instance 2 cost 20000
#
interface 10GE1/0/3
description TO-HOSTD
port default vlan 5
stp edged-port enable
#
return
```

1.3 Configuring a Data Center Network Based on VS and VRRP

Applicable Products and Versions

- CE16800, CE12800, and CE12800E series switches (excluding the CE12800E configured with FD-X series cards)
- V100R002C00 and later versions
- For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

On the data center network shown in [Figure 1-3](#), a CE12800 connects to multiple access switches. VSs need to be created on the CE12800 to function as core and aggregation switches and set up a three-layer network structure with one core switch, two aggregation switches, and multiple access switches. On the three-layer network, an access switch needs to be dual-homed to two aggregation switches, and VRRP needs to be configured on aggregation switches for gateway backup.

Figure 1-3 Networking of a VS- and VRRP-based data center network

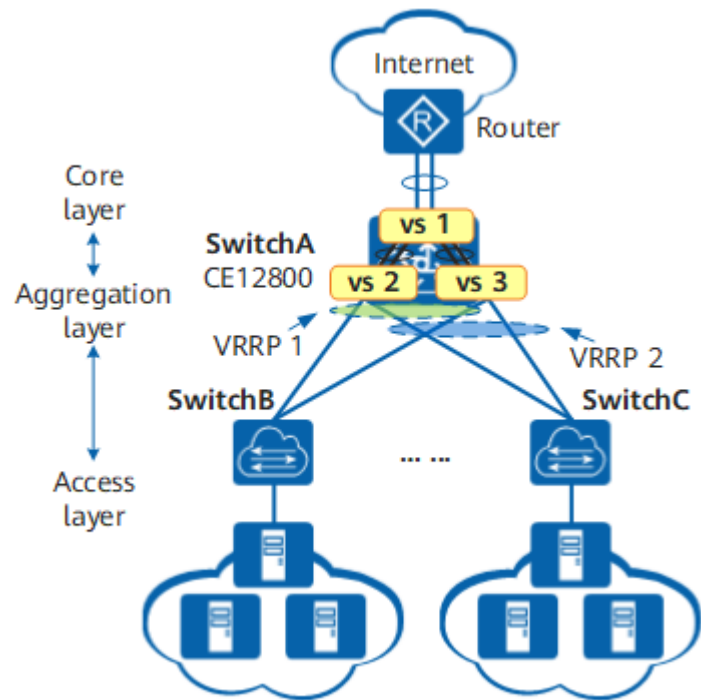


Table 1-3 Data Preparation

Device Name	Interface Number	IP Address	Interconnected Device and Interface Number
Router	Eth-Trunk1 10GE1/0/1 10GE1/0/2	10.10.10.1/24	vs1: Eth-Trunk1
vs1	Eth-Trunk1 10GE1/0/1 10GE1/0/2	10.10.10.2/24	Router: Eth-Trunk1
	Eth-Trunk2 10GE1/0/3 10GE1/0/4	10.10.20.1/24	vs2: Eth-Trunk2

Device Name	Interface Number	IP Address	Interconnected Device and Interface Number
	Eth-Trunk3 • 10GE1/0/5 • 10GE1/0/6	10.10.30.1/24	vs3: Eth-Trunk3
vs2	Eth-Trunk2 • 10GE2/0/1 • 10GE2/0/2	10.10.20.2/24	vs1: Eth-Trunk2
	10GE2/0/3	VLANIF 10: 10.10.40.1/24	SwitchB: 10GE1/0/1
	10GE2/0/4	VLANIF 20: 10.10.50.1/24	SwitchC: 10GE1/0/1
vs3	Eth-Trunk3 • 10GE3/0/1 • 10GE3/0/2	10.10.30.2/24	vs1: Eth-Trunk3
	10GE3/0/3	VLANIF 10: 10.10.40.2/24	SwitchB: 10GE1/0/2
	10GE3/0/4	VLANIF 20: 10.10.50.2/24	SwitchC: 10GE1/0/2
SwitchB	10GE1/0/1	-	vs2: 10GE2/0/3
	10GE1/0/2	-	vs3: 10GE3/0/3
SwitchC	10GE1/0/1	-	vs2: 10GE2/0/4
	10GE1/0/2	-	vs3: 10GE3/0/4

Configuration Roadmap

The configuration roadmap is as follows:

1. Create VSs as core and aggregation switches.
2. Configure VRRP on aggregation switches VSs for gateway backup.
3. Configure routes between upstream and downstream devices to implement Layer 2 and Layer 3 interconnection.

Procedure

Step 1 On Switch A, create vs1 as a core switch, and create vs2 and vs3 as aggregation switches.

NOTE

The VS function is controlled by a license and is disabled on a new device by default. To use the VS function, apply for and purchase the license from the Huawei agent or Huawei local office.

Create VS1, VS2, and VS3 in group mode, and assign interfaces in slots 1, 2, and 3 to VSs 1, 2, and 3 respectively.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchA
[*HUAWEI] commit
[~SwitchA] admin
[~SwitchA-admin] virtual-system vs1
[*SwitchA-admin-vs:vs1] port-mode group
[*SwitchA-admin-vs:vs1] assign interface 10ge 1/0/0
Warning: All configurations of the interfaces will be deleted. Interfaces 10GE1/0/0-23 of the same group
will be assigned. Continue? [Y/N]: y
[*SwitchA-admin-vs:vs1] quit
[*SwitchA-admin] virtual-system vs2
[*SwitchA-admin-vs:vs2] port-mode group
[*SwitchA-admin-vs:vs2] assign interface 10ge 2/0/0
Warning: All configurations of the interfaces will be deleted. Interfaces 10GE2/0/0-23 of the same group
will be assigned. Continue? [Y/N]: y
[*SwitchA-admin-vs:vs2] quit
[*SwitchA-admin] virtual-system vs3
[*SwitchA-admin-vs:vs3] port-mode group
[*SwitchA-admin-vs:vs3] assign interface 10ge 3/0/0
Warning: All configurations of the interfaces will be deleted. Interfaces 10GE3/0/0-23 of the same group
will be assigned. Continue? [Y/N]: y
[*SwitchA-admin-vs:vs3] commit
[~SwitchA-admin-vs:vs3] return
```

Configure management IP addresses and management accounts for the VSs.

```
<SwitchA> switch virtual-system vs1
<SwitchA-vs1> system-view
[~SwitchA-vs1] interface MEth 0/0/0
[~SwitchA-vs1-METh0/0/0] ip address 10.1.1.10 24
[*SwitchA-vs1-METh0/0/0] quit
[*SwitchA-vs1] user-interface vty 0 4
[*SwitchA-vs1-ui-vty0-4] authentication-mode aaa
[*SwitchA-vs1-ui-vty0-4] user privilege level 3
[*SwitchA-vs1-ui-vty0-4] quit
[*SwitchA-vs1] aaa
[*SwitchA-vs1-aaa] local-user vs1_user password cipher VS1_password
[*SwitchA-vs1-aaa] local-user vs1_user service-type telnet
[*SwitchA-vs1-aaa] local-user vs1_user level 3
[*SwitchA-vs1-aaa] quit
[*SwitchA-vs1] undo telnet server disable
[*SwitchA-vs1] commit
[~SwitchA-vs1] quit
<SwitchA-vs1> quit
```

The configurations of vs2 and vs3 are similar to that of vs1 and are not mentioned here.

Step 2 Configure VLANs on the access and aggregation layers for Layer 2 interconnection.

On vs2 and vs3, configure the interfaces that connect to SwitchB to allow packets from VLAN 10 to pass through.

```
<SwitchA> switch virtual-system vs2
<SwitchA-vs2> system-view
[~SwitchA-vs2] vlan batch 10
[*SwitchA-vs2] interface 10ge 2/0/3
[*SwitchA-vs2-10GE2/0/3] description To_SwitchB
[*SwitchA-vs2-10GE2/0/3] port link-type trunk
[*SwitchA-vs2-10GE2/0/3] undo port trunk allow-pass vlan 1
```

```
[*SwitchA-vs2-10GE2/0/3] port trunk allow-pass vlan 10
[*SwitchA-vs2-10GE2/0/3] quit
[*SwitchA-vs2] interface vlanif 10
[*SwitchA-vs2-Vlanif10] ip address 10.10.40.1 24
[*SwitchA-vs2-Vlanif10] commit
[~SwitchA-vs2-Vlanif10] return
<SwitchA-vs2> quit
<SwitchA> switch virtual-system vs3
<SwitchA-vs3> system-view
[~SwitchA-vs3] vlan batch 10
[*SwitchA-vs3] interface 10ge 3/0/3
[*SwitchA-vs3-10GE2/0/3] description To_SwitchB
[*SwitchA-vs3-10GE2/0/3] port link-type trunk
[*SwitchA-vs3-10GE2/0/3] undo port trunk allow-pass vlan 1
[*SwitchA-vs3-10GE2/0/3] port trunk allow-pass vlan 10
[*SwitchA-vs3] interface vlanif 10
[*SwitchA-vs3-Vlanif10] ip address 10.10.40.2 24
[*SwitchA-vs3-Vlanif10] commit
[~SwitchA-vs3-Vlanif10] return
<SwitchA-vs3> quit
```

On SwitchB, configure the interfaces that connect to vs2 and vs3 to allow packets from VLAN 10 to pass through.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchB
[*HUAWEI] commit
[~SwitchB] vlan batch 10
[*SwitchB] interface 10ge 1/0/1
[*SwitchB-10GE1/0/1] description To_vs2
[*SwitchB-10GE1/0/1] port link-type trunk
[*SwitchB-10GE1/0/1] undo port trunk allow-pass vlan 1
[*SwitchB-10GE1/0/1] port trunk allow-pass vlan 10
[*SwitchB-10GE1/0/1] quit
[*SwitchB] interface 10ge 1/0/2
[*SwitchB-10GE1/0/2] description To_vs3
[*SwitchB-10GE1/0/2] port link-type trunk
[*SwitchB-10GE1/0/2] undo port trunk allow-pass vlan 1
[*SwitchB-10GE1/0/2] port trunk allow-pass vlan 10
[*SwitchB-10GE1/0/2] quit
[*SwitchB] commit
```

On vs2 and vs3, configure the interfaces that connect to SwitchC to allow packets from VLAN 20 to pass through according to the preceding configuration. The configuration is not mentioned here.

Step 3 Configure VRRP on vs2 and vs3 at the aggregation layer.

On vs2 and vs3, create VRRP group 1. Set the priority of vs2 to 120 and priority of vs3 to 100 so that vs2 becomes the master device to transmit SwitchB uplink traffic.

```
<SwitchA> switch virtual-system vs2
<SwitchA-vs2> system-view
[~SwitchA-vs2] interface vlanif 10
[~SwitchA-vs2-Vlanif10] vrrp vrid 1 virtual-ip 10.10.40.100
[*SwitchA-vs2-Vlanif10] vrrp vrid 1 priority 120
[*SwitchA-vs2-Vlanif10] commit
[~SwitchA-vs2-Vlanif10] return
<SwitchA-vs2> quit
<SwitchA> switch virtual-system vs3
<SwitchA-vs3> system-view
[~SwitchA-vs3] interface vlanif 10
[~SwitchA-vs3-Vlanif10] vrrp vrid 1 virtual-ip 10.10.40.100
[*SwitchA-vs3-Vlanif10] commit
[~SwitchA-vs3-Vlanif10] return
<SwitchA-vs3> quit
```


On vs2 and vs3, create VRRP group 2. Set the priority of vs3 to 120 and priority of vs2 to 100 so that vs3 becomes the master device to transmit SwitchC uplink traffic.

```
<SwitchA> switch virtual-system vs2
<SwitchA-vs2> system-view
[~SwitchA-vs2] interface vlanif 20
[~SwitchA-vs2-Vlanif20] vrrp vrid 2 virtual-ip 10.10.50.100
[*SwitchA-vs2-Vlanif20] commit
[~SwitchA-vs2-Vlanif20] return
<SwitchA-vs2> quit
<SwitchA> switch virtual-system vs3
<SwitchA-vs3> system-view
[~SwitchA-vs3] interface vlanif 20
[~SwitchA-vs3-Vlanif20] vrrp vrid 2 virtual-ip 10.10.50.100
[*SwitchA-vs3-Vlanif20] vrrp vrid 2 priority 120
[*SwitchA-vs3-Vlanif20] commit
[~SwitchA-vs3-Vlanif20] return
<SwitchA-vs3> quit
```

Step 4 Configure routes between the router, aggregation switches, and core switch to implement Layer 3 interconnection.

```
<SwitchA> switch virtual-system vs1
<SwitchA-vs1> system-view
[~SwitchA-vs1] ospf 1
[*SwitchA-vs1-ospf-1] area 0
[*SwitchA-vs1-ospf-1-area-0.0.0.0] network 10.10.10.0 0.0.0.255
[*SwitchA-vs1-ospf-1-area-0.0.0.0] network 10.10.20.0 0.0.0.255
[*SwitchA-vs1-ospf-1-area-0.0.0.0] network 10.10.30.0 0.0.0.255
[*SwitchA-vs1-ospf-1-area-0.0.0.0] commit
[~SwitchA-vs1-ospf-1-area-0.0.0.0] return
<SwitchA-vs1> quit
```

Configure OSPF on the router, vs2, and vs3. The configurations of the router, vs2, and vs3 are similar to the configuration of vs1 and are not mentioned here.

----End

Checking the Configuration

After the preceding configurations are complete, check whether servers can ping the router successfully (you need to set the default gateway for the servers). In this example, the servers ping the router.

```
PC> ping 10.10.10.1

Ping 10.10.10.1: 32 data bytes, Press Ctrl_C to break
From 10.10.10.1: bytes=32 seq=1 ttl=253 time=93 ms
From 10.10.10.1: bytes=32 seq=2 ttl=253 time=125 ms
From 10.10.10.1: bytes=32 seq=3 ttl=253 time=94 ms
From 10.10.10.1: bytes=32 seq=4 ttl=253 time=47 ms
From 10.10.10.1: bytes=32 seq=5 ttl=253 time=109 ms

--- 10.10.10.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 47/93/125 ms
```

Configuration Files

- Configuration file of Router

```
#
sysname Router
```

```
#
interface Eth-Trunk1
ip address 10.10.10.1 255.255.255.0
#
interface XGigabitEthernet1/0/1
eth-trunk 1
#
interface XGigabitEthernet1/0/2
eth-trunk 1
#
ospf 1
area 0.0.0.0
network 10.10.10.0 0.0.0.255
#
return
```

- Configuration file of SwitchA

```
#
sysname SwitchA
#
admin
virtual-system vs1
port-mode group
resource u4route upper-limit 60000
resource m4route upper-limit 1000
resource u6route upper-limit 16000
resource m6route upper-limit 100
resource vlan upper-limit 4063
resource mpls enable
resource trill enable
resource mcast enable
resource vpn-instance upper-limit 4096
resource cpu weight 5
resource memory ratio-threshold 100
assign interface 10GE1/0/0
assign interface 10GE1/0/1
assign interface 10GE1/0/2
assign interface 10GE1/0/3
assign interface 10GE1/0/4
assign interface 10GE1/0/5
assign interface 10GE1/0/6
assign interface 10GE1/0/7
assign interface 10GE1/0/8
assign interface 10GE1/0/9
assign interface 10GE1/0/10
assign interface 10GE1/0/11
assign interface 10GE1/0/12
assign interface 10GE1/0/13
assign interface 10GE1/0/14
assign interface 10GE1/0/15
assign interface 10GE1/0/16
assign interface 10GE1/0/17
assign interface 10GE1/0/18
assign interface 10GE1/0/19
assign interface 10GE1/0/20
assign interface 10GE1/0/21
assign interface 10GE1/0/22
assign interface 10GE1/0/23
virtual-system vs2
port-mode group
resource u4route upper-limit 60000
resource m4route upper-limit 1000
resource u6route upper-limit 16000
resource m6route upper-limit 100
resource vlan upper-limit 4063
resource mpls enable
resource trill enable
resource mcast enable
resource vpn-instance upper-limit 4096
resource cpu weight 5
```

```
resource memory ratio-threshold 100
assign interface 10GE2/0/0
assign interface 10GE2/0/1
assign interface 10GE2/0/2
assign interface 10GE2/0/3
assign interface 10GE2/0/4
assign interface 10GE2/0/5
assign interface 10GE2/0/6
assign interface 10GE2/0/7
assign interface 10GE2/0/8
assign interface 10GE2/0/9
assign interface 10GE2/0/10
assign interface 10GE2/0/11
assign interface 10GE2/0/12
assign interface 10GE2/0/13
assign interface 10GE2/0/14
assign interface 10GE2/0/15
assign interface 10GE2/0/16
assign interface 10GE2/0/17
assign interface 10GE2/0/18
assign interface 10GE2/0/19
assign interface 10GE2/0/20
assign interface 10GE2/0/21
assign interface 10GE2/0/22
assign interface 10GE2/0/23
virtual-system vs3
port-mode group
resource u4route upper-limit 60000
resource m4route upper-limit 1000
resource u6route upper-limit 16000
resource m6route upper-limit 100
resource vlan upper-limit 4063
resource mpls enable
resource trill enable
resource mcast enable
resource vpn-instance upper-limit 4096
resource cpu weight 5
resource memory ratio-threshold 100
assign interface 10GE3/0/0
assign interface 10GE3/0/1
assign interface 10GE3/0/2
assign interface 10GE3/0/3
assign interface 10GE3/0/4
assign interface 10GE3/0/5
assign interface 10GE3/0/6
assign interface 10GE3/0/7
assign interface 10GE3/0/8
assign interface 10GE3/0/9
assign interface 10GE3/0/10
assign interface 10GE3/0/11
assign interface 10GE3/0/12
assign interface 10GE3/0/13
assign interface 10GE3/0/14
assign interface 10GE3/0/15
assign interface 10GE3/0/16
assign interface 10GE3/0/17
assign interface 10GE3/0/18
assign interface 10GE3/0/19
assign interface 10GE3/0/20
assign interface 10GE3/0/21
assign interface 10GE3/0/22
assign interface 10GE3/0/23
#
return
```

- Configuration file of vs1

```
#
sysname vs1
#
aaa
```

```

local-user vs1_user password cipher %^%#TT4N+w)%[C+rM>)S8Ti!p1iV@RZh(3MO7QGd96Z%^%#
local-user vs1_user service-type telnet
local-user vs1_user level 3
#
interface MEth0/0/0
ip address 10.1.1.10 255.255.255.0
#
interface Eth-Trunk1
undo portswitch
description To_Router
ip address 10.10.10.2 255.255.255.0
#
interface Eth-Trunk2
undo portswitch
description To_vs2
ip address 10.10.20.1 255.255.255.0
#
interface Eth-Trunk3
undo portswitch
description To_vs3
ip address 10.10.30.1 255.255.255.0
#
interface 10GE1/0/1
eth-trunk 1
#
interface 10GE1/0/2
eth-trunk 1
#
interface 10GE1/0/3
eth-trunk 2
#
interface 10GE1/0/4
eth-trunk 2
#
interface 10GE1/0/5
eth-trunk 3
#
interface 10GE1/0/6
eth-trunk 3
#
ospf 1
area 0.0.0.0
network 10.10.10.0 0.0.0.255
network 10.10.20.0 0.0.0.255
network 10.10.30.0 0.0.0.255
#
user-interface vty 0 4
authentication-mode aaa
user privilege level 3
#
return

```

- Configuration file of vs2

```

#
sysname vs2
#
vlan batch 10 20
#
aaa
local-user vs2_user password cipher %^%#%j]%Lv%StAqlveR"YfuS<{^%;~p*$D<&&@X/Xs(/%^%#
local-user vs2_user service-type telnet
local-user vs2_user level 3
#
interface Vlanif10
ip address 10.10.40.1 24
vrp vrid 1 virtual-ip 10.10.40.100
vrp vrid 1 priority 120
#
interface Vlanif20
ip address 10.10.50.1 24

```

```
vrrp vrid 2 virtual-ip 10.10.50.100
#
interface MEth0/0/0
ip address 10.1.1.20 255.255.255.0
#
interface Eth-Trunk2
undo portswitch
description To_vs1
ip address 10.10.20.2 255.255.255.0
#
interface 10GE2/0/1
eth-trunk 2
#
interface 10GE2/0/2
eth-trunk 2
#
interface 10GE2/0/3
description To_SwitchB
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 10
#
interface 10GE2/0/4
description To_SwitchC
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 20
#
ospf 1
area 0.0.0.0
network 10.10.20.0 0.0.0.255
network 10.10.40.0 0.0.0.255
network 10.10.50.0 0.0.0.255
#
user-interface vty 0 4
authentication-mode aaa
user privilege level 3
#
return
```

- Configuration file of vs3

```
#
sysname vs3
#
vlan batch 10 20
#
aaa
local-user vs3_user password cipher %^%#Ucmwl^>OxIF$(.XyM&%lOJ}ASW#3:OxGvM;^yF!%^%#
local-user vs3_user service-type telnet
local-user vs3_user level 3
#
interface Vlanif10
ip address 10.10.40.2 24
vrrp vrid 1 virtual-ip 10.10.40.100
#
interface Vlanif20
ip address 10.10.50.2 24
vrrp vrid 2 virtual-ip 10.10.50.100
vrrp vrid 2 priority 120
#
interface MEth0/0/0
ip address 10.1.1.30 255.255.255.0
#
interface Eth-Trunk3
undo portswitch
description To_vs1
ip address 10.10.30.2 255.255.255.0
#
interface 10GE3/0/1
eth-trunk 3
```

```
#
interface 10GE3/0/2
eth-trunk 3
#
interface 10GE3/0/3
description To_SwitchB
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 10
#
interface 10GE3/0/4
description To_SwitchC
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 20
#
ospf 1
area 0.0.0.0
network 10.10.30.0 0.0.0.255
network 10.10.40.0 0.0.0.255
network 10.10.50.0 0.0.0.255
#
user-interface vty 0 4
authentication-mode aaa
user privilege level 3
#
return
```

- Configuration file of SwitchB at the access layer

```
#
sysname SwitchB
#
vlan batch 10
#
interface 10GE1/0/1
description To_vs2
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 10
#
interface 10GE1/0/2
description To_vs3
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 10
#
return
```

- Configuration file of SwitchC at the access layer

```
#
sysname SwitchC
#
vlan batch 20
#
interface 10GE1/0/1
description To_vs2
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 20
#
interface 10GE1/0/2
description To_vs3
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 20
#
return
```

1.4 Example for Configuring IPv4 VXLAN in Distributed Gateway Deployment Mode

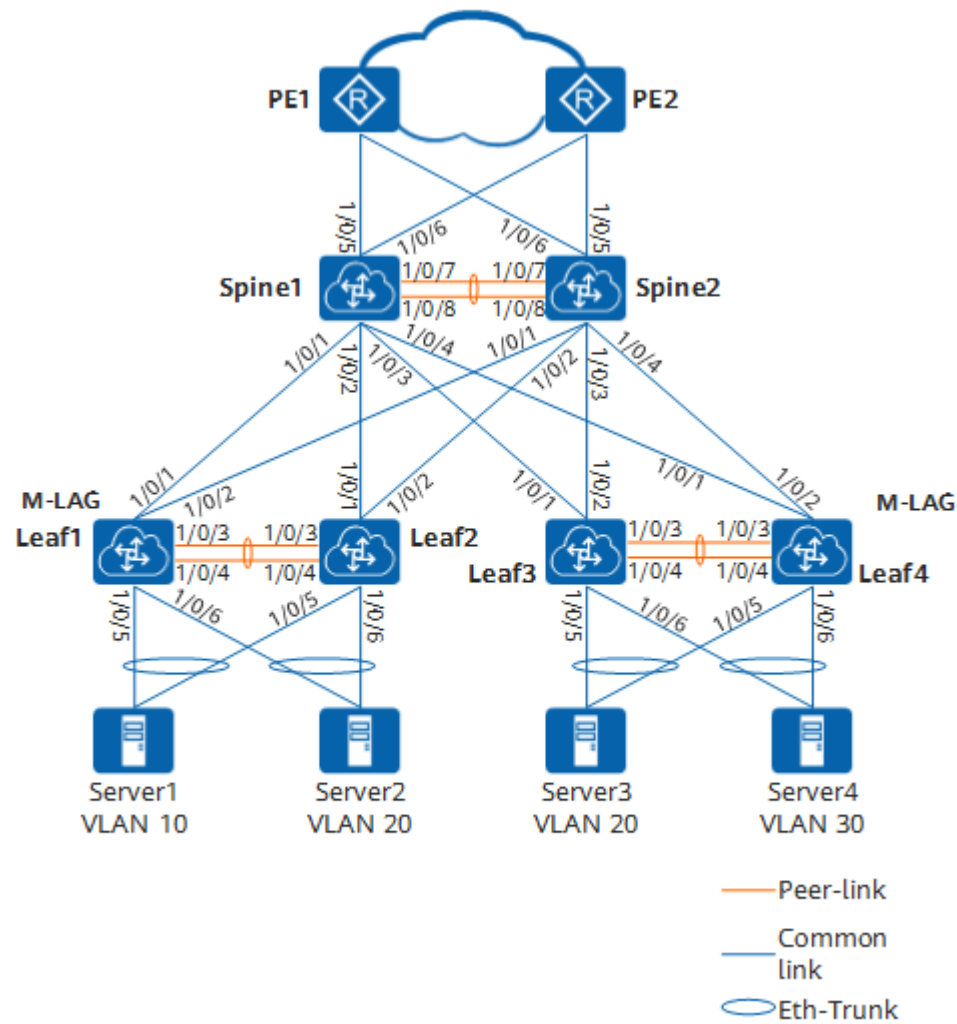
Applicable Products and Versions

- This example applies to CE16800, CE12800, CE12800E, CE8800, CE7800, CE6800 (excluding the CE6820, CE6850EI, CE6810EI, and CE6810LI), and CE5880EI series switches running V200R002C50 or later versions.
- For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

In [Figure 1-4](#), the new data center network of an enterprise uses the distributed gateway deployment mode. The underlay network is an IPv4 network, and the overlay network is an IPv4 or IPv6 network. Leaf nodes function as Layer 3 gateways and connect to servers. Spine nodes function as aggregation devices for east-to-west traffic and egress gateways. To ensure high reliability, spine and leaf nodes establish M-LAGs.

Figure 1-4 VXLAN in distributed gateway deployment mode



NOTE

In the preceding figure, 1/0/1 indicates an interface number and the interface rate is 10 Gbit/s. That is, 1/0/1 indicates 10GE1/0/1. Other interfaces are specified in a similar way.

Table 1-4 IP addresses of interfaces

Device Name	Interface	IP Address	Device Name	Interface	IP Address
Spine 1	10GE1/0/1	192.168.1.1/24	Spine 2	10GE1/0/1	192.168.5.1/24
	10GE1/0/2	192.168.2.1/24		10GE1/0/2	192.168.6.1/24
	10GE1/0/3	192.168.3.1/24		10GE1/0/3	192.168.7.1/24

Device Name	Interface	IP Address	Device Name	Interface	IP Address
	10GE1/0/4	192.168.4.1/24		10GE1/0/4	192.168.8.1/24
	10GE1/0/5	IPv4: 10.1.10.1/24 IPv6: fc00:10::1/64		10GE1/0/5	IPv4: 10.1.30.1/24 IPv6: fc00:30::1/64
	10GE1/0/6	IPv4: 10.1.20.1/24 IPv6: fc00:20::1/64		10GE1/0/6	IPv4: 10.1.40.1/24 IPv6: fc00:40::1/64
	Loopback0	4.4.4.4/32		Loopback0	5.5.5.5/32
	Loopback1	1.1.1.1/32		Loopback1	1.1.1.1/32
Leaf1	10GE1/0/1	192.168.1.2/24	Leaf2	10GE1/0/1	192.168.2.2/24
	10GE1/0/2	192.168.5.2/24		10GE1/0/2	192.168.6.2/24
	Loopback0	6.6.6.6/32		Loopback0	7.7.7.7/32
	Loopback1	2.2.2.2/32		Loopback1	2.2.2.2/32
Leaf3	10GE1/0/1	192.168.3.2/24	Leaf4	10GE1/0/1	192.168.4.2/24
	10GE1/0/2	192.168.7.2/24		10GE1/0/2	192.168.8.2/24
	Loopback0	8.8.8.8/32		Loopback0	9.9.9.9/32
	Loopback1	3.3.3.3/32		Loopback1	3.3.3.3/32

Configuration Roadmap

- The configuration roadmap is as follows:
1. Configure a routing protocol to ensure Layer 3 connectivity on the underlay network.
 2. Configure M-LAG to implement active-active server access.
 3. Configure BGP EVPN to establish a VXLAN tunnel.

Configuration Notes

In addition to the configuration in the procedure, you may also need to run the following commands for the CE12800 based on actual requirements:

- Run the **assign forward nvo3 service extend enable** command in the system view to enable the NVO3 service extension function.

 NOTE

By default, the NVO3 service extension function is disabled on the switch. After the NVO3 service is configured on the switch, there is a high probability that other ACL resource-consuming services such as MQC, simplified ACL, traffic policing, BD traffic statistics collection, and DHCP fail to be configured on the switch. You can run this command on a switch configured with the NVO3 service to reduce the configuration failure possibility on EC (except the CE-L48GT-EC and CE-L48GS-EC cards), ED, EF, and EG series cards.

- Run the **assign forward nvo3 f-linecard compatibility enable** command in the system view to ensure that VXLAN traffic can be forwarded when the card interoperability mode is non-enhanced mode.

 NOTE

If VXLAN traffic is forwarded between cards when the card interoperability mode is non-enhanced mode, the VXLAN traffic may fail to be forwarded. To use the VXLAN function, you must configure the **assign forward nvo3 f-linecard compatibility enable** command when the card interoperability mode is non-enhanced mode.

Procedure

- Step 1** Configure a routing protocol to implement Layer 3 connectivity on the underlay network.

Configure Leaf1. The configurations of other devices are similar to the configuration of Leaf1. For details, see [Configuration Files](#).

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf1
[*HUAWEI] commit
[~Leaf1] interface 10ge 1/0/1
[~Leaf1-10GE1/0/1] undo portswitch
[*Leaf1-10GE1/0/1] ip address 192.168.1.2 24
[*Leaf1-10GE1/0/1] ospf network-type p2p
[*Leaf1-10GE1/0/1] quit
[*Leaf1] interface 10ge 1/0/2
[*Leaf1-10GE1/0/2] undo portswitch
[*Leaf1-10GE1/0/2] ip address 192.168.5.2 24
[*Leaf1-10GE1/0/2] ospf network-type p2p
[*Leaf1-10GE1/0/2] quit
[*Leaf1] interface loopback 0
[*Leaf1-LoopBack0] ip address 6.6.6.6 32
[*Leaf1-LoopBack0] quit
[*Leaf1] interface loopback 1
[*Leaf1-LoopBack1] ip address 2.2.2.2 32
[*Leaf1-LoopBack1] quit
[*Leaf1] ospf
[*Leaf1-ospf-1] area 0
[*Leaf1-ospf-1-area-0.0.0.0] network 192.168.1.0 0.0.0.255
[*Leaf1-ospf-1-area-0.0.0.0] network 192.168.5.0 0.0.0.255
[*Leaf1-ospf-1-area-0.0.0.0] network 6.6.6.6 0.0.0.0
[*Leaf1-ospf-1-area-0.0.0.0] network 2.2.2.2 0.0.0.0
[*Leaf1-ospf-1-area-0.0.0.0] quit
[*Leaf1-ospf-1] quit
[*Leaf1] commit
```

After OSPF is configured, leaf and spine nodes can learn the IP address of each other's loopback interface using OSPF and successfully ping each other.

Step 2 Configure leaf nodes to establish M-LAGs. In this example, Leaf1 and Leaf2 establish an M-LAG, and Leaf3 and Leaf4 establish an M-LAG. The configurations of Leaf3 and Leaf4 are similar to those of Leaf1 and Leaf2.

Configure Leaf1.

```
[~Leaf1] stp mode rstp
[*Leaf1] stp v-stp enable
[*Leaf1] dfs-group 1
[*Leaf1-dfs-group-1] source ip 6.6.6.6
[*Leaf1-dfs-group-1] priority 150
[*Leaf1-dfs-group-1] quit
[*Leaf1] interface eth-trunk 1
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/3
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/4
[*Leaf1-Eth-Trunk1] mode lacp-static
[*Leaf1-Eth-Trunk1] peer-link 1
[*Leaf1-Eth-Trunk1] quit
[*Leaf1] interface eth-trunk 2
[*Leaf1-Eth-Trunk2] trunkport 10ge 1/0/5
[*Leaf1-Eth-Trunk2] mode lacp-static
[*Leaf1-Eth-Trunk2] dfs-group 1 m-lag 1
[*Leaf1-Eth-Trunk2] stp edged-port enable
[*Leaf1-Eth-Trunk2] quit
[*Leaf1] interface eth-trunk 3
[*Leaf1-Eth-Trunk3] trunkport 10ge 1/0/6
[*Leaf1-Eth-Trunk3] mode lacp-static
[*Leaf1-Eth-Trunk3] dfs-group 1 m-lag 2
[*Leaf1-Eth-Trunk3] stp edged-port enable
[*Leaf1-Eth-Trunk3] quit
[*Leaf1] commit
[~Leaf1] monitor-link group 1
[*Leaf1-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf1-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf1-mtlk-group1] port eth-trunk 2 downlink 1
[*Leaf1-mtlk-group1] port eth-trunk 3 downlink 2
[*Leaf1-mtlk-group1] quit
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] stp mode rstp
[*Leaf2] stp v-stp enable
[*Leaf2] dfs-group 1
[*Leaf2-dfs-group-1] source ip 7.7.7.7
[*Leaf2-dfs-group-1] quit
[*Leaf2] interface eth-trunk 1
[*Leaf2-Eth-Trunk1] trunkport 10ge 1/0/3
[*Leaf2-Eth-Trunk1] trunkport 10ge 1/0/4
[*Leaf2-Eth-Trunk1] mode lacp-static
[*Leaf2-Eth-Trunk1] peer-link 1
[*Leaf2-Eth-Trunk1] quit
[*Leaf2] interface eth-trunk 2
[*Leaf2-Eth-Trunk2] mode lacp-static
[*Leaf2-Eth-Trunk2] trunkport 10ge 1/0/5
[*Leaf2-Eth-Trunk2] dfs-group 1 m-lag 1
[*Leaf2-Eth-Trunk2] stp edged-port enable
[*Leaf2-Eth-Trunk2] quit
[*Leaf2] interface eth-trunk 3
[*Leaf2-Eth-Trunk3] mode lacp-static
[*Leaf2-Eth-Trunk3] trunkport 10ge 1/0/6
[*Leaf2-Eth-Trunk3] dfs-group 1 m-lag 2
[*Leaf2-Eth-Trunk3] stp edged-port enable
[*Leaf2-Eth-Trunk3] quit
[*Leaf2] commit
[~Leaf2] monitor-link group 1
[*Leaf2-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf2-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf2-mtlk-group1] port eth-trunk 2 downlink 1
```

```
[*Leaf2-mtlk-group1] port eth-trunk 3 downlink 2
[*Leaf2-mtlk-group1] quit
[*Leaf2] commit
```

Step 3 Configure Spine1 and Spine2 to establish an M-LAG.

Configure Spine1.

```
[~Spine1] stp mode rstp
[*Spine1] stp v-stp enable
[*Spine1] dfs-group 1
[*Spine1-dfs-group-1] source ip 4.4.4.4
[*Spine1-dfs-group-1] priority 150
[*Spine1-dfs-group-1] quit
[*Spine1] interface eth-trunk 1
[*Spine1-Eth-Trunk1] trunkport 10ge 1/0/7
[*Spine1-Eth-Trunk1] trunkport 10ge 1/0/8
[*Spine1-Eth-Trunk1] mode lacp-static
[*Spine1-Eth-Trunk1] peer-link 1
[*Spine1-Eth-Trunk1] quit
[*Spine1] commit
```

Configure Spine2.

```
[~Spine2] stp mode rstp
[*Spine2] stp v-stp enable
[*Spine2] dfs-group 1
[*Spine2-dfs-group-1] source ip 5.5.5.5
[*Spine2-dfs-group-1] quit
[*Spine2] interface eth-trunk 1
[*Spine2-Eth-Trunk1] trunkport 10ge 1/0/7
[*Spine2-Eth-Trunk1] trunkport 10ge 1/0/8
[*Spine2-Eth-Trunk1] mode lacp-static
[*Spine2-Eth-Trunk1] peer-link 1
[*Spine2-Eth-Trunk1] quit
[*Spine2] commit
```

Step 4 Configure BGP EVPN to establish a VXLAN tunnel.

1. Configure the VXLAN tunnel mode and enable the VXLAN ACL extension function. (This step is required only on the CE12800, CE16800 equipped with A series cards, CE6870EI, and CE6875EI.)

Configure Spine1. The configurations of other devices are similar to the configuration of Spine1.

```
[~Spine1] ip tunnel mode vxlan
[*Spine1] assign forward nvo3 acl extend enable
[*Spine1] commit
```

NOTE

After configuring the VXLAN tunnel mode and enabling the VXLAN ACL extension function, save the configuration and restart the device to make the configuration take effect. You can restart the device immediately or after completing all the configurations.

2. Configure service access points.

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1.

```
[~Leaf1] bridge-domain 10
[*Leaf1-bd10] quit
[*Leaf1] bridge-domain 20
[*Leaf1-bd20] quit
[*Leaf1] interface eth-trunk 2.10 mode l2
[*Leaf1-Eth-Trunk2.1] encapsulation dot1q vid 10
[*Leaf1-Eth-Trunk2.1] bridge-domain 10
[*Leaf1-Eth-Trunk2.1] quit
```

```
[*Leaf1] interface eth-trunk 3.20 mode l2
[*Leaf1-Eth-Trunk3.1] encapsulation dot1q vid 20
[*Leaf1-Eth-Trunk3.1] bridge-domain 20
[*Leaf1-Eth-Trunk3.1] quit
[*Leaf1] commit
```

3. Establish BGP EVPN peer relationships. Spine1 and Spine2 function as route reflectors (RRs).

Configure Spine1. The configuration of Spine2 is similar to that of Spine1. For details, see [Configuration Files](#).

```
[~Spine1] evpn-overlay enable
[*Spine1] bgp 100
[*Spine1-bgp] peer 6.6.6.6 as-number 100
[*Spine1-bgp] peer 6.6.6.6 connect-interface LoopBack0
[*Spine1-bgp] peer 7.7.7.7 as-number 100
[*Spine1-bgp] peer 7.7.7.7 connect-interface LoopBack0
[*Spine1-bgp] peer 8.8.8.8 as-number 100
[*Spine1-bgp] peer 8.8.8.8 connect-interface LoopBack0
[*Spine1-bgp] peer 9.9.9.9 as-number 100
[*Spine1-bgp] peer 9.9.9.9 connect-interface LoopBack0
[*Spine1-bgp] l2vpn-family evpn
[*Spine1-bgp-af-evpn] peer 6.6.6.6 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 6.6.6.6 advertise irb
[*Spine1-bgp-af-evpn] peer 6.6.6.6 advertise irbv6
[*Spine1-bgp-af-evpn] peer 6.6.6.6 reflect-client
[*Spine1-bgp-af-evpn] peer 7.7.7.7 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 7.7.7.7 advertise irb
[*Spine1-bgp-af-evpn] peer 7.7.7.7 advertise irbv6
[*Spine1-bgp-af-evpn] peer 7.7.7.7 reflect-client
[*Spine1-bgp-af-evpn] peer 8.8.8.8 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 8.8.8.8 advertise irb
[*Spine1-bgp-af-evpn] peer 8.8.8.8 advertise irbv6
[*Spine1-bgp-af-evpn] peer 8.8.8.8 reflect-client
[*Spine1-bgp-af-evpn] peer 9.9.9.9 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 9.9.9.9 advertise irb
[*Spine1-bgp-af-evpn] peer 9.9.9.9 advertise irbv6
[*Spine1-bgp-af-evpn] peer 9.9.9.9 reflect-client
[*Spine1-bgp-af-evpn] undo policy vpn-target
[*Spine1-bgp-af-evpn] quit
[*Spine1-bgp] quit
[*Spine1] commit
```

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1. For details, see [Configuration Files](#).

```
[~Leaf1] evpn-overlay enable
[*Leaf1] bgp 100
[*Leaf1-bgp] peer 4.4.4.4 as-number 100
[*Leaf1-bgp] peer 4.4.4.4 connect-interface LoopBack0
[*Leaf1-bgp] peer 5.5.5.5 as-number 100
[*Leaf1-bgp] peer 5.5.5.5 connect-interface LoopBack0
[*Leaf1-bgp] l2vpn-family evpn
[*Leaf1-bgp-af-evpn] peer 4.4.4.4 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 4.4.4.4 advertise irb
[*Leaf1-bgp-af-evpn] peer 4.4.4.4 advertise irbv6
[*Leaf1-bgp-af-evpn] peer 5.5.5.5 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 5.5.5.5 advertise irb
[*Leaf1-bgp-af-evpn] peer 5.5.5.5 advertise irbv6
[*Leaf1-bgp-af-evpn] quit
[*Leaf1-bgp] quit
[*Leaf1] commit
```

4. Configure VPN and EVPN instances.

Configure Spine1. The configuration of Spine2 is similar to that of Spine1.
For details, see [Configuration Files](#).

```
[~Spine1] ip vpn-instance vpn1
[*Spine1-vpn-instance-vpn1] vxlan vni 5000
[*Spine1-vpn-instance-vpn1] ipv4-family
[*Spine1-vpn-instance-vpn1-af-ipv4] route-distinguisher 4.4.4.4:1
[*Spine1-vpn-instance-vpn1-af-ipv4] vpn-target 0:1 evpn
[*Spine1-vpn-instance-vpn1-af-ipv4] quit
[*Spine1-vpn-instance-vpn1] ipv6-family
[*Spine1-vpn-instance-vpn1-af-ipv6] route-distinguisher 4.4.4.4:1
[*Spine1-vpn-instance-vpn1-af-ipv6] vpn-target 0:1 evpn
[*Spine1-vpn-instance-vpn1-af-ipv6] quit
[*Spine1-vpn-instance-vpn1] quit
[*Spine1] bgp 100
[*Spine1-bgp] ipv4-family vpn-instance vpn1
[*Spine1-bgp-vpn1] import-route static
[*Spine1-bgp-vpn1] advertise l2vpn evpn
[*Spine1-bgp-vpn1] quit
[*Spine1-bgp] ipv6-family vpn-instance vpn1
[*Spine1-bgp-6-vpn1] import-route static
[*Spine1-bgp-6-vpn1] advertise l2vpn evpn
[*Spine1-bgp-6-vpn1] quit
[*Spine1-bgp] quit
[*Spine1] interface nve 1 //Configure an NVE interface.
[*Spine1-Nve1] source 1.1.1.1 //Spine1 and Spine2 establish an M-LAG active-active system. The IP
addresses and MAC addresses of NVE interfaces on the two devices must be the same.
[*Spine1-Nve1] mac-address 0000-5e00-0101
[*Spine1-Nve1] quit
[*Spine1] commit
```

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to
the configuration of Leaf1. For details, see [Configuration Files](#).

```
[~Leaf1] ip vpn-instance vpn1
[*Leaf1-vpn-instance-vpn1] vxlan vni 5000
[*Leaf1-vpn-instance-vpn1] ipv4-family
[*Leaf1-vpn-instance-vpn1-af-ipv4] route-distinguisher 6.6.6.6:1
[*Leaf1-vpn-instance-vpn1-af-ipv4] vpn-target 0:1 evpn
[*Leaf1-vpn-instance-vpn1-af-ipv4] quit
[*Leaf1-vpn-instance-vpn1] ipv6-family
[*Leaf1-vpn-instance-vpn1-af-ipv6] route-distinguisher 6.6.6.6:1
[*Leaf1-vpn-instance-vpn1-af-ipv6] vpn-target 0:1 evpn
[*Leaf1-vpn-instance-vpn1-af-ipv6] quit
[*Leaf1-vpn-instance-vpn1] quit
[*Leaf1] bridge-domain 10
[*Leaf1-bd10] vxlan vni 10
[*Leaf1-bd10] evpn
[*Leaf1-bd10-evpn] route-distinguisher 6.6.6.6:10
[*Leaf1-bd10-evpn] vpn-target 0:10
[*Leaf1-bd10-evpn] vpn-target 0:1 export-extcommunity
[*Leaf1-bd10-evpn] quit
[*Leaf1-bd10] quit
[*Leaf1] bridge-domain 20
[*Leaf1-bd20] vxlan vni 20
[*Leaf1-bd20] evpn
[*Leaf1-bd20-evpn] route-distinguisher 6.6.6.6:20
[*Leaf1-bd20-evpn] vpn-target 0:20
[*Leaf1-bd20-evpn] vpn-target 0:1 export-extcommunity
[*Leaf1-bd20-evpn] quit
[*Leaf1-bd20] quit
[*Leaf1] bgp 100
[*Leaf1-bgp] ipv4-family vpn-instance vpn1
[*Leaf1-bgp-vpn1] import-route direct
[*Leaf1-bgp-vpn1] advertise l2vpn evpn
[*Leaf1-bgp-vpn1] quit
[*Leaf1-bgp] ipv6-family vpn-instance vpn1
[*Leaf1-bgp-6-vpn1] import-route direct
[*Leaf1-bgp-6-vpn1] advertise l2vpn evpn
[*Leaf1-bgp-6-vpn1] quit
[*Leaf1-bgp] quit
```

```
[*Leaf1] interface nve 1 //Configure an NVE interface.
[*Leaf1-Nve1] source 2.2.2.2 //Leaf1 and Leaf2 establish an M-LAG active-active system. The IP
addresses and MAC addresses of NVE interfaces on the two devices must be the same.
[*Leaf1-Nve1] mac-address 0000-5e00-0102
[*Leaf1-Nve1] vni 10 head-end peer-list protocol bgp
[*Leaf1-Nve1] vni 20 head-end peer-list protocol bgp
[*Leaf1-Nve1] quit
[*Leaf1] commit
```

5. Configure service loopback interfaces on Leaf1, Leaf2, Leaf3, and Leaf4. (This step is required only on the CE6850HI, CE6850U-HI, CE6851HI, CE6860EI, CE7850EI, CE8850EI, and CE8860EI.)

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1.

```
[~Leaf1] interface eth-trunk 10
[*Leaf1-Eth-Trunk10] trunkport 10ge 1/0/10 to 1/0/11
[*Leaf1-Eth-Trunk10] service type tunnel
[*Leaf1-Eth-Trunk10] quit
[*Leaf1] commit
```

6. Configure Leaf1, Leaf2, Leaf3, and Leaf4 as Layer 3 gateways.

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1. For details, see [Configuration Files](#).

```
[~Leaf1] interface vbdif 10
[*Leaf1-Vbdif10] ip binding vpn-instance vpn1
[*Leaf1-Vbdif10] ip address 10.1.1.1 24 //Leaf1 and Leaf2 establish an M-LAG active-active system.
The IP addresses and MAC addresses of VBDIF interfaces on the two devices must be the same.
[*Leaf1-Vbdif10] ipv6 enable
[*Leaf1-Vbdif10] ipv6 address fc00:1::1 64
[*Leaf1-Vbdif10] mac-address 0000-5e00-0105
[*Leaf1-Vbdif10] vxlan anycast-gateway enable
[*Leaf1-Vbdif10] arp collect host enable
[*Leaf1-Vbdif10] ipv6 nd collect host enable
[*Leaf1-Vbdif10] quit
[*Leaf1] interface vbdif 20
[*Leaf1-Vbdif20] ip binding vpn-instance vpn1
[*Leaf1-Vbdif20] ip address 10.1.2.1 24
[*Leaf1-Vbdif20] ipv6 enable
[*Leaf1-Vbdif20] ipv6 address fc00:2::1 64
[*Leaf1-Vbdif20] mac-address 0000-5e00-0106
[*Leaf1-Vbdif20] vxlan anycast-gateway enable
[*Leaf1-Vbdif20] arp collect host enable
[*Leaf1-Vbdif20] ipv6 nd collect host enable
[*Leaf1-Vbdif20] quit
[*Leaf1] commit
```

- Step 5** Configure static routes on Spine1 and Spine2 to implement north-south traffic transmission.

Configure Spine1. The configuration of Spine2 is similar to that of Spine1. For details, see [Configuration Files](#).

```
[~Spine1] interface 10ge 1/0/5
[~Spine1-10GE1/0/5] undo portswitch
[*Spine1-10GE1/0/5] ip address 10.1.10.1 24
[*Spine1-10GE1/0/5] ipv6 enable
[*Spine1-10GE1/0/5] ipv6 address fc00:10::1 64
[*Spine1-10GE1/0/5] quit
[~Spine1] interface 10ge 1/0/6
[~Spine1-10GE1/0/6] undo portswitch
[*Spine1-10GE1/0/6] ip address 10.1.20.1 24
[*Spine1-10GE1/0/6] ipv6 enable
[*Spine1-10GE1/0/6] ipv6 address fc00:20::1 64
[*Spine1-10GE1/0/6] quit
[*Spine1] ip route-static 0.0.0.0 0.0.0.0 10.1.10.2 //Configure an IPv4 static route to a PE on the public
network.
[*Spine1] ip route-static 0.0.0.0 0.0.0.0 10.1.20.2
```

```
[*Spine1] ip route-static 10.1.1.0 24 vpn-instance vpn1 //Configure an IPv4 static route to the server
network segment with the next hop being the VPN instance.
[*Spine1] ip route-static 10.1.2.0 24 vpn-instance vpn1
[*Spine1] ip route-static 10.1.3.0 24 vpn-instance vpn1
[*Spine1] ip route-static vpn-instance vpn1 0.0.0.0 0.0.0.0 public //Configure an IPv4 static route of the
VPN instance with the next hop being the public network instance.
[*Spine1] ipv6 route-static :: 0 fc00:10::2 //Configure an IPv6 static route to a PE on the public network.
[*Spine1] ipv6 route-static :: 0 fc00:20::2
[*Spine1] ipv6 route-static fc00:1:: 64 vpn-instance vpn1 //Configure an IPv6 static route to the server
network segment with the next hop being the VPN instance.
[*Spine1] ipv6 route-static fc00:2:: 64 vpn-instance vpn1
[*Spine1] ipv6 route-static fc00:3:: 64 vpn-instance vpn1
[*Spine1] ipv6 route-static vpn-instance vpn1 :: 0 public //Configure an IPv6 static route of the VPN
instance with the next hop being the public network instance.
[*Spine1] interface eth-trunk 10 //If a static route of which only a VPN instance is specified as the next
hop (without any outbound interface or next-hop address specified) is configured on switches excluding the
CE16800, CE12800, CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI,
CE6875EI, CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI, you need to
configure a service loopback interface.
[*Spine1-Eth-Trunk10] trunkport 10ge 1/0/10 to 1/0/11
[*Spine1-Eth-Trunk10] service type tunnel
[*Spine1-Eth-Trunk10] quit
[*Spine1] commit
```

NOTE

When the CE6855HI, CE6856HI, and CE7855EI function as Layer 3 VXLAN gateways, they need to use VBDIF interfaces to connect to external networks.

When the CE6855HI, CE6856HI, and CE7855EI function as Layer 3 VXLAN gateways, you cannot run the **ip route-static vpn-instance vpn-source-name destination-address { mask | mask-length } { public | vpn-instance vpn-destination-name }** command to configure a static route with only a VPN instance specified as the next hop (no outbound interface or next-hop address is specified) to implement inter-VPN communication within a VXLAN network or between a VXLAN network and a non-VXLAN network.

----End

Verifying the Configuration

After the preceding configurations are complete, run the **display vxlan tunnel** command to check VXLAN tunnel information. The following example shows the command output on Spine1.

```
[~Spine1] display vxlan tunnel
Number of vxlan tunnel : 2
Tunnel ID   Source           Destination      State Type    Uptime
-----
4026531841  1.1.1.1          2.2.2.2          up   dynamic  0035h21m
4026531842  1.1.1.1          3.3.3.3          up   dynamic  0036h21m
```

The preceding command output indicates that servers can communicate.

Configuration Files

- Spine1 configuration file

```
#
sysname Spine1
#
assign forward nvo3 acl extend enable //This step is required only on the CE12800, CE16800
equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
priority 150
source ip 4.4.4.4
#
```



```
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
 route-distinguisher 4.4.4.4:1
 vpn-target 0:1 export-extcommunity evpn
 vpn-target 0:1 import-extcommunity evpn
ipv6-family
 route-distinguisher 4.4.4.4:1
 vpn-target 0:1 export-extcommunity evpn
 vpn-target 0:1 import-extcommunity evpn
vxlan vni 5000
#
interface Eth-Trunk1
 mode lacp-static
 peer-link 1
#
interface Eth-Trunk10 //This step is required only on switches excluding the CE16800, CE12800,
 CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI, CE6875EI,
 CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI.
 service type tunnel
#
interface 10GE1/0/1
 undo portswitch
 ip address 192.168.1.1 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/2
 undo portswitch
 ip address 192.168.2.1 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/3
 undo portswitch
 ip address 192.168.3.1 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/4
 undo portswitch
 ip address 192.168.4.1 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/5
 undo portswitch
 ipv6 enable
 ip address 10.1.10.1 255.255.255.0
 ipv6 address FC00:10::1/64
#
interface 10GE1/0/6
 undo portswitch
 ipv6 enable
 ip address 10.1.20.1 255.255.255.0
 ipv6 address FC00:20::1/64
#
interface 10GE1/0/7
 eth-trunk 1
#
interface 10GE1/0/8
 eth-trunk 1
#
interface 10GE1/0/10 //This step is required only on switches excluding the CE16800, CE12800,
 CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI, CE6875EI,
 CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI.
 eth-trunk 10
#
interface 10GE1/0/11 //This step is required only on switches excluding the CE16800, CE12800,
```

CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI, CE6875EI, CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI.

```
eth-trunk 10
#
interface LoopBack0
 ip address 4.4.4.4 255.255.255.255
#
interface LoopBack1
 ip address 1.1.1.1 255.255.255.255
#
interface Nve1
 source 1.1.1.1
 mac-address 0000-5e00-0101
#
bgp 100
 peer 6.6.6.6 as-number 100
 peer 6.6.6.6 connect-interface LoopBack0
 peer 7.7.7.7 as-number 100
 peer 7.7.7.7 connect-interface LoopBack0
 peer 8.8.8.8 as-number 100
 peer 8.8.8.8 connect-interface LoopBack0
 peer 9.9.9.9 as-number 100
 peer 9.9.9.9 connect-interface LoopBack0
#
ipv4-family unicast
 peer 6.6.6.6 enable
 peer 7.7.7.7 enable
 peer 8.8.8.8 enable
 peer 9.9.9.9 enable
#
ipv4-family vpn-instance vpn1
 import-route static
 advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
 import-route static
 advertise l2vpn evpn
#
l2vpn-family evpn
 undo policy vpn-target
 peer 6.6.6.6 enable
 peer 6.6.6.6 advertise irb
 peer 6.6.6.6 advertise irbv6
 peer 6.6.6.6 reflect-client
 peer 7.7.7.7 enable
 peer 7.7.7.7 advertise irb
 peer 7.7.7.7 advertise irbv6
 peer 7.7.7.7 reflect-client
 peer 8.8.8.8 enable
 peer 8.8.8.8 advertise irb
 peer 8.8.8.8 advertise irbv6
 peer 8.8.8.8 reflect-client
 peer 9.9.9.9 enable
 peer 9.9.9.9 advertise irb
 peer 9.9.9.9 advertise irbv6
 peer 9.9.9.9 reflect-client
#
ospf 1
 area 0.0.0.0
 network 1.1.1.1 0.0.0.0
 network 4.4.4.4 0.0.0.0
 network 192.168.1.0 0.0.0.255
 network 192.168.2.0 0.0.0.255
 network 192.168.3.0 0.0.0.255
 network 192.168.4.0 0.0.0.255
#
ip route-static 0.0.0.0 0.0.0.0 10.1.10.2
ip route-static 0.0.0.0 0.0.0.0 10.1.20.2
ip route-static 10.1.1.0 255.255.255.0 vpn-instance vpn1
```

```
ip route-static 10.1.2.0 255.255.255.0 vpn-instance vpn1
ip route-static 10.1.3.0 255.255.255.0 vpn-instance vpn1
ip route-static vpn-instance vpn1 0.0.0.0 0.0.0.0 public
#
ipv6 route-static :: 0 FC00:10::2
ipv6 route-static :: 0 FC00:20::2
ipv6 route-static FC00:1:: 64 vpn-instance vpn1
ipv6 route-static FC00:2:: 64 vpn-instance vpn1
ipv6 route-static FC00:3:: 64 vpn-instance vpn1
ipv6 route-static vpn-instance vpn1 :: 0 public
#
return
```

- Spine2 configuration file

```
#
sysname Spine2
#
assign forward nvo3 acl extend enable //This step is required only on the CE12800, CE16800
equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
source ip 5.5.5.5
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 5.5.5.5:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 5.5.5.5:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 5000
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk10 //This step is required only on switches excluding the CE16800, CE12800,
CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI, CE6875EI,
CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI.
service type tunnel
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.5.1 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.6.1 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/3
undo portswitch
ip address 192.168.7.1 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/4
undo portswitch
ip address 192.168.8.1 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/5
undo portswitch
```

```
ipv6 enable
ip address 10.1.30.1 255.255.255.0
ipv6 address FC00:30::1/64
#
interface 10GE1/0/6
undo portswitch
ipv6 enable
ip address 10.1.40.1 255.255.255.0
ipv6 address FC00:40::1/64
#
interface 10GE1/0/7
eth-trunk 1
#
interface 10GE1/0/8
eth-trunk 1
#
interface 10GE1/0/10 //This step is required only on switches excluding the CE16800, CE12800,
CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI, CE6875EI,
CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI.
eth-trunk 10
#
interface 10GE1/0/11 //This step is required only on switches excluding the CE16800, CE12800,
CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI, CE6875EI,
CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI.
eth-trunk 10
#
interface LoopBack0
ip address 5.5.5.5 255.255.255.255
#
interface LoopBack1
ip address 1.1.1.1 255.255.255.255
#
interface Nve1
source 1.1.1.1
mac-address 0000-5e00-0101
#
bgp 100
peer 6.6.6.6 as-number 100
peer 6.6.6.6 connect-interface LoopBack0
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack0
peer 8.8.8.8 as-number 100
peer 8.8.8.8 connect-interface LoopBack0
peer 9.9.9.9 as-number 100
peer 9.9.9.9 connect-interface LoopBack0
#
ipv4-family unicast
peer 6.6.6.6 enable
peer 7.7.7.7 enable
peer 8.8.8.8 enable
peer 9.9.9.9 enable
#
ipv4-family vpn-instance vpn1
import-route static
advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
import-route static
advertise l2vpn evpn
#
l2vpn-family evpn
undo policy vpn-target
peer 6.6.6.6 enable
peer 6.6.6.6 advertise irb
peer 6.6.6.6 advertise irbv6
peer 6.6.6.6 reflect-client
peer 7.7.7.7 enable
peer 7.7.7.7 advertise irb
peer 7.7.7.7 advertise irbv6
```

```
peer 7.7.7.7 reflect-client
peer 8.8.8.8 enable
peer 8.8.8.8 advertise irb
peer 8.8.8.8 advertise irbv6
peer 8.8.8.8 reflect-client
peer 9.9.9.9 enable
peer 9.9.9.9 advertise irb
peer 9.9.9.9 advertise irbv6
peer 9.9.9.9 reflect-client
#
ospf 1
area 0.0.0.0
network 1.1.1.1 0.0.0.0
network 5.5.5.5 0.0.0.0
network 192.168.5.0 0.0.0.255
network 192.168.6.0 0.0.0.255
network 192.168.7.0 0.0.0.255
network 192.168.8.0 0.0.0.255
#
ip route-static 0.0.0.0 0.0.0.0 10.1.30.2
ip route-static 0.0.0.0 0.0.0.0 10.1.40.2
ip route-static 10.1.1.0 255.255.255.0 vpn-instance vpn1
ip route-static 10.1.2.0 255.255.255.0 vpn-instance vpn1
ip route-static 10.1.3.0 255.255.255.0 vpn-instance vpn1
ip route-static vpn-instance vpn1 0.0.0.0 0.0.0.0 public
#
ipv6 route-static :: 0 FC00:30::2
ipv6 route-static :: 0 FC00:40::2
ipv6 route-static FC00:1:: 64 vpn-instance vpn1
ipv6 route-static FC00:2:: 64 vpn-instance vpn1
ipv6 route-static FC00:3:: 64 vpn-instance vpn1
ipv6 route-static vpn-instance vpn1 :: 0 public
#
return
```

- Leaf1 configuration file

```
#
sysname Leaf1
#
assign forward nvo3 acl extend enable //This step is required only on the CE12800, CE16800
equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
priority 150
source ip 6.6.6.6
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 6.6.6.6:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 6.6.6.6:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 5000
#
bridge-domain 10
vxlan vni 10
evpn
route-distinguisher 6.6.6.6:10
vpn-target 0:10 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:10 import-extcommunity
#
```

```
bridge-domain 20
vxlan vni 20
evpn
 route-distinguisher 6.6.6.6:20
 vpn-target 0:20 export-extcommunity
 vpn-target 0:1 export-extcommunity
 vpn-target 0:20 import-extcommunity
#
interface Vbdif10
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.1.1.1 255.255.255.0
 ipv6 address FC00:1::1/64
 mac-address 0000-5e00-0105
 ipv6 nd collect host enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Vbdif20
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.1.2.1 255.255.255.0
 ipv6 address FC00:2::1/64
 mac-address 0000-5e00-0106
 ipv6 nd collect host enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Eth-Trunk1
 mode lacp-static
 peer-link 1
#
interface Eth-Trunk2
 stp edged-port enable
 mode lacp-static
 dfs-group 1 m-lag 1
#
interface Eth-Trunk2.10 mode l2
 encapsulation dot1q vid 10
 bridge-domain 10
#
interface Eth-Trunk3
 stp edged-port enable
 mode lacp-static
 dfs-group 1 m-lag 2
#
interface Eth-Trunk3.20 mode l2
 encapsulation dot1q vid 20
 bridge-domain 20
#
interface Eth-Trunk10 //This step is required only on the CE6850HI, CE6850U-HI, CE6851HI,
CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
 service type tunnel
#
interface 10GE1/0/1
 undo portswitch
 ip address 192.168.1.2 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/2
 undo portswitch
 ip address 192.168.5.2 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/3
 eth-trunk 1
#
interface 10GE1/0/4
 eth-trunk 1
```

```
#
interface 10GE1/0/5
eth-trunk 2
#
interface 10GE1/0/6
eth-trunk 3
#
interface 10GE1/0/10 //This step is required only on the CE6850HI, CE6850U-HI, CE6851HI,
CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
eth-trunk 10
#
interface 10GE1/0/11 //This step is required only on the CE6850HI, CE6850U-HI, CE6851HI,
CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
eth-trunk 10
#
interface LoopBack0
ip address 6.6.6.6 255.255.255.255
#
interface LoopBack1
ip address 2.2.2.2 255.255.255.255
#
interface Nve1
source 2.2.2.2
vni 10 head-end peer-list protocol bgp
vni 20 head-end peer-list protocol bgp
mac-address 0000-5e00-0102
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk2 downlink 1
port Eth-Trunk3 downlink 2
#
bgp 100
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack0
peer 5.5.5.5 as-number 100
peer 5.5.5.5 connect-interface LoopBack0
#
ipv4-family unicast
peer 4.4.4.4 enable
peer 5.5.5.5 enable
#
ipv4-family vpn-instance vpn1
import-route direct
advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
import-route direct
advertise l2vpn evpn
#
l2vpn-family evpn
policy vpn-target
peer 4.4.4.4 enable
peer 4.4.4.4 advertise irb
peer 4.4.4.4 advertise irbv6
peer 5.5.5.5 enable
peer 5.5.5.5 advertise irb
peer 5.5.5.5 advertise irbv6
#
ospf 1
area 0.0.0.0
network 2.2.2.2 0.0.0.0
network 6.6.6.6 0.0.0.0
network 192.168.1.0 0.0.0.255
network 192.168.5.0 0.0.0.255
#
return
```

- Leaf2 configuration file

```
#
sysname Leaf2
#
assign forward nvo3 acl extend enable //This step is required only on the CE12800, CE16800
equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
source ip 7.7.7.7
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 7.7.7.7:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 7.7.7.7:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 5000
#
bridge-domain 10
vxlan vni 10
evpn
route-distinguisher 7.7.7.7:10
vpn-target 0:10 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:10 import-extcommunity
#
bridge-domain 20
vxlan vni 20
evpn
route-distinguisher 7.7.7.7:20
vpn-target 0:20 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:20 import-extcommunity
#
interface Vbdif10
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.1.1.1 255.255.255.0
ipv6 address FC00:1::1/64
mac-address 0000-5e00-0105
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Vbdif20
ip binding vpn-instance vpn1
ipv6 enable
ipv6 address FC00:2::1/64
ip address 10.1.2.1 255.255.255.0
mac-address 0000-5e00-0106
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk2
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 1
```



```
#
interface Eth-Trunk2.10 mode l2
 encapsulation dot1q vid 10
 bridge-domain 10
#
interface Eth-Trunk3
 stp edged-port enable
 mode lacp-static
 dfs-group 1 m-lag 2
#
interface Eth-Trunk3.20 mode l2
 encapsulation dot1q vid 20
 bridge-domain 20
#
interface Eth-Trunk10 //This step is required only on the CE6850HI, CE6850U-HI, CE6851HI,
CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
 service type tunnel
#
interface 10GE1/0/1
 undo portswitch
 ip address 192.168.2.2 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/2
 undo portswitch
 ip address 192.168.6.2 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/3
 eth-trunk 1
#
interface 10GE1/0/4
 eth-trunk 1
#
interface 10GE1/0/5
 eth-trunk 2
#
interface 10GE1/0/6
 eth-trunk 3
#
interface 10GE1/0/10 //This step is required only on the CE6850HI, CE6850U-HI, CE6851HI,
CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
 eth-trunk 10
#
interface 10GE1/0/11 //This step is required only on the CE6850HI, CE6850U-HI, CE6851HI,
CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
 eth-trunk 10
#
interface LoopBack0
 ip address 7.7.7.7 255.255.255.255
#
interface LoopBack1
 ip address 2.2.2.2 255.255.255.255
#
interface Nve1
 source 2.2.2.2
 vni 10 head-end peer-list protocol bgp
 vni 20 head-end peer-list protocol bgp
 mac-address 0000-5e00-0102
#
monitor-link group 1
 port 10GE1/0/1 uplink
 port 10GE1/0/2 uplink
 port Eth-Trunk2 downlink 1
 port Eth-Trunk3 downlink 2
#
bgp 100
 peer 4.4.4.4 as-number 100
 peer 4.4.4.4 connect-interface LoopBack0
```

```
peer 5.5.5.5 as-number 100
peer 5.5.5.5 connect-interface LoopBack0
#
ipv4-family unicast
peer 4.4.4.4 enable
peer 5.5.5.5 enable
#
ipv4-family vpn-instance vpn1
import-route direct
advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
import-route direct
advertise l2vpn evpn
#
l2vpn-family evpn
policy vpn-target
peer 4.4.4.4 enable
peer 4.4.4.4 advertise irb
peer 4.4.4.4 advertise irbv6
peer 5.5.5.5 enable
peer 5.5.5.5 advertise irb
peer 5.5.5.5 advertise irbv6
#
ospf 1
area 0.0.0.0
network 2.2.2.2 0.0.0.0
network 7.7.7.7 0.0.0.0
network 192.168.2.0 0.0.0.255
network 192.168.6.0 0.0.0.255
#
return
```

- Leaf3 configuration file

```
#
sysname Leaf3
#
assign forward nvo3 acl extend enable //This step is required only on the CE12800, CE16800
equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
priority 150
source ip 8.8.8.8
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 8.8.8.8:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 8.8.8.8:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 5000
#
bridge-domain 20
vxlan vni 20
evpn
route-distinguisher 8.8.8.8:20
vpn-target 0:20 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:20 import-extcommunity
#
bridge-domain 30
vxlan vni 30
```

```
evpn
 route-distinguisher 8.8.8.8:30
 vpn-target 0:30 export-extcommunity
 vpn-target 0:1 export-extcommunity
 vpn-target 0:30 import-extcommunity
#
interface Vbdfif20
 ip binding vpn-instance vpn1
 ipv6 enable
 ipv6 address FC00:2::1/64
 ip address 10.1.2.1 255.255.255.0
 mac-address 0000-5e00-0106
 ipv6 nd collect host enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Vbdfif30
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.1.3.1 255.255.255.0
 ipv6 address FC00:3::1/64
 mac-address 0000-5e00-0107
 ipv6 nd collect host enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Eth-Trunk1
 mode lacp-static
 peer-link 1
#
interface Eth-Trunk2
 stp edged-port enable
 mode lacp-static
 dfs-group 1 m-lag 1
#
interface Eth-Trunk2.20 mode l2
 encapsulation dot1q vid 20
 bridge-domain 20
#
interface Eth-Trunk3
 stp edged-port enable
 mode lacp-static
 dfs-group 1 m-lag 2
#
interface Eth-Trunk3.30 mode l2
 encapsulation dot1q vid 30
 bridge-domain 30
#
interface Eth-Trunk10 //This step is required only on the CE6850HI, CE6850U-HI, CE6851HI,
CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
 service type tunnel
#
interface 10GE1/0/1
 undo portswitch
 ip address 192.168.3.2 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/2
 undo portswitch
 ip address 192.168.7.2 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/3
 eth-trunk 1
#
interface 10GE1/0/4
 eth-trunk 1
#
interface 10GE1/0/5
```

```
eth-trunk 2
#
interface 10GE1/0/6
eth-trunk 3
#
interface 10GE1/0/10 //This step is required only on the CE6850HI, CE6850U-HI, CE6851HI,
CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
eth-trunk 10
#
interface 10GE1/0/11 //This step is required only on the CE6850HI, CE6850U-HI, CE6851HI,
CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
eth-trunk 10
#
interface LoopBack0
ip address 8.8.8.8 255.255.255.255
#
interface LoopBack1
ip address 3.3.3.3 255.255.255.255
#
interface Nve1
source 3.3.3.3
vni 20 head-end peer-list protocol bgp
vni 30 head-end peer-list protocol bgp
mac-address 0000-5e00-0103
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk2 downlink 1
port Eth-Trunk3 downlink 2
#
bgp 100
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack0
peer 5.5.5.5 as-number 100
peer 5.5.5.5 connect-interface LoopBack0
#
ipv4-family unicast
peer 4.4.4.4 enable
peer 5.5.5.5 enable
#
ipv4-family vpn-instance vpn1
import-route direct
advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
import-route direct
advertise l2vpn evpn
#
l2vpn-family evpn
policy vpn-target
peer 4.4.4.4 enable
peer 4.4.4.4 advertise irb
peer 4.4.4.4 advertise irbv6
peer 5.5.5.5 enable
peer 5.5.5.5 advertise irb
peer 5.5.5.5 advertise irbv6
#
ospf 1
area 0.0.0.0
network 3.3.3.3 0.0.0.0
network 8.8.8.8 0.0.0.0
network 192.168.3.0 0.0.0.255
network 192.168.7.0 0.0.0.255
#
return
```

- Leaf4 configuration file

```
#
sysname Leaf4
```

```
#
assign forward nvo3 acl extend enable //This step is required only on the CE12800, CE16800
equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
source ip 9.9.9.9
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 9.9.9.9:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 9.9.9.9:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 5000
#
bridge-domain 20
vxlan vni 20
evpn
route-distinguisher 9.9.9.9:20
vpn-target 0:20 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:20 import-extcommunity
#
bridge-domain 30
vxlan vni 30
evpn
route-distinguisher 9.9.9.9:30
vpn-target 0:30 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:30 import-extcommunity
#
interface Vbdif20
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.1.2.1 255.255.255.0
ipv6 address FC00:2::1/64
mac-address 0000-5e00-0106
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Vbdif30
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.1.3.1 255.255.255.0
ipv6 address FC00:3::1/64
mac-address 0000-5e00-0107
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk2
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 1
#
interface Eth-Trunk2.20 mode l2
```

```
encapsulation dot1q vid 20
bridge-domain 20
#
interface Eth-Trunk3
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 2
#
interface Eth-Trunk3.30 mode l2
encapsulation dot1q vid 30
bridge-domain 30
#
interface Eth-Trunk10 //This step is required only on the CE6850HI, CE6850U-HI, CE6851HI,
CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
service type tunnel
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.4.2 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.8.2 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/3
eth-trunk 1
#
interface 10GE1/0/4
eth-trunk 1
#
interface 10GE1/0/5
eth-trunk 2
#
interface 10GE1/0/6
eth-trunk 3
#
interface 10GE1/0/10 //This step is required only on the CE6850HI, CE6850U-HI, CE6851HI,
CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
eth-trunk 10
#
interface 10GE1/0/11 //This step is required only on the CE6850HI, CE6850U-HI, CE6851HI,
CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
eth-trunk 10
#
interface LoopBack0
ip address 9.9.9.9 255.255.255.255
#
interface LoopBack1
ip address 3.3.3.3 255.255.255.255
#
interface Nve1
source 3.3.3.3
vni 20 head-end peer-list protocol bgp
vni 30 head-end peer-list protocol bgp
mac-address 0000-5e00-0103
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk2 downlink 1
port Eth-Trunk3 downlink 2
#
bgp 100
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack0
peer 5.5.5.5 as-number 100
peer 5.5.5.5 connect-interface LoopBack0
```

```
#
ipv4-family unicast
peer 4.4.4.4 enable
peer 5.5.5.5 enable
#
ipv4-family vpn-instance vpn1
import-route direct
advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
import-route direct
advertise l2vpn evpn
#
l2vpn-family evpn
policy vpn-target
peer 4.4.4.4 enable
peer 4.4.4.4 advertise irb
peer 4.4.4.4 advertise irbv6
peer 5.5.5.5 enable
peer 5.5.5.5 advertise irb
peer 5.5.5.5 advertise irbv6
#
ospf 1
area 0.0.0.0
network 3.3.3.3 0.0.0.0
network 9.9.9.9 0.0.0.0
network 192.168.4.0 0.0.0.255
network 192.168.8.0 0.0.0.255
#
return
```

1.5 Example for Configuring IPv6 VXLAN in Distributed Gateway Deployment Mode

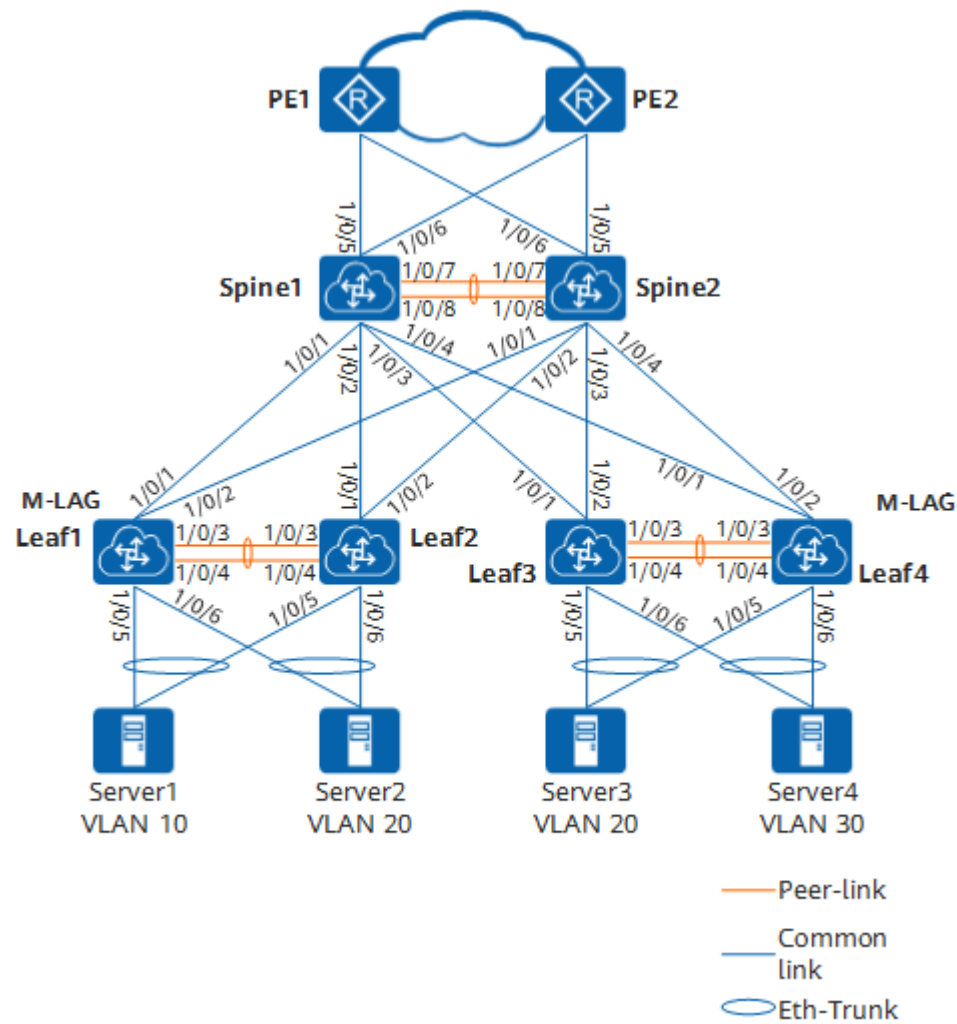
Applicable Products and Versions

- This example applies to CE16800 equipped with G/GK series cards, CE6863, CE6863E, CE6863K, CE6881, CE6881K, CE6881E, and CE5881 series switches running V200R020C00 or later versions.
- For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

In [Figure 1-5](#), the new data center network of an enterprise uses the distributed gateway deployment mode. The underlay network is an IPv6 network, and the overlay network is an IPv4 or IPv6 network. Leaf nodes function as Layer 3 gateways and connect to servers. Spine nodes function as aggregation devices for east-to-west traffic and egress gateways. To ensure high reliability, spine and leaf nodes establish M-LAGs.

Figure 1-5 VXLAN in distributed gateway deployment mode



NOTE

In the preceding figure, 1/0/1 indicates an interface number and the interface rate is 10 Gbit/s. That is, 1/0/1 indicates 10GE1/0/1. Other interfaces are specified in a similar way.

Table 1-5 IP addresses of interfaces

Device Name	Interface	IP Address	Device Name	Interface	IP Address
Spine 1	10GE1/0/1	2001:db8:1:d301::1/64	Spine 2	10GE1/0/1	2001:db8:1:d305::1/64
	10GE1/0/2	2001:db8:1:d302::1/64		10GE1/0/2	2001:db8:1:d306::1/64
	10GE1/0/3	2001:db8:1:d303::1/64		10GE1/0/3	2001:db8:1:d307::1/64

Device Name	Interface	IP Address	Device Name	Interface	IP Address
	10GE1/0/4	2001:db8:1:d304::1/64		10GE1/0/4	2001:db8:1:d308::1/64
	10GE1/0/5	IPv4: 10.1.10.1/24 IPv6: fc00:10::1/64		10GE1/0/5	IPv4: 10.1.30.1/24 IPv6: fc00:30::1/64
	10GE1/0/6	IPv4: 10.1.20.1/24 IPv6: fc00:20::1/64		10GE1/0/6	IPv4: 10.1.40.1/24 IPv6: fc00:40::1/64
	Loopback0	4.4.4.4/32		Loopback0	5.5.5.5/32
	Loopback1	2001:db8:1:1301::1/128		Loopback1	2001:db8:1:1301::1/128
	Loopback2	2001:db8:1:4301::1/128		Loopback2	2001:db8:1:5301::1/128
Leaf1	10GE1/0/1	2001:db8:1:d301::2/64	Leaf2	10GE1/0/1	2001:db8:1:d302::2/64
	10GE1/0/2	2001:db8:1:d305::2/64		10GE1/0/2	2001:db8:1:d306::2/64
	Loopback0	6.6.6.6/32		Loopback0	7.7.7.7/32
	Loopback1	2001:db8:1:2301::1/128		Loopback1	2001:db8:1:2301::1/128
	Loopback2	2001:db8:1:6301::1/128		Loopback2	2001:db8:1:7301::1/128
Leaf3	10GE1/0/1	2001:db8:1:d303::2/64	Leaf4	10GE1/0/1	2001:db8:1:d304::2/64
	10GE1/0/2	2001:db8:1:d307::2/64		10GE1/0/2	2001:db8:1:d308::2/64
	Loopback0	8.8.8.8/32		Loopback0	9.9.9.9/32
	Loopback1	2001:db8:1:3301::1/128		Loopback1	2001:db8:1:3301::1/128
	Loopback2	2001:db8:1:8301::1/128		Loopback2	2001:db8:1:9301::1/128

Configuration Roadmap

The configuration roadmap is as follows:

1. Configure a routing protocol to ensure Layer 3 connectivity on the underlay network.
2. Configure M-LAG to implement active-active server access.
3. Configure BGP EVPN to establish a VXLAN tunnel.

Procedure

Step 1 Configure a routing protocol to implement Layer 3 connectivity on the underlay network.

Configure Leaf1. The configurations of other devices are similar to the configuration of Leaf1. For details, see [Configuration Files](#).

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf1
[*HUAWEI] commit
[~Leaf1] ospfv3
[*Leaf1-ospfv3-1] router-id 6.6.6.6
[*Leaf1-ospfv3-1] quit
[*Leaf1] interface 10ge 1/0/1
[*Leaf1-10GE1/0/1] undo portswitch
[*Leaf1-10GE1/0/1] ipv6 enable
[*Leaf1-10GE1/0/1] ipv6 address 2001:db8:1:d301::2 64
[*Leaf1-10GE1/0/1] ospfv3 1 area 0
[*Leaf1-10GE1/0/1] ospfv3 network-type p2p
[*Leaf1-10GE1/0/1] quit
[*Leaf1] interface 10ge 1/0/2
[*Leaf1-10GE1/0/2] undo portswitch
[*Leaf1-10GE1/0/2] ipv6 enable
[*Leaf1-10GE1/0/2] ipv6 address 2001:db8:1:d305::2 64
[*Leaf1-10GE1/0/2] ospfv3 1 area 0
[*Leaf1-10GE1/0/2] ospfv3 network-type p2p
[*Leaf1-10GE1/0/2] quit
[*Leaf1] interface loopback 0
[*Leaf1-LoopBack0] ip address 6.6.6.6 32
[*Leaf1-LoopBack0] quit
[*Leaf1] interface loopback 1
[*Leaf1-LoopBack1] ipv6 enable
[*Leaf1-LoopBack1] ipv6 address 2001:db8:1:2301::1 128
[*Leaf1-LoopBack1] ospfv3 1 area 0
[*Leaf1-LoopBack1] quit
[*Leaf1] interface loopback 2
[*Leaf1-LoopBack2] ipv6 enable
[*Leaf1-LoopBack2] ipv6 address 2001:db8:1:6301::1 128
[*Leaf1-LoopBack2] ospfv3 1 area 0
[*Leaf1-LoopBack2] quit
[*Leaf1] commit
```

After OSPFv3 is configured, leaf and spine nodes can learn the IPv6 address of each other's loopback interface using OSPFv3 and successfully ping each other.

Step 2 Configure leaf nodes to establish M-LAGs. In this example, Leaf1 and Leaf2 establish an M-LAG. Leaf3 and Leaf4 establish an M-LAG. The configurations of Leaf3 and Leaf4 are similar to those of Leaf1 and Leaf2.

Configure Leaf1.

```
[~Leaf1] stp mode rstp
[*Leaf1] stp v-stp enable
[*Leaf1] dfs-group 1
```

```
[*Leaf1-dfs-group-1] source ipv6 2001:db8:1:6301::1
[*Leaf1-dfs-group-1] priority 150
[*Leaf1-dfs-group-1] quit
[*Leaf1] interface eth-trunk 1
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/3
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/4
[*Leaf1-Eth-Trunk1] mode lacp-static
[*Leaf1-Eth-Trunk1] peer-link 1
[*Leaf1-Eth-Trunk1] quit
[*Leaf1] interface eth-trunk 2
[*Leaf1-Eth-Trunk2] trunkport 10ge 1/0/5
[*Leaf1-Eth-Trunk2] mode lacp-static
[*Leaf1-Eth-Trunk2] dfs-group 1 m-lag 1
[*Leaf1-Eth-Trunk2] stp edged-port enable
[*Leaf1-Eth-Trunk2] quit
[*Leaf1] interface eth-trunk 3
[*Leaf1-Eth-Trunk3] trunkport 10ge 1/0/6
[*Leaf1-Eth-Trunk3] mode lacp-static
[*Leaf1-Eth-Trunk3] dfs-group 1 m-lag 2
[*Leaf1-Eth-Trunk3] stp edged-port enable
[*Leaf1-Eth-Trunk3] quit
[*Leaf1] commit
[~Leaf1] monitor-link group 1
[*Leaf1-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf1-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf1-mtlk-group1] port eth-trunk 2 downlink 1
[*Leaf1-mtlk-group1] port eth-trunk 3 downlink 2
[*Leaf1-mtlk-group1] quit
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] stp mode rstp
[*Leaf2] stp v-stp enable
[*Leaf2] dfs-group 1
[*Leaf2-dfs-group-1] source ipv6 2001:db8:1:7301::1
[*Leaf2-dfs-group-1] quit
[*Leaf2] interface eth-trunk 1
[*Leaf2-Eth-Trunk1] trunkport 10ge 1/0/3
[*Leaf2-Eth-Trunk1] trunkport 10ge 1/0/4
[*Leaf2-Eth-Trunk1] mode lacp-static
[*Leaf2-Eth-Trunk1] peer-link 1
[*Leaf2-Eth-Trunk1] quit
[*Leaf2] interface eth-trunk 2
[*Leaf2-Eth-Trunk2] mode lacp-static
[*Leaf2-Eth-Trunk2] trunkport 10ge 1/0/5
[*Leaf2-Eth-Trunk2] dfs-group 1 m-lag 1
[*Leaf2-Eth-Trunk2] stp edged-port enable
[*Leaf2-Eth-Trunk2] quit
[*Leaf2] interface eth-trunk 3
[*Leaf2-Eth-Trunk3] mode lacp-static
[*Leaf2-Eth-Trunk3] trunkport 10ge 1/0/6
[*Leaf2-Eth-Trunk3] dfs-group 1 m-lag 2
[*Leaf2-Eth-Trunk3] stp edged-port enable
[*Leaf2-Eth-Trunk3] quit
[*Leaf2] commit
[~Leaf2] monitor-link group 1
[*Leaf2-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf2-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf2-mtlk-group1] port eth-trunk 2 downlink 1
[*Leaf2-mtlk-group1] port eth-trunk 3 downlink 2
[*Leaf2-mtlk-group1] quit
[*Leaf2] commit
```

Step 3 Configure Spine1 and Spine2 to establish an M-LAG.

Configure Spine1.

```
[~Spine1] stp mode rstp
[*Spine1] stp v-stp enable
```

```
[*Spine1] dfs-group 1
[*Spine1-dfs-group-1] source ipv6 2001:db8:1:4301::1
[*Spine1-dfs-group-1] priority 150
[*Spine1-dfs-group-1] quit
[*Spine1] interface eth-trunk 1
[*Spine1-Eth-Trunk1] trunkport 10ge 1/0/7
[*Spine1-Eth-Trunk1] trunkport 10ge 1/0/8
[*Spine1-Eth-Trunk1] mode lacp-static
[*Spine1-Eth-Trunk1] peer-link 1
[*Spine1-Eth-Trunk1] quit
[*Spine1] commit
```

Configure Spine2.

```
[~Spine2] stp mode rstp
[*Spine2] stp v-stp enable
[*Spine2] dfs-group 1
[*Spine2-dfs-group-1] source ipv6 2001:db8:1:5301::1
[*Spine2-dfs-group-1] quit
[*Spine2] interface eth-trunk 1
[*Spine2-Eth-Trunk1] trunkport 10ge 1/0/7
[*Spine2-Eth-Trunk1] trunkport 10ge 1/0/8
[*Spine2-Eth-Trunk1] mode lacp-static
[*Spine2-Eth-Trunk1] peer-link 1
[*Spine2-Eth-Trunk1] quit
[*Spine2] commit
```

Step 4 Configure BGP EVPN to establish a VXLAN tunnel.

1. Configure service access points.

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1.

```
[~Leaf1] bridge-domain 10
[*Leaf1-bd10] quit
[*Leaf1] bridge-domain 20
[*Leaf1-bd20] quit
[*Leaf1] interface eth-trunk 2.10 mode l2
[*Leaf1-Eth-Trunk2.1] encapsulation dot1q vid 10
[*Leaf1-Eth-Trunk2.1] bridge-domain 10
[*Leaf1-Eth-Trunk2.1] quit
[*Leaf1] interface eth-trunk 3.20 mode l2
[*Leaf1-Eth-Trunk3.1] encapsulation dot1q vid 20
[*Leaf1-Eth-Trunk3.1] bridge-domain 20
[*Leaf1-Eth-Trunk3.1] quit
[*Leaf1] commit
```

2. Establish BGP EVPN peer relationships. Spine1 and Spine2 function as route reflectors (RRs).

Configure Spine1. The configuration of Spine2 is similar to that of Spine1. For details, see [Configuration Files](#).

```
[~Spine1] evpn-overlay enable
[*Spine1] bgp 100
[*Spine1-bgp] peer 2001:db8:1:6301::1 as-number 100
[*Spine1-bgp] peer 2001:db8:1:6301::1 connect-interface LoopBack2
[*Spine1-bgp] peer 2001:db8:1:7301::1 as-number 100
[*Spine1-bgp] peer 2001:db8:1:7301::1 connect-interface LoopBack2
[*Spine1-bgp] peer 2001:db8:1:8301::1 as-number 100
[*Spine1-bgp] peer 2001:db8:1:8301::1 connect-interface LoopBack2
[*Spine1-bgp] peer 2001:db8:1:9301::1 as-number 100
[*Spine1-bgp] peer 2001:db8:1:9301::1 connect-interface LoopBack2
[*Spine1-bgp] l2vpn-family evpn
[*Spine1-bgp-af-evpn] peer 2001:db8:1:6301::1 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 2001:db8:1:6301::1 advertise irb
[*Spine1-bgp-af-evpn] peer 2001:db8:1:6301::1 advertise irbv6
[*Spine1-bgp-af-evpn] peer 2001:db8:1:6301::1 reflect-client
[*Spine1-bgp-af-evpn] peer 2001:db8:1:7301::1 enable
```

```
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 2001:db8:1:7301::1 advertise irb
[*Spine1-bgp-af-evpn] peer 2001:db8:1:7301::1 advertise irbv6
[*Spine1-bgp-af-evpn] peer 2001:db8:1:7301::1 reflect-client
[*Spine1-bgp-af-evpn] peer 2001:db8:1:8301::1 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 2001:db8:1:8301::1 advertise irb
[*Spine1-bgp-af-evpn] peer 2001:db8:1:8301::1 advertise irbv6
[*Spine1-bgp-af-evpn] peer 2001:db8:1:8301::1 reflect-client
[*Spine1-bgp-af-evpn] peer 2001:db8:1:9301::1 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 2001:db8:1:9301::1 advertise irb
[*Spine1-bgp-af-evpn] peer 2001:db8:1:9301::1 advertise irbv6
[*Spine1-bgp-af-evpn] peer 2001:db8:1:9301::1 reflect-client
[*Spine1-bgp-af-evpn] undo policy vpn-target
[*Spine1-bgp-af-evpn] quit
[*Spine1-bgp] quit
[*Spine1] commit
```

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1. For details, see [Configuration Files](#).

```
[~Leaf1] evpn-overlay enable
[*Leaf1] bgp 100
[*Leaf1-bgp] peer 2001:db8:1:4301::1 as-number 100
[*Leaf1-bgp] peer 2001:db8:1:4301::1 connect-interface LoopBack2
[*Leaf1-bgp] peer 2001:db8:1:5301::1 as-number 100
[*Leaf1-bgp] peer 2001:db8:1:5301::1 connect-interface LoopBack2
[*Leaf1-bgp] l2vpn-family evpn
[*Leaf1-bgp-af-evpn] peer 2001:db8:1:4301::1 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 2001:db8:1:4301::1 advertise irb
[*Leaf1-bgp-af-evpn] peer 2001:db8:1:4301::1 advertise irbv6
[*Leaf1-bgp-af-evpn] peer 2001:db8:1:5301::1 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 2001:db8:1:5301::1 advertise irb
[*Leaf1-bgp-af-evpn] peer 2001:db8:1:5301::1 advertise irbv6
[*Leaf1-bgp-af-evpn] quit
[*Leaf1-bgp] quit
[*Leaf1] commit
```

3. Configure VPN and EVPN instances.

Configure Spine1. The configuration of Spine2 is similar to that of Spine1. For details, see [Configuration Files](#).

```
[~Spine1] ip vpn-instance vpn1
[*Spine1-vpn-instance-vpn1] vxlan vni 5000
[*Spine1-vpn-instance-vpn1] ipv4-family
[*Spine1-vpn-instance-vpn1-af-ipv4] route-distinguisher 4.4.4.4:1
[*Spine1-vpn-instance-vpn1-af-ipv4] vpn-target 0:1 evpn
[*Spine1-vpn-instance-vpn1-af-ipv4] quit
[*Spine1-vpn-instance-vpn1] ipv6-family
[*Spine1-vpn-instance-vpn1-af-ipv6] route-distinguisher 4.4.4.4:1
[*Spine1-vpn-instance-vpn1-af-ipv6] vpn-target 0:1 evpn
[*Spine1-vpn-instance-vpn1-af-ipv6] quit
[*Spine1-vpn-instance-vpn1] quit
[*Spine1] bgp 100
[*Spine1-bgp] ipv4-family vpn-instance vpn1
[*Spine1-bgp-vpn1] import-route static
[*Spine1-bgp-vpn1] advertise l2vpn evpn
[*Spine1-bgp-vpn1] quit
[*Spine1-bgp] ipv6-family vpn-instance vpn1
[*Spine1-bgp-6-vpn1] import-route static
[*Spine1-bgp-6-vpn1] advertise l2vpn evpn
[*Spine1-bgp-6-vpn1] quit
[*Spine1-bgp] quit
[*Spine1] interface nve 1 //Configure an NVE interface.
[*Spine1-Nve1] source 2001:db8:1:1301::1 //Spine1 and Spine2 establish an M-LAG active-active
system. The IP addresses and MAC addresses of NVE interfaces on the two devices must be the same.
[*Spine1-Nve1] mac-address 0000-5e00-0101
```

```
[*Spine1-Nve1] quit
[*Spine1] commit
```

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1. For details, see [Configuration Files](#).

```
[~Leaf1] ip vpn-instance vpn1
[*Leaf1-vpn-instance-vpn1] vxlan vni 5000
[*Leaf1-vpn-instance-vpn1] ipv4-family
[*Leaf1-vpn-instance-vpn1-af-ipv4] route-distinguisher 6.6.6.6:1
[*Leaf1-vpn-instance-vpn1-af-ipv4] vpn-target 0:1 evpn
[*Leaf1-vpn-instance-vpn1-af-ipv4] quit
[*Leaf1-vpn-instance-vpn1] ipv6-family
[*Leaf1-vpn-instance-vpn1-af-ipv6] route-distinguisher 6.6.6.6:1
[*Leaf1-vpn-instance-vpn1-af-ipv6] vpn-target 0:1 evpn
[*Leaf1-vpn-instance-vpn1-af-ipv6] quit
[*Leaf1-vpn-instance-vpn1] quit
[*Leaf1] bridge-domain 10
[*Leaf1-bd10] vxlan vni 10
[*Leaf1-bd10] evpn
[*Leaf1-bd10-evpn] route-distinguisher 6.6.6.6:10
[*Leaf1-bd10-evpn] vpn-target 0:10
[*Leaf1-bd10-evpn] vpn-target 0:1 export-extcommunity
[*Leaf1-bd10-evpn] quit
[*Leaf1-bd10] quit
[*Leaf1] bridge-domain 20
[*Leaf1-bd20] vxlan vni 20
[*Leaf1-bd20] evpn
[*Leaf1-bd20-evpn] route-distinguisher 6.6.6.6:20
[*Leaf1-bd20-evpn] vpn-target 0:20
[*Leaf1-bd20-evpn] vpn-target 0:1 export-extcommunity
[*Leaf1-bd20-evpn] quit
[*Leaf1-bd20] quit
[*Leaf1] bgp 100
[*Leaf1-bgp] ipv4-family vpn-instance vpn1
[*Leaf1-bgp-vpn1] import-route direct
[*Leaf1-bgp-vpn1] advertise l2vpn evpn
[*Leaf1-bgp-vpn1] quit
[*Leaf1-bgp] ipv6-family vpn-instance vpn1
[*Leaf1-bgp-6-vpn1] import-route direct
[*Leaf1-bgp-6-vpn1] advertise l2vpn evpn
[*Leaf1-bgp-6-vpn1] quit
[*Leaf1-bgp] quit
[*Leaf1] interface nve 1 //Configure an NVE interface.
[*Leaf1-Nve1] source 2001:db8:1:2301::1 //Leaf1 and Leaf2 establish an M-LAG active-active
system. The IP addresses and MAC addresses of NVE interfaces on the two devices must be the same.
[*Leaf1-Nve1] mac-address 0000-5e00-0102
[*Leaf1-Nve1] vni 10 head-end peer-list protocol bgp
[*Leaf1-Nve1] vni 20 head-end peer-list protocol bgp
[*Leaf1-Nve1] quit
[*Leaf1] commit
```

4. Configure Leaf1, Leaf2, Leaf3, and Leaf4 as Layer 3 gateways.

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1. For details, see [Configuration Files](#).

```
[~Leaf1] interface vbdif 10
[*Leaf1-Vbdif10] ip binding vpn-instance vpn1
[*Leaf1-Vbdif10] ip address 10.1.1.1 24 //Leaf1 and Leaf2 establish an M-LAG active-active system.
The IP addresses and MAC addresses of VBDIF interfaces on the two devices must be the same.
[*Leaf1-Vbdif10] ipv6 enable
[*Leaf1-Vbdif10] ipv6 address fc00:1::1 64
[*Leaf1-Vbdif10] mac-address 0000-5e00-0105
[*Leaf1-Vbdif10] vxlan anycast-gateway enable
[*Leaf1-Vbdif10] arp collect host enable
[*Leaf1-Vbdif10] ipv6 nd collect host enable
[*Leaf1-Vbdif10] quit
[*Leaf1] interface vbdif 20
[*Leaf1-Vbdif20] ip binding vpn-instance vpn1
[*Leaf1-Vbdif20] ip address 10.1.2.1 24
[*Leaf1-Vbdif20] ipv6 enable
```

```
[*Leaf1-Vbdif20] ipv6 address fc00:2::1 64
[*Leaf1-Vbdif20] mac-address 0000-5e00-0106
[*Leaf1-Vbdif20] vxlan anycast-gateway enable
[*Leaf1-Vbdif20] arp collect host enable
[*Leaf1-Vbdif20] ipv6 nd collect host enable
[*Leaf1-Vbdif20] quit
[*Leaf1] commit
```

Step 5 Configure static routes on Spine1 and Spine2 to implement north-south traffic transmission.

Configure Spine1. The configuration of Spine2 is similar to that of Spine1. For details, see [Configuration Files](#).

```
[~Spine1] interface 10ge 1/0/5
[~Spine1-10GE1/0/5] undo portswitch
[*Spine1-10GE1/0/5] ip address 10.1.10.1 24
[*Spine1-10GE1/0/5] ipv6 enable
[*Spine1-10GE1/0/5] ipv6 address fc00:10::1 64
[*Spine1-10GE1/0/5] quit
[~Spine1] interface 10ge 1/0/6
[~Spine1-10GE1/0/6] undo portswitch
[~Spine1-10GE1/0/6] ip address 10.1.20.1 24
[~Spine1-10GE1/0/6] ipv6 enable
[*Spine1-10GE1/0/6] ipv6 address fc00:20::1 64
[*Spine1-10GE1/0/6] quit
[*Spine1] ip route-static 0.0.0.0 0.0.0.0 10.1.10.2 //Configure an IPv4 static route to a PE on the public network.
[*Spine1] ip route-static 0.0.0.0 0.0.0.0 10.1.20.2
[*Spine1] ip route-static 10.1.1.0 24 vpn-instance vpn1 //Configure an IPv4 static route to the server network segment with the next hop being the VPN instance.
[*Spine1] ip route-static 10.1.2.0 24 vpn-instance vpn1
[*Spine1] ip route-static 10.1.3.0 24 vpn-instance vpn1
[*Spine1] ip route-static vpn-instance vpn1 0.0.0.0 0.0.0.0 public //Configure an IPv4 static route of the VPN instance with the next hop being the public network instance.
[*Spine1] ipv6 route-static :: 0 fc00:10::2 //Configure an IPv6 static route to a PE on the public network.
[*Spine1] ipv6 route-static :: 0 fc00:20::2
[*Spine1] ipv6 route-static fc00:1:: 64 vpn-instance vpn1 //Configure an IPv6 static route to the server network segment with the next hop being the VPN instance.
[*Spine1] ipv6 route-static fc00:2:: 64 vpn-instance vpn1
[*Spine1] ipv6 route-static fc00:3:: 64 vpn-instance vpn1
[*Spine1] ipv6 route-static vpn-instance vpn1 :: 0 public //Configure an IPv6 static route of the VPN instance with the next hop being the public network instance.
[*Spine1] commit
```

----End

Verifying the Configuration

After the preceding configurations are complete, run the **display vxlan tunnel** command to check VXLAN tunnel information. The following example shows the command output on Spine1.

```
[~Spine1] display vxlan tunnel
Number of vxlan tunnel : 2
Tunnel ID   Source           Destination      State Type   Uptime
-----
4026531841  2001:db8:1:1301::1  2001:db8:1:2301::1  up    dynamic 0035h21m
4026531842  2001:db8:1:1301::1  2001:db8:1:3301::1  up    dynamic 0036h21m
```

The preceding command output indicates that servers can communicate.

Configuration Files

- Spine1 configuration file

sysname Spine1

```
#
dfs-group 1
priority 150
source ipv6 2001:db8:1:4301::1
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 4.4.4.4:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 4.4.4.4:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 5000
#
ospfv3 1
router-id 4.4.4.4
area 0.0.0.0
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface 10GE1/0/1
undo portswitch
ipv6 enable
ip address 2001:db8:1:D301::1/64
ospfv3 1 area 0.0.0.0
ospfv3 network-type p2p
#
interface 10GE1/0/2
undo portswitch
ipv6 enable
ip address 2001:db8:1:D302::1/64
ospfv3 1 area 0.0.0.0
ospfv3 network-type p2p
#
interface 10GE1/0/3
undo portswitch
ipv6 enable
ip address 2001:db8:1:D303::1/64
ospfv3 1 area 0.0.0.0
ospfv3 network-type p2p
#
interface 10GE1/0/4
undo portswitch
ipv6 enable
ip address 2001:db8:1:D304::1/64
ospfv3 1 area 0.0.0.0
ospfv3 network-type p2p
#
interface 10GE1/0/5
undo portswitch
ipv6 enable
ip address 10.1.10.1 255.255.255.0
ipv6 address FC00:10::1/64
#
interface 10GE1/0/6
undo portswitch
ipv6 enable
ip address 10.1.20.1 255.255.255.0
ipv6 address FC00:20::1/64
#
```



```
interface 10GE1/0/7
eth-trunk 1
#
interface 10GE1/0/8
eth-trunk 1
#
interface LoopBack0
ip address 4.4.4.4 255.255.255.255
#
interface LoopBack1
ipv6 enable
ipv6 address 2001:db8:1:1301::1/128
ospfv3 1 area 0.0.0.0
#
interface LoopBack2
ipv6 enable
ipv6 address 2001:db8:1:4301::1/128
ospfv3 1 area 0.0.0.0
#
interface Nve1
source 2001:DB8:1:1301::1
mac-address 0000-5e00-0101
#
bgp 100
peer 2001:DB8:1:6301::1 as-number 100
peer 2001:DB8:1:6301::1 connect-interface LoopBack2
peer 2001:DB8:1:7301::1 as-number 100
peer 2001:DB8:1:7301::1 connect-interface LoopBack2
peer 2001:DB8:1:8301::1 as-number 100
peer 2001:DB8:1:8301::1 connect-interface LoopBack2
peer 2001:DB8:1:9301::1 as-number 100
peer 2001:DB8:1:9301::1 connect-interface LoopBack2
#
ipv4-family unicast
#
ipv4-family vpn-instance vpn1
import-route static
advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
import-route static
advertise l2vpn evpn
#
l2vpn-family evpn
undo policy vpn-target
peer 2001:DB8:1:6301::1 enable
peer 2001:DB8:1:6301::1 advertise irb
peer 2001:DB8:1:6301::1 advertise irbv6
peer 2001:DB8:1:6301::1 reflect-client
peer 2001:DB8:1:7301::1 enable
peer 2001:DB8:1:7301::1 advertise irb
peer 2001:DB8:1:7301::1 advertise irbv6
peer 2001:DB8:1:7301::1 reflect-client
peer 2001:DB8:1:8301::1 enable
peer 2001:DB8:1:8301::1 advertise irb
peer 2001:DB8:1:8301::1 advertise irbv6
peer 2001:DB8:1:8301::1 reflect-client
peer 2001:DB8:1:9301::1 enable
peer 2001:DB8:1:9301::1 advertise irb
peer 2001:DB8:1:9301::1 advertise irbv6
peer 2001:DB8:1:9301::1 reflect-client
#
ip route-static 0.0.0.0 0.0.0.0 10.1.10.2
ip route-static 0.0.0.0 0.0.0.0 10.1.20.2
ip route-static 10.1.1.0 255.255.255.0 vpn-instance vpn1
ip route-static 10.1.2.0 255.255.255.0 vpn-instance vpn1
ip route-static 10.1.3.0 255.255.255.0 vpn-instance vpn1
ip route-static vpn-instance vpn1 0.0.0.0 0.0.0.0 public
#
```

```
ipv6 route-static :: 0 FC00:10::2
ipv6 route-static :: 0 FC00:20::2
ipv6 route-static FC00:1:: 64 vpn-instance vpn1
ipv6 route-static FC00:2:: 64 vpn-instance vpn1
ipv6 route-static FC00:3:: 64 vpn-instance vpn1
ipv6 route-static vpn-instance vpn1 :: 0 public
#
return
```

- Spine2 configuration file

```
#
sysname Spine2
#
dfs-group 1
 source ipv6 2001:DB8:1:5301::1
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
 ipv4-family
  route-distinguisher 5.5.5.5:1
  vpn-target 0:1 export-extcommunity evpn
  vpn-target 0:1 import-extcommunity evpn
 ipv6-family
  route-distinguisher 5.5.5.5:1
  vpn-target 0:1 export-extcommunity evpn
  vpn-target 0:1 import-extcommunity evpn
 vxlan vni 5000
#
ospfv3 1
 router-id 5.5.5.5
 area 0.0.0.0
#
interface Eth-Trunk1
 mode lacp-static
 peer-link 1
#
interface 10GE1/0/1
 undo portswitch
 ipv6 enable
 ipv6 address 2001:DB8:1:D305::1/64
 ospfv3 1 area 0.0.0.0
 ospfv3 network-type p2p
#
interface 10GE1/0/2
 undo portswitch
 ipv6 enable
 ipv6 address 2001:DB8:1:D306::1/64
 ospfv3 1 area 0.0.0.0
 ospfv3 network-type p2p
#
interface 10GE1/0/3
 undo portswitch
 ipv6 enable
 ipv6 address 2001:DB8:1:D307::1/64
 ospfv3 1 area 0.0.0.0
 ospfv3 network-type p2p
#
interface 10GE1/0/4
 undo portswitch
 ipv6 enable
 ipv6 address 2001:DB8:1:D308::1/64
 ospfv3 1 area 0.0.0.0
 ospfv3 network-type p2p
#
interface 10GE1/0/5
 undo portswitch
```

```
ipv6 enable
ip address 10.1.30.1 255.255.255.0
ipv6 address FC00:30::1/64
#
interface 10GE1/0/6
undo portswitch
ipv6 enable
ip address 10.1.40.1 255.255.255.0
ipv6 address FC00:40::1/64
#
interface 10GE1/0/7
eth-trunk 1
#
interface 10GE1/0/8
eth-trunk 1
#
interface LoopBack0
ip address 5.5.5.5 255.255.255.255
#
interface LoopBack1
ipv6 enable
ipv6 address 2001:DB8:1:1301::1/128
ospfv3 1 area 0.0.0.0
#
interface LoopBack2
ipv6 enable
ipv6 address 2001:DB8:1:5301::1/128
ospfv3 1 area 0.0.0.0
#
interface Nve1
source 2001:DB8:1:1301::1
mac-address 0000-5e00-0101
#
bgp 100
peer 2001:DB8:1:6301::1 as-number 100
peer 2001:DB8:1:6301::1 connect-interface LoopBack2
peer 2001:DB8:1:7301::1 as-number 100
peer 2001:DB8:1:7301::1 connect-interface LoopBack2
peer 2001:DB8:1:8301::1 as-number 100
peer 2001:DB8:1:8301::1 connect-interface LoopBack2
peer 2001:DB8:1:9301::1 as-number 100
peer 2001:DB8:1:9301::1 connect-interface LoopBack2
#
ipv4-family unicast
#
ipv4-family vpn-instance vpn1
import-route static
advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
import-route static
advertise l2vpn evpn
#
l2vpn-family evpn
undo policy vpn-target
peer 2001:DB8:1:6301::1 enable
peer 2001:DB8:1:6301::1 advertise irb
peer 2001:DB8:1:6301::1 advertise irbv6
peer 2001:DB8:1:6301::1 reflect-client
peer 2001:DB8:1:7301::1 enable
peer 2001:DB8:1:7301::1 advertise irb
peer 2001:DB8:1:7301::1 advertise irbv6
peer 2001:DB8:1:7301::1 reflect-client
peer 2001:DB8:1:8301::1 enable
peer 2001:DB8:1:8301::1 advertise irb
peer 2001:DB8:1:8301::1 advertise irbv6
peer 2001:DB8:1:8301::1 reflect-client
peer 2001:DB8:1:9301::1 enable
peer 2001:DB8:1:9301::1 advertise irb
```

```
peer 2001:DB8:1:9301::1 advertise irbv6
peer 2001:DB8:1:9301::1 reflect-client
#
ip route-static 0.0.0.0 0.0.0.0 10.1.30.2
ip route-static 0.0.0.0 0.0.0.0 10.1.40.2
ip route-static 10.1.1.0 255.255.255.0 vpn-instance vpn1
ip route-static 10.1.2.0 255.255.255.0 vpn-instance vpn1
ip route-static 10.1.3.0 255.255.255.0 vpn-instance vpn1
ip route-static vpn-instance vpn1 0.0.0.0 0.0.0.0 public
#
ipv6 route-static :: 0 FC00:30::2
ipv6 route-static :: 0 FC00:40::2
ipv6 route-static FC00:1:: 64 vpn-instance vpn1
ipv6 route-static FC00:2:: 64 vpn-instance vpn1
ipv6 route-static FC00:3:: 64 vpn-instance vpn1
ipv6 route-static vpn-instance vpn1 :: 0 public
#
return
```

- Leaf1 configuration file

```
#
sysname Leaf1
#
dfs-group 1
priority 150
source ipv6 2001:DB8:1:6301::1
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 6.6.6.6:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 6.6.6.6:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 5000
#
bridge-domain 10
vxlan vni 10
evpn
route-distinguisher 6.6.6.6:10
vpn-target 0:10 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:10 import-extcommunity
#
bridge-domain 20
vxlan vni 20
evpn
route-distinguisher 6.6.6.6:20
vpn-target 0:20 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:20 import-extcommunity
#
ospfv3 1
router-id 6.6.6.6
area 0.0.0.0
#
interface Vbdf10
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.1.1.1 255.255.255.0
ipv6 address FC00:1::1/64
mac-address 0000-5e00-0105
ipv6 nd collect host enable
```

```
vxlan anycast-gateway enable
arp collect host enable
#
interface Vbdi20
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.1.2.1 255.255.255.0
ipv6 address FC00:2::1/64
mac-address 0000-5e00-0106
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk2
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 1
#
interface Eth-Trunk2.10 mode l2
encapsulation dot1q vid 10
bridge-domain 10
#
interface Eth-Trunk3
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 2
#
interface Eth-Trunk3.20 mode l2
encapsulation dot1q vid 20
bridge-domain 20
#
interface 10GE1/0/1
undo portswitch
ipv6 enable
ipv6 address 2001:DB8:1:D301::2/64
ospfv3 1 area 0.0.0.0
ospfv3 network-type p2p
#
interface 10GE1/0/2
undo portswitch
ipv6 enable
ipv6 address 2001:DB8:1:D305::2/64
ospfv3 1 area 0.0.0.0
ospfv3 network-type p2p
#
interface 10GE1/0/3
eth-trunk 1
#
interface 10GE1/0/4
eth-trunk 1
#
interface 10GE1/0/5
eth-trunk 2
#
interface 10GE1/0/6
eth-trunk 3
#
interface LoopBack0
ip address 6.6.6.6 255.255.255.255
#
interface LoopBack1
ipv6 enable
ipv6 address 2001:DB8:1:2301::1/128
ospfv3 1 area 0.0.0.0
#
```

```
interface LoopBack2
  ipv6 enable
  ip address 6.6.6.6 255.255.255.255
  ipv6 address 2001:DB8:1:6301::1/128
  ospfv3 1 area 0.0.0.0
#
interface Nve1
  source 2001:DB8:1:2301::1
  vni 10 head-end peer-list protocol bgp
  vni 20 head-end peer-list protocol bgp
  mac-address 0000-5e00-0102
#
monitor-link group 1
  port 10GE1/0/1 uplink
  port 10GE1/0/2 uplink
  port Eth-Trunk2 downlink 1
  port Eth-Trunk3 downlink 2
#
bgp 100
  peer 2001:DB8:1:4301::1 as-number 100
  peer 2001:DB8:1:4301::1 connect-interface LoopBack2
  peer 2001:DB8:1:5301::1 as-number 100
  peer 2001:DB8:1:5301::1 connect-interface LoopBack2
#
ipv4-family unicast
#
ipv4-family vpn-instance vpn1
  import-route direct
  advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
  import-route direct
  advertise l2vpn evpn
#
l2vpn-family evpn
  policy vpn-target
  peer 2001:DB8:1:4301::1 enable
  peer 2001:DB8:1:4301::1 advertise irb
  peer 2001:DB8:1:4301::1 advertise irbv6
  peer 2001:DB8:1:5301::1 enable
  peer 2001:DB8:1:5301::1 advertise irb
  peer 2001:DB8:1:5301::1 advertise irbv6
#
return
```

- Leaf2 configuration file

```
#
sysname Leaf2
#
dfs-group 1
  source ipv6 2001:DB8:1:7301::1
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
  ipv4-family
    route-distinguisher 7.7.7.7:1
    vpn-target 0:1 export-extcommunity evpn
    vpn-target 0:1 import-extcommunity evpn
  ipv6-family
    route-distinguisher 7.7.7.7:1
    vpn-target 0:1 export-extcommunity evpn
    vpn-target 0:1 import-extcommunity evpn
  vxlan vni 5000
#
bridge-domain 10
  vxlan vni 10
```

```
evpn
 route-distinguisher 7.7.7.7:10
 vpn-target 0:10 export-extcommunity
 vpn-target 0:1 export-extcommunity
 vpn-target 0:10 import-extcommunity
#
bridge-domain 20
vxlan vni 20
evpn
 route-distinguisher 7.7.7.7:20
 vpn-target 0:20 export-extcommunity
 vpn-target 0:1 export-extcommunity
 vpn-target 0:20 import-extcommunity
#
ospfv3 1
 router-id 7.7.7.7
 area 0.0.0.0
#
interface Vbdif10
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address ip address 10.1.1.1 255.255.255.0
 ipv6 address FC00:1::1/64
 mac-address 0000-5e00-0105
 ipv6 nd collect host enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Vbdif20
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.1.2.1 255.255.255.0
 ipv6 address FC00:2::1/64
 mac-address 0000-5e00-0106
 ipv6 nd collect host enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Eth-Trunk1
 mode lacp-static
 peer-link 1
#
interface Eth-Trunk2
 stp edged-port enable
 mode lacp-static
 dfs-group 1 m-lag 1
#
interface Eth-Trunk2.10 mode l2
 encapsulation dot1q vid 10
 bridge-domain 10
#
interface Eth-Trunk3
 stp edged-port enable
 mode lacp-static
 dfs-group 1 m-lag 2
#
interface Eth-Trunk3.20 mode l2
 encapsulation dot1q vid 20
 bridge-domain 20
#
interface 10GE1/0/1
 undo portswitch
 ipv6 enable
 ipv6 address 2001:DB8:1:D302::2/64
 ospfv3 1 area 0.0.0.0
 ospfv3 network-type p2p
#
interface 10GE1/0/2
 undo portswitch
```

```
ipv6 enable
ipv6 address 2001:DB8:1:D306::2/64
ospfv3 1 area 0.0.0.0
ospfv3 network-type p2p
#
interface 10GE1/0/3
eth-trunk 1
#
interface 10GE1/0/4
eth-trunk 1
#
interface 10GE1/0/5
eth-trunk 2
#
interface 10GE1/0/6
eth-trunk 3
#
interface LoopBack0
ip address 7.7.7.7 255.255.255.255
#
interface LoopBack1
ipv6 enable
ipv6 address 2001:DB8:1:2301::1/128
ospfv3 1 area 0.0.0.0
#
interface LoopBack2
ipv6 enable
ipv6 address 2001:DB8:1:7301::1/128
ospfv3 1 area 0.0.0.0
#
interface Nve1
source 2001:DB8:1:2301::1
vni 10 head-end peer-list protocol bgp
vni 20 head-end peer-list protocol bgp
mac-address 0000-5e00-0102
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk2 downlink 1
port Eth-Trunk3 downlink 2
#
bgp 100
peer 2001:DB8:1:4301::1 as-number 100
peer 2001:DB8:1:4301::1 connect-interface LoopBack2
peer 2001:DB8:1:5301::1 as-number 100
peer 2001:DB8:1:5301::1 connect-interface LoopBack2
#
ipv4-family unicast
#
ipv4-family vpn-instance vpn1
import-route direct
advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
import-route direct
advertise l2vpn evpn
#
l2vpn-family evpn
policy vpn-target
peer 2001:DB8:1:4301::1 enable
peer 2001:DB8:1:4301::1 advertise irb
peer 2001:DB8:1:4301::1 advertise irbv6
peer 2001:DB8:1:5301::1 enable
peer 2001:DB8:1:5301::1 advertise irb
peer 2001:DB8:1:5301::1 advertise irbv6
#
return
```

- Leaf3 configuration file


```
#
sysname Leaf3
#
dfs-group 1
priority 150
source ipv6 2001:DB8:1:8301::1
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 8.8.8:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 8.8.8:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 5000
#
bridge-domain 20
vxlan vni 20
evpn
route-distinguisher 8.8.8:20
vpn-target 0:20 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:20 import-extcommunity
#
bridge-domain 30
vxlan vni 30
evpn
route-distinguisher 8.8.8:30
vpn-target 0:30 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:30 import-extcommunity
#
ospfv3 1
router-id 8.8.8.8
area 0.0.0.0
#
interface Vbdif20
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.1.2.1 255.255.255.0
ipv6 address FC00:2::1/64
mac-address 0000-5e00-0106
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Vbdif30
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.1.3.1 255.255.255.0
ipv6 address FC00:3::1/64
mac-address 0000-5e00-0107
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk2
stp edged-port enable
```

```
mode lacp-static
dfs-group 1 m-lag 1
#
interface Eth-Trunk2.20 mode l2
encapsulation dot1q vid 20
bridge-domain 20
#
interface Eth-Trunk3
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 2
#
interface Eth-Trunk3.30 mode l2
encapsulation dot1q vid 30
bridge-domain 30
#
interface 10GE1/0/1
undo portswitch
ipv6 enable
ipv6 address 2001:DB8:1:D303::2/64
ospfv3 1 area 0.0.0.0
ospfv3 network-type p2p
#
interface 10GE1/0/2
undo portswitch
ipv6 enable
ipv6 address 2001:DB8:1:D307::2/64
ospfv3 1 area 0.0.0.0
ospfv3 network-type p2p
#
interface 10GE1/0/3
eth-trunk 1
#
interface 10GE1/0/4
eth-trunk 1
#
interface 10GE1/0/5
eth-trunk 2
#
interface 10GE1/0/6
eth-trunk 3
#
interface LoopBack0
ip address 8.8.8.8 255.255.255.255
#
interface LoopBack1
ipv6 enable
ipv6 address 2001:DB8:1:3301::1/128
ospfv3 1 area 0.0.0.0
#
interface LoopBack2
ipv6 enable
ipv6 address 2001:DB8:1:8301::1/128
ospfv3 1 area 0.0.0.0
#
interface Nve1
source 2001:DB8:1:3301::1
vni 20 head-end peer-list protocol bgp
vni 30 head-end peer-list protocol bgp
mac-address 0000-5e00-0103
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk2 downlink 1
port Eth-Trunk3 downlink 2
#
bgp 100
peer 2001:DB8:1:4301::1 as-number 100
```

```
peer 2001:DB8:1:4301::1 connect-interface LoopBack2
peer 2001:DB8:1:5301::1 as-number 100
peer 2001:DB8:1:5301::1 connect-interface LoopBack2
#
ipv4-family unicast
#
ipv4-family vpn-instance vpn1
import-route direct
advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
import-route direct
advertise l2vpn evpn
#
l2vpn-family evpn
policy vpn-target
peer 2001:DB8:1:4301::1 enable
peer 2001:DB8:1:4301::1 advertise irb
peer 2001:DB8:1:4301::1 advertise irbv6
peer 2001:DB8:1:5301::1 enable
peer 2001:DB8:1:5301::1 advertise irb
peer 2001:DB8:1:5301::1 advertise irbv6
#
return
```

- Leaf4 configuration file

```
#
sysname Leaf4
#
dfs-group 1
source ipv6 2001:DB8:1:9301::1
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 9.9.9.9:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 9.9.9.9:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 5000
#
bridge-domain 20
vxlan vni 20
evpn
route-distinguisher 9.9.9.9:20
vpn-target 0:20 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:20 import-extcommunity
#
bridge-domain 30
vxlan vni 30
evpn
route-distinguisher 9.9.9.9:30
vpn-target 0:30 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:30 import-extcommunity
#
ospfv3 1
router-id 9.9.9.9
area 0.0.0.0
#
interface Vbdif20
ip binding vpn-instance vpn1
```

```
ipv6 enable
ip address 10.1.2.1 255.255.255.0
ipv6 address FC00:2::1/64
mac-address 0000-5e00-0106
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Vbdf30
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.1.3.1 255.255.255.0
ipv6 address FC00:3::1/64
mac-address 0000-5e00-0107
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk2
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 1
#
interface Eth-Trunk2.20 mode l2
encapsulation dot1q vid 20
bridge-domain 20
#
interface Eth-Trunk3
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 2
#
interface Eth-Trunk3.30 mode l2
encapsulation dot1q vid 30
bridge-domain 30
#
interface 10GE1/0/1
undo portswitch
ipv6 enable
ipv6 address 2001:DB8:1:D304::2/64
ospfv3 1 area 0.0.0.0
ospfv3 network-type p2p
#
interface 10GE1/0/2
undo portswitch
ipv6 enable
ipv6 address 2001:DB8:1:D308::2/64
ospfv3 1 area 0.0.0.0
ospfv3 network-type p2p
#
interface 10GE1/0/3
eth-trunk 1
#
interface 10GE1/0/4
eth-trunk 1
#
interface 10GE1/0/5
eth-trunk 2
#
interface 10GE1/0/6
eth-trunk 3
#
interface LoopBack0
ip address 9.9.9.9 255.255.255.255
#
```

```
interface LoopBack1
  ipv6 enable
  ipv6 address 2001:DB8:1:3301::1/128
  ospfv3 1 area 0.0.0.0
#
interface LoopBack2
  ipv6 enable
  ipv6 address 2001:DB8:1:9301::1/128
  ospfv3 1 area 0.0.0.0
#
interface Nve1
  source 2001:DB8:1:3301::1
  vni 20 head-end peer-list protocol bgp
  vni 30 head-end peer-list protocol bgp
  mac-address 0000-5e00-0103
#
monitor-link group 1
  port 10GE1/0/1 uplink
  port 10GE1/0/2 uplink
  port Eth-Trunk2 downlink 1
  port Eth-Trunk3 downlink 2
#
bgp 100
  peer 2001:DB8:1:4301::1 as-number 100
  peer 2001:DB8:1:4301::1 connect-interface LoopBack2
  peer 2001:DB8:1:5301::1 as-number 100
  peer 2001:DB8:1:5301::1 connect-interface LoopBack2
#
ipv4-family unicast
#
ipv4-family vpn-instance vpn1
  import-route direct
  advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
  import-route direct
  advertise l2vpn evpn
#
l2vpn-family evpn
  policy vpn-target
  peer 2001:DB8:1:4301::1 enable
  peer 2001:DB8:1:4301::1 advertise irb
  peer 2001:DB8:1:4301::1 advertise irbv6
  peer 2001:DB8:1:5301::1 enable
  peer 2001:DB8:1:5301::1 advertise irb
  peer 2001:DB8:1:5301::1 advertise irbv6
#
return
```

1.6 Example for Configuring VXLAN in Centralized Gateway Mode

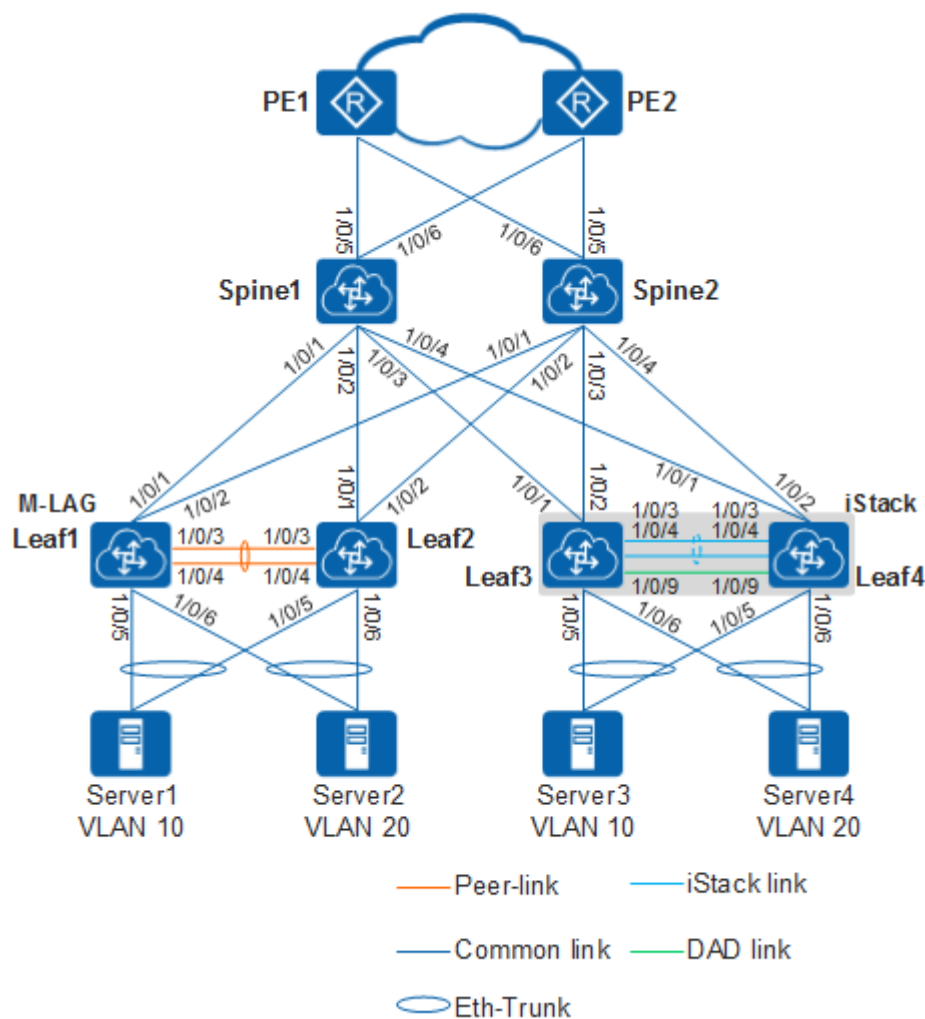
Applicable Products and Versions

- This example applies to CE16800, CE12800, CE12800E, CE8800, CE7800, CE6800 (excluding the CE6820, CE6850EI, CE6810EI, and CE6810LI), and CE5880EI series switches running V200R001C00 or later versions.
- For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

In **Figure 1-6**, a new data center network of an enterprise uses the centralized gateway deployment. Spine switches function as Layer 3 gateways, and leaf switches function as Layer 2 access devices to connect to servers. To ensure high reliability, spine switches use centralized all-active gateways, and leaf switches are deployed in stack mode or using M-LAG.

Figure 1-6 VXLAN in centralized gateway deployment mode



NOTE

In the preceding figure, 1/0/1 indicates an interface number and the interface rate is 10 Gbit/s. That is, 1/0/1 indicates 10GE1/0/1. Other interfaces are specified in a similar way.

Table 1-6 Interface IP address list

Devic e Nam e	Interface	IP Address	Devic e Nam e	Interface	IP Address
Spine 1	10GE1/0/1	192.168.1.1/24	Spine 2	10GE1/0/1	192.168.5.1/24
	10GE1/0/2	192.168.2.1/24		10GE1/0/2	192.168.6.1/24
	10GE1/0/3	192.168.3.1/24		10GE1/0/3	192.168.7.1/24
	10GE1/0/4	192.168.4.1/24		10GE1/0/4	192.168.8.1/24
	10GE1/0/5	10.1.10.1/24		10GE1/0/5	10.1.30.1/24
	10GE1/0/6	10.1.20.1/24		10GE1/0/6	10.1.40.1/24
	Loopback1	1.1.1.1/32		Loopback1	1.1.1.1/32
	Loopback2	4.4.4.4/32		Loopback2	5.5.5.5/32
Leaf1	10GE1/0/1	192.168.1.2/24	Leaf2	10GE1/0/1	192.168.2.2/24
	10GE1/0/2	192.168.5.2/24		10GE1/0/2	192.168.6.2/24
	Loopback1	2.2.2.2/32		Loopback1	2.2.2.2/32
	Loopback2	6.6.6.6/32		Loopback2	7.7.7.7/32
Leaf3	10GE1/0/1	192.168.3.2/24	Leaf4	10GE1/0/1	192.168.4.2/24
	10GE1/0/2	192.168.7.2/24		10GE1/0/2	192.168.8.2/24
	Loopback1	3.3.3.3/32		Loopback1	3.3.3.3/32

Configuration Roadmap

The configuration roadmap is as follows:

1. Configure a routing protocol to ensure Layer 3 connectivity on the underlay network.
2. Configure an M-LAG or a stack system to implement active-active server access.
3. Configure BGP EVPN to establish VXLAN tunnels between leaf and spine switches.

Configuration Notes

In addition to the configuration in the procedure, you may also need to run the following commands for the CE12800 based on actual requirements:

- Run the **assign forward nvo3 service extend enable** command in the system view to enable the NVO3 service extension function.

NOTE

By default, the NVO3 service extension function is disabled on the switch. After the NVO3 service is configured on the switch, there is a high probability that other ACL resource-consuming services such as MQC, simplified ACL, traffic policing, BD traffic statistics collection, and DHCP fail to be configured on the switch. You can run this command on a switch configured with the NVO3 service to reduce the configuration failure possibility on EC (except the CE-L48GT-EC and CE-L48GS-EC cards), ED, EF, and EG series cards.

- Run the **assign forward nvo3 f-linecard compatibility enable** command in the system view to ensure that VXLAN traffic can be forwarded when the card interoperability mode is non-enhanced mode.

NOTE

If VXLAN traffic is forwarded between cards when the card interoperability mode is non-enhanced mode, the VXLAN traffic may fail to be forwarded. To use the VXLAN function, you must configure the **assign forward nvo3 f-linecard compatibility enable** command when the card interoperability mode is non-enhanced mode.

Procedure

- Step 1** Set up a stack between leaf switches. In this example, Leaf3 and Leaf4 set up a stack, and Leaf3 is the master switch.

Configure Leaf3. After the configuration is complete, save the configuration and then restart Leaf3.

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf3
[*HUAWEI] commit
[~Leaf3] stack
[~Leaf3-stack] stack member 1 priority 150
[*Leaf3-stack] stack member 1 domain 10
[*Leaf3-stack] quit
[*Leaf3] interface stack-port 1/1
[*Leaf3-Stack-Port1/1] port member-group interface 10ge 1/0/3 to 1/0/4
Warning: After the configuration is complete,
1.The interface(s) (10GE1/0/3-1/0/4) will be converted to stack mode and be configured with the port crc-
statistics trigger error-down command if the configuration does not exist.
2.The interface(s) may go Error-Down (crc-statistics) because there is no shutdown configuration on the
interfaces.Continue? [Y/N]: y
[*Leaf3-Stack-Port1/1] quit
[*Leaf3] commit
[~Leaf3] quit
<Leaf3> save
Warning: The current configuration will be written to the device. Continue? [Y/N]: y
<Leaf3> reboot
Warning: The system will reboot. Continue? [Y/N]: y
```

Configure Leaf4. After the configuration is complete, save the configuration and then restart Leaf4.

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf4
[*HUAWEI] commit
```



```
[~Leaf4] stack
[*Leaf4-stack] stack member 1 domain 10
[*Leaf4-stack] stack member 1 renumber 2 inherit-config
Warning: The stack configuration of member ID 1 will be inherited to member ID 2 after the device resets.
Continue? [Y/N]: y
[*Leaf4-stack] quit
[*Leaf4] interface stack-port 1/1
[*Leaf4-Stack-Port1/1] port member-group interface 10ge 1/0/3 to 1/0/4
Warning: After the configuration is complete,
1.The interface(s) (10GE1/0/3-1/0/4) will be converted to stack mode and be configured with the port crc-
statistics trigger error-down command if the configuration does not exist.
2.The interface(s) may go Error-Down (crc-statistics) because there is no shutdown configuration on the
interfaces.Continue? [Y/N]: y
[*Leaf4-Stack-Port1/1] quit
[*Leaf4] commit
[~Leaf4] quit
<Leaf4> save
Warning: The current configuration will be written to the device. Continue? [Y/N]: y
<Leaf4> reboot
Warning: The system will reboot. Continue? [Y/N]: y
```

Connect Leaf3 and Leaf4 using stack cables. Wait for several minutes and run the **display stack** command on Leaf3. After the stack is set up successfully, configure the DAD function and save the stack configuration.

```
<Leaf3> display stack
-----
MemberID Role   MAC             Priority DeviceType      Description
-----
1      Master  0004-9f31-d520  150      CE6850-48S6Q-HI
2      Standby 0004-9f62-1f40  100      CE6850-48S6Q-HI
-----

<Leaf3> system-view
[~Leaf3] sysname iStack
[*Leaf3] commit
[~iStack] interface 10ge 1/0/9
[~iStack-10GE1/0/9] dual-active detect mode direct
Warning: The interface will block common data packets, except BPDU packets. Continue? [Y/N]: y
[*iStack-10GE1/0/9] quit
[*iStack] interface 10ge 2/0/9
[*iStack-10GE2/0/9] dual-active detect mode direct
Warning: The interface will block common data packets, except BPDU packets. Continue? [Y/N]: y
[*iStack-10GE2/0/9] commit
[~iStack-10GE2/0/9] return
<iStack> save
Warning: The current configuration will be written to the device. Continue? [Y/N]: y
```

Step 2 Configure a routing protocol to implement Layer 3 connectivity on the underlay network.

Configure Leaf1. Repeat this step for other switches. For details, see [Configuration Files](#).

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf1
[*HUAWEI] commit
[~Leaf1] interface 10ge 1/0/1
[~Leaf1-10GE1/0/1] undo portswitch
[*Leaf1-10GE1/0/1] ip address 192.168.1.2 24
[*Leaf1-10GE1/0/1] ospf network-type p2p
[*Leaf1-10GE1/0/1] quit
[*Leaf1] interface 10ge 1/0/2
[*Leaf1-10GE1/0/2] undo portswitch
[*Leaf1-10GE1/0/2] ip address 192.168.5.2 24
[*Leaf1-10GE1/0/2] ospf network-type p2p
[*Leaf1-10GE1/0/2] quit
[*Leaf1] interface loopback 1
[*Leaf1-LoopBack1] ip address 2.2.2.2 32
[*Leaf1-LoopBack1] quit
```

```
[*Leaf1] interface loopback 2
[*Leaf1-LoopBack2] ip address 6.6.6.6 32
[*Leaf1-LoopBack2] quit
[*Leaf1] ospf
[*Leaf1-ospf-1] area 0
[*Leaf1-ospf-1-area-0.0.0.0] network 192.168.1.0 0.0.0.255
[*Leaf1-ospf-1-area-0.0.0.0] network 192.168.5.0 0.0.0.255
[*Leaf1-ospf-1-area-0.0.0.0] network 2.2.2.2 0.0.0.0
[*Leaf1-ospf-1-area-0.0.0.0] network 6.6.6.6 0.0.0.0
[*Leaf1-ospf-1-area-0.0.0.0] quit
[*Leaf1-ospf-1] quit
[*Leaf1] commit
```

After OSPF is configured, leaf and spine switches can learn the IP addresses of loopback interfaces of each other using OSPF and successfully ping each other.

Step 3 Set up an M-LAG between leaf switches. In this example, Leaf1 and Leaf2 set up an M-LAG.

Configure Leaf1.

```
[~Leaf1] stp mode rstp
[*Leaf1] stp v-stp enable
[*Leaf1] dfs-group 1
[*Leaf1-dfs-group-1] source ip 6.6.6.6
[*Leaf1-dfs-group-1] priority 150
[*Leaf1-dfs-group-1] quit
[*Leaf1] interface eth-trunk 1
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/3
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/4
[*Leaf1-Eth-Trunk1] mode lacp-static
[*Leaf1-Eth-Trunk1] peer-link 1
[*Leaf1-Eth-Trunk1] quit
[*Leaf1] interface eth-trunk 2
[*Leaf1-Eth-Trunk2] trunkport 10ge 1/0/5
[*Leaf1-Eth-Trunk2] mode lacp-static
[*Leaf1-Eth-Trunk2] dfs-group 1 m-lag 1
[*Leaf1-Eth-Trunk2] stp edged-port enable
[*Leaf1-Eth-Trunk2] quit
[*Leaf1] interface eth-trunk 3
[*Leaf1-Eth-Trunk3] trunkport 10ge 1/0/6
[*Leaf1-Eth-Trunk3] mode lacp-static
[*Leaf1-Eth-Trunk3] dfs-group 1 m-lag 2
[*Leaf1-Eth-Trunk3] stp edged-port enable
[*Leaf1-Eth-Trunk3] quit
[*Leaf1] commit
[~Leaf1] monitor-link group 1
[*Leaf1-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf1-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf1-mtlk-group1] port eth-trunk 2 downlink 1
[*Leaf1-mtlk-group1] port eth-trunk 3 downlink 2
[*Leaf1-mtlk-group1] quit
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] stp mode rstp
[*Leaf2] stp v-stp enable
[*Leaf2] dfs-group 1
[*Leaf2-dfs-group-1] source ip 7.7.7.7
[*Leaf2-dfs-group-1] quit
[*Leaf2] interface eth-trunk 1
[*Leaf2-Eth-Trunk1] trunkport 10ge 1/0/3
[*Leaf2-Eth-Trunk1] trunkport 10ge 1/0/4
[*Leaf2-Eth-Trunk1] mode lacp-static
[*Leaf2-Eth-Trunk1] peer-link 1
[*Leaf2-Eth-Trunk1] quit
[*Leaf2] interface eth-trunk 2
[*Leaf2-Eth-Trunk2] mode lacp-static
```

```
[*Leaf2-Eth-Trunk2] trunkport 10ge 1/0/5
[*Leaf2-Eth-Trunk2] dfs-group 1 m-lag 1
[*Leaf2-Eth-Trunk2] stp edged-port enable
[*Leaf2-Eth-Trunk2] quit
[*Leaf2] interface eth-trunk 3
[*Leaf2-Eth-Trunk3] mode lacp-static
[*Leaf2-Eth-Trunk3] trunkport 10ge 1/0/6
[*Leaf2-Eth-Trunk3] dfs-group 1 m-lag 2
[*Leaf2-Eth-Trunk3] stp edged-port enable
[*Leaf2-Eth-Trunk3] quit
[*Leaf2] commit
[~Leaf2] monitor-link group 1
[*Leaf2-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf2-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf2-mtlk-group1] port eth-trunk 2 downlink 1
[*Leaf2-mtlk-group1] port eth-trunk 3 downlink 2
[*Leaf2-mtlk-group1] quit
[*Leaf2] commit
```

Step 4 Configure BGP EVPN to establish a VXLAN tunnel.

1. Configure the VXLAN tunnel mode and enable the VXLAN ACL extension function. (This step only needs to be performed on the CE12800, CE16800 equipped with A series cards, CE6870EI, and CE6875EI.)

Configure Spine1. Repeat this step for other switches.

```
[~Spine1] ip tunnel mode vxlan
[*Spine1] assign forward nvo3 acl extend enable
[*Spine1] commit
```

NOTE

After configuring the VXLAN tunnel mode and enabling the VXLAN ACL extension function, you need to save the configuration and restart the switch to make the configuration take effect. You can restart the switch immediately or after all the configurations are complete.

2. Configure a service access point.

Configure Leaf1. Repeat this step for Leaf2.

```
[~Leaf1] bridge-domain 10
[*Leaf1-bd10] quit
[*Leaf1] bridge-domain 20
[*Leaf1-bd20] quit
[*Leaf1] interface eth-trunk 2.10 mode l2
[*Leaf1-Eth-Trunk2.10] encapsulation dot1q vid 10
[*Leaf1-Eth-Trunk2.10] bridge-domain 10
[*Leaf1-Eth-Trunk2.10] quit
[*Leaf1] interface eth-trunk 3.20 mode l2
[*Leaf1-Eth-Trunk3.20] encapsulation dot1q vid 20
[*Leaf1-Eth-Trunk3.20] bridge-domain 20
[*Leaf1-Eth-Trunk3.20] quit
[*Leaf1] commit
```

Configure iStack.

```
[~iStack] bridge-domain 10
[*iStack-bd10] quit
[*iStack] bridge-domain 20
[*iStack-bd20] quit
[*iStack] interface eth-trunk 2
[*iStack-Eth-Trunk2] trunkport 10ge 1/0/5 2/0/5
[*iStack-Eth-Trunk2] mode lacp-static
[*iStack-Eth-Trunk2] stp edged-port enable
[*iStack-Eth-Trunk2] quit
[*iStack] interface eth-trunk 2.10 mode l2
[*iStack-Eth-Trunk2.10] encapsulation dot1q vid 10
[*iStack-Eth-Trunk2.10] bridge-domain 10
[*iStack-Eth-Trunk2.10] quit
[*iStack] interface eth-trunk 3
```

```
[*iStack-Eth-Trunk3] trunkport 10ge 1/0/6 2/0/6
[*iStack-Eth-Trunk3] mode lacp-static
[*iStack-Eth-Trunk3] stp edged-port enable
[*iStack-Eth-Trunk3] quit
[*iStack] interface eth-trunk 3.20 mode l2
[*iStack-Eth-Trunk3.20] encapsulation dot1q vid 20
[*iStack-Eth-Trunk3.20] bridge-domain 20
[*iStack-Eth-Trunk3.20] quit
[*iStack] commit
```

3. Establish BGP EVPN peer relationships. Spine1 and Spine2 function as RRs.

Configure Spine1. Repeat this step for Spine2. For details, see [Configuration Files](#).

```
[~Spine1] evpn-overlay enable
[*Spine1] bgp 100
[*Spine1-bgp] peer 3.3.3.3 as-number 100
[*Spine1-bgp] peer 3.3.3.3 connect-interface LoopBack2
[*Spine1-bgp] peer 6.6.6.6 as-number 100
[*Spine1-bgp] peer 6.6.6.6 connect-interface LoopBack2
[*Spine1-bgp] peer 7.7.7.7 as-number 100
[*Spine1-bgp] peer 7.7.7.7 connect-interface LoopBack2
[*Spine1-bgp] l2vpn-family evpn
[*Spine1-bgp-af-evpn] peer 3.3.3.3 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 3.3.3.3 reflect-client
[*Spine1-bgp-af-evpn] peer 6.6.6.6 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 6.6.6.6 reflect-client
[*Spine1-bgp-af-evpn] peer 7.7.7.7 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 7.7.7.7 reflect-client
[*Spine1-bgp-af-evpn] undo policy vpn-target
[*Spine1-bgp-af-evpn] quit
[*Spine1-bgp] quit
[*Spine1] commit
```

Configure Leaf1. Repeat this step for Leaf2 and iStack. For details, see [Configuration Files](#).

```
[~Leaf1] evpn-overlay enable
[*Leaf1] bgp 100
[*Leaf1-bgp] peer 4.4.4.4 as-number 100
[*Leaf1-bgp] peer 4.4.4.4 connect-interface LoopBack2
[*Leaf1-bgp] peer 5.5.5.5 as-number 100
[*Leaf1-bgp] peer 5.5.5.5 connect-interface LoopBack2
[*Leaf1-bgp] l2vpn-family evpn
[*Leaf1-bgp-af-evpn] peer 4.4.4.4 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 5.5.5.5 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] quit
[*Leaf1-bgp] quit
[*Leaf1] commit
```

4. Configure EVPN instances and the ingress replication function.

Configure Spine1. Repeat this step for other switches. For details, see [Configuration Files](#).

```
[~Spine1] bridge-domain 10
[~Spine1-bd10] vxlan vni 10
[*Spine1-bd10] evpn
[*Spine1-bd10-evpn] route-distinguisher 4.4.4.4:10
[*Spine1-bd10-evpn] vpn-target 0:10
[*Spine1-bd10-evpn] quit
[*Spine1-bd10] quit
[*Spine1] bridge-domain 20
[*Spine1-bd20] vxlan vni 20
[*Spine1-bd20] evpn
[*Spine1-bd20-evpn] route-distinguisher 4.4.4.4:20
[*Spine1-bd20-evpn] vpn-target 0:20
```

```
[*Spine1-bd20-evpn] quit
[*Spine1-bd20] quit
[*Spine1] interface nve 1
[*Spine1-Nve1] source 1.1.1.1 //Leaf1 and Leaf2 set up an M-LAG active-active system. The IP
addresses of NVE interfaces on the two devices must be the same.
[*Spine1-Nve1] vni 10 head-end peer-list protocol bgp
[*Spine1-Nve1] vni 20 head-end peer-list protocol bgp
[*Spine1-Nve1] quit
[*Spine1] commit
```

5. Configure service loopback interfaces on Spine1 and Spine2. (This step only needs to be performed on the CE6850HI, CE6850U-HI, CE6851HI, CE6860EI, CE7850EI, CE8850EI, and CE8860EI.)

```
[~Spine1] interface eth-trunk 10
[*Spine1-Eth-Trunk10] trunkport 10ge 1/0/10 to 1/0/11
[*Spine1-Eth-Trunk10] service type tunnel
[*Spine1-Eth-Trunk10] quit
[*Spine1] commit
```

6. Configure all-active gateways on Spine1 and Spine2.

Configure Spine1.

```
[~Spine1] ip vpn-instance vpn1
[*Spine1-vpn-instance-vpn1] ipv4-family
[*Spine1-vpn-instance-vpn1-af-ipv4] route-distinguisher 4.4.4.4:1
[*Spine1-vpn-instance-vpn1-af-ipv4] vpn-target 0:1
[*Spine1-vpn-instance-vpn1-af-ipv4] quit
[*Spine1-vpn-instance-vpn1] quit
[*Spine1] interface vbdif 10
[*Spine1-Vbdif10] ip binding vpn-instance vpn1
[*Spine1-Vbdif10] ip address 10.1.1.1 24 //Spine1 and Spine2 function as all-active gateways. The IP
addresses and MAC addresses of the VBDIF interfaces on the two devices must be the same.
[*Spine1-Vbdif10] mac-address 0000-5e00-0101
[*Spine1-Vbdif10] quit
[*Spine1] interface vbdif 20
[*Spine1-Vbdif20] ip binding vpn-instance vpn1
[*Spine1-Vbdif20] ip address 10.1.2.1 24
[*Spine1-Vbdif20] mac-address 0000-5e00-0102
[*Spine1-Vbdif20] quit
[*Spine1] dfs-group 1
[*Spine1-dfs-group-1] source ip 4.4.4.4
[*Spine1-dfs-group-1] active-active-gateway
[*Spine1-dfs-group-1-active-active-gateway] peer 5.5.5.5
[*Spine1-dfs-group-1-active-active-gateway] quit
[*Spine1-dfs-group-1] quit
[*Spine1] commit
```

- Step 5** Configure static routes on Spine1 and Spine2 to implement north-south traffic transmission.

Configure Spine1. Repeat this step for Spine2. For details, see [Configuration Files](#).

```
[~Spine1] interface 10ge 1/0/5
[~Spine1-10GE1/0/5] undo portswitch
[*Spine1-10GE1/0/5] ip address 10.1.10.1 24
[*Spine1-10GE1/0/5] quit
[~Spine1] interface 10ge 1/0/6
[~Spine1-10GE1/0/6] undo portswitch
[*Spine1-10GE1/0/6] ip address 10.1.20.1 24
[*Spine1-10GE1/0/6] quit
[*Spine1] ip route-static 0.0.0.0 0.0.0.0 10.1.10.2 //Configure a static route to a PE on the public network.
[*Spine1] ip route-static 0.0.0.0 0.0.0.0 10.1.20.2
[*Spine1] ip route-static 10.1.1.0 24 vpn-instance vpn1 //Configure a static route to the server network
segment with the next hop being the VPN instance.
[*Spine1] ip route-static 10.1.2.0 24 vpn-instance vpn1
[*Spine1] ip route-static vpn-instance vpn1 0.0.0.0 0.0.0.0 public //Configure a static route of the VPN
instance with the next hop being the public network instance.
[*Spine1] interface eth-trunk 10 //If a static route of which only a VPN instance is specified as the next
```

hop (without any outbound interface or next-hop address specified) is configured on switches excluding the CE16800, CE12800, CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI, CE6875EI, CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI, you need to configure a service loopback interface.

```
[*Spine1-Eth-Trunk10] trunkport 10ge 1/0/10 to 1/0/11
[*Spine1-Eth-Trunk10] service type tunnel
[*Spine1-Eth-Trunk10] quit
[*Spine1] commit
```

NOTE

When the CE6855HI, CE6856HI, and CE7855EI function as Layer 3 VXLAN gateways, they need to use VBDIF interfaces to connect to external networks.

When the CE6855HI, CE6856HI, and CE7855EI function as Layer 3 VXLAN gateways, you cannot run the **ip route-static vpn-instance vpn-source-name destination-address { mask | mask-length } { public | vpn-instance vpn-destination-name }** command to configure a static route with only a VPN instance specified as the next hop (no outbound interface or next-hop address is specified) to implement inter-VPN communication within a VXLAN network or between a VXLAN network and a non-VXLAN network.

----End

Verifying the Configuration

Run the **display vxlan tunnel** command to check the VXLAN tunnel information. The following example shows the command output on Spine1.

```
[~Spine1] display vxlan tunnel
Number of vxlan tunnel : 2

```

Tunnel ID	Source	Destination	State	Type	Uptime
4026531841	1.1.1.1	2.2.2.2	up	dynamic	0035h21m
4026531842	1.1.1.1	3.3.3.3	up	dynamic	0036h21m

The preceding information indicates that servers can communicate.

Configuration Files

- Spine1 configuration file

```
#
sysname Spine1
#
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800,
CE16800 equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
source ip 4.4.4.4
#
active-active-gateway
peer 5.5.5.5
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 4.4.4.4:1
vpn-target 0:1 export-extcommunity
vpn-target 0:1 import-extcommunity
#
bridge-domain 10
vxlan vni 10
evpn
route-distinguisher 4.4.4.4:10
vpn-target 0:10 export-extcommunity
vpn-target 0:10 import-extcommunity
#
```

```
bridge-domain 20
vxlان vni 20
evpn
route-distinguisher 4.4.4:20
vpn-target 0:20 export-extcommunity
vpn-target 0:20 import-extcommunity
#
interface Vbdif10
ip binding vpn-instance vpn1
ip address 10.1.1.1 255.255.255.0
mac-address 0000-5e00-0101
#
interface Vbdif20
ip binding vpn-instance vpn1
ip address 10.1.2.1 255.255.255.0
mac-address 0000-5e00-0102
#
interface Eth-Trunk10 //This step only needs to be performed on switches excluding the CE16800,
CE12800, CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI,
CE6875EI, CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI.
service type tunnel
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.1.1 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.2.1 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/3
undo portswitch
ip address 192.168.3.1 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/4
undo portswitch
ip address 192.168.4.1 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/5
undo portswitch
ip address 10.1.10.1 255.255.255.0
#
interface 10GE1/0/6
undo portswitch
ip address 10.1.20.1 255.255.255.0
#
interface 10GE1/0/10 //This step only needs to be performed on switches excluding the CE16800,
CE12800, CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI,
CE6875EI, CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI.
eth-trunk 10
#
interface 10GE1/0/11 //This step only needs to be performed on switches excluding the CE16800,
CE12800, CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI,
CE6875EI, CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI.
eth-trunk 10
#
interface LoopBack1
ip address 1.1.1.1 255.255.255.255
#
interface LoopBack2
ip address 4.4.4.4 255.255.255.255
#
interface Nve1
source 1.1.1.1
vni 10 head-end peer-list protocol bgp
```

```
vni 20 head-end peer-list protocol bgp
#
bgp 100
peer 3.3.3.3 as-number 100
peer 3.3.3.3 connect-interface LoopBack2
peer 6.6.6.6 as-number 100
peer 6.6.6.6 connect-interface LoopBack2
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack2
#
ipv4-family unicast
peer 3.3.3.3 enable
peer 6.6.6.6 enable
peer 7.7.7.7 enable
#
l2vpn-family evpn
undo policy vpn-target
peer 3.3.3.3 enable
peer 3.3.3.3 reflect-client
peer 6.6.6.6 enable
peer 6.6.6.6 reflect-client
peer 7.7.7.7 enable
peer 7.7.7.7 reflect-client
#
ospf 1
area 0.0.0.0
network 1.1.1.1 0.0.0.0
network 4.4.4.4 0.0.0.0
network 192.168.1.0 0.0.0.255
network 192.168.2.0 0.0.0.255
network 192.168.3.0 0.0.0.255
network 192.168.4.0 0.0.0.255
#
ip route-static 0.0.0.0 0.0.0.0 10.1.10.2
ip route-static 0.0.0.0 0.0.0.0 10.1.20.2
ip route-static 10.1.1.0 255.255.255.0 vpn-instance vpn1
ip route-static 10.1.2.0 255.255.255.0 vpn-instance vpn1
ip route-static vpn-instance vpn1 0.0.0.0 0.0.0.0 public
#
return
```

- Spine2 configuration file

```
#
sysname Spine2
#
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800, CE16800 equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
source ip 5.5.5.5
#
active-active-gateway
peer 4.4.4.4
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 5.5.5.5:1
vpn-target 0:1 export-extcommunity
vpn-target 0:1 import-extcommunity
#
bridge-domain 10
vxlan vni 10
evpn
route-distinguisher 5.5.5.5:10
vpn-target 0:10 export-extcommunity
vpn-target 0:10 import-extcommunity
#
bridge-domain 20
```



```
vxlan vni 20
evpn
 route-distinguisher 5.5.5.5:20
 vpn-target 0:20 export-extcommunity
 vpn-target 0:20 import-extcommunity
#
interface Vbdif10
 ip binding vpn-instance vpn1
 ip address 10.1.1.1 255.255.255.0
 mac-address 0000-5e00-0101
#
interface Vbdif20
 ip binding vpn-instance vpn1
 ip address 10.1.2.1 255.255.255.0
 mac-address 0000-5e00-0102
#
interface Eth-Trunk10 //This step only needs to be performed on switches excluding the CE16800,
CE12800, CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI,
CE6875EI, CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI.
 service type tunnel
#
interface 10GE1/0/1
 undo portswitch
 ip address 192.168.5.1 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/2
 undo portswitch
 ip address 192.168.6.1 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/3
 undo portswitch
 ip address 192.168.7.1 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/4
 undo portswitch
 ip address 192.168.8.1 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/5
 undo portswitch
 ip address 10.1.30.1 255.255.255.0
#
interface 10GE1/0/6
 undo portswitch
 ip address 10.1.40.1 255.255.255.0
#
interface 10GE1/0/10 //This step only needs to be performed on switches excluding the CE16800,
CE12800, CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI,
CE6875EI, CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI.
 eth-trunk 10
#
interface 10GE1/0/11 //This step only needs to be performed on switches excluding the CE16800,
CE12800, CE12800E (configured with ED-E, EG-E, and EGA-E series cards), CE5880EI, CE6870EI,
CE6875EI, CE6881, CE5881, CE6881K, CE6881E, CE6863, CE6863E, CE6863K, and CE6880EI.
 eth-trunk 10
#
interface LoopBack1
 ip address 1.1.1.1 255.255.255.255
#
interface LoopBack2
 ip address 5.5.5.5 255.255.255.255
#
interface Nve1
 source 1.1.1.1
 vni 10 head-end peer-list protocol bgp
 vni 20 head-end peer-list protocol bgp
```

```
#
bgp 100
peer 3.3.3.3 as-number 100
peer 3.3.3.3 connect-interface LoopBack2
peer 6.6.6.6 as-number 100
peer 6.6.6.6 connect-interface LoopBack2
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack2
#
ipv4-family unicast
peer 3.3.3.3 enable
peer 6.6.6.6 enable
peer 7.7.7.7 enable
#
l2vpn-family evpn
undo policy vpn-target
peer 3.3.3.3 enable
peer 3.3.3.3 reflect-client
peer 6.6.6.6 enable
peer 6.6.6.6 reflect-client
peer 7.7.7.7 enable
peer 7.7.7.7 reflect-client
#
ospf 1
area 0.0.0.0
network 1.1.1.1 0.0.0.0
network 5.5.5.5 0.0.0.0
network 192.168.5.0 0.0.0.255
network 192.168.6.0 0.0.0.255
network 192.168.7.0 0.0.0.255
network 192.168.8.0 0.0.0.255
#
ip route-static 0.0.0.0 0.0.0.0 10.1.30.2
ip route-static 0.0.0.0 0.0.0.0 10.1.40.2
ip route-static 10.1.1.0 255.255.255.0 vpn-instance vpn1
ip route-static 10.1.2.0 255.255.255.0 vpn-instance vpn1
ip route-static vpn-instance vpn1 0.0.0.0 0.0.0.0 public
#
return
```

- Leaf1 configuration file

```
#
sysname Leaf1
#
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800,
CE16800 equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
priority 150
source ip 6.6.6.6
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
bridge-domain 10
vxlan vni 10
evpn
route-distinguisher 6.6.6.6:10
vpn-target 0:10 export-extcommunity
vpn-target 0:10 import-extcommunity
#
bridge-domain 20
vxlan vni 20
evpn
route-distinguisher 6.6.6.6:20
vpn-target 0:20 export-extcommunity
vpn-target 0:20 import-extcommunity
#
```

```

interface Eth-Trunk1
 mode lacp-static
 peer-link 1
#
interface Eth-Trunk2
 stp edged-port enable
 mode lacp-static
 dfs-group 1 m-lag 1
#
interface Eth-Trunk2.10 mode l2
 encapsulation dot1q vid 10
 bridge-domain 10
#
interface Eth-Trunk3
 stp edged-port enable
 mode lacp-static
 dfs-group 1 m-lag 2
#
interface Eth-Trunk3.20 mode l2
 encapsulation dot1q vid 20
 bridge-domain 20
#
interface 10GE1/0/1
 undo portswitch
 ip address 192.168.1.2 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/2
 undo portswitch
 ip address 192.168.5.2 255.255.255.0
 ospf network-type p2p
#
interface 10GE1/0/3
 eth-trunk 1
#
interface 10GE1/0/4
 eth-trunk 1
#
interface 10GE1/0/5
 eth-trunk 2
#
interface 10GE1/0/6
 eth-trunk 3
#
interface LoopBack1
 ip address 2.2.2.2 255.255.255.255
#
interface LoopBack2
 ip address 6.6.6.6 255.255.255.255
#
interface Nve1
 source 2.2.2.2
 vni 10 head-end peer-list protocol bgp
 vni 20 head-end peer-list protocol bgp
#
monitor-link group 1
 port 10GE1/0/1 uplink
 port 10GE1/0/2 uplink
 port Eth-Trunk2 downlink 1
 port Eth-Trunk3 downlink 2
#
bgp 100
 peer 4.4.4.4 as-number 100
 peer 4.4.4.4 connect-interface LoopBack2
 peer 5.5.5.5 as-number 100
 peer 5.5.5.5 connect-interface LoopBack2
#
ipv4-family unicast
 peer 4.4.4.4 enable

```

```
peer 5.5.5.5 enable
#
l2vpn-family evpn
policy vpn-target
peer 4.4.4.4 enable
peer 5.5.5.5 enable
#
ospf 1
area 0.0.0.0
network 2.2.2.2 0.0.0.0
network 6.6.6.6 0.0.0.0
network 192.168.1.0 0.0.0.255
network 192.168.5.0 0.0.0.255
#
return
```

- Leaf2 configuration file

```
#
sysname Leaf2
#
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800, CE16800 equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
source ip 7.7.7.7
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
bridge-domain 10
vxlan vni 10
evpn
route-distinguisher 7.7.7.7:10
vpn-target 0:10 export-extcommunity
vpn-target 0:10 import-extcommunity
#
bridge-domain 20
vxlan vni 20
evpn
route-distinguisher 7.7.7.7:20
vpn-target 0:20 export-extcommunity
vpn-target 0:20 import-extcommunity
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk2
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 1
#
interface Eth-Trunk2.10 mode l2
encapsulation dot1q vid 10
bridge-domain 10
#
interface Eth-Trunk3
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 2
#
interface Eth-Trunk3.20 mode l2
encapsulation dot1q vid 20
bridge-domain 20
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.2.2 255.255.255.0
```

```
ospf network-type p2p
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.6.2 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/3
eth-trunk 1
#
interface 10GE1/0/4
eth-trunk 1
#
interface 10GE1/0/5
eth-trunk 2
#
interface 10GE1/0/6
eth-trunk 3
#
interface LoopBack1
ip address 2.2.2.2 255.255.255.255
#
interface LoopBack2
ip address 7.7.7.7 255.255.255.255
#
interface Nve1
source 2.2.2.2
vni 10 head-end peer-list protocol bgp
vni 20 head-end peer-list protocol bgp
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk2 downlink 1
port Eth-Trunk3 downlink 2
#
bgp 100
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack2
peer 5.5.5.5 as-number 100
peer 5.5.5.5 connect-interface LoopBack2
#
ipv4-family unicast
peer 4.4.4.4 enable
peer 5.5.5.5 enable
#
l2vpn-family evpn
policy vpn-target
peer 4.4.4.4 enable
peer 5.5.5.5 enable
#
ospf 1
area 0.0.0.0
network 2.2.2.2 0.0.0.0
network 7.7.7.7 0.0.0.0
network 192.168.2.0 0.0.0.255
network 192.168.6.0 0.0.0.255
#
return
```

- iStack configuration file

```
#
sysname iStack
#
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800,
CE16800 equipped with A series cards, CE6870EI, and CE6875EI.
#
evpn-overlay enable
#
bridge-domain 10
```

```

vxlan vni 10
evpn
 route-distinguisher 3.3.3.3:10
 vpn-target 0:10 export-extcommunity
 vpn-target 0:10 import-extcommunity
#
bridge-domain 20
vxlan vni 20
evpn
 route-distinguisher 3.3.3.3:20
 vpn-target 0:20 export-extcommunity
 vpn-target 0:20 import-extcommunity
#
stack
#
stack member 1 domain 10
stack member 1 priority 150
#
stack member 2 domain 10
#
interface Eth-Trunk2
stp edged-port enable
mode lacp-static
#
interface Eth-Trunk2.10 mode l2
encapsulation dot1q vid 10
bridge-domain 10
#
interface Eth-Trunk3
stp edged-port enable
mode lacp-static
#
interface Eth-Trunk3.20 mode l2
encapsulation dot1q vid 20
bridge-domain 20
#
interface Stack-Port1/1
#
interface Stack-Port2/1
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.3.2 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.7.2 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/3
port mode stack
stack-port 1/1
port crc-statistics trigger error-down
#
interface 10GE1/0/4
port mode stack
stack-port 1/1
port crc-statistics trigger error-down
#
interface 10GE1/0/5
eth-trunk 2
#
interface 10GE1/0/6
eth-trunk 3
#
interface 10GE1/0/9
dual-active detect mode direct
#

```

```
interface 10GE2/0/1
 undo portswitch
 ip address 192.168.4.2 255.255.255.0
 ospf network-type p2p
#
interface 10GE2/0/2
 undo portswitch
 ip address 192.168.8.2 255.255.255.0
 ospf network-type p2p
#
interface 10GE2/0/3
 port mode stack
 stack-port 2/1
 port crc-statistics trigger error-down
#
interface 10GE2/0/4
 port mode stack
 stack-port 2/1
 port crc-statistics trigger error-down
#
interface 10GE2/0/5
 eth-trunk 2
#
interface 10GE2/0/6
 eth-trunk 3
#
interface 10GE2/0/9
 dual-active detect mode direct
#
interface LoopBack1
 ip address 3.3.3.3 255.255.255.255
#
interface Nve1
 source 3.3.3.3
 vni 10 head-end peer-list protocol bgp
 vni 20 head-end peer-list protocol bgp
#
bgp 100
 peer 4.4.4.4 as-number 100
 peer 4.4.4.4 connect-interface LoopBack1
 peer 5.5.5.5 as-number 100
 peer 5.5.5.5 connect-interface LoopBack1
#
ipv4-family unicast
 peer 4.4.4.4 enable
 peer 5.5.5.5 enable
#
l2vpn-family evpn
 policy vpn-target
 peer 4.4.4.4 enable
 peer 5.5.5.5 enable
#
ospf 1
 area 0.0.0.0
 network 3.3.3.3 0.0.0.0
 network 192.168.3.0 0.0.0.255
 network 192.168.4.0 0.0.0.255
 network 192.168.7.0 0.0.0.255
 network 192.168.8.0 0.0.0.255
#
return
```

1.7 Example for Configuring a VXLAN Network Using a Two-Layer Architecture in Distributed Gateway Deployment Mode in a DC

Applicable Products and Versions

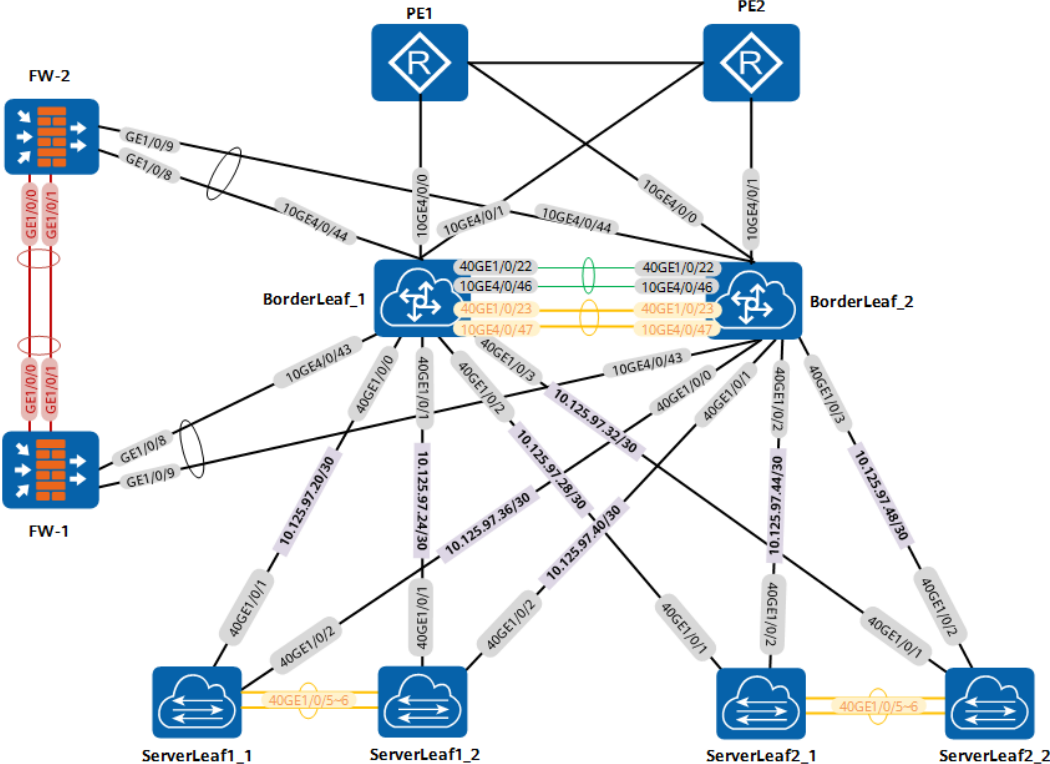
- This example applies to CloudEngine16800 series switches running V200R005C00 or later versions, CE9800 series switches running V200R020C00 or later versions, CE8800 series switches running V100R006C00 or later versions, CE7800 series switches running V100R003C00 or later versions, and CE12800, CE6800, and CE5800 series switches running V100R001C00 or later versions.

Networking Requirements

In [Figure 1-7](#), spine, border leaf, and service leaf nodes are combined. The server leaf nodes and the combined spine/border leaf/service leaf nodes form a two-layer architecture in the physical topology.

- Border leaf layer: Border leaf nodes function as egress nodes of the distributed overlay network. In the southbound direction, they connect to server leaf nodes through Layer 3 routed interfaces, implementing ECMP forwarding. In the northbound direction, they connect to egress routers (PEs).
- Server leaf layer: Server leaf nodes are configured with M-LAG and connected to spine nodes through Layer 3 routed interfaces in the northbound direction.

Figure 1-7 Networking for configuring a VXLAN network using a two-layer architecture in distributed gateway deployment mode in a DC



Plan two types of loopback interface addresses for switches. The suggestions are as follows:

- Loopback0: Its IP address is used as the VTEP IP address. The members in an active-active device group must use the same VTEP IP address.
- Loopback1:
 - Its IP address is used as the router ID.
 - Its IP address is used as the DFS group IP address of an M-LAG.
 - It is used as the source interface that sends BGP packets when a BGP EVPN peer relationship is established.

Table 1-7 lists the loopback interface address plan for each switch.

Table 1-7 Data plan (loopback interface address plan)

Device Name	IP Address of Loopback0	IP Address of Loopback1
BorderLeaf _1	10.125.99.1/32 (virtual MAC address: 0000-5e00-0101)	10.125.98.1/32
BorderLeaf _2	10.125.99.1/32 (virtual MAC address: 0000-5e00-0101)	10.125.98.2/32

Device Name	IP Address of Loopback0	IP Address of Loopback1
ServerLeaf1_1	10.125.99.2/32	10.125.98.3/32
ServerLeaf1_2	10.125.99.2/32	10.125.98.4/32
ServerLeaf2_1	10.125.99.3/32	10.125.98.5/32
ServerLeaf2_2	10.125.99.3/32	10.125.98.6/32

Table 1-8 Interconnection IP address plan

Device Name	Interface Number	IP Address	Connected Device and Interface Number	Description
BorderLeaf1	Eth-Trunk 20	10.125.97.17/30	BorderLeaf2: Eth-Trunk 20	Egress bypass link
	10GE4/0/0	10.125.97.1/30	PE1	-
	10GE4/0/1	10.125.97.5/30	PE2	-
	40GE1/0/0	10.125.97.21/30	ServerLeaf1_1: 40GE1/0/1	-
	40GE1/0/1	10.125.97.25/30	ServerLeaf1_2: 40GE1/0/1	-
	40GE1/0/2	10.125.97.29/30	ServerLeaf2_1: 40GE1/0/1	-
	40GE1/0/3	10.125.97.33/30	ServerLeaf2_2: 40GE1/0/1	-
	10GE4/0/43	-	FW-1	-
	10GE4/0/44	-	FW-2	-
	VLANIF 11	10.125.97.57/29	-	IP address of the interface on the management link for interconnection with a firewall

Device Name	Interface Number	IP Address	Connected Device and Interface Number	Description
BorderLeaf_2	Eth-Trunk 20	10.125.97.18/30	BorderLeaf_1: Eth-Trunk 20	Egress bypass link
	10GE4/0/0	10.125.97.9/30	PE1	-
	10GE4/0/1	10.125.97.13/30	PE2	-
	40GE1/0/0	10.125.97.37/30	ServerLeaf1_1: 40GE1/0/2	-
	40GE1/0/1	10.125.97.41/30	ServerLeaf1_2: 40GE1/0/2	-
	40GE1/0/2	10.125.97.45/30	ServerLeaf2_1: 40GE1/0/2	-
	40GE1/0/3	10.125.97.49/30	ServerLeaf2_2: 40GE1/0/2	-
	10GE4/0/43	-	FW-1	-
	10GE4/0/44	-	FW-2	-
	VLANIF 11	10.125.97.57/29	-	IP address of the interface on the management link for interconnection with a firewall
ServerLeaf1_1	40GE1/0/1	10.125.97.22/30	BorderLeaf_1: 40GE1/0/0	-
	40GE1/0/2	10.125.97.38/30	BorderLeaf_2: 40GE1/0/0	-
ServerLeaf1_2	40GE1/0/1	10.125.97.26/30	BorderLeaf_1: 40GE1/0/1	-
	40GE1/0/2	10.125.97.42/30	BorderLeaf_2: 40GE1/0/1	-
ServerLeaf2_1	40GE1/0/1	10.125.97.30/30	BorderLeaf_1: 40GE1/0/2	-
	40GE1/0/2	10.125.97.46/30	BorderLeaf_2: 40GE1/0/2	-
ServerLeaf2_2	40GE1/0/1	10.125.97.34/30	BorderLeaf_1: 40GE1/0/3	-

Device Name	Interface Number	IP Address	Connected Device and Interface Number	Description
	40GE1/0/2	10.125.97.50/30	BorderLeaf_2: 40GE1/0/3	-
FW-1	Eth-Trunk 0 (GigabitEthernet 1/0/0 and 1/0/1)	10.125.97.73/30	FW2: Eth-Trunk 0	Heartbeat interface of the firewall
	Eth-Trunk 11 (GigabitEthernet 1/0/8 and 1/0/9)	-	-	Interfaces connected to BorderLeaf_1 and BorderLeaf_2
	VLANIF 3004	10.125.97.242/30	-	vsys_1
FW-2	Eth-Trunk 0	10.125.97.74/30	FW1: Eth-Trunk 0	Heartbeat interface of the firewall
	Eth-Trunk 11 (GigabitEthernet 1/0/8 and 1/0/9)	10.125.97.242/30	-	Interfaces connected to BorderLeaf_1 and BorderLeaf_2

Table 1-9 External network IP address

IP address for accessing the external network
1.2.3.4/24

Table 1-10 RDs and RTs of devices

Device Name	VLAN ID	BD ID	VNI	EVPN Instance			VPN Instance			
				RD	RT		VPN Instance Name	VNI	RD	RT
					ERT/IRT	ERT				ERT/IRT (EVPN)
ServerLeaf 1_1	10	10	10	10:2	100:10	100:5010	vpn1	5010	20:2	100:5010

ServerLeaf 1_2				1 0: 4			vpn2		20:4	
ServerLeaf 2_1	2 0	20	20	1 0: 3	10 0:2 0		vpn3		20:3	
ServerLeaf 2_2				1 0: 5			vpn4		20:5	

Configuration Roadmap

The configuration roadmap is as follows:

1. Configure VXLAN optimization commands.
2. Configure the underlay network.
 - a. Configure border leaf nodes.
 - i. Configure IP addresses for Layer 3 interconnection with server leaf nodes, management IP addresses for interconnection with firewalls, Loopback0 IP addresses (used as VTEP IP addresses), Loopback1 IP addresses (used as router IDs and DFS group IP addresses), and VTEP IP addresses of NVE interfaces.
 - ii. Configure the M-LAG global mode, DFS group, peer-link, and M-LAG interfaces connected to firewalls.
 - iii. Configure OSPF, set the network type of OSPF interfaces to P2P, advertise loopback interface addresses and management IP addresses for interconnection with firewalls, and configure BGP EVPN as the VXLAN control plane.
 - b. Configure server leaf nodes.
The configuration roadmap is the same as that of border leaf nodes.
 - c. Configure firewalls.
 - i. Configure basic information about firewalls.
 - ii. Disable active and standby firewalls from backing up the running configurations locally.
 - iii. Configure interfaces for connecting firewalls to border leaf nodes or service leaf nodes.
 - iv. Configure heartbeat interfaces between the two firewalls.
 - v. Configure the active/standby mirroring mode for the two firewalls.
 - vi. Configure security zones and the default security policy. The configuration only needs to be performed on FW-1, and FW-2 will automatically synchronize the configuration.
 - vii. Enable the vSYS function on the firewalls. The configuration only needs to be performed on FW-1, and FW-2 will automatically synchronize the configuration.

3. Configure the overlay network.
 - a. Configure border leaf nodes.
 - b. Configure server leaf nodes.
 - c. Configure firewalls.

Procedure

Step 1 Configure VXLAN optimization commands.

Before configuring VXLAN on a CE switch, configure VXLAN optimization commands, service loopback, and a reserved VLAN for a Layer 3 interface based on the device model to ensure stable service running. The configuration commands vary according to the device model. When CE16800, CE6881, CE6863, CE6863E, or CE6863K switches are used as border leaf, service leaf, and server leaf nodes, you need to perform the following configurations. The following uses BorderLeaf_1 as an example.

Configure BorderLeaf_1. The configurations of other devices are similar to the configuration of BorderLeaf_1.

```
<HUAWEI> system-view
[~HUAWEI] sysname BorderLeaf_1
[*HUAWEI] commit
[*BorderLeaf_1] system resource large-route //Set the system resource mode to the large-route mode.
The configuration takes effect after the device is restarted.
[*BorderLeaf_1] vxlan tunnel-status track exact-route //Enable subscription to the status of the exact
route to a VXLAN tunnel destination to optimize network convergence performance.
[*BorderLeaf_1] port high-performance mode mode2 //When a CE6863, CE6863E, or CE6863K switch
functions as a border leaf node, configure the high-performance mode to adjust the bandwidth of
internally connected interfaces.
[*BorderLeaf_1] commit
```

Step 2 Configure the underlay network.

1. Configure border leaf nodes.
 - a. Configure IP addresses.
 - i. Configure IP addresses for interfaces connecting border leaf nodes to other devices.

Configure IP addresses for interfaces connecting BorderLeaf_1 to server leaf nodes.

```
[~BorderLeaf_1] interface 40GE 1/0/0
[~BorderLeaf_1-40GE1/0/0] description to ServerLeaf1_1
[*BorderLeaf_1-40GE1/0/0] undo portswitch
[*BorderLeaf_1-40GE1/0/0] ip address 10.125.97.21 255.255.255.252
[*BorderLeaf_1-40GE1/0/0] ospf network-type p2p //Set the network type of the OSPF
interface connected to a server leaf node to P2P.
[*BorderLeaf_1-40GE1/0/0] quit
[*BorderLeaf_1] interface 40GE 1/0/1
[*BorderLeaf_1-40GE1/0/1] description to ServerLeaf1_2
[*BorderLeaf_1-40GE1/0/1] undo portswitch
[*BorderLeaf_1-40GE1/0/1] ip address 10.125.97.25 255.255.255.252
[*BorderLeaf_1-40GE1/0/1] ospf network-type p2p
[*BorderLeaf_1-40GE1/0/1] quit
[*BorderLeaf_1] interface 40GE 1/0/2
[*BorderLeaf_1-40GE1/0/2] description to ServerLeaf2_1
[*BorderLeaf_1-40GE1/0/2] undo portswitch
[*BorderLeaf_1-40GE1/0/2] ip address 10.125.97.29 255.255.255.252
[*BorderLeaf_1-40GE1/0/2] ospf network-type p2p
[*BorderLeaf_1-40GE1/0/2] quit
[*BorderLeaf_1] interface 40GE 1/0/3
```

```
[*BorderLeaf_1-40GE1/0/3] description to ServerLeaf2_2
[*BorderLeaf_1-40GE1/0/3] undo portswitch
[*BorderLeaf_1-40GE1/0/3] ip address 10.125.97.33 255.255.255.252
[*BorderLeaf_1-40GE1/0/3] ospf network-type p2p
[*BorderLeaf_1-40GE1/0/3] quit
[*BorderLeaf_1-40GE1/0/3] commit
```

Configure IP addresses for interfaces connecting BorderLeaf_2 to server leaf nodes.

```
[~BorderLeaf_2] interface 40GE 1/0/0
[~BorderLeaf_2-40GE1/0/0] description to ServerLeaf1_1
[*BorderLeaf_2-40GE1/0/0] undo portswitch
[*BorderLeaf_2-40GE1/0/0] ip address 10.125.97.37 255.255.255.252
[*BorderLeaf_2-40GE1/0/0] ospf network-type p2p
[*BorderLeaf_2-40GE1/0/0] quit
[*BorderLeaf_2] interface 40GE 1/0/1
[*BorderLeaf_2-40GE1/0/1] description to ServerLeaf1_2
[*BorderLeaf_2-40GE1/0/1] undo portswitch
[*BorderLeaf_2-40GE1/0/1] ip address 10.125.97.41 255.255.255.252
[*BorderLeaf_2-40GE1/0/1] ospf network-type p2p
[*BorderLeaf_2-40GE1/0/1] quit
[*BorderLeaf_2] interface 40GE 1/0/2
[*BorderLeaf_2-40GE1/0/2] description to ServerLeaf2_1
[*BorderLeaf_2-40GE1/0/2] undo portswitch
[*BorderLeaf_2-40GE1/0/2] ip address 10.125.97.45 255.255.255.252
[*BorderLeaf_2-40GE1/0/2] ospf network-type p2p
[*BorderLeaf_2-40GE1/0/2] quit
[*BorderLeaf_2] interface 40GE 1/0/3
[*BorderLeaf_2-40GE1/0/3] description to ServerLeaf2_2
[*BorderLeaf_2-40GE1/0/3] undo portswitch
[*BorderLeaf_2-40GE1/0/3] ip address 10.125.97.49 255.255.255.252
[*BorderLeaf_2-40GE1/0/3] ospf network-type p2p
[*BorderLeaf_2-40GE1/0/3] quit
[*BorderLeaf_2-40GE1/0/3] commit
```

Configure IP addresses for interfaces connecting BorderLeaf_1 to PEs.

```
[~BorderLeaf_1] interface 10GE 4/0/0
[*BorderLeaf_1-10GE4/0/0] description to PE1
[*BorderLeaf_1-10GE4/0/0] undo portswitch
[*BorderLeaf_1-10GE4/0/0] ip address 10.125.97.1 255.255.255.252
[*BorderLeaf_1-10GE4/0/0] quit
[*BorderLeaf_1] interface 10GE 4/0/1
[*BorderLeaf_1-10GE4/0/1] description to PE2
[*BorderLeaf_1-10GE4/0/1] undo portswitch
[*BorderLeaf_1-10GE4/0/1] ip address 10.125.97.5 255.255.255.252
[*BorderLeaf_1-10GE4/0/1] quit
[*BorderLeaf_1] commit
```

Configure IP addresses for interfaces connecting BorderLeaf_2 to PEs.

```
[~BorderLeaf_2] interface 10GE 4/0/0
[*BorderLeaf_2-10GE4/0/0] description to PE1
[*BorderLeaf_2-10GE4/0/0] undo portswitch
[*BorderLeaf_2-10GE4/0/0] ip address 10.125.97.9 255.255.255.252
[*BorderLeaf_2-10GE4/0/0] quit
[*BorderLeaf_2] interface 10GE 4/0/1
[*BorderLeaf_2-10GE4/0/1] description to PE2
[*BorderLeaf_2-10GE4/0/1] undo portswitch
[*BorderLeaf_2-10GE4/0/1] ip address 10.125.97.13 255.255.255.252
[*BorderLeaf_2-10GE4/0/1] quit
[*BorderLeaf_2] commit
```

- ii. Configure IP addresses for loopback interfaces on border leaf nodes.

Configure IP addresses for loopback interfaces on BorderLeaf_1.

```
[~BorderLeaf_1] interface LoopBack 0
[*BorderLeaf_1-LoopBack0] description VTEP
[*BorderLeaf_1-LoopBack0] ipv6 enable //Enable IPv6 if it is required.
```

```
[*BorderLeaf_1-LoopBack0]ip address 10.125.99.1 255.255.255.255
[*BorderLeaf_1-LoopBack0]quit
[*BorderLeaf_1]interface LoopBack 1
[*BorderLeaf_1-LoopBack1]description DFS-GROUP/ROUTER-ID
[*BorderLeaf_1-LoopBack1]ip address 10.125.98.1 255.255.255.255
[*BorderLeaf_1-LoopBack1]quit
[*BorderLeaf_1-LoopBack1]commit
```

Configure IP addresses for loopback interfaces on BorderLeaf_2.

```
[~BorderLeaf_2]interface LoopBack 0
[*BorderLeaf_2-LoopBack0]description VTEP
[*BorderLeaf_2-LoopBack0]ipv6 enable //Enable IPv6 if it is required.
[*BorderLeaf_2-LoopBack0]ip address 10.125.99.1 255.255.255.255
[*BorderLeaf_2-LoopBack0]quit
[*BorderLeaf_2]interface LoopBack 1
[*BorderLeaf_2-LoopBack1]description DFS-GROUP/ROUTER-ID
[*BorderLeaf_2-LoopBack1]ip address 10.125.98.2 255.255.255.255
[*BorderLeaf_2-LoopBack1]quit
[*BorderLeaf_2-LoopBack1]commit
```

- iii. Configure VTEP IP addresses and virtual MAC addresses for NVE interfaces.

Configure an NVE interface on BorderLeaf_1.

```
[~BorderLeaf_1]interface nve 1
[*BorderLeaf_1-Nve1]source 10.125.99.1
[*BorderLeaf_1-Nve1]mac-address 0000-5e00-0101
[*BorderLeaf_1-Nve1]commit
```

Configure an NVE interface on BorderLeaf_2.

```
[~BorderLeaf_2]interface nve 1
[*BorderLeaf_2-Nve1]source 10.125.99.1
[*BorderLeaf_2-Nve1]mac-address 0000-5e00-0101
[*BorderLeaf_2-Nve1]commit
```

- b. Configure M-LAG.

- i. Configure the M-LAG mode.

Configure the M-LAG mode on BorderLeaf_1.

```
[~BorderLeaf_1]stp mode rstp
[*BorderLeaf_1]stp v-stp enable //Configure M-LAG in V-STP mode.
[*BorderLeaf_1]commit
```

Configure the M-LAG mode on BorderLeaf_2.

```
[~BorderLeaf_2]stp mode rstp
[*BorderLeaf_2]stp v-stp enable //Configure M-LAG in V-STP mode.
[*BorderLeaf_2]commit
```

- ii. Configure a DFS group.

Configure a DFS group on BorderLeaf_1.

```
[~BorderLeaf_1]dfs-group 1
[*BorderLeaf_1-dfs-group-1]priority 150 //Set the DFS priority to be higher than that of
the remote device. The default priority is 100.
[*BorderLeaf_1-dfs-group-1]dual-active detection enhanced enable
[*BorderLeaf_1-dfs-group-1]source ip 10.125.98.1 peer 10.125.98.2
[*BorderLeaf_1-dfs-group-1]consistency-check enable mode loose //Enable M-LAG
configuration consistency check and specify the loose mode.
[*BorderLeaf_1-dfs-group-1]quit
[*BorderLeaf_1-dfs-group-1]commit
```

Configure a DFS group on BorderLeaf_2.

```
[~BorderLeaf_2]dfs-group 1
[*BorderLeaf_2-dfs-group-1]dual-active detection enhanced enable
[*BorderLeaf_2-dfs-group-1]source ip 10.125.98.2 peer 10.125.98.1
[*BorderLeaf_2-dfs-group-1]consistency-check enable mode loose
[*BorderLeaf_2-dfs-group-1]quit
[*BorderLeaf_2-dfs-group-1]commit
```


iii. Configure a peer-link.

Configure a peer-link on BorderLeaf_1.

```
[~BorderLeaf_1]interface Eth-Trunk 0
[*BorderLeaf_1-Eth-Trunk0]trunkport 10GE 4/0/47
[*BorderLeaf_1-Eth-Trunk0]trunkport 40GE 1/0/23
[*BorderLeaf_1-Eth-Trunk0]mode lacp-static
[*BorderLeaf_1-Eth-Trunk0]peer-link 1
[*BorderLeaf_1-Eth-Trunk0]lacp mixed-rate link enable //Enable interfaces operating at
different rates to forward packets after the interfaces are added to an Eth-Trunk in LACP
mode.
[*BorderLeaf_1-Eth-Trunk0]quit
[*BorderLeaf_1]interface 40GE 4/0/23
[*BorderLeaf_1-40GE4/0/23]distribute-weight 4 //Set the load balancing weight of
40GE member interfaces to 4 and retain the default weight 1 of 10GE member interfaces.
[*BorderLeaf_1-40GE4/0/23]commit
```

Configure a peer-link on BorderLeaf_2.

```
[~BorderLeaf_1]interface Eth-Trunk 0
[*BorderLeaf_1-Eth-Trunk0]trunkport 10GE 4/0/47
[*BorderLeaf_1-Eth-Trunk0]trunkport 40GE 1/0/23
[*BorderLeaf_1-Eth-Trunk0]mode lacp-static
[*BorderLeaf_1-Eth-Trunk0]peer-link 1
[*BorderLeaf_1-Eth-Trunk0]lacp mixed-rate link enable //Enable interfaces operating at
different rates to forward packets after the interfaces are added to an Eth-Trunk in LACP
mode.
[*BorderLeaf_1-Eth-Trunk0]quit
[*BorderLeaf_1]interface 40GE 4/0/23
[*BorderLeaf_1-40GE4/0/23]distribute-weight 4 //Set the load balancing weight of
40GE member interfaces to 4 and retain the default weight 1 of 10GE member interfaces.
[*BorderLeaf_1-40GE4/0/23]quit
[*BorderLeaf_1-40GE4/0/23]commit
```

iv. Configure M-LAG interfaces.

Configure service links for interconnection with firewalls.

Configure interfaces for connecting BorderLeaf_1 to firewalls.

```
[~BorderLeaf_1]interface Eth-Trunk 11 //Configure a service interface for connecting to
the active firewall (FW-1).
[*BorderLeaf_1-Eth-Trunk11]trunkport 10GE 4/0/43
[*BorderLeaf_1-Eth-Trunk11]port link-type trunk
[*BorderLeaf_1-Eth-Trunk11]undo port trunk allow-pass vlan 1
[*BorderLeaf_1-Eth-Trunk11]stp edged-port enable
[*BorderLeaf_1-Eth-Trunk11]mode lacp-static
[*BorderLeaf_1-Eth-Trunk11]dfs-group 1 m-lag 3
[*BorderLeaf_1-Eth-Trunk11]quit
[*BorderLeaf_1]interface Eth-Trunk12 //Configure a service interface for connecting to
the standby firewall (FW-2).
[*BorderLeaf_1-Eth-Trunk11]trunkport 10GE 4/0/44
[*BorderLeaf_1-Eth-Trunk11]port link-type trunk
[*BorderLeaf_1-Eth-Trunk11]undo port trunk allow-pass vlan 1
[*BorderLeaf_1-Eth-Trunk11]stp edged-port enable
[*BorderLeaf_1-Eth-Trunk11]mode lacp-static
[*BorderLeaf_1-Eth-Trunk11]dfs-group 1 m-lag 4
[*BorderLeaf_1-Eth-Trunk11]quit
[*BorderLeaf_1-Eth-Trunk11]commit
```

Configure interconnection between BorderLeaf_2 and firewalls.

```
[~BorderLeaf_2]interface Eth-Trunk 11 //Configure a service interface for connecting to
the active firewall (FW-1).
[*BorderLeaf_2-Eth-Trunk11]trunkport 10GE 4/0/43
[*BorderLeaf_2-Eth-Trunk11]port link-type trunk
[*BorderLeaf_2-Eth-Trunk11]undo port trunk allow-pass vlan 1
[*BorderLeaf_2-Eth-Trunk11]stp edged-port enable
[*BorderLeaf_2-Eth-Trunk11]mode lacp-static
[*BorderLeaf_2-Eth-Trunk11]dfs-group 1 m-lag 3
[*BorderLeaf_2-Eth-Trunk11]quit
[*BorderLeaf_2]interface Eth-Trunk12 //Configure a service interface for connecting to
the standby firewall (FW-2).
```

```
[*BorderLeaf_2-Eth-Trunk11]trunkport 10GE 4/0/44
[*BorderLeaf_2-Eth-Trunk11]port link-type trunk
[*BorderLeaf_2-Eth-Trunk11]undo port trunk allow-pass vlan 1
[*BorderLeaf_2-Eth-Trunk11]stp edged-port enable
[*BorderLeaf_2-Eth-Trunk11]mode lacp-static
[*BorderLeaf_2-Eth-Trunk11]dfs-group 1 m-lag 4
[*BorderLeaf_2-Eth-Trunk11]quit
[*BorderLeaf_2-Eth-Trunk11]commit
```

c. Configure a routing protocol.

i. Configure OSPF to ensure connectivity on the VXLAN underlay network.

Configure OSPF on BorderLeaf_1.

```
[~BorderLeaf_1]bfd //Enable BFD globally.
[*BorderLeaf_1-bfd]quit
[*BorderLeaf_1]ospf
[*BorderLeaf_1]ospf 1 router-id 10.125.98.1
[*BorderLeaf_1-ospf-1]bfd all-interfaces enable
[*BorderLeaf_1-ospf-1]bfd all-interfaces min-tx-interval 500 min-rx-interval 500
detect-multiplier 3 //Set the interval for receiving and sending BFD packets to 500 ms
and the detection multiplier to 3 only when all devices on the network support hardware-
based BFD. In other scenarios, retain the default BFD parameter settings. That is, the
interval for receiving and sending BFD packets is 1000 ms and the detection multiplier is
3. The models that support hardware-based BFD include the CE6857, CE6865, CE8861,
CE8868, CE6875, CE5880, CE8850-64CQ-EI, CE6880, CE6881, CE6863, CE6863E, CE6863K,
CE12800E, CE16800, and CE12800 (configured with only the CE-L48XS-FG, CE-L48XS-FD1,
CE-L24LQ-FD, CE-L36LQ-FD, CE-L08CF-FG1, CE-L12CQ-FD, CE-L16CQ-FD, CE-L36CQ-FD,
CE-L36CQ-FD1, CE-L36CQ-FG, or CE-L36CQ-SD cards).
[*BorderLeaf_1-ospf-1]lsa-arrival-interval intelligent-timer 50 50 50 //Set the
interval for receiving OSPF LSAs to shorten the convergence time.
[*BorderLeaf_1-ospf-1]area 0.0.0.0
[*BorderLeaf_1-ospf-1-area-0.0.0.0]network 10.125.97.20 0.0.0.3 //Configure the
border leaf node to establish neighbor relationships with the four server leaf nodes.
[*BorderLeaf_1-ospf-1-area-0.0.0.0]network 10.125.97.24 0.0.0.3
[*BorderLeaf_1-ospf-1-area-0.0.0.0]network 10.125.97.28 0.0.0.3
[*BorderLeaf_1-ospf-1-area-0.0.0.0]network 10.125.97.32 0.0.0.3
[*BorderLeaf_1-ospf-1-area-0.0.0.0]network 10.125.98.1 0.0.0.0 //Advertise the
loopback interface address.
[*BorderLeaf_1-ospf-1-area-0.0.0.0]network 10.125.99.1 0.0.0.0
[*BorderLeaf_1-ospf-1-area-0.0.0.0]quit
[*BorderLeaf_1-ospf-1]quit
[*BorderLeaf_1-ospf-1]commit
```

Configure OSPF on BorderLeaf_2.

```
[~BorderLeaf_2]bfd //Enable BFD globally.
[*BorderLeaf_2-bfd]quit
[*BorderLeaf_2]ospf
[*BorderLeaf_2]ospf 1 router-id 10.125.98.2
[*BorderLeaf_2-ospf-1]bfd all-interfaces enable
[*BorderLeaf_2-ospf-1]bfd all-interfaces min-tx-interval 500 min-rx-interval 500
detect-multiplier 3 //Set the interval for receiving and sending BFD packets to 500 ms
and the detection multiplier to 3 only when all devices on the network support hardware-
based BFD. In other scenarios, retain the default BFD parameter settings. That is, the
interval for receiving and sending BFD packets is 1000 ms and the detection multiplier is 3.
[*BorderLeaf_2-ospf-1]lsa-arrival-interval intelligent-timer 50 50 50 //Set the
interval for receiving OSPF LSAs to shorten the convergence time.
[*BorderLeaf_2-ospf-1]area 0.0.0.0
[*BorderLeaf_2-ospf-1-area-0.0.0.0]network 10.125.97.36 0.0.0.3 //Configure the
border leaf node to establish neighbor relationships with the four server leaf nodes.
[*BorderLeaf_2-ospf-1-area-0.0.0.0]network 10.125.97.40 0.0.0.3
[*BorderLeaf_2-ospf-1-area-0.0.0.0]network 10.125.97.44 0.0.0.3
[*BorderLeaf_2-ospf-1-area-0.0.0.0]network 10.125.97.48 0.0.0.3
[*BorderLeaf_2-ospf-1-area-0.0.0.0]network 10.125.98.2 0.0.0.0 //Advertise the
loopback interface address.
[*BorderLeaf_2-ospf-1-area-0.0.0.0]network 10.125.99.1 0.0.0.0
[*BorderLeaf_2-ospf-1-area-0.0.0.0]quit
[*BorderLeaf_2-ospf-1]quit
[*BorderLeaf_2-ospf-1]commit
```

- ii. Perform either of the following configurations to optimize OSPF network convergence performance upon faults based on the device version.

- For V200R005C00 and earlier versions

Configure OSPF network convergence performance optimization triggered upon faults on BorderLeaf_1.

```
[~BorderLeaf_1]ospf 1
[*BorderLeaf_1-ospf-1]stub-router on-startup 3000 include-stub //Configure the
border leaf node as a stub router to improve network convergence performance
upon faults. If the border leaf node is a CE12800 series switch, set the interval
specified by on-startup to 3000s. If the border leaf node is a fixed switch, set the
interval specified by on-startup to 600s.
[*BorderLeaf_1-ospf-1]quit
[*BorderLeaf_1-ospf-1]commit
```

Configure OSPF network convergence performance optimization triggered upon faults on BorderLeaf_2. The configuration procedure and data are the same as those of BorderLeaf_1.

- For V200R005C10 and later versions

Configure OSPF network convergence performance optimization triggered upon faults on BorderLeaf_1.

```
[~BorderLeaf_1]interface 40GE 1/0/0
[*BorderLeaf_1-40GE1/0/0]ospf peer hold-max-cost timer 300000 //Set the
interval at which the maximum cost is stored in local LSAs to 300s after OSPF
neighbor relationships are established between all border leaf nodes and server leaf
nodes. The value is the sum of the delay (240s) after which an M-LAG interface
goes Up (overlay routes are converged during this period) plus the device entry
synchronization period (60s).
[*BorderLeaf_1-40GE1/0/0]quit
[*BorderLeaf_1]interface 40GE 1/0/1
[*BorderLeaf_1-40GE1/0/1]ospf peer hold-max-cost timer 300000
[*BorderLeaf_1-40GE1/0/1]quit
[*BorderLeaf_1]interface 40GE 1/0/2
[*BorderLeaf_1-40GE1/0/2]ospf peer hold-max-cost timer 300000
[*BorderLeaf_1-40GE1/0/2]quit
[*BorderLeaf_1]interface 40GE 1/0/3
[*BorderLeaf_1-40GE1/0/3]ospf peer hold-max-cost timer 300000
[*BorderLeaf_1-40GE1/0/3]quit
[*BorderLeaf_1-40GE1/0/3]commit
```

Configure OSPF network convergence performance optimization triggered upon faults on BorderLeaf_2. The configuration procedure and data are the same as those of BorderLeaf_1.

- iii. Configure BGP EVPN.

Configure BorderLeaf_1.

```
[~BorderLeaf_1] evpn-overlay enable //Configure EVPN as the VXLAN control plane.
[*BorderLeaf_1] bgp 100
[*BorderLeaf_1-bgp] router-id 10.125.98.1
[*BorderLeaf_1-bgp] advertise lowest-priority all-address-family peer-up delay 360 //
When the peer status changes from Down to Up, the priority of BGP routes is changed to
be the lowest. Route advertisement is delayed to prevent packet loss during traffic
switchback.
[*BorderLeaf_1-bgp] undo default ipv4-unicast //Disable the IPv4 unicast address
family for BGP peers to reduce the device load.
[*BorderLeaf_1-bgp] group ServerLeaf internal //Configure a peer group named
ServerLeaf and add peers to this group.
[*BorderLeaf_1-bgp] peer 10.125.98.3 group ServerLeaf
[*BorderLeaf_1-bgp] peer 10.125.98.4 group ServerLeaf
[*BorderLeaf_1-bgp] peer 10.125.98.5 group ServerLeaf
[*BorderLeaf_1-bgp] peer 10.125.98.6 group ServerLeaf
```

```
[*BorderLeaf_1-bgp] peer ServerLeaf connect-interface LoopBack1 //Specify the
source interface that sends BGP packets.
[*BorderLeaf_1-bgp] l2vpn-family evpn //Enable the BGP-EVPN address family and
enter the BGP-EVPN address family view.
[*BorderLeaf_1-bgp-af-evpn] undo policy vpn-target //Disable the border leaf node
from performing VPN target-based filtering for received EVPN routes.
[*BorderLeaf_1-bgp-af-evpn] peer ServerLeaf enable
[*BorderLeaf_1-bgp-af-evpn] peer 10.125.98.3 group ServerLeaf
[*BorderLeaf_1-bgp-af-evpn] peer 10.125.98.4 group ServerLeaf
[*BorderLeaf_1-bgp-af-evpn] peer 10.125.98.5 group ServerLeaf
[*BorderLeaf_1-bgp-af-evpn] peer 10.125.98.6 group ServerLeaf
[*BorderLeaf_1-bgp-af-evpn] peer ServerLeaf advertise irb //Enable the border leaf
node to advertise IRB and IRBv6 routes to the BGP EVPN peer group ServerLeaf.
[*BorderLeaf_1-bgp-af-evpn] peer ServerLeaf advertise irbv6
[*BorderLeaf_1-bgp-af-evpn] peer ServerLeaf reflect-client //Configure the border leaf
node as a route reflector (RR).
[*BorderLeaf_1-bgp-af-evpn] quit
[*BorderLeaf_1-bgp] quit
[*BorderLeaf_1-bgp] commit
```

Configure BorderLeaf_2.

```
[~BorderLeaf_2]evpn-overlay enable
[*BorderLeaf_2]bgp 100
[*BorderLeaf_2-bgp]router-id 10.125.98.2
[*BorderLeaf_2-bgp]advertise lowest-priority all-address-family peer-up delay 360 //
When the peer status changes from Down to Up, the priority of BGP routes is changed to
be the lowest. Route advertisement is delayed to prevent packet loss during traffic
switchback.
[*BorderLeaf_2-bgp]undo default ipv4-unicast //Disable the IPv4 unicast address
family for BGP peers to reduce the device load.
[*BorderLeaf_2-bgp]group ServerLeaf internal //Configure a peer group named
ServerLeaf and add peers to this group.
[*BorderLeaf_2-bgp]peer 10.125.98.3 group ServerLeaf
[*BorderLeaf_2-bgp]peer 10.125.98.4 group ServerLeaf
[*BorderLeaf_2-bgp]peer 10.125.98.5 group ServerLeaf
[*BorderLeaf_2-bgp]peer 10.125.98.6 group ServerLeaf
[*BorderLeaf_2-bgp]peer ServerLeaf connect-interface LoopBack1 //Specify the source
interface that sends BGP packets.
[*BorderLeaf_2-bgp]l2vpn-family evpn //Enable the BGP-EVPN address family and
enter the BGP-EVPN address family view.
[*BorderLeaf_2-bgp-af-evpn]undo policy vpn-target //Disable the border leaf node
from performing VPN target-based filtering for received EVPN routes.
[*BorderLeaf_2-bgp-af-evpn]peer ServerLeaf enable
[*BorderLeaf_2-bgp-af-evpn]peer 10.125.98.3 group ServerLeaf
[*BorderLeaf_2-bgp-af-evpn]peer 10.125.98.4 group ServerLeaf
[*BorderLeaf_2-bgp-af-evpn]peer 10.125.98.5 group ServerLeaf
[*BorderLeaf_2-bgp-af-evpn]peer 10.125.98.6 group ServerLeaf
[*BorderLeaf_2-bgp-af-evpn]peer ServerLeaf advertise irb
[*BorderLeaf_2-bgp-af-evpn]peer ServerLeaf advertise irbv6
[*BorderLeaf_2-bgp-af-evpn]peer ServerLeaf reflect-client
[*BorderLeaf_2-bgp-af-evpn]quit
[*BorderLeaf_2-bgp]quit
[*BorderLeaf_2-bgp]commit
```

2. Configure server leaf nodes.

a. Configure IP addresses.

- i. Configure IP addresses for interfaces connecting server leaf nodes to border leaf nodes.

Configure IP addresses for interfaces connecting ServerLeaf1_1 to border leaf nodes.

```
[~ServerLeaf1_1] interface 40GE 1/0/1
[*ServerLeaf1_1-40GE1/0/1] description to BorderLeaf_1
[*ServerLeaf1_1-40GE1/0/1] undo portswitch
[*ServerLeaf1_1-40GE1/0/1] ip address 10.125.97.22 255.255.255.252
[*ServerLeaf1_1-40GE1/0/1] ospf network-type p2p //Set the network type of the OSPF
interface connected to a border leaf node to P2P.
```

```
[*ServerLeaf1_1-40GE1/0/1] quit
[*ServerLeaf1_1] interface 40GE 1/0/2
[*ServerLeaf1_1-40GE1/0/2] description to BorderLeaf_2
[*ServerLeaf1_1-40GE1/0/2] undo portswitch
[*ServerLeaf1_1-40GE1/0/2] ip address 10.125.97.38 255.255.255.252
[*ServerLeaf1_1-40GE1/0/2] ospf network-type p2p
[*ServerLeaf1_1-40GE1/0/2] quit
[*ServerLeaf1_1-40GE1/0/2] commit
```

Configure IP addresses for interfaces connecting ServerLeaf1_2 to border leaf nodes.

```
[~ServerLeaf1_2] interface 40GE 1/0/1
[*ServerLeaf1_2-40GE1/0/1] description to BorderLeaf_1
[*ServerLeaf1_2-40GE1/0/1] undo portswitch
[*ServerLeaf1_2-40GE1/0/1] ip address 10.125.97.26 255.255.255.252
[*ServerLeaf1_2-40GE1/0/1] ospf network-type p2p
[*ServerLeaf1_2-40GE1/0/1] quit
[*ServerLeaf1_2] interface 40GE 1/0/2
[*ServerLeaf1_2-40GE1/0/2] description to BorderLeaf_2
[*ServerLeaf1_2-40GE1/0/2] undo portswitch
[*ServerLeaf1_2-40GE1/0/2] ip address 10.125.97.42 255.255.255.252
[*ServerLeaf1_2-40GE1/0/2] ospf network-type p2p
[*ServerLeaf1_2-40GE1/0/2] quit
[*ServerLeaf1_2-40GE1/0/2] commit
```

Configure IP addresses for interfaces connecting ServerLeaf2_1 to border leaf nodes.

```
[~ServerLeaf2_1] interface 40GE 1/0/1
[*ServerLeaf2_1-40GE1/0/1] description to BorderLeaf_1
[*ServerLeaf2_1-40GE1/0/1] undo portswitch
[*ServerLeaf2_1-40GE1/0/1] ip address 10.125.97.30 255.255.255.252
[*ServerLeaf2_1-40GE1/0/1] ospf network-type p2p
[*ServerLeaf2_1-40GE1/0/1] quit
[*ServerLeaf2_1] interface 40GE 1/0/2
[*ServerLeaf2_1-40GE1/0/2] description to BorderLeaf_2
[*ServerLeaf2_1-40GE1/0/2] undo portswitch
[*ServerLeaf2_1-40GE1/0/2] ip address 10.125.97.46 255.255.255.252
[*ServerLeaf2_1-40GE1/0/2] ospf network-type p2p
[*ServerLeaf2_1-40GE1/0/2] quit
[*ServerLeaf2_1-40GE1/0/2] commit
```

Configure IP addresses for interfaces connecting ServerLeaf2_2 to border leaf nodes.

```
[~ServerLeaf2_2] interface 40GE 1/0/1
[*ServerLeaf2_2-40GE1/0/1] description to BorderLeaf_1
[*ServerLeaf2_2-40GE1/0/1] undo portswitch
[*ServerLeaf2_2-40GE1/0/1] ip address 10.125.97.34 255.255.255.252
[*ServerLeaf2_2-40GE1/0/1] ospf network-type p2p
[*ServerLeaf2_2-40GE1/0/1] quit
[*ServerLeaf2_2] interface 40GE 1/0/2
[*ServerLeaf2_2-40GE1/0/2] description to BorderLeaf_2
[*ServerLeaf2_2-40GE1/0/2] undo portswitch
[*ServerLeaf2_2-40GE1/0/2] ip address 10.125.97.50 255.255.255.252
[*ServerLeaf2_2-40GE1/0/2] ospf network-type p2p
[*ServerLeaf2_2-40GE1/0/2] quit
[*ServerLeaf2_2-40GE1/0/2] commit
```

- ii. Configure IP addresses for loopback interfaces on server leaf nodes.

Configure IP addresses for loopback interfaces on ServerLeaf1_1.

```
[~ServerLeaf1_1] interface LoopBack 0
[*ServerLeaf1_1-LoopBack0] description VTEP
[*ServerLeaf1_1-LoopBack0] ipv6 enable //Enable IPv6 if it is required.
[*ServerLeaf1_1-LoopBack0] ip address 10.125.99.2 255.255.255.255
[*ServerLeaf1_1-LoopBack0] quit
[*ServerLeaf1_1] interface LoopBack 1
[*ServerLeaf1_1-LoopBack1] description DFS-GROUP/ROUTER-ID
[*ServerLeaf1_1-LoopBack1] ip address 10.125.98.3 255.255.255.255
```

```
[*ServerLeaf1_1-LoopBack1] quit
[*ServerLeaf1_1] commit
```

Configure IP addresses for loopback interfaces on ServerLeaf1_2.

```
[~ServerLeaf1_2] interface LoopBack 0
[*ServerLeaf1_2-LoopBack0] description VTEP
[*ServerLeaf1_2-LoopBack0] ipv6 enable //Enable IPv6 if it is required.
[*ServerLeaf1_2-LoopBack0] ip address 10.125.99.2 255.255.255.255
[*ServerLeaf1_2-LoopBack0] quit
[*ServerLeaf1_2] interface LoopBack 1
[*ServerLeaf1_2-LoopBack1] description DFS-GROUP/ROUTER-ID
[*ServerLeaf1_2-LoopBack1] ip address 10.125.98.4 255.255.255.255
[*ServerLeaf1_2-LoopBack1] quit
[*ServerLeaf1_2] commit
```

Configure IP addresses for loopback interfaces on ServerLeaf2_1.

```
[~ServerLeaf2_1] interface LoopBack 0
[*ServerLeaf2_1-LoopBack0] description VTEP
[*ServerLeaf2_1-LoopBack0] ipv6 enable //Enable IPv6 if it is required.
[*ServerLeaf2_1-LoopBack0] ip address 10.125.99.3 255.255.255.255
[*ServerLeaf2_1-LoopBack0] quit
[*ServerLeaf2_1] interface LoopBack 1
[*ServerLeaf2_1-LoopBack1] description DFS-GROUP/ROUTER-ID
[*ServerLeaf2_1-LoopBack1] ip address 10.125.98.5 255.255.255.255
[*ServerLeaf2_1-LoopBack1] quit
[*ServerLeaf2_1] commit
```

Configure IP addresses for loopback interfaces on ServerLeaf2_2.

```
[~ServerLeaf2_2] interface LoopBack 0
[*ServerLeaf2_2-LoopBack0] description VTEP
[*ServerLeaf2_2-LoopBack0] ipv6 enable //Enable IPv6 if it is required.
[*ServerLeaf2_2-LoopBack0] ip address 10.125.99.3 255.255.255.255
[*ServerLeaf2_2-LoopBack0] quit
[*ServerLeaf2_2] interface LoopBack 1
[*ServerLeaf2_2-LoopBack1] description DFS-GROUP/ROUTER-ID
[*ServerLeaf2_2-LoopBack1] ip address 10.125.98.6 255.255.255.255
[*ServerLeaf2_2-LoopBack1] quit
[*ServerLeaf2_2] commit
```

- iii. Configure VTEP IP addresses and virtual MAC addresses for NVE interfaces.

Configure an NVE interface on ServerLeaf1_1.

```
[~ServerLeaf1_1] interface nve 1
[*ServerLeaf1_1-Nve1] source 10.125.99.2
[*ServerLeaf1_1-Nve1] mac-address 0000-5e00-0102
[*ServerLeaf1_1-Nve1] commit
```

Configure an NVE interface on ServerLeaf1_2.

```
[~ServerLeaf1_2] interface nve 1
[*ServerLeaf1_2-Nve1] source 10.125.99.2
[*ServerLeaf1_2-Nve1] mac-address 0000-5e00-0102
[*ServerLeaf1_2-Nve1] commit
```

Configure an NVE interface on ServerLeaf2_1.

```
[~ServerLeaf2_1] interface nve 1
[*ServerLeaf2_1-Nve1] source 10.125.99.3
[*ServerLeaf2_1-Nve1] mac-address 0000-5e00-0103
[*ServerLeaf2_1-Nve1] commit
```

Configure an NVE interface on ServerLeaf2_2.

```
[~ServerLeaf2_2] interface nve 1
[*ServerLeaf2_2-Nve1] source 10.125.99.3
[*ServerLeaf2_2-Nve1] mac-address 0000-5e00-0103
[*ServerLeaf2_2-Nve1] commit
```

- b. Configure M-LAG.

- i. Configure the M-LAG mode.

Configure the M-LAG mode on ServerLeaf1_1.

```
[~ServerLeaf1_1] stp mode rstp
[*ServerLeaf1_1] stp v-stp enable           //Configure M-LAG in V-STP mode.
[*ServerLeaf1_1] stp tc-protection          //Enable TC BPDU protection on the device.
[*ServerLeaf1_1] stp bpdu-protection       //Enable BPDU protection on the device.
[*ServerLeaf1_1] commit
```

Configure the M-LAG mode on ServerLeaf1_2, ServerLeaf2_1, and ServerLeaf2_2. The configuration procedure and data are the same as those of ServerLeaf1_1.

ii. Configure a DFS group.

Configure a DFS group on ServerLeaf1_1.

```
[~ServerLeaf1_1] dfs-group 1
[*ServerLeaf1_1-dfs-group-1] priority 150 //Set the DFS priority to be higher than that
of the remote device. The default priority is 100.
[*ServerLeaf1_1-dfs-group-1] dual-active detection enhanced enable
[*ServerLeaf1_1-dfs-group-1] source ip 10.125.98.3 peer 10.125.98.4
[*ServerLeaf1_1-dfs-group-1] consistency-check enable mode loose //Enable M-LAG
configuration consistency check and specify the loose mode.
[*ServerLeaf1_1-dfs-group-1] quit
[*ServerLeaf1_1-dfs-group-1] commit
```

Configure a DFS group on ServerLeaf1_2.

```
[~ServerLeaf1_2] dfs-group 1
[*ServerLeaf1_2-dfs-group-1] dual-active detection enhanced enable
[*ServerLeaf1_2-dfs-group-1] source ip 10.125.98.4 peer 10.125.98.3
[*ServerLeaf1_2-dfs-group-1] consistency-check enable mode loose //Enable M-LAG
configuration consistency check and specify the loose mode.
[*ServerLeaf1_2-dfs-group-1] quit
[*ServerLeaf1_2-dfs-group-1] commit
```

Configure a DFS group on ServerLeaf2_1.

```
[~ServerLeaf2_1] dfs-group 1
[*ServerLeaf2_1-dfs-group-1] priority 150 //Set the DFS priority to be higher than that
of the remote device. The default priority is 100.
[*ServerLeaf2_1-dfs-group-1] dual-active detection enhanced enable
[*ServerLeaf2_1-dfs-group-1] source ip 10.125.98.5 peer 10.125.98.6
[*ServerLeaf2_1-dfs-group-1] consistency-check enable mode loose //Enable M-LAG
configuration consistency check and specify the loose mode.
[*ServerLeaf2_1-dfs-group-1] quit
[*ServerLeaf2_1-dfs-group-1] commit
```

Configure a DFS group on ServerLeaf2_2.

```
[~ServerLeaf2_2] dfs-group 1
[*ServerLeaf2_2-dfs-group-1] dual-active detection enhanced enable
[*ServerLeaf2_2-dfs-group-1] source ip 10.125.98.6 peer 10.125.98.5
[*ServerLeaf2_2-dfs-group-1] consistency-check enable mode loose //Enable M-LAG
configuration consistency check and specify the loose mode.
[*ServerLeaf2_2-dfs-group-1] quit
[*ServerLeaf2_2-dfs-group-1] commit
```

iii. Configure a peer-link.

Configure a peer-link on ServerLeaf1_1.

```
[~ServerLeaf1_1] interface Eth-Trunk 0
[*ServerLeaf1_1-Eth-Trunk0] trunkport 40GE 1/0/5 to 1/0/6
[*ServerLeaf1_1-Eth-Trunk0] mode lacp-static
[*ServerLeaf1_1-Eth-Trunk0] peer-link 1
[*ServerLeaf1_1-Eth-Trunk0] commit
```

Configure a peer-link on ServerLeaf1_2.

```
[~ServerLeaf1_2] interface Eth-Trunk 0
[*ServerLeaf1_2-Eth-Trunk0] trunkport 40GE 1/0/5 to 1/0/6
[*ServerLeaf1_2-Eth-Trunk0] mode lacp-static
[*ServerLeaf1_2-Eth-Trunk0] peer-link 1
[*ServerLeaf1_2-Eth-Trunk0] commit
```

Configure a peer-link on ServerLeaf2_1.

```
[~ServerLeaf2_1] interface Eth-Trunk 0
[*ServerLeaf2_1-Eth-Trunk0] trunkport 40GE 1/0/5 to 1/0/6
[*ServerLeaf2_1-Eth-Trunk0] mode lacp-static
[*ServerLeaf2_1-Eth-Trunk0] peer-link 1
[*ServerLeaf2_1-Eth-Trunk0] commit
```

Configure a peer-link on ServerLeaf2_2.

```
[~ServerLeaf2_2] interface Eth-Trunk 0
[*ServerLeaf2_2-Eth-Trunk0] trunkport 40GE 1/0/5 to 1/0/6
[*ServerLeaf2_2-Eth-Trunk0] mode lacp-static
[*ServerLeaf2_2-Eth-Trunk0] peer-link 1
[*ServerLeaf2_2-Eth-Trunk0] commit
```

- iv. Configure server leaf nodes so that service servers can connect to the server leaf nodes in load balancing mode.

Configure ServerLeaf1_1 to connect to service servers.

```
[~ServerLeaf1_1] interface Eth-Trunk 10
[*ServerLeaf1_1-Eth-Trunk10] port link-type trunk
[*ServerLeaf1_1-Eth-Trunk10] undo port trunk allow-pass vlan 1
[*ServerLeaf1_1-Eth-Trunk10] trunkport 10ge 1/0/1
[*ServerLeaf1_1-Eth-Trunk10] dfs-group 1 m-lag 10
[*ServerLeaf1_1-Eth-Trunk10] mode lacp-static
[*ServerLeaf1_1-Eth-Trunk10] stp edged-port enable //Configure an edge port.
[*ServerLeaf1_1-Eth-Trunk10] quit
[*ServerLeaf1_1] interface 10GE 1/0/1 //Configure the server access interface.
[*ServerLeaf1_1-10GE1/0/1] storm suppression unknown-unicast 5 //Configure
unknown unicast traffic suppression. The empirical value is 5% of the bandwidth of the
10GE interface. It is recommended that unknown unicast traffic suppression be configured
on all service interfaces.
[*ServerLeaf1_1-10GE1/0/1] storm suppression multicast packets 1000 //Configure
multicast traffic suppression. The empirical value is 1000 pps.
[*ServerLeaf1_1-10GE1/0/1] storm suppression broadcast packets 1000 //Configure
broadcast traffic suppression. The empirical value is 1000 pps. It is recommended that
broadcast traffic suppression be configured on all service interfaces.
[*ServerLeaf1_1-10GE1/0/1] commit
```

Configure ServerLeaf1_2 to connect to service servers.

```
[~ServerLeaf1_2] interface Eth-Trunk 10
[*ServerLeaf1_2-Eth-Trunk10] port link-type trunk
[*ServerLeaf1_2-Eth-Trunk10] undo port trunk allow-pass vlan 1
[*ServerLeaf1_2-Eth-Trunk10] trunkport 10ge 1/0/1
[*ServerLeaf1_2-Eth-Trunk10] dfs-group 1 m-lag 10
[*ServerLeaf1_2-Eth-Trunk10] mode lacp-static
[*ServerLeaf1_2-Eth-Trunk10] stp edged-port enable //Configure an edge port.
[*ServerLeaf1_2-Eth-Trunk10] quit
[*ServerLeaf1_2] interface 10GE 1/0/1 //Configure the server access interface.
[*ServerLeaf1_2-10GE1/0/1] storm suppression unknown-unicast 5 //Configure
unknown unicast traffic suppression. The empirical value is 5% of the bandwidth of the
10GE interface. It is recommended that unknown unicast traffic suppression be configured
on all service interfaces.
[*ServerLeaf1_2-10GE1/0/1] storm suppression multicast packets 1000 //Configure
multicast traffic suppression. The empirical value is 1000 pps.
[*ServerLeaf1_2-10GE1/0/1] storm suppression broadcast packets 1000 //Configure
broadcast traffic suppression. The empirical value is 1000 pps. It is recommended that
broadcast traffic suppression be configured on all service interfaces.
[*ServerLeaf1_2-10GE1/0/1] commit
```

Configure ServerLeaf2_1 and ServerLeaf2_2 to connect to service servers. The configuration procedures are similar to the preceding configuration procedures.

- v. Configure server leaf nodes so that service servers can connect to the server leaf nodes in active/standby mode.

Configure ServerLeaf1_1 to connect to service servers.

```
[~ServerLeaf1_1] interface 10GE 1/0/2
[*ServerLeaf1_1-10GE1/0/2] port link-type trunk
```



```
[*ServerLeaf1_1-10GE1/0/2] undo port trunk allow-pass vlan 1 //Configure the
interface to deny packets from VLAN 1 to prevent loops.
[*ServerLeaf1_1-10GE1/0/2] storm suppression unknown-unicast 5 //Configure
unknown unicast traffic suppression. The empirical value is 5% of the bandwidth of the
10GE interface. It is recommended that unknown unicast traffic suppression be configured
on all service interfaces.
[*ServerLeaf1_1-10GE1/0/2] storm suppression multicast packets 1000 //Configure
multicast traffic suppression. The empirical value is 1000 pps.
[*ServerLeaf1_1-10GE1/0/2] storm suppression broadcast packets 1000 //Configure
broadcast traffic suppression. The empirical value is 1000 pps. It is recommended that
broadcast traffic suppression be configured on all service interfaces.
[*ServerLeaf1_1-10GE1/0/2] stp edged-port enable
[*ServerLeaf1_1-10GE1/0/2] commit
```

Configure ServerLeaf1_2 to connect to service servers.

```
[~ServerLeaf1_2] interface 10GE 1/0/2
[*ServerLeaf1_2-10GE1/0/2] port link-type trunk
[*ServerLeaf1_2-10GE1/0/2] undo port trunk allow-pass vlan 1 //Configure the
interface to deny packets from VLAN 1 to prevent loops.
[*ServerLeaf1_2-10GE1/0/2] storm suppression unknown-unicast 5 //Configure
unknown unicast traffic suppression. The empirical value is 5% of the bandwidth of the
10GE interface. It is recommended that unknown unicast traffic suppression be configured
on all service interfaces.
[*ServerLeaf1_2-10GE1/0/2] storm suppression multicast packets 1000 //Configure
multicast traffic suppression. The empirical value is 1000 pps.
[*ServerLeaf1_2-10GE1/0/2] storm suppression broadcast packets 1000 //Configure
broadcast traffic suppression. The empirical value is 1000 pps. It is recommended that
broadcast traffic suppression be configured on all service interfaces.
[*ServerLeaf1_2-10GE1/0/2] stp edged-port enable
[*ServerLeaf1_2-10GE1/0/2] commit
```

Configure ServerLeaf2_1 and ServerLeaf2_2 to connect to service servers. The configuration procedures are similar to the preceding configuration procedures.

- vi. Configure a Monitor Link group to associate uplink interfaces with downlink interfaces. This prevents a failure to forward user-side traffic when all uplinks of a device fail.

In this example, only one downlink interface is configured. Configure downlink interfaces based on actual requirements.

Configure a Monitor Link group on ServerLeaf1_1.

```
[~ServerLeaf1_1] monitor-link group 1
[*ServerLeaf1_1-mtlk-group1] port 40GE1/0/1 uplink
[*ServerLeaf1_1-mtlk-group1] port 40GE1/0/2 uplink
[*ServerLeaf1_1-mtlk-group1] port Eth-Trunk10 downlink 1
[*ServerLeaf1_1-mtlk-group1] timer recover-time 60 //Configure the WTR time of
the Monitor Link group. This prevents packet loss during traffic switchback upon an uplink
fault.
[*ServerLeaf1_1-mtlk-group1] commit
```

Configure Monitor Link groups on ServerLeaf1_2, ServerLeaf2_1, and ServerLeaf2_2. The configuration procedure and data are the same as those of ServerLeaf1_1.

- c. Configure a routing protocol.
 - i. Configure OSPF to ensure connectivity on the VXLAN underlay network.

Configure OSPF on ServerLeaf1_1.

```
[~ServerLeaf1_1] bfd //Enable BFD globally.
[*ServerLeaf1_1-bfd] quit
[*ServerLeaf1_1] ospf 1 router-id 10.125.98.3
[*ServerLeaf1_1-ospf-1] bfd all-interfaces enable
[*ServerLeaf1_1-ospf-1] bfd all-interfaces min-tx-interval 500 min-rx-interval 500
detect-multiplier 3 //Set the interval for receiving and sending BFD packets to 500 ms
```

and the detection multiplier to 3 only when all devices on the network support hardware-based BFD. In other scenarios, retain the default BFD parameter settings. That is, the interval for receiving and sending BFD packets is 1000 ms and the detection multiplier is 3. The models that support hardware-based BFD include the CE6857, CE6865, CE8861, CE8868, CE6875, CE5880, CE8850-64CQ-EI, CE6880, CE6881, CE6863, CE6863K, CE12800E, CE16800, CE12800 (configured with only the CE-L48XS-FG, CE-L48XS-FD1, CE-L24LQ-FD, CE-L36LQ-FD, CE-L08CF-FG1, CE-L12CQ-FD, CE-L16CQ-FD, CE-L36CQ-FD, CE-L36CQ-FD1, CE-L36CQ-FG, or CE-L36CQ-SD cards.)

```
[*ServerLeaf1_1-ospf-1] lsa-arrival-interval intelligent-timer 50 50 50 //Set the
interval for receiving OSPF LSAs to shorten the convergence time.
[*ServerLeaf1_1-ospf-1] area 0.0.0.0
[*ServerLeaf1_1-ospf-1-area-0.0.0.0] network 10.125.97.20 0.0.0.3 //Configure the
server leaf node to establish neighbor relationships with the two border leaf nodes.
[*ServerLeaf1_1-ospf-1-area-0.0.0.0] network 10.125.97.36 0.0.0.3
[*ServerLeaf1_1-ospf-1-area-0.0.0.0] network 10.125.98.3 0.0.0.0 //Advertise the
loopback interface address.
[*ServerLeaf1_1-ospf-1-area-0.0.0.0] network 10.125.99.2 0.0.0.0
[*ServerLeaf1_1-ospf-1-area-0.0.0.0] quit
[*ServerLeaf1_1-ospf-1] quit
[*ServerLeaf1_1-ospf-1] commit
```

Configure OSPF on ServerLeaf1_2.

```
[~ServerLeaf1_2] bfd
[*ServerLeaf1_2-bfd] quit
[*ServerLeaf1_2] ospf 1 router-id 10.125.98.4
[*ServerLeaf1_2-ospf-1] bfd all-interfaces enable
[*ServerLeaf1_2-ospf-1] bfd all-interfaces min-tx-interval 500 min-rx-interval 500
detect-multiplier 3
[*ServerLeaf1_2-ospf-1] lsa-arrival-interval intelligent-timer 50 50 50 //Set the
interval for receiving OSPF LSAs to shorten the convergence time.
[*ServerLeaf1_2-ospf-1] area 0.0.0.0
[*ServerLeaf1_2-ospf-1-area-0.0.0.0] network 10.125.97.24 0.0.0.3 //Configure the
server leaf node to establish neighbor relationships with the two border leaf nodes.
[*ServerLeaf1_2-ospf-1-area-0.0.0.0] network 10.125.97.40 0.0.0.3
[*ServerLeaf1_2-ospf-1-area-0.0.0.0] network 10.125.98.4 0.0.0.0 //Advertise the
loopback interface address.
[*ServerLeaf1_2-ospf-1-area-0.0.0.0] network 10.125.99.2 0.0.0.0
[*ServerLeaf1_2-ospf-1-area-0.0.0.0] quit
[*ServerLeaf1_2-ospf-1] quit
[*ServerLeaf1_2-ospf-1] commit
```

Configure OSPF on ServerLeaf2_1.

```
[~ServerLeaf2_1] bfd
[*ServerLeaf2_1-bfd] quit
[*ServerLeaf2_1] ospf 1 router-id 10.125.98.5
[*ServerLeaf2_1-ospf-1] bfd all-interfaces enable
[*ServerLeaf2_1-ospf-1] bfd all-interfaces min-tx-interval 500 min-rx-interval 500
detect-multiplier 3
[*ServerLeaf2_1-ospf-1] lsa-arrival-interval intelligent-timer 50 50 50 //Set the
interval for receiving OSPF LSAs to shorten the convergence time.
[*ServerLeaf2_1-ospf-1] area 0.0.0.0
[*ServerLeaf2_1-ospf-1-area-0.0.0.0] network 10.125.97.28 0.0.0.3 //Configure the
server leaf node to establish neighbor relationships with the two border leaf nodes.
[*ServerLeaf2_1-ospf-1-area-0.0.0.0] network 10.125.97.44 0.0.0.3
[*ServerLeaf2_1-ospf-1-area-0.0.0.0] network 10.125.98.5 0.0.0.0 //Advertise the
loopback interface address.
[*ServerLeaf2_1-ospf-1-area-0.0.0.0] network 10.125.99.3 0.0.0.0
[*ServerLeaf2_1-ospf-1-area-0.0.0.0] quit
[*ServerLeaf2_1-ospf-1] quit
[*ServerLeaf2_1-ospf-1] commit
```

Configure OSPF on ServerLeaf2_2.

```
[~ServerLeaf2_2] bfd
[*ServerLeaf2_2-bfd] quit
[*ServerLeaf2_2] ospf 1 router-id 10.125.98.6
[*ServerLeaf2_2-ospf-1] bfd all-interfaces enable
[*ServerLeaf2_2-ospf-1] bfd all-interfaces min-tx-interval 500 min-rx-interval 500
detect-multiplier 3
[*ServerLeaf2_2-ospf-1] lsa-arrival-interval intelligent-timer 50 50 50 //Set the
interval for receiving OSPF LSAs to shorten the convergence time.
```

```
[*ServerLeaf2_2-ospf-1] area 0.0.0.0
[*ServerLeaf2_2-ospf-1-area-0.0.0.0] network 10.125.97.32 0.0.0.3 //Configure the
server leaf node to establish neighbor relationships with the two border leaf nodes.
[*ServerLeaf2_2-ospf-1-area-0.0.0.0] network 10.125.97.48 0.0.0.3
[*ServerLeaf2_2-ospf-1-area-0.0.0.0] network 10.125.98.6 0.0.0.0 //Advertise the
loopback interface address.
[*ServerLeaf2_2-ospf-1-area-0.0.0.0] network 10.125.99.3 0.0.0.0
[*ServerLeaf2_2-ospf-1-area-0.0.0.0] quit
[*ServerLeaf2_2-ospf-1] quit
[*ServerLeaf2_2-ospf-1] commit
```

ii. Perform either of the following configurations to optimize OSPF network convergence performance upon faults based on the device version.

- For V200R005C00 and earlier versions

Configure OSPF network convergence performance optimization triggered upon faults on ServerLeaf1_1.

```
[~ServerLeaf1_1] ospf 1
[*ServerLeaf1_1-ospf-1] stub-router on-startup 720 include-stub //Configure the
server leaf node as a stub router to improve network convergence performance
upon faults. If the server leaf node is a CE12800 series switch, set the interval
specified by on-startup to 3000s. If the server leaf node is a fixed switch, set the
interval specified by on-startup to 720s.
[*ServerLeaf1_1-ospf-1] quit
[*ServerLeaf1_1-ospf-1] commit
```

Configure OSPF network convergence performance optimization triggered upon faults on ServerLeaf1_2, ServerLeaf2_1, and ServerLeaf2_2. The configuration procedure and data are the same as those of ServerLeaf1_1.

- For V200R005C10 and later versions

Configure OSPF network convergence performance optimization triggered upon faults on ServerLeaf1_1.

```
[~ServerLeaf1_1] interface 40GE 1/0/1
[*ServerLeaf1_1-40GE1/0/1] ospf peer hold-max-cost timer 300000 //Set the
interval at which the maximum cost is stored in local LSAs to 300s after OSPF
neighbor relationships are established between all border leaf nodes and server leaf
nodes. The value is the sum of the delay (240s) after which an M-LAG interface
goes Up (overlay routes are converged during this period) plus the device entry
synchronization period (60s).
[*ServerLeaf1_1-40GE1/0/1] quit
[*ServerLeaf1_1] interface 40GE 1/0/2
[*ServerLeaf1_1-40GE1/0/2] ospf peer hold-max-cost timer 300000
[*ServerLeaf1_1-40GE1/0/2] quit
[*ServerLeaf1_1-40GE1/0/2] commit
```

Configure OSPF network convergence performance optimization triggered upon faults on ServerLeaf1_2, ServerLeaf2_1, and ServerLeaf2_2. The configuration procedure and data are the same as those of ServerLeaf1_1.

iii. Configure BGP EVPN.

Configure BGP EVPN on ServerLeaf1_1.

```
[~ServerLeaf1_1] evpn-overlay enable //Configure EVPN as the VXLAN control plane.
[*ServerLeaf1_1] bgp 100
[*ServerLeaf1_1-bgp] router-id 10.125.98.3
[*ServerLeaf1_1-bgp] advertise lowest-priority all-address-family peer-up delay 360 //
When the peer status changes from Down to Up, the priority of BGP routes is changed to
be the lowest. Route advertisement is delayed to prevent packet loss during traffic
switchback.
[*ServerLeaf1_1-bgp] undo default ipv4-unicast //Disable the IPv4 unicast address
family for BGP peers to reduce the device load.
[*ServerLeaf1_1-bgp] group BorderLeaf internal //Configure a peer group named
BorderLeaf and add peers to this group.
```

```
[*ServerLeaf1_1-bgp] peer 10.125.98.1 group BorderLeaf
[*ServerLeaf1_1-bgp] peer 10.125.98.2 group BorderLeaf
[*ServerLeaf1_1-bgp] peer ServerLeaf connect-interface LoopBack1 //Specify the
source interface that sends BGP packets.
[*ServerLeaf1_1-bgp] l2vpn-family evpn //Enable the BGP-EVPN address family
and enter the BGP-EVPN address family view.
[*ServerLeaf1_1-bgp-af-evpn] undo policy vpn-target //Disable the server leaf node
from performing VPN target-based filtering for received EVPN routes.
[*ServerLeaf1_1-bgp-af-evpn] peer BorderLeaf enable
[*ServerLeaf1_1-bgp-af-evpn] peer 10.125.98.1 group BorderLeaf
[*ServerLeaf1_1-bgp-af-evpn] peer 10.125.98.2 group BorderLeaf
[*ServerLeaf1_1-bgp-af-evpn] peer BorderLeaf advertise irb //Enable the server leaf
node to advertise IRB and IRBv6 routes to the BGP EVPN peer group BorderLeaf.
[*ServerLeaf1_1-bgp-af-evpn] peer BorderLeaf advertise irbv6
[*ServerLeaf1_1-bgp-af-evpn] quit
[*ServerLeaf1_1-bgp] quit
[*ServerLeaf1_1-bgp] commit
```

Configure BGP EVPN on ServerLeaf1_2.

```
[~ServerLeaf1_2] evpn-overlay enable //Configure EVPN as the VXLAN control plane.
[*ServerLeaf1_2] bgp 100
[*ServerLeaf1_2-bgp] router-id 10.125.98.4
[*ServerLeaf1_2-bgp] advertise lowest-priority all-address-family peer-up delay 360 //
When the peer status changes from Down to Up, the priority of BGP routes is changed to
be the lowest. Route advertisement is delayed to prevent packet loss during traffic
switchback.
[*ServerLeaf1_2-bgp] undo default ipv4-unicast //Disable the IPv4 unicast address
family for BGP peers to reduce the device load.
[*ServerLeaf1_2-bgp] group BorderLeaf internal //Configure a peer group named
BorderLeaf and add peers to this group.
[*ServerLeaf1_2-bgp] peer 10.125.98.1 group BorderLeaf
[*ServerLeaf1_2-bgp] peer 10.125.98.2 group BorderLeaf
[*ServerLeaf1_2-bgp] peer ServerLeaf connect-interface LoopBack1 //Specify the
source interface that sends BGP packets.
[*ServerLeaf1_2-bgp] l2vpn-family evpn //Enable the BGP-EVPN address family
and enter the BGP-EVPN address family view.
[*ServerLeaf1_2-bgp-af-evpn] undo policy vpn-target //Disable the server leaf node
from performing VPN target-based filtering for received EVPN routes.
[*ServerLeaf1_2-bgp-af-evpn] peer BorderLeaf enable
[*ServerLeaf1_2-bgp-af-evpn] peer 10.125.98.1 group BorderLeaf
[*ServerLeaf1_2-bgp-af-evpn] peer 10.125.98.2 group BorderLeaf
[*ServerLeaf1_2-bgp-af-evpn] peer BorderLeaf advertise irb //Enable the server leaf
node to advertise IRB and IRBv6 routes to the BGP EVPN peer group BorderLeaf.
[*ServerLeaf1_2-bgp-af-evpn] peer BorderLeaf advertise irbv6
[*ServerLeaf1_2-bgp-af-evpn] quit
[*ServerLeaf1_2-bgp] quit
[*ServerLeaf1_2-bgp] commit
```

Configure BGP EVPN on ServerLeaf2_1 and ServerLeaf2_2. The configuration procedures are similar to the preceding configuration procedures.

3. Configure firewalls.

a. Configure basic information about firewalls.

i. Configure device names of firewalls.

Configure the device name of FW-1.

```
<HUAWEI> system-view
[HUAWEI] sysname FW-1
```

Configure the device name of FW-2.

```
<HUAWEI> system-view
[HUAWEI] sysname FW-2
```

ii. Configure IP addresses of management interfaces on the firewalls.

Configure the management IP address of FW-1.

```
[FW-1] interface GigabitEthernet 0/0/0
[FW-1-GigabitEthernet0/0/0] ip address 192.168.39.50 24
[FW-1-GigabitEthernet0/0/0] service-manage http permit
[FW-1-GigabitEthernet0/0/0] service-manage https permit
[FW-1-GigabitEthernet0/0/0] service-manage ping permit
[FW-1-GigabitEthernet0/0/0] quit
```

Configure the management IP address of FW-2.

```
[FW-2] interface GigabitEthernet 0/0/0
[FW-2-GigabitEthernet0/0/0] ip address 192.168.39.51 24
[FW-2-GigabitEthernet0/0/0] service-manage http permit
[FW-2-GigabitEthernet0/0/0] service-manage https permit
[FW-2-GigabitEthernet0/0/0] service-manage ping permit
[FW-2-GigabitEthernet0/0/0] quit
```

- b. Disable active and standby firewalls from backing up the running configurations locally.

Disable FW-1 from backing up the running configurations locally.

```
[FW-1] configuration backup local disable
```

Disable FW2 from backing up the running configurations locally.

```
[FW-2] configuration backup local disable
```

- c. Configure interfaces for connecting firewalls to border leaf nodes.

Configure a service interface on FW-1.

```
[FW-1] interface Eth-Trunk11
[FW-1-Eth-Trunk11] portswitch
[FW-1-Eth-Trunk11] port link-type trunk
[FW-1-Eth-Trunk11] undo port trunk allow-pass vlan 1
[FW-1-Eth-Trunk11] trunkport GigabitEthernet 1/0/8 to 1/0/9
[FW-1-Eth-Trunk11] mode lacp-static
[FW-1-Eth-Trunk11] quit
[FW-1] interface GigabitEthernet1/0/8 //Enable the interface.
[FW-1-GigabitEthernet1/0/8] undo shutdown
[FW-1-GigabitEthernet1/0/8] quit
[FW-1] interface GigabitEthernet1/0/9
[FW-1-GigabitEthernet1/0/9] undo shutdown
[FW-1-GigabitEthernet1/0/9] quit
```

Configure a service interface on FW-2. The configuration procedure and data are the same as those on FW-1.

- d. Configure heartbeat interfaces between the two firewalls.

Configure a heartbeat interface on FW-1.

```
[FW-1] interface Eth-Trunk0
[FW-1-Eth-Trunk0] description HRP
[FW-1-Eth-Trunk0] ip address 10.125.97.73 255.255.255.252
[FW-1-Eth-Trunk0] trunkport GigabitEthernet 1/0/0 to 1/0/1
[FW-1-Eth-Trunk0] mode lacp-static
[FW-1-Eth-Trunk0] quit
[FW-1] interface GigabitEthernet1/0/0 //Enable the interface.
[FW-1-GigabitEthernet1/0/0] undo shutdown
[FW-1-GigabitEthernet1/0/0] quit
[FW-1] interface GigabitEthernet1/0/1
[FW-1-GigabitEthernet1/0/1] undo shutdown
[FW-1-GigabitEthernet1/0/1] quit
```

Configure a heartbeat interface on FW-2.

```
[FW-2] interface Eth-Trunk0
[FW-2-Eth-Trunk0] description HRP
[FW-2-Eth-Trunk0] ip address 10.125.97.74 255.255.255.252
[FW-2-Eth-Trunk0] trunkport GigabitEthernet 1/0/0 to 1/0/1
[FW-2-Eth-Trunk0] mode lacp-static
[FW-2-Eth-Trunk0] quit
[FW-2] interface GigabitEthernet1/0/0 //Enable the interface.
[FW-2-GigabitEthernet1/0/0] undo shutdown
[FW-2-GigabitEthernet1/0/0] quit
```

```
[FW-2] interface GigabitEthernet1/0/1
[FW-2-GigabitEthernet1/0/1] undo shutdown
[FW-2-GigabitEthernet1/0/1] quit
```

- e. Configure the active/standby mirroring mode for the two firewalls.

Configure FW-1.

```
[FW-1] hrp interface Eth-Trunk0 remote 10.125.97.74 //Specify the heartbeat interface.
[FW-1] hrp mirror config enable //Configure the mirroring mode.
[FW-1] hrp enable //Enable the hot standby function.
[FW-1] hrp track interface Eth-Trunk1 //Configure a VGMP group to monitor
uplink and downlink service interfaces.
[FW-1] undo hrp track trunk-member enable //Disable the HRP module from
monitoring the status of Eth-Trunk member interfaces.
[FW-1] hrp mgt-interface Eth-Trunk1 //Configure the hot standby management
interface.
[FW-1] hrp mirror session enable //Enable the quick session backup function.
[FW-1] hrp standby config enable //Enable certain configuration functions on
the standby firewall.
[FW-1] hrp base config enable //Configure the firewall to start with basic
configurations and synchronize other configurations from the remote device.
[FW-1] undo hrp preempt //Disable HRP preemption in mirroring mode.
```

Configure FW-2.

```
[FW-2] hrp interface Eth-Trunk0 remote 10.125.97.73
[FW-2] hrp mirror config enable
[FW-2] hrp enable //After the mirroring mode is configured on the standby firewall and
the hot standby function is enabled, other configurations can be synchronized from the active
firewall.
[FW-2] hrp track interface Eth-Trunk1
[FW-2] undo hrp track trunk-member enable
[FW-2] hrp mgt-interface Eth-Trunk1
[FW-2] hrp mirror session enable
[FW-2] hrp standby config enable
[FW-2] hrp base config enable
[FW-2] undo hrp preempt
```

- f. Configure security zones and the default security policy. You only need to perform the following configurations on FW-1, and FW-2 will automatically synchronize the configurations.

Add Virtual-if0, the management interface, and the heartbeat interface to security zones.

```
[FW-1] firewall zone untrust
[FW-1-zone-untrust] add interface Virtual-if0
[FW-1-zone-untrust] quit
[FW-1] firewall zone dmz
[FW-1-zone-dmz] add interface Eth-Trunk1
[FW-1-zone-dmz] add interface Eth-Trunk0
[FW-1-zone-dmz] quit
```

Configure the default security policy and set the action to permit.

```
[FW-1] security-policy
[FW-1-policy-security] default action permit
[FW-1-policy-security] quit
```

- g. Enable the vSYS function on the firewalls. You only need to perform the following configurations on FW-1, and FW-2 will automatically synchronize the configurations.

```
[FW-1] vsys enable //Enable the vSYS function on the firewall.
[FW-1] interface Virtual-if api transform //Enable the Virtual-if name conversion function of
the northbound interface.
[FW-1] firewall forward cross-vsys extended //Configure the firewall to work in extended
mode so that packets can be forwarded across the vSYS twice. This configuration is mandatory
when different VPNs communicate with each other through the EIP at the egress vSYS.
```

Step 3 Configure the overlay network.

1. Configure server leaf nodes.

- a. Configure the VXLAN tunnel mode and enable the VXLAN ACL extension function. (This step is required only on the CE12800, CE16800 equipped with A series cards, CE6870EI, and CE6875EI.)

Configure ServerLeaf1_1.

```
[~ServerLeaf1_1] ip tunnel mode vxlan
[*ServerLeaf1_1] assign forward nvo3 acl extend enable
[*ServerLeaf1_1] commit
```

 NOTE

After configuring the VXLAN tunnel mode or enabling the VXLAN ACL extension function, save the configuration and restart the device to make the configuration take effect. You can restart the device immediately or after completing all the configurations.

Configure ServerLeaf1_2, ServerLeaf2_1, and ServerLeaf2_2. The configuration procedure and data are the same as those of ServerLeaf1_1.

- b. Configure service access points.

Configure a service access point on ServerLeaf1_1.

```
[~ServerLeaf1_1] bridge-domain 10
[*ServerLeaf1_1-bd10] quit
[*ServerLeaf1_1] interface eth-trunk 10.1 mode l2
[*ServerLeaf1_1-Eth-Trunk10.1] encapsulation dot1q vid 10
[*ServerLeaf1_1-Eth-Trunk10.1] bridge-domain 10
[*ServerLeaf1_1-Eth-Trunk10.1] quit
[*ServerLeaf1_1-Eth-Trunk10.1] commit
```

Configure a service access point on ServerLeaf1_2.

```
[~ServerLeaf1_2] bridge-domain 10
[*ServerLeaf1_2-bd10] quit
[*ServerLeaf1_2] interface eth-trunk 10.1 mode l2
[*ServerLeaf1_2-Eth-Trunk10.1] encapsulation dot1q vid 10
[*ServerLeaf1_2-Eth-Trunk10.1] bridge-domain 10
[*ServerLeaf1_2-Eth-Trunk10.1] quit
[*ServerLeaf1_2-Eth-Trunk10.1] commit
```

Configure service access points on ServerLeaf2_1 and ServerLeaf2_2. The configuration procedures are similar to the preceding configuration procedures except that you need to configure BD 20 instead of BD 10.

- c. Configure VPN and EVPN instances.

Configure ServerLeaf1_1.

```
[~ServerLeaf1_1] ip vpn-instance vpn1
[*ServerLeaf1_1-vpn-instance-vpn1] vxlan vni 5010
[*ServerLeaf1_1-vpn-instance-vpn1] ipv4-family
[*ServerLeaf1_1-vpn-instance-vpn1-af-ipv4] route-distinguisher 20:2
[*ServerLeaf1_1-vpn-instance-vpn1-af-ipv4] vpn-target 100:5010 evpn
[*ServerLeaf1_1-vpn-instance-vpn1-af-ipv4] quit
[*ServerLeaf1_1-vpn-instance-vpn1] quit
[*ServerLeaf1_1] bridge-domain 10
[*ServerLeaf1_1-bd10] vxlan vni 10
[*ServerLeaf1_1-bd10] evpn
[*ServerLeaf1_1-bd10-evpn] route-distinguisher 10:2
[*ServerLeaf1_1-bd10-evpn] vpn-target 100:10
[*ServerLeaf1_1-bd10-evpn] vpn-target 100:5010 export-extcommunity
[*ServerLeaf1_1-bd10-evpn] quit
[*ServerLeaf1_1-bd10] quit
[*ServerLeaf1_1] commit
```

Configure ServerLeaf1_2.

```
[~ServerLeaf1_2] ip vpn-instance vpn2
[*ServerLeaf1_2-vpn-instance-vpn1] vxlan vni 5010
[*ServerLeaf1_2-vpn-instance-vpn1] ipv4-family
```

```
[*ServerLeaf1_2-vpn-instance-vpn1-af-ipv4] route-distinguisher 20:4
[*ServerLeaf1_2-vpn-instance-vpn1-af-ipv4] vpn-target 100:5010 evpn
[*ServerLeaf1_2-vpn-instance-vpn1-af-ipv4] quit
[*ServerLeaf1_2-vpn-instance-vpn1] quit
[*ServerLeaf1_2] bridge-domain 10
[*ServerLeaf1_2-bd10] vxlan vni 10
[*ServerLeaf1_2-bd10] evpn
[*ServerLeaf1_2-bd10-evpn] route-distinguisher 10:4
[*ServerLeaf1_2-bd10-evpn] vpn-target 100:10
[*ServerLeaf1_2-bd10-evpn] vpn-target 100:5010 export-extcommunity
[*ServerLeaf1_2-bd10-evpn] quit
[*ServerLeaf1_2-bd10] quit
[*ServerLeaf1_2] commit
```

Configure ServerLeaf2_1 and ServerLeaf2_2. The configuration procedures are similar to the preceding configuration procedures.

- d. Configure an ingress replication list corresponding to a VNI.

Configure an ingress replication list on ServerLeaf1_1.

```
[~ServerLeaf1_1] interface nve 1
[*ServerLeaf1_1-Nve1] vni 10 head-end peer-list protocol bgp
```

Configure an ingress replication list on ServerLeaf1_2.

```
[~ServerLeaf1_2] interface nve 1
[*ServerLeaf1_2-Nve1] vni 10 head-end peer-list protocol bgp
```

The configurations of ServerLeaf2_1 and ServerLeaf2_2 are similar to the preceding configurations.

- e. Configure server leaf nodes as Layer 3 VXLAN gateways.

Configure a service loopback interface on ServerLeaf1_1. The configurations of ServerLeaf1_2, ServerLeaf2_1, and ServerLeaf2_2 are similar to the configuration of ServerLeaf1_1. (This step is required only on the CE8850EI.)

```
[~ServerLeaf1_1] interface eth-trunk 2
[*ServerLeaf1_1-Eth-Trunk2] service type tunnel
[*ServerLeaf1_1-Eth-Trunk2] quit
[*ServerLeaf1_1] interface 10ge 1/0/5
[*ServerLeaf1_1-10GE1/0/5] eth-trunk 2
[*ServerLeaf1_1-10GE1/0/5] quit
[*Device2] commit
```

NOTE

- A member interface must be an idle physical interface that does not transmit any service. There is no requirement on the interface status.
- Ensure that the bandwidth of the Eth-Trunk service loopback interface is at least twice the bandwidth required for transmitting traffic on the Layer 3 VXLAN gateway. For example, if traffic is sent from users to the gateway across the VXLAN network at a rate of 10 Gbit/s, add two 10GE interfaces to the Eth-Trunk used as the service loopback interface.

Configure ServerLeaf1_1 as a Layer 3 VXLAN gateway.

```
[~ServerLeaf1_1] interface vbdif10
[*ServerLeaf1_1-Vbdif10] ip binding vpn-instance vpn1
[*ServerLeaf1_1-Vbdif10] ip address 10.1.1.1 255.255.255.0
[*ServerLeaf1_1-Vbdif10] mac-address 0000-5e00-0102
[*ServerLeaf1_1-Vbdif10] vxlan anycast-gateway enable
[*ServerLeaf1_1-Vbdif10] arp collect host enable
[*ServerLeaf1_1-Vbdif10] quit
[*ServerLeaf1_1] commit
```

The configurations of ServerLeaf1_2, ServerLeaf2_1, and ServerLeaf2_2 are similar to the configuration of ServerLeaf1_1. Note that VBDIF

interfaces on ServerLeaf1_1 and ServerLeaf1_2 must be configured with the same IP address and MAC address.

- f. Configure server leaf nodes to advertise IP prefix routes to BGP peers.

Configure ServerLeaf1_1. The configurations of ServerLeaf1_2, ServerLeaf2_1, and ServerLeaf2_2 are similar to the configuration of ServerLeaf1_1.

```
[~ServerLeaf1_1] bgp 100  
[~ServerLeaf1_1-bgp] ipv4-family vpn-instance vpn1  
[*ServerLeaf1_1-bgp] import-route direct  
[*ServerLeaf1_1-bgp] import-route static //Import routes from firewalls.  
[*ServerLeaf1_1-bgp] advertise l2vpn evpn  
[*ServerLeaf1_1-bgp] quit  
[*ServerLeaf1_1-bgp] quit  
[*ServerLeaf1_1] commit
```

2. Configure border leaf nodes.

Configure a static route for accessing the external network through border leaf nodes.

Configure Border_Leaf1_1.

```
[~Border_Leaf1_1] ip route-static 1.2.3.4 255.255.255.0 10.125.97.242
```

Configure Border_Leaf1_2.

```
[~Border_Leaf1_2] ip route-static 1.2.3.4 255.255.255.0 10.125.97.242
```

3. Configure the firewalls.

The services associated with the external network are forwarded by the firewalls, but the source IP address remains unchanged. To enable tenants to use public IP addresses to communicate with devices on the Internet, configure SNAT to perform translation between private and public source IP addresses.

- a. Configure vSYS key parameters.

Configure FW-1.

```
[FW-1] vlan 3004  
[FW-1-vlan3004] quit  
[FW-1] interface Vlanif3004  
[FW-1-Vlanif3004] ip binding vpn-instance vsys_1  
[FW-1-Vlanif3004] ip address 10.125.97.242 255.255.255.252  
[FW-1-Vlanif3004] quit  
[FW-1] ip route-static 0.0.0.0 0.0.0.0 public  
[FW-1] ip route-static 10.132.1.0 255.255.255.0 10.125.97.241
```

Configure FW-2.

```
[FW-2] vlan 3004  
[FW-2-vlan3004] quit  
[FW-2] interface Vlanif3004  
[FW-2-Vlanif3004] ip binding vpn-instance vsys_1  
[FW-2-Vlanif3004] ip address 10.125.97.242 255.255.255.252  
[FW-2-Vlanif3004] quit  
[FW-2] ip route-static 0.0.0.0 0.0.0.0 public  
[FW-2] ip route-static 10.132.1.0 255.255.255.0 10.125.97.241
```

- b. Configure public key parameters.

Configure FW-1.

```
[FW-1] vsys name vsys_1 1  
[FW-1-vsys_1] assign vlan 3004  
[FW-1-vsys_1] quit  
[FW-1] ip vpn-instance vsys_1
```

```
[FW-1-vsys_1] ipv4-family
[FW-1-vsys_1] quit
[FW-1] vlan 3005
[FW-1-vlan3005] quit
[FW-1] interface Vlanif3005
[FW-1-Vlanif3005] ip address 10.125.97.242 255.255.255.252
[FW-1-Vlanif3005] quit
[FW-1] interface Eth-Trunk11
[FW-1-Eth-Trunk11] portswitch
[FW-1-Eth-Trunk11] port link-type trunk
[FW-1-Eth-Trunk11] undo port trunk allow-pass vlan 1
[FW-1-Eth-Trunk11] port trunk allow-pass vlan 3004 to 3005
[FW-1-Eth-Trunk11] quit
[FW-1] ip route-static 0.0.0.0 0.0.0.0 10.125.97.241
[FW-1] ip route-static 10.132.1.0 255.255.255.0 vpn-instance vsys_1
```

Configure FW-2.

```
[FW-2] vsys name vsys_1 1
[FW-2-vsys_1] assign vlan 3004
[FW-2-vsys_1] quit
[FW-2] ip vpn-instance vsys_1
[FW-2-vsys_1] ipv4-family
[FW-2-vsys_1] quit
[FW-2] vlan 3005
[FW-2-vlan3005] quit
[FW-2] interface Vlanif3005
[FW-2-Vlanif3005] ip address 10.125.97.242 255.255.255.252
[FW-2-Vlanif3005] quit
[FW-2] interface Eth-Trunk11
[FW-2-Eth-Trunk11] portswitch
[FW-2-Eth-Trunk11] port link-type trunk
[FW-2-Eth-Trunk11] undo port trunk allow-pass vlan 1
[FW-2-Eth-Trunk11] port trunk allow-pass vlan 3004 to 3005
[FW-2-Eth-Trunk11] quit
[FW-2] ip route-static 0.0.0.0 0.0.0.0 10.125.97.241
[FW-2] ip route-static 10.132.1.0 255.255.255.0 vpn-instance vsys_1
```

c. Configure SNAT.

Configure FW_1.

```
[FW-1] nat address-group addgrp 0
[FW-1-address-group-addgrp] mode pat
[FW-1-address-group-addgrp] section 0 1.2.3.4 1.2.3.4
[FW-1-address-group-addgrp] quit
[FW-1] security-policy
[FW-1-policy-security] rule name rule1
[FW-1-policy-security-rule-rule1] source-address 10.132.1.0 mask 255.255.255.0
[FW-1-policy-security-rule-rule1] action permit
[FW-1] nat-policy
[FW-1-policy-nat] rule name rule1
[FW-1-policy-nat-rule-rule1] description SNAT_01
[FW-1-policy-nat-rule-rule1] source-zone trust
[FW-1-policy-nat-rule-rule1] destination-zone untrust
[FW-1-policy-nat-rule-rule1] source-address 10.132.1.0 mask 255.255.255.0
[FW-1-policy-nat-rule-rule1] action source-nat address-group addgrp
[FW-1-policy-nat-rule-rule1] quit
[FW-1-policy-nat] quit
[FW-1] firewall import-flow public 1.2.3.4 1.2.3.4 vpn-instance vsys_1
```

Configure FW-2. The configuration procedure and parameters are the same as those of FW_1.

----End

Verifying the Configuration

1. After the configurations on the underlay network are complete, perform the following steps to check whether the configurations are correct.

- a. Check the OSPF neighbor status on the underlay network and whether the loopback interface addresses can ping each other. The following uses the command output on BorderLeaf_1 as an example.

OSPF neighbor relationships have been established between BorderLeaf_1 and server leaf nodes.

```
<BorderLeaf_1> display ospf peer brief
OSPF Process 1 with Router ID 10.125.98.1
Peer Statistic Information
```

```
Total number of peer(s): 4
Peer(s) in full state: 4
```

Area Id	Interface	Neighbor id	State
0.0.0.0	100GE1/0/0	10.125.98.3	Full
0.0.0.0	100GE1/0/1	10.125.98.4	Full
0.0.0.0	100GE1/0/2	10.125.98.5	Full
0.0.0.0	100GE1/0/3	10.125.98.6	Full

- b. Check the BGP EVPN peer status. The following uses the command output on BorderLeaf_1 as an example. BGP EVPN peer relationships have been established between BorderLeaf_1 and server leaf nodes.

```
<BorderLeaf_1> display bgp evpn peer
BGP local router ID : 10.125.98.1
Local AS number : 100
Total number of peers : 4
Peers in established state : 4
```

Peer	V	AS	MsgRcvd	MsgSent	OutQ	Up/Down	State	PrefRcv
10.125.98.3	4	100	646	2973	0	08:44:07	Established	3
10.125.98.4	4	100	651	2983	0	08:43:53	Established	3
10.125.98.5	4	100	605	2729	0	08:43:50	Established	0
10.125.98.6	4	100	607	2733	0	08:44:21	Established	0

2. After the configurations on the overlay network are complete, perform the following steps to check whether the configurations are correct.

- a. Run the **display vxlan tunnel** command on ServerLeaf1_2, ServerLeaf2_1, and ServerLeaf2_2 to check VXLAN tunnel information. The following uses the command output on ServerLeaf1_1 as an example.

```
[~ServerLeaf1_1] display vxlan tunnel
```

```
Number of vxlan tunnel : 1
```

Tunnel ID	Source	Destination	State	Type	Uptime
4026531841	10.125.99.2	10.125.99.3	up	dynamic	0032h21m

NOTE

The status of tunnels on ServerLeaf1_2, ServerLeaf2_1, and ServerLeaf2_2 changes to Up only after servers are connected to Layer 2 sub-interfaces on the server leaf nodes. When no server is connected to the Layer 2 sub-interfaces, the VXLAN tunnel status cannot be checked because no IRB route is advertised.

- b. Check the BGP EVPN peer status. The following uses the command output on BorderLeaf_1 as an example. BGP EVPN peer relationships have been established between BorderLeaf_1 and server leaf nodes.

```
<BorderLeaf_1> display bgp evpn peer
BGP local router ID : 10.125.98.1
Local AS number : 100
Total number of peers : 4
Peers in established state : 4
```

Peer	V	AS	MsgRcvd	MsgSent	OutQ	Up/Down	State	PrefRcv
10.125.98.3	4	100	646	2973	0	08:44:07	Established	3

10.125.98.4	4	100	651	2983	0 08:43:53	Established	3
10.125.98.5	4	100	605	2729	0 08:43:50	Established	0
10.125.98.6	4	100	607	2733	0 08:44:21	Established	0

Configuration Files

- BorderLeaf_1 configuration file

```
# -----IP addresses of interfaces connecting BorderLeaf_1 to server leaf nodes
interface 40GE1/0/0
description to ServerLeaf1_1
undo portswitch
ip address 10.125.97.21 255.255.255.252
#
interface 40GE1/0/1
description to ServerLeaf1_2
undo portswitch
ip address 10.125.97.25 255.255.255.252
#
interface 40GE1/0/2
description to ServerLeaf2_1
undo portswitch
ip address 10.125.97.29 255.255.255.252
#
interface 40GE1/0/3
description to ServerLeaf2_2
undo portswitch
ip address 10.125.97.33 255.255.255.252
#

# -----IP address of the interface connecting BorderLeaf_1 to firewalls
vlan 11
interface Vlanif11
description to FW1-2
ip address 10.125.97.57 255.255.255.248
mac-address 0000-5e00-0101
#

# -----IP addresses of loopback interfaces on BorderLeaf_1
interface LoopBack0
description VTEP
ipv6 enable //Enable IPv6 if it is required.
ip address 10.125.99.1 255.255.255.255
#
interface LoopBack1
description AC-MGMT/DFS-GROUP/ROUTER-ID
ip address 10.125.98.1 255.255.255.255
#

# -----NVE interface of BorderLeaf_1
interface Nve1
source 10.125.99.1
mac-address 0000-5e00-0101
#

# -----M-LAG mode of BorderLeaf_1
stp mode rstp
stp v-stp enable //Configure M-LAG in V-STP mode.
#

# -----M-LAG DFS group
dfs-group 1
dual-active detection enhanced enable
priority 150 //Set the DFS priority to be higher than that of the remote device. The default priority
is 100.
source ip 10.125.98.1 peer 10.125.98.2
consistency-check enable mode loose //Enable M-LAG configuration consistency check and specify
the loose mode.
#
```

```
#-----peer-link
interface Eth-Trunk0
trunkport 10GE 4/0/47
trunkport 40GE 1/0/23
mode lacp-static
peer-link 1
lacp mixed-rate link enable //Enable interfaces operating at different rates to forward packets after
the interfaces are added to an Eth-Trunk in LACP mode.
#
interface 40GE1/0/23
distribute-weight 4 //Set the load balancing weight of 40GE member interfaces to 4 and retain
the default weight 1 of 10GE member interfaces.
#

#-----M-LAG interface connecting BorderLeaf_1 to firewalls
interface Eth-Trunk1 //Configure a management interface for connecting to the active firewall.
description FW_1_MGMT
trunkport 10GE 4/0/41
port default vlan 11
mode lacp-static
dfs-group 1 m-lag 1
#
interface Eth-Trunk2 //Configure a management interface for connecting to the standby firewall.
description FW_2_MGMT
trunkport 10GE 4/0/42
port default vlan 11
mode lacp-static
dfs-group 1 m-lag 2
#
interface Eth-Trunk11 //Configure a service interface for connecting to the active firewall.
trunkport 10GE 4/0/43
port link-type trunk
undo port trunk allow-pass vlan 1
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 3
#
interface Eth-Trunk12 //Configure a service interface for connecting to the standby firewall.
trunkport 10GE 4/0/44
port link-type trunk
undo port trunk allow-pass vlan 1
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 4
#

#-----OSPF configurations that ensure connectivity on the VXLAN underlay network
bfd //Enable BFD globally.
#
ospf 1 router-id 10.125.98.1
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 500 min-rx-interval 500 detect-multiplier 3 //Set the interval for
receiving and sending BFD packets to 500 ms and the detection multiplier to 3 only when all devices
on the network support hardware-based BFD. In other scenarios, retain the default BFD parameter
settings. That is, the interval for receiving and sending BFD packets is 1000 ms and the detection
multiplier is 3. The models that support hardware-based BFD include the CE6857, CE6865, CE8861,
CE8868, CE6875, CE5880, CE8850-64CQ-EI, CE6880, CE6881, CE6863, CE6863E, CE6863K, CE12800E,
CE16800, and CE12800 (configured with only the CE-L48XS-FG, CE-L48XS-FD1, CE-L24LQ-FD, CE-
L36LQ-FD, CE-L08CF-FG1, CE-L12CQ-FD, CE-L16CQ-FD, CE-L36CQ-FD, CE-L36CQ-FD1, CE-L36CQ-FG,
or CE-L36CQ-SD cards).
lsa-arrival-interval intelligent-timer 50 50 50 //Set the interval for receiving OSPF LSAs to shorten
the convergence time.
area 0.0.0.0
network 10.125.97.20 0.0.0.3
network 10.125.97.24 0.0.0.3
network 10.125.97.28 0.0.0.3
network 10.125.97.32 0.0.0.3 //Configure the border leaf node to establish neighbor relationships
with the four server leaf nodes.
network 10.125.97.56 0.0.0.7 //Advertise the in-band management address of the firewall to
```

```
ensure connectivity.
 network 10.125.98.1 0.0.0.0
 network 10.125.99.1 0.0.0.0 //Advertise the loopback interface address.
#
interface 40GE1/0/0
 description to ServerLeaf1_1
 undo portswitch
 ip address 10.125.97.21 255.255.255.252
 ospf network-type p2p //Set the network type of the OSPF interface connected to a server leaf
 node to P2P.
#
interface 40GE1/0/1
 description to ServerLeaf1_2
 undo portswitch
 ip address 10.125.97.25 255.255.255.252
 ospf network-type p2p
#
interface 40GE1/0/2
 description to ServerLeaf2_1
 undo portswitch
 ip address 10.125.97.29 255.255.255.252
 ospf network-type p2p
#
interface 40GE1/0/3
 description to ServerLeaf2_2
 undo portswitch
 ip address 10.125.97.33 255.255.255.252
 ospf network-type p2p
#
#-----OSPF network convergence performance optimization triggered upon faults
interface 40GE1/0/0
 ospf peer hold-max-cost timer 300000 //Set the interval at which the maximum cost is stored in
 local LSAs to 300s after OSPF neighbor relationships are established between all border leaf nodes
 and server leaf nodes. The value is the sum of the delay (240s) after which an M-LAG interface goes
 Up (overlay routes are converged during this period) plus the device entry synchronization period
 (60s).
#
interface 40GE1/0/1
 ospf peer hold-max-cost timer 300000
#
interface 40GE1/0/2
 ospf peer hold-max-cost timer 300000
#
interface 40GE1/0/3
 ospf peer hold-max-cost timer 300000
#
#-----BGP EVPN
evpn-overlay enable //Configure EVPN as the VXLAN control plane.
#
bgp 100
 router-id 10.125.98.1
 advertise lowest-priority all-address-family peer-up delay 360 //When the peer status changes from
 Down to Up, the priority of BGP routes is changed to be the lowest. Route advertisement is delayed
 to prevent packet loss during traffic switchback.
 undo default ipv4-unicast //Disable the IPv4 unicast address family for BGP peers to reduce the
 device load.
 group ServerLeaf internal //Configure a peer group named ServerLeaf and add peers to this
 group.
 peer 10.125.98.3 group ServerLeaf
 peer 10.125.98.4 group ServerLeaf
 peer 10.125.98.5 group ServerLeaf
 peer 10.125.98.6 group ServerLeaf
 peer ServerLeaf connect-interface LoopBack1 //Specify the source interface that sends BGP packets.
#
 l2vpn-family evpn //Enable the BGP-EVPN address family and enter the BGP-EVPN address
 family view.
 undo policy vpn-target //Disable the border leaf node from performing VPN target-based
```

```
filtering for received EVPN routes.
 peer ServerLeaf enable
 peer 10.125.98.3 group ServerLeaf
 peer 10.125.98.4 group ServerLeaf
 peer 10.125.98.5 group ServerLeaf
 peer 10.125.98.6 group ServerLeaf
 peer ServerLeaf advertise irb //Enable the border leaf node to advertise IRB and IRBv6 routes to
the BGP EVPN peer group ServerLeaf.
 peer ServerLeaf advertise irbv6
 peer ServerLeaf reflect-client //Configure the border leaf node as an RR.
#

#
ip route-static 1.2.3.4 255.255.255.0 10.125.97.242
#
```

- **BorderLeaf_2 configuration file**

```
# -----IP addresses of interfaces connecting BorderLeaf_2 to server leaf nodes
interface 40GE1/0/0
 description to ServerLeaf1_1
 undo portswitch
 ip address 10.125.97.37 255.255.255.252
#
interface 40GE1/0/1
 description to ServerLeaf1_2
 undo portswitch
 ip address 10.125.97.41 255.255.255.252
#
interface 40GE1/0/2
 description to ServerLeaf2_1
 undo portswitch
 ip address 10.125.97.45 255.255.255.252
#
interface 40GE1/0/3
 description to ServerLeaf2_2
 undo portswitch
 ip address 10.125.97.49 255.255.255.252
#

# -----IP address of the interface connecting BorderLeaf_2 to firewalls
vlan 11
 interface Vlanif11
  description to FW1-2
  ip address 10.125.97.57 255.255.255.248
  mac-address 0000-5e00-0101
#

# -----IP addresses of loopback interfaces on BorderLeaf_2
interface LoopBack0
 description VTEP
 ipv6 enable //Enable IPv6 if it is required.
 ip address 10.125.99.1 255.255.255.255
#
interface LoopBack1
 description AC-MGMT/DFS-GROUP/ROUTER-ID
 ip address 10.125.98.2 255.255.255.255
#

# -----NVE interface of BorderLeaf_2
interface Nve1
 source 10.125.99.1
 mac-address 0000-5e00-0101
#

# -----M-LAG mode of BorderLeaf_2
stp mode rstp
 stp v-stp enable
#

# -----M-LAG DFS group
```

```
dfs-group 1
 dual-active detection enhanced enable
 source ip 10.125.98.2 peer 10.125.98.1
 consistency-check enable mode loose
#

#-----peer-link
interface Eth-Trunk0
 trunkport 10GE 4/0/47
 trunkport 40GE 1/0/23
 mode lacp-static
 peer-link 1
 lacp mixed-rate link enable
#
interface 40GE1/0/23
 distribute-weight 4
#

#-----M-LAG interface connecting BorderLeaf_2 to firewalls
interface Eth-Trunk1
 description FW_1_MGMT
 trunkport 10GE 4/0/41
 port default vlan 11
 mode lacp-static
 dfs-group 1 m-lag 1
#
interface Eth-Trunk2
 description FW_2_MGMT
 trunkport 10GE 4/0/42
 port default vlan 11
 mode lacp-static
 dfs-group 1 m-lag 2
#
interface Eth-Trunk11
 trunkport 10GE 4/0/43
 port link-type trunk
 undo port trunk allow-pass vlan 1
 stp edged-port enable
 mode lacp-static
 dfs-group 1 m-lag 3
#
interface Eth-Trunk12
 trunkport 10GE 4/0/44
 port link-type trunk
 undo port trunk allow-pass vlan 1
 stp edged-port enable
 mode lacp-static
 dfs-group 1 m-lag 4
#

#-----OSPF configurations that ensure connectivity on the VXLAN underlay network
bfd
#
ospf 1 router-id 10.125.98.2
 bfd all-interfaces enable
 bfd all-interfaces min-tx-interval 500 min-rx-interval 500 detect-multiplier 3 //Set the interval for
receiving and sending BFD packets to 500 ms and the detection multiplier to 3 only when all devices
on the network support hardware-based BFD. In other scenarios, retain the default BFD parameter
settings. That is, the interval for receiving and sending BFD packets is 1000 ms and the detection
multiplier is 3.
 lsa-arrival-interval intelligent-timer 50 50 50
 area 0.0.0.0
 network 10.125.97.36 0.0.0.3
 network 10.125.97.40 0.0.0.3
 network 10.125.97.44 0.0.0.3
 network 10.125.97.48 0.0.0.3
 network 10.125.97.56 0.0.0.7
 network 10.125.98.2 0.0.0.0
 network 10.125.99.1 0.0.0.0
```



```
#
interface 40GE1/0/0
description to ServerLeaf1_1
undo portswitch
ip address 10.125.97.37 255.255.255.252
ospf network-type p2p
#
interface 40GE1/0/1
description to ServerLeaf1_2
undo portswitch
ip address 10.125.97.41 255.255.255.252
ospf network-type p2p
#
interface 40GE1/0/2
description to ServerLeaf2_1
undo portswitch
ip address 10.125.97.45 255.255.255.252
ospf network-type p2p
#
interface 40GE1/0/3
description to ServerLeaf2_2
undo portswitch
ip address 10.125.97.49 255.255.255.252
ospf network-type p2p
#
#-----OSPF network convergence performance optimization triggered upon faults
interface 40GE1/0/0
ospf peer hold-max-cost timer 300000 //Set the interval at which the maximum cost is stored in
local LSAs to 300s after OSPF neighbor relationships are established between all border leaf nodes
and server leaf nodes. The value is the sum of the delay (240s) after which an M-LAG interface goes
Up (overlay routes are converged during this period) plus the device entry synchronization period
(60s).
#
interface 40GE1/0/1
ospf peer hold-max-cost timer 300000
#
interface 40GE1/0/2
ospf peer hold-max-cost timer 300000
#
interface 40GE1/0/3
ospf peer hold-max-cost timer 300000
#
#-----BGP EVPN
evpn-overlay enable
#
bgp 100
router-id 10.125.98.2
advertise lowest-priority all-address-family peer-up delay 360
undo default ipv4-unicast
group ServerLeaf internal
peer 10.125.98.3 group ServerLeaf
peer 10.125.98.4 group ServerLeaf
peer 10.125.98.5 group ServerLeaf
peer 10.125.98.6 group ServerLeaf
peer ServerLeaf connect-interface LoopBack1
#
l2vpn-family evpn
undo policy vpn-target
peer ServerLeaf enable
peer 10.125.98.3 group ServerLeaf
peer 10.125.98.4 group ServerLeaf
peer 10.125.98.5 group ServerLeaf
peer 10.125.98.6 group ServerLeaf
peer ServerLeaf advertise irb
peer ServerLeaf advertise irbv6
peer ServerLeaf reflect-client
#
```

```
#
ip route-static 1.2.3.4 255.255.255.0 10.125.97.242
#
```

- **ServerLeaf1_1 configuration file**

```
# -----IP addresses of interfaces connecting ServerLeaf1_1 to border leaf nodes
interface 40GE1/0/1
description to BorderLeaf_1
undo portswitch
ip address 10.125.97.22 255.255.255.252
ospf network-type p2p
#
interface 40GE1/0/2
description to BorderLeaf_2
undo portswitch
ip address 10.125.97.38 255.255.255.252
ospf network-type p2p
#

#-----IP addresses of loopback interfaces
interface LoopBack0
description VTEP
ipv6 enable //Enable IPv6 if it is required.
ip address 10.125.99.2 255.255.255.255
#
interface LoopBack1
description AC-MGMT/DFS-GROUP/ROUTER-ID
ip address 10.125.98.3 255.255.255.255
#

#-----VTEP IP addresses and virtual MAC addresses for NVE interfaces
interface Nve1
source 10.125.99.2
mac-address 0000-5e00-0102
#

#-----M-LAG mode
stp mode rstp
stp v-stp enable //Configure M-LAG in V-STP mode.
stp tc-protection //Enable TC BPDU protection on the device.
stp bpdu-protection //Enable BPDU protection on the device.
arp ip-conflict-detect enable //Enable IP address conflict detection on the device.
#

#-----M-LAG DFS group
dfs-group 1
dual-active detection enhanced enable
priority 150 //Set the DFS priority to be higher than that of the remote device. The default priority
is 100.
source ip 10.125.98.3 peer 10.125.98.4
consistency-check enable mode loose //Enable M-LAG configuration consistency check and specify
the loose mode.
#

#-----peer-link
interface Eth-Trunk0
trunkport 40GE 1/0/5 to 1/0/6
mode lacp-static
peer-link 1
#

#-----Server leaf node configurations ensuring that service servers can connect to the server leaf
nodes in load balancing mode
interface eth-trunk 10
port link-type trunk
undo port trunk allow-pass vlan 1
trunkport 10ge 1/0/1
dfs-group 1 m-lag 10
mode lacp-static
```

```
stp edged-port enable //Configure an edge interface.
#
interface 10GE1/0/1 //Configure the server access interface.
storm suppression unknown-unicast 5 //Configure unknown unicast traffic suppression. The
empirical value is 5% of the bandwidth of the 10GE interface. It is recommended that unknown
unicast traffic suppression be configured on all service interfaces.
storm suppression multicast packets 1000 //Configure multicast traffic suppression. The empirical
value is 1000 pps.
storm suppression broadcast packets 1000 //Configure broadcast traffic suppression. The empirical
value is 1000 pps. It is recommended that broadcast traffic suppression be configured on all service
interfaces.
#

#-----Server leaf node configurations ensuring that servers can connect to the server leaf nodes in
active/standby mode
interface 10GE1/0/2
port link-type trunk
undo port trunk allow-pass vlan 1 //Configure the interface to deny packets from VLAN 1 to prevent
loops.
storm suppression unknown-unicast 5 //Configure unknown unicast traffic suppression. The
empirical value is 5% of the bandwidth of the 10GE interface. It is recommended that unknown
unicast traffic suppression be configured on all service interfaces.
storm suppression multicast packets 1000 //Configure multicast traffic suppression. The empirical
value is 1000 pps.
storm suppression broadcast packets 1000 //Configure broadcast traffic suppression. The empirical
value is 1000 pps. It is recommended that broadcast traffic suppression be configured on all service
interfaces.
stp edged-port enable
#

#-----Monitor Link group that associates uplink interfaces with downlink interfaces
monitor-link group 1
port 40GE1/0/1 uplink
port 40GE1/0/2 uplink
port Eth-Trunk10 downlink 1
timer recover-time 60 //Configure the WTR time of the Monitor Link group. This prevents packet
loss during traffic switchback upon an uplink fault.
#

# -----OSPF
bfd //Enable BFD globally.
#
ospf 1 router-id 10.125.98.3
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 500 min-rx-interval 500 detect-multiplier 3 //Set the interval for
receiving and sending BFD packets to 500 ms and the detection multiplier to 3 only when all devices
on the network support hardware-based BFD. In other scenarios, retain the default BFD parameter
settings. That is, the interval for receiving and sending BFD packets is 1000 ms and the detection
multiplier is 3. The models that support hardware-based BFD include the CE6857, CE6865, CE8861,
CE8868, CE6875, CE5880, CE8850-64CQ-EI, CE6880, CE6881, CE6863, CE6863E, CE6863K, CE12800E,
CE16800, and CE12800 (configured with only the CE-L48XS-FG, CE-L48XS-FD1, CE-L24LQ-FD, CE-
L36LQ-FD, CE-L08CF-FG1, CE-L12CQ-FD, CE-L16CQ-FD, CE-L36CQ-FD, CE-L36CQ-FD1, CE-L36CQ-FG,
or CE-L36CQ-SD cards).
lsa-arrival-interval intelligent-timer 50 50 50 //Set the interval for receiving OSPF LSAs to shorten
the convergence time.
area 0.0.0.0
network 10.125.97.20 0.0.0.3
network 10.125.97.36 0.0.0.3 //Configure the server leaf node to establish neighbor relationships
with the two border leaf nodes.
network 10.125.98.3 0.0.0.0
network 10.125.99.2 0.0.0.0 //Advertise the loopback interface address.
#

#-----Network convergence performance optimization triggered upon faults
interface 40GE1/0/2
ospf peer hold-max-cost timer 300000 //Set the interval at which the maximum cost is stored in
local LSAs to 300s after OSPF neighbor relationships are established between all border leaf nodes
and server leaf nodes. The value is the sum of the delay (240s) after which an M-LAG interface goes
Up (overlay routes are converged during this period) plus the device entry synchronization period
```

```
(60s).
#
interface 40GE1/0/3
 ospf peer hold-max-cost timer 300000
#
#-----BGP EVPN
evpn-overlay enable //Configure EVPN as the VXLAN control plane.
#
bgp 100
 router-id 10.125.98.3
 undo default ipv4-unicast //Disable the IPv4 unicast address family for BGP peers to reduce the
device load.
 group BorderLeaf internal //Configure a peer group named BorderLeaf and add peers to the peer
group.
 peer 10.125.98.1 group BorderLeaf
 peer 10.125.98.2 group BorderLeaf
 peer Spine connect-interface LoopBack1 //Specify the source interface that sends BGP packets.
#
l2vpn-family evpn
 policy vpn-target
 peer BorderLeaf enable
 peer 10.125.98.1 group BorderLeaf
 peer 10.125.98.2 group BorderLeaf
 peer Spine advertise irb
 peer Spine advertise irbv6
#
#-----Overlay network configuration
ip vpn-instance vpn1
 ipv4-family
  route-distinguisher 20:2
  vpn-target 100:5010 export-extcommunity evpn
  vpn-target 100:5010 import-extcommunity evpn
 vxlan vni 5010
#
bridge-domain 10
 vxlan vni 10
 evpn
  route-distinguisher 10:2
  vpn-target 100:10 export-extcommunity
  vpn-target 100:5010 export-extcommunity
  vpn-target 100:10 import-extcommunity
#
interface Vbdif10
 ip binding vpn-instance vpn1
 ip address 10.1.1.1 255.255.255.0
 mac-address 0000-5e00-0102
 vxlan anycast-gateway enable
 arp collect host enable
#
```

- ServerLeaf1_2 configuration file

```
# -----IP addresses of interfaces connecting ServerLeaf1_2 to border leaf nodes
interface 40GE1/0/1
 description to BorderLeaf_1
 undo portswitch
 ip address 10.125.97.26 255.255.255.252
 ospf network-type p2p
#
interface 40GE1/0/2
 description to BorderLeaf_2
 undo portswitch
 ip address 10.125.97.42 255.255.255.252
 ospf network-type p2p
#
#-----IP addresses of loopback interfaces
interface LoopBack0
 description VTEP
```

```
ipv6 enable //Enable IPv6 if it is required.
ip address 10.125.99.2 255.255.255.255
#
interface LoopBack1
description AC-MGMT/DFS-GROUP/ROUTER-ID
ip address 10.125.98.4 255.255.255.0
#

#-----VTEP IP addresses and virtual MAC addresses for NVE interfaces
interface Nve1
source 10.125.99.2
mac-address 0000-5e00-0102
#

#-----M-LAG mode
stp mode rstp
stp v-stp enable //Configure M-LAG in V-STP mode.
stp tc-protection //Enable TC BPDU protection on the device.
stp bpdu-protection //Enable BPDU protection on the device.
arp ip-conflict-detect enable //Enable IP address conflict detection on the device.
#

#-----M-LAG DFS group
dfs-group 1
dual-active detection enhanced enable
source ip 10.125.98.4 peer 10.125.98.3
consistency-check enable mode loose
#

#-----peer-link
interface Eth-Trunk0
trunkport 40GE 1/0/5 to 1/0/6
mode lacp-static
peer-link 1
#

#-----Server leaf node configurations ensuring that service servers can connect to the server leaf
nodes in load balancing mode
interface eth-trunk 10
port link-type trunk
undo port trunk allow-pass vlan 1
trunkport 10ge 1/0/1
dfs-group 1 m-lag 10
mode lacp-static
stp edged-port enable
#
interface 10GE1/0/1
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
#

#-----Server leaf node configurations ensuring that servers can connect to the server leaf nodes in
active/standby mode
interface 10GE1/0/2
port link-type trunk
undo port trunk allow-pass vlan 1
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
stp edged-port enable
#

#-----Monitor Link group that associates uplink interfaces with downlink interfaces
monitor-link group 1
port 40GE1/0/1 uplink
port 40GE1/0/2 uplink
port Eth-Trunk10 downlink 1
timer recover-time 60
```

```
#

# -----OSPF
bfd
#
ospf 1 router-id 10.125.98.4
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 500 min-rx-interval 500 detect-multiplier 3 //Set the interval for
receiving and sending BFD packets to 500 ms and the detection multiplier to 3 only when all devices
on the network support hardware-based BFD. In other scenarios, retain the default BFD parameter
settings. That is, the interval for receiving and sending BFD packets is 1000 ms and the detection
multiplier is 3.
lsa-arrival-interval intelligent-timer 50 50 50 //Optimize the three-layer architecture and set the
OSPF convergence time when there are multiple ECMP paths between two physical devices.
area 0.0.0.0
network 10.125.97.24 0.0.0.3
network 10.125.97.40 0.0.0.3
network 10.125.98.4 0.0.0.0
network 10.125.99.2 0.0.0.0
#

#-----Network convergence performance optimization triggered upon faults
interface 40GE1/0/2
ospf peer hold-max-cost timer 300000 //Set the interval at which the maximum cost is stored in
local LSAs to 300s after OSPF neighbor relationships are established between all border leaf nodes
and server leaf nodes. The value is the sum of the delay (240s) after which an M-LAG interface goes
Up (overlay routes are converged during this period) plus the device entry synchronization period
(60s).
#
interface 40GE1/0/3
ospf peer hold-max-cost timer 300000
#

#-----BGP EVPN
evpn-overlay enable
#
bgp 100
router-id 10.125.98.4
undo default ipv4-unicast
group Spine internal
peer 10.125.98.1 group Spine
peer 10.125.98.2 group Spine
peer Spine connect-interface LoopBack1
#
l2vpn-family evpn
policy vpn-target
peer Spine enable
peer 10.125.98.1 group Spine
peer 10.125.98.2 group Spine
peer Spine advertise irb
peer Spine advertise irbv6
#

#
assign forward nvo3 acl extend enable //This step is required only on the CE12800, CE16800
equipped with A series cards, CE6870EI, and CE6875EI.
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 20:4
vpn-target 100:5010 export-extcommunity evpn
vpn-target 100:5010 import-extcommunity evpn
vxlan vni 5010
#
bridge-domain 10
vxlan vni 10
evpn
route-distinguisher 10:4
vpn-target 100:10 export-extcommunity
```

```
vpn-target 100:5010 export-extcommunity
vpn-target 100:10 import-extcommunity
#
interface Vbdi10
ip binding vpn-instance vpn1
ip address 10.1.1.1 255.255.255.0
mac-address 0000-5e00-0102
vxlan anycast-gateway enable
arp collect host enable
#
```

- The configuration files of ServerLeaf2_1 and ServerLeaf2_2 are similar to those of ServerLeaf1_1 and ServerLeaf1_2, and are not mentioned here.
- FW-1 configuration file

```
#-----vSYS key configurations
interface Vlanif3004
ip binding vpn-instance vsys_1
ip address 10.125.97.242
255.255.255.252
#
ip route-static 0.0.0.0 0.0.0.0 public
ip route-static 10.132.1.0 255.255.255.0
10.125.97.241
#
#-----Public key configurations
vsys name vsys_1 1
assign vlan 3004
#
ip vpn-instance vsys_1
ipv4-family
#
interface Vlanif3005
ip address 10.125.97.242
255.255.255.252
#
interface Eth-Trunk11
portswitch
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 3004 to
3005
#
ip route-static 0.0.0.0 0.0.0.0 10.125.97.241
ip route-static 10.132.1.0 255.255.255.0 vpn-instance
vsys_78b845f994647ad9
#
#-----SNAT configurations
nat address-group addgrp 0
mode pat
section 0 1.2.3.4 1.2.3.4
#
security-policy
rule name 20191228113827
source-address 10.132.1.0 mask 255.255.255.0
action permit
#
nat-policy
rule name rule1
description SNAT_01
source-zone trust
destination-zone untrust
source-address 10.132.1.0 mask 255.255.255.0
action source-nat address-group addgrp
#
firewall import-flow public 1.2.3.4 1.2.3.4 vpn-instance vsys_1
#
```

- FW-2 configuration file

```
#-----vSYS key configurations
interface Vlanif3004
 ip binding vpn-instance vsys_1
 ip address 10.125.97.242
 255.255.255.252
#
ip route-static 0.0.0.0 0.0.0.0 public
ip route-static 10.132.1.0 255.255.255.0
10.125.97.241
#
#-----Public key configurations
#
vsys name vsys_1 1
assign vlan 3004
#
ip vpn-instance vsys_1
 ipv4-family
#
interface Vlanif3005
 ip address 10.125.97.242
 255.255.255.252
#
interface Eth-Trunk11
 portswitch
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 3004 to
 3005
#
ip route-static 0.0.0.0 0.0.0.0 10.125.97.241
ip route-static 10.132.1.0 255.255.255.0 vpn-instance vsys_1
#
# -----SNAT configurations
nat address-group addgrp 0
 mode pat
 section 0 1.2.3.4 1.2.3.4
#
security-policy
 rule name 20191228113827
  source-address 10.132.1.0 mask 255.255.255.0
  action permit
#
nat-policy
 rule name rule1
  description SNAT_01
  source-zone trust
  destination-zone untrust
  source-address 10.132.1.0 mask 255.255.255.0
  action source-nat address-group addgrp
#
firewall import-flow public 1.2.3.4 1.2.3.4 vpn-instance vsys_1
#
```

1.8 Example for Configuring NFVI Distributed Gateways (Asymmetric Mode)

Applicable Products and Versions

- V200R005C10: The CE6865EI or CE8861EI can be deployed as a leaf switch on the network. The CE12800 can be deployed as a spine switch on the network.

- V200R005C20: The CE6881 or CE6863 can be deployed as a leaf switch on the network. The CE16800 can be deployed as a spine switch on the network.
- V200R019C00: The CE6865EI, CE8861EI, CE6881, or CE6863 can be deployed as a leaf switch on the network. The CE12800 or CE16800 can be deployed as a spine switch on the network.
- V200R019C10:
 - The CE6857F, CE6857S, CE6865EI, or CE8861EI can be deployed as a leaf switch on the network. The CE12800 can be deployed as a spine switch on the network.
 - The CE6881 or CE6863 can be deployed as a leaf switch on the network. The CE16800 can be deployed as a spine switch on the network.
- V200R020C00: The CE6881 or CE6863 can be deployed as a leaf switch on the network. The CE16800 can be deployed as a spine switch on the network.
- V200R020C10:
 - The CE6857F, CE6857S, CE6865EI, or CE8861EI can be deployed as a leaf switch on the network. The CE12800 can be deployed as a spine switch on the network.
 - The CE6881 or CE6863 can be deployed as a leaf switch on the network. The CE16800 can be deployed as a spine switch on the network.
- V200R021C00: The CE6881, CE6863, CE6857F, CE6857S, CE6865E, or CE8861EI can be deployed as a leaf switch on the network. The CE16800 can be deployed as a spine switch on the network.
- V200R021C10, later versions: The CE6863E, CE6865E, CE6857F, CE6857S, or CE6881-48S6CQ can be deployed as a leaf switch on the network. The CE16800 equipped with G/GK series board can be deployed as a spine switch on the network.

Networking Requirements

Huawei's NFVI telecommunications (telco) cloud is a networking solution that consists of Data Center Interconnect (DCI) and data communication network (DCN) technologies. Mobile phone traffic enters the DCN and is sent to its virtualized unified gateway (vUGW) and virtual multiservice engine (vMSE). After being processed by vUGW and vMSE, the phone traffic is forwarded over the Internet through the DCN to the destination devices. Similarly, response traffic sent over the Internet from the destination devices to the mobile phones also undergoes this process. To meet the preceding requirements and ensure that the phone traffic is load balanced within the DCN, you need to deploy the NFVI distributed gateway function on devices in the DCN.

In [Figure 1-8](#), DCGW1 and DCGW2 are the DCN's border gateways, and they can exchange Internet routes with devices on the external network. Leaf switches provide network access for Virtualized Network Functions (VNFs). As virtualized NEs, VNF1 and VNF2 can be deployed separately to implement the functions of the vUGW and vMSE. VNF1 and VNF2 are connected to leaf switches through respective interface process units (IPUs).

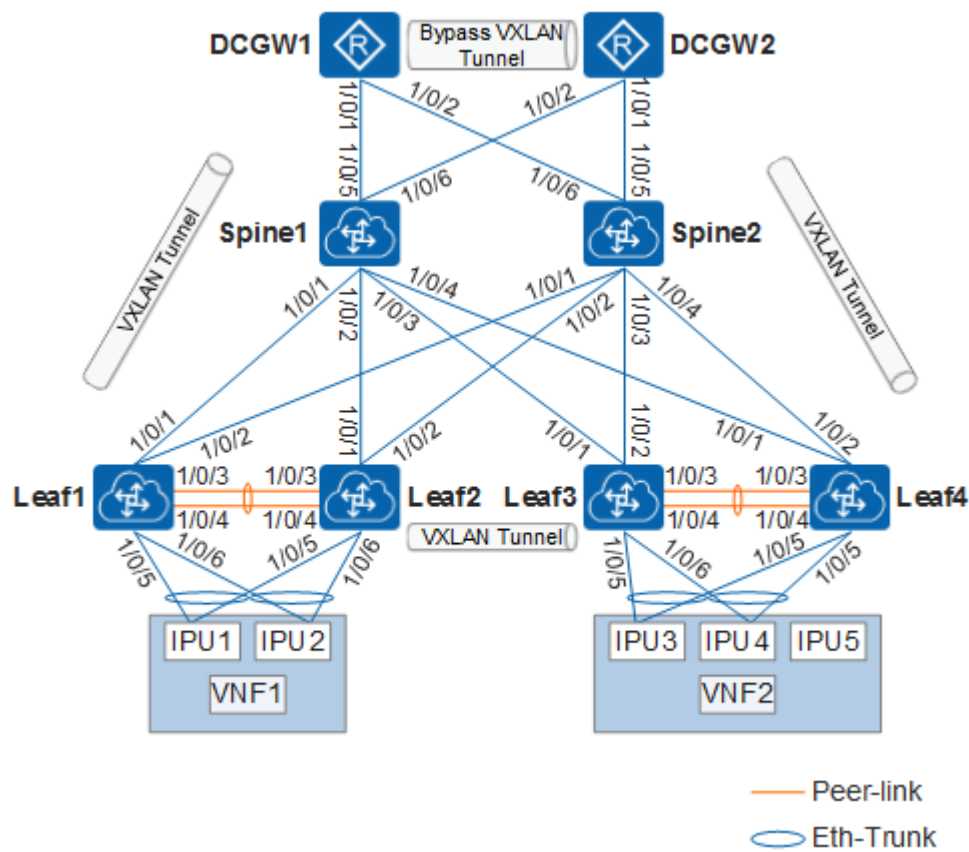
This networking combines the distributed gateway function and the VXLAN active-active gateway function:

- The VXLAN active-active gateway function is configured on DCGW1 and DCGW2. Specifically, a bypass VXLAN tunnel is set up between DCGW1 and

DCGW2. In addition, they use a virtual anycast VTEP address to establish VXLAN tunnels with leaf switches.

- Spine switches transparently transmit packets and connect to DCGW devices and leaf switches.
- The distributed gateway function is configured on leaf switches and a VXLAN tunnel is established between the two M-LAG systems.

Figure 1-8 NFVI distributed gateway networking



NOTE

In the preceding figure, 1/0/1 indicates an interface number and the interface rate is 10 Gbit/s. That is, 1/0/1 indicates 10GE1/0/1. Other interfaces are specified in a similar way.

Table 1-11 Interface IP address list

Device Name	Interface	IP Address	Device Name	Interface	IP Address
DCGW1	10GE1/0/1	192.168.9.2/24	DCGW2	10GE1/0/1	192.168.11.2/24
	10GE1/0/2	192.168.12.2/24		10GE1/0/2	192.168.10.2/24

Device Name	Interface	IP Address	Device Name	Interface	IP Address
	Loopback0	9.9.9.9/32		Loopback0	10.10.10.10/32
	Loopback1	11.11.11.11/32		Loopback1	11.11.11.11/32
Spine 1	10GE1/0/1	192.168.1.1/24	Spine 2	10GE1/0/1	192.168.5.1/24
	10GE1/0/2	192.168.2.1/24		10GE1/0/2	192.168.6.1/24
	10GE1/0/3	192.168.3.1/24		10GE1/0/3	192.168.7.1/24
	10GE1/0/4	192.168.4.1/24		10GE1/0/4	192.168.8.1/24
	10GE1/0/5	192.168.9.1/24		10GE1/0/5	192.168.11.1/24
	10GE1/0/6	192.168.10.1/24		10GE1/0/6	192.168.12.1/24
	Loopback0	7.7.7.7/32		Loopback0	8.8.8.8/32
Leaf1	10GE1/0/1	192.168.1.2/24	Leaf2	10GE1/0/1	192.168.2.2/24
	10GE1/0/2	192.168.5.2/24		10GE1/0/2	192.168.6.2/24
	Loopback0	1.1.1.1/32		Loopback0	2.2.2.2/32
	Loopback1	5.5.5.5/32		Loopback1	5.5.5.5/32
Leaf3	10GE1/0/1	192.168.3.2/24	Leaf4	10GE1/0/1	192.168.4.2/24
	10GE1/0/2	192.168.7.2/24		10GE1/0/2	192.168.8.2/24
	Loopback0	3.3.3.3/32		Loopback0	4.4.4.4/32
	Loopback1	6.6.6.6/32		Loopback1	6.6.6.6/32
IPU1	-	IPv4: 10.1.1.2/24 IPv6: fc00:1::2/64	IPU2	-	IPv4: 10.2.1.2/24 IPv6: fc00:2::2/64

Device Name	Interface	IP Address	Device Name	Interface	IP Address
IPU3	-	IPv4: 10.3.1.2/24 IPv6: fc00:3::2/64	IPU4	-	IPv4: 10.4.1.2/24 IPv6: fc00:4::2/64
VNF1	-	IPv4: 201.1.1.1/32 IPv6: 201::1/128	VNF2	-	IPv4: 202.1.1.1/32 IPv6: 202::1/128

Configuration Roadmap

The configuration roadmap is as follows:

1. Configure a routing protocol to ensure Layer 3 connectivity on the network.
2. Configure M-LAG on leaf switches.
3. Configure BGP EVPN to establish a VXLAN tunnel.
4. Configure static routes from leaf switches to VNFs and advertise the routes using BGP EVPN.
5. Configure load balancing among routes.

Procedure

- Step 1** Configure IP addresses for physical interfaces and loopback interfaces, and configure a routing protocol to ensure Layer 3 connectivity. OSPF is used in this example.

Configure Leaf1. Repeat this step for other switches. For details, see [Configuration Files](#).

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf1
[*HUAWEI] commit
[~Leaf1] bfd //Enable BFD
[*Leaf1-bfd] quit
[*Leaf1] interface 10ge 1/0/1
[*Leaf1-10GE1/0/1] undo portswitch
[*Leaf1-10GE1/0/1] ip address 192.168.1.2 24
[*Leaf1-10GE1/0/1] ospf network-type p2p
[*Leaf1-10GE1/0/1] ospf peer hold-max-cost timer 800000
[*Leaf1-10GE1/0/1] port crc-statistics trigger error-down
[*Leaf1-10GE1/0/1] trap-threshold crc-statistics 100 interval 10
[*Leaf1-10GE1/0/1] quit
[*Leaf1] interface 10ge 1/0/2
[*Leaf1-10GE1/0/2] undo portswitch
[*Leaf1-10GE1/0/2] ip address 192.168.5.2 24
[*Leaf1-10GE1/0/2] ospf network-type p2p
[*Leaf1-10GE1/0/2] ospf peer hold-max-cost timer 800000
[*Leaf1-10GE1/0/2] port crc-statistics trigger error-down
[*Leaf1-10GE1/0/2] trap-threshold crc-statistics 100 interval 10
[*Leaf1-10GE1/0/2] quit
```

```
[*Leaf1] interface loopback 1
[*Leaf1-LoopBack1] ip address 5.5.5.5 32
[*Leaf1-LoopBack1] quit
[*Leaf1] interface loopback 0
[*Leaf1-LoopBack0] ip address 1.1.1.1 32
[*Leaf1-LoopBack0] quit
[*Leaf1] ospf 1 router-id 1.1.1.1
[*Leaf1-ospf-1] spf-schedule-interval intelligent-timer 50 50 50
[*Leaf1-ospf-1] lsa-originate-interval intelligent-timer 500 50 100
[*Leaf1-ospf-1] lsa-arrival-interval intelligent-timer 50 50 50
[*Leaf1-ospf-1] bfd all-interfaces enable
[*Leaf1-ospf-1] bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
[*Leaf1-ospf-1] area 0
[*Leaf1-ospf-1-area-0.0.0.0] network 192.168.1.0 0.0.0.255
[*Leaf1-ospf-1-area-0.0.0.0] network 192.168.5.0 0.0.0.255
[*Leaf1-ospf-1-area-0.0.0.0] network 1.1.1.1 0.0.0.0
[*Leaf1-ospf-1-area-0.0.0.0] network 5.5.5.5 0.0.0.0
[*Leaf1-ospf-1-area-0.0.0.0] quit
[*Leaf1-ospf-1] quit
[*Leaf1] commit
```

Step 2 (For the CE8861EI, CE6857EI, and CE6865EI) Configure the resource allocation mode on leaf switches.

Configure Leaf1. Repeat this step for Leaf2, Leaf3, and Leaf4.

```
[~Leaf1] assign forward layer-3 resource large-overlay //Set the Layer 3 resource allocation mode to
large-overlay so that the switch supports more VXLAN overlay entries.
[*Leaf1] assign forward ipv6 longer-mask resource share-mode //Set the resource allocation mode of
IPv6 addresses and routes with prefixes longer than 64 bits to shared mode. In this mode, IPv4 addresses
and routes share chip resources with IPv6 addresses and routes.
[*Leaf1] commit
```

NOTE

After configuring the resource allocation mode, save the configuration and restart the switch to make the configuration take effect.

Step 3 Configure M-LAG on leaf switches.

Configure Leaf1 and Leaf2 to establish an M-LAG. Repeat this step for Leaf3 and Leaf4.

```
[~Leaf1] stp mode rstp
[*Leaf1] stp v-stp enable
[*Leaf1] dfs-group 1
[*Leaf1-dfs-group-1] source ip 1.1.1.1
[*Leaf1-dfs-group-1] priority 150
[*Leaf1-dfs-group-1] m-lag up-delay 240 auto-recovery interval 10 //Configure the delay for an M-LAG
member interface to report the Up event and the automatic recovery interval.
[*Leaf1-dfs-group-1] dual-active detection delay 0
[*Leaf1-dfs-group-1] quit
[*Leaf1] interface eth-trunk 1 //Configure the peer-link.
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/3
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/4
[*Leaf1-Eth-Trunk1] mode lacp-static
[*Leaf1-Eth-Trunk1] peer-link 1
[*Leaf1-Eth-Trunk1] quit
[*Leaf1] interface 10ge 1/0/3
[*Leaf1-10GE1/0/3] port crc-statistics trigger error-down
[*Leaf1-10GE1/0/3] trap-threshold crc-statistics 100 interval 10
[*Leaf1-10GE1/0/3] quit
[*Leaf1] interface 10ge 1/0/4
[*Leaf1-10GE1/0/4] port crc-statistics trigger error-down
[*Leaf1-10GE1/0/4] trap-threshold crc-statistics 100 interval 10
[*Leaf1-10GE1/0/4] quit
[*Leaf1] interface eth-trunk 10 //Configure an M-LAG member interface connected to a VNF.
[*Leaf1-Eth-Trunk10] trunkport 10ge 1/0/5
[*Leaf1-Eth-Trunk10] mode lacp-static
```

```
[*Leaf1-Eth-Trunk10] lacp timeout fast
[*Leaf1-Eth-Trunk10] dfs-group 1 m-lag 1
[*Leaf1-Eth-Trunk10] arp anti-attack rate-limit 200 //Set the maximum number of ARP packets that are
allowed to pass through per second to 200.
[*Leaf1-Eth-Trunk10] stp edged-port enable
[*Leaf1-Eth-Trunk10] quit
[*Leaf1] interface eth-trunk 11
[*Leaf1-Eth-Trunk11] trunkport 10ge 1/0/6
[*Leaf1-Eth-Trunk11] mode lacp-static
[*Leaf1-Eth-Trunk11] lacp timeout fast
[*Leaf1-Eth-Trunk11] dfs-group 1 m-lag 2
[*Leaf1-Eth-Trunk11] arp anti-attack rate-limit 200 //Set the maximum number of ARP packets that are
allowed to pass through per second to 200.
[*Leaf1-Eth-Trunk11] stp edged-port enable
[*Leaf1-Eth-Trunk11] quit
[*Leaf1] commit
[~Leaf1] monitor-link group 1
[*Leaf1-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf1-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf1-mtlk-group1] port eth-trunk 10 downlink 1
[*Leaf1-mtlk-group1] port eth-trunk 11 downlink 2
[*Leaf1-mtlk-group1] timer recover-time 60
[*Leaf1-mtlk-group1] quit
[*Leaf1] interface 10ge 1/0/5
[*Leaf1-10GE1/0/5] storm suppression unknown-unicast 5 //Configure unknown unicast suppression on
10GE1/0/5 of the access switch.
[*Leaf1-10GE1/0/5] storm suppression multicast packets 1000 //Configure unknown multicast
suppression on 10GE1/0/5 of the access switch.
[*Leaf1-10GE1/0/5] storm suppression broadcast packets 1000 //Configure unknown broadcast
suppression on 10GE1/0/5 of the access switch.
[*Leaf1-10GE1/0/5] port crc-statistics trigger error-down
[*Leaf1-10GE1/0/5] trap-threshold crc-statistics 100 interval 10
[*Leaf1-10GE1/0/5] quit
[*Leaf1] interface 10ge 1/0/6
[*Leaf1-10GE1/0/6] storm suppression unknown-unicast 5
[*Leaf1-10GE1/0/6] storm suppression multicast packets 1000
[*Leaf1-10GE1/0/6] storm suppression broadcast packets 1000
[*Leaf1-10GE1/0/6] port crc-statistics trigger error-down
[*Leaf1-10GE1/0/6] trap-threshold crc-statistics 100 interval 10
[*Leaf1-10GE1/0/6] quit
[*Leaf1] commit
[~Leaf2] stp mode rstp
[*Leaf2] stp v-stp enable
[*Leaf2] dfs-group 1
[*Leaf2-dfs-group-1] source ip 2.2.2.2
[*Leaf2-dfs-group-1] m-lag up-delay 240 auto-recovery interval 10
[*Leaf2-dfs-group-1] dual-active detection delay 0
[*Leaf2-dfs-group-1] quit
[*Leaf2] interface eth-trunk 1
[*Leaf2-Eth-Trunk1] trunkport 10ge 1/0/3
[*Leaf2-Eth-Trunk1] trunkport 10ge 1/0/4
[*Leaf2-Eth-Trunk1] mode lacp-static
[*Leaf2-Eth-Trunk1] peer-link 1
[*Leaf2-Eth-Trunk1] quit
[*Leaf2] interface eth-trunk 10
[*Leaf2-Eth-Trunk10] trunkport 10ge 1/0/5
[*Leaf2-Eth-Trunk10] mode lacp-static
[*Leaf2-Eth-Trunk10] lacp timeout fast
[*Leaf2-Eth-Trunk10] dfs-group 1 m-lag 1
[*Leaf2-Eth-Trunk10] arp anti-attack rate-limit 200 //Set the maximum number of ARP packets that are
allowed to pass through per second to 200.
[*Leaf2-Eth-Trunk10] stp edged-port enable
[*Leaf2-Eth-Trunk10] quit
[*Leaf2] interface eth-trunk 11
[*Leaf2-Eth-Trunk11] trunkport 10ge 1/0/6
[*Leaf2-Eth-Trunk11] mode lacp-static
[*Leaf2-Eth-Trunk11] lacp timeout fast
[*Leaf2-Eth-Trunk11] dfs-group 1 m-lag 2
[*Leaf2-Eth-Trunk11] arp anti-attack rate-limit 200 //Set the maximum number of ARP packets that are
```

```
allowed to pass through per second to 200.
[*Leaf2-Eth-Trunk11] stp edged-port enable
[*Leaf2-Eth-Trunk11] quit
[*Leaf2] commit
[~Leaf2] monitor-link group 1
[*Leaf2-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf2-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf2-mtlk-group1] port eth-trunk 10 downlink 1
[*Leaf2-mtlk-group1] port eth-trunk 11 downlink 2
[*Leaf2-mtlk-group1] timer recover-time 60
[*Leaf2-mtlk-group1] quit
[*Leaf2] interface 10ge 1/0/5
[*Leaf2-10GE1/0/5] storm suppression unknown-unicast 5 //Configure unknown unicast suppression on
10GE1/0/5 of the access switch.
[*Leaf2-10GE1/0/5] storm suppression multicast packets 1000 //Configure unknown multicast
suppression on 10GE1/0/5 of the access switch.
[*Leaf2-10GE1/0/5] storm suppression broadcast packets 1000 //Configure unknown broadcast
suppression on 10GE1/0/5 of the access switch.
[*Leaf2-10GE1/0/5] quit
[*Leaf2] interface 10ge 1/0/6
[*Leaf2-10GE1/0/6] storm suppression unknown-unicast 5
[*Leaf2-10GE1/0/6] storm suppression multicast packets 1000
[*Leaf2-10GE1/0/6] storm suppression broadcast packets 1000
[*Leaf2-10GE1/0/6] quit
[*Leaf2] commit
```

Step 4 Configure BGP EVPN to establish a VXLAN tunnel.

1. (For the CE12800 and CE16800 equipped with A series cards) Configure the VXLAN tunnel mode and enable the VXLAN ACL extension function on spine switches.

Configure Spine1. Repeat this step for Spine2.

```
[~Spine1] ip tunnel mode vxlan
[*Spine1] assign forward nvo3 acl extend enable
[*Spine1] commit
```

NOTE

After configuring the VXLAN tunnel mode and enabling the VXLAN ACL extension function, save the configuration and restart the switch to make the configuration take effect.

2. Establish BGP EVPN peer relationships. Spine1 and Spine2 function as RRs of leaf nodes.

Configure Spine1. Repeat this step for Spine2.

```
[~Spine1] evpn-overlay enable
[*Spine1] bgp 100
[*Spine1] router-id 7.7.7.7
[*Spine1-bgp] peer 1.1.1.1 as-number 100
[*Spine1-bgp] peer 1.1.1.1 connect-interface LoopBack0
[*Spine1-bgp] peer 2.2.2.2 as-number 100
[*Spine1-bgp] peer 2.2.2.2 connect-interface LoopBack0
[*Spine1-bgp] peer 3.3.3.3 as-number 100
[*Spine1-bgp] peer 3.3.3.3 connect-interface LoopBack0
[*Spine1-bgp] peer 4.4.4.4 as-number 100
[*Spine1-bgp] peer 4.4.4.4 connect-interface LoopBack0
[*Spine1-bgp] peer 9.9.9.9 as-number 100
[*Spine1-bgp] peer 9.9.9.9 connect-interface LoopBack0
[*Spine1-bgp] peer 10.10.10.10 as-number 100
[*Spine1-bgp] peer 10.10.10.10 connect-interface LoopBack0
[*Spine1-bgp] l2vpn-family evpn
[*Spine1-bgp-af-evpn] bestroute add-path path-number 64 //Enable the BGP ADD-PATH function.
[*Spine1-bgp-af-evpn] peer 1.1.1.1 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 1.1.1.1 reflect-client
[*Spine1-bgp-af-evpn] peer 1.1.1.1 advertise irb
```

```
[*Spine1-bgp-af-evpn] peer 1.1.1.1 advertise irbv6
[*Spine1-bgp-af-evpn] peer 1.1.1.1 capability-advertise add-path both
[*Spine1-bgp-af-evpn] peer 1.1.1.1 advertise add-path path-number 64
[*Spine1-bgp-af-evpn] peer 2.2.2.2 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 2.2.2.2 advertise irb
[*Spine1-bgp-af-evpn] peer 2.2.2.2 advertise irbv6
[*Spine1-bgp-af-evpn] peer 2.2.2.2 reflect-client
[*Spine1-bgp-af-evpn] peer 2.2.2.2 capability-advertise add-path both
[*Spine1-bgp-af-evpn] peer 2.2.2.2 advertise add-path path-number 64
[*Spine1-bgp-af-evpn] peer 3.3.3.3 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 3.3.3.3 advertise irb
[*Spine1-bgp-af-evpn] peer 3.3.3.3 advertise irbv6
[*Spine1-bgp-af-evpn] peer 3.3.3.3 reflect-client
[*Spine1-bgp-af-evpn] peer 3.3.3.3 capability-advertise add-path both
[*Spine1-bgp-af-evpn] peer 3.3.3.3 advertise add-path path-number 64
[*Spine1-bgp-af-evpn] peer 4.4.4.4 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 4.4.4.4 advertise irb
[*Spine1-bgp-af-evpn] peer 4.4.4.4 advertise irbv6
[*Spine1-bgp-af-evpn] peer 4.4.4.4 reflect-client
[*Spine1-bgp-af-evpn] peer 4.4.4.4 capability-advertise add-path both
[*Spine1-bgp-af-evpn] peer 4.4.4.4 advertise add-path path-number 64
[*Spine1-bgp-af-evpn] peer 9.9.9.9 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 9.9.9.9 advertise irb
[*Spine1-bgp-af-evpn] peer 9.9.9.9 advertise irbv6
[*Spine1-bgp-af-evpn] peer 9.9.9.9 capability-advertise add-path both
[*Spine1-bgp-af-evpn] peer 9.9.9.9 advertise add-path path-number 64
[*Spine1-bgp-af-evpn] peer 10.10.10.10 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 10.10.10.10 advertise irb
[*Spine1-bgp-af-evpn] peer 10.10.10.10 advertise irbv6
[*Spine1-bgp-af-evpn] peer 10.10.10.10 capability-advertise add-path both
[*Spine1-bgp-af-evpn] peer 10.10.10.10 advertise add-path path-number 64
[*Spine1-bgp-af-evpn] undo policy vpn-target
[*Spine1-bgp-af-evpn] quit
[*Spine1-bgp] quit
[*Spine1] commit
```

Configure Leaf1. Repeat this step for Leaf2, Leaf3, and Leaf4.

```
[~Leaf1] evpn-overlay enable
[*Leaf1] bgp 100
[*Leaf1] router-id 1.1.1.1
[*Leaf1-bgp] peer 7.7.7.7 as-number 100
[*Leaf1-bgp] peer 7.7.7.7 connect-interface LoopBack0
[*Leaf1-bgp] peer 8.8.8.8 as-number 100
[*Leaf1-bgp] peer 8.8.8.8 connect-interface LoopBack0
[*Leaf1-bgp] l2vpn-family evpn
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 advertise irb
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 advertise irbv6
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 advertise irb
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 advertise irbv6
[*Leaf1-bgp-af-evpn] quit
[*Leaf1-bgp] quit
[*Leaf1] commit
```

3. Configure VPN and EVPN instances.

Configure Leaf1. Repeat this step for Leaf2, Leaf3, and Leaf4.

```
[~Leaf1] ip vpn-instance vpn1
[*Leaf1-vpn-instance-vpn1] vxlan vni 200
[*Leaf1-vpn-instance-vpn1] ipv4-family
[*Leaf1-vpn-instance-vpn1-af-ipv4] route-distinguisher 1.1.1.1:1
[*Leaf1-vpn-instance-vpn1-af-ipv4] vpn-target 0:1 evpn
```



```
[*Leaf1-vpn-instance-vpn1-af-ipv4] quit
[*Leaf1-vpn-instance-vpn1] ipv6-family
[*Leaf1-vpn-instance-vpn1-af-ipv6] route-distinguisher 1.1.1.1:1
[*Leaf1-vpn-instance-vpn1-af-ipv6] vpn-target 0:1 evpn
[*Leaf1-vpn-instance-vpn1-af-ipv6] quit
[*Leaf1-vpn-instance-vpn1] quit
[*Leaf1] bridge-domain 10
[*Leaf1-bd10] vxlan vni 110
[*Leaf1-bd10] evpn
[*Leaf1-bd10-evpn] route-distinguisher 1.1.1.1:110
[*Leaf1-bd10-evpn] vpn-target 0:110
[*Leaf1-bd10-evpn] vpn-target 0:1 export-extcommunity
[*Leaf1-bd10-evpn] quit
[*Leaf1-bd10] quit
[*Leaf1] bridge-domain 20
[*Leaf1-bd20] vxlan vni 120
[*Leaf1-bd20] evpn
[*Leaf1-bd20-evpn] route-distinguisher 1.1.1.1:120
[*Leaf1-bd20-evpn] vpn-target 0:120
[*Leaf1-bd20-evpn] vpn-target 0:1 export-extcommunity
[*Leaf1-bd20-evpn] quit
[*Leaf1-bd20] quit
[*Leaf1] bridge-domain 30
[*Leaf1-bd30] vxlan vni 130
[*Leaf1-bd30] evpn
[*Leaf1-bd30-evpn] route-distinguisher 1.1.1.1:130
[*Leaf1-bd30-evpn] vpn-target 0:130
[*Leaf1-bd30-evpn] vpn-target 0:1 export-extcommunity
[*Leaf1-bd30-evpn] quit
[*Leaf1-bd30] quit
[*Leaf1] bridge-domain 40
[*Leaf1-bd40] vxlan vni 140
[*Leaf1-bd40] evpn
[*Leaf1-bd40-evpn] route-distinguisher 1.1.1.1:140
[*Leaf1-bd40-evpn] vpn-target 0:140
[*Leaf1-bd40-evpn] vpn-target 0:1 export-extcommunity
[*Leaf1-bd40-evpn] quit
[*Leaf1-bd40] quit
[*Leaf1] interface nve 1
[*Leaf1-Nve1] source 5.5.5.5 //The IP addresses and MAC addresses of NVE interfaces on the two
devices that establish an M-LAG must be the same.
[*Leaf1-Nve1] mac-address 0000-5e00-0111
[*Leaf1-Nve1] vni 110 head-end peer-list protocol bgp
[*Leaf1-Nve1] vni 120 head-end peer-list protocol bgp
[*Leaf1-Nve1] vni 130 head-end peer-list protocol bgp
[*Leaf1-Nve1] vni 140 head-end peer-list protocol bgp
[*Leaf1-Nve1] quit
[*Leaf1] commit
```

4. Configure Layer 3 gateways.

Configure Leaf1. Repeat this step for Leaf2, Leaf3, and Leaf4.

```
[~Leaf1] interface vbdif10 //The IP addresses and MAC addresses of the same VBDIF interface on a
leaf switch and a DCGW must be the same.
[*Leaf1-Vbdif10] ip binding vpn-instance vpn1
[*Leaf1-Vbdif10] ip address 10.1.1.1 24
[*Leaf1-Vbdif10] ipv6 enable
[*Leaf1-Vbdif10] ipv6 address fc00:1::1 64
[*Leaf1-Vbdif10] arp generate-rd-table enable
[*Leaf1-Vbdif10] ipv6 nd generate-rd-table enable
[*Leaf1-Vbdif10] arp broadcast-detect enable
[*Leaf1-Vbdif10] vxlan anycast-gateway enable
[*Leaf1-Vbdif10] arp collect host enable
[*Leaf1-Vbdif10] ipv6 nd collect host enable
[*Leaf1-Vbdif10] mac-address 0000-5e00-0101
[*Leaf1-Vbdif10] quit
[*Leaf1] interface vbdif20
[*Leaf1-Vbdif20] ip binding vpn-instance vpn1
[*Leaf1-Vbdif20] ip address 10.2.1.1 24
[*Leaf1-Vbdif20] ipv6 enable
```

```
[*Leaf1-Vbdif20] ipv6 address fc00:2::1 64
[*Leaf1-Vbdif20] arp generate-rd-table enable
[*Leaf1-Vbdif20] ipv6 nd generate-rd-table enable
[*Leaf1-Vbdif20] arp broadcast-detect enable
[*Leaf1-Vbdif20] vxlan anycast-gateway enable
[*Leaf1-Vbdif20] arp collect host enable
[*Leaf1-Vbdif20] ipv6 nd collect host enable
[*Leaf1-Vbdif20] mac-address 0000-5e00-0102
[*Leaf1-Vbdif20] quit
[*Leaf1] interface vbdif30
[*Leaf1-Vbdif30] ip binding vpn-instance vpn1
[*Leaf1-Vbdif30] ip address 10.3.1.1 24
[*Leaf1-Vbdif30] ipv6 enable
[*Leaf1-Vbdif30] ipv6 address fc00:3::1 64
[*Leaf1-Vbdif30] arp generate-rd-table enable
[*Leaf1-Vbdif30] ipv6 nd generate-rd-table enable
[*Leaf1-Vbdif30] arp broadcast-detect enable
[*Leaf1-Vbdif30] vxlan anycast-gateway enable
[*Leaf1-Vbdif30] arp collect host enable
[*Leaf1-Vbdif30] ipv6 nd collect host enable
[*Leaf1-Vbdif30] mac-address 0000-5e00-0103
[*Leaf1-Vbdif30] quit
[*Leaf1] interface vbdif40
[*Leaf1-Vbdif40] ip binding vpn-instance vpn1
[*Leaf1-Vbdif40] ip address 10.4.1.1 24
[*Leaf1-Vbdif40] ipv6 enable
[*Leaf1-Vbdif40] ipv6 address fc00:4::1 64
[*Leaf1-Vbdif40] arp generate-rd-table enable
[*Leaf1-Vbdif40] ipv6 nd generate-rd-table enable
[*Leaf1-Vbdif40] arp broadcast-detect enable
[*Leaf1-Vbdif40] vxlan anycast-gateway enable
[*Leaf1-Vbdif40] arp collect host enable
[*Leaf1-Vbdif40] ipv6 nd collect host enable
[*Leaf1-Vbdif40] mac-address 0000-5e00-0104
[*Leaf1-Vbdif40] quit
[*Leaf1] commit
```

5. Configure service access points.

Configure Leaf1. Repeat this step for Leaf2, Leaf3, and Leaf4.

```
[~Leaf1] interface eth-trunk 10.10 mode l2
[*Leaf1-Eth-Trunk10.10] encapsulation dot1q vid 10
[*Leaf1-Eth-Trunk10.10] bridge-domain 10
[*Leaf1-Eth-Trunk10.10] quit
[*Leaf1] interface eth-trunk 11.20 mode l2
[*Leaf1-Eth-Trunk11.20] encapsulation dot1q vid 20
[*Leaf1-Eth-Trunk11.20] bridge-domain 20
[*Leaf1-Eth-Trunk11.20] quit
[*Leaf1] commit
```

Step 5 Configure BFD on leaf switches to detect links between leaf switches and VNFs.

Configure Leaf1. Repeat this step for Leaf2, Leaf3, and Leaf4.

```
[~Leaf1] bfd toipu1_v4 bind peer-ip 10.1.1.2 vpn-instance vpn1 interface vbdif10 source-ip 10.1.1.1
one-arm-echo
[*Leaf1-bfd-session-toipu1_v4] discriminator local 10
[*Leaf1-bfd-session-toipu1_v4] detect-multiplier 4
[*Leaf1-bfd-session-toipu1_v4] min-echo-rx-interval 500
[*Leaf1-bfd-session-toipu1_v4] quit
[*Leaf1] bfd toipu2_v4 bind peer-ip 10.2.1.2 vpn-instance vpn1 interface vbdif20 source-ip 10.2.1.1 one-
arm-echo
[*Leaf1-bfd-session-toipu2_v4] discriminator local 30
[*Leaf1-bfd-session-toipu2_v4] detect-multiplier 4
[*Leaf1-bfd-session-toipu2_v4] min-echo-rx-interval 500
[*Leaf1-bfd-session-toipu2_v4] quit
[*Leaf1] bfd toipu1_v6 bind peer-ipv6 fc00:1::2 vpn-instance vpn1 interface vbdif10 source-ipv6
fc00:1::1 one-arm-echo
[*Leaf1-bfd-session-toipu1_v6] discriminator local 50
[*Leaf1-bfd-session-toipu1_v6] detect-multiplier 4
```

```
[*Leaf1-bfd-session-toipu1_v6] min-echo-rx-interval 500
[*Leaf1-bfd-session-toipu1_v6] quit
[*Leaf1] bfd toipu2_v6 bind peer-ipv6 fc00:2::2 vpn-instance vpn1 interface vbdif20 source-ipv6
fc00:2::1 one-arm-echo
[*Leaf1-bfd-session-toipu2_v6] discriminator local 70
[*Leaf1-bfd-session-toipu2_v6] detect-multiplier 4
[*Leaf1-bfd-session-toipu2_v6] min-echo-rx-interval 500
[*Leaf1-bfd-session-toipu2_v6] quit
[*Leaf1] bfd
[*Leaf1-bfd] bfd forwarding match remote-discriminator 20 //Specify the remote discriminator of the
BFD for M-LAG.
[*Leaf1-bfd] bfd forwarding match remote-discriminator 40
[*Leaf1-bfd] bfd forwarding match remote-discriminator 60
[*Leaf1-bfd] bfd forwarding match remote-discriminator 80
[*Leaf1-bfd] quit
[*Leaf1] commit
```

NOTE

In the NFVI telco cloud solution deployed in traditional mode, the BFD detection interval can be set to 500 x 4.

In the NFVI telco cloud solution deployed in SDN mode, the BFD detection interval can be set to 500 x 4. To improve the detection performance, the BFD detection interval can be set to 300 x 6.

Step 6 Configure static VPN routes from leaf switches to VNFs, configure BGP EVPN to import static VPN routes, and configure a routing policy for the L3VPN instance to keep the original next hops of the static VPN routes.

Configure Leaf1. Repeat this step for Leaf2, Leaf3, and Leaf4.

```
[~Leaf1] route-policy import_static_policy permit node 100 //Configure a routing policy to add
community attributes when importing static routes to BGP.
[*Leaf1-route-policy] if-match tag 82345
[*Leaf1-route-policy] apply community 82345 additive
[*Leaf1-route-policy] quit
[*Leaf1] route-policy import_static_policy permit node 999
[*Leaf1-route-policy] quit
[*Leaf1] ip community-filter basic apply_gwip_route index 10 permit 82345 //Create a community filter.
[*Leaf1] ip community-filter basic suppress_route index 10 permit 82346
[*Leaf1] route-policy export_policy deny node 10 //Configure a routing policy to filter out or permit
routes.
[*Leaf1-route-policy] if-match community-filter suppress_route
[*Leaf1-route-policy] quit
[*Leaf1] route-policy export_policy permit node 20
[*Leaf1-route-policy] if-match community-filter apply_gwip_route
[*Leaf1-route-policy] apply gateway-ip origin-nexthop //Configure the next-hop address of the route as
the gateway IP address. When configuring an L3VPN instance on a Layer 2 or Layer 3 gateway to advertise
static VPN routes that can reach a VNF to an EVPN, you need to create a routing policy to filter out the
static VPN routes that can reach the VNF in the L3VPN instance.
[*Leaf1-route-policy] apply ipv6 gateway-ip origin-nexthop
[*Leaf1-route-policy] quit
[*Leaf1] route-policy export_policy permit node 999
[*Leaf1-route-policy] quit
[*Leaf1] ip route-static vpn-instance vpn1 201.1.1.1 32 10.1.1.2 preference 255 tag 82345 track bfd-
session toipu1_v4 inter-protocol-ecmp //The configured tag must match the routing policy so that the
next-hop address of the route advertised by the leaf switch to the DCGW is the gateway IP address,
implementing asymmetric forwarding.
[*Leaf1] ip route-static vpn-instance vpn1 201.1.1.1 32 10.2.1.2 preference 255 tag 82345 track bfd-
session toipu2_v4 inter-protocol-ecmp
[*Leaf1] ipv6 route-static vpn-instance vpn1 201::1 128 fc00:1::2 preference 255 tag 82345 track bfd-
session toipu1_v6 inter-protocol-ecmp
[*Leaf1] ipv6 route-static vpn-instance vpn1 201::1 128 fc00:2::2 preference 255 tag 82345 track bfd-
session toipu2_v6 inter-protocol-ecmp
[*Leaf1] bgp 100
[*Leaf1-bgp] ipv4-family vpn-instance vpn1
[*Leaf1-bgp-vpn1] import-route static route-policy import_static_policy
[*Leaf1-bgp-vpn1] advertise l2vpn evpn import-route-multipath //Advertise all routes with the same
```

```
destination address.
[*Leaf1-bgp-vpn1] irb asymmetric //Enable the asymmetric mode for IRB routes.
[*Leaf1-bgp-vpn1] quit
[*Leaf1-bgp] ipv6-family vpn-instance vpn1
[*Leaf1-bgp-6-vpn1] import-route static route-policy import_static_policy
[*Leaf1-bgp-6-vpn1] advertise l2vpn evpn import-route-multipath
[*Leaf1-bgp-6-vpn1] irb asymmetric
[*Leaf1-bgp-6-vpn1] quit
[*Leaf1-bgp] quit
[*Leaf1] commit
```

Step 7 Configure load balancing among routes.

Configure Leaf1. Repeat this step for Leaf2, Leaf3, and Leaf4.

```
[~Leaf1] bgp 100
[*Leaf1-bgp] ipv4-family vpn-instance vpn1
[*Leaf1-bgp-vpn1] maximum load-balancing 64
[*Leaf1-bgp-vpn1] quit
[*Leaf1-bgp] ipv6-family vpn-instance vpn1
[*Leaf1-bgp-6-vpn1] maximum load-balancing 64
[*Leaf1-bgp-6-vpn1] quit
[*Leaf1-bgp] l2vpn-family evpn
[*Leaf1-bgp-af-evpn] bestroute add-path path-number 64 //Enable the BGP ADD-PATH function.
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 capability-advertise add-path both
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 advertise add-path path-number 64
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 route-policy export_policy export //Configure a routing policy for the
RR.
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 capability-advertise add-path both
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 advertise add-path path-number 64
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 route-policy export_policy export
[*Leaf1-bgp-af-evpn] quit
[*Leaf1-bgp] quit
[*Leaf1] commit
```

----End

Verifying the Configuration

After completing the configurations, run the **display ip routing-table vpn-instance vpn1** command on a leaf switch. The command output shows that the leaf switch has received routes from the VPN instance on a DCGW and the next hop is the anycast VTEP address. The following uses the command output on Leaf 1 as an example.

```
[~Leaf1] display ip routing-table vpn-instance vpn1
Proto: Protocol      Pre: Preference
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
Routing Table : vpn1
Destinations : 16      Routes : 18

Destination/Mask  Proto  Pre  Cost    Flags NextHop      Interface
-----
0.0.0.0/0        IBGP   255  0        RD 11.11.11.11  VXLAN
10.1.1.0/24       Direct 0    0        D 10.1.1.1    Vbdif10
10.1.1.1/32       Direct 0    0        D 127.0.0.1   Vbdif10
10.1.1.255/32     Direct 0    0        D 127.0.0.1   Vbdif10
10.2.1.0/24       Direct 0    0        D 10.2.1.1    Vbdif20
10.2.1.1/32       Direct 0    0        D 127.0.0.1   Vbdif20
10.2.1.255/32     Direct 0    0        D 127.0.0.1   Vbdif20
10.3.1.0/24       Direct 0    0        D 10.3.1.1    Vbdif30
10.3.1.1/32       Direct 0    0        D 127.0.0.1   Vbdif30
10.3.1.255/32     Direct 0    0        D 127.0.0.1   Vbdif30
10.4.1.0/24       Direct 0    0        D 10.4.1.1    Vbdif40
10.4.1.1/32       Direct 0    0        D 127.0.0.1   Vbdif40
10.4.1.255/32     Direct 0    0        D 127.0.0.1   Vbdif40
```

```
201.1.1.1/32 Static 255 0 RD 10.1.1.2 Vbdf10
Static 255 0 RD 10.2.1.2 Vbdf20
202.1.1.1/32 IBGP 255 0 RD 10.3.1.2 Vbdf30
IBGP 255 0 RD 10.4.1.2 Vbdf40
255.255.255.255/32 Direct 0 0 D 127.0.0.1 InLoopBack0
```

On a DCGW, you can view ARP information about IPUs. The information is not provided here.

Configuration Files

- Spine1 configuration file

```
#
sysname Spine1
#
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800 and
CE16800 equipped with A series cards.
#
evpn-overlay enable
#
bfd
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.1.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.2.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/3
undo portswitch
ip address 192.168.3.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/4
undo portswitch
ip address 192.168.4.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/5
undo portswitch
ip address 192.168.9.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
#
interface 10GE1/0/6
undo portswitch
ip address 192.168.10.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
#
interface LoopBack0
ip address 7.7.7.7 255.255.255.255
#
```

```
bgp 100
router-id 7.7.7.7
peer 1.1.1.1 as-number 100
peer 1.1.1.1 connect-interface LoopBack0
peer 2.2.2.2 as-number 100
peer 2.2.2.2 connect-interface LoopBack0
peer 3.3.3.3 as-number 100
peer 3.3.3.3 connect-interface LoopBack0
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack0
peer 9.9.9.9 as-number 100
peer 9.9.9.9 connect-interface LoopBack0
peer 10.10.10.10 as-number 100
peer 10.10.10.10 connect-interface LoopBack0
#
ipv4-family unicast
peer 1.1.1.1 enable
peer 2.2.2.2 enable
peer 3.3.3.3 enable
peer 4.4.4.4 enable
peer 9.9.9.9 enable
peer 10.10.10.10 enable
#
l2vpn-family evpn
undo policy vpn-target
bestroute add-path path-number 64
peer 1.1.1.1 enable
peer 1.1.1.1 advertise irb
peer 1.1.1.1 advertise irbv6
peer 1.1.1.1 reflect-client
peer 1.1.1.1 capability-advertise add-path both
peer 1.1.1.1 advertise add-path path-number 64
peer 2.2.2.2 enable
peer 2.2.2.2 advertise irb
peer 2.2.2.2 advertise irbv6
peer 2.2.2.2 reflect-client
peer 2.2.2.2 capability-advertise add-path both
peer 2.2.2.2 advertise add-path path-number 64
peer 3.3.3.3 enable
peer 3.3.3.3 advertise irb
peer 3.3.3.3 advertise irbv6
peer 3.3.3.3 reflect-client
peer 3.3.3.3 capability-advertise add-path both
peer 3.3.3.3 advertise add-path path-number 64
peer 4.4.4.4 enable
peer 4.4.4.4 advertise irb
peer 4.4.4.4 advertise irbv6
peer 4.4.4.4 reflect-client
peer 4.4.4.4 capability-advertise add-path both
peer 4.4.4.4 advertise add-path path-number 64
peer 9.9.9.9 enable
peer 9.9.9.9 advertise irb
peer 9.9.9.9 advertise irbv6
peer 9.9.9.9 capability-advertise add-path both
peer 9.9.9.9 advertise add-path path-number 64
peer 10.10.10.10 enable
peer 10.10.10.10 advertise irb
peer 10.10.10.10 advertise irbv6
peer 10.10.10.10 capability-advertise add-path both
peer 10.10.10.10 advertise add-path path-number 64
#
ospf 1 router-id 7.7.7.7
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
spf-schedule-interval intelligent-timer 50 50 50
lsa-originate-interval intelligent-timer 500 50 100
lsa-arrival-interval intelligent-timer 50 50 50
area 0.0.0.0
network 7.7.7.7 0.0.0.0
```

```
network 192.168.1.0 0.0.0.255
network 192.168.2.0 0.0.0.255
network 192.168.3.0 0.0.0.255
network 192.168.4.0 0.0.0.255
network 192.168.9.0 0.0.0.255
network 192.168.10.0 0.0.0.255
#
return
```

- Spine2 configuration file

```
#
sysname Spine2
#
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800 and
CE16800 equipped with A series cards.
#
evpn-overlay enable
#
bfd
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.5.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.6.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/3
undo portswitch
ip address 192.168.7.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/4
undo portswitch
ip address 192.168.8.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/5
undo portswitch
ip address 192.168.11.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
#
interface 10GE1/0/6
undo portswitch
ip address 192.168.12.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
#
interface LoopBack0
ip address 8.8.8.8 255.255.255.255
#
bgp 100
router-id 8.8.8.8
peer 1.1.1.1 as-number 100
```

```
peer 1.1.1.1 connect-interface LoopBack0
peer 2.2.2.2 as-number 100
peer 2.2.2.2 connect-interface LoopBack0
peer 3.3.3.3 as-number 100
peer 3.3.3.3 connect-interface LoopBack0
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack0
peer 9.9.9.9 as-number 100
peer 9.9.9.9 connect-interface LoopBack0
peer 10.10.10.10 as-number 100
peer 10.10.10.10 connect-interface LoopBack0
#
ipv4-family unicast
peer 1.1.1.1 enable
peer 2.2.2.2 enable
peer 3.3.3.3 enable
peer 4.4.4.4 enable
peer 9.9.9.9 enable
peer 10.10.10.10 enable
#
l2vpn-family evpn
undo policy vpn-target
bestroute add-path path-number 64
peer 1.1.1.1 enable
peer 1.1.1.1 advertise irb
peer 1.1.1.1 advertise irbv6
peer 1.1.1.1 reflect-client
peer 1.1.1.1 capability-advertise add-path both
peer 1.1.1.1 advertise add-path path-number 64
peer 2.2.2.2 enable
peer 2.2.2.2 advertise irb
peer 2.2.2.2 advertise irbv6
peer 2.2.2.2 reflect-client
peer 2.2.2.2 capability-advertise add-path both
peer 2.2.2.2 advertise add-path path-number 64
peer 3.3.3.3 enable
peer 3.3.3.3 advertise irb
peer 3.3.3.3 advertise irbv6
peer 3.3.3.3 reflect-client
peer 3.3.3.3 capability-advertise add-path both
peer 3.3.3.3 advertise add-path path-number 64
peer 4.4.4.4 enable
peer 4.4.4.4 advertise irb
peer 4.4.4.4 advertise irbv6
peer 4.4.4.4 reflect-client
peer 4.4.4.4 capability-advertise add-path both
peer 4.4.4.4 advertise add-path path-number 64
peer 9.9.9.9 enable
peer 9.9.9.9 advertise irb
peer 9.9.9.9 advertise irbv6
peer 9.9.9.9 capability-advertise add-path both
peer 9.9.9.9 advertise add-path path-number 64
peer 10.10.10.10 enable
peer 10.10.10.10 advertise irb
peer 10.10.10.10 advertise irbv6
peer 10.10.10.10 capability-advertise add-path both
peer 10.10.10.10 advertise add-path path-number 64
#
ospf 1 router-id 8.8.8.8
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
spf-schedule-interval intelligent-timer 50 50 50
lsa-originate-interval intelligent-timer 500 50 100
lsa-arrival-interval intelligent-timer 50 50 50
area 0.0.0.0
network 8.8.8.8 0.0.0.0
network 192.168.5.0 0.0.0.255
network 192.168.6.0 0.0.0.255
network 192.168.7.0 0.0.0.255
```



```
network 192.168.8.0 0.0.0.255
network 192.168.11.0 0.0.0.255
network 192.168.12.0 0.0.0.255
#
return
```

- Leaf1 configuration file

```
#
sysname Leaf1
#
dfs-group 1
priority 150
source ip 1.1.1.1
m-lag up-delay 240 auto-recovery interval 10
dual-active detection delay 0
#
assign forward layer-3 resource large-overlay //This step only needs to be performed on the
CE8861EI, CE6857EI, and CE6865EI.
#
assign forward ipv6 longer-mask resource share-mode //This step only needs to be performed on the
CE8861EI, CE6857EI, and CE6865EI.
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 1.1.1.1:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 1.1.1.1:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 200
#
bfd
bfd forwarding match remote-discriminator 20
bfd forwarding match remote-discriminator 40
bfd forwarding match remote-discriminator 60
bfd forwarding match remote-discriminator 80
#
bridge-domain 10
vxlan vni 110
evpn
route-distinguisher 1.1.1.1:110
vpn-target 0:110 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:110 import-extcommunity
#
bridge-domain 20
vxlan vni 120
evpn
route-distinguisher 1.1.1.1:120
vpn-target 0:120 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:120 import-extcommunity
#
bridge-domain 30
vxlan vni 130
evpn
route-distinguisher 1.1.1.1:130
vpn-target 0:130 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:130 import-extcommunity
#
bridge-domain 40
vxlan vni 140
```

```
evpn
 route-distinguisher 1.1.1.1:140
 vpn-target 0:140 export-extcommunity
 vpn-target 0:1 export-extcommunity
 vpn-target 0:140 import-extcommunity
#
interface Vbdif10
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.1.1.1 255.255.255.0
 ipv6 address FC00:1::1/64
 arp generate-rd-table enable
 arp broadcast-detect enable
 mac-address 0000-5e00-0101
 ipv6 nd collect host enable
 ipv6 nd generate-rd-table enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Vbdif20
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.2.1.1 255.255.255.0
 ipv6 address FC00:2::1/64
 arp generate-rd-table enable
 arp broadcast-detect enable
 mac-address 0000-5e00-0102
 ipv6 nd collect host enable
 ipv6 nd generate-rd-table enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Vbdif30
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.3.1.1 255.255.255.0
 ipv6 address FC00:3::1/64
 arp generate-rd-table enable
 arp broadcast-detect enable
 mac-address 0000-5e00-0103
 ipv6 nd collect host enable
 ipv6 nd generate-rd-table enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Vbdif40
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.4.1.1 255.255.255.0
 ipv6 address FC00:4::1/64
 arp generate-rd-table enable
 arp broadcast-detect enable
 mac-address 0000-5e00-0104
 ipv6 nd collect host enable
 ipv6 nd generate-rd-table enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Eth-Trunk1
 mode lacp-static
 peer-link 1
#
interface Eth-Trunk10
 stp edged-port enable
 mode lacp-static
 lacp timeout fast
 dfs-group 1 m-lag 1
 arp anti-attack rate-limit 200
#
```

```
interface Eth-Trunk10.10 mode l2
 encapsulation dot1q vid 10
 bridge-domain 10
#
interface Eth-Trunk11
 stp edged-port enable
 mode lacp-static
 lacp timeout fast
 dfs-group 1 m-lag 2
 arp anti-attack rate-limit 200
#
interface Eth-Trunk11.20 mode l2
 encapsulation dot1q vid 20
 bridge-domain 20
#
interface 10GE1/0/1
 undo portswitch
 ip address 192.168.1.2 255.255.255.0
 ospf network-type p2p
 ospf peer hold-max-cost timer 800000
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/2
 undo portswitch
 ip address 192.168.5.2 255.255.255.0
 ospf network-type p2p
 ospf peer hold-max-cost timer 800000
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/3
 eth-trunk 1
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/4
 eth-trunk 1
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/5
 eth-trunk 10
 storm suppression unknown-unicast 5
 storm suppression multicast packets 1000
 storm suppression broadcast packets 1000
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/6
 eth-trunk 11
 storm suppression unknown-unicast 5
 storm suppression multicast packets 1000
 storm suppression broadcast packets 1000
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface LoopBack0
 ip address 1.1.1.1 255.255.255.255
#
interface LoopBack1
 ip address 5.5.5.5 255.255.255.255
#
interface Nve1
 source 5.5.5.5
 vni 110 head-end peer-list protocol bgp
 vni 120 head-end peer-list protocol bgp
 vni 130 head-end peer-list protocol bgp
 vni 140 head-end peer-list protocol bgp
```

```
mac-address 0000-5e00-0111
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk11 downlink 2
timer recover-time 60
#
bfd toipu1_v4 bind peer-ip 10.1.1.2 vpn-instance vpn1 interface Vbdif10 source-ip 10.1.1.1 one-arm-echo
discriminator local 10
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu1_v6 bind peer-ipv6 FC00:1::2 vpn-instance vpn1 interface Vbdif10 source-ipv6 FC00:1::1 one-arm-echo
discriminator local 50
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu2_v4 bind peer-ip 10.2.1.2 vpn-instance vpn1 interface Vbdif20 source-ip 10.2.1.1 one-arm-echo
discriminator local 30
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu2_v6 bind peer-ipv6 FC00:2::2 vpn-instance vpn1 interface Vbdif20 source-ipv6 FC00:2::1 one-arm-echo
discriminator local 70
detect-multiplier 4
min-echo-rx-interval 500
#
bgp 100
router-id 1.1.1.1
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack0
peer 8.8.8.8 as-number 100
peer 8.8.8.8 connect-interface LoopBack0
#
ipv4-family unicast
peer 7.7.7.7 enable
peer 8.8.8.8 enable
#
ipv4-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing 64
irb asymmetric
advertise l2vpn evpn import-route-multipath
#
ipv6-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing 64
irb asymmetric
advertise l2vpn evpn import-route-multipath
#
l2vpn-family evpn
policy vpn-target
bestroute add-path path-number 64
peer 7.7.7.7 enable
peer 7.7.7.7 route-policy export_policy export
peer 7.7.7.7 advertise irb
peer 7.7.7.7 advertise irbv6
peer 7.7.7.7 capability-advertise add-path both
peer 7.7.7.7 advertise add-path path-number 64
peer 8.8.8.8 enable
peer 8.8.8.8 route-policy export_policy export
peer 8.8.8.8 advertise irb
peer 8.8.8.8 advertise irbv6
```

```
peer 8.8.8.8 capability-advertise add-path both
peer 8.8.8.8 advertise add-path path-number 64
#
ospf 1 router-id 1.1.1.1
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
spf-schedule-interval intelligent-timer 50 50 50
lsa-originate-interval intelligent-timer 500 50 100
lsa-arrival-interval intelligent-timer 50 50 50
area 0.0.0.0
network 1.1.1.1 0.0.0.0
network 5.5.5.5 0.0.0.0
network 192.168.1.0 0.0.0.255
network 192.168.5.0 0.0.0.255
#
route-policy export_policy deny node 10
if-match community-filter suppress_route
#
route-policy export_policy permit node 20
if-match community-filter apply_gwip_route
apply gateway-ip origin-nexthop
apply ipv6 gateway-ip origin-nexthop
#
route-policy export_policy permit node 999
#
route-policy import_static_policy permit node 100
if-match tag 82345
apply community 82345 additive
#
route-policy import_static_policy permit node 999
#
ip community-filter basic apply_gwip_route index 10 permit 82345
ip community-filter basic suppress_route index 10 permit 82346
#
ip route-static vpn-instance vpn1 201.1.1.1 255.255.255.255 10.1.1.2 preference 255 tag 82345 track
bfd-session toipu1_v4 inter-protocol-ecmp
ip route-static vpn-instance vpn1 201.1.1.1 255.255.255.255 10.2.1.2 preference 255 tag 82345 track
bfd-session toipu2_v4 inter-protocol-ecmp
#
ipv6 route-static vpn-instance vpn1 201::1 128 FC00:1::2 preference 255 tag 82345 track bfd-session
toipu1_v6 inter-protocol-ecmp
ipv6 route-static vpn-instance vpn1 201::1 128 FC00:2::2 preference 255 tag 82345 track bfd-session
toipu2_v6 inter-protocol-ecmp
#
return
```

- Leaf2 configuration file

```
#
sysname Leaf2
#
dfs-group 1
source ip 2.2.2.2
m-lag up-delay 240 auto-recovery interval 10
dual-active detection delay 0
#
assign forward layer-3 resource large-overlay //This step only needs to be performed on the
CE8861EI, CE6857EI, and CE6865EI.
#
assign forward ipv6 longer-mask resource share-mode //This step only needs to be performed on the
CE8861EI, CE6857EI, and CE6865EI.
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 2.2.2.2:1
vpn-target 0:1 export-extcommunity evpn
```

```
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 2.2.2.2:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 200
#
bfd
bfd forwarding match remote-discriminator 10
bfd forwarding match remote-discriminator 30
bfd forwarding match remote-discriminator 50
bfd forwarding match remote-discriminator 70
#
bridge-domain 10
vxlan vni 110
evpn
route-distinguisher 2.2.2.2:110
vpn-target 0:110 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:110 import-extcommunity
#
bridge-domain 20
vxlan vni 120
evpn
route-distinguisher 2.2.2.2:120
vpn-target 0:120 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:120 import-extcommunity
#
bridge-domain 30
vxlan vni 130
evpn
route-distinguisher 2.2.2.2:130
vpn-target 0:130 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:130 import-extcommunity
#
bridge-domain 40
vxlan vni 140
evpn
route-distinguisher 2.2.2.2:140
vpn-target 0:140 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:140 import-extcommunity
#
interface Vbdif10
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.1.1.1 255.255.255.0
ipv6 address FC00:1::1/64
arp generate-rd-table enable
arp broadcast-detect enable
mac-address 0000-5e00-0101
ipv6 nd collect host enable
ipv6 nd generate-rd-table enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Vbdif20
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.2.1.1 255.255.255.0
ipv6 address FC00:2::1/64
arp generate-rd-table enable
arp broadcast-detect enable
mac-address 0000-5e00-0102
ipv6 nd collect host enable
ipv6 nd generate-rd-table enable
vxlan anycast-gateway enable
```

```
arp collect host enable
#
interface Vbdf30
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.3.1.1 255.255.255.0
ipv6 address FC00:3::1/64
arp generate-rd-table enable
arp broadcast-detect enable
mac-address 0000-5e00-0103
ipv6 nd collect host enable
ipv6 nd generate-rd-table enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Vbdf40
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.4.1.1 255.255.255.0
ipv6 address FC00:4::1/64
arp generate-rd-table enable
arp broadcast-detect enable
mac-address 0000-5e00-0104
ipv6 nd collect host enable
ipv6 nd generate-rd-table enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk10
stp edged-port enable
mode lacp-static
lacp timeout fast
dfs-group 1 m-lag 1
arp anti-attack rate-limit 200
#
interface Eth-Trunk10.10 mode l2
encapsulation dot1q vid 10
bridge-domain 10
#
interface Eth-Trunk11
stp edged-port enable
mode lacp-static
lacp timeout fast
dfs-group 1 m-lag 2
arp anti-attack rate-limit 200
#
interface Eth-Trunk11.20 mode l2
encapsulation dot1q vid 20
bridge-domain 20
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.2.2 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 800000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.6.2 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 800000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
```

```
#
interface 10GE1/0/3
eth-trunk 1
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/4
eth-trunk 1
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/5
eth-trunk 10
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/6
eth-trunk 11
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface LoopBack0
ip address 2.2.2.2 255.255.255.255
#
interface LoopBack1
ip address 5.5.5.5 255.255.255.255
#
interface Nve1
source 5.5.5.5
vni 110 head-end peer-list protocol bgp
vni 120 head-end peer-list protocol bgp
vni 130 head-end peer-list protocol bgp
vni 140 head-end peer-list protocol bgp
mac-address 0000-5e00-0111
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk11 downlink 2
timer recover-time 60
#
bfd toipu1_v4 bind peer-ip 10.1.1.2 vpn-instance vpn1 interface Vbdif10 source-ip 10.1.1.1 one-arm-echo
discriminator local 20
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu1_v6 bind peer-ipv6 FC00:1::2 vpn-instance vpn1 interface Vbdif10 source-ipv6 FC00:1::1 one-arm-echo
discriminator local 60
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu2_v4 bind peer-ip 10.2.1.2 vpn-instance vpn1 interface Vbdif20 source-ip 10.2.1.1 one-arm-echo
discriminator local 40
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu2_v6 bind peer-ipv6 FC00:2::2 vpn-instance vpn1 interface Vbdif20 source-ipv6 FC00:2::1 one-arm-echo
discriminator local 80
```



```
detect-multiplier 4
min-echo-rx-interval 500
#
bgp 100
router-id 2.2.2.2
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack0
peer 8.8.8.8 as-number 100
peer 8.8.8.8 connect-interface LoopBack0
#
ipv4-family unicast
peer 7.7.7.7 enable
peer 8.8.8.8 enable
#
ipv4-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing 64
irb asymmetric
advertise l2vpn evpn import-route-multipath
#
ipv6-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing 64
irb asymmetric
advertise l2vpn evpn import-route-multipath
#
l2vpn-family evpn
policy vpn-target
bestroute add-path path-number 64
peer 7.7.7.7 enable
peer 7.7.7.7 route-policy export_policy export
peer 7.7.7.7 advertise irb
peer 7.7.7.7 advertise irbv6
peer 7.7.7.7 capability-advertise add-path both
peer 7.7.7.7 advertise add-path path-number 64
peer 8.8.8.8 enable
peer 8.8.8.8 route-policy export_policy export
peer 8.8.8.8 advertise irb
peer 8.8.8.8 advertise irbv6
peer 8.8.8.8 capability-advertise add-path both
peer 8.8.8.8 advertise add-path path-number 64
#
ospf 1 router-id 2.2.2.2
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
spf-schedule-interval intelligent-timer 50 50 50
lsa-originate-interval intelligent-timer 500 50 100
lsa-arrival-interval intelligent-timer 50 50 50
area 0.0.0.0
network 2.2.2.2 0.0.0.0
network 5.5.5.5 0.0.0.0
network 192.168.2.0 0.0.0.255
network 192.168.6.0 0.0.0.255
#
route-policy export_policy deny node 10
if-match community-filter suppress_route
#
route-policy export_policy permit node 20
if-match community-filter apply_gwip_route
apply gateway-ip origin-nexthop
apply ipv6 gateway-ip origin-nexthop
#
route-policy export_policy permit node 999
#
route-policy import_static_policy permit node 100
if-match tag 82345
apply community 82345 additive
#
route-policy import_static_policy permit node 999
```

```
#
ip community-filter basic apply_gwip_route index 10 permit 82345
ip community-filter basic suppress_route index 10 permit 82346
#
ip route-static vpn-instance vpn1 201.1.1.1 255.255.255.255 10.1.1.2 preference 255 tag 82345 track
bfd-session toipu1_v4 inter-protocol-ecmp
ip route-static vpn-instance vpn1 201.1.1.1 255.255.255.255 10.2.1.2 preference 255 tag 82345 track
bfd-session toipu2_v4 inter-protocol-ecmp
#
ipv6 route-static vpn-instance vpn1 201::1 128 FC00:1::2 preference 255 tag 82345 track bfd-session
toipu1_v6 inter-protocol-ecmp
ipv6 route-static vpn-instance vpn1 201::1 128 FC00:2::2 preference 255 tag 82345 track bfd-session
toipu2_v6 inter-protocol-ecmp
#
return
```

- Leaf3 configuration file

```
#
sysname Leaf3
#
dfs-group 1
priority 150
source ip 3.3.3.3
m-lag up-delay 240 auto-recovery interval 10
dual-active detection delay 0
#
assign forward layer-3 resource large-overlay //This step only needs to be performed on the
CE8861EI, CE6857EI, and CE6865EI.
#
assign forward ipv6 longer-mask resource share-mode //This step only needs to be performed on the
CE8861EI, CE6857EI, and CE6865EI.
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 3.3.3.3:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 1.1.1.1:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 200
#
bfd
bfd forwarding match remote-discriminator 20
bfd forwarding match remote-discriminator 40
bfd forwarding match remote-discriminator 60
bfd forwarding match remote-discriminator 80
#
bridge-domain 10
vxlan vni 110
evpn
route-distinguisher 3.3.3.3:110
vpn-target 0:110 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:110 import-extcommunity
#
bridge-domain 20
vxlan vni 120
evpn
route-distinguisher 3.3.3.3:120
vpn-target 0:120 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:120 import-extcommunity
#
```

```
bridge-domain 30
vxlan vni 130
evpn
 route-distinguisher 3.3.3.3:130
 vpn-target 0:130 export-extcommunity
 vpn-target 0:1 export-extcommunity
 vpn-target 0:130 import-extcommunity
#
bridge-domain 40
vxlan vni 140
evpn
 route-distinguisher 3.3.3.3:140
 vpn-target 0:140 export-extcommunity
 vpn-target 0:1 export-extcommunity
 vpn-target 0:140 import-extcommunity
#
interface Vbdif10
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.1.1.1 255.255.255.0
 ipv6 address FC00:1::1/64
 arp generate-rd-table enable
 arp broadcast-detect enable
 mac-address 0000-5e00-0101
 ipv6 nd collect host enable
 ipv6 nd generate-rd-table enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Vbdif20
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.2.1.1 255.255.255.0
 ipv6 address FC00:2::1/64
 arp generate-rd-table enable
 arp broadcast-detect enable
 mac-address 0000-5e00-0102
 ipv6 nd collect host enable
 ipv6 nd generate-rd-table enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Vbdif30
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.3.1.1 255.255.255.0
 ipv6 address FC00:3::1/64
 arp generate-rd-table enable
 arp broadcast-detect enable
 mac-address 0000-5e00-0103
 ipv6 nd collect host enable
 ipv6 nd generate-rd-table enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Vbdif40
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.4.1.1 255.255.255.0
 ipv6 address FC00:4::1/64
 arp generate-rd-table enable
 arp broadcast-detect enable
 mac-address 0000-5e00-0104
 ipv6 nd collect host enable
 ipv6 nd generate-rd-table enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Eth-Trunk1
```

```
mode lacp-static
peer-link 1
#
interface Eth-Trunk10
stp edged-port enable
mode lacp-static
lacp timeout fast
dfs-group 1 m-lag 1
arp anti-attack rate-limit 200
#
interface Eth-Trunk10.30 mode l2
encapsulation dot1q vid 30
bridge-domain 30
#
interface Eth-Trunk11
stp edged-port enable
mode lacp-static
lacp timeout fast
dfs-group 1 m-lag 2
arp anti-attack rate-limit 200
#
interface Eth-Trunk11.40 mode l2
encapsulation dot1q vid 40
bridge-domain 40
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.3.2 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 800000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.7.2 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 800000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/3
eth-trunk 1
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/4
eth-trunk 1
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/5
eth-trunk 10
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/6
eth-trunk 11
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface LoopBack0
ip address 3.3.3.3 255.255.255.255
```

```
#
interface LoopBack1
ip address 6.6.6.6 255.255.255.255
#
interface Nve1
source 6.6.6.6
vni 110 head-end peer-list protocol bgp
vni 120 head-end peer-list protocol bgp
vni 130 head-end peer-list protocol bgp
vni 140 head-end peer-list protocol bgp
mac-address 0000-5e00-0112
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk11 downlink 2
timer recover-time 60
#
bfd toipu3_v4 bind peer-ip 10.3.1.2 vpn-instance vpn1 interface Vbdif30 source-ip 10.3.1.1 one-arm-echo
discriminator local 10
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu3_v6 bind peer-ipv6 FC00:3::2 vpn-instance vpn1 interface Vbdif30 source-ipv6 FC00:3::1 one-arm-echo
discriminator local 50
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu4_v4 bind peer-ip 10.4.1.2 vpn-instance vpn1 interface Vbdif40 source-ip 10.4.1.1 one-arm-echo
discriminator local 30
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu4_v6 bind peer-ipv6 FC00:4::2 vpn-instance vpn1 interface Vbdif40 source-ipv6 FC00:4::1 one-arm-echo
discriminator local 70
detect-multiplier 4
min-echo-rx-interval 500
#
bgp 100
router-id 3.3.3.3
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack0
peer 8.8.8.8 as-number 100
peer 8.8.8.8 connect-interface LoopBack0
#
ipv4-family unicast
peer 7.7.7.7 enable
peer 8.8.8.8 enable
#
ipv4-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing 64
irb asymmetric
advertise l2vpn evpn import-route-multipath
#
ipv6-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing 64
irb asymmetric
advertise l2vpn evpn import-route-multipath
#
l2vpn-family evpn
policy vpn-target
bestroute add-path path-number 64
```

```
peer 7.7.7.7 enable
peer 7.7.7.7 route-policy export_policy export
peer 7.7.7.7 advertise irb
peer 7.7.7.7 advertise irbv6
peer 7.7.7.7 capability-advertise add-path both
peer 7.7.7.7 advertise add-path path-number 64
peer 8.8.8.8 enable
peer 8.8.8.8 route-policy export_policy export
peer 8.8.8.8 advertise irb
peer 8.8.8.8 advertise irbv6
peer 8.8.8.8 capability-advertise add-path both
peer 8.8.8.8 advertise add-path path-number 64
#
ospf 1 router-id 3.3.3.3
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
spf-schedule-interval intelligent-timer 50 50 50
lsa-originate-interval intelligent-timer 500 50 100
lsa-arrival-interval intelligent-timer 50 50 50
area 0.0.0.0
network 3.3.3.3 0.0.0.0
network 6.6.6.6 0.0.0.0
network 192.168.3.0 0.0.0.255
network 192.168.7.0 0.0.0.255
#
route-policy export_policy deny node 10
if-match community-filter suppress_route
#
route-policy export_policy permit node 20
if-match community-filter apply_gwip_route
apply gateway-ip origin-nexthop
apply ipv6 gateway-ip origin-nexthop
#
route-policy export_policy permit node 999
#
route-policy import_static_policy permit node 100
if-match tag 82345
apply community 82345 additive
#
route-policy import_static_policy permit node 999
#
ip community-filter basic apply_gwip_route index 10 permit 82345
ip community-filter basic suppress_route index 10 permit 82346
#
ip route-static vpn-instance vpn1 202.1.1.1 255.255.255.255 10.3.1.2 preference 255 tag 82345 track
bfd-session toipu3_v4 inter-protocol-ecmp
ip route-static vpn-instance vpn1 202.1.1.1 255.255.255.255 10.4.1.2 preference 255 tag 82345 track
bfd-session toipu4_v4 inter-protocol-ecmp
#
ipv6 route-static vpn-instance vpn1 202::1 128 FC00:3::2 preference 255 tag 82345 track bfd-session
toipu3_v6 inter-protocol-ecmp
ipv6 route-static vpn-instance vpn1 202::1 128 FC00:4::2 preference 255 tag 82345 track bfd-session
toipu4_v6 inter-protocol-ecmp
#
return
```

- Leaf4 configuration file

```
#
sysname Leaf4
#
dfs-group 1
source ip 4.4.4.4
m-lag up-delay 240 auto-recovery interval 10
dual-active detection delay 0
#
assign forward layer-3 resource large-overlay //This step only needs to be performed on the
CE8861EI, CE6857EI, and CE6865EI.
#
assign forward ipv6 longer-mask resource share-mode //This step only needs to be performed on the
CE8861EI, CE6857EI, and CE6865EI.
```

```
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
  ipv4-family
    route-distinguisher 4.4.4.4:1
    vpn-target 0:1 export-extcommunity evpn
    vpn-target 0:1 import-extcommunity evpn
  ipv6-family
    route-distinguisher 1.1.1.1:1
    vpn-target 0:1 export-extcommunity evpn
    vpn-target 0:1 import-extcommunity evpn
  vxlan vni 200
#
bfd
bfd forwarding match remote-discriminator 10
bfd forwarding match remote-discriminator 30
bfd forwarding match remote-discriminator 50
bfd forwarding match remote-discriminator 70
#
bridge-domain 10
  vxlan vni 110
  evpn
    route-distinguisher 4.4.4.4:110
    vpn-target 0:110 export-extcommunity
    vpn-target 0:1 export-extcommunity
    vpn-target 0:110 import-extcommunity
#
bridge-domain 20
  vxlan vni 120
  evpn
    route-distinguisher 4.4.4.4:120
    vpn-target 0:120 export-extcommunity
    vpn-target 0:1 export-extcommunity
    vpn-target 0:120 import-extcommunity
#
bridge-domain 30
  vxlan vni 130
  evpn
    route-distinguisher 4.4.4.4:130
    vpn-target 0:130 export-extcommunity
    vpn-target 0:1 export-extcommunity
    vpn-target 0:130 import-extcommunity
#
bridge-domain 40
  vxlan vni 140
  evpn
    route-distinguisher 4.4.4.4:140
    vpn-target 0:140 export-extcommunity
    vpn-target 0:1 export-extcommunity
    vpn-target 0:140 import-extcommunity
#
interface Vbdif10
  ip binding vpn-instance vpn1
  ipv6 enable
  ip address 10.1.1.1 255.255.255.0
  ipv6 address FC00:1::1/64
  arp generate-rd-table enable
  arp broadcast-detect enable
  mac-address 0000-5e00-0101
  ipv6 nd collect-host enable
  ipv6 nd generate-rd-table enable
  vxlan anycast-gateway enable
  arp collect-host enable
#
interface Vbdif20
```

```
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.2.1.1 255.255.255.0
ipv6 address FC00:2::1/64
arp generate-rd-table enable
arp broadcast-detect enable
mac-address 0000-5e00-0102
ipv6 nd collect host enable
ipv6 nd generate-rd-table enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Vbdf30
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.3.1.1 255.255.255.0
ipv6 address FC00:3::1/64
arp generate-rd-table enable
arp broadcast-detect enable
mac-address 0000-5e00-0103
ipv6 nd collect host enable
ipv6 nd generate-rd-table enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Vbdf40
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.4.1.1 255.255.255.0
ipv6 address FC00:4::1/64
arp generate-rd-table enable
arp broadcast-detect enable
mac-address 0000-5e00-0104
ipv6 nd collect host enable
ipv6 nd generate-rd-table enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk10
stp edged-port enable
mode lacp-static
lacp timeout fast
dfs-group 1 m-lag 1
arp anti-attack rate-limit 200
#
interface Eth-Trunk10.30 mode l2
encapsulation dot1q vid 30
bridge-domain 30
#
interface Eth-Trunk11
stp edged-port enable
mode lacp-static
lacp timeout fast
dfs-group 1 m-lag 2
arp anti-attack rate-limit 200
#
interface Eth-Trunk11.40 mode l2
encapsulation dot1q vid 40
bridge-domain 40
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.4.2 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 800000
```



```
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.8.2 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 800000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/3
eth-trunk 1
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/4
eth-trunk 1
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/5
eth-trunk 10
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/6
eth-trunk 11
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface LoopBack0
ip address 4.4.4.4 255.255.255.255
#
interface LoopBack1
ip address 6.6.6.6 255.255.255.255
#
interface Nve1
source 6.6.6.6
vni 110 head-end peer-list protocol bgp
vni 120 head-end peer-list protocol bgp
vni 130 head-end peer-list protocol bgp
vni 140 head-end peer-list protocol bgp
mac-address 0000-5e00-0112
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk11 downlink 2
timer recover-time 60
#
bfd toipu3_v4 bind peer-ip 10.3.1.2 vpn-instance vpn1 interface Vbdif30 source-ip 10.3.1.1 one-arm-echo
discriminator local 20
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu3_v6 bind peer-ipv6 FC00:3::2 vpn-instance vpn1 interface Vbdif30 source-ipv6 FC00:3::1 one-arm-echo
discriminator local 60
detect-multiplier 4
min-echo-rx-interval 500
```

```
#
bfd toipu4_v4 bind peer-ip 10.4.1.2 vpn-instance vpn1 interface Vbdif40 source-ip 10.4.1.1 one-arm-
echo
discriminator local 40
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu4_v6 bind peer-ipv6 FC00:4::2 vpn-instance vpn1 interface Vbdif40 source-ipv6 FC00:4::1 one-
arm-echo
discriminator local 80
detect-multiplier 4
min-echo-rx-interval 500
#
bgp 100
router-id 4.4.4.4
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack0
peer 8.8.8.8 as-number 100
peer 8.8.8.8 connect-interface LoopBack0
#
ipv4-family unicast
peer 7.7.7.7 enable
peer 8.8.8.8 enable
#
ipv4-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing 64
irb asymmetric
advertise l2vpn evpn import-route-multipath
#
ipv6-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing 64
irb asymmetric
advertise l2vpn evpn import-route-multipath
#
l2vpn-family evpn
policy vpn-target
bestroute add-path path-number 64
peer 7.7.7.7 enable
peer 7.7.7.7 route-policy export_policy export
peer 7.7.7.7 advertise irb
peer 7.7.7.7 advertise irbv6
peer 7.7.7.7 capability-advertise add-path both
peer 7.7.7.7 advertise add-path path-number 64
peer 8.8.8.8 enable
peer 8.8.8.8 route-policy export_policy export
peer 8.8.8.8 advertise irb
peer 8.8.8.8 advertise irbv6
peer 8.8.8.8 capability-advertise add-path both
peer 8.8.8.8 advertise add-path path-number 64
#
ospf 1 router-id 4.4.4.4
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
spf-schedule-interval intelligent-timer 50 50 50
lsa-originate-interval intelligent-timer 500 50 100
lsa-arrival-interval intelligent-timer 50 50 50
area 0.0.0.0
network 4.4.4.4 0.0.0.0
network 6.6.6.6 0.0.0.0
network 192.168.4.0 0.0.0.255
network 192.168.8.0 0.0.0.255
#
route-policy export_policy deny node 10
if-match community-filter suppress_route
#
route-policy export_policy permit node 20
if-match community-filter apply_gwip_route
```

```
apply gateway-ip origin-nexthop
apply ipv6 gateway-ip origin-nexthop
#
route-policy export_policy permit node 999
#
route-policy import_static_policy permit node 100
if-match tag 82345
apply community 82345 additive
#
route-policy import_static_policy permit node 999
#
ip community-filter basic apply_gwip_route index 10 permit 82345
ip community-filter basic suppress_route index 10 permit 82346
#
ip route-static vpn-instance vpn1 202.1.1.1 255.255.255.255 10.3.1.2 preference 255 tag 82345 track
bfd-session toipu3_v4 inter-protocol-ecmp
ip route-static vpn-instance vpn1 202.1.1.1 255.255.255.255 10.4.1.2 preference 255 tag 82345 track
bfd-session toipu4_v4 inter-protocol-ecmp
#
ipv6 route-static vpn-instance vpn1 202::1 128 FC00:3::2 preference 255 tag 82345 track bfd-session
toipu3_v6 inter-protocol-ecmp
ipv6 route-static vpn-instance vpn1 202::1 128 FC00:4::2 preference 255 tag 82345 track bfd-session
toipu4_v6 inter-protocol-ecmp
#
return
```

1.9 Example for Configuring NFVI Distributed Gateways (Symmetric Mode)

Applicable Products and Versions

- V200R020C10:
 - The CE6857F, CE6857S, CE6865EI, or CE8861EI can be deployed as a leaf switch on the network. The CE12800 can be deployed as a spine switch on the network.
 - The CE6881 or CE6863 can be deployed as a leaf switch on the network. The CE16800 can be deployed as a spine switch on the network.
- V200R021C00: The CE6881, CE6863, CE6857F, CE6857S, CE6865E, or CE8861EI can be deployed as a leaf switch on the network. The CE16800 can be deployed as a spine switch on the network.
- V200R021C10, later versions: The CE6863E, CE6865E, CE6857F, CE6857S, or CE6881-48S6CQ can be deployed as a leaf switch on the network. The CE16800 equipped with G/GK series board can be deployed as a spine switch on the network.

Networking Requirements

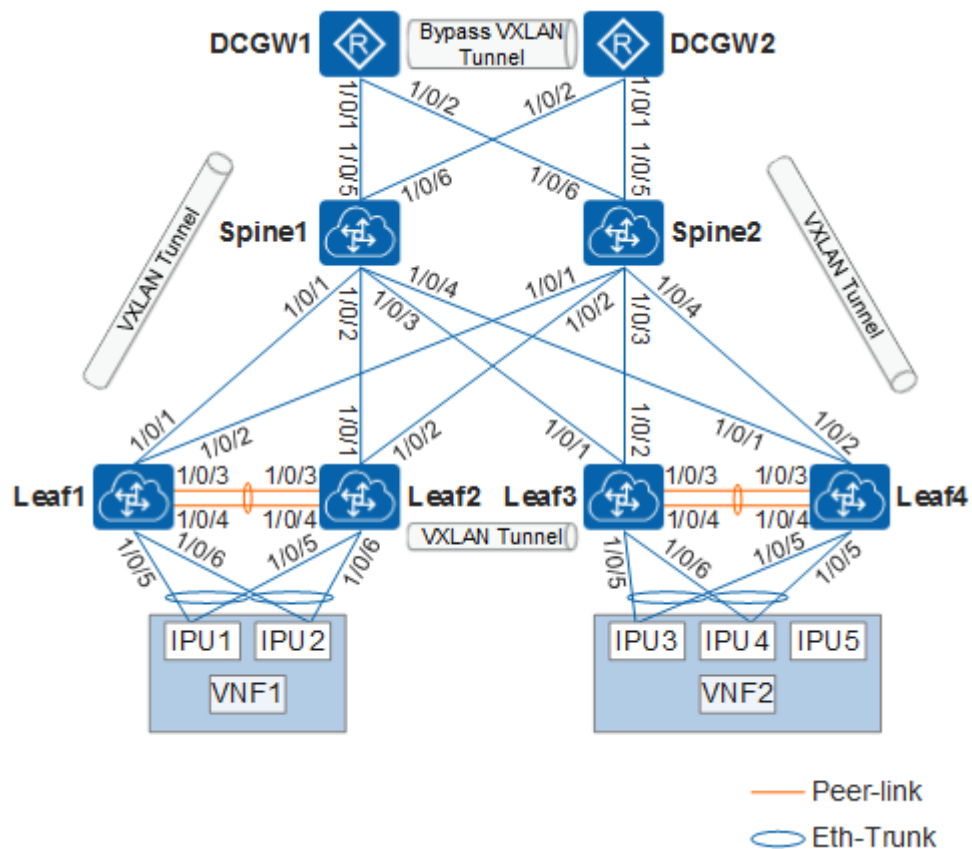
Huawei's NFVI telco cloud solution incorporates DCI and DCN solutions. A large volume of mobile phone traffic enters the DCN and accesses the vUGW and vMSE on the DCN. After being processed by the vUGW and vMSE, the mobile phone traffic is forwarded over the Internet through the DCN to the destination devices. Similarly, response traffic sent over the Internet from the destination devices to the mobile phones also undergoes this process. To meet the preceding requirements and ensure that the mobile phone traffic is load balanced within the DCN, you need to deploy the NFVI distributed gateway function on devices in the DCN.

In **Figure 1-9**, DCGW1 and DCGW2 are the DCN's border gateways, and they can exchange Internet routes with devices on the external network. Leaf switches provide network access for Virtualized Network Functions (VNFs). As virtualized NEs, VNF1 and VNF2 can be deployed separately to implement the functions of the vUGW and vMSE. VNF1 and VNF2 are connected to leaf switches through interface processing units (IPUs).

This networking combines the distributed gateway function and the VXLAN active-active gateway function:

- The VXLAN active-active gateway function is configured on DCGW1 and DCGW2. Specifically, a bypass VXLAN tunnel is established between DCGW1 and DCGW2. In addition, they use a virtual anycast VTEP address to establish VXLAN tunnels with leaf switches.
- Spine switches transparently transmit packets and connect to DCGW devices and leaf switches.
- The distributed gateway function is configured on leaf switches and a VXLAN tunnel is established between the two M-LAG systems.

Figure 1-9 NFVI distributed gateway networking



NOTE

In the preceding figure, 1/0/1 indicates an interface number and the interface rate is 10 Gbit/s. That is, 1/0/1 indicates 10GE1/0/1. Other interfaces are specified in a similar way.

Table 1-12 IP addresses of interfaces

Device Name	Interface	IP Address	Device Name	Interface	IP Address
DCG W1	10GE1/0/1	192.168.9.2/24	DCG W2	10GE1/0/1	192.168.11.2/24
	10GE1/0/2	192.168.12.2/24		10GE1/0/2	192.168.10.2/24
	Loopback0	9.9.9.9/32		Loopback0	10.10.10.10/32
	Loopback1	11.11.11.11/32		Loopback1	11.11.11.11/32
Spine 1	10GE1/0/1	192.168.1.1/24	Spine 2	10GE1/0/1	192.168.5.1/24
	10GE1/0/2	192.168.2.1/24		10GE1/0/2	192.168.6.1/24
	10GE1/0/3	192.168.3.1/24		10GE1/0/3	192.168.7.1/24
	10GE1/0/4	192.168.4.1/24		10GE1/0/4	192.168.8.1/24
	10GE1/0/5	192.168.9.1/24		10GE1/0/5	192.168.11.1/24
	10GE1/0/6	192.168.10.1/24		10GE1/0/6	192.168.12.1/24
	Loopback0	7.7.7.7/32		Loopback0	8.8.8.8/32
Leaf1	10GE1/0/1	192.168.1.2/24	Leaf2	10GE1/0/1	192.168.2.2/24
	10GE1/0/2	192.168.5.2/24		10GE1/0/2	192.168.6.2/24
	Loopback0	1.1.1.1/32		Loopback0	2.2.2.2/32
	Loopback1	5.5.5.5/32		Loopback1	5.5.5.5/32
Leaf3	10GE1/0/1	192.168.3.2/24	Leaf4	10GE1/0/1	192.168.4.2/24
	10GE1/0/2	192.168.7.2/24		10GE1/0/2	192.168.8.2/24
	Loopback0	3.3.3.3/32		Loopback0	4.4.4.4/32
	Loopback1	6.6.6.6/32		Loopback1	6.6.6.6/32

Device Name	Interface	IP Address	Device Name	Interface	IP Address
IPU1	-	IPv4: 10.1.1.2/24 IPv6: fc00:1::2/64	IPU2	-	IPv4: 10.2.1.2/24 IPv6: fc00:2::2/64
IPU3	-	IPv4: 10.3.1.2/24 IPv6: fc00:3::2/64	IPU4	-	IPv4: 10.4.1.2/24 IPv6: fc00:4::2/64
VNF1	-	IPv4: 201.1.1.1/32 IPv6: 201::1/128	VNF2	-	IPv4: 202.1.1.1/32 IPv6: 202::1/128

Configuration Roadmap

The configuration roadmap is as follows:

1. Configure a routing protocol to ensure Layer 3 connectivity on the network.
2. Configure M-LAG on leaf switches.
3. Configure BGP EVPN to establish VXLAN tunnels.
4. Configure static routes from leaf switches to VNFs and advertise the routes using BGP EVPN.
5. Configure load balancing among routes.

Procedure

- Step 1** Configure IP addresses for physical interfaces and loopback interfaces, and configure a routing protocol to ensure Layer 3 connectivity. OSPF is used in this example.

Configure Leaf1. The configurations of other devices are similar to the configuration of Leaf1. For details, see [Configuration Files](#).

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf1
[*HUAWEI] commit
[~Leaf1] bfd //Enable BFD
[*Leaf1-bfd] quit
[*Leaf1] interface 10ge 1/0/1
[*Leaf1-10GE1/0/1] undo portswitch
[*Leaf1-10GE1/0/1] ip address 192.168.1.2 24
[*Leaf1-10GE1/0/1] ospf network-type p2p
[*Leaf1-10GE1/0/1] ospf peer hold-max-cost timer 800000
[*Leaf1-10GE1/0/1] port crc-statistics trigger error-down
[*Leaf1-10GE1/0/1] trap-threshold crc-statistics 100 interval 10
[*Leaf1-10GE1/0/1] quit
[*Leaf1] interface 10ge 1/0/2
```

```
[*Leaf1-10GE1/0/2] undo portswitch
[*Leaf1-10GE1/0/2] ip address 192.168.5.2 24
[*Leaf1-10GE1/0/2] ospf network-type p2p
[*Leaf1-10GE1/0/2] ospf peer hold-max-cost timer 800000
[*Leaf1-10GE1/0/2] port crc-statistics trigger error-down
[*Leaf1-10GE1/0/2] trap-threshold crc-statistics 100 interval 10
[*Leaf1-10GE1/0/2] quit
[*Leaf1] interface loopback 1
[*Leaf1-LoopBack1] ip address 5.5.5.5 32
[*Leaf1-LoopBack1] quit
[*Leaf1] interface loopback 0
[*Leaf1-LoopBack0] ip address 1.1.1.1 32
[*Leaf1-LoopBack0] quit
[*Leaf1] ospf 1 router-id 1.1.1.1
[*Leaf1-ospf-1] spf-schedule-interval intelligent-timer 50 50 50
[*Leaf1-ospf-1] lsa-originate-interval intelligent-timer 500 50 100
[*Leaf1-ospf-1] lsa-arrival-interval intelligent-timer 50 50 50
[*Leaf1-ospf-1] bfd all-interfaces enable
[*Leaf1-ospf-1] bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
[*Leaf1-ospf-1] area 0
[*Leaf1-ospf-1-area-0.0.0.0] network 192.168.1.0 0.0.0.255
[*Leaf1-ospf-1-area-0.0.0.0] network 192.168.5.0 0.0.0.255
[*Leaf1-ospf-1-area-0.0.0.0] network 1.1.1.1 0.0.0.0
[*Leaf1-ospf-1-area-0.0.0.0] network 5.5.5.5 0.0.0.0
[*Leaf1-ospf-1-area-0.0.0.0] quit
[*Leaf1-ospf-1] quit
[*Leaf1] commit
```

Step 2 (For the CE8861EI, CE6857EI, CE6865E, CE8850E-32CQ-EI, and CE6865EI)
Configure the resource allocation mode on leaf switches.

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1.

```
[~Leaf1] assign forward layer-3 resource large-overlay //Set the Layer 3 resource allocation mode to
large-overlay so that the switch supports more VXLAN overlay entries.
[*Leaf1] assign forward ipv6 longer-mask resource share-mode //Set the resource allocation mode of
IPv6 addresses and routes with prefixes longer than 64 bits to shared mode. In this mode, IPv4 addresses
and routes share chip resources with IPv6 addresses and routes.
[*Leaf1] commit
```

NOTE

After configuring the resource allocation mode, save the configuration and restart the switch to make the configuration take effect.

Step 3 Configure M-LAG on leaf switches.

Configure Leaf1 and Leaf2 to establish an M-LAG. The configurations of Leaf3 and Leaf4 are similar to those of Leaf1 and Leaf2.

```
[~Leaf1] stp mode rstp
[*Leaf1] stp v-stp enable
[*Leaf1] dfs-group 1
[*Leaf1-dfs-group-1] source ip 1.1.1.1
[*Leaf1-dfs-group-1] priority 150
[*Leaf1-dfs-group-1] m-lag up-delay 240 auto-recovery interval 10 //Configure the delay for an M-LAG
member interface to report the Up event and the automatic recovery interval.
[*Leaf1-dfs-group-1] dual-active detection delay 0
[*Leaf1-dfs-group-1] quit
[*Leaf1] interface eth-trunk 1 //Configure the peer-link.
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/3
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/4
[*Leaf1-Eth-Trunk1] mode lacp-static
[*Leaf1-Eth-Trunk1] peer-link 1
[*Leaf1-Eth-Trunk1] quit
[*Leaf1] interface 10ge 1/0/3
[*Leaf1-10GE1/0/3] port crc-statistics trigger error-down
```

```
[*Leaf1-10GE1/0/3] trap-threshold crc-statistics 100 interval 10
[*Leaf1-10GE1/0/3] quit
[*Leaf1] interface 10ge 1/0/4
[*Leaf1-10GE1/0/4] port crc-statistics trigger error-down
[*Leaf1-10GE1/0/4] trap-threshold crc-statistics 100 interval 10
[*Leaf1-10GE1/0/4] quit
[*Leaf1] interface eth-trunk 10 //Configure an M-LAG member interface for connecting to a VNF.
[*Leaf1-Eth-Trunk10] trunkport 10ge 1/0/5
[*Leaf1-Eth-Trunk10] mode lacp-static
[*Leaf1-Eth-Trunk10] lacp timeout fast
[*Leaf1-Eth-Trunk10] dfs-group 1 m-lag 1
[*Leaf1-Eth-Trunk10] arp anti-attack rate-limit 200 //Set the maximum number of ARP packets that are
allowed to pass through per second to 200.
[*Leaf1-Eth-Trunk10] stp edged-port enable
[*Leaf1-Eth-Trunk10] quit
[*Leaf1] interface eth-trunk 11
[*Leaf1-Eth-Trunk11] trunkport 10ge 1/0/6
[*Leaf1-Eth-Trunk11] mode lacp-static
[*Leaf1-Eth-Trunk11] lacp timeout fast
[*Leaf1-Eth-Trunk11] dfs-group 1 m-lag 2
[*Leaf1-Eth-Trunk11] arp anti-attack rate-limit 200 //Set the maximum number of ARP packets that are
allowed to pass through per second to 200.
[*Leaf1-Eth-Trunk11] stp edged-port enable
[*Leaf1-Eth-Trunk11] quit
[*Leaf1] commit
[~Leaf1] monitor-link group 1
[*Leaf1-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf1-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf1-mtlk-group1] port eth-trunk 10 downlink 1
[*Leaf1-mtlk-group1] port eth-trunk 11 downlink 2
[*Leaf1-mtlk-group1] timer recover-time 60
[*Leaf1-mtlk-group1] quit
[*Leaf1] interface 10ge 1/0/5
[*Leaf1-10GE1/0/5] storm suppression unknown-unicast 5 //Configure unknown unicast suppression on
10GE1/0/5 of the access switch.
[*Leaf1-10GE1/0/5] storm suppression multicast packets 1000 //Configure unknown multicast
suppression on 10GE1/0/5 of the access switch.
[*Leaf1-10GE1/0/5] storm suppression broadcast packets 1000 //Configure unknown broadcast
suppression on 10GE1/0/5 of the access switch.
[*Leaf1-10GE1/0/5] port crc-statistics trigger error-down
[*Leaf1-10GE1/0/5] trap-threshold crc-statistics 100 interval 10
[*Leaf1-10GE1/0/5] quit
[*Leaf1] interface 10ge 1/0/6
[*Leaf1-10GE1/0/6] storm suppression unknown-unicast 5
[*Leaf1-10GE1/0/6] storm suppression multicast packets 1000
[*Leaf1-10GE1/0/6] storm suppression broadcast packets 1000
[*Leaf1-10GE1/0/6] port crc-statistics trigger error-down
[*Leaf1-10GE1/0/6] trap-threshold crc-statistics 100 interval 10
[*Leaf1-10GE1/0/6] quit
[*Leaf1] commit
[~Leaf2] stp mode rstp
[*Leaf2] stp v-stp enable
[*Leaf2] dfs-group 1
[*Leaf2-dfs-group-1] source ip 2.2.2.2
[*Leaf2-dfs-group-1] m-lag up-delay 240 auto-recovery interval 10
[*Leaf2-dfs-group-1] dual-active detection delay 0
[*Leaf2-dfs-group-1] quit
[*Leaf2] interface eth-trunk 1
[*Leaf2-Eth-Trunk1] trunkport 10ge 1/0/3
[*Leaf2-Eth-Trunk1] trunkport 10ge 1/0/4
[*Leaf2-Eth-Trunk1] mode lacp-static
[*Leaf2-Eth-Trunk1] peer-link 1
[*Leaf2-Eth-Trunk1] quit
[*Leaf2] interface eth-trunk 10
[*Leaf2-Eth-Trunk10] trunkport 10ge 1/0/5
[*Leaf2-Eth-Trunk10] mode lacp-static
[*Leaf2-Eth-Trunk10] lacp timeout fast
[*Leaf2-Eth-Trunk10] dfs-group 1 m-lag 1
[*Leaf2-Eth-Trunk10] arp anti-attack rate-limit 200 //Set the maximum number of ARP packets that are
```



```
allowed to pass through per second to 200.
[*Leaf2-Eth-Trunk10] stp edged-port enable
[*Leaf2-Eth-Trunk10] quit
[*Leaf2] interface eth-trunk 11
[*Leaf2-Eth-Trunk11] trunkport 10ge 1/0/6
[*Leaf2-Eth-Trunk11] mode lacp-static
[*Leaf2-Eth-Trunk11] lacp timeout fast
[*Leaf2-Eth-Trunk11] dfs-group 1 m-lag 2
[*Leaf2-Eth-Trunk11] arp anti-attack rate-limit 200 //Set the maximum number of ARP packets that are
allowed to pass through per second to 200.
[*Leaf2-Eth-Trunk11] stp edged-port enable
[*Leaf2-Eth-Trunk11] quit
[*Leaf2] commit
[~Leaf2] monitor-link group 1
[*Leaf2-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf2-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf2-mtlk-group1] port eth-trunk 10 downlink 1
[*Leaf2-mtlk-group1] port eth-trunk 11 downlink 2
[*Leaf2-mtlk-group1] timer recover-time 60
[*Leaf2-mtlk-group1] quit
[*Leaf2] interface 10ge 1/0/5
[*Leaf2-10GE1/0/5] storm suppression unknown-unicast 5 //Configure unknown unicast suppression on
10GE1/0/5 of the access switch.
[*Leaf2-10GE1/0/5] storm suppression multicast packets 1000 //Configure unknown multicast
suppression on 10GE1/0/5 of the access switch.
[*Leaf2-10GE1/0/5] storm suppression broadcast packets 1000 //Configure unknown broadcast
suppression on 10GE1/0/5 of the access switch.
[*Leaf2-10GE1/0/5] quit
[*Leaf2] interface 10ge 1/0/6
[*Leaf2-10GE1/0/6] storm suppression unknown-unicast 5
[*Leaf2-10GE1/0/6] storm suppression multicast packets 1000
[*Leaf2-10GE1/0/6] storm suppression broadcast packets 1000
[*Leaf2-10GE1/0/6] quit
[*Leaf2] commit
```

Step 4 Configure BGP EVPN to establish VXLAN tunnels.

1. (For the CE12800 and CE16800 equipped with A series cards) Configure the VXLAN tunnel mode and enable the VXLAN ACL extension function on spine switches.

Configure Spine1. The configuration of Spine2 is similar to that of Spine1.

```
[~Spine1] ip tunnel mode vxlan
[*Spine1] assign forward nvo3 acl extend enable
[*Spine1] commit
```

NOTE

After configuring the VXLAN tunnel mode and enabling the VXLAN ACL extension function, save the configuration and restart the switch to make the configuration take effect.

2. Configure BGP EVPN peer relationships. Spine1 and Spine2 function as route reflectors (RRs) of leaf switches.

Configure Spine1. The configuration of Spine2 is similar to that of Spine1.

```
[~Spine1] evpn-overlay enable
[*Spine1] bgp 100
[*Spine1] router-id 7.7.7.7
[*Spine1-bgp] peer 1.1.1.1 as-number 100
[*Spine1-bgp] peer 1.1.1.1 connect-interface LoopBack0
[*Spine1-bgp] peer 2.2.2.2 as-number 100
[*Spine1-bgp] peer 2.2.2.2 connect-interface LoopBack0
[*Spine1-bgp] peer 3.3.3.3 as-number 100
[*Spine1-bgp] peer 3.3.3.3 connect-interface LoopBack0
[*Spine1-bgp] peer 4.4.4.4 as-number 100
[*Spine1-bgp] peer 4.4.4.4 connect-interface LoopBack0
[*Spine1-bgp] peer 9.9.9.9 as-number 100
```

```
[*Spine1-bgp] peer 9.9.9.9 connect-interface LoopBack0
[*Spine1-bgp] peer 10.10.10.10 as-number 100
[*Spine1-bgp] peer 10.10.10.10 connect-interface LoopBack0
[*Spine1-bgp] l2vpn-family evpn
[*Spine1-bgp-af-evpn] bestroute add-path path-number 64 //Enable the BGP ADD-PATH function.
[*Spine1-bgp-af-evpn] peer 1.1.1.1 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 1.1.1.1 reflect-client
[*Spine1-bgp-af-evpn] peer 1.1.1.1 advertise irb
[*Spine1-bgp-af-evpn] peer 1.1.1.1 advertise irbv6
[*Spine1-bgp-af-evpn] peer 1.1.1.1 capability-advertise add-path both
[*Spine1-bgp-af-evpn] peer 1.1.1.1 advertise add-path path-number 64
[*Spine1-bgp-af-evpn] peer 2.2.2.2 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 2.2.2.2 advertise irb
[*Spine1-bgp-af-evpn] peer 2.2.2.2 advertise irbv6
[*Spine1-bgp-af-evpn] peer 2.2.2.2 reflect-client
[*Spine1-bgp-af-evpn] peer 2.2.2.2 capability-advertise add-path both
[*Spine1-bgp-af-evpn] peer 2.2.2.2 advertise add-path path-number 64
[*Spine1-bgp-af-evpn] peer 3.3.3.3 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 3.3.3.3 advertise irb
[*Spine1-bgp-af-evpn] peer 3.3.3.3 advertise irbv6
[*Spine1-bgp-af-evpn] peer 3.3.3.3 reflect-client
[*Spine1-bgp-af-evpn] peer 3.3.3.3 capability-advertise add-path both
[*Spine1-bgp-af-evpn] peer 3.3.3.3 advertise add-path path-number 64
[*Spine1-bgp-af-evpn] peer 4.4.4.4 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 4.4.4.4 advertise irb
[*Spine1-bgp-af-evpn] peer 4.4.4.4 advertise irbv6
[*Spine1-bgp-af-evpn] peer 4.4.4.4 reflect-client
[*Spine1-bgp-af-evpn] peer 4.4.4.4 capability-advertise add-path both
[*Spine1-bgp-af-evpn] peer 4.4.4.4 advertise add-path path-number 64
[*Spine1-bgp-af-evpn] peer 9.9.9.9 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 9.9.9.9 advertise irb
[*Spine1-bgp-af-evpn] peer 9.9.9.9 advertise irbv6
[*Spine1-bgp-af-evpn] peer 9.9.9.9 capability-advertise add-path both
[*Spine1-bgp-af-evpn] peer 9.9.9.9 advertise add-path path-number 64
[*Spine1-bgp-af-evpn] peer 10.10.10.10 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 10.10.10.10 advertise irb
[*Spine1-bgp-af-evpn] peer 10.10.10.10 advertise irbv6
[*Spine1-bgp-af-evpn] peer 10.10.10.10 capability-advertise add-path both
[*Spine1-bgp-af-evpn] peer 10.10.10.10 advertise add-path path-number 64
[*Spine1-bgp-af-evpn] undo policy vpn-target
[*Spine1-bgp-af-evpn] quit
[*Spine1-bgp] quit
[*Spine1] commit
```

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1.

```
[~Leaf1] evpn-overlay enable
[*Leaf1] bgp 100
[*Leaf1] router-id 1.1.1.1
[*Leaf1-bgp] peer 7.7.7.7 as-number 100
[*Leaf1-bgp] peer 7.7.7.7 connect-interface LoopBack0
[*Leaf1-bgp] peer 8.8.8.8 as-number 100
[*Leaf1-bgp] peer 8.8.8.8 connect-interface LoopBack0
[*Leaf1-bgp] l2vpn-family evpn
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 advertise irb
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 advertise irbv6
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 advertise irb
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 advertise irbv6
[*Leaf1-bgp-af-evpn] quit
```

```
[*Leaf1-bgp] quit
[*Leaf1] commit
```

3. Configure VPN and EVPN instances.

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1.

```
[~Leaf1] ip vpn-instance vpn1
[*Leaf1-vpn-instance-vpn1] vxlan vni 200
[*Leaf1-vpn-instance-vpn1] ipv4-family
[*Leaf1-vpn-instance-vpn1-af-ipv4] route-distinguisher 1.1.1.1:1
[*Leaf1-vpn-instance-vpn1-af-ipv4] vpn-target 0:1 evpn
[*Leaf1-vpn-instance-vpn1-af-ipv4] quit
[*Leaf1-vpn-instance-vpn1] ipv6-family
[*Leaf1-vpn-instance-vpn1-af-ipv6] route-distinguisher 1.1.1.1:1
[*Leaf1-vpn-instance-vpn1-af-ipv6] vpn-target 0:1 evpn
[*Leaf1-vpn-instance-vpn1-af-ipv6] quit
[*Leaf1-vpn-instance-vpn1] quit
[*Leaf1] bridge-domain 10
[*Leaf1-bd10] vxlan vni 110
[*Leaf1-bd10] evpn
[*Leaf1-bd10-evpn] route-distinguisher 1.1.1.1:110
[*Leaf1-bd10-evpn] vpn-target 0:110
[*Leaf1-bd10-evpn] vpn-target 0:1 export-extcommunity
[*Leaf1-bd10-evpn] quit
[*Leaf1-bd10] quit
[*Leaf1] bridge-domain 20
[*Leaf1-bd20] vxlan vni 120
[*Leaf1-bd20] evpn
[*Leaf1-bd20-evpn] route-distinguisher 1.1.1.1:120
[*Leaf1-bd20-evpn] vpn-target 0:120
[*Leaf1-bd20-evpn] vpn-target 0:1 export-extcommunity
[*Leaf1-bd20-evpn] quit
[*Leaf1-bd20] quit
[*Leaf1] interface nve 1
[*Leaf1-Nve1] source 5.5.5.5 //The IP addresses and MAC addresses of NVE interfaces on the two
devices that establish an M-LAG must be the same.
[*Leaf1-Nve1] mac-address 0000-5e00-0111
[*Leaf1-Nve1] vni 110 head-end peer-list protocol bgp
[*Leaf1-Nve1] vni 120 head-end peer-list protocol bgp
[*Leaf1-Nve1] quit
[*Leaf1] commit
```

4. Configure Layer 3 gateways.

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1.

```
[~Leaf1] interface vbdif10 //The IP addresses and MAC addresses of the same VBDIF interface on a
leaf switch and a DCGW must be the same.
[*Leaf1-Vbdif10] ip binding vpn-instance vpn1
[*Leaf1-Vbdif10] ip address 10.1.1.1 24
[*Leaf1-Vbdif10] ipv6 enable
[*Leaf1-Vbdif10] ipv6 address fc00:1::1 64
[*Leaf1-Vbdif10] arp broadcast-detect enable
[*Leaf1-Vbdif10] vxlan anycast-gateway enable
[*Leaf1-Vbdif10] arp collect host enable
[*Leaf1-Vbdif10] ipv6 nd collect host enable
[*Leaf1-Vbdif10] mac-address 0000-5e00-0101
[*Leaf1-Vbdif10] quit
[*Leaf1] interface vbdif20
[*Leaf1-Vbdif20] ip binding vpn-instance vpn1
[*Leaf1-Vbdif20] ip address 10.2.1.1 24
[*Leaf1-Vbdif20] ipv6 enable
[*Leaf1-Vbdif20] ipv6 address fc00:2::1 64
[*Leaf1-Vbdif20] arp broadcast-detect enable
[*Leaf1-Vbdif20] vxlan anycast-gateway enable
[*Leaf1-Vbdif20] arp collect host enable
[*Leaf1-Vbdif20] ipv6 nd collect host enable
[*Leaf1-Vbdif20] mac-address 0000-5e00-0102
```

```
[*Leaf1-Vbdif20] quit
[*Leaf1] commit
```

5. Configure service access points.

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1.

```
[~Leaf1] interface eth-trunk 10.10 mode l2
[*Leaf1-Eth-Trunk10.10] encapsulation dot1q vid 10
[*Leaf1-Eth-Trunk10.10] bridge-domain 10
[*Leaf1-Eth-Trunk10.10] quit
[*Leaf1] interface eth-trunk 11.20 mode l2
[*Leaf1-Eth-Trunk11.20] encapsulation dot1q vid 20
[*Leaf1-Eth-Trunk11.20] bridge-domain 20
[*Leaf1-Eth-Trunk11.20] quit
[*Leaf1] commit
```

Step 5 Configure BFD on leaf switches to detect links between leaf switches and VNFs.

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1.

```
[~Leaf1] bfd toipu1_v4 bind peer-ip 10.1.1.2 vpn-instance vpn1 interface vbdif10 source-ip 10.1.1.1
one-arm-echo
[*Leaf1-bfd-session-toipu1_v4] discriminator local 10
[*Leaf1-bfd-session-toipu1_v4] detect-multiplier 4
[*Leaf1-bfd-session-toipu1_v4] min-echo-rx-interval 500
[*Leaf1-bfd-session-toipu1_v4] quit
[*Leaf1] bfd toipu2_v4 bind peer-ip 10.2.1.2 vpn-instance vpn1 interface vbdif20 source-ip 10.2.1.1 one-
arm-echo
[*Leaf1-bfd-session-toipu2_v4] discriminator local 30
[*Leaf1-bfd-session-toipu2_v4] detect-multiplier 4
[*Leaf1-bfd-session-toipu2_v4] min-echo-rx-interval 500
[*Leaf1-bfd-session-toipu2_v4] quit
[*Leaf1] bfd toipu1_v6 bind peer-ipv6 fc00:1::2 vpn-instance vpn1 interface vbdif10 source-ipv6
fc00:1::1 one-arm-echo
[*Leaf1-bfd-session-toipu1_v6] discriminator local 50
[*Leaf1-bfd-session-toipu1_v6] detect-multiplier 4
[*Leaf1-bfd-session-toipu1_v6] min-echo-rx-interval 500
[*Leaf1-bfd-session-toipu1_v6] quit
[*Leaf1] bfd toipu2_v6 bind peer-ipv6 fc00:2::2 vpn-instance vpn1 interface vbdif20 source-ipv6
fc00:2::1 one-arm-echo
[*Leaf1-bfd-session-toipu2_v6] discriminator local 70
[*Leaf1-bfd-session-toipu2_v6] detect-multiplier 4
[*Leaf1-bfd-session-toipu2_v6] min-echo-rx-interval 500
[*Leaf1-bfd-session-toipu2_v6] quit
[*Leaf1] bfd
[*Leaf1-bfd] bfd forwarding match remote-discriminator 20 //Specify the remote discriminator of the
BFD for M-LAG.
[*Leaf1-bfd] bfd forwarding match remote-discriminator 40
[*Leaf1-bfd] bfd forwarding match remote-discriminator 60
[*Leaf1-bfd] bfd forwarding match remote-discriminator 80
[*Leaf1-bfd] quit
[*Leaf1] commit
```

 NOTE

In the NFVI telco cloud solution deployed in traditional mode, the BFD detection interval can be set to 500 x 4.

In the NFVI telco cloud solution deployed in SDN mode, the BFD detection interval can be set to 500 x 4. To improve the detection performance, the BFD detection interval can be set to 300 x 6.

Step 6 Configure static VPN routes from leaf switches to VNFs, configure BGP EVPN to import static VPN routes, and configure a routing policy for the L3VPN instance to keep the original next hops of the static VPN routes.

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1.

```
[~Leaf1] route-policy import_static_policy permit node 100 //Configure a routing policy to add the
community attribute when importing static routes to BGP.
[*Leaf1-route-policy] if-match tag 82345
[*Leaf1-route-policy] apply community 82345 additive
[*Leaf1-route-policy] quit
[*Leaf1] route-policy import_static_policy permit node 999
[*Leaf1-route-policy] quit
[*Leaf1] ip community-filter basic apply_gwip_route index 10 permit 82345 //Create a community filter.
[*Leaf1] ip community-filter basic suppress_route index 10 permit 82346
[*Leaf1] route-policy export_policy deny node 10 //Configure a routing policy to filter out or permit
routes.
[*Leaf1-route-policy] if-match community-filter suppress_route
[*Leaf1-route-policy] quit
[*Leaf1] route-policy export_policy permit node 20
[*Leaf1-route-policy] if-match community-filter apply_gwip_route
[*Leaf1-route-policy] apply gateway-ip origin-nexthop //Configure the next-hop address of the route as
the gateway IP address. When configuring an L3VPN instance on a Layer 2 or Layer 3 gateway to advertise
static VPN routes that can reach a VNF to an EVPN, you need to create a routing policy to filter out the
static VPN routes that can reach the VNF in the L3VPN instance.
[*Leaf1-route-policy] apply ipv6 gateway-ip origin-nexthop
[*Leaf1-route-policy] quit
[*Leaf1] route-policy export_policy permit node 999
[*Leaf1-route-policy] quit
[*Leaf1] ip route-static vpn-instance vpn1 201.1.1.1 32 10.1.1.2 preference 255 tag 82345 track bfd-
session toipu1_v4 inter-protocol-ecmp //The configured tag must match the routing policy so that the
next-hop address of the route advertised by the leaf switch to the DCGW is the gateway IP address,
implementing asymmetric forwarding.
[*Leaf1] ip route-static vpn-instance vpn1 201.1.1.1 32 10.2.1.2 preference 255 tag 82345 track bfd-
session toipu2_v4 inter-protocol-ecmp
[*Leaf1] ipv6 route-static vpn-instance vpn1 201::1 128 fc00:1::2 preference 255 tag 82345 track bfd-
session toipu1_v6 inter-protocol-ecmp
[*Leaf1] ipv6 route-static vpn-instance vpn1 201::1 128 fc00:2::2 preference 255 tag 82345 track bfd-
session toipu2_v6 inter-protocol-ecmp
[*Leaf1] bgp 100
[*Leaf1-bgp] ipv4-family vpn-instance vpn1
[*Leaf1-bgp-vpn1] import-route static route-policy import_static_policy
[*Leaf1-bgp-vpn1] advertise l2vpn evpn import-route-multipath //Advertise all routes with the same
destination address.
[*Leaf1-bgp-vpn1] quit
[*Leaf1-bgp] ipv6-family vpn-instance vpn1
[*Leaf1-bgp-6-vpn1] import-route static route-policy import_static_policy
[*Leaf1-bgp-6-vpn1] advertise l2vpn evpn import-route-multipath
[*Leaf1-bgp-6-vpn1] quit
[*Leaf1-bgp] quit
[*Leaf1] commit
```

Step 7 Configure load balancing among routes.

Configure Leaf1. The configurations of Leaf2, Leaf3, and Leaf4 are similar to the configuration of Leaf1.

```
[~Leaf1] bgp 100
[*Leaf1-bgp] ipv4-family vpn-instance vpn1
[*Leaf1-bgp-vpn1] maximum load-balancing mixed 64 //Set the maximum number of equal-cost routes
for load balancing between BGP routes and routes of other protocols.
[*Leaf1-bgp-vpn1] quit
[*Leaf1-bgp] ipv6-family vpn-instance vpn1
[*Leaf1-bgp-6-vpn1] maximum load-balancing mixed 64
[*Leaf1-bgp-6-vpn1] quit
[*Leaf1-bgp] l2vpn-family evpn
[*Leaf1-bgp-af-evpn] bestroute add-path path-number 64 //Enable the BGP ADD-PATH function.
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 capability-advertise add-path both
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 advertise add-path path-number 64
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 route-policy export_policy export //Configure a routing policy for the
RR.
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 capability-advertise add-path both
```

```
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 advertise add-path path-number 64
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 route-policy export_policy export
[*Leaf1-bgp-af-evpn] quit
[*Leaf1-bgp] quit
[*Leaf1] vxlan-overlay network local-preference enable //Configure the device to forward VXLAN
network-side traffic only through a local outbound interface (non-VXLAN outbound interface) after the
traffic matches a route with multiple next hops.
[*Leaf1] commit
```

----End

Verifying the Configuration

After completing the configurations, run the **display ip routing-table vpn-instance vpn1** command on a leaf switch. The command output shows that the leaf switch has received routes from the VPN instance on a DCGW and the next hop is the anycast VTEP address. The following uses the command output on Leaf1 as an example.

```
[~Leaf1] display ip routing-table vpn-instance vpn1
Proto: Protocol      Pre: Preference
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
Routing Table : vpn1
Destinations : 16      Routes : 18

Destination/Mask  Proto  Pre  Cost    Flags NextHop      Interface
-----
0.0.0.0/0        IBGP   255  0        RD 11.11.11.11  VXLAN
10.1.1.0/24       Direct 0    0        D 10.1.1.1     Vbdif10
10.1.1.1/32       Direct 0    0        D 127.0.0.1    Vbdif10
10.1.1.255/32     Direct 0    0        D 127.0.0.1    Vbdif10
10.2.1.0/24       Direct 0    0        D 10.2.1.1     Vbdif20
10.2.1.1/32       Direct 0    0        D 127.0.0.1    Vbdif20
10.2.1.255/32     Direct 0    0        D 127.0.0.1    Vbdif20
10.3.1.0/24       Direct 0    0        D 10.3.1.1     Vbdif30
10.3.1.1/32       Direct 0    0        D 127.0.0.1    Vbdif30
10.3.1.255/32     Direct 0    0        D 127.0.0.1    Vbdif30
10.4.1.0/24       Direct 0    0        D 10.4.1.1     Vbdif40
10.4.1.1/32       Direct 0    0        D 127.0.0.1    Vbdif40
10.4.1.255/32     Direct 0    0        D 127.0.0.1    Vbdif40
201.1.1.1/32      Static 255  0        RD 10.1.1.2     Vbdif10
                  Static 255  0        RD 10.2.1.2     Vbdif20
202.1.1.1/32      IBGP   255  0        RD 10.3.1.2     VXLAN
                  IBGP   255  0        RD 10.4.1.2     VXLAN
255.255.255.255/32 Direct 0    0        D 127.0.0.1    InLoopBack0
```

The VRF routing table information can be viewed on DCGW devices, and the next hops are the IP addresses of the IPUs.

Configuration Files

- Spine1 configuration file

```
#
sysname Spine1
#
assign forward nvo3 acl extend enable //This step is required only on the CE12800 and CE16800
equipped with A series cards.
#
evpn-overlay enable
#
bfd
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.1.1 255.255.255.0
```

```
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.2.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/3
undo portswitch
ip address 192.168.3.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/4
undo portswitch
ip address 192.168.4.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/5
undo portswitch
ip address 192.168.9.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
#
interface 10GE1/0/6
undo portswitch
ip address 192.168.10.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
#
interface LoopBack0
ip address 7.7.7.7 255.255.255.255
#
bgp 100
router-id 7.7.7.7
peer 1.1.1.1 as-number 100
peer 1.1.1.1 connect-interface LoopBack0
peer 2.2.2.2 as-number 100
peer 2.2.2.2 connect-interface LoopBack0
peer 3.3.3.3 as-number 100
peer 3.3.3.3 connect-interface LoopBack0
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack0
peer 9.9.9.9 as-number 100
peer 9.9.9.9 connect-interface LoopBack0
peer 10.10.10.10 as-number 100
peer 10.10.10.10 connect-interface LoopBack0
#
ipv4-family unicast
peer 1.1.1.1 enable
peer 2.2.2.2 enable
peer 3.3.3.3 enable
peer 4.4.4.4 enable
peer 9.9.9.9 enable
peer 10.10.10.10 enable
#
l2vpn-family evpn
undo policy vpn-target
```

```
bestroute add-path path-number 64
peer 1.1.1.1 enable
peer 1.1.1.1 advertise irb
peer 1.1.1.1 advertise irbv6
peer 1.1.1.1 reflect-client
peer 1.1.1.1 capability-advertise add-path both
peer 1.1.1.1 advertise add-path path-number 64
peer 2.2.2.2 enable
peer 2.2.2.2 advertise irb
peer 2.2.2.2 advertise irbv6
peer 2.2.2.2 reflect-client
peer 2.2.2.2 capability-advertise add-path both
peer 2.2.2.2 advertise add-path path-number 64
peer 3.3.3.3 enable
peer 3.3.3.3 advertise irb
peer 3.3.3.3 advertise irbv6
peer 3.3.3.3 reflect-client
peer 3.3.3.3 capability-advertise add-path both
peer 3.3.3.3 advertise add-path path-number 64
peer 4.4.4.4 enable
peer 4.4.4.4 advertise irb
peer 4.4.4.4 advertise irbv6
peer 4.4.4.4 reflect-client
peer 4.4.4.4 capability-advertise add-path both
peer 4.4.4.4 advertise add-path path-number 64
peer 9.9.9.9 enable
peer 9.9.9.9 advertise irb
peer 9.9.9.9 advertise irbv6
peer 9.9.9.9 capability-advertise add-path both
peer 9.9.9.9 advertise add-path path-number 64
peer 10.10.10.10 enable
peer 10.10.10.10 advertise irb
peer 10.10.10.10 advertise irbv6
peer 10.10.10.10 capability-advertise add-path both
peer 10.10.10.10 advertise add-path path-number 64
#
ospf 1 router-id 7.7.7.7
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
spf-schedule-interval intelligent-timer 50 50 50
lsa-originate-interval intelligent-timer 500 50 100
lsa-arrival-interval intelligent-timer 50 50 50
area 0.0.0.0
network 7.7.7.7 0.0.0.0
network 192.168.1.0 0.0.0.255
network 192.168.2.0 0.0.0.255
network 192.168.3.0 0.0.0.255
network 192.168.4.0 0.0.0.255
network 192.168.9.0 0.0.0.255
network 192.168.10.0 0.0.0.255
#
return
```

- Spine2 configuration file

```
#
sysname Spine2
#
assign forward nvo3 acl extend enable //This step is required only on the CE12800 and CE16800
equipped with A series cards.
#
evpn-overlay enable
#
bfd
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.5.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
```



```
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.6.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/3
undo portswitch
ip address 192.168.7.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/4
undo portswitch
ip address 192.168.8.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/5
undo portswitch
ip address 192.168.11.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
#
interface 10GE1/0/6
undo portswitch
ip address 192.168.12.1 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 780000
#
interface LoopBack0
ip address 8.8.8.8 255.255.255.255
#
bgp 100
router-id 8.8.8.8
peer 1.1.1.1 as-number 100
peer 1.1.1.1 connect-interface LoopBack0
peer 2.2.2.2 as-number 100
peer 2.2.2.2 connect-interface LoopBack0
peer 3.3.3.3 as-number 100
peer 3.3.3.3 connect-interface LoopBack0
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack0
peer 9.9.9.9 as-number 100
peer 9.9.9.9 connect-interface LoopBack0
peer 10.10.10.10 as-number 100
peer 10.10.10.10 connect-interface LoopBack0
#
ipv4-family unicast
peer 1.1.1.1 enable
peer 2.2.2.2 enable
peer 3.3.3.3 enable
peer 4.4.4.4 enable
peer 9.9.9.9 enable
peer 10.10.10.10 enable
#
l2vpn-family evpn
undo policy vpn-target
bestroute add-path path-number 64
peer 1.1.1.1 enable
peer 1.1.1.1 advertise irb
```

```
peer 1.1.1.1 advertise irbv6
peer 1.1.1.1 reflect-client
peer 1.1.1.1 capability-advertise add-path both
peer 1.1.1.1 advertise add-path path-number 64
peer 2.2.2.2 enable
peer 2.2.2.2 advertise irb
peer 2.2.2.2 advertise irbv6
peer 2.2.2.2 reflect-client
peer 2.2.2.2 capability-advertise add-path both
peer 2.2.2.2 advertise add-path path-number 64
peer 3.3.3.3 enable
peer 3.3.3.3 advertise irb
peer 3.3.3.3 advertise irbv6
peer 3.3.3.3 reflect-client
peer 3.3.3.3 capability-advertise add-path both
peer 3.3.3.3 advertise add-path path-number 64
peer 4.4.4.4 enable
peer 4.4.4.4 advertise irb
peer 4.4.4.4 advertise irbv6
peer 4.4.4.4 reflect-client
peer 4.4.4.4 capability-advertise add-path both
peer 4.4.4.4 advertise add-path path-number 64
peer 9.9.9.9 enable
peer 9.9.9.9 advertise irb
peer 9.9.9.9 advertise irbv6
peer 9.9.9.9 capability-advertise add-path both
peer 9.9.9.9 advertise add-path path-number 64
peer 10.10.10.10 enable
peer 10.10.10.10 advertise irb
peer 10.10.10.10 advertise irbv6
peer 10.10.10.10 capability-advertise add-path both
peer 10.10.10.10 advertise add-path path-number 64
#
ospf 1 router-id 8.8.8.8
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
spf-schedule-interval intelligent-timer 50 50 50
lsa-originate-interval intelligent-timer 500 50 100
lsa-arrival-interval intelligent-timer 50 50 50
area 0.0.0.0
network 8.8.8.8 0.0.0.0
network 192.168.5.0 0.0.0.255
network 192.168.6.0 0.0.0.255
network 192.168.7.0 0.0.0.255
network 192.168.8.0 0.0.0.255
network 192.168.11.0 0.0.0.255
network 192.168.12.0 0.0.0.255
#
return
```

- Leaf1 configuration file

```
#
sysname Leaf1
#
dfs-group 1
priority 150
source ip 1.1.1.1
m-lag up-delay 240 auto-recovery interval 10
dual-active detection delay 0
#
vxlan-overlay network local-preference enable
#
assign forward layer-3 resource large-overlay //This step is required only on the CE8861EI,
CE6857EI, CE6865E, CE8850E-32CQ-EI, and CE6865EI.
#
assign forward ipv6 longer-mask resource share-mode //This step is required only on the CE8861EI,
CE6857EI, CE6865E, CE8850E-32CQ-EI, and CE6865EI.
#
stp mode rstp
stp v-stp enable
```

```
#
evpn-overlay enable
#
ip vpn-instance vpn1
  ipv4-family
    route-distinguisher 1.1.1.1:1
    vpn-target 0:1 export-extcommunity evpn
    vpn-target 0:1 import-extcommunity evpn
  ipv6-family
    route-distinguisher 1.1.1.1:1
    vpn-target 0:1 export-extcommunity evpn
    vpn-target 0:1 import-extcommunity evpn
  vxlan vni 200
#
bfd
bfd forwarding match remote-discriminator 20
bfd forwarding match remote-discriminator 40
bfd forwarding match remote-discriminator 60
bfd forwarding match remote-discriminator 80
#
bridge-domain 10
  vxlan vni 110
  evpn
    route-distinguisher 1.1.1.1:110
    vpn-target 0:110 export-extcommunity
    vpn-target 0:1 export-extcommunity
    vpn-target 0:110 import-extcommunity
#
bridge-domain 20
  vxlan vni 120
  evpn
    route-distinguisher 1.1.1.1:120
    vpn-target 0:120 export-extcommunity
    vpn-target 0:1 export-extcommunity
    vpn-target 0:120 import-extcommunity
#
interface Vbdif10
  ip binding vpn-instance vpn1
  ipv6 enable
  ip address 10.1.1.1 255.255.255.0
  ipv6 address FC00:1::1/64
  arp broadcast-detect enable
  mac-address 0000-5e00-0101
  ipv6 nd collect host enable
  vxlan anycast-gateway enable
  arp collect host enable
#
interface Vbdif20
  ip binding vpn-instance vpn1
  ipv6 enable
  ip address 10.2.1.1 255.255.255.0
  ipv6 address FC00:2::1/64
  arp broadcast-detect enable
  mac-address 0000-5e00-0102
  ipv6 nd collect host enable
  vxlan anycast-gateway enable
  arp collect host enable
#
interface Eth-Trunk1
  mode lacp-static
  peer-link 1
#
interface Eth-Trunk10
  stp edged-port enable
  mode lacp-static
  lacp timeout fast
  dfs-group 1 m-lag 1
  arp anti-attack rate-limit 200
#
```

```
interface Eth-Trunk10.10 mode l2
 encapsulation dot1q vid 10
 bridge-domain 10
#
interface Eth-Trunk11
 stp edged-port enable
 mode lacp-static
 lacp timeout fast
 dfs-group 1 m-lag 2
 arp anti-attack rate-limit 200
#
interface Eth-Trunk11.20 mode l2
 encapsulation dot1q vid 20
 bridge-domain 20
#
interface 10GE1/0/1
 undo portswitch
 ip address 192.168.1.2 255.255.255.0
 ospf network-type p2p
 ospf peer hold-max-cost timer 800000
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/2
 undo portswitch
 ip address 192.168.5.2 255.255.255.0
 ospf network-type p2p
 ospf peer hold-max-cost timer 800000
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/3
 eth-trunk 1
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/4
 eth-trunk 1
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/5
 eth-trunk 10
 storm suppression unknown-unicast 5
 storm suppression multicast packets 1000
 storm suppression broadcast packets 1000
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/6
 eth-trunk 11
 storm suppression unknown-unicast 5
 storm suppression multicast packets 1000
 storm suppression broadcast packets 1000
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface LoopBack0
 ip address 1.1.1.1 255.255.255.255
#
interface LoopBack1
 ip address 5.5.5.5 255.255.255.255
#
interface Nve1
 source 5.5.5.5
 vni 110 head-end peer-list protocol bgp
 vni 120 head-end peer-list protocol bgp
 mac-address 0000-5e00-0111
#
```

```
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk11 downlink 2
timer recover-time 60
#
bfd toipu1_v4 bind peer-ip 10.1.1.2 vpn-instance vpn1 interface Vbdif10 source-ip 10.1.1.1 one-arm-
echo
discriminator local 10
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu1_v6 bind peer-ipv6 FC00:1::2 vpn-instance vpn1 interface Vbdif10 source-ipv6 FC00:1::1 one-
arm-echo
discriminator local 50
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu2_v4 bind peer-ip 10.2.1.2 vpn-instance vpn1 interface Vbdif20 source-ip 10.2.1.1 one-arm-
echo
discriminator local 30
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu2_v6 bind peer-ipv6 FC00:2::2 vpn-instance vpn1 interface Vbdif20 source-ipv6 FC00:2::1 one-
arm-echo
discriminator local 70
detect-multiplier 4
min-echo-rx-interval 500
#
bgp 100
router-id 1.1.1.1
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack0
peer 8.8.8.8 as-number 100
peer 8.8.8.8 connect-interface LoopBack0
#
ipv4-family unicast
peer 7.7.7.7 enable
peer 8.8.8.8 enable
#
ipv4-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing mixed 64
advertise l2vpn evpn import-route-multipath
#
ipv6-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing mixed 64
advertise l2vpn evpn import-route-multipath
#
l2vpn-family evpn
policy vpn-target
bestroute add-path path-number 64
peer 7.7.7.7 enable
peer 7.7.7.7 route-policy export_policy export
peer 7.7.7.7 advertise irb
peer 7.7.7.7 advertise irbv6
peer 7.7.7.7 capability-advertise add-path both
peer 7.7.7.7 advertise add-path path-number 64
peer 8.8.8.8 enable
peer 8.8.8.8 route-policy export_policy export
peer 8.8.8.8 advertise irb
peer 8.8.8.8 advertise irbv6
peer 8.8.8.8 capability-advertise add-path both
peer 8.8.8.8 advertise add-path path-number 64
#
ospf 1 router-id 1.1.1.1
```

```
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
spf-schedule-interval intelligent-timer 50 50 50
lsa-originate-interval intelligent-timer 500 50 100
lsa-arrival-interval intelligent-timer 50 50 50
area 0.0.0.0
 network 1.1.1.1 0.0.0.0
 network 5.5.5.5 0.0.0.0
 network 192.168.1.0 0.0.0.255
 network 192.168.5.0 0.0.0.255
#
route-policy export_policy deny node 10
 if-match community-filter suppress_route
#
route-policy export_policy permit node 20
 if-match community-filter apply_gwip_route
 apply gateway-ip origin-nexthop
 apply ipv6 gateway-ip origin-nexthop
#
route-policy export_policy permit node 999
#
route-policy import_static_policy permit node 100
 if-match tag 82345
 apply community 82345 additive
#
route-policy import_static_policy permit node 999
#
ip community-filter basic apply_gwip_route index 10 permit 82345
ip community-filter basic suppress_route index 10 permit 82346
#
ip route-static vpn-instance vpn1 201.1.1.1 255.255.255.255 10.1.1.2 preference 255 tag 82345 track
bfd-session toipu1_v4 inter-protocol-ecmp
ip route-static vpn-instance vpn1 201.1.1.1 255.255.255.255 10.2.1.2 preference 255 tag 82345 track
bfd-session toipu2_v4 inter-protocol-ecmp
#
ipv6 route-static vpn-instance vpn1 201::1 128 FC00:1::2 preference 255 tag 82345 track bfd-session
toipu1_v6 inter-protocol-ecmp
ipv6 route-static vpn-instance vpn1 201::1 128 FC00:2::2 preference 255 tag 82345 track bfd-session
toipu2_v6 inter-protocol-ecmp
#
return
```

- Leaf2 configuration file

```
#
sysname Leaf2
#
dfs-group 1
 source ip 2.2.2.2
 m-lag up-delay 240 auto-recovery interval 10
 dual-active detection delay 0
#
vxlan-overlay network local-preference enable
#
assign forward layer-3 resource large-overlay //This step is required only on the CE8861EI,
CE6857EI, CE6865E, CE8850E-32CQ-EI, and CE6865EI.
#
assign forward ipv6 longer-mask resource share-mode //This step is required only on the CE8861EI,
CE6857EI, CE6865E, CE8850E-32CQ-EI, and CE6865EI.
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
 ipv4-family
  route-distinguisher 2.2.2.2:1
  vpn-target 0:1 export-extcommunity evpn
  vpn-target 0:1 import-extcommunity evpn
 ipv6-family
```

```
route-distinguisher 2.2.2.2:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 200
#
bfd
bfd forwarding match remote-discriminator 10
bfd forwarding match remote-discriminator 30
bfd forwarding match remote-discriminator 50
bfd forwarding match remote-discriminator 70
#
bridge-domain 10
vxlan vni 110
evpn
route-distinguisher 2.2.2.2:110
vpn-target 0:110 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:110 import-extcommunity
#
bridge-domain 20
vxlan vni 120
evpn
route-distinguisher 2.2.2.2:120
vpn-target 0:120 export-extcommunity
vpn-target 0:1 export-extcommunity
vpn-target 0:120 import-extcommunity
#
interface Vbdif10
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.1.1.1 255.255.255.0
ipv6 address FC00:1::1/64
arp broadcast-detect enable
mac-address 0000-5e00-0101
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Vbdif20
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.2.1.1 255.255.255.0
ipv6 address FC00:2::1/64
arp broadcast-detect enable
mac-address 0000-5e00-0102
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk10
stp edged-port enable
mode lacp-static
lacp timeout fast
dfs-group 1 m-lag 1
arp anti-attack rate-limit 200
#
interface Eth-Trunk10.10 mode l2
encapsulation dot1q vid 10
bridge-domain 10
#
interface Eth-Trunk11
stp edged-port enable
mode lacp-static
lacp timeout fast
dfs-group 1 m-lag 2
```

```
arp anti-attack rate-limit 200
#
interface Eth-Trunk11.20 mode l2
encapsulation dot1q vid 20
bridge-domain 20
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.2.2 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 800000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.6.2 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 800000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/3
eth-trunk 1
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/4
eth-trunk 1
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/5
eth-trunk 10
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/6
eth-trunk 11
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface LoopBack0
ip address 2.2.2.2 255.255.255.255
#
interface LoopBack1
ip address 5.5.5.5 255.255.255.255
#
interface Nve1
source 5.5.5.5
vni 110 head-end peer-list protocol bgp
vni 120 head-end peer-list protocol bgp
mac-address 0000-5e00-0111
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk11 downlink 2
timer recover-time 60
#
bfd toipu1_v4 bind peer-ip 10.1.1.2 vpn-instance vpn1 interface Vbdif10 source-ip 10.1.1.1 one-arm-echo
```



```
discriminator local 20
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu1_v6 bind peer-ipv6 FC00:1::2 vpn-instance vpn1 interface Vbdif10 source-ipv6 FC00:1::1 one-
arm-echo
discriminator local 60
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu2_v4 bind peer-ip 10.2.1.2 vpn-instance vpn1 interface Vbdif20 source-ip 10.2.1.1 one-arm-
echo
discriminator local 40
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu2_v6 bind peer-ipv6 FC00:2::2 vpn-instance vpn1 interface Vbdif20 source-ipv6 FC00:2::1 one-
arm-echo
discriminator local 80
detect-multiplier 4
min-echo-rx-interval 500
#
bgp 100
router-id 2.2.2.2
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack0
peer 8.8.8.8 as-number 100
peer 8.8.8.8 connect-interface LoopBack0
#
ipv4-family unicast
peer 7.7.7.7 enable
peer 8.8.8.8 enable
#
ipv4-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing mixed 64
advertise l2vpn evpn import-route-multipath
#
ipv6-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing mixed 64
advertise l2vpn evpn import-route-multipath
#
l2vpn-family evpn
policy vpn-target
bestroute add-path path-number 64
peer 7.7.7.7 enable
peer 7.7.7.7 route-policy export_policy export
peer 7.7.7.7 advertise irb
peer 7.7.7.7 advertise irbv6
peer 7.7.7.7 capability-advertise add-path both
peer 7.7.7.7 advertise add-path path-number 64
peer 8.8.8.8 enable
peer 8.8.8.8 route-policy export_policy export
peer 8.8.8.8 advertise irb
peer 8.8.8.8 advertise irbv6
peer 8.8.8.8 capability-advertise add-path both
peer 8.8.8.8 advertise add-path path-number 64
#
ospf 1 router-id 2.2.2.2
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
spf-schedule-interval intelligent-timer 50 50 50
lsa-originate-interval intelligent-timer 500 50 100
lsa-arrival-interval intelligent-timer 50 50 50
area 0.0.0.0
network 2.2.2.2 0.0.0.0
network 5.5.5.5 0.0.0.0
network 192.168.2.0 0.0.0.255
```

```
network 192.168.6.0 0.0.0.255
#
route-policy export_policy deny node 10
if-match community-filter suppress_route
#
route-policy export_policy permit node 20
if-match community-filter apply_gwip_route
apply gateway-ip origin-nexthop
apply ipv6 gateway-ip origin-nexthop
#
route-policy export_policy permit node 999
#
route-policy import_static_policy permit node 100
if-match tag 82345
apply community 82345 additive
#
route-policy import_static_policy permit node 999
#
ip community-filter basic apply_gwip_route index 10 permit 82345
ip community-filter basic suppress_route index 10 permit 82346
#
ip route-static vpn-instance vpn1 201.1.1.1 255.255.255.255 10.1.1.2 preference 255 tag 82345 track
bfd-session toipu1_v4 inter-protocol-ecmp
ip route-static vpn-instance vpn1 201.1.1.1 255.255.255.255 10.2.1.2 preference 255 tag 82345 track
bfd-session toipu2_v4 inter-protocol-ecmp
#
ipv6 route-static vpn-instance vpn1 201::1 128 FC00:1::2 preference 255 tag 82345 track bfd-session
toipu1_v6 inter-protocol-ecmp
ipv6 route-static vpn-instance vpn1 201::1 128 FC00:2::2 preference 255 tag 82345 track bfd-session
toipu2_v6 inter-protocol-ecmp
#
return
```

- Leaf3 configuration file

```
#
sysname Leaf3
#
dfs-group 1
priority 150
source ip 3.3.3.3
m-lag up-delay 240 auto-recovery interval 10
dual-active detection delay 0
#
vxlan-overlay network local-preference enable
#
assign forward layer-3 resource large-overlay //This step is required only on the CE8861EI,
CE6857EI, CE6865E, CE8850E-32CQ-EI, and CE6865EI.
#
assign forward ipv6 longer-mask resource share-mode //This step is required only on the CE8861EI,
CE6857EI, CE6865E, CE8850E-32CQ-EI, and CE6865EI.
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 3.3.3.3:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 1.1.1.1:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 200
#
bfd
bfd forwarding match remote-discriminator 20
bfd forwarding match remote-discriminator 40
```

```
bfd forwarding match remote-discriminator 60
bfd forwarding match remote-discriminator 80
#
bridge-domain 30
vxlan vni 130
evpn
 route-distinguisher 3.3.3.3:130
 vpn-target 0:130 export-extcommunity
 vpn-target 0:1 export-extcommunity
 vpn-target 0:130 import-extcommunity
#
bridge-domain 40
vxlan vni 140
evpn
 route-distinguisher 3.3.3.3:140
 vpn-target 0:140 export-extcommunity
 vpn-target 0:1 export-extcommunity
 vpn-target 0:140 import-extcommunity
#
interface Vbdif30
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.3.1.1 255.255.255.0
 ipv6 address FC00:3::1/64
 arp broadcast-detect enable
 mac-address 0000-5e00-0103
 ipv6 nd collect host enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Vbdif40
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.4.1.1 255.255.255.0
 ipv6 address FC00:4::1/64
 arp broadcast-detect enable
 mac-address 0000-5e00-0104
 ipv6 nd collect host enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Eth-Trunk1
 mode lacp-static
 peer-link 1
#
interface Eth-Trunk10
 stp edged-port enable
 mode lacp-static
 lacp timeout fast
 dfs-group 1 m-lag 1
 arp anti-attack rate-limit 200
#
interface Eth-Trunk10.30 mode l2
 encapsulation dot1q vid 30
 bridge-domain 30
#
interface Eth-Trunk11
 stp edged-port enable
 mode lacp-static
 lacp timeout fast
 dfs-group 1 m-lag 2
 arp anti-attack rate-limit 200
#
interface Eth-Trunk11.40 mode l2
 encapsulation dot1q vid 40
 bridge-domain 40
#
interface 10GE1/0/1
 undo portswitch
```

```
ip address 192.168.3.2 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 800000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.7.2 255.255.255.0
ospf network-type p2p
ospf peer hold-max-cost timer 800000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/3
eth-trunk 1
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/4
eth-trunk 1
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/5
eth-trunk 10
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/6
eth-trunk 11
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface LoopBack0
ip address 3.3.3.3 255.255.255.255
#
interface LoopBack1
ip address 6.6.6.6 255.255.255.255
#
interface Nve1
source 6.6.6.6
vni 130 head-end peer-list protocol bgp
vni 140 head-end peer-list protocol bgp
mac-address 0000-5e00-0112
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk11 downlink 2
timer recover-time 60
#
bfd toipu3_v4 bind peer-ip 10.3.1.2 vpn-instance vpn1 interface Vbdif30 source-ip 10.3.1.1 one-arm-echo
discriminator local 10
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu3_v6 bind peer-ipv6 FC00:3::2 vpn-instance vpn1 interface Vbdif30 source-ipv6 FC00:3::1 one-arm-echo
discriminator local 50
detect-multiplier 4
```

```
min-echo-rx-interval 500
#
bfd toipu4_v4 bind peer-ip 10.4.1.2 vpn-instance vpn1 interface Vbdif40 source-ip 10.4.1.1 one-arm-
echo
discriminator local 30
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu4_v6 bind peer-ipv6 FC00:4::2 vpn-instance vpn1 interface Vbdif40 source-ipv6 FC00:4::1 one-
arm-echo
discriminator local 70
detect-multiplier 4
min-echo-rx-interval 500
#
bgp 100
router-id 3.3.3.3
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack0
peer 8.8.8.8 as-number 100
peer 8.8.8.8 connect-interface LoopBack0
#
ipv4-family unicast
peer 7.7.7.7 enable
peer 8.8.8.8 enable
#
ipv4-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing mixed 64
advertise l2vpn evpn import-route-multipath
#
ipv6-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing mixed 64
advertise l2vpn evpn import-route-multipath
#
l2vpn-family evpn
policy vpn-target
bestroute add-path path-number 64
peer 7.7.7.7 enable
peer 7.7.7.7 route-policy export_policy export
peer 7.7.7.7 advertise irb
peer 7.7.7.7 advertise irbv6
peer 7.7.7.7 capability-advertise add-path both
peer 7.7.7.7 advertise add-path path-number 64
peer 8.8.8.8 enable
peer 8.8.8.8 route-policy export_policy export
peer 8.8.8.8 advertise irb
peer 8.8.8.8 advertise irbv6
peer 8.8.8.8 capability-advertise add-path both
peer 8.8.8.8 advertise add-path path-number 64
#
ospf 1 router-id 3.3.3.3
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
spf-schedule-interval intelligent-timer 50 50 50
lsa-originate-interval intelligent-timer 500 50 100
lsa-arrival-interval intelligent-timer 50 50 50
area 0.0.0.0
network 3.3.3.3 0.0.0.0
network 6.6.6.6 0.0.0.0
network 192.168.3.0 0.0.0.255
network 192.168.7.0 0.0.0.255
#
route-policy export_policy deny node 10
if-match community-filter suppress_route
#
route-policy export_policy permit node 20
if-match community-filter apply_gwip_route
apply gateway-ip origin-nexthop
```

```
apply ipv6 gateway-ip origin-nexthop
#
route-policy export_policy permit node 999
#
route-policy import_static_policy permit node 100
if-match tag 82345
apply community 82345 additive
#
route-policy import_static_policy permit node 999
#
ip community-filter basic apply_gwip_route index 10 permit 82345
ip community-filter basic suppress_route index 10 permit 82346
#
ip route-static vpn-instance vpn1 202.1.1.1 255.255.255.255 10.3.1.2 preference 255 tag 82345 track
bfd-session toipu3_v4 inter-protocol-ecmp
ip route-static vpn-instance vpn1 202.1.1.1 255.255.255.255 10.4.1.2 preference 255 tag 82345 track
bfd-session toipu4_v4 inter-protocol-ecmp
#
ipv6 route-static vpn-instance vpn1 202::1 128 FC00:3::2 preference 255 tag 82345 track bfd-session
toipu3_v6 inter-protocol-ecmp
ipv6 route-static vpn-instance vpn1 202::1 128 FC00:4::2 preference 255 tag 82345 track bfd-session
toipu4_v6 inter-protocol-ecmp
#
return
```

- Leaf4 configuration file

```
#
sysname Leaf4
#
dfs-group 1
source ip 4.4.4.4
m-lag up-delay 240 auto-recovery interval 10
dual-active detection delay 0
#
vxlan-overlay network local-preference enable
#
assign forward layer-3 resource large-overlay //This step is required only on the CE8861EI,
CE6857EI, CE6865E, CE8850E-32CQ-EI, and CE6865EI.
#
assign forward ipv6 longer-mask resource share-mode //This step is required only on the CE8861EI,
CE6857EI, CE6865E, CE8850E-32CQ-EI, and CE6865EI.
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 4.4.4.4:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
ipv6-family
route-distinguisher 1.1.1.1:1
vpn-target 0:1 export-extcommunity evpn
vpn-target 0:1 import-extcommunity evpn
vxlan vni 200
#
bfd
bfd forwarding match remote-discriminator 10
bfd forwarding match remote-discriminator 30
bfd forwarding match remote-discriminator 50
bfd forwarding match remote-discriminator 70
#
bridge-domain 30
vxlan vni 130
evpn
route-distinguisher 4.4.4.4:130
vpn-target 0:130 export-extcommunity
vpn-target 0:1 export-extcommunity
```

```
vpn-target 0:130 import-extcommunity
#
bridge-domain 40
vxlan vni 140
evpn
 route-distinguisher 4.4.4.4:140
 vpn-target 0:140 export-extcommunity
 vpn-target 0:1 export-extcommunity
 vpn-target 0:140 import-extcommunity
#
interface Vbdf30
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.3.1.1 255.255.255.0
 ipv6 address FC00:3::1/64
 arp broadcast-detect enable
 mac-address 0000-5e00-0103
 ipv6 nd collect host enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Vbdf40
 ip binding vpn-instance vpn1
 ipv6 enable
 ip address 10.4.1.1 255.255.255.0
 ipv6 address FC00:4::1/6
 arp broadcast-detect enable
 mac-address 0000-5e00-0104
 ipv6 nd collect host enable
 vxlan anycast-gateway enable
 arp collect host enable
#
interface Eth-Trunk1
 mode lacp-static
 peer-link 1
#
interface Eth-Trunk10
 stp edged-port enable
 mode lacp-static
 lacp timeout fast
 dfs-group 1 m-lag 1
 arp anti-attack rate-limit 200
#
interface Eth-Trunk10.30 mode l2
 encapsulation dot1q vid 30
 bridge-domain 30
#
interface Eth-Trunk11
 stp edged-port enable
 mode lacp-static
 lacp timeout fast
 dfs-group 1 m-lag 2
 arp anti-attack rate-limit 200
#
interface Eth-Trunk11.40 mode l2
 encapsulation dot1q vid 40
 bridge-domain 40
#
interface 10GE1/0/1
 undo portswitch
 ip address 192.168.4.2 255.255.255.0
 ospf network-type p2p
 ospf peer hold-max-cost timer 800000
 port crc-statistics trigger error-down
 trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/2
 undo portswitch
 ip address 192.168.8.2 255.255.255.0
```

```
ospf network-type p2p
ospf peer hold-max-cost timer 800000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/3
eth-trunk 1
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/4
eth-trunk 1
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/5
eth-trunk 10
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface 10GE1/0/6
eth-trunk 11
storm suppression unknown-unicast 5
storm suppression multicast packets 1000
storm suppression broadcast packets 1000
port crc-statistics trigger error-down
trap-threshold crc-statistics 100 interval 10
#
interface LoopBack0
ip address 4.4.4.4 255.255.255.255
#
interface LoopBack1
ip address 6.6.6.6 255.255.255.255
#
interface Nve1
source 6.6.6.6
vni 130 head-end peer-list protocol bgp
vni 140 head-end peer-list protocol bgp
mac-address 0000-5e00-0112
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk11 downlink 2
timer recover-time 60
#
bfd toipu3_v4 bind peer-ip 10.3.1.2 vpn-instance vpn1 interface Vbdif30 source-ip 10.3.1.1 one-arm-
echo
discriminator local 20
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu3_v6 bind peer-ipv6 FC00:3::2 vpn-instance vpn1 interface Vbdif30 source-ipv6 FC00:3::1 one-
arm-echo
discriminator local 60
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu4_v4 bind peer-ip 10.4.1.2 vpn-instance vpn1 interface Vbdif40 source-ip 10.4.1.1 one-arm-
echo
discriminator local 40
detect-multiplier 4
min-echo-rx-interval 500
#
bfd toipu4_v6 bind peer-ipv6 FC00:4::2 vpn-instance vpn1 interface Vbdif40 source-ipv6 FC00:4::1 one-
```



```
arm-echo
discriminator local 80
detect-multiplier 4
min-echo-rx-interval 500
#
bgp 100
router-id 4.4.4.4
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack0
peer 8.8.8.8 as-number 100
peer 8.8.8.8 connect-interface LoopBack0
#
ipv4-family unicast
peer 7.7.7.7 enable
peer 8.8.8.8 enable
#
ipv4-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing mixed 64
advertise l2vpn evpn import-route-multipath
#
ipv6-family vpn-instance vpn1
import-route static route-policy import_static_policy
maximum load-balancing mixed 64
advertise l2vpn evpn import-route-multipath
#
l2vpn-family evpn
policy vpn-target
bestroute add-path path-number 64
peer 7.7.7.7 enable
peer 7.7.7.7 route-policy export_policy export
peer 7.7.7.7 advertise irb
peer 7.7.7.7 advertise irbv6
peer 7.7.7.7 capability-advertise add-path both
peer 7.7.7.7 advertise add-path path-number 64
peer 8.8.8.8 enable
peer 8.8.8.8 route-policy export_policy export
peer 8.8.8.8 advertise irb
peer 8.8.8.8 advertise irbv6
peer 8.8.8.8 capability-advertise add-path both
peer 8.8.8.8 advertise add-path path-number 64
#
ospf 1 router-id 4.4.4.4
bfd all-interfaces enable
bfd all-interfaces min-tx-interval 300 min-rx-interval 300 detect-multiplier 6
spf-schedule-interval intelligent-timer 50 50 50
lsa-originate-interval intelligent-timer 500 50 100
lsa-arrival-interval intelligent-timer 50 50 50
area 0.0.0.0
network 4.4.4.4 0.0.0.0
network 6.6.6.6 0.0.0.0
network 192.168.4.0 0.0.0.255
network 192.168.8.0 0.0.0.255
#
route-policy export_policy deny node 10
if-match community-filter suppress_route
#
route-policy export_policy permit node 20
if-match community-filter apply_gwip_route
apply gateway-ip origin-nexthop
apply ipv6 gateway-ip origin-nexthop
#
route-policy export_policy permit node 999
#
route-policy import_static_policy permit node 100
if-match tag 82345
apply community 82345 additive
#
route-policy import_static_policy permit node 999
```

```
#
ip community-filter basic apply_gwip_route index 10 permit 82345
ip community-filter basic suppress_route index 10 permit 82346
#
ip route-static vpn-instance vpn1 202.1.1.1 255.255.255.255 10.3.1.2 preference 255 tag 82345 track
bfd-session toipu3_v4 inter-protocol-ecmp
ip route-static vpn-instance vpn1 202.1.1.1 255.255.255.255 10.4.1.2 preference 255 tag 82345 track
bfd-session toipu4_v4 inter-protocol-ecmp
#
ipv6 route-static vpn-instance vpn1 202::1 128 FC00:3::2 preference 255 tag 82345 track bfd-session
toipu3_v6 inter-protocol-ecmp
ipv6 route-static vpn-instance vpn1 202::1 128 FC00:4::2 preference 255 tag 82345 track bfd-session
toipu4_v6 inter-protocol-ecmp
#
return
```

1.10 Example for Configuring M-LAG Lite

Applicable Products and Versions

CloudEngine 16800, 12800, 12800E, 9800, 8800, 7800, 6800, and 5800 series switches

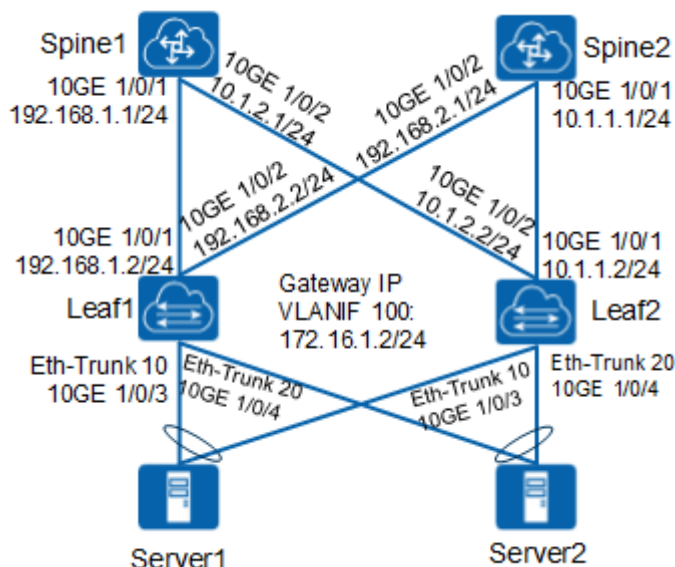
V200R003C00 or later versions

For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

On the network shown in [Figure 1-10](#), in the data center, customers want to implement device independence at the access layer, control plane isolation and device fault isolation, and the version upgrade cannot be mutually affected. Inter-device link aggregation (single-machine) configuration ensures the decoupling of control planes between multiple devices. The device does not need to be configured with a heartbeat link to synchronize protocol packets to achieve dual-active forwarding.

Figure 1-10 M-LAG lite networks



Configuration Roadmap

The configuration roadmap is as follows:

1. Configure the Eth-Trunk on Leaf switches connected to servers with the M-LAG lite way.

To ensure successful Eth-Trunk negotiation, ensure that Leaf1 and Leaf2 have the same LACP system priority and Eth-Trunks have the same interface ID and LACP system ID. In addition, the Eth-Trunk member interfaces on the two leaf switches must have different numbers in the LACP protocol to prevent LACP negotiation failures.

NOTE

If a device that supports STP is connected to leaf switches in M-LAG Lite mode, disable STP on the device and run the **stp edged-port enable** command on the downlink interfaces of the leaf switches. Otherwise, STP flapping occurs on the device.

2. Configure IP addresses on each interface.
3. Create VLANIF interfaces on leaf switches and configure the same IP addresses and MAC address for the VLANIF interfaces to provide dual-active gateways.
 - A leaf switch must be a Layer 3 gateway but not a Layer 2 transparent transmission device. If a spine switch serves as a Layer 3 gateway and servers connect to leaf switches in M-LAG Lite mode, server ARP entries learned by the spine device will have two egresses and MAC address flapping occurs. Therefore, the leaf switch to which servers connect in M-LAG Lite mode must be a Layer 3 gateway.
 - To make leaf switches function as Layer 3 gateways, that is, all traffic from servers to leaf switches be forwarded at Layer 3 (including the traffic forwarded in the same Layer 2 domain), enable proxy ARP on leaf switches.
 - Configure the same IP addresses for gateways on Leaf 1 and Leaf2 so that ECMP load balancing is performed on the traffic from spine switches to leaf switches. If a link between a server and leaf switch fails, for example, a link between Server1 and Leaf1 becomes faulty, only one link is available for traffic transmission and the traffic from Server1 to Leaf1 may be interrupted due to hash-based ECMP. Therefore, you need to ensure that the gateway interface of a leaf switch can advertise the ARP Vlink direct route to its neighbors and advertise the host route translated by the ARP to spine switches. Then you need to configure a corresponding dynamic routing protocol on leaf switches to import the direct route. If the preceding fault occurs, the ARP entry mapping Server1 on Leaf1 is deleted, the ARP Vlink direct route is unavailable, and the traffic from spine switches to Server1 is forwarded to Leaf2 to ensure normal traffic forwarding.
 - To ensure normal communication between Server1 and Server2, after learning the ARP Vlink direct route advertised by leaf switches, spine switches advertise the route to other leaf switches. In normal cases, both the host routes translated by the ARP (not delivered to the forwarding entry) and the host routes advertised by spine switches are available on leaf switches. You need to set a higher priority for the host routes translated by the ARP, to ensure that the local ARP entries are forwarded. If a link between a server and leaf switch fails, for example, a link

between Server1 and Leaf1 fails, the ARP entry mapping Server1 on Leaf1 is deleted, and the ARP Vlink direct route becomes unavailable. The traffic from Server2 to Server1 starts from Server2, Leaf1, Spine1 or Spine2, Leaf2, and finally arrives at Server1.

4. Configure the routing protocol between the Leaf layer and the Spine layer switches to establish a neighbor relationship.
5. On leaf layer switches, associate uplink and downlink interfaces with the Monitor Link group to prevent a user-side traffic forwarding failure and traffic loss due to the uplink fault.

Procedure

Step 1 Configure Inter-device link aggregation in LACP mode.

Configure Leaf1

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf1 //Modify the device name to Leaf1.
[*HUAWEI] commit
[~Leaf1] vlan batch 100
[*Leaf1] interface eth-trunk 10
[*Leaf1-Eth-Trunk10] trunkport 10ge 1/0/3
[*Leaf1-Eth-Trunk10] mode lacp-static
[*Leaf1-Eth-Trunk10] lacp system-id 00e0-cf00-0000 //Configure the same LACP system ID on Leaf1 and Leaf2.
[*Leaf1-Eth-Trunk10] port link-type trunk
[*Leaf1-Eth-Trunk10] port trunk pvid vlan 100
[*Leaf1-Eth-Trunk10] port trunk allow-pass vlan 100
[*Leaf1-Eth-Trunk10] quit
[*Leaf1] interface eth-trunk 20
[*Leaf1-Eth-Trunk20] trunkport 10ge 1/0/4
[*Leaf1-Eth-Trunk20] mode lacp-static
[*Leaf1-Eth-Trunk20] lacp system-id 00e0-cf00-0001 //Configure the same LACP system ID on Leaf1 and Leaf2.
[*Leaf1-Eth-Trunk20] port link-type trunk
[*Leaf1-Eth-Trunk20] port trunk pvid vlan 100
[*Leaf1-Eth-Trunk20] port trunk allow-pass vlan 100
[*Leaf1-Eth-Trunk20] quit
[*Leaf1] lacp priority 100 //Configure the same LACP priority on Leaf1 and Leaf2.
[*Leaf1] commit
```

Configure Leaf2

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf2 //Modify the device name to Leaf2.
[*HUAWEI] commit
[~Leaf2] vlan batch 100
[*Leaf2] interface eth-trunk 10
[*Leaf2-Eth-Trunk10] trunkport 10ge 1/0/3
[*Leaf2-Eth-Trunk10] mode lacp-static
[*Leaf2-Eth-Trunk10] lacp system-id 00e0-cf00-0000 //Configure the same LACP system ID on Leaf1 and Leaf2.
[*Leaf2-Eth-Trunk10] lacp port-id-extension enable //Configure numbers of member interfaces of an Eth-Trunk in LACP mode on the Leaf2 to increase by 32768.
[*Leaf2-Eth-Trunk10] port link-type trunk
[*Leaf2-Eth-Trunk10] port trunk pvid vlan 100
[*Leaf2-Eth-Trunk10] port trunk allow-pass vlan 100
[*Leaf2-Eth-Trunk10] quit
[*Leaf2] interface eth-trunk 20
[*Leaf2-Eth-Trunk20] trunkport 10ge 1/0/4
[*Leaf2-Eth-Trunk20] mode lacp-static
[*Leaf2-Eth-Trunk20] lacp system-id 00e0-cf00-0001 //Configure the same LACP system ID on Leaf1 and Leaf2.
[*Leaf2-Eth-Trunk20] lacp port-id-extension enable //Configure numbers of member interfaces of an Eth-Trunk in LACP mode on the Leaf2 to increase by 32768.
```

```
[*Leaf2-Eth-Trunk20] port link-type trunk
[*Leaf2-Eth-Trunk20] port trunk pvid vlan 100
[*Leaf2-Eth-Trunk20] port trunk allow-pass vlan 100
[*Leaf2-Eth-Trunk20] quit
[*Leaf2] lacp priority 100 //Configure the same LACP priority on Leaf1 and Leaf2.
[*Leaf2] commit
```

Step 2 Configure IP addresses on each interface.

Configure Spine1

```
<HUAWEI> system-view
[~HUAWEI] sysname Spine1 //Modify the device name to Spine1.
[*HUAWEI] commit
[~Spine1] interface 10ge 1/0/1
[~Spine1-10GE1/0/1] undo portswitch
[*Spine1-10GE1/0/1] port-isolate l3 enable //Enable Layer 3 port isolation. The CE5880EI and CE6880EI
do not support Layer 3 port isolation.
[*Spine1-10GE1/0/1] ip address 192.168.1.1 24
[*Spine1-10GE1/0/1] quit
[*Spine1] interface 10ge 1/0/2
[*Spine1-10GE1/0/2] undo portswitch
[*Spine1-10GE1/0/2] port-isolate l3 enable //Enable Layer 3 port isolation. The CE5880EI and CE6880EI
do not support Layer 3 port isolation.
[*Spine1-10GE1/0/2] ip address 10.1.2.1 24
[*Spine1-10GE1/0/2] quit
[*Spine1] commit
```

Configure Spine2

```
<HUAWEI> system-view
[~HUAWEI] sysname Spine2 //Modify the device name to Spine2.
[*HUAWEI] commit
[~Spine2] interface 10ge 1/0/1
[~Spine2-10GE1/0/1] undo portswitch
[*Spine2-10GE1/0/1] port-isolate l3 enable //Enable Layer 3 port isolation. The CE5880EI and CE6880EI
do not support Layer 3 port isolation.
[*Spine2-10GE1/0/1] ip address 10.10.1.1 24
[*Spine2-10GE1/0/1] quit
[*Spine2] interface 10ge 1/0/2
[*Spine2-10GE1/0/2] undo portswitch
[*Spine2-10GE1/0/2] port-isolate l3 enable //Enable Layer 3 port isolation. The CE5880EI and CE6880EI
do not support Layer 3 port isolation.
[*Spine2-10GE1/0/2] ip address 192.168.2.1 24
[*Spine2-10GE1/0/2] quit
[*Spine2] commit
```

Configure Leaf1

```
[~Leaf1] interface 10ge 1/0/1
[~Leaf1-10GE1/0/1] undo portswitch
[*Leaf1-10GE1/0/1] port-isolate l3 enable //Enable Layer 3 port isolation. The CE5880EI and CE6880EI do
not support Layer 3 port isolation.
[*Leaf1-10GE1/0/1] ip address 192.168.1.2 24
[*Leaf1-10GE1/0/1] quit
[*Leaf1] interface 10ge 1/0/2
[*Leaf1-10GE1/0/2] undo portswitch
[*Leaf1-10GE1/0/2] port-isolate l3 enable //Enable Layer 3 port isolation. The CE5880EI and CE6880EI do
not support Layer 3 port isolation.
[*Leaf1-10GE1/0/2] ip address 192.168.2.2 24
[*Leaf1-10GE1/0/2] quit
[*Leaf1] commit
```

Configure Leaf2

```
[~Leaf2] interface 10ge 1/0/1
[~Leaf2-10GE1/0/1] undo portswitch
[*Leaf2-10GE1/0/1] port-isolate l3 enable //Enable Layer 3 port isolation. The CE5880EI and CE6880EI do
not support Layer 3 port isolation.
[*Leaf2-10GE1/0/1] ip address 10.1.1.2 24
```

```
[*Leaf2-10GE1/0/1] quit
[*Leaf2] interface 10ge 1/0/2
[*Leaf2-10GE1/0/2] undo portswitch
[*Leaf2-10GE1/0/2] port-isolate l3 enable //Enable Layer 3 port isolation. The CE5880EI and CE6880EI do
not support Layer 3 port isolation.
[*Leaf2-10GE1/0/2] ip address 10.1.2.2 24
[*Leaf2-10GE1/0/2] quit
[*Leaf2] commit
```

Step 3 Create VLANIF interfaces on leaf switches and configure the same IP addresses and MAC address for the VLANIF interfaces to provide dual-active gateways.

Configure Leaf1

```
[~Leaf1] interface vlanif 100
[*Leaf1-Vlanif100] ip address 172.16.1.2 24
[*Leaf1-Vlanif100] mac-address 0000-5e00-0101
[*Leaf1-Vlanif100] arp timeout 90 //Set the aging time of dynamic ARP entries to 90 seconds.
[*Leaf1-Vlanif100] arp proxy anyway enable //Enable proxy ARP anyway on an interface, after the ARP
request packet between the servers is captured on the Leaf layer, the ARP reply is performed on the
interface MAC/IP of the Layer 3 gateway interface. After the upstream traffic of the server reaches the Leaf
layer, Layer 3 forwarding is performed.
[*Leaf1-Vlanif100] arp delete trigger link-down enable //Enables the device to delete ARP entries
learned by an interface immediately after the interface link goes Down.
[*Leaf1-Vlanif100] arp direct-route enable //Enable the ARP Vlink direct route function on the interface,
and then redistribute the route to the neighbor through the routing protocol to prevent the route blackhole
from being formed when the link is faulty.
[*Leaf1-Vlanif100] arp direct-route preference 1 //Set the priority of ARP Vlink direct routes to 1.
[*Leaf1-Vlanif100] arp direct-route delay 120 //(Optional) In V200R019C00 and later versions, you can
configure a delay in advertising ARP Vlink direct routes to prevent traffic loss caused by slow ARP Vlink
direct route establishment but fast route diversion. Only CE16800, CE12800, CE12800E, CE5880EI,
CE6850HI, CE6850U-HI, CE6851HI, CE6855HI, CE6856HI, CE6857EI, CE6860EI, CE6865EI, CE6865SI,
CE6870EI, CE6875EI, CE6880EI, CE6881, CE5881, CE6881K, CE6820, CE6863, CE6863K, CE6881E, CE7850EI,
CE7855EI, CE8850EI, CE8861EI, CE6857E, CE6857F, CE6857S, CE8850E-32CQ-EI, CE9860EI, CE8868EI, and
CE8860EI support this command.
[*Leaf1-Vlanif100] quit
[*Leaf1] commit
```

Configure Leaf2

```
[~Leaf2] interface vlanif 100
[*Leaf2-Vlanif100] ip address 172.16.1.2 24
[*Leaf2-Vlanif100] mac-address 0000-5e00-0101
[*Leaf2-Vlanif100] arp timeout 90 //Set the aging time of dynamic ARP entries to 90 seconds.
[*Leaf2-Vlanif100] arp proxy anyway enable //Enable proxy ARP anyway on an interface, after the ARP
request packet between the servers is captured on the Leaf layer, the ARP reply is performed on the
interface MAC/IP of the Layer 3 gateway interface. After the upstream traffic of the server reaches the Leaf
layer, Layer 3 forwarding is performed.
[*Leaf2-Vlanif100] arp delete trigger link-down enable //Enables the device to delete ARP entries
learned by an interface immediately after the interface link goes Down.
[*Leaf2-Vlanif100] arp direct-route enable //Enable the ARP Vlink direct route function on the interface,
and then redistribute the route to the neighbor through the routing protocol to prevent the route blackhole
from being formed when the link is faulty.
[*Leaf2-Vlanif100] arp direct-route preference 1 //Set the priority of ARP Vlink direct routes to 1.
[*Leaf2-Vlanif100] arp direct-route delay 120 //(Optional) In V200R019C00 and later versions, you can
configure a delay in advertising ARP Vlink direct routes to prevent traffic loss caused by slow ARP Vlink
direct route establishment but fast route diversion. Only CE16800, CE12800, CE12800E, CE5880EI,
CE6850HI, CE6850U-HI, CE6851HI, CE6855HI, CE6856HI, CE6857EI, CE6860EI, CE6865EI, CE6865SI,
CE6870EI, CE6875EI, CE6880EI, CE6881, CE5881, CE6881K, CE6820, CE6863, CE6863K, CE6881E, CE7850EI,
CE7855EI, CE8850EI, CE8861EI, CE6857E, CE6857F, CE6857S, CE8850E-32CQ-EI, CE9860EI, CE8868EI, and
CE8860EI support this command.
[*Leaf2-Vlanif100] quit
[*Leaf2] commit
```

NOTE

For a load balancing server connected to a leaf layer switch, the ARP reply or ARP request packet sent by the server is sent to the leaf1 device only because the HASH route is selected. The leaf 2 device cannot receive the ARP packet sent by the server. As a result, the ARP entry of the server is not learned on the interface. Therefore, the server can send ARP request packets on the interfaces of the two NICs periodically. If the server does not have the function, you can run the **arp smart-discover enable** command on the leaf layer switch to enable ARP active detection. The CE5810EI, CE5850EI, CE5850HI, CE5855EI, CE5855E, CE5855F, CE5880EI, CE6810EI, CE6850EI, CE6881K, CE6863K, CE6881E, CE6820, CE6881, CE5881, CE6863, CE6863E, and CE6880EI do not support this function.

Step 4 (Optional) Configure the traffic policy in the outbound direction to prevent the other servers from learning the ARP entries on the access switch.

NOTE

In V200R005C00 and earlier versions, the ARP entry can be learned by the hosts in the same domain. After the ARP entry is enabled, the hosts in the same domain can learn from each other. If the server server1 is changed from dual-homed to single-homed due to a link failure (link failure to the Leaf1 device), the traffic of the server accessing Server1 on the same network segment as the server Server1 is HASH to Leaf1, which will be different due to the Layer 2 domain. And cut off. In this scenario, you need to configure the global outbound application flow policy to prevent the connected service devices from learning each other's ARP entries.

Configure Leaf1

```
[~Leaf1] arp fast-reply disable //Disable the fast ARP reply function to ensure that the upstream ARP
packets are not blocked by the traffic policy.
[*Leaf1] acl 4000
[*Leaf1-acl-L2-4000] rule 5 permit type arp
[*Leaf1-acl-L2-4000] quit
[*Leaf1] traffic classifier arp-no-forwarding type and
[*Leaf1-classifier-arp-no-forwarding] if-match acl 4000
[*Leaf1-classifier-arp-no-forwarding] quit
[*Leaf1] traffic classifier arp-forwarding type and
[*Leaf1-classifier-arp-forwarding] if-match acl 4000
[*Leaf1-classifier-arp-forwarding] if-match source-mac 0000-5e00-0101 ffff-ffff-ffff
[*Leaf1] quit
[*Leaf1] traffic behavior arp-forwarding
[*Leaf1-behavior-arp-forwarding] quit
[*Leaf1] traffic behavior arp-no-forwarding
[*Leaf1-behavior-arp-no-forwarding] deny
[*Leaf1-behavior-arp-no-forwarding] quit
[*Leaf1] traffic policy arp-no-forwarding
[*Leaf1-trafficpolicy-arp-no-forwarding] classifier arp-forwarding behavior arp-forwarding precedence 5
[*Leaf1-trafficpolicy-arp-no-forwarding] classifier arp-no-forwarding behavior arp-no-forwarding
precedence 10
[*Leaf1-trafficpolicy-arp-no-forwarding] quit
[*Leaf1] traffic-policy arp-no-forwarding global outbound
[*Leaf1] commit
```

Configure Leaf2

```
[~Leaf2] arp fast-reply disable //Disable the fast ARP reply function to ensure that the upstream ARP
packets are not blocked by the traffic policy.
[*Leaf2] acl 4000
[*Leaf2-acl-L2-4000] rule 5 permit type arp
[*Leaf2-acl-L2-4000] quit
[*Leaf2] traffic classifier arp-no-forwarding type and
[*Leaf2-classifier-arp-no-forwarding] if-match acl 4000
[*Leaf2-classifier-arp-no-forwarding] quit
[*Leaf2] traffic classifier arp-forwarding type and
[*Leaf2-classifier-arp-forwarding] if-match acl 4000
[*Leaf2-classifier-arp-forwarding] if-match source-mac 0000-5e00-0101 ffff-ffff-ffff
[*Leaf2] quit
```

```
[*Leaf2] traffic behavior arp-forwarding
[*Leaf2-behavior-arp-forwarding] quit
[*Leaf2] traffic behavior arp-no-forwarding
[*Leaf2-behavior-arp-no-forwarding] deny
[*Leaf2-behavior-arp-no-forwarding] quit
[*Leaf2] traffic policy arp-no-forwarding
[*Leaf2-trafficpolicy-arp-no-forwarding] classifier arp-forwarding behavior arp-forwarding precedence 5
[*Leaf2-trafficpolicy-arp-no-forwarding] classifier arp-no-forwarding behavior arp-no-forwarding
precedence 10
[*Leaf2-trafficpolicy-arp-no-forwarding] quit
[*Leaf2] traffic-policy arp-no-forwarding global outbound
[*Leaf2] commit
```

Step 5 Configure BGP between the aggregation layer and the access layer switch to implement Layer 3 connectivity.

Configure Spine1

```
[~Spine1] bgp 65009
[*Spine1-bgp] group leaf external
[*Spine1-bgp] peer 192.168.1.2 as-number 65020
[*Spine1-bgp] peer 192.168.1.2 group leaf
[*Spine1-bgp] peer 10.1.2.2 as-number 65021
[*Spine1-bgp] peer 10.1.2.2 group leaf
[*Spine1-bgp] load-balancing as-path-relax //Disable the device from comparing the AS_Path attributes
of the same length of the routes for load balancing. This ensures that the routes sent from different leaf
switches to the spine switch can implement load balancing.
[*Spine1-bgp] timer keepalive 10 hold 30
[*Spine1-bgp] preference 20 200 10
[*Spine1-bgp] quit
[*Spine1] commit
```

Configure Spine2

```
[~Spine2] bgp 65009
[*Spine2-bgp] group leaf external
[*Spine2-bgp] peer 192.168.2.2 as-number 65020
[*Spine2-bgp] peer 192.168.2.2 group leaf
[*Spine2-bgp] peer 10.1.1.2 as-number 65021
[*Spine2-bgp] peer 10.1.1.2 group leaf
[*Spine2-bgp] load-balancing as-path-relax //Disable the device from comparing the AS_Path attributes
of the same length of the routes for load balancing. This ensures that the routes sent from different leaf
switches to the spine switch can implement load balancing.
[*Spine2-bgp] timer keepalive 10 hold 30
[*Spine2-bgp] preference 20 200 10
[*Spine2-bgp] quit
[*Spine2] commit
```

Configure Leaf1

```
[~Leaf1] bgp 65020
[*Leaf1-bgp] group spine external
[*Leaf1-bgp] peer spine as-number 65009
[*Leaf1-bgp] peer 192.168.1.1 as-number 65009
[*Leaf1-bgp] peer 192.168.1.1 group spine
[*Leaf1-bgp] peer 192.168.2.1 as-number 65009
[*Leaf1-bgp] peer 192.168.2.1 group spine
[*Leaf1-bgp] timer keepalive 10 hold 30
[*Leaf1-bgp] preference 20 200 10
[*Leaf1-bgp] import-route direct //Configure the switch to import direct routes. You can configure a
routing policy to filter unnecessary routes.
[*Leaf1-bgp] quit
[*Leaf1] commit
```

Configure Leaf2

```
[~Leaf2] bgp 65021
[*Leaf2-bgp] group spine external
[*Leaf2-bgp] peer spine as-number 65009
[*Leaf2-bgp] peer 10.1.2.1 as-number 65009
```



```
[*Leaf2-bgp] peer 10.1.2.1 group spine
[*Leaf2-bgp] peer 10.1.1.1 as-number 65009
[*Leaf2-bgp] peer 10.1.1.1 group spine
[*Leaf2-bgp] timer keepalive 10 hold 30
[*Leaf2-bgp] preference 20 200 10
[*Leaf2-bgp] import-route direct //Configure the switch to import direct routes. You can configure a
routing policy to filter unnecessary routes.
[*Leaf2-bgp] quit
[*Leaf2] commit
```

NOTE

This example configures different ASs on Leaf1 and Leaf2. If the same AS is configured on Leaf1 and Leaf2, the **peer allow-as-loop** command needs to be added to ensure that the routes of different leaf switches can be advertised to each other after passing through the spine switch.

- Step 6** On leaf layer switches, associate uplink and downlink interfaces with the Monitor Link group to prevent a user-side traffic forwarding failure and traffic loss due to the uplink fault.

Configure Leaf1

```
[~Leaf1] monitor-link group 1
[*Leaf1-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf1-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf1-mtlk-group1] port eth-trunk 10 downlink 1
[*Leaf1-mtlk-group1] port eth-trunk 20 downlink 2
[*Leaf1-mtlk-group1] quit
[*Leaf1] commit
```

Configure Leaf2

```
[~Leaf2] monitor-link group 1
[*Leaf2-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf2-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf2-mtlk-group1] port eth-trunk 10 downlink 1
[*Leaf2-mtlk-group1] port eth-trunk 20 downlink 2
[*Leaf2-mtlk-group1] quit
[*Leaf2] commit
```

----End

Verifying the Configuration

- Check the Eth-Trunk information of the device and check that the M-LAG Lite has been successfully negotiated with the server. The port status is Up.

```
[~Leaf1] display eth-trunk 10
Eth-Trunk10's state information is:
(h): high priority
(r): reference port
Local:
LAG ID: 10                Working Mode: Static
Preempt Delay: Disabled   Hash Arithmetic: profile default
System Priority: 100       System ID: 00e0-cf00-0000
Least Active-linknumber: 1 Max Active-linknumber: 32
Operating Status: up      Number Of Up Ports In Trunk: 1
Timeout Period: Slow
PortKeyMode: Auto
-----
ActorPortName      Status  PortType PortPri PortNo PortKey PortState Weight
10GE1/0/3(hr)      Selected 10GE    32768   1    2625  10111100 1
Partner:
-----
ActorPortName      SysPri  SystemID   PortPri PortNo PortKey PortState
10GE1/0/3          32768   00e0-fc12-2401 32768   1    2625  10111100
```

```
[~Leaf1] display eth-trunk 20
Eth-Trunk20's state information is:
(h): high priority
(r): reference port
Local:
LAG ID: 20                Working Mode: Static
Preempt Delay: Disabled   Hash Arithmetic: profile default
System Priority: 100       System ID: 00e0-cf00-0001
Least Active-linknumber: 1 Max Active-linknumber: 32
Operating Status: up      Number Of Up Ports In Trunk: 1
Timeout Period: Slow
PortKeyMode: Auto

-----
ActorPortName      Status  PortType PortPri PortNo PortKey PortState Weight
10GE1/0/4(hr)      Selected 10GE    32768  2    5185  10111100 1

Partner:
-----
ActorPortName      SysPri  SystemID  PortPri PortNo PortKey PortState
10GE1/0/4          32768   00e0-fc39-8b01 32768  1    5185  10111100
```

- Check the device routing relationship establishment.

```
[~Spine1] display bgp routing-table
BGP Local router ID is 10.1.1.190
Status codes: * - valid, > - best, d - damped, h - history,
               i - internal, s - suppressed, S - Stale
Origin        : i - IGP, e - EGP, ? - incomplete

Total Number of Routes: 6
  Network      NextHop      MED      LocPrf  PrefVal Path/Ogn
*> 10.1.2.0/24  10.1.2.2          0          0   65021?
*   172.16.1.0/24 10.1.2.2          0          0   65021?
*   192.168.1.2    10.1.2.2          0          0   65021?
*> 10.1.1.0/24  10.1.2.2          0          0   65021?
*   192.168.1.2    192.168.1.2       0          0   65021?
   192.168.1.0/24 192.168.1.2       0          0   65021?

[~Spine1] display ip routing-table
Proto: Protocol      Pre: Preference
Route Flags: R - relay, D - download to fib, T - to vpn-instance, B - black hole route
-----
Routing Table : _public_
Destinations : 15    Routes : 15

Destination/Mask  Proto  Pre  Cost    Flags NextHop        Interface
-----
0.0.0.0/0        Static 60   0        RD 10.1.1.1    MEth0/0/0
10.1.2.0/24      Direct 0    0        D 10.1.2.1    10GE1/0/2
10.1.2.1/32      Direct 0    0        D 127.0.0.1   10GE1/0/2
10.1.2.255/32    Direct 0    0        D 127.0.0.1   10GE1/0/2
127.0.0.0/8      Direct 0    0        D 127.0.0.1   InLoopBack0
127.0.0.1/32     Direct 0    0        D 127.0.0.1   InLoopBack0
127.255.255.255/32 Direct 0    0        D 127.0.0.1   InLoopBack0
172.16.1.0/24    EBGp   20   0        RD 10.1.2.2    10GE1/0/2
10.1.1.0/24      Direct 0    0        D 10.1.1.190  MEth0/0/0
10.1.1.190/32    Direct 0    0        D 127.0.0.1   MEth0/0/0
10.1.1.255/32    Direct 0    0        D 127.0.0.1   MEth0/0/0
192.168.1.0/24   Direct 0    0        D 192.168.1.1 10GE1/0/1
192.168.1.1/32   Direct 0    0        D 127.0.0.1   10GE1/0/1
192.168.1.255/32 Direct 0    0        D 127.0.0.1   10GE1/0/1
255.255.255.255/32 Direct 0    0        D 127.0.0.1   InLoopBack0?
```

Configuration Files

- Spine1 configuration file

```
#
sysname Spine1
#
interface 10GE1/0/1
```

```
undo portswitch
ip address 192.168.1.1 255.255.255.0
port-isolate l3 enable
#
interface 10GE1/0/2
undo portswitch
ip address 10.1.2.1 255.255.255.0
port-isolate l3 enable
#
bgp 65009
timer keepalive 10 hold 30
group leaf external
peer 10.1.2.2 as-number 65021
peer 10.1.2.2 group leaf
peer 192.168.1.2 as-number 65020
peer 192.168.1.2 group leaf
load-balancing as-path-relax
#
ipv4-family unicast
preference 20 200 10
peer leaf enable
peer 10.1.2.2 enable
peer 10.1.2.2 group leaf
peer 192.168.1.2 enable
peer 192.168.1.2 group leaf
#
return
```

- Spine2 configuration file

```
#
sysname Spine2
#
interface 10GE1/0/1
undo portswitch
ip address 10.1.1.1 255.255.255.0
port-isolate l3 enable
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.2.1 255.255.255.0
port-isolate l3 enable
#
bgp 65009
timer keepalive 10 hold 30
group leaf external
peer 10.1.1.2 as-number 65021
peer 10.1.1.2 group leaf
peer 192.168.2.2 as-number 65020
peer 192.168.2.2 group leaf
load-balancing as-path-relax
#
ipv4-family unicast
preference 20 200 10
peer leaf enable
peer 10.1.1.2 enable
peer 10.1.1.2 group leaf
peer 192.168.2.2 enable
peer 192.168.2.2 group leaf
#
return
```

- Leaf1 configuration file (V200R005C00 and earlier versions)

```
#
sysname Leaf1
#
arp fast-reply disable
#
vlan batch 100
#
lacp priority 100
```

```
#
traffic-policy arp-no-forwarding global outbound
#
acl number 4000
 rule 5 permit type arp
#
traffic classifier arp-forwarding type and
 if-match acl 4000
 if-match source-mac 0000-5e00-0101 ffff-ffff-ffff
#
traffic classifier arp-no-forwarding type and
 if-match acl 4000
#
traffic behavior arp-forwarding
#
traffic behavior arp-no-forwarding
 deny
#
traffic policy arp-no-forwarding
 classifier arp-forwarding behavior arp-forwarding precedence 5
 classifier arp-no-forwarding behavior arp-no-forwarding precedence 10
#
interface Vlanif100
 ip address 172.16.1.2 255.255.255.0
 arp timeout 90
 arp proxy anyway enable
 mac-address 0000-5e00-0101
 arp delete trigger link-down enable
 arp direct-route enable
 arp direct-route preference 1
#
interface Eth-Trunk10
 port link-type trunk
 port trunk pvid vlan 100
 port trunk allow-pass vlan 100
 mode lacp-static
 lacp system-id 00e0-cf00-0000
#
interface Eth-Trunk20
 port link-type trunk
 port trunk pvid vlan 100
 port trunk allow-pass vlan 100
 mode lacp-static
 lacp system-id 00e0-cf00-0001
#
interface 10GE1/0/1
 undo portswitch
 ip address 192.168.1.2 255.255.255.0
 port-isolate l3 enable
#
interface 10GE1/0/2
 undo portswitch
 ip address 192.168.2.2 255.255.255.0
 port-isolate l3 enable
#
interface 10GE1/0/3
 eth-trunk 10
#
interface 10GE1/0/4
 eth-trunk 20
#
monitor-link group 1
 port 10GE1/0/1 uplink
 port 10GE1/0/2 uplink
 port Eth-Trunk10 downlink 1
 port Eth-Trunk20 downlink 2
#
bgp 65020
 timer keepalive 10 hold 30
```

```
group spine external
peer spine as-number 65009
peer 192.168.1.1 as-number 65009
peer 192.168.1.1 group spine
peer 192.168.2.1 as-number 65009
peer 192.168.2.1 group spine
#
ipv4-family unicast
preference 20 200 10
import-route direct
peer spine enable
peer 192.168.1.1 enable
peer 192.168.1.1 group spine
peer 192.168.2.1 enable
peer 192.168.2.1 group spine
#
return
```

- Leaf2 configuration file (V200R005C00 and earlier versions)

```
#
sysname Leaf2
#
arp fast-reply disable
#
vlan batch 100
#
lacp priority 100
#
traffic-policy arp-no-forwarding global outbound
#
acl number 4000
rule 5 permit type arp
#
traffic classifier arp-forwarding type and
if-match acl 4000
if-match source-mac 0000-5e00-0101 ffff-ffff-ffff
#
traffic classifier arp-no-forwarding type and
if-match acl 4000
#
traffic behavior arp-forwarding
#
traffic behavior arp-no-forwarding
deny
#
traffic policy arp-no-forwarding
classifier arp-forwarding behavior arp-forwarding precedence 5
classifier arp-no-forwarding behavior arp-no-forwarding precedence 10
#
interface Vlanif100
ip address 172.16.1.2 255.255.255.0
arp timeout 90
arp proxy anyway enable
mac-address 0000-5e00-0101
arp delete trigger link-down enable
arp direct-route enable
arp direct-route preference 1
#
interface Eth-Trunk10
port link-type trunk
port trunk pvid vlan 100
port trunk allow-pass vlan 100
mode lacp-static
lacp system-id 00e0-cf00-0000
lacp port-id-extension enable
#
interface Eth-Trunk20
port link-type trunk
port trunk pvid vlan 100
port trunk allow-pass vlan 100
```

```
mode lacp-static
lacp system-id 00e0-cf00-0001
lacp port-id-extension enable
#
interface 10GE1/0/1
undo portswitch
ip address 10.1.1.2 255.255.255.0
port-isolate l3 enable
#
interface 10GE1/0/2
undo portswitch
ip address 10.1.2.2 255.255.255.0
port-isolate l3 enable
#
interface 10GE1/0/3
eth-trunk 10
#
interface 10GE1/0/4
eth-trunk 20
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk20 downlink 2
#
bgp 65021
timer keepalive 10 hold 30
group spine external
peer spine as-number 65009
peer 10.1.1.1 as-number 65009
peer 10.1.1.1 group spine
peer 10.1.2.1 as-number 65009
peer 10.1.2.1 group spine
#
ipv4-family unicast
preference 20 200 10
import-route direct
peer spine enable
peer 10.1.1.1 enable
peer 10.1.1.1 group spine
peer 10.1.2.1 enable
peer 10.1.2.1 group spine
#
return
```

- Leaf1 configuration file (V200R005C10 and later versions)

```
#
sysname Leaf1
#
vlan batch 100
#
lacp priority 100
#
interface Vlanif100
ip address 172.16.1.2 255.255.255.0
arp timeout 90
arp proxy anyway enable
mac-address 0000-5e00-0101
arp delete trigger link-down enable
arp direct-route enable
arp direct-route preference 1
arp direct-route delay 120
#
interface Eth-Trunk10
port link-type trunk
port trunk pvid vlan 100
port trunk allow-pass vlan 100
mode lacp-static
lacp system-id 00e0-cf00-0000
```

```
#
interface Eth-Trunk20
port link-type trunk
port trunk pvid vlan 100
port trunk allow-pass vlan 100
mode lacp-static
lacp system-id 00e0-cf00-0001
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.1.2 255.255.255.0
port-isolate l3 enable
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.2.2 255.255.255.0
port-isolate l3 enable
#
interface 10GE1/0/3
eth-trunk 10
#
interface 10GE1/0/4
eth-trunk 20
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk20 downlink 2
#
bgp 65020
timer keepalive 10 hold 30
group spine external
peer spine as-number 65009
peer 192.168.1.1 as-number 65009
peer 192.168.1.1 group spine
peer 192.168.2.1 as-number 65009
peer 192.168.2.1 group spine
#
ipv4-family unicast
preference 20 200 10
import-route direct
peer spine enable
peer 192.168.1.1 enable
peer 192.168.1.1 group spine
peer 192.168.2.1 enable
peer 192.168.2.1 group spine
#
return
```

- Leaf2 configuration file (V200R005C10 and later versions)

```
#
sysname Leaf2
#
vlan batch 100
#
lacp priority 100
#
interface Vlanif100
ip address 172.16.1.2 255.255.255.0
arp timeout 90
arp proxy anyway enable
mac-address 0000-5e00-0101
arp delete trigger link-down enable
arp direct-route enable
arp direct-route preference 1
arp direct-route delay 120
#
interface Eth-Trunk10
port link-type trunk
```

```
port trunk pvid vlan 100
port trunk allow-pass vlan 100
mode lacp-static
lacp system-id 00e0-cf00-0000
lacp port-id-extension enable
#
interface Eth-Trunk20
port link-type trunk
port trunk pvid vlan 100
port trunk allow-pass vlan 100
mode lacp-static
lacp system-id 00e0-cf00-0001
lacp port-id-extension enable
#
interface 10GE1/0/1
undo portswitch
ip address 10.1.1.2 255.255.255.0
port-isolate l3 enable
#
interface 10GE1/0/2
undo portswitch
ip address 10.1.2.2 255.255.255.0
port-isolate l3 enable
#
interface 10GE1/0/3
eth-trunk 10
#
interface 10GE1/0/4
eth-trunk 20
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk20 downlink 2
#
bgp 65021
timer keepalive 10 hold 30
group spine external
peer spine as-number 65009
peer 10.1.1.1 as-number 65009
peer 10.1.1.1 group spine
peer 10.1.2.1 as-number 65009
peer 10.1.2.1 group spine
#
ipv4-family unicast
preference 20 200 10
import-route direct
peer spine enable
peer 10.1.1.1 enable
peer 10.1.1.1 group spine
peer 10.1.2.1 enable
peer 10.1.2.1 group spine
#
return
```

1.11 Example for Configuring IPv6 M-LAG Lite

Applicable Products and Versions

The following products running V200R019C00 or later versions:

- CE12800E (configured with FD-X series cards)

- CE6850HI, CE6850U-HI, CE6851HI, CE6855HI, CE6856HI, CE6857EI, CE6857E, CE6857S, CE6857F, CE6860EI, CE6865EI, CE6865E, CE8850E-32CQ-EI, CE7850EI, CE7855EI, CE8850EI, CE8860EI, CE8861EI, and CE8868EI

The following products running V200R005C20 or later versions:

- CloudEngine 16800, CloudEngine 6881, and CloudEngine 6863

The following products running V200R019C10 or later versions:

- CE6881K, CE6881E, and CE6863K

The following products running V200R020C00 or later versions:

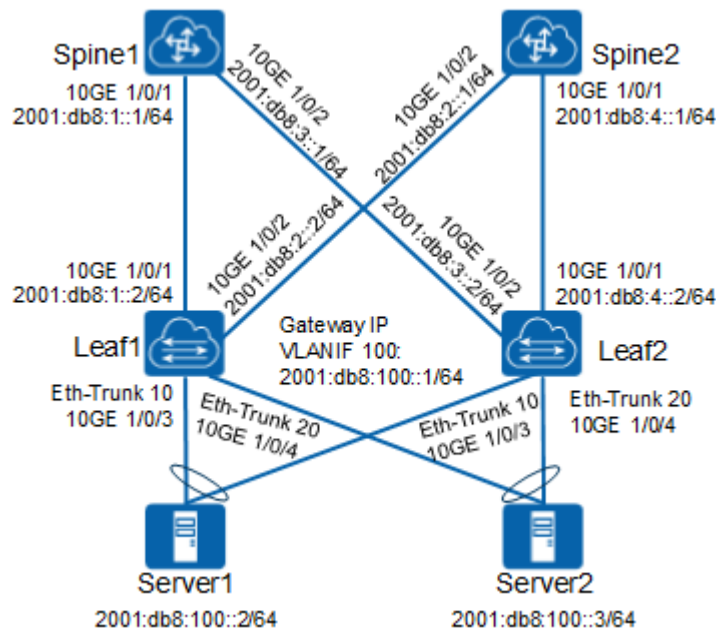
- CE9860EI and CE5881

For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

As shown in [Figure 1-11](#), on an IPv6 network, data center customers require device isolation at the access layer, control plane isolation, and device fault isolation, packet forwarding in active-active mode without being affected by upgrade. Multichassis Link Aggregation Group (M-LAG) Lite decouples the control planes of multiple devices, and achieves packet active-active forwarding in active-active mode among the devices, without the need to configure heartbeat links to synchronize data through negotiation packets.

Figure 1-11 IPv6 M-LAG Lite networking



Configuration Roadmap

NOTE

This example only explains the configuration of IPv6 M-LAG Lite. If you need to configure M-LAG Lite over the IPv4/IPv6 dual stack, see [1.10 Example for Configuring M-LAG Lite](#) and this example.

The configuration roadmap is as follows:

1. Configure servers to connect to leaf switches in M-LAG Lite mode.

To ensure successful Eth-Trunk negotiation, ensure that Leaf1 and Leaf2 have the same LACP system priority and Eth-Trunks have the same interface ID and LACP system ID. In addition, the Eth-Trunk member interfaces on the two leaf switches must have different numbers in the LACP protocol to prevent LACP negotiation failures.

 NOTE

If a device that supports STP is connected to leaf switches in M-LAG Lite mode, disable STP on the device and run the **stp edged-port enable** command on the downlink interfaces of the leaf switches. Otherwise, STP flapping occurs on the device.

2. Configure an IPv6 address for each interface.
3. Configure the same IPv6 address and MAC address for VLANIF interfaces on leaf switches. The leaf switches function as the active-active gateways of servers.
 - A leaf switch must be a Layer 3 gateway but not a Layer 2 transparent transmission device. If a spine switch serves as a Layer 3 gateway and servers connect to leaf switches in M-LAG Lite mode, server ND entries learned by the spine device will have two egresses and MAC address flapping occurs. Therefore, the leaf switch to which servers connect in M-LAG Lite mode must be a Layer 3 gateway.
 - To make leaf switches function as Layer 3 gateways, that is, all traffic from servers to leaf switches be forwarded at Layer 3 (including the traffic forwarded in the same Layer 2 domain), enable proxy ND on leaf switches.
 - Configure the same IPv6 addresses for gateways on Leaf 1 and Leaf2 so that ECMP load balancing is performed on the traffic from spine switches to leaf switches. If a link between a server and leaf switch fails, for example, a link between Server1 and Leaf1 becomes faulty, only one link is available for traffic transmission and the traffic from Server1 to Leaf1 may be interrupted due to hash-based ECMP. Therefore, you need to ensure that the gateway interface of a leaf switch can advertise the NDP Vlink direct route to its neighbors and advertise the host route translated by the ND to spine switches. Then you need to configure a corresponding dynamic routing protocol on leaf switches to import the direct route. If the preceding fault occurs, the ND entry mapping Server1 on Leaf1 is deleted, the NDP Vlink direct route is unavailable, and the traffic from spine switches to Server1 is forwarded to Leaf2 to ensure normal traffic forwarding.
 - To ensure normal communication between Server1 and Server2, after learning the NDP Vlink direct route advertised by leaf switches, spine switches advertise the route to other leaf switches. In normal cases, both the host routes translated by the ND (not delivered to the forwarding entry) and the host routes advertised by spine switches are available on leaf switches. You need to set a higher priority for the host routes translated by the ND, to ensure that the local ND entries are forwarded. If a link between a server and leaf switch fails, for example, a link between Server1 and Leaf1 fails, the ND entry mapping Server1 on Leaf1 is deleted, and the NDP Vlink direct route becomes unavailable. The

traffic from Server2 to Server1 starts from Server2, Leaf1, Spine1 or Spine2, Leaf2, and finally arrives at Server1.

4. Configure a routing protocol between leaf and spine switches to establish neighbor relationships.
5. Associate uplink and downlink interfaces in the Monitor Link group on the leaf switch to prevent user-side traffic from being discarded due to uplink faults.

Procedure

Step 1 Configure inter-device link aggregation in LACP mode.

Configure Leaf1.

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf1 //Change the device name to Leaf1.
[*HUAWEI] commit
[~Leaf1] vlan batch 100
[*Leaf1] interface eth-trunk 10
[*Leaf1-Eth-Trunk10] trunkport 10ge 1/0/3
[*Leaf1-Eth-Trunk10] mode lacp-static
[*Leaf1-Eth-Trunk10] lacp system-id 00e0-cf00-0000 //Configure the same LACP system ID on Leaf1 and Leaf2.
[*Leaf1-Eth-Trunk10] port link-type trunk
[*Leaf1-Eth-Trunk10] port trunk pvid vlan 100
[*Leaf1-Eth-Trunk10] port trunk allow-pass vlan 100
[*Leaf1-Eth-Trunk10] quit
[*Leaf1] interface eth-trunk 20
[*Leaf1-Eth-Trunk20] trunkport 10ge 1/0/4
[*Leaf1-Eth-Trunk20] mode lacp-static
[*Leaf1-Eth-Trunk20] lacp system-id 00e0-cf00-0001 //Configure the same LACP system ID on Leaf1 and Leaf2.
[*Leaf1-Eth-Trunk20] port link-type trunk
[*Leaf1-Eth-Trunk20] port trunk pvid vlan 100
[*Leaf1-Eth-Trunk20] port trunk allow-pass vlan 100
[*Leaf1-Eth-Trunk20] quit
[*Leaf1] lacp priority 100 //Configure the same LACP system priority on Leaf1 and Leaf2.
[*Leaf1] commit
```

Configure Leaf2.

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf2 //Change the device name to Leaf2.
[*HUAWEI] commit
[~Leaf2] vlan batch 100
[*Leaf2] interface eth-trunk 10
[*Leaf2-Eth-Trunk10] trunkport 10ge 1/0/3
[*Leaf2-Eth-Trunk10] mode lacp-static
[*Leaf2-Eth-Trunk10] lacp system-id 00e0-cf00-0000 //Configure the same LACP system ID on Leaf1 and Leaf2.
[*Leaf2-Eth-Trunk10] lacp port-id-extension enable //Enable the extension of Eth-Trunk member interface IDs on leaf2. Increase IDs of each Eth-Trunk member interface in LACP mode by 32768 to prevent the same interface ID of Eth-Trunk member interfaces on the two leaf switches.
[*Leaf2-Eth-Trunk10] port link-type trunk
[*Leaf2-Eth-Trunk10] port trunk pvid vlan 100
[*Leaf2-Eth-Trunk10] port trunk allow-pass vlan 100
[*Leaf2-Eth-Trunk10] quit
[*Leaf2] interface eth-trunk 20
[*Leaf2-Eth-Trunk20] trunkport 10ge 1/0/4
[*Leaf2-Eth-Trunk20] mode lacp-static
[*Leaf2-Eth-Trunk20] lacp system-id 00e0-cf00-0001 //Configure the same LACP system ID on Leaf1 and Leaf2.
[*Leaf2-Eth-Trunk20] lacp port-id-extension enable //Enable the extension of Eth-Trunk member interface IDs on leaf2. Increase IDs of each Eth-Trunk member interface in LACP mode by 32768 to prevent the same interface ID of Eth-Trunk member interfaces on the two leaf switches.
[*Leaf2-Eth-Trunk20] port link-type trunk
```

```
[*Leaf2-Eth-Trunk20] port trunk pvid vlan 100
[*Leaf2-Eth-Trunk20] port trunk allow-pass vlan 100
[*Leaf2-Eth-Trunk20] quit
[*Leaf2] lacp priority 100 //Configure the same LACP system priority on Leaf1 and Leaf2.
[*Leaf2] commit
```

Step 2 Configure interfaces and IPv6 addresses for the interfaces.

Configure Spine1.

```
<HUAWEI> system-view
[~HUAWEI] sysname Spine1 //Change the device name to Spine1.
[*HUAWEI] commit
[~Spine1] interface 10ge 1/0/1
[~Spine1-10GE1/0/1] undo portswitch
[*Spine1-10GE1/0/1] port-isolate l3 enable
[*Spine1-10GE1/0/1] ipv6 enable
[*Spine1-10GE1/0/1] ipv6 address 2001:db8:1::1 64
[*Spine1-10GE1/0/1] quit
[*Spine1] interface 10ge 1/0/2
[*Spine1-10GE1/0/2] undo portswitch
[*Spine1-10GE1/0/2] port-isolate l3 enable
[*Spine1-10GE1/0/2] ipv6 enable
[*Spine1-10GE1/0/2] ipv6 address 2001:db8:3::1 64
[*Spine1-10GE1/0/2] quit
[*Spine1] commit
```

Configure Spine2.

```
<HUAWEI> system-view
[~HUAWEI] sysname Spine2 //Change the device name to Spine2.
[*HUAWEI] commit
[~Spine2] interface 10ge 1/0/1
[~Spine2-10GE1/0/1] undo portswitch
[*Spine2-10GE1/0/1] port-isolate l3 enable
[*Spine2-10GE1/0/1] ipv6 enable
[*Spine2-10GE1/0/1] ipv6 address 2001:db8:4::1 64
[*Spine2-10GE1/0/1] quit
[*Spine2] interface 10ge 1/0/2
[*Spine2-10GE1/0/2] undo portswitch
[*Spine2-10GE1/0/2] port-isolate l3 enable
[*Spine2-10GE1/0/2] ipv6 enable
[*Spine2-10GE1/0/2] ipv6 address 2001:db8:2::1 64
[*Spine2-10GE1/0/2] quit
[*Spine2] commit
```

Configure Leaf1.

```
[~Leaf1] interface 10ge 1/0/1
[~Leaf1-10GE1/0/1] undo portswitch
[*Leaf1-10GE1/0/1] port-isolate l3 enable
[*Leaf1-10GE1/0/1] ipv6 enable
[*Leaf1-10GE1/0/1] ipv6 address 2001:db8:1::2 64
[*Leaf1-10GE1/0/1] quit
[*Leaf1] interface 10ge 1/0/2
[*Leaf1-10GE1/0/2] undo portswitch
[*Leaf1-10GE1/0/2] port-isolate l3 enable
[*Leaf1-10GE1/0/2] ipv6 enable
[*Leaf1-10GE1/0/2] ipv6 address 2001:db8:2::2 64
[*Leaf1-10GE1/0/2] quit
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] interface 10ge 1/0/1
[~Leaf2-10GE1/0/1] undo portswitch
[*Leaf2-10GE1/0/1] port-isolate l3 enable
[*Leaf2-10GE1/0/1] ipv6 enable
[*Leaf2-10GE1/0/1] ipv6 address 2001:db8:4::2 64
[*Leaf2-10GE1/0/1] quit
```

```
[*Leaf2] interface 10ge 1/0/2
[*Leaf2-10GE1/0/2] undo portswitch
[*Leaf2-10GE1/0/2] port-isolate l3 enable
[*Leaf2-10GE1/0/2] ipv6 enable
[*Leaf2-10GE1/0/2] ipv6 address 2001:db8:3::2 64
[*Leaf2-10GE1/0/2] quit
[*Leaf2] commit
```

Step 3 Configure the same IPv6 address and MAC address for VLANIF interfaces on leaf switches. The leaf switches function as the active-active gateways of access servers.

Configure Leaf1.

```
[~Leaf1] interface vlanif 100
[*Leaf1-Vlanif100] ipv6 enable
[*Leaf1-Vlanif100] ipv6 address 2001:db8:100::1 64
[*Leaf1-Vlanif100] ipv6 address FE80::300:3EFF:FE11:985 link-local
[*Leaf1-Vlanif100] mac-address 0000-5e00-0101
[*Leaf1-Vlanif100] ipv6 nd proxy anyway enable //Enable any proxy ND response. After obtaining NS
request packets sent by servers, leaf1 responds with the MAC address of its Layer 3 gateway interface. In
this way, uplink traffic of servers is forwarded at Layer 3 after reaching leaf1.
[*Leaf1-Vlanif100] ipv6 nd delete trigger link-down enable //Enable fast deletion of ND entries if a link
is Down. If a member interface goes Down, ND entries on the interface are deleted immediately.
[*Leaf1-Vlanif100] ipv6 nd direct-route enable //Enable interfaces to advertise NDP Vlink direct routes
and import direct routes through the routing protocol. This enables the device to advertise ND-translated
host routes to neighbors and prevents route black holes caused by link faults.
[*Leaf1-Vlanif100] ipv6 nd direct-route preference 1 //Set the priority of an NDP Vlink direct route to 1.
[*Leaf1-Vlanif100] ipv6 nd direct-route delay 120 //(Optional) In V200R019C00 and later versions, you
can configure a delay in advertising IPv6 NDP Vlink direct routes to prevent traffic loss caused by slow IPv6
NDP Vlink direct route establishment but fast route diversion. Only CE16800, CE12800, CE12800E,
CE5880EI, CE6850HI, CE6850U-HI, CE6851HI, CE6855HI, CE6856HI, CE6857EI, CE6860EI, CE6865EI,
CE6865SI, CE6870EI, CE6875EI, CE6880EI, CE6881, CE5881, CE6881K, CE6820, CE6863, CE6863K, CE6881E,
CE7850EI, CE7855EI, CE8850EI, CE8861EI, CE6857E, CE6857F, CE8850E-32CQ-EI, CE9860EI, CE8868EI, and
CE8860EI support this command.
[*Leaf1-Vlanif100] quit
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] interface vlanif 100
[*Leaf2-Vlanif100] ipv6 enable
[*Leaf2-Vlanif100] ipv6 address 2001:db8:100::1 64
[*Leaf2-Vlanif100] ipv6 address FE80::300:3EFF:FE11:985 link-local
[*Leaf2-Vlanif100] mac-address 0000-5e00-0101
[*Leaf2-Vlanif100] ipv6 nd proxy anyway enable //Enable any proxy ND response. After obtaining NS
request packets sent by servers, leaf2 responds with the MAC address of its Layer 3 gateway interface. In
this way, uplink traffic of servers is forwarded at Layer 3 after reaching leaf2.
[*Leaf2-Vlanif100] ipv6 nd delete trigger link-down enable //Enable fast deletion of ND entries if a link
is Down. If a member interface goes Down, ND entries on the interface are deleted immediately.
[*Leaf2-Vlanif100] ipv6 nd direct-route enable // Enable interfaces to advertise NDP Vlink direct routes
and import direct routes through the routing protocol. This enables the device to advertise ND-translated
host routes to neighbors and prevents route black holes caused by link faults.
[*Leaf2-Vlanif100] ipv6 nd direct-route preference 1 //Set the priority of an NDP Vlink direct route to 1.
[*Leaf2-Vlanif100] ipv6 nd direct-route delay 120 //(Optional) In V200R019C00 and later versions, you
can configure a delay in advertising IPv6 NDP Vlink direct routes to prevent traffic loss caused by slow IPv6
NDP Vlink direct route establishment but fast route diversion. Only CE16800, CE12800, CE12800E,
CE5880EI, CE6850HI, CE6850U-HI, CE6851HI, CE6855HI, CE6856HI, CE6857EI, CE6860EI, CE6865EI,
CE6865SI, CE6870EI, CE6875EI, CE6880EI, CE6881, CE5881, CE6881K, CE6820, CE6863, CE6863K, CE6881E,
CE7850EI, CE7855EI, CE8850EI, CE8861EI, CE6857E, CE6857F, CE8850E-32CQ-EI, CE9860EI, CE8868EI, and
CE8860EI support this command.
[*Leaf2-Vlanif100] quit
[*Leaf2] commit
```

Step 4 Configure BGP on the aggregation and access switches to implement Layer 3 interworking.

Configure Spine1.

```
[~Spine1] bgp 65009
[*Spine1-bgp] router-id 172.16.1.1
[*Spine1-bgp] group leaf external
[*Spine1-bgp] peer 2001:db8:1::2 as-number 65020
[*Spine1-bgp] peer 2001:db8:1::2 group leaf
[*Spine1-bgp] peer 2001:db8:3::2 as-number 65021
[*Spine1-bgp] peer 2001:db8:3::2 group leaf
[*Spine1-bgp] timer keepalive 10 hold 30
[*Spine1-bgp] ipv6-family unicast
[*Spine1-bgp-af-ipv6] load-balancing as-path-relax //Disable the device from comparing the AS_Path
attributes of the same length of the routes for load balancing. This ensures that the routes sent from
different leaf switches to the spine switch can implement load balancing.
[*Spine1-bgp-af-ipv6] preference 20 200 10
[*Spine1-bgp-af-ipv6] peer 2001:db8:1::2 enable
[*Spine1-bgp-af-ipv6] peer 2001:db8:3::2 enable
[*Spine1-bgp-af-ipv6] network 2001:db8:1:: 64
[*Spine1-bgp-af-ipv6] network 2001:db8:3:: 64
[*Spine1-bgp-af-ipv6] maximum load-balancing 32
[*Spine1-bgp-af-ipv6] quit
[*Spine1-bgp] quit
[*Spine1] commit
```

Configure Spine2.

```
[~Spine2] bgp 65009
[*Spine2-bgp] router-id 172.16.2.1
[*Spine2-bgp] group leaf external
[*Spine2-bgp] peer 2001:db8:2::2 as-number 65020
[*Spine2-bgp] peer 2001:db8:2::2 group leaf
[*Spine2-bgp] peer 2001:db8:4::2 as-number 65021
[*Spine2-bgp] peer 2001:db8:4::2 group leaf
[*Spine2-bgp] timer keepalive 10 hold 30
[*Spine2-bgp] ipv6-family unicast
[*Spine2-bgp-af-ipv6] load-balancing as-path-relax //Disable the device from comparing the AS_Path
attributes of the same length of the routes for load balancing. This ensures that the routes sent from
different leaf switches to the spine switch can implement load balancing.
[*Spine2-bgp-af-ipv6] preference 20 200 10
[*Spine2-bgp-af-ipv6] peer 2001:db8:2::2 enable
[*Spine2-bgp-af-ipv6] peer 2001:db8:4::2 enable
[*Spine2-bgp-af-ipv6] network 2001:db8:2:: 64
[*Spine2-bgp-af-ipv6] network 2001:db8:4:: 64
[*Spine2-bgp-af-ipv6] maximum load-balancing 32
[*Spine2-bgp-af-ipv6] quit
[*Spine2-bgp] quit
[*Spine2] commit
```

Configure Leaf1.

```
[~Leaf1] bgp 65020
[*Leaf1-bgp] router-id 172.16.3.1
[*Leaf1-bgp] group spine external
[*Leaf1-bgp] peer spine as-number 65009
[*Leaf1-bgp] peer 2001:db8:1::1 as-number 65009
[*Leaf1-bgp] peer 2001:db8:1::1 group spine
[*Leaf1-bgp] peer 2001:db8:2::1 as-number 65009
[*Leaf1-bgp] peer 2001:db8:2::1 group spine
[*Leaf1-bgp] timer keepalive 10 hold 30
[*Leaf1-bgp] ipv6-family unicast
[*Leaf1-bgp-af-ipv6] preference 20 200 10
[*Leaf1-bgp-af-ipv6] peer 2001:db8:1::1 enable
[*Leaf1-bgp-af-ipv6] peer 2001:db8:2::1 enable
[*Leaf1-bgp-af-ipv6] network 2001:db8:1:: 64
[*Leaf1-bgp-af-ipv6] network 2001:db8:2:: 64
[*Leaf1-bgp-af-ipv6] network 2001:db8:100:: 64
[*Leaf1-bgp-af-ipv6] import-route direct // Import the routes learned through direct routes. You can
configure a routing policy based on the actual networking to filter out unnecessary routes.
[*Leaf1-bgp-af-ipv6] maximum load-balancing 32
[*Leaf1-bgp-af-ipv6] quit
[*Leaf1-bgp] quit
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] bgp 65021
[*Leaf2-bgp] router-id 172.16.4.1
[*Leaf2-bgp] group spine external
[*Leaf2-bgp] peer spine as-number 65009
[*Leaf2-bgp] peer 2001:db8:3::1 as-number 65009
[*Leaf2-bgp] peer 2001:db8:3::1 group spine
[*Leaf2-bgp] peer 2001:db8:4::1 as-number 65009
[*Leaf2-bgp] peer 2001:db8:4::1 group spine
[*Leaf2-bgp] timer keepalive 10 hold 30
[*Leaf2-bgp] ipv6-family unicast
[*Leaf2-bgp-af-ipv6] preference 20 200 10
[*Leaf2-bgp-af-ipv6] peer 2001:db8:3::1 enable
[*Leaf2-bgp-af-ipv6] peer 2001:db8:4::1 enable
[*Leaf2-bgp-af-ipv6] network 2001:db8:3:: 64
[*Leaf2-bgp-af-ipv6] network 2001:db8:4:: 64
[*Leaf2-bgp-af-ipv6] network 2001:db8:100:: 64
[*Leaf2-bgp-af-ipv6] import-route direct //Import the routes learned through direct routes. You can
configure a routing policy based on the actual networking to filter out unnecessary routes.
[*Leaf2-bgp-af-ipv6] maximum load-balancing 32
[*Leaf2-bgp-af-ipv6] quit
[*Leaf2-bgp] quit
[*Leaf2] commit
```

NOTE

This example configures different ASs on Leaf1 and Leaf2. If the same AS is configured on Leaf1 and Leaf2, the **peer allow-as-loop** command needs to be added to ensure that the routes of different leaf switches can be advertised to each other after passing through the spine switch.

Step 5 Configure the uplink and downlink interfaces associated with Monitor Link on leaf switches.

Configure Leaf1.

```
[~Leaf1] monitor-link group 1
[*Leaf1-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf1-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf1-mtlk-group1] port eth-trunk 10 downlink 1
[*Leaf1-mtlk-group1] port eth-trunk 20 downlink 2
[*Leaf1-mtlk-group1] timer recover-time 60
[*Leaf1-mtlk-group1] quit
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] monitor-link group 1
[*Leaf2-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf2-mtlk-group1] port 10ge 1/0/2 uplink
[*Leaf2-mtlk-group1] port eth-trunk 10 downlink 1
[*Leaf2-mtlk-group1] port eth-trunk 20 downlink 2
[*Leaf2-mtlk-group1] timer recover-time 60
[*Leaf2-mtlk-group1] quit
[*Leaf2] commit
```

----End

Verifying the Configuration

- Check the Eth-Trunk information on Leaf1. The information shows that M-LAG Lite has successfully negotiated with the server and interfaces are Up.

```
[~Leaf1] display eth-trunk 10
Eth-Trunk10's state information is:
(h): high priority
(r): reference port
```

```

Local:
LAG ID: 10                      Working Mode: Static
Preempt Delay: Disabled        Hash Arithmetic: profile default
System Priority: 100            System ID: 00e0-cf00-0000
Least Active-linknumber: 1     Max Active-linknumber: 32
Operating Status: up           Number Of Up Ports In Trunk: 1
Timeout Period: Slow
PortKeyMode: Auto
-----
ActorPortName      Status  PortType PortPri PortNo PortKey PortState Weight
10GE1/0/3(hr)      Selected 10GE    32768  1    2625  10111100 1

Partner:
-----
ActorPortName      SysPri SystemID      PortPri PortNo PortKey PortState
10GE1/0/3          32768 00e0-fc12-2401 32768  1    2625  10111100
[~Leaf1] display eth-trunk 20
Eth-Trunk20's state information is:
(h): high priority
(r): reference port
Local:
LAG ID: 20                      Working Mode: Static
Preempt Delay: Disabled        Hash Arithmetic: profile default
System Priority: 100            System ID: 00e0-cf00-0001
Least Active-linknumber: 1     Max Active-linknumber: 32
Operating Status: up           Number Of Up Ports In Trunk: 1
Timeout Period: Slow
PortKeyMode: Auto
-----
ActorPortName      Status  PortType PortPri PortNo PortKey PortState Weight
10GE1/0/4(hr)      Selected 10GE    32768  2    5185  10111100 1

Partner:
-----
ActorPortName      SysPri SystemID      PortPri PortNo PortKey PortState
10GE1/0/4          32768 00e0-fc39-8b01 32768  1    5185  10111100

```

- Check whether Server1 and Server2 can communicate with each other.
Set the IPv6 addresses of Server1 and Server2 to 2001:db8:100::2/64 and 2001:db8:100::3/64 respectively.
After the configuration is complete, Server1 and Server2 can ping each other.

Configuration Files

- Spine1 configuration file

sysname Spine1

interface 10GE1/0/1
undo portswitch
ipv6 enable
ipv6 address 2001:DB8:1::1/64
port-isolate l3 enable

interface 10GE1/0/2
undo portswitch
ipv6 enable
ipv6 address 2001:DB8:3::1/64
port-isolate l3 enable

bgp 65009
router-id 172.16.1.1
timer keepalive 10 hold 30
group leaf external
peer 2001:DB8:1::2 as-number 65020
peer 2001:DB8:1::2 group leaf


```
peer 2001:DB8:3::2 as-number 65021
peer 2001:DB8:3::2 group leaf
#
ipv6-family unicast
 preference 20 200 10
 network 2001:DB8:1:: 64
 network 2001:DB8:3:: 64
 maximum load-balancing 32
 peer 2001:DB8:1::2 enable
 peer 2001:DB8:3::2 enable
 load-balancing as-path-relax
#
return
```

- Spine2 configuration file

```
#
sysname Spine2
#
interface 10GE1/0/1
 undo portswitch
 ipv6 enable
 ipv6 address 2001:DB8:4::1/64
 port-isolate l3 enable
#
interface 10GE1/0/2
 undo portswitch
 ipv6 enable
 ipv6 address 2001:DB8:2::1/64
 port-isolate l3 enable
#
bgp 65009
 router-id 172.16.2.1
 timer keepalive 10 hold 30
 group leaf external
 peer 2001:DB8:2::2 as-number 65020
 peer 2001:DB8:2::2 group leaf
 peer 2001:DB8:4::2 as-number 65021
 peer 2001:DB8:4::2 group leaf
#
ipv6-family unicast
 preference 20 200 10
 network 2001:DB8:2:: 64
 network 2001:DB8:4:: 64
 maximum load-balancing 32
 peer 2001:DB8:2::2 enable
 peer 2001:DB8:4::2 enable
 load-balancing as-path-relax
#
return
```

- Leaf1 configuration file

```
#
sysname Leaf1
#
vlan batch 100
#
lACP priority 100
#
interface Vlanif100
 ipv6 enable
 ipv6 address 2001:DB8:100::1/64
 ipv6 address FE80::300:3EFF:FE11:985 link-local
 mac-address 0000-5e00-0101
 ipv6 nd proxy anyway enable
 ipv6 nd delete trigger link-down enable
 ipv6 nd direct-route enable
 ipv6 nd direct-route delay 120
 ipv6 nd direct-route preference 1
#
interface Eth-Trunk10
```

```
port link-type trunk
port trunk pvid vlan 100
port trunk allow-pass vlan 100
mode lacp-static
lacp system-id 00e0-cf00-0000
#
interface Eth-Trunk20
port link-type trunk
port trunk pvid vlan 100
port trunk allow-pass vlan 100
mode lacp-static
lacp system-id 00e0-cf00-0001
#
interface 10GE1/0/1
undo portswitch
ipv6 enable
ipv6 address 2001:DB8:1::2/64
port-isolate l3 enable
#
interface 10GE1/0/2
undo portswitch
ipv6 enable
ipv6 address 2001:DB8:2::2/64
port-isolate l3 enable
#
interface 10GE1/0/3
eth-trunk 10
#
interface 10GE1/0/4
eth-trunk 20
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk20 downlink 2
timer recover-time 60
#
bgp 65020
router-id 172.16.3.1
timer keepalive 10 hold 30
group spine external
peer spine as-number 65009
peer 2001:DB8:1::1 as-number 65009
peer 2001:DB8:1::1 group spine
peer 2001:DB8:2::1 as-number 65009
peer 2001:DB8:2::1 group spine
#
ipv6-family unicast
preference 20 200 10
network 2001:DB8:1:: 64
network 2001:DB8:2:: 64
network 2001:DB8:100:: 64
import-route direct
maximum load-balancing 32
peer 2001:DB8:1::1 enable
peer 2001:DB8:2::1 enable
#
return
```

- Leaf2 configuration file

```
#
sysname Leaf2
#
vlan batch 100
#
lacp priority 100
#
interface Vlanif100
ipv6 enable
```

```
ipv6 address 2001:DB8:100::1/64
ipv6 address FE80::300:3EFF:FE11:985 link-local
mac-address 0000-5e00-0101
ipv6 nd proxy anyway enable
ipv6 nd delete trigger link-down enable
ipv6 nd direct-route enable
ipv6 nd direct-route delay 120
ipv6 nd direct-route preference 1
#
interface Eth-Trunk10
port link-type trunk
port trunk pvid vlan 100
port trunk allow-pass vlan 100
mode lacp-static
lacp system-id 00e0-cf00-0000
lacp port-id-extension enable
#
interface Eth-Trunk20
port link-type trunk
port trunk pvid vlan 100
port trunk allow-pass vlan 100
mode lacp-static
lacp system-id 00e0-cf00-0001
lacp port-id-extension enable
#
interface 10GE1/0/1
undo portswitch
ipv6 enable
ipv6 address 2001:DB8:4::2/64
port-isolate l3 enable
#
interface 10GE1/0/2
undo portswitch
ipv6 enable
ipv6 address 2001:DB8:3::2/64
port-isolate l3 enable
#
interface 10GE1/0/3
eth-trunk 10
#
interface 10GE1/0/4
eth-trunk 20
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/2 uplink
port Eth-Trunk10 downlink 1
port Eth-Trunk20 downlink 2
timer recover-time 60
#
bgp 65021
router-id 172.16.4.1
timer keepalive 10 hold 30
group spine external
peer spine as-number 65009
peer 2001:DB8:3::1 as-number 65009
peer 2001:DB8:3::1 group spine
peer 2001:DB8:4::1 as-number 65009
peer 2001:DB8:4::1 group spine
#
ipv6-family unicast
preference 20 200 10
network 2001:DB8:3:: 64
network 2001:DB8:4:: 64
network 2001:DB8:100:: 64
import-route direct
maximum load-balancing 32
peer 2001:DB8:3::1 enable
peer 2001:DB8:4::1 enable
```

```
#  
return
```

1.12 Example for Configuring VRRP over VXLAN with Inter-DC Master and Backup Gateways

Applicable Products and Versions

- This example applies to CE16800, CE12800, CE12800E, CE8800, CE7800, CE6800 (excluding the CE6820, CE6850EI, CE6810EI, and CE6810LI), and CE5880EI series switches running V200R002C50 or later versions.
- For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

On the network shown in [Figure 1-12](#), an enterprise has two DCs deployed in different regions. It is required that the two DCs back up each other and active-active gateways be deployed in each DC to improve reliability. To achieve this, configure VRRP over VXLAN to implement inter-DC master/backup gateway.

Figure 1-12 VRRP over VXLAN with inter-DC master and backup gateways on different VXLAN networks

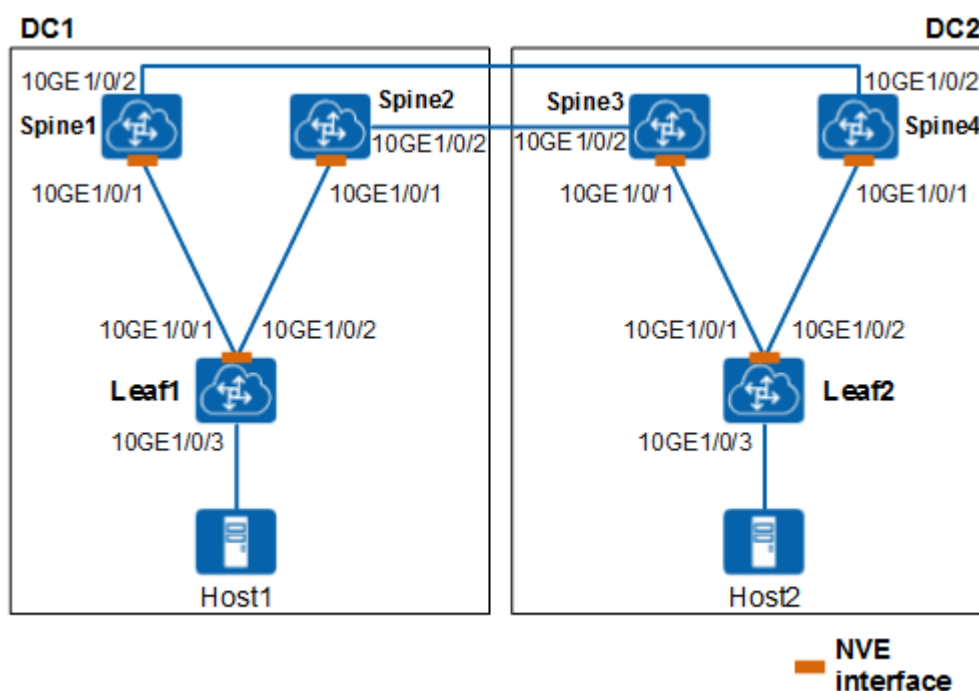


Table 1-13 Interface IP Addresses

Device	Interface	IP Address
Spine1	10GE1/0/1	192.168.1.1/24
	10GE1/0/2	192.168.5.1/24
	Loopback1	1.1.1.1/32
	Loopback2	5.5.5.5/32
Spine2	10GE1/0/1	192.168.2.1/24
	10GE1/0/2	192.168.6.1/24
	Loopback1	1.1.1.1/32
	Loopback2	6.6.6.6/32
Leaf1	10GE1/0/1	192.168.1.2/24
	10GE1/0/2	192.168.2.2/24
	Loopback1	3.3.3.3/32
Spine3	10GE1/0/1	192.168.3.1/24
	10GE1/0/2	192.168.6.2/24
	Loopback1	2.2.2.2/32
	Loopback2	7.7.7.7/32
Spine4	10GE1/0/1	192.168.4.1/24
	10GE1/0/2	192.168.5.2/24
	Loopback1	2.2.2.2/32
	Loopback2	8.8.8.8/32
Leaf2	10GE1/0/1	192.168.3.2/24
	10GE1/0/2	192.168.4.2/24
	Loopback1	4.4.4.4/32

Configuration Roadmap

- The configuration roadmap is as follows:
1. Configure a routing protocol for VXLAN tunnel establishment. In this example, OSPF is used.
 2. Establish a VXLAN tunnel between spine switches in different DCs, and configure a VNI and bind it to a BD.
 3. Establish a VXLAN tunnel between leaf switches and between a leaf switch and a spine switch, and configure a VNI and bind it to a BD.

4. Create a VBDIF interface on each Spine and configure VRRP for master/backup gateway negotiation.
5. Deploy a DFS group on the two spine switches in each DC for device pairing, allowing active-active gateway implementation.
6. Configure Layer 2 sub-interfaces on the Leaf in each DC for host access.
7. (Optional) Perform either of the following configurations to prevent packet loss during traffic switchback after the master devices recover.
 - After VRRP status becomes stable, configure the two master VRRP devices to advertise VTEP routes after a delay that is longer than the time taken by VRRP-becoming-master. If the two master devices need to be restarted due to failures, manually power off one master device and do not restore it until the other master device recovers.
 - Configure a VTEP direct route on the loopback interfaces of the two master VRRP devices and associate the direct route with the VRRP status, so that the cost of the direct route can be adjusted according to the VRRP status. Specifically, if VRRP is not in the Master state, the cost increases, and the route priority decreases. Configure a dynamic routing protocol to import the direct route, to ensure that the VRRP status affects route selection of the dynamic routing protocol and traffic is imported to the recovery link only after VRRP status becomes Master. In addition, enable the backup VRRP devices to forward service traffic through VBDIF interfaces. If multiple VRRP backup groups need to be configured, configure the VRRP backup group that is associated with the loopback interface route as the mVRRP backup group and the rest as service VRRP backup groups, and bind the service VRRP backup groups to the mVRRP backup group.

Data Preparation

To complete the configuration, you need the following data:

- Interface IP addresses (For details, see [Table 1-13](#).)
- OSPF area (0)
- BD ID (10)
- VNI ID (10)

Procedure

1. Configure a routing protocol.
Configure Spine1. Repeat this step for Leaf1, Leaf2, Spine2, Spine3, and Spine4. For configuration details, see [Configuration Files](#).

```
<HUAWEI> system-view
[~HUAWEI] sysname Spine1
[*HUAWEI] commit
[~Spine1] interface loopback 1
[*Spine1-LoopBack1] ip address 1.1.1.1 32
[*Spine1-LoopBack1] quit
[*Spine1] interface loopback 2
[*Spine1-LoopBack2] ip address 5.5.5.5 32
[*Spine1-LoopBack2] quit
[*Spine1] interface 10ge 1/0/1
[*Spine1-10GE1/0/1] undo portswitch
[*Spine1-10GE1/0/1] ip address 192.168.1.1 24
[*Spine1-10GE1/0/1] quit
```

```
[*Spine1] interface 10ge 1/0/2
[*Spine1-10GE1/0/2] undo portswitch
[*Spine1-10GE1/0/2] ip address 192.168.5.1 24
[*Spine1-10GE1/0/2] quit
[*Spine1] ospf
[*Spine1-ospf-1] area 0
[*Spine1-ospf-1-area-0.0.0.0] network 1.1.1.1 0.0.0.0
[*Spine1-ospf-1-area-0.0.0.0] network 5.5.5.5 0.0.0.0
[*Spine1-ospf-1-area-0.0.0.0] network 192.168.1.0 0.0.0.255
[*Spine1-ospf-1-area-0.0.0.0] network 192.168.5.0 0.0.0.255
[*Spine1-ospf-1-area-0.0.0.0] quit
[*Spine1-ospf-1] quit
[*Spine1] commit
```

After OSPF is configured, leaf and spine switches can learn the IP addresses of loopback interfaces of each other using OSPF and successfully ping each other.

2. Configure the VXLAN tunnel mode and enable the VXLAN ACL extension function. (This step only needs to be performed on the CE12800, CE16800 equipped with A series cards, CE6870EI, and CE6875EI.)

Configure Spine1. Repeat this step for Leaf1, Leaf2, Spine2, Spine3, and Spine4.

```
[~Spine1] ip tunnel mode vxlan
[*Spine1] assign forward nvo3 acl extend enable
[*Spine1] commit
```

NOTE

After configuring the VXLAN tunnel mode and enabling the VXLAN ACL extension function, save the configuration and restart the switch to make the configuration take effect.

3. Configure a service access point on Leaf1 and Leaf2, respectively.

Configure Leaf1. Repeat this step for Leaf2.

```
[~Leaf1] bridge-domain 10
[*Leaf1-bd10] quit
[*Leaf1] interface 10ge 1/0/3.1 mode l2
[*Leaf1-10GE1/0/3.1] encapsulation dot1q vid 10
[*Leaf1-10GE1/0/3.1] bridge-domain 10
[*Leaf1-10GE1/0/3.1] quit
[*Leaf1] commit
```

4. Configure VXLAN tunnels in DCs and between DC1 and DC2.

Configure Spine1. Repeat this step for Leaf1, Leaf2, Spine2, Spine3, and Spine4. For configuration details, see [Configuration Files](#).

```
[~Spine1] evpn-overlay enable
[*Spine1] commit
[~Spine1] bgp 100
[*Spine1-bgp] peer 3.3.3.3 as-number 100
[*Spine1-bgp] peer 3.3.3.3 connect-interface loopback 2
[*Spine1-bgp] peer 4.4.4.4 as-number 100
[*Spine1-bgp] peer 4.4.4.4 connect-interface loopback 2
[*Spine1-bgp] peer 6.6.6.6 as-number 100
[*Spine1-bgp] peer 6.6.6.6 connect-interface loopback 2
[*Spine1-bgp] peer 7.7.7.7 as-number 100
[*Spine1-bgp] peer 7.7.7.7 connect-interface loopback 2
[*Spine1-bgp] peer 8.8.8.8 as-number 100
[*Spine1-bgp] peer 8.8.8.8 connect-interface loopback 2
[*Spine1-bgp] l2vpn-family evpn
[*Spine1-bgp-af-evpn] peer 3.3.3.3 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 4.4.4.4 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 6.6.6.6 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
```

```
[*Spine1-bgp-af-evpn] peer 7.7.7.7 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] peer 8.8.8.8 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Spine1-bgp-af-evpn] quit
[*Spine1-bgp] quit
[*Spine1] commit
[~Spine1] bridge-domain 10
[*Spine1-bd10] vxlan vni 10
[*Spine1-bd10] evpn
[*Spine1-bd10-evpn] route-distinguisher 10:1
[*Spine1-bd10-evpn] vpn-target 11:1
[*Spine1-bd10-evpn] quit
[*Spine1-bd10] quit
[*Spine1] commit
[~Spine1] interface nve 1
[*Spine1-Nve1] source 1.1.1.1
[*Spine1-Nve1] vni 10 head-end peer-list protocol bgp
[*Spine1-Nve1] quit
[*Spine1] commit
```

Configure Leaf1. Repeat this step for Leaf2. For configuration details, see [Configuration Files](#).

```
[~Leaf1] evpn-overlay enable
[*Leaf1] commit
[~Leaf1] bgp 100
[*Leaf1-bgp] peer 4.4.4.4 as-number 100
[*Leaf1-bgp] peer 4.4.4.4 connect-interface loopback 1
[*Leaf1-bgp] peer 5.5.5.5 as-number 100
[*Leaf1-bgp] peer 5.5.5.5 connect-interface loopback 1
[*Leaf1-bgp] peer 6.6.6.6 as-number 100
[*Leaf1-bgp] peer 6.6.6.6 connect-interface loopback 1
[*Leaf1-bgp] peer 7.7.7.7 as-number 100
[*Leaf1-bgp] peer 7.7.7.7 connect-interface loopback 1
[*Leaf1-bgp] peer 8.8.8.8 as-number 100
[*Leaf1-bgp] peer 8.8.8.8 connect-interface loopback 1
[*Leaf1-bgp] l2vpn-family evpn
[*Leaf1-bgp-af-evpn] peer 4.4.4.4 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 5.5.5.5 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 6.6.6.6 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 7.7.7.7 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] peer 8.8.8.8 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Leaf1-bgp-af-evpn] quit
[*Leaf1-bgp] quit
[*Leaf1] commit
[~Leaf1] bridge-domain 10
[~Leaf1-bd10] vxlan vni 10
[*Leaf1-bd10] evpn
[*Leaf1-bd10-evpn] route-distinguisher 50:1
[*Leaf1-bd10-evpn] vpn-target 11:1
[*Leaf1-bd10-evpn] quit
[*Leaf1-bd10] quit
[*Leaf1] commit
[~Leaf1] interface nve 1
[*Leaf1-Nve1] source 3.3.3.3
[*Leaf1-Nve1] vni 10 head-end peer-list protocol bgp
[*Leaf1-Nve1] quit
[*Leaf1] commit
```

5. Configure all-active gateways in the DC on spine switches.

Configure service loopback interfaces on spine switches. (This step only needs to be performed on the CE6850HI, CE6850U-HI, CE6851HI, CE6860EI, CE7850EI, CE8850EI, and CE8860EI.)

Configure Spine1. Repeat this step for Spine2, Spine3 and Spine4. For configuration details, see [Configuration Files](#).

```
[~Spine1] interface eth-trunk 1
[*Spine1-Eth-Trunk1] service type tunnel
[*Spine1-Eth-Trunk1] trunkport 10ge 1/0/5
[*Spine1-Eth-Trunk1] quit
[*Spine1] commit
```

Configure Spine1.

```
[~Spine1] interface vbdif 10
[*Spine1-Vbdif10] ip address 10.1.1.2 24
[*Spine1-Vbdif10] mac-address 0000-5e00-0105
[*Spine1-Vbdif10] quit
[*Spine1] dfs-group 1
[*Spine1-dfs-group-1] source ip 5.5.5.5
[*Spine1-dfs-group-1] active-active-gateway
[*Spine1-dfs-group-1-active-active-gateway] peer 6.6.6.6
[*Spine1-dfs-group-1-active-active-gateway] quit
[*Spine1-dfs-group-1] quit
[*Spine1] commit
```

Configure Spine2.

```
[~Spine2] interface vbdif 10
[*Spine2-Vbdif10] ip address 10.1.1.2 24
[*Spine2-Vbdif10] mac-address 0000-5e00-0105
[*Spine2-Vbdif10] quit
[*Spine2] dfs-group 1
[*Spine2-dfs-group-1] source ip 6.6.6.6
[*Spine2-dfs-group-1] active-active-gateway
[*Spine2-dfs-group-1-active-active-gateway] peer 5.5.5.5
[*Spine2-dfs-group-1-active-active-gateway] quit
[*Spine2-dfs-group-1] quit
[*Spine2] commit
```

Repeat this step for Spine3 and Spine4. For configuration details, see [Configuration Files](#).

6. Configure VRRP.

Configure Spine1. Repeat this step for Spine2. For configuration details, see [Configuration Files](#).

```
[~Spine1] interface vbdif 10
[*Spine1-Vbdif10] vrrp vrid 1 virtual-ip 10.1.1.1
[*Spine1-Vbdif10] vrrp vrid 1 priority 130
[*Spine1-Vbdif10] quit
[*Spine1] commit
```

Configure Spine3. Repeat this step for Spine4. For configuration details, see [Configuration Files](#).

```
[~Spine3] interface vbdif 10
[*Spine3-Vbdif10] vrrp vrid 1 virtual-ip 10.1.1.1
[*Spine3-Vbdif10] vrrp vrid 1 track ip route 1.1.1.1 255.255.255.255 increase 100
[*Spine3-Vbdif10] quit
[*Spine3] commit
```

7. (Optional) Perform either of the following configurations to prevent packet loss during traffic switchback after the master devices recover.

- Configure the VTEPs on the two VRRP master devices to advertise routes in delayed mode. (The mode is used in this example.)

Configure Spine1. Repeat this step for Spine2. For configuration details, see [Configuration Files](#).

```
[~Spine1] ospf
[*Spine1-ospf-1] stub-router on-startup 180
[*Spine1-ospf-1] quit
[*Spine1] commit
```

- Configure association between VTEP direct routes and VRRP states on the loopback interfaces of the two VRRP master devices. (The mode is not used in this example, and is for reference only.)

Configure Spine1. Repeat this step for Spine2. For configuration details, see [Configuration Files](#).

```
[~Spine1] interface loopback 1
[~Spine1-LoopBack1] direct-route track vrrp interface vbdif 10 vrid 1 degrade-cost 20
[*Spine1-LoopBack1] quit
[*Spine1] interface vbdif 10
[*Spine1-Vbdif10] vrrp vrid 1 backup-forward
[*Spine1-Vbdif10] quit
[*Spine1] ospf
[*Spine1-ospf-1] default cost inherit-metric
[*Spine1-ospf-1] quit
[*Spine1] commit
```

Verifying the Configuration

After completing the configurations, run the **display vxlan tunnel** command on spine and leaf switches of the two DCs to view VXLAN tunnel information and run the **display vrrp interface interface-type interface-number verbose** command on spine switches to view mVRRP status information. The following example shows the command output on Spine1.

```
[~Spine1] display vxlan tunnel
Number of vxlan tunnel : 3

```

Tunnel ID	Source	Destination	State	Type	Uptime
4026531841	1.1.1.1	2.2.2.2	up	dynamic	0035h21m
4026531842	1.1.1.1	3.3.3.3	up	dynamic	0035h22m
4026531843	1.1.1.1	4.4.4.4	up	dynamic	0035h23m

```
[~Spine1] display vrrp interface vbdif 10 verbose
Vbdif 10 | Virtual Router 1
State           : Master
Virtual IP      : 10.1.1.1
Master IP       : 10.1.1.2
PriorityRun      : 130
PriorityConfig   : 130
MasterPriority   : 130
Preempt         : YES   Delay Time : 0s   Remain : --
Hold Multiplier : 3
TimerRun        : 1s
TimerConfig     : 1s
Auth Type       : NONE
Virtual MAC     : 0000-5e00-0101
Check TTL       : YES
Config Type     : Normal
Create Time     : 2016-10-29 05:41:23
Last Change Time : 2016-10-29 05:41:33
```

Configuration Files

- Spine1 configuration file

```
#
sysname Spine1
#
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800,
CE16800 equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
source ip 5.5.5.5
#
active-active-gateway
peer 6.6.6.6
```

```
#
evpn-overlay enable
#
bridge-domain 10
vxlan vni 10
evpn
  route-distinguisher 10:1
  vpn-target 11:1 export-extcommunity
  vpn-target 11:1 import-extcommunity
#
interface Vbdif10
ip address 10.1.1.2 255.255.255.0
vrrp vrid 1 virtual-ip 10.1.1.1
vrrp vrid 1 priority 130
mac-address 0000-5e00-0105
#
interface Eth-Trunk1 //This step only needs to be performed on the CE6850HI, CE6850U-HI,
CE6851HI, CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
service type tunnel
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.1.1 255.255.255.0
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.5.1 255.255.255.0
#
interface 10GE1/0/5 //This step only needs to be performed on the CE6850HI, CE6850U-HI,
CE6851HI, CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
eth-trunk 1
#
interface LoopBack1
ip address 1.1.1.1 255.255.255.255
#
interface LoopBack2
ip address 5.5.5.5 255.255.255.255
#
interface Nve1
source 1.1.1.1
vni 10 head-end peer-list protocol bgp
#
bgp 100
peer 3.3.3.3 as-number 100
peer 3.3.3.3 connect-interface LoopBack2
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack2
peer 6.6.6.6 as-number 100
peer 6.6.6.6 connect-interface LoopBack2
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack2
peer 8.8.8.8 as-number 100
peer 8.8.8.8 connect-interface LoopBack2
#
ipv4-family unicast
peer 3.3.3.3 enable
peer 4.4.4.4 enable
peer 6.6.6.6 enable
peer 7.7.7.7 enable
peer 8.8.8.8 enable
#
l2vpn-family evpn
policy vpn-target
peer 3.3.3.3 enable
peer 4.4.4.4 enable
peer 6.6.6.6 enable
peer 7.7.7.7 enable
peer 8.8.8.8 enable
#
```

```
ospf 1
 stub-router on-startup 180
 area 0.0.0.0
  network 1.1.1.1 0.0.0.0
  network 5.5.5.5 0.0.0.0
  network 192.168.1.0 0.0.0.255
  network 192.168.5.0 0.0.0.255
#
return
```

- Spine2 configuration file

```
#
sysname Spine2
#
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800, CE16800 equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
source ip 6.6.6.6
#
active-active-gateway
peer 5.5.5.5
#
evpn-overlay enable
#
bridge-domain 10
vxlan vni 10
evpn
 route-distinguisher 20:1
 vpn-target 11:1 export-extcommunity
 vpn-target 11:1 import-extcommunity
#
interface Vbdi10
 ip address 10.1.1.2 255.255.255.0
 vrrp vrid 1 virtual-ip 10.1.1.1
 vrrp vrid 1 priority 130
 mac-address 0000-5e00-0105
#
interface Eth-Trunk1 //This step only needs to be performed on the CE6850HI, CE6850U-HI, CE6851HI, CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
 service type tunnel
#
interface 10GE1/0/1
 undo portswitch
 ip address 192.168.2.1 255.255.255.0
#
interface 10GE1/0/2
 undo portswitch
 ip address 192.168.6.1 255.255.255.0
#
interface 10GE1/0/5 //This step only needs to be performed on the CE6850HI, CE6850U-HI, CE6851HI, CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
 eth-trunk 1
#
interface LoopBack1
 ip address 1.1.1.1 255.255.255.255
#
interface LoopBack2
 ip address 6.6.6.6 255.255.255.255
#
interface Nve1
 source 1.1.1.1
 vni 10 head-end peer-list protocol bgp
#
bgp 100
 peer 3.3.3.3 as-number 100
 peer 3.3.3.3 connect-interface LoopBack2
 peer 4.4.4.4 as-number 100
 peer 4.4.4.4 connect-interface LoopBack2
 peer 5.5.5.5 as-number 100
```

```
peer 5.5.5.5 connect-interface LoopBack2
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack2
peer 8.8.8.8 as-number 100
peer 8.8.8.8 connect-interface LoopBack2
#
ipv4-family unicast
peer 3.3.3.3 enable
peer 4.4.4.4 enable
peer 5.5.5.5 enable
peer 7.7.7.7 enable
peer 8.8.8.8 enable
#
l2vpn-family evpn
policy vpn-target
peer 3.3.3.3 enable
peer 4.4.4.4 enable
peer 5.5.5.5 enable
peer 7.7.7.7 enable
peer 8.8.8.8 enable
#
ospf 1
stub-router on-startup 180
area 0.0.0.0
network 1.1.1.1 0.0.0.0
network 6.6.6.6 0.0.0.0
network 192.168.2.0 0.0.0.255
network 192.168.6.0 0.0.0.255
#
return
```

- Spine3 configuration file

```
#
sysname Spine3
#
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800,
CE16800 equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
source ip 7.7.7.7
#
active-active-gateway
peer 8.8.8.8
#
evpn-overlay enable
#
bridge-domain 10
vxlan vni 10
evpn
route-distinguisher 30:1
vpn-target 11:1 export-extcommunity
vpn-target 11:1 import-extcommunity
#
interface Vbdif10
ip address 10.1.1.3 255.255.255.0
vrp vrid 1 virtual-ip 10.1.1.1
vrp vrid 1 track ip route 1.1.1.1 255.255.255.255 increase 100
mac-address 0000-5e00-0106
#
interface Eth-Trunk1 //This step only needs to be performed on the CE6850HI, CE6850U-HI,
CE6851HI, CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
service type tunnel
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.3.1 255.255.255.0
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.6.2 255.255.255.0
```

```
#
interface 10GE1/0/5 //This step only needs to be performed on the CE6850HI, CE6850U-HI,
CE6851HI, CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
eth-trunk 1
#
interface LoopBack1
ip address 2.2.2.2 255.255.255.255
#
interface LoopBack2
ip address 7.7.7.7 255.255.255.255
#
interface Nve1
source 2.2.2.2
vni 10 head-end peer-list protocol bgp
#
bgp 100
peer 3.3.3.3 as-number 100
peer 3.3.3.3 connect-interface LoopBack2
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack2
peer 5.5.5.5 as-number 100
peer 5.5.5.5 connect-interface LoopBack2
peer 6.6.6.6 as-number 100
peer 6.6.6.6 connect-interface LoopBack2
peer 8.8.8.8 as-number 100
peer 8.8.8.8 connect-interface LoopBack2
#
ipv4-family unicast
peer 3.3.3.3 enable
peer 4.4.4.4 enable
peer 5.5.5.5 enable
peer 6.6.6.6 enable
peer 8.8.8.8 enable
#
l2vpn-family evpn
policy vpn-target
peer 3.3.3.3 enable
peer 4.4.4.4 enable
peer 5.5.5.5 enable
peer 6.6.6.6 enable
peer 8.8.8.8 enable
#
ospf 1
area 0.0.0.0
network 2.2.2.2 0.0.0.0
network 7.7.7.7 0.0.0.0
network 192.168.3.0 0.0.0.255
network 192.168.6.0 0.0.0.255
#
return
```

- Spine4 configuration file

```
#
sysname Spine4
#
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800,
CE16800 equipped with A series cards, CE6870EI, and CE6875EI.
#
dfs-group 1
source ip 8.8.8.8
#
active-active-gateway
peer 7.7.7.7
#
evpn-overlay enable
#
bridge-domain 10
vxlan vni 10
evpn
route-distinguisher 40:1
```

```
vpn-target 11:1 export-extcommunity
vpn-target 11:1 import-extcommunity
#
interface Vbdi10
ip address 10.1.1.3 255.255.255.0
vrrp vrid 1 virtual-ip 10.1.1.1
vrrp vrid 1 track ip route 1.1.1.1 255.255.255.255 increase 100
mac-address 0000-5e00-0106
#
interface Eth-Trunk1 //This step only needs to be performed on the CE6850HI, CE6850U-HI,
CE6851HI, CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
service type tunnel
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.4.1 255.255.255.0
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.5.2 255.255.255.0
#
interface 10GE1/0/5 //This step only needs to be performed on the CE6850HI, CE6850U-HI,
CE6851HI, CE6860EI, CE7850EI, CE8850EI, and CE8860EI.
eth-trunk 1
#
interface LoopBack1
ip address 2.2.2.2 255.255.255.255
#
interface LoopBack2
ip address 8.8.8.8 255.255.255.255
#
interface Nve1
source 2.2.2.2
vni 10 head-end peer-list protocol bgp
#
bgp 100
peer 3.3.3.3 as-number 100
peer 3.3.3.3 connect-interface LoopBack2
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack2
peer 5.5.5.5 as-number 100
peer 5.5.5.5 connect-interface LoopBack2
peer 6.6.6.6 as-number 100
peer 6.6.6.6 connect-interface LoopBack2
peer 7.7.7.7 as-number 100
peer 7.7.7.7 connect-interface LoopBack2
#
ipv4-family unicast
peer 3.3.3.3 enable
peer 4.4.4.4 enable
peer 5.5.5.5 enable
peer 6.6.6.6 enable
peer 7.7.7.7 enable
#
l2vpn-family evpn
policy vpn-target
peer 3.3.3.3 enable
peer 4.4.4.4 enable
peer 5.5.5.5 enable
peer 6.6.6.6 enable
peer 7.7.7.7 enable
#
ospf 1
area 0.0.0.0
network 2.2.2.2 0.0.0.0
network 8.8.8.8 0.0.0.0
network 192.168.4.0 0.0.0.255
network 192.168.6.0 0.0.0.255
```

```
#  
return
```

- Leaf1 configuration file

```
#  
sysname Leaf1  
#  
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800,  
CE16800 equipped with A series cards, CE6870EI, and CE6875EI.  
#  
evpn-overlay enable  
#  
bridge-domain 10  
vxlan vni 10  
evpn  
route-distinguisher 50:1  
vpn-target 11:1 export-extcommunity  
vpn-target 11:1 import-extcommunity  
#  
interface 10GE1/0/1  
undo portswitch  
ip address 192.168.1.2 255.255.255.0  
#  
interface 10GE1/0/2  
undo portswitch  
ip address 192.168.2.2 255.255.255.0  
#  
interface LoopBack1  
ip address 3.3.3.3 255.255.255.255  
#  
interface Nve1  
source 3.3.3.3  
vni 10 head-end peer-list protocol bgp  
#  
bgp 100  
peer 4.4.4.4 as-number 100  
peer 4.4.4.4 connect-interface LoopBack1  
peer 5.5.5.5 as-number 100  
peer 5.5.5.5 connect-interface LoopBack1  
peer 6.6.6.6 as-number 100  
peer 6.6.6.6 connect-interface LoopBack1  
peer 7.7.7.7 as-number 100  
peer 7.7.7.7 connect-interface LoopBack1  
peer 8.8.8.8 as-number 100  
peer 8.8.8.8 connect-interface LoopBack1  
#  
ipv4-family unicast  
peer 4.4.4.4 enable  
peer 5.5.5.5 enable  
peer 6.6.6.6 enable  
peer 7.7.7.7 enable  
peer 8.8.8.8 enable  
#  
l2vpn-family evpn  
policy vpn-target  
peer 4.4.4.4 enable  
peer 5.5.5.5 enable  
peer 6.6.6.6 enable  
peer 7.7.7.7 enable  
peer 8.8.8.8 enable  
#  
ospf 1  
area 0.0.0.0  
network 3.3.3.3 0.0.0.0  
network 192.168.1.0 0.0.0.255  
network 192.168.2.0 0.0.0.255  
#  
return
```

- Leaf2 configuration file


```
#
sysname Leaf2
#
assign forward nvo3 acl extend enable //This step only needs to be performed on the CE12800,
CE16800 equipped with A series cards, CE6870EI, and CE6875EI.
#
evpn-overlay enable
#
bridge-domain 10
vxlan vni 10
evpn
  route-distinguisher 60:1
  vpn-target 11:1 export-extcommunity
  vpn-target 11:1 import-extcommunity
#
interface 10GE1/0/1
  undo portswitch
  ip address 192.168.3.2 255.255.255.0
#
interface 10GE1/0/2
  undo portswitch
  ip address 192.168.4.2 255.255.255.0
#
interface LoopBack1
  ip address 4.4.4.4 255.255.255.255
#
interface Nve1
  source 4.4.4.4
  vni 10 head-end peer-list protocol bgp
#
bgp 100
  peer 3.3.3.3 as-number 100
  peer 3.3.3.3 connect-interface LoopBack1
  peer 5.5.5.5 as-number 100
  peer 5.5.5.5 connect-interface LoopBack1
  peer 6.6.6.6 as-number 100
  peer 6.6.6.6 connect-interface LoopBack1
  peer 7.7.7.7 as-number 100
  peer 7.7.7.7 connect-interface LoopBack1
  peer 8.8.8.8 as-number 100
  peer 8.8.8.8 connect-interface LoopBack1
#
ipv4-family unicast
  peer 3.3.3.3 enable
  peer 5.5.5.5 enable
  peer 6.6.6.6 enable
  peer 7.7.7.7 enable
  peer 8.8.8.8 enable
#
l2vpn-family evpn
  policy vpn-target
  peer 3.3.3.3 enable
  peer 5.5.5.5 enable
  peer 6.6.6.6 enable
  peer 7.7.7.7 enable
  peer 8.8.8.8 enable
#
ospf 1
  area 0.0.0.0
  network 4.4.4.4 0.0.0.0
  network 192.168.3.0 0.0.0.255
  network 192.168.4.0 0.0.0.255
#
return
```

1.13 Configuring M-LAG Dual-Active Distributed Gateways as DHCPv4/DHCPv6 Relay Agents

Applicable Products and Versions

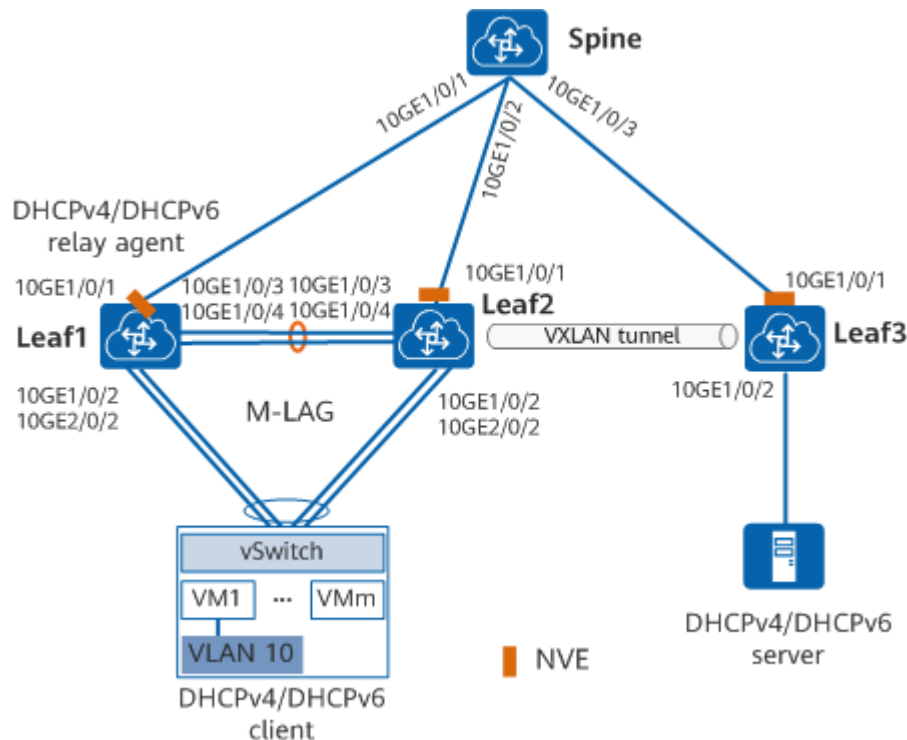
This example applies to CE12800 series switches running V100R006C00 or later versions, CE8800, CE7800, CE6800, and CE5800 series switches running V200R002C50 or later versions, CE9800 series switches running V200R020C00 or later versions, CE12800E series switches running V200R005C00 or later versions, and CloudEngine 16800 series switches running V200R005C20 or later versions.

For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

An enterprise deploys a VXLAN network with distributed gateways, as shown in [Figure 1-13](#). DHCPv4 and DHCPv6 services need to be deployed so that a DHCPv4/DHCPv6 server can allocate IP addresses to users. Leaf1 and Leaf2 function as VXLAN gateways and are configured as DHCPv4/DHCPv6 relay agents, and the DHCPv4/DHCPv6 server function is deployed on a server. To ensure high reliability, Leaf1 and Leaf2 establish an M-LAG.

Figure 1-13 Networking for configuring M-LAG dual-active distributed gateways as DHCPv4/DHCPv6 relay agents



Configuration Roadmap

The configuration roadmap is as follows:

1. Configure VXLAN in distributed gateway mode using BGP EVPN on Leaf1, Leaf2, Leaf3, and Spine, respectively. Leaf1, Leaf2, and Leaf3 function as VXLAN gateways.
2. Configure Leaf1 and Leaf2 to establish an M-LAG.
 - Configure a DFS group on Leaf1 and Leaf2, respectively.
 - Configure the link between Leaf1 and Leaf2 as the peer-link.
 - Configure an M-LAG member interface on Leaf1 and Leaf2, respectively.
3. Configure a DHCPv4 VPN instance and a DHCPv6 VPN instance on Leaf1, Leaf2, and Leaf3.
4. Configure the DHCPv4 and DHCPv6 relay functions on Leaf1 and Leaf2. The DHCPv4/DHCPv6 server function is deployed on a server.

Data Preparation

To complete the configuration, you need the following data:

- IP addresses of interfaces connecting devices
- Routing protocol: BGP
- BD ID: 10
- VNI: 20

Procedure

Step 1 Configure M-LAG.

Configure V-STP, DAD link, DFS group, peer-link, and M-LAG member interfaces on Leaf1 and Leaf2.

1. Configure V-STP.

Configure Leaf1.

```
[~Leaf1] stp mode rstp
[*Leaf1] stp v-stp enable
[*Leaf1] stp flush disable
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] stp mode rstp
[*Leaf2] stp v-stp enable
[*Leaf2] stp flush disable
[~Leaf2] commit
```

2. Configure a DFS group.

Configure Leaf1. In this example, the IP addresses of management interfaces are used. The interfaces on both ends of the DAD link can also be service interfaces. The IP addresses bound to the DFS group must be reachable at Layer 3.

```
[~Leaf1] dfs-group 1
[*Leaf1-dfs-group-1] source ip 10.100.0.1 peer 10.101.0.1 //Bind the DAD link address to the DFS
```

```
group.  
[*Leaf1-dfs-group-1] dual-active detection enhanced enable  
[*Leaf1-dfs-group-1] priority 150  
[*Leaf1-dfs-group-1] quit  
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] dfs-group 1  
[*Leaf2-dfs-group-1] source ip 10.101.0.1 peer 10.100.0.1 //Bind the DAD link address to the DFS  
group.  
[*Leaf2-dfs-group-1] dual-active detection enhanced enable  
[*Leaf2-dfs-group-1] priority 120  
[*Leaf2-dfs-group-1] quit  
[*Leaf2] commit
```

3. Configure a peer-link.

Configure Leaf1. For modular switches, it is recommended that Eth-Trunk member interfaces of the peer-link be deployed on different cards to prevent a peer-link fault caused by a single point of failure on a card.

```
[~Leaf1] interface eth-trunk 1  
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/3  
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/4  
[*Leaf1-Eth-Trunk1] mode lacp-static  
[*Leaf1-Eth-Trunk1] peer-link 1  
[*Leaf1-Eth-Trunk1] quit  
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] interface eth-trunk 1  
[*Leaf2-Eth-Trunk1] trunkport 10ge 1/0/3  
[*Leaf2-Eth-Trunk1] trunkport 10ge 1/0/4  
[*Leaf2-Eth-Trunk1] mode lacp-static  
[*Leaf2-Eth-Trunk1] peer-link 1  
[*Leaf2-Eth-Trunk1] quit  
[*Leaf2] commit
```

4. Configure M-LAG member interfaces.

Configure Leaf1. The uplink interfaces that connect SwitchA to Leaf1 and Leaf2 must be added to an Eth-Trunk interface, and the working mode of the Eth-Trunk interface must be the same as that of the Eth-Trunk interfaces on Leaf1 and Leaf2.

```
[~Leaf1] interface eth-trunk 10  
[*Leaf1-Eth-Trunk10] trunkport 10ge 1/0/2  
[*Leaf1-Eth-Trunk10] trunkport 10ge 2/0/2  
[*Leaf1-Eth-Trunk10] mode lacp-static  
[*Leaf1-Eth-Trunk10] dfs-group 1 m-lag 1  
[*Leaf1-Eth-Trunk10] stp edged-port enable  
[*Leaf1-Eth-Trunk10] quit  
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] interface eth-trunk 10  
[*Leaf2-Eth-Trunk10] trunkport 10ge 1/0/2  
[*Leaf2-Eth-Trunk10] trunkport 10ge 2/0/2  
[*Leaf2-Eth-Trunk10] mode lacp-static  
[*Leaf2-Eth-Trunk10] dfs-group 1 m-lag 1  
[*Leaf2-Eth-Trunk10] stp edged-port enable  
[*Leaf2-Eth-Trunk10] quit  
[*Leaf2] commit
```

Step 2 Configure a service access point on Leaf1, Leaf2, and Leaf3, respectively.

Configure Leaf1. The configuration of Leaf2 is similar to that of Leaf1.

```
[~Leaf1] bridge-domain 10
[*Leaf1-bd10] quit
[*Leaf1] interface eth-Trunk 10.10 mode l2
[*Leaf1-Eth-Trunk10.10] encapsulation dot1q vid 10
[*Leaf1-Eth-Trunk10.10] bridge-domain 10
[*Leaf1-Eth-Trunk10.10] quit
[*Leaf1] interface eth-trunk 10
[*Leaf1-Eth-Trunk10] port link-type trunk
[*Leaf1-Eth-Trunk10] quit
[*Leaf1] commit
```

Configure Leaf3.

```
[~Leaf3] bridge-domain 5090
[*Leaf3-bd5090] quit
[*Leaf3] interface 10GE1/0/2.2010 mode l2
[*Leaf3-10GE1/0/2.2010] encapsulation dot1q vid 3421
[*Leaf3-10GE1/0/2.2010] bridge-domain 5090
[*Leaf3-10GE1/0/2.2010] quit
[*Leaf3] commit
```

Step 3 Configure EVPN as the VXLAN control plane on Leaf1, Leaf2, Leaf3, and Spine, respectively.

Configure Leaf1. The configurations of Leaf2, Leaf3, and Spine are similar to the configuration of Leaf1.

```
[~Leaf1] evpn-overlay enable
[*Leaf1] commit
```

Step 4 Establish BGP EVPN peer relationships.

Configure Leaf1. The configuration of Leaf2 is similar to that of Leaf1.

```
[~Leaf1] bgp 100
[*Leaf1-bgp] group spine external
[*Leaf1-bgp] peer 10.51.3.2 as-number 65001
[*Leaf1-bgp] peer 10.51.3.2 group spine
[*Leaf1-bgp] quit
[*Leaf1] interface LoopBack 0
[*Leaf1-LoopBack0] ip address 192.168.1.2 32
[*Leaf1-LoopBack0] quit
[*Leaf1] bgp 200 instance evpn
[*Leaf1-bgp-instance-evpn] peer 10.10.11.33 as-number 200
[*Leaf1-bgp-instance-evpn] peer 10.10.11.33 connect-interface LoopBack0
[*Leaf1-bgp-instance-evpn] l2vpn-family evpn
[*Leaf1-bgp-instance-evpn-af-evpn] policy vpn-target
[*Leaf1-bgp-instance-evpn-af-evpn] peer 10.10.11.33 enable
[*Leaf1-bgp-instance-evpn-af-evpn] peer 10.10.11.33 advertise irb
[*Leaf1-bgp-instance-evpn-af-evpn] peer 10.10.11.33 advertise irbv6
[*Leaf1-bgp-instance-evpn-af-evpn] quit
[*Leaf1-bgp-instance-evpn] quit
[*Leaf1] commit
```

Configure Spine. Establish BGP EVPN peer relationships with Leaf1, Leaf2, and Leaf3 on Spine.

```
[~Spine] bgp 65001
[*Spine-bgp] group leaf external
[*Spine-bgp] peer 10.51.3.1 as-number 100
[*Spine-bgp] peer 10.51.3.1 group leaf
[*Spine-bgp] peer 10.51.1.1 as-number 100
[*Spine-bgp] peer 10.51.1.1 group leaf
[*Spine-bgp] group leaf3 external
[*Spine-bgp] peer 10.56.1.1 as-number 100
[*Spine-bgp] peer 10.56.1.1 group leaf3
[*Spine-bgp] quit
```

```
[*Spine] interface LoopBack 0
[*Spine-LoopBack0] ip address 10.10.11.33 32
[*Spine-LoopBack0] quit
[*Spine] bgp 200 instance evpn
[*Spine-bgp-instance-evpn] peer 192.168.1.2 as-number 200
[*Spine-bgp-instance-evpn] peer 192.168.1.2 connect-interface LoopBack0
[*Spine-bgp-instance-evpn] peer 192.168.1.3 as-number 200
[*Spine-bgp-instance-evpn] peer 192.168.1.3 connect-interface LoopBack0
[*Spine-bgp-instance-evpn] peer 192.168.2.2 as-number 200
[*Spine-bgp-instance-evpn] peer 192.168.2.2 connect-interface LoopBack0
[*Spine-bgp-instance-evpn] l2vpn-family evpn
[*Spine-bgp-instance-evpn-af-evpn] undo policy vpn-target
[*Spine-bgp-instance-evpn-af-evpn] peer 192.168.1.2 enable
[*Spine-bgp-instance-evpn-af-evpn] peer 192.168.1.2 advertise irb
[*Spine-bgp-instance-evpn-af-evpn] peer 192.168.1.2 advertise irbv6
[*Spine-bgp-instance-evpn-af-evpn] peer 192.168.1.2 reflect-client
[*Spine-bgp-instance-evpn-af-evpn] peer 192.168.1.3 enable
[*Spine-bgp-instance-evpn-af-evpn] peer 192.168.1.3 advertise irb
[*Spine-bgp-instance-evpn-af-evpn] peer 192.168.1.3 advertise irbv6
[*Spine-bgp-instance-evpn-af-evpn] peer 192.168.1.3 reflect-client
[*Spine-bgp-instance-evpn-af-evpn] peer 192.168.2.2 enable
[*Spine-bgp-instance-evpn-af-evpn] peer 192.168.2.2 advertise irb
[*Spine-bgp-instance-evpn-af-evpn] peer 192.168.2.2 advertise irbv6
[*Spine-bgp-instance-evpn-af-evpn] peer 192.168.2.2 reflect-client
[*Spine-bgp-instance-evpn-af-evpn] quit
[*Spine-bgp-instance-evpn] quit
[*Spine] commit
```

Step 5 Configure VPN and EVPN instances.

Configure Leaf1. The configuration of Leaf2 is similar to that of Leaf1.

```
[~Leaf1] ip vpn-instance vpn1
[*Leaf1-vpn-instance-vpn1] vxlan vni 20
[*Leaf1-vpn-instance-vpn1] ipv4-family
[*Leaf1-vpn-instance-vpn1-af-ipv4] route-distinguisher 21:85
[*Leaf1-vpn-instance-vpn1-af-ipv4] vpn-target 68:1 evpn
[*Leaf1-vpn-instance-vpn1-af-ipv4] vpn-target 68:1
[*Leaf1-vpn-instance-vpn1-af-ipv4] quit
[*Leaf1-vpn-instance-vpn1] ipv6-family
[*Leaf1-vpn-instance-vpn1-af-ipv6] route-distinguisher 21:85
[*Leaf1-vpn-instance-vpn1-af-ipv6] vpn-target 68:1 evpn
[*Leaf1-vpn-instance-vpn1-af-ipv6] vpn-target 68:1
[*Leaf1-vpn-instance-vpn1-af-ipv6] quit
[*Leaf1-vpn-instance-vpn1] quit
[*Leaf1] bridge-domain 10
[*Leaf1-bd10] vxlan vni 10
[*Leaf1-bd10] evpn
[*Leaf1-bd10-evpn] route-distinguisher 21:10
[*Leaf1-bd10-evpn] vpn-target 68:1 export-extcommunity
[*Leaf1-bd10-evpn] vpn-target 0:10 export-extcommunity
[*Leaf1-bd10-evpn] vpn-target 0:10 import-extcommunity
[*Leaf1-bd10-evpn] quit
[*Leaf1-bd10] quit
[*Leaf1] commit
```

Configure Leaf3.

```
[~Leaf3] ip vpn-instance vpn1
[*Leaf3-vpn-instance-vpn1] vxlan vni 20
[*Leaf3-vpn-instance-vpn1] ipv4-family
[*Leaf3-vpn-instance-vpn1-af-ipv4] route-distinguisher 21:6
[*Leaf3-vpn-instance-vpn1-af-ipv4] vpn-target 68:1 evpn
[*Leaf3-vpn-instance-vpn1-af-ipv4] vpn-target 68:1
[*Leaf3-vpn-instance-vpn1-af-ipv4] quit
[*Leaf3-vpn-instance-vpn1] ipv6-family
[*Leaf3-vpn-instance-vpn1-af-ipv6] route-distinguisher 21:6
[*Leaf3-vpn-instance-vpn1-af-ipv6] vpn-target 68:1 evpn
[*Leaf3-vpn-instance-vpn1-af-ipv6] vpn-target 68:1
[*Leaf3-vpn-instance-vpn1-af-ipv6] quit
```

```
[*Leaf3-vpn-instance-vpn1] quit
[*Leaf3] bridge-domain 5090
[*Leaf3-bd5090] vxlan vni 5154
[*Leaf3-bd5090] evpn
[*Leaf3-bd5090-evpn] route-distinguisher 20:5154
[*Leaf3-bd5090-evpn] vpn-target 68:1 export-extcommunity
[*Leaf3-bd5090-evpn] vpn-target 0:5154 export-extcommunity
[*Leaf3-bd5090-evpn] vpn-target 0:5154 import-extcommunity
[*Leaf3-bd5090-evpn] quit
[*Leaf3-bd5090] quit
[*Leaf3] commit
```

Step 6 Configure ingress replication.

Because Leaf1 and Leaf2 are dual-active devices, ensure that the IP addresses and MAC addresses of NVE interfaces on the two devices are the same.

Configure Leaf1. The configuration of Leaf2 is similar to that of Leaf1.

```
[~Leaf1] interface nve1
[*Leaf1-Nve1] source 192.168.1.1
[*Leaf1-Nve1] mac-address 0000-5e00-0103
[*Leaf1-Nve1] quit
[*Leaf1] commit
```

Advertise the NVE interface IP address in BGP. The configuration of Leaf2 is similar to that of Leaf1.

```
[~Leaf1] bgp 100
[~Leaf1-bgp] network 192.168.1.1 32
```

Configure Leaf3.

```
[~Leaf3] interface nve1
[*Leaf3-Nve1] vni 5154 head-end peer-list protocol bgp
[*Leaf3-Nve1] quit
[*Leaf3] commit
```

Step 7 Configure Leaf1 and Leaf2 as Layer 3 VXLAN gateways.

Configure Leaf1 as a Layer 3 VXLAN gateway. The configuration of Leaf2 is similar to that of Leaf1.

```
[~Leaf1] interface vbdif10
[*Leaf1-vbdif10] ip binding vpn-instance vpn1
[*Leaf1-vbdif10] ip address 10.202.0.1 255.255.0.0
[*Leaf1-vbdif10] mac-address 0000-5e00-0112
[*Leaf1-vbdif10] ipv6 enable
[*Leaf1-vbdif10] ipv6 address 2001:db8:3::1 64
[*Leaf1-vbdif10] vxlan anycast-gateway enable
[*Leaf1-vbdif10] arp collect host enable
[*Leaf1-vbdif10] ipv6 nd collect host enable
[*Leaf1-vbdif10] quit
[*Leaf1] commit
```

Step 8 Configure Leaf3 as the gateway for the DHCPv4/DHCPv6 server.

Configure Leaf3.

```
[~Leaf3] interface vbdif5090
[*Leaf3-vbdif20] ip binding vpn-instance vpn1
[*Leaf3-vbdif20] ip address 10.209.0.1 255.255.0.0
[*Leaf3-vbdif20] mac-address 0000-5e00-0102
[*Leaf3-vbdif20] ipv6 enable
[*Leaf3-vbdif20] ipv6 address 2001:db8:1::1/64
[*Leaf3-vbdif20] vxlan anycast-gateway enable
[*Leaf3-vbdif20] arp collect host enable
[*Leaf3-vbdif20] ipv6 nd collect host enable
```

```
[*Leaf3-vbdif20] quit
[*Leaf3] commit
```

Step 9 Configure Leaf1, Leaf2, and Leaf3 to advertise IP prefix routes to BGP peers.

Configure Leaf1. The configurations of Leaf2 and Leaf3 are similar to the configuration of Leaf1.

```
[~Leaf1] bgp 100
[~Leaf1-bgp] ipv6-family vpn-instance vpn1
[*Leaf1-bgp-6-vpn1] default-route imported
[*Leaf1-bgp-6-vpn1] import-route static
[*Leaf1-bgp-6-vpn1] import-route direct
[*Leaf1-bgp-6-vpn1] advertise l2vpn evpn
[*Leaf1-bgp-6-vpn1] quit
[*Leaf1-bgp] ipv4-family vpn-instance vpn1
[*Leaf1-bgp-4-vpn1] default-route imported
[*Leaf1-bgp-4-vpn1] import-route static
[*Leaf1-bgp-4-vpn1] import-route direct
[*Leaf1-bgp-4-vpn1] advertise l2vpn evpn
[*Leaf1-bgp-4-vpn1] quit
[*Leaf1-bgp] quit
[*Leaf1] commit
```

Step 10 Configure the DHCP relay function on Leaf1 and Leaf2.

Configure a VBDIF interface as the source interface when the DHCPv4/DHCPv6 relay agents forward packets. The configuration of Leaf2 is similar to that of Leaf1.

```
[~Leaf1] interface vbdif5013
[*Leaf1-Vbdif5013] ip binding vpn-instance vpn1
[*Leaf1-Vbdif5013] ipv6 enable
[*Leaf1-Vbdif5013] ip address 10.202.10.1 255.255.0.0
[*Leaf1-Vbdif5013] ipv6 address 2001:db8:10::1/64
[*Leaf1-Vbdif5013] ipv6 nd collect host enable
[*Leaf1-Vbdif5013] vxlan anycast-gateway enable
[*Leaf1-Vbdif5013] arp collect host enable
[*Leaf1-Vbdif5013] quit
[*Leaf1] commit
```

Configure Leaf1. The configuration of Leaf2 is similar to that of Leaf1.

```
[~Leaf1] interface vbdif10
[*Leaf1-Vbdif10] dhcp select relay
[*Leaf1-Vbdif10] dhcp relay binding server ip 10.209.0.1
[*Leaf1-Vbdif10] dhcp relay source-interface vbdif5013
[*Leaf1-Vbdif10] dhcp option82 vendor-specific insert enable
[*Leaf1-Vbdif10] dhcp option82 vendor-specific format vendor-sub-option 10 source-ip-address 10.202.10.1
[*Leaf1-Vbdif10] dhcpv6 relay destination 2001:db8:1::1
[*Leaf1-Vbdif10] dhcpv6 relay source-interface vbdif5013
[*Leaf1-Vbdif10] ipv6 nd ra prefix 2001:db8:3::/64 0 0 no-autoconfig off-link
[*Leaf1-Vbdif10] ipv6 nd ra halt disable
[*Leaf1-Vbdif10] ipv6 nd autoconfig managed-address-flag
[*Leaf1-Vbdif10] ipv6 nd autoconfig other-flag
[*Leaf1-Vbdif10] quit
[*Leaf1] commit
```

Step 11 Configure a DHCPv4/DHCPv6 server. To ensure reliability, it is recommended that the DHCPv4/DHCPv6 server function be deployed on a downstream server.

The DHCPv4/DHCPv6 server must meet the following conditions:

- Configure an address pool on the DHCPv4/DHCPv6 server so that the server can allocate correct IP addresses to clients.
- It is recommended that a lease for IP addresses in the address pool be configured to improve IP address utilization.

Step 12 Verify the configuration.

```
[~Leaf1]display dhcp relay interface Vbdif 10
DHCP relay agent running information of interface Vbdif10 :
Server IP address/VPN Name : 10.209.0.1      vpn1
Gateway address in use    : 10.202.0.1
[~Leaf1]display dhcpv6 relay interface Vbdif 10
DHCPv6 relay agent running information of interface Vbdif10 :
Gateway address in use    : 2001:db8:3::1
Gateway switch enable
```

After the preceding configurations are complete, run the **display vxlan tunnel** command on Leaf1 to check VXLAN tunnel information.

```
[~Leaf1] display vxlan tunnel
Number of vxlan tunnel : 1
Tunnel ID   Source           Destination      State Type    Uptime
-----
4026531844  192.168.1.1      192.168.2.1    up   dynamic  02:52:23
```

Run the **display dfs-group 1 m-lag** command on Leaf1 and Leaf2 to check M-LAG information. The following example shows the command output on Leaf1.

```
[~Leaf1] display dfs-group 1 m-lag
*          : Local node
Heart beat state : OK
Node 1 *
  Dfs-Group ID : 1
  Priority     : 150
  Address      : ip address 10.100.0.1
  State        : Master
  Causation    : -
  System ID    : 0052-7512-2401
  SysName      : Leaf1
  Version      : V200R020C00SPC200
  Device Type  : CE16800
Node 2
  Dfs-Group ID : 1
  Priority     : 120
  Address      : ip address 10.101.0.1
  State        : Backup
  Causation    : -
  System ID    : 0034-2354-9901
  SysName      : Leaf2
  Version      : V200R020C00SPC200
  Device Type  : CE16800
```

Check M-LAG information on Leaf1.

```
[~Leaf1] display dfs-group 1 node 1 m-lag brief
* - Local node

M-Lag ID   Interface   Port State   Status           Consistency-check
  1      Eth-Trunk 10   Up           active(*)-active  -

Failed reason:
  1 -- Relationship between vlan and port is inconsistent
  2 -- STP configuration under the port is inconsistent
  3 -- STP port priority configuration is inconsistent
  4 -- LACP mode of M-LAG is inconsistent
  5 -- M-LAG configuration is inconsistent
  6 -- The number of M-LAG members is inconsistent
```

DHCPv4/DHCPv6 clients can obtain addresses from the DHCPv4/DHCPv6 server through the DHCPv4/DHCPv6 relay agents and go online, indicating that the configuration is successful.

----End

Configuration Files

- Leaf1 configuration file

```
#
sysname Leaf1
#
dfs-group 1
priority 150
source ip 10.100.0.1 peer 10.101.0.1
dual-active detection enhanced enable
#
stp mode rstp
stp v-stp enable
stp flush disable
#
dhcp enable
#
dhcpv6 enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 21:85
vpn-target 68:1 export-extcommunity
vpn-target 68:1 export-extcommunity evpn
vpn-target 68:1 import-extcommunity
vpn-target 68:1 import-extcommunity evpn
ipv6-family
route-distinguisher 21:85
vpn-target 68:1 export-extcommunity
vpn-target 68:1 export-extcommunity evpn
vpn-target 68:1 import-extcommunity
vpn-target 68:1 import-extcommunity evpn
vxlan vni 20
#
bridge-domain 10
vxlan vni 10
#
evpn
route-distinguisher 21:10
vpn-target 68:1 export-extcommunity
vpn-target 0:10 export-extcommunity
vpn-target 0:10 import-extcommunity
#
interface Vbdif10
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.202.0.1 255.255.0.0
ipv6 address 2001:db8:3::1/64
ipv6 nd ra prefix 2001:db8:3::/64 0 0 no-autoconfig off-link
mac-address 0000-5e00-0112
ipv6 nd ra halt disable
ipv6 nd autoconfig managed-address-flag
ipv6 nd autoconfig other-flag
ipv6 nd collect host enable
dhcp select relay
dhcp relay binding server ip 10.209.0.1
dhcp relay source-interface Vbdif5013
dhcp option82 vendor-specific insert enable
dhcp option82 vendor-specific format vendor-sub-option 10 source-ip-address 10.202.10.1
vxlan anycast-gateway enable
arp collect host enable
dhcpv6 relay destination 2001:db8:1::1
dhcpv6 relay source-ip-address 2001:db8:10::1
#
interface Vbdif5013
ip binding vpn-instance vpn1
ipv6 enable
```

```
ip address 10.202.10.1 255.255.0.0
ipv6 address 2001:db8:10::1/64
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface MEth0/0/0
ip address 10.100.0.1 255.255.255.0
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk10
port link-type trunk
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 1
#
interface Eth-Trunk10.10 mode l2
encapsulation dot1q vid 10
bridge-domain 10
#
interface 10GE1/0/1
undo portswitch
ip address 10.51.3.1 255.255.255.0
#
interface 10GE1/0/2
eth-trunk 10
#
interface 10GE1/0/3
eth-trunk 1
#
interface 10GE2/0/2
eth-trunk 10
#
interface 10GE1/0/4
eth-trunk 1
#
interface LoopBack0
ip address 192.168.1.2 255.255.255.255
#
interface LoopBack1
ip address 192.168.1.1 255.255.255.255
#
interface Nve1
source 192.168.1.1
mac-address 0000-5e00-0103
#
bgp 100
group spine external
peer 10.51.3.2 as-number 65001
peer 10.51.3.2 group spine
#
ipv4-family unicast
import-route direct
peer spine enable
peer spine allow-as-loop
peer 10.51.3.2 enable
peer 10.51.3.2 group spine
network 192.168.1.1 255.255.255.255
#
ipv4-family vpn-instance vpn1
default-route imported
import-route direct
import-route static
advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
```

```
default-route imported
import-route direct
import-route static
advertise l2vpn evpn
#
bgp 200 instance evpn
peer 10.10.11.33 as-number 200
peer 10.10.11.33 connect-interface LoopBack0
#
l2vpn-family evpn
policy vpn-target
peer 10.10.11.33 enable
peer 10.10.11.33 advertise irb
peer 10.10.11.33 advertise irbv6
#
return
```

- Leaf2 configuration file

```
#
sysname Leaf2
#
dfs-group 1
priority 120
source ip 10.101.0.1 peer 10.100.0.1
dual-active detection enhanced enable
#
stp mode rstp
stp v-stp enable
stp flush disable
#
dhcp enable
#
dhcpv6 enable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 21:85
vpn-target 68:1 export-extcommunity
vpn-target 68:1 export-extcommunity evpn
vpn-target 68:1 import-extcommunity
vpn-target 68:1 import-extcommunity evpn
ipv6-family
route-distinguisher 21:85
vpn-target 68:1 export-extcommunity
vpn-target 68:1 export-extcommunity evpn
vpn-target 68:1 import-extcommunity
vpn-target 68:1 import-extcommunity evpn
vxlan vni 20
#
bridge-domain 10
vxlan vni 10
#
evpn
route-distinguisher 21:10
vpn-target 68:1 export-extcommunity
vpn-target 0:10 export-extcommunity
vpn-target 0:10 import-extcommunity
#
interface Vbdif10
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.202.0.1 255.255.0.0
ipv6 address 2001:db8:3::1/64
ipv6 nd ra prefix 2001:db8:3::/64 0 0 no-autoconfig off-link
mac-address 0000-5e00-0112
ipv6 nd ra halt disable
ipv6 nd autoconfig managed-address-flag
ipv6 nd autoconfig other-flag
```

```

ipv6 nd collect host enable
dhcp select relay
dhcp relay binding server ip 10.209.0.1
dhcp relay source-interface Vbdif5013
dhcp option82 vendor-specific insert enable
dhcp option82 vendor-specific format vendor-sub-option 10 source-ip-address 10.202.10.2
vxlan anycast-gateway enable
arp collect host enable
dhcpv6 relay destination 2001:db8:1::1
dhcpv6 relay source-ip-address 2001:db8:10::1
#
interface Vbdif5013
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.202.10.2 255.255.0.0
ipv6 address 2001:db8:10::2/64
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface MEth0/0/0
ip address 10.101.0.1 255.255.255.0
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk10
port link-type trunk
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 1
#
interface Eth-Trunk10.10 mode l2
encapsulation dot1q vid 10
bridge-domain 10
#
interface 10GE1/0/1
undo portswitch
ip address 10.51.1.1 255.255.255.0
#
interface 10GE1/0/2
eth-trunk 10
#
interface 10GE1/0/3
eth-trunk 1
#
interface 10GE2/0/2
eth-trunk 10
#
interface 10GE1/0/4
eth-trunk 1
#
interface LoopBack0
ip address 192.168.1.3 255.255.255.255
#
interface LoopBack1
ip address 192.168.1.1 255.255.255.255
#
interface Nve1
source 192.168.1.1
mac-address 0000-5e00-0103
#
bgp 100
group spine external
peer 10.51.3.2 as-number 65001
peer 10.51.3.2 group spine
#
ipv4-family unicast

```

```
import-route direct
peer spine enable
peer spine allow-as-loop
peer 10.51.3.2 enable
peer 10.51.3.2 group spine
network 192.168.1.1 255.255.255.255
#
ipv4-family vpn-instance vpn1
default-route imported
import-route direct
import-route static
advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
default-route imported
import-route direct
import-route static
advertise l2vpn evpn
#
bgp 200 instance evpn
peer 10.10.11.33 as-number 200
peer 10.10.11.33 connect-interface LoopBack0
#
l2vpn-family evpn
policy vpn-target
peer 10.10.11.33 enable
peer 10.10.11.33 advertise irb
peer 10.10.11.33 advertise irbv6
#
return
```

- Leaf3 configuration file

```
#
sysname Leaf3
#
vlan batch 2030
#
stp disable
#
evpn-overlay enable
#
ip vpn-instance vpn1
ipv4-family
route-distinguisher 21:6
vpn-target 68:1 export-extcommunity
vpn-target 68:1 export-extcommunity evpn
vpn-target 68:1 import-extcommunity
vpn-target 68:1 import-extcommunity evpn
ipv6-family
route-distinguisher 21:6
vpn-target 68:1 export-extcommunity
vpn-target 68:1 export-extcommunity evpn
vpn-target 68:1 import-extcommunity
vpn-target 68:1 import-extcommunity evpn
vxlan vni 20
#
bridge-domain 5090
vxlan vni 5154
#
evpn
route-distinguisher 20:5154
vpn-target 68:1 export-extcommunity
vpn-target 0:5154 export-extcommunity
vpn-target 0:5154 import-extcommunity
#
interface Vbdi5090
ip binding vpn-instance vpn1
ipv6 enable
ip address 10.209.0.1 255.255.0.0
ipv6 address 2001:db8:1::1/64
```

```
mac-address 0000-5e00-0102
ipv6 nd collect host enable
vxlan anycast-gateway enable
arp collect host enable
#
interface 10GE1/0/1
undo portswitch
ip address 10.56.1.1 255.255.255.0
#
interface 10GE1/0/2
#
interface 10GE1/0/2.2010 mode l2
encapsulation dot1q vid 3421
bridge-domain 5090
#
interface LoopBack0
ip address 192.168.2.2 255.255.255.255
#
interface LoopBack1
ip address 192.168.2.1 255.255.255.255
#
interface Nve1
vni 5154 head-end peer-list protocol bgp
#
bgp 100
group spine external
peer 10.56.1.2 as-number 65001
peer 10.56.1.2 group spine
#
ipv4-family unicast
import-route direct
peer spine enable
peer spine allow-as-loop
peer 10.56.1.2 enable
peer 10.56.1.2 group spine
#
ipv4-family vpn-instance vpn1
default-route imported
import-route direct
import-route static
advertise l2vpn evpn
#
ipv6-family vpn-instance vpn1
default-route imported
import-route direct
import-route static
advertise l2vpn evpn
#
bgp 200 instance evpn
peer 10.10.11.12 as-number 200
peer 10.10.11.12 connect-interface LoopBack0
peer 10.10.11.33 as-number 200
peer 10.10.11.33 connect-interface LoopBack0
#
l2vpn-family evpn
policy vpn-target
peer 10.10.11.12 enable
peer 10.10.11.12 advertise irb
peer 10.10.11.12 advertise irbv6
peer 10.10.11.33 enable
peer 10.10.11.33 advertise irb
peer 10.10.11.33 advertise irbv6
#
return
```

- Spine configuration file

```
#
sysname Spine
#
stp disable
```

```
#
evpn-overlay enable
#
dhcp server request-packet all-interface disable
#
interface 10GE1/0/1
undo portswitch
ip address 10.51.3.2 255.255.255.0
#
interface 10GE1/0/2
undo portswitch
ip address 10.51.1.2 255.255.255.0
#
interface 10GE1/0/3
undo portswitch
ip address 10.56.1.2 255.255.255.0
#
interface LoopBack0
ip address 10.10.11.33 255.255.255.255
#
bgp 65001
group leaf external
peer 10.51.1.1 as-number 100
peer 10.51.1.1 group leaf
peer 10.51.3.1 as-number 100
peer 10.51.3.1 group leaf
group leaf3 external
peer 10.56.1.1 as-number 100
peer 10.56.1.1 group leaf3
#
ipv4-family unicast
network 10.10.11.33 255.255.255.255
peer leaf enable
peer 10.51.1.1 enable
peer 10.51.1.1 group leaf
peer 10.51.3.1 enable
peer 10.51.3.1 group leaf
peer leaf3 enable
peer 10.56.1.1 enable
peer 10.56.1.1 group leaf3
#
bgp 200 instance evpn
peer 192.168.1.2 as-number 200
peer 192.168.1.2 connect-interface LoopBack0
peer 192.168.1.3 as-number 200
peer 192.168.1.3 connect-interface LoopBack0
peer 192.168.2.2 as-number 200
peer 192.168.2.2 connect-interface LoopBack0
#
l2vpn-family evpn
undo policy vpn-target
peer 192.168.1.2 enable
peer 192.168.1.2 advertise irb
peer 192.168.1.2 advertise irbv6
peer 192.168.1.2 reflect-client
peer 192.168.1.3 enable
peer 192.168.1.3 advertise irb
peer 192.168.1.3 advertise irbv6
peer 192.168.1.3 reflect-client
peer 192.168.2.2 enable
peer 192.168.2.2 advertise irb
peer 192.168.2.2 advertise irbv6
peer 192.168.2.2 reflect-client
#
return
```

1.14 Configuring M-LAG and Transparent Firewalls

Applicable Products and Versions

This example applies to CE12800, CE7800, CE6800, and CE5800 series switches running V100R005C10 or later versions, CE8800 series switches running V100R006C00 or later versions, CE9800 series switches running V200R020C00 or later versions, CloudEngine 16800 series switches running V200R005C20 or later versions, and CE12800E series switches running V200R005C00 or later versions.

For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

On the data center network shown in [Figure 1-14](#):

- Two core switches are connected through 10GE link aggregation.
- Firewalls at the aggregation layer connect to upstream and downstream devices through GE interfaces.
- Aggregation switches connect to upstream and downstream devices through 10GE interfaces.
- Multiple devices are deployed at the access layer and access devices connect to devices at the aggregation layer through 10GE interfaces.

This example uses the CE12804 switch and USG9520 gateway.

Figure 1-14 Networking for configuring M-LAG and transparent firewalls

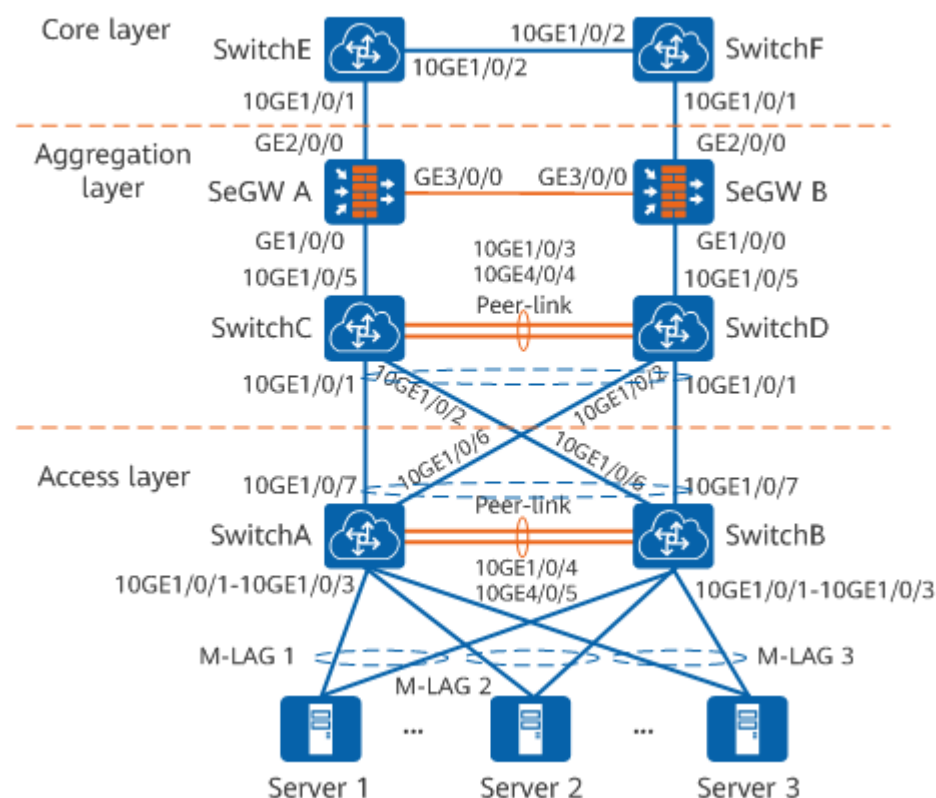


Table 1-14 Data plan

Device Name	Interface	IP Address	Virtual MAC Address
SwitchA	Management interface	10.1.1.1/24	-
SwitchB	Management interface	10.1.1.2/24	-
SwitchC	Management interface	10.2.1.1/24	-
	VLAN IF 11	10.3.1.1/24	0000-5e00-0101
	VLAN IF 200	10.4.1.1/24	-
	VLAN IF 300	10.6.1.1/24	-
SwitchD	Management interface	10.2.1.2/24	-
	VLAN IF 11	10.3.1.1/24	0000-5e00-0101
	VLAN IF 200	10.5.1.1/24	-
	VLAN IF 300	10.6.1.2/24	-
SwitchE	VLAN IF 200	10.4.1.2/24	-

Device Name	Interface	IP Address	Virtual MAC Address
	VLAN IF 400	10.7.1.1/24	-
SwitchF	VLAN IF 200	10.5.1.2/24	-
	VLAN IF 400	10.7.1.2/24	-
SeGW A	GigabitEthernet 3/0/0	10.10.0.1/24	-
SeGW B	GigabitEthernet 3/0/0	10.10.0.2/24	-

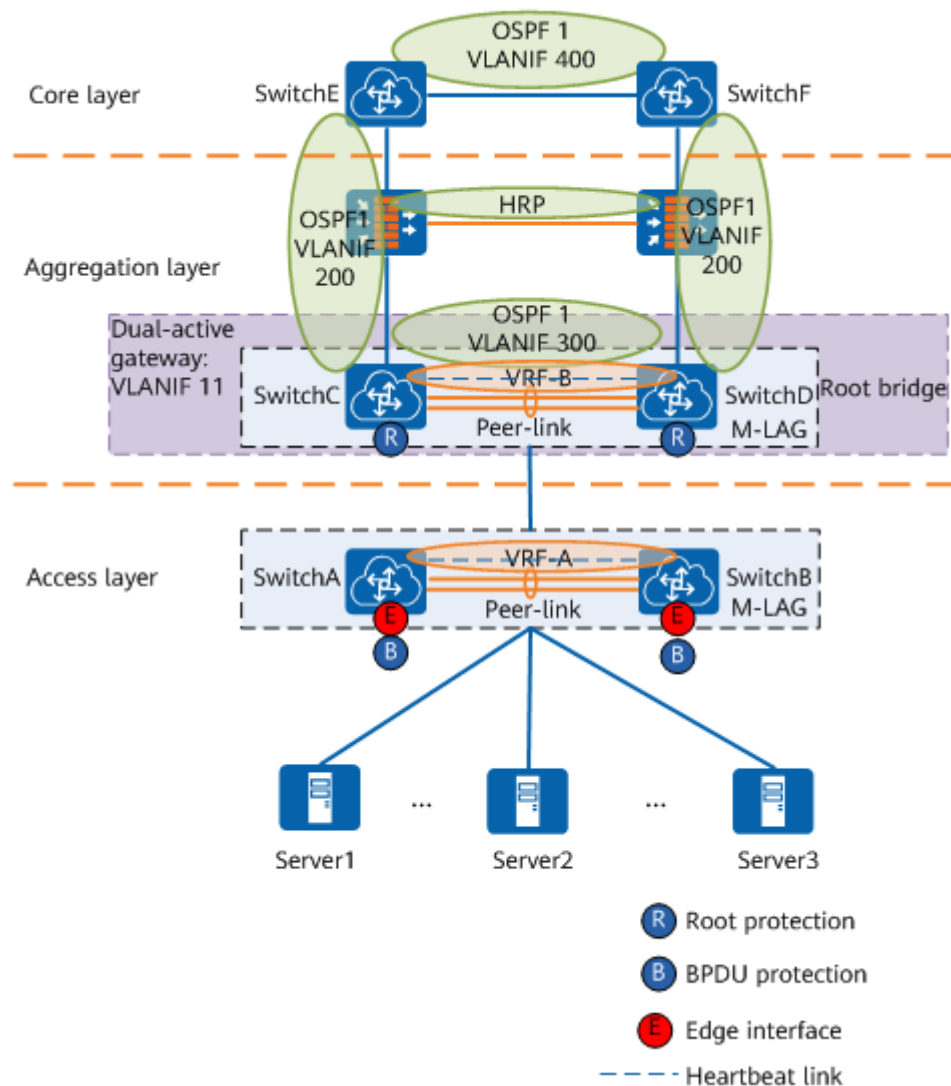
Requirement Analysis

The customer wants to build a stable large Layer 2 network where the dual-homing mode is used to ensure reliability and links load balance traffic to improve the link use efficiency. To ensure the server service security, switches connect to transparent firewalls (SeGWs) in inline mode to provide security defense.

- Devices at the core and aggregation layers constitute a square-shaped network so that traffic from gateways is sent and received along the same path.
- Security gateways at the aggregation layer use the transparent mode, are enabled with the Hot Redundancy Protocol (HRP), and work in load balance mode to enhance network robustness.
- M-LAG is deployed at the aggregation and access layers to form a loop-free topology.

Figure 1-15 shows the logical networking after M-LAG and transparent firewalls are deployed.

Figure 1-15 Networking for configuring M-LAG and transparent firewalls



Configuration Roadmap

The configuration roadmap is as follows:

1. Configure M-LAG between SwitchA and SwitchB and between SwitchC and SwitchD to implement dual-homing access. When access and aggregation switches work normally, links load balance traffic and a fault of any aggregation switch does not affect services. High service reliability is therefore ensured.
 - Configure SwitchC and SwitchD as root bridges and enable root protection on downstream interfaces to ensure that the interfaces can forward traffic normally. Configure interfaces on SwitchA and SwitchB connected to user terminals as edge interfaces to accelerate route convergence and enable BPDU protection to enhance network stability.
 - Create VLANIF interfaces on SwitchC and SwitchD and configure the same IP addresses and MAC address for the VLANIF interfaces to provide dual-active gateways.

2. Configure security gateways at the aggregation layer to use the transparent mode, enable HRP, and configure them to work in load balance mode to enhance network robustness.
3. Enable OSPF on SwitchC, SwitchD, SwitchE, and SwitchF to implement Layer 3 connectivity.

Procedure

Step 1 Configure M-LAG.

1. Configure dual-active detection links, V-STP, DFS groups, peer-links, and M-LAG member interfaces on SwitchA, SwitchB, SwitchC, and SwitchD.

The dual-active detection links are connected to management interfaces to implement interworking, DFS groups must be bound to IP addresses of management interfaces to ensure communication, and management interfaces are bound to VPN instances to implement isolation.

It is recommended that Eth-Trunk member interfaces of the peer-link be deployed on different cards to prevent the peer-link fault caused by a single-point failure.

Configure SwitchA.

Configure the Eth-Trunk on SwitchA connected to servers as the edge interface and enable BPDU protection on the Eth-Trunk. The Eth-Trunk on the access switch connected to Server 1 is used as an example.

The uplink interface of the server connected to the switch needs to be bound to an aggregated link, and the link aggregation modes on the server and switch must be consistent.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchA
[*HUAWEI] commit
[~SwitchA] stp mode rstp
[*SwitchA] stp v-stp enable
[*SwitchA] stp flush disable
[*SwitchA] ip vpn-instance VRF-A //Create VRF-A.
[SwitchA-vpn-instance-VRF-A] ipv4-family
[SwitchA-vpn-instance-VRF-A-af-ipv4] route-distinguisher 100:1
[SwitchA-vpn-instance-VRF-A-af-ipv4] vpn-target 111:1 both
[SwitchA-vpn-instance-VRF-A-af-ipv4] quit
[SwitchA-vpn-instance-VRF-A] quit
[*SwitchA] interface meth 0/0/0
[SwitchA-Meth0/0/0] ip binding vpn-instance VRF-A //Bind the management interface to VRF-A.
[SwitchA-Meth0/0/0] ip address 10.1.1.1 24
[SwitchA-Meth0/0/0] quit
[*SwitchA] dfs-group 1
[SwitchA-dfs-group-1] source ip 10.1.1.1 vpn-instance VRF-A peer 10.1.1.2 //Configure the IPv4
address and VPN instance bound to the DFS group.
[SwitchA-dfs-group-1] dual-active detection enhanced enable
[SwitchA-dfs-group-1] priority 150
[SwitchA-dfs-group-1] quit
[*SwitchA] interface eth-trunk 0
[SwitchA-Eth-Trunk0] trunkport 10ge 1/0/4
[SwitchA-Eth-Trunk0] trunkport 10ge 4/0/5 //Configure interfaces on different cards as Eth-Trunk
member interfaces of the peer-link.
[SwitchA-Eth-Trunk0] mode lacp-static
[SwitchA-Eth-Trunk0] peer-link 1
[SwitchA-Eth-Trunk0] port vlan exclude 1
[SwitchA-Eth-Trunk0] quit
[*SwitchA] vlan batch 11
```

```
[*SwitchA] interface eth-trunk 10
[*SwitchA-Eth-Trunk10] mode lacp-dynamic
[*SwitchA-Eth-Trunk10] port link-type access
[*SwitchA-Eth-Trunk10] port default vlan 11
[*SwitchA-Eth-Trunk10] trunkport 10ge 1/0/1
[*SwitchA-Eth-Trunk10] dfs-group 1 m-lag 1
[*SwitchA-Eth-Trunk10] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[*SwitchA-Eth-Trunk10] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchA-Eth-Trunk10] quit
[*SwitchA] interface eth-trunk 20
[*SwitchA-Eth-Trunk20] mode lacp-dynamic
[*SwitchA-Eth-Trunk20] port link-type access
[*SwitchA-Eth-Trunk20] port default vlan 11
[*SwitchA-Eth-Trunk20] trunkport 10ge 1/0/2
[*SwitchA-Eth-Trunk20] dfs-group 1 m-lag 2
[*SwitchA-Eth-Trunk20] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[*SwitchA-Eth-Trunk20] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchA-Eth-Trunk20] quit
[*SwitchA] interface eth-trunk 30
[*SwitchA-Eth-Trunk30] mode lacp-dynamic
[*SwitchA-Eth-Trunk30] port link-type access
[*SwitchA-Eth-Trunk30] port default vlan 11
[*SwitchA-Eth-Trunk30] trunkport 10ge 1/0/3
[*SwitchA-Eth-Trunk30] dfs-group 1 m-lag 3
[*SwitchA-Eth-Trunk30] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[*SwitchA-Eth-Trunk30] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (On the CE8800, CE7800, CE6800 (except the CE6870EI and
CE6875EI), and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view.)
[*SwitchA-Eth-Trunk30] quit
[*SwitchA] stp bpdu-protection //Enable BPDU protection on the edge interface.
[*SwitchA] interface eth-trunk 40
[*SwitchA-Eth-Trunk40] mode lacp-static
[*SwitchA-Eth-Trunk40] port link-type trunk
[*SwitchA-Eth-Trunk40] undo port trunk allow-pass vlan 1
[*SwitchA-Eth-Trunk40] port trunk allow-pass vlan 11
[*SwitchA-Eth-Trunk40] trunkport 10ge 1/0/6 to 1/0/7
[*SwitchA-Eth-Trunk40] dfs-group 1 m-lag 4
[*SwitchA-Eth-Trunk40] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (On the CE8800, CE7800, CE6800 (except the CE6870EI and
CE6875EI), and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view.)
[*SwitchA-Eth-Trunk40] quit
[*SwitchA] lacp m-lag priority 10
[*SwitchA] lacp m-lag system-id 00e0-fc12-3456
[*SwitchA] interface 10ge 1/0/9
[*SwitchA-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchA-10GE1/0/9] quit
[*SwitchA] commit
```

Configure SwitchB.

Configure the Eth-Trunk on SwitchB connected to servers as the edge interface and enable BPDU protection on the Eth-Trunk. The Eth-Trunk on the access switch connected to Server 1 is used as an example.

The uplink interface of the server connected to the switch needs to be bound to an aggregated link, and the link aggregation modes on the server and switch must be consistent.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchB
```

```
[*HUAWEI] commit
[~SwitchB] stp mode rstp
[SwitchB] stp v-stp enable
[SwitchB] stp flush disable
[SwitchB] ip vpn-instance VRF-A //Create VRF-A.
[SwitchB-vpn-instance-VRF-A] ipv4-family
[SwitchB-vpn-instance-VRF-A-af-ipv4] route-distinguisher 100:2
[SwitchB-vpn-instance-VRF-A-af-ipv4] vpn-target 111:1 both
[SwitchB-vpn-instance-VRF-A-af-ipv4] quit
[SwitchB-vpn-instance-VRF-A] quit
[SwitchB] interface meth 0/0/0
[SwitchB-MEth0/0/0] ip binding vpn-instance VRF-A //Bind the management interface to VRF-A.
[SwitchB-MEth0/0/0] ip address 10.1.1.2 24
[SwitchB-MEth0/0/0] quit
[SwitchB] dfs-group 1
[SwitchB-dfs-group-1] source ip 10.1.1.2 vpn-instance VRF-A peer 10.1.1.1 //Configure the IPv4
address and VPN instance bound to the DFS group.
[SwitchB-dfs-group-1] dual-active detection enhanced enable
[SwitchB-dfs-group-1] priority 120
[SwitchB-dfs-group-1] quit
[SwitchB] interface eth-trunk 0
[SwitchB-Eth-Trunk0] trunkport 10ge 1/0/4
[SwitchB-Eth-Trunk0] trunkport 10ge 4/0/5 //Configure inter-card member interfaces of the Eth-
Trunk of the peer-link.
[SwitchB-Eth-Trunk0] mode lacp-static
[SwitchB-Eth-Trunk0] peer-link 1
[SwitchB-Eth-Trunk0] port vlan exclude 1
[SwitchB-Eth-Trunk0] quit
[SwitchB] vlan batch 11
[SwitchB] interface eth-trunk 10
[SwitchB-Eth-Trunk10] mode lacp-dynamic
[SwitchB-Eth-Trunk10] port link-type access
[SwitchB-Eth-Trunk10] port default vlan 11
[SwitchB-Eth-Trunk10] trunkport 10ge 1/0/1
[SwitchB-Eth-Trunk10] dfs-group 1 m-lag 1
[SwitchB-Eth-Trunk10] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[SwitchB-Eth-Trunk10] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (On the CE8800, CE7800, CE6800 (except the CE6870EI and
CE6875EI), and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view.)
[SwitchB-Eth-Trunk10] quit
[SwitchB] interface eth-trunk 20
[SwitchB-Eth-Trunk20] mode lacp-dynamic
[SwitchB-Eth-Trunk20] port link-type access
[SwitchB-Eth-Trunk20] port default vlan 11
[SwitchB-Eth-Trunk20] trunkport 10ge 1/0/2
[SwitchB-Eth-Trunk20] dfs-group 1 m-lag 2
[SwitchB-Eth-Trunk20] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[SwitchB-Eth-Trunk20] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (On the CE8800, CE7800, CE6800 (except the CE6870EI and
CE6875EI), and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view.)
[SwitchB-Eth-Trunk20] quit
[SwitchB] interface eth-trunk 30
[SwitchB-Eth-Trunk30] mode lacp-dynamic
[SwitchB-Eth-Trunk30] port link-type access
[SwitchB-Eth-Trunk30] port default vlan 11
[SwitchB-Eth-Trunk30] trunkport 10ge 1/0/3
[SwitchB-Eth-Trunk30] dfs-group 1 m-lag 3
[SwitchB-Eth-Trunk30] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[SwitchB-Eth-Trunk30] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (On the CE8800, CE7800, CE6800 (except the CE6870EI and
CE6875EI), and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view.)
[SwitchB-Eth-Trunk30] quit
[SwitchB] stp bpdu-protection //Enable BPDU protection on the edge interface.
[SwitchB] interface eth-trunk 40
[SwitchB-Eth-Trunk40] mode lacp-static
[SwitchB-Eth-Trunk40] port link-type trunk
```

```
[*SwitchB-Eth-Trunk40] undo port trunk allow-pass vlan 1
[*SwitchB-Eth-Trunk40] port trunk allow-pass vlan 11
[*SwitchB-Eth-Trunk40] trunkport 10ge 1/0/6 to 1/0/7
[*SwitchB-Eth-Trunk40] dfs-group 1 m-lag 4
[*SwitchB-Eth-Trunk40] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (On the CE8800, CE7800, CE6800 (except the CE6870EI and
CE6875EI), and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view.)
[*SwitchB-Eth-Trunk40] quit
[*SwitchB] lacp m-lag priority 10
[*SwitchB] lacp m-lag system-id 00e0-fc12-3456
[*SwitchB] interface 10ge 1/0/9
[*SwitchB-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchB-10GE1/0/9] quit
[*SwitchB] commit
```

Configure SwitchC.

Configure SwitchC as the root bridge of the STP network, and enable root protection on the Eth-Trunk of SwitchC connected to the access switch so that the Eth-Trunk can forward traffic normally.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchC
[HUAWEI] commit
[~SwitchC] stp mode rstp
[*SwitchC] stp root primary //Configure the aggregation device as the root bridge of the STP
network.
[*SwitchC] stp bridge-address 00e0-fc39-1300 //Configure the MAC address of the root bridge
(MAC address of the master device).
[*SwitchC] stp v-stp enable
[*SwitchC] stp flush disable
[*SwitchC] ip vpn-instance VRF-B //Create VRF-B.
[*SwitchC-vpn-instance-VRF-B] ipv4-family
[*SwitchC-vpn-instance-VRF-B-af-ipv4] route-distinguisher 101:1
[*SwitchC-vpn-instance-VRF-B-af-ipv4] vpn-target 111:1 both
[*SwitchC-vpn-instance-VRF-B-af-ipv4] quit
[*SwitchC-vpn-instance-VRF-B] quit
[*SwitchC] interface meth 0/0/0
[*SwitchC-MEth0/0/0] ip binding vpn-instance VRF-B //Bind the management interface to VRF-B.
[*SwitchC-MEth0/0/0] ip address 10.2.1.1 24
[*SwitchC-MEth0/0/0] quit
[*SwitchC] dfs-group 1
[*SwitchC-dfs-group-1] source ip 10.2.1.1 vpn-instance VRF-B peer 10.2.1.2 //Configure the IPv4
address and VPN instance bound to the DFS group.
[*SwitchC-dfs-group-1] dual-active detection enhanced enable
[*SwitchC-dfs-group-1] priority 150
[*SwitchC-dfs-group-1] quit
[*SwitchC] interface eth-trunk 0
[*SwitchC-Eth-Trunk0] trunkport 10ge 1/0/3
[*SwitchC-Eth-Trunk0] trunkport 10ge 4/0/4 //Configure inter-card member interfaces of the Eth-
Trunk of the peer-link.
[*SwitchC-Eth-Trunk0] mode lacp-static
[*SwitchC-Eth-Trunk0] peer-link 1
[*SwitchC-Eth-Trunk0] port vlan exclude 1
[*SwitchC-Eth-Trunk0] quit
[*SwitchC] vlan batch 11
[*SwitchC] interface eth-trunk 30
[*SwitchC-Eth-Trunk30] mode lacp-static
[*SwitchC-Eth-Trunk30] port link-type trunk
[*SwitchC-Eth-Trunk30] undo port trunk allow-pass vlan 1
[*SwitchC-Eth-Trunk30] port trunk allow-pass vlan 11
[*SwitchC-Eth-Trunk30] trunkport 10ge 1/0/1 to 1/0/2
[*SwitchC-Eth-Trunk30] dfs-group 1 m-lag 1
[*SwitchC-Eth-Trunk30] stp root-protection //Enable root protection.
[*SwitchC-Eth-Trunk30] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (On the CE8800, CE7800, CE6800 (except the CE6870EI and
CE6875EI), and CE5800 series switches, this command can be run only in the GE interface view, 10GE
```



```
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view.)  
[*SwitchC-Eth-Trunk30] quit  
[*SwitchC] lacp m-lag priority 10  
[*SwitchC] lacp m-lag system-id 00e0-fc12-3451  
[*SwitchC] commit
```

Configure SwitchD.

Configure SwitchD as the root bridge of the STP network, and enable root protection on the Eth-Trunk of SwitchD connected to the access switch so that the Eth-Trunk can forward traffic normally.

```
<HUAWEI> system-view  
[~HUAWEI] sysname SwitchD  
[*HUAWEI] commit  
[~SwitchD] stp mode rstp  
[*SwitchD] stp root primary //Configure the aggregation device as the root bridge of the STP network.  
[*SwitchD] stp bridge-address 00e0-fc39-1300 //Configure the MAC address of the root bridge (MAC address of the master device).  
[*SwitchD] stp v-stp enable  
[*SwitchD] stp flush disable  
[*SwitchD] ip vpn-instance VRF-B //Create VRF-B.  
[*SwitchD-vpn-instance-VRF-B] ipv4-family  
[*SwitchD-vpn-instance-VRF-B-af-ipv4] route-distinguisher 101:2  
[*SwitchD-vpn-instance-VRF-B-af-ipv4] vpn-target 111:1 both  
[*SwitchD-vpn-instance-VRF-B-af-ipv4] quit  
[*SwitchD-vpn-instance-VRF-B] quit  
[*SwitchD] interface meth 0/0/0  
[*SwitchD-MEth0/0/0] ip binding vpn-instance VRF-B //Bind the management interface to VRF-B.  
[*SwitchD-MEth0/0/0] ip address 10.2.1.2 24  
[*SwitchD-MEth0/0/0] quit  
[*SwitchD] dfs-group 1  
[*SwitchD-dfs-group-1] source ip 10.2.1.2 vpn-instance VRF-B peer 10.2.1.1 //Configure the IPv4 address and VPN instance bound to the DFS group.  
[*SwitchD-dfs-group-1] dual-active detection enhanced enable  
[*SwitchD-dfs-group-1] priority 120  
[*SwitchD-dfs-group-1] quit  
[*SwitchD] interface eth-trunk 0  
[*SwitchD-Eth-Trunk0] trunkport 10ge 1/0/3  
[*SwitchD-Eth-Trunk0] trunkport 10ge 4/0/4 //Configure inter-card member interfaces of the Eth-Trunk of the peer-link.  
[*SwitchD-Eth-Trunk0] mode lacp-static  
[*SwitchD-Eth-Trunk0] peer-link 1  
[*SwitchD-Eth-Trunk0] port vlan exclude 1  
[*SwitchD-Eth-Trunk0] quit  
[*SwitchD] vlan batch 11  
[*SwitchD] interface eth-trunk 30  
[*SwitchD-Eth-Trunk30] mode lacp-static  
[*SwitchD-Eth-Trunk30] port link-type trunk  
[*SwitchD-Eth-Trunk30] undo port trunk allow-pass vlan 1  
[*SwitchD-Eth-Trunk30] port trunk allow-pass vlan 11  
[*SwitchD-Eth-Trunk30] trunkport 10ge 1/0/1 to 1/0/2  
[*SwitchD-Eth-Trunk30] dfs-group 1 m-lag 1  
[*SwitchD-Eth-Trunk30] stp root-protection //Enable root protection.  
[*SwitchD-Eth-Trunk30] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast packets to 10 Mbit/s on this interface. (On the CE8800, CE7800, CE6800 (except the CE6870EI and CE6875EI), and CE5800 series switches, this command can be run only in the GE interface view, 10GE interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view.)  
[*SwitchD-Eth-Trunk30] quit  
[*SwitchD] lacp m-lag priority 10  
[*SwitchD] lacp m-lag system-id 00e0-fc12-3451  
[*SwitchD] commit
```

2. Create VLANIF interfaces on SwitchC and SwitchD and configure IP addresses and virtual MAC addresses for the VLANIF interfaces to configure active-active gateways.

Configure SwitchC.

```
[~SwitchC] interface vlanif 11
[*SwitchC-Vlanif11] ip address 10.3.1.1 24
[*SwitchC-Vlanif11] mac-address 0000-5e00-0101
[*SwitchC-Vlanif11] quit
[*SwitchC] commit
```

Configure SwitchD.

```
[~SwitchD] interface vlanif 11
[*SwitchD-Vlanif11] ip address 10.3.1.1 24
[*SwitchD-Vlanif11] mac-address 0000-5e00-0101
[*SwitchD-Vlanif11] quit
[*SwitchD] commit
```

Step 2 Configure SeGWA and SeGWB to work in transparent mode and enable HRP.

1. Switch service interfaces of SeGWA and SeGWB to Layer 2 interfaces and add them to the same VLAN.

Configure SeGWA.

```
<USG9000> system-view
[USG9000] sysname SeGWA
[SeGWA] interface GigabitEthernet 1/0/0
[SeGWA-GigabitEthernet1/0/0] portswitch
[SeGWA-GigabitEthernet1/0/0] quit
[SeGWA] interface GigabitEthernet 2/0/0
[SeGWA-GigabitEthernet2/0/0] portswitch
[SeGWA-GigabitEthernet2/0/0] quit
[SeGWA] vlan 200
[SeGWA-vlan200] port GigabitEthernet 1/0/0
[SeGWA-vlan200] port GigabitEthernet 2/0/0
[SeGWA-vlan200] quit
```

Configure SeGWB.

```
<USG9000> system-view
[USG9000] sysname SeGWB
[SeGWB] interface GigabitEthernet 1/0/0
[SeGWB-GigabitEthernet1/0/0] portswitch
[SeGWB-GigabitEthernet1/0/0] quit
[SeGWB] interface GigabitEthernet 2/0/0
[SeGWB-GigabitEthernet2/0/0] portswitch
[SeGWB-GigabitEthernet2/0/0] quit
[SeGWB] vlan 200
[SeGWB-vlan200] port GigabitEthernet 1/0/0
[SeGWB-vlan200] port GigabitEthernet 2/0/0
[SeGWB-vlan200] quit
```

2. Configure IP addresses for heartbeat interfaces of SeGWA and SeGWB.

Configure SeGWA.

```
[SeGWA] interface GigabitEthernet 3/0/0
[SeGWA-GigabitEthernet3/0/0] ip address 10.10.0.1 24
[SeGWA-GigabitEthernet3/0/0] quit
```

Configure SeGWB.

```
[SeGWB] interface GigabitEthernet 3/0/0
[SeGWB-GigabitEthernet3/0/0] ip address 10.10.0.2 24
[SeGWB-GigabitEthernet3/0/0] quit
```

3. Add uplink interfaces of SeGWA and SeGWB to the untrusted zone, downlink interfaces to the trusted zone, and heartbeat interfaces to the DMZ.

Configure SeGWA.

```
[SeGWA] firewall zone untrust
[SeGWA-zone-untrust] add interface GigabitEthernet 2/0/0
```

```
[SeGWA-zone-untrust] quit
[SeGWA] firewall zone trust
[SeGWA-zone-trust] add interface GigabitEthernet 1/0/0
[SeGWA-zone-trust] quit
[SeGWA] firewall zone dmz
[SeGWA-zone-dmz] add interface GigabitEthernet 3/0/0
[SeGWA-zone-dmz] quit
```

Configure SeGWB.

```
[SeGWB] firewall zone untrust
[SeGWB-zone-untrust] add interface GigabitEthernet 2/0/0
[SeGWB-zone-untrust] quit
[SeGWB] firewall zone trust
[SeGWB-zone-trust] add interface GigabitEthernet 1/0/0
[SeGWB-zone-trust] quit
[SeGWB] firewall zone dmz
[SeGWB-zone-dmz] add interface GigabitEthernet 3/0/0
[SeGWB-zone-dmz] quit
```

4. Configure the VLAN for the service interface associated with the VGMP group, specify the heartbeat interface, and enable HRP.

Configure SeGWA.

```
[SeGWA] hrp track vlan 200
[SeGWA] hrp interface GigabitEthernet 3/0/0 remote 10.10.0.2
[SeGWA] hrp enable
[SeGWA] hrp mirror session enable
```

Configure SeGWB.

```
[SeGWB] hrp track vlan 200
[SeGWB] hrp interface GigabitEthernet 3/0/0 remote 10.10.0.1
[SeGWB] hrp enable
[SeGWB] hrp mirror session enable
```

5. Configure security functions such as the security policy, IPS, and attack defense on SeGWA. The configuration of SeGWA is automatically backed up to SeGWB. For details, see the security gateway documentation.

Step 3 Enable OSPF on SwitchC, SwitchD, SwitchE, and SwitchF.

1. Add interfaces on SwitchC, SwitchD, SwitchE, and SwitchF to VLANs and configure IP addresses for VLANIF interfaces.

Configure SwitchC.

```
[~SwitchC] vlan batch 200 300
[*SwitchC] interface 10ge 1/0/5
[*SwitchC-10GE1/0/5] port link-type trunk
[*SwitchC-10GE1/0/5] undo port trunk allow-pass vlan 1
[*SwitchC-10GE1/0/5] port trunk allow-pass vlan 200
[*SwitchC-10GE1/0/5] storm suppression broadcast 1 //Configure the percentage of bandwidth
occupied by broadcast packets on the interface, and the percentage rate limit is 1%.
[*SwitchC-10GE1/0/5] quit
[*SwitchC] interface eth-trunk 0
[*SwitchC-Eth-Trunk0] port vlan exclude 200 //Configure the peer-link interface not to allow
packets from VLAN 200.
[*SwitchC-Eth-Trunk0] quit
[*SwitchC] interface vlanif 200
[*SwitchC-Vlanif200] ospf network-type p2p
[*SwitchC-Vlanif200] ip address 10.4.1.1 24
[*SwitchC-Vlanif200] quit
[*SwitchC] interface vlanif 300
[*SwitchC-Vlanif300] ospf network-type p2p
[*SwitchC-Vlanif300] ip address 10.6.1.1 24
[*SwitchC-Vlanif300] quit
[*SwitchC] interface 10ge 1/0/9
```

```
[*SwitchC-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchC-10GE1/0/9] quit
[*SwitchC] commit
```

Configure SwitchD.

```
[~SwitchD] vlan batch 200 300
[*SwitchD] interface 10ge 1/0/5
[*SwitchD-10GE1/0/5] port link-type trunk
[*SwitchD-10GE1/0/5] undo port trunk allow-pass vlan 1
[*SwitchD-10GE1/0/5] port trunk allow-pass vlan 200
[*SwitchD-10GE1/0/5] storm suppression broadcast 1 //Configure the percentage of bandwidth
occupied by broadcast packets on the interface, and the percentage rate limit is 1%.
[*SwitchD-10GE1/0/5] quit
[*SwitchD] interface eth-trunk 0
[*SwitchD-Eth-Trunk0] port vlan exclude 200 //Configure the peer-link interface not to allow
packets from VLAN 200.
[*SwitchD-Eth-Trunk0] quit
[*SwitchD] interface vlanif 200
[*SwitchD-Vlanif200] ospf network-type p2p
[*SwitchD-Vlanif200] ip address 10.5.1.1 24
[*SwitchD-Vlanif200] quit
[*SwitchD] interface vlanif 300
[*SwitchD-Vlanif300] ospf network-type p2p
[*SwitchD-Vlanif300] ip address 10.6.1.2 24
[*SwitchD-Vlanif300] quit
[*SwitchD] interface 10ge 1/0/9
[*SwitchD-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchD-10GE1/0/9] quit
[*SwitchD] commit
```

Configure SwitchE.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchE
[*HUAWEI] commit
[~SwitchE] vlan batch 200 400
[*SwitchE] interface 10ge 1/0/1
[*SwitchE-10GE1/0/1] port link-type trunk
[*SwitchE-10GE1/0/1] undo port trunk allow-pass vlan 1
[*SwitchE-10GE1/0/1] port trunk allow-pass vlan 200
[*SwitchE-10GE1/0/1] storm suppression broadcast 1 //Configure the percentage of bandwidth
occupied by broadcast packets on the interface, and the percentage rate limit is 1%.
[*SwitchE-10GE1/0/1] quit
[*SwitchE] interface 10ge 1/0/2
[*SwitchE-10GE1/0/2] port link-type trunk
[*SwitchE-10GE1/0/2] undo port trunk allow-pass vlan 1
[*SwitchE-10GE1/0/2] port trunk allow-pass vlan 400
[*SwitchE-10GE1/0/2] storm suppression broadcast 1 //Configure the percentage of bandwidth
occupied by broadcast packets on the interface, and the percentage rate limit is 1%.
[*SwitchE-10GE1/0/2] quit
[*SwitchE] interface vlanif 200
[*SwitchE-Vlanif200] ospf network-type p2p
[*SwitchE-Vlanif200] ip address 10.4.1.2 24
[*SwitchE-Vlanif200] quit
[*SwitchE] interface vlanif 400
[*SwitchE-Vlanif400] ospf network-type p2p
[*SwitchE-Vlanif400] ip address 10.7.1.1 24
[*SwitchE-Vlanif400] quit
[*SwitchE] interface 10ge 1/0/9
[*SwitchE-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchE-10GE1/0/9] quit
[*SwitchE] commit
```

Configure SwitchF.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchF
```

```
[*HUAWEI] commit
[~SwitchF] vlan batch 200 400
[*SwitchF] interface 10ge 1/0/1
[*SwitchF-10GE1/0/1] port link-type trunk
[*SwitchF-10GE1/0/1] undo port trunk allow-pass vlan 1
[*SwitchF-10GE1/0/1] port trunk allow-pass vlan 200
[*SwitchF-10GE1/0/1] storm suppression broadcast 1 //Configure the percentage of bandwidth
occupied by broadcast packets on the interface, and the percentage rate limit is 1%.
[*SwitchF-10GE1/0/1] quit
[*SwitchF] interface 10ge 1/0/2
[*SwitchF-10GE1/0/2] port link-type trunk
[*SwitchF-10GE1/0/2] undo port trunk allow-pass vlan 1
[*SwitchF-10GE1/0/2] port trunk allow-pass vlan 400
[*SwitchF-10GE1/0/2] storm suppression broadcast 1 //Configure the percentage of bandwidth
occupied by broadcast packets on the interface, and the percentage rate limit is 1%.
[*SwitchF-10GE1/0/2] quit
[*SwitchF] interface vlanif 200
[*SwitchF-Vlanif200] ospf network-type p2p
[*SwitchF-Vlanif200] ip address 10.5.1.2 24
[*SwitchF-Vlanif200] quit
[*SwitchF] interface vlanif 400
[*SwitchF-Vlanif400] ospf network-type p2p
[*SwitchF-Vlanif400] ip address 10.7.1.2 24
[*SwitchF-Vlanif400] quit
[*SwitchF] interface 10ge 1/0/9
[*SwitchF-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchF-10GE1/0/9] quit
[*SwitchF] commit
```

2. Configure OSPF on SwitchC, SwitchD, SwitchE, and SwitchF to ensure Layer 3 connectivity.

Configure SwitchC.

```
[~SwitchC] ospf 1
[~SwitchC-ospf-1] import-route direct //Configure the switch to import direct routes. You can
configure a routing policy to filter unnecessary routes.
[*SwitchC-ospf-1] area 0
[*SwitchC-ospf-1-area-0.0.0.0] network 10.4.1.0 0.0.0.255
[*SwitchC-ospf-1-area-0.0.0.0] network 10.6.1.0 0.0.0.255
[*SwitchC-ospf-1-area-0.0.0.0] quit
[*SwitchC-ospf-1] quit
[*SwitchC] commit
```

Configure SwitchD.

```
[~SwitchD] ospf 1
[~SwitchD-ospf-1] import-route direct //Configure the switch to import direct routes. You can
configure a routing policy to filter unnecessary routes.
[*SwitchD-ospf-1] area 0
[*SwitchD-ospf-1-area-0.0.0.0] network 10.5.1.0 0.0.0.255
[*SwitchD-ospf-1-area-0.0.0.0] network 10.6.1.0 0.0.0.255
[*SwitchD-ospf-1-area-0.0.0.0] quit
[*SwitchD-ospf-1] quit
[*SwitchD] commit
```

Configure SwitchE.

```
[~SwitchE] ospf 1
[*SwitchE-ospf-1] area 0
[*SwitchE-ospf-1-area-0.0.0.0] network 10.4.1.0 0.0.0.255
[*SwitchE-ospf-1-area-0.0.0.0] network 10.7.1.0 0.0.0.255
[*SwitchE-ospf-1-area-0.0.0.0] quit
[*SwitchE-ospf-1] quit
[*SwitchE] commit
```

Configure SwitchF.

```
[~SwitchF] ospf 1
[*SwitchF-ospf-1] area 0
```

```
[*SwitchF-ospf-1-area-0.0.0.0] network 10.5.1.0 0.0.0.255
[*SwitchF-ospf-1-area-0.0.0.0] network 10.7.1.0 0.0.0.255
[*SwitchF-ospf-1-area-0.0.0.0] quit
[*SwitchF-ospf-1] quit
[*SwitchF] commit
```

Step 4 Verify the configuration.

- Run the **display dfs-group** command to check M-LAG information.

Check information about the M-LAG with DFS group 1.

```
[~SwitchA] display dfs-group 1 m-lag
```

```
*          : Local node
Heart beat state : OK
Node 1 *
  Dfs-Group ID   : 1
  Priority        : 150
  Address         : ip address 10.1.1.1
  State          : Master
  Causation      : -
  System ID      : 00e0-fc95-7c01
  SysName        : SwitchA
  Version        : V100R006C00
  Device Type    : CE12800
```

```
Node 2
  Dfs-Group ID   : 1
  Priority        : 120
  Address         : ip address 10.1.1.2
  State          : Backup
  Causation      : -
  System ID      : 00e0-fc95-7c11
  SysName        : SwitchB
  Version        : V100R006C00
  Device Type    : CE12800
```

```
[~SwitchC] display dfs-group 1 m-lag
```

```
*          : Local node
Heart beat state : OK
Node 1 *
  Dfs-Group ID   : 1
  Priority        : 150
  Address         : ip address 10.2.1.1
  State          : Master
  Causation      : -
  System ID      : 00e0-fc39-1300
  SysName        : SwitchC
  Version        : V100R006C00
  Device Type    : CE12800
```

```
Node 2
  Dfs-Group ID   : 1
  Priority        : 120
  Address         : ip address 10.2.1.2
  State          : Backup
  Causation      : -
  System ID      : 00e0-fc39-1311
  SysName        : SwitchD
  Version        : V100R006C00
  Device Type    : CE12800
```

Check M-LAG information on SwitchA.

```
[~SwitchA] display dfs-group 1 node 1 m-lag brief
```

```
* - Local node
```

M-Lag ID	Interface	Port State	Status
1	Eth-Trunk 10	Up	active(*)-active
2	Eth-Trunk 20	Up	active(*)-active
3	Eth-Trunk 30	Up	active(*)-active
4	Eth-Trunk 40	Up	active(*)-active

Check M-LAG information on SwitchC.

```
[~SwitchC] display dfs-group 1 node 2 m-lag brief
* - Local node
```

M-Lag ID	Interface	Port State	Status
1	Eth-Trunk 30	Up	active-active(*)

In the preceding information, the value of **Heart beat state** is **OK**, indicating that the dual-active detection status is normal. SwitchA and SwitchC are used as Node 1, the priority is 150, and the value of **State** is **Master**. SwitchB and SwitchD are used as Node 2, the priority is 120, and the value of **State** is **Backup**. The value of **Causation** is -, the values of **Port State** of Node 1 and Node 2 are both **Up**, and the M-LAG status of both Node 1 and Node 2 is **active**, indicating that the M-LAG configuration is correct.

- Run the **display hrp state** command on SeGW A to check the HRP status. The following information indicates that the HRP is set up successfully.

```
HRP_M[SeGWA] display hrp state
Role: active, peer: active
Running priority: 51008, peer: 51008
Core state: normal, peer: normal
Backup channel usage: 0%
Stable time: 0 days, 18 hours, 41 minutes
```

----End

Configuration Files

- SwitchA configuration file

```
#
sysname SwitchA
#
dfs-group 1
priority 150
source ip 10.1.1.1 vpn-instance VRF-A peer 10.1.1.2
dual-active detection enhanced enable
#
vlan batch 11
#
stp mode rstp
stp v-stp enable
stp bpdu-protection
stp flush disable
#
lacp m-lag system-id 00e0-fc12-3456
lacp m-lag priority 10
#
ip vpn-instance VRF-A
ipv4-family
route-distinguisher 100:1
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface MEth0/0/0
ip binding vpn-instance VRF-A
ip address 10.1.1.1 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk10
port default vlan 11
stp edged-port enable
mode lacp-dynamic
dfs-group 1 m-lag 1
```

```
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk20
port default vlan 11
stp edged-port enable
mode lacp-dynamic
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk30
port default vlan 11
stp edged-port enable
mode lacp-dynamic
dfs-group 1 m-lag 3
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk40
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
mode lacp-static
dfs-group 1 m-lag 4
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 10
#
interface 10GE1/0/2
eth-trunk 20
#
interface 10GE1/0/3
eth-trunk 30
#
interface 10GE1/0/4
eth-trunk 0
#
interface 10GE1/0/6
eth-trunk 40
#
interface 10GE1/0/7
eth-trunk 40
#
interface 10GE1/0/9
shutdown
#
interface 10GE4/0/5
eth-trunk 0
#
return
```

- SwitchB configuration file

```
#
sysname SwitchB
#
dfs-group 1
priority 120
source ip 10.1.1.2 vpn-instance VRF-A peer 10.1.1.1
dual-active detection enhanced enable
#
vlan batch 11
#
stp mode rstp
stp v-stp enable
stp bpdu-protection
stp flush disable
#
lacp m-lag system-id 00e0-fc12-3456
lacp m-lag priority 10
#
ip vpn-instance VRF-A
```



```
ipv4-family
 route-distinguisher 100:2
 vpn-target 111:1 export-extcommunity
 vpn-target 111:1 import-extcommunity
#
interface MEth0/0/0
 ip binding vpn-instance VRF-A
 ip address 10.1.1.2 255.255.255.0
#
interface Eth-Trunk0
 mode lacp-static
 peer-link 1
 port vlan exclude 1
#
interface Eth-Trunk10
 port default vlan 11
 stp edged-port enable
 mode lacp-dynamic
 dfs-group 1 m-lag 1
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk20
 port default vlan 11
 stp edged-port enable
 mode lacp-dynamic
 dfs-group 1 m-lag 2
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk30
 port default vlan 11
 stp edged-port enable
 mode lacp-dynamic
 dfs-group 1 m-lag 3
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk40
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 11
 mode lacp-static
 dfs-group 1 m-lag 4
 storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
 eth-trunk 10
#
interface 10GE1/0/2
 eth-trunk 20
#
interface 10GE1/0/3
 eth-trunk 30
#
interface 10GE1/0/4
 eth-trunk 0
#
interface 10GE1/0/6
 eth-trunk 40
#
interface 10GE1/0/7
 eth-trunk 40
#
interface 10GE1/0/9
 shutdown
#
interface 10GE4/0/5
 eth-trunk 0
#
return
```

- SwitchC configuration file

```
#
sysname SwitchC
#
dfs-group 1
priority 150
source ip 10.2.1.1 vpn-instance VRF-B peer 10.2.1.2
dual-active detection enhanced enable
#
vlan batch 11 200 300
#
stp bridge-address 00e0-fc39-1300
stp mode rstp
stp v-stp enable
stp instance 0 root primary
stp flush disable
#
lACP m-lag system-id 00e0-fc12-3451
lACP m-lag priority 10
#
ip vpn-instance VRF-B
ipv4-family
route-distinguisher 101:1
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface Vlanif11
ip address 10.3.1.1 255.255.255.0
mac-address 0000-5e00-0101
#
interface Vlanif200
ip address 10.4.1.1 255.255.255.0
ospf network-type p2p
#
interface Vlanif300
ip address 10.6.1.1 255.255.255.0
ospf network-type p2p
#
interface MEth0/0/0
ip binding vpn-instance VRF-B
ip address 10.2.1.1 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1 200
#
interface Eth-Trunk30
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 30
#
interface 10GE1/0/2
eth-trunk 30
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 200
storm suppression broadcast 1
```

```
#
interface 10GE1/0/9
shutdown
#
interface 10GE4/0/4
eth-trunk 0
#
ospf 1
import-route direct
area 0.0.0.0
 network 10.4.1.0 0.0.0.255
 network 10.6.1.0 0.0.0.255
#
return
```

- SwitchD configuration file

```
#
sysname SwitchD
#
dfs-group 1
 priority 120
 source ip 10.2.1.2 vpn-instance VRF-B peer 10.2.1.1
 dual-active detection enhanced enable
#
vlan batch 11 200 300
#
stp bridge-address 00e0-fc39-1300
stp mode rstp
stp v-stp enable
stp instance 0 root primary
stp flush disable
#
lacp m-lag system-id 00e0-fc12-3451
lacp m-lag priority 10
#
ip vpn-instance VRF-B
 ipv4-family
  route-distinguisher 101:2
  vpn-target 111:1 export-extcommunity
  vpn-target 111:1 import-extcommunity
#
interface Vlanif11
 ip address 10.3.1.1 255.255.255.0
 mac-address 0000-5e00-0101
#
interface Vlanif200
 ip address 10.5.1.1 255.255.255.0
 ospf network-type p2p
#
interface Vlanif300
 ip address 10.6.1.2 255.255.255.0
 ospf network-type p2p
#
interface MEth0/0/0
 ip binding vpn-instance VRF-B
 ip address 10.2.1.2 255.255.255.0
#
interface Eth-Trunk0
 mode lacp-static
 peer-link 1
 port vlan exclude 1 200
#
interface Eth-Trunk30
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 11
 stp root-protection
 mode lacp-static
 dfs-group 1 m-lag 1
 storm suppression broadcast cir 10 mbps
```

```
#
interface 10GE1/0/1
eth-trunk 30
#
interface 10GE1/0/2
eth-trunk 30
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 200
storm suppression broadcast 1
#
interface 10GE1/0/9
shutdown
#
interface 10GE4/0/4
eth-trunk 0
#
ospf 1
import-route direct
area 0.0.0.0
network 10.5.1.0 0.0.0.255
network 10.6.1.0 0.0.0.255
#
return
```

- SwitchE configuration file

```
#
sysname SwitchE
#
vlan batch 200 400
#
interface Vlanif200
ip address 10.4.1.2 255.255.255.0
ospf network-type p2p
#
interface Vlanif400
ip address 10.7.1.1 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/1
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 200
storm suppression broadcast 1
#
interface 10GE1/0/2
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 400
storm suppression broadcast 1
#
interface 10GE1/0/9
shutdown
#
ospf 1
area 0.0.0.0
network 10.4.1.0 0.0.0.255
network 10.7.1.0 0.0.0.255
#
return
```

- SwitchF configuration file

```
#
sysname SwitchF
#
```

```
vlan batch 200 400
#
interface Vlanif200
ip address 10.5.1.2 255.255.255.0
ospf network-type p2p
#
interface Vlanif400
ip address 10.7.1.2 255.255.255.0
ospf network-type p2p
#
interface 10GE1/0/1
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 200
storm suppression broadcast 1
#
interface 10GE1/0/2
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 400
storm suppression broadcast 1
#
interface 10GE1/0/9
shutdown
#
ospf 1
area 0.0.0.0
network 10.5.1.0 0.0.0.255
network 10.7.1.0 0.0.0.255
#
return
```

- SeGW A configuration file

```
#
sysname SeGWA
#
hrp enable
hrp track vlan 200
hrp mirror session enable
hrp interface GigabitEthernet 3/0/0 remote 10.10.0.2
#
vlan 200
port GigabitEthernet 1/0/0
port GigabitEthernet 2/0/0
#
interface GigabitEthernet 1/0/0
portswitch
#
interface GigabitEthernet 2/0/0
portswitch
#
interface GigabitEthernet3/0/0
ip address 10.10.0.1 24
#
firewall zone trust
set priority 85
add interface GigabitEthernet 1/0/0
#
firewall zone dmz
set priority 50
add interface GigabitEthernet 3/0/0
#
firewall zone untrust
set priority 5
add interface GigabitEthernet 2/0/0
#
return
```

- SeGW B configuration file

```
#
sysname SeGWB
#
hrp enable
hrp track vlan 200
hrp mirror session enable
hrp interface GigabitEthernet 3/0/0 remote 10.10.0.1
#
vlan 200
port GigabitEthernet 1/0/0
port GigabitEthernet 2/0/0
#
interface GigabitEthernet 1/0/0
portswitch
#
interface GigabitEthernet 2/0/0
portswitch
#
interface GigabitEthernet3/0/0
ip address 10.10.0.2 24
#
firewall zone trust
set priority 85
add interface GigabitEthernet 1/0/0
#
firewall zone dmz
set priority 50
add interface GigabitEthernet 3/0/0
#
firewall zone untrust
set priority 5
add interface GigabitEthernet 2/0/0
#
return
```

1.15 Configuring M-LAG, VSs, and Firewalls in Bypass Mode

Applicable Products and Versions

This example applies to CE12800, CE7800, CE6800, and CE5800 series switches running V100R005C10 or later versions, CE8800 series switches running V100R006C00 or later versions, CE9800 series switches running V200R020C00 or later versions, CloudEngine 16800 series switches running V200R005C20 or later versions, and CE12800E series switches running V200R005C00 or later versions. (CE12800E series switches do not support the VS function after FD-X series cards are installed.)

For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

On the data center network shown in [Figure 1-16](#):

- Core switches are connected to aggregation switches through 10GE interfaces.
- Firewalls at the aggregation layer connect to upstream and downstream devices through GE interfaces.

- Aggregation switches connect to upstream and downstream devices through 10GE interfaces.
- Multiple devices are deployed at the access layer and access devices connect to devices at the aggregation layer through 10GE interfaces.

This example uses the CE12804 switch and USG9520 gateway.

Figure 1-16 Networking for configuring M-LAG, VSs, and firewalls in bypass mode

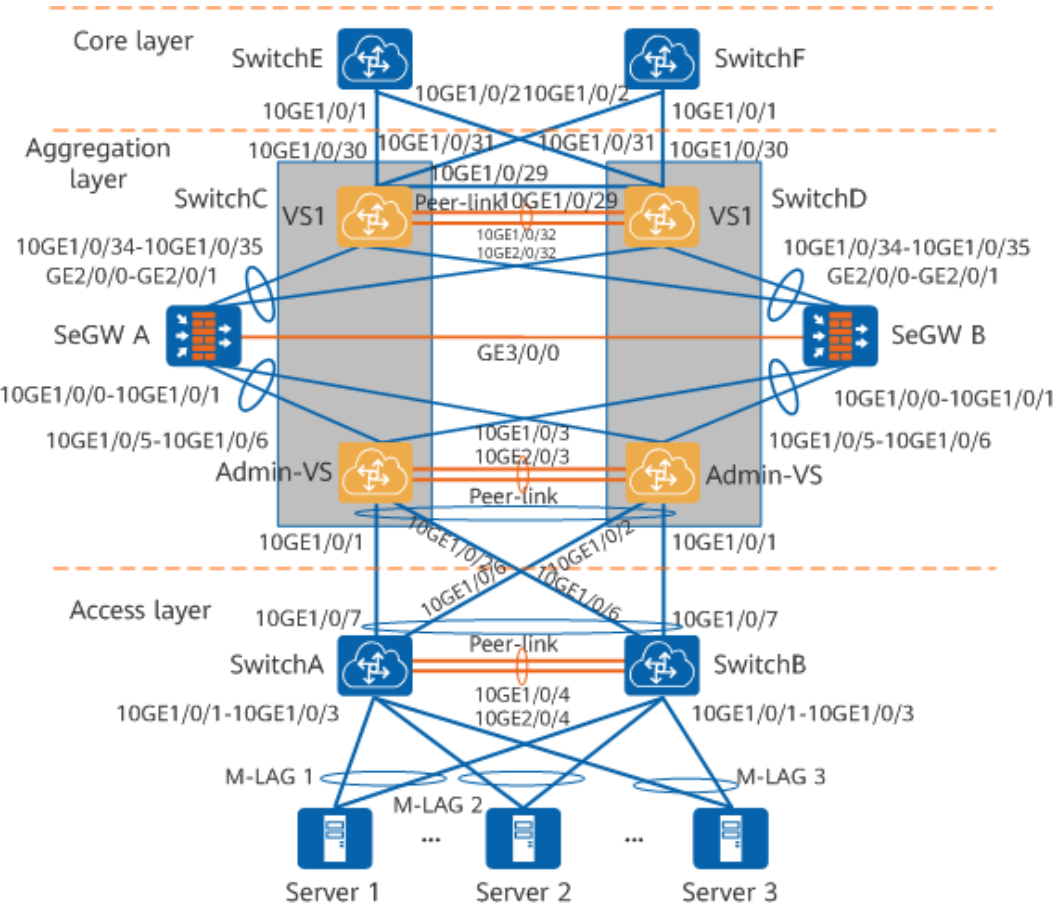


Table 1-15 Data plan

Device Name		Interface	IP Address	Virtual MAC Address
SwitchA		Management interface	10.1.1.1/24	-
SwitchB		Management interface	10.1.1.2/24	-
SwitchC	Admin-VS	Management interface	10.2.1.1/24	-

Device Name		Interface	IP Address	Virtual MAC Address
		VLANIF 11	10.4.1.1/24	0000-5e00-0102
		VLANIF 20	10.5.1.1/24	0000-5e00-0103
	VS1	Management interface	10.3.1.1/24	-
		VLANIF 30	10.6.1.1/24	0000-5e00-0104
		VLANIF 40	10.7.2.1/24	-
		10GE1/0/29	10.7.1.1/24	-
		10GE1/0/30	10.8.1.1/24	-
		10GE1/0/31	10.8.2.1/24	-
SwitchD	Admin-VS	Management interface	10.2.1.2/24	-
		VLANIF 11	10.4.1.1/24	0000-5e00-0102
		VLANIF 20	10.5.1.1/24	0000-5e00-0103
	VS1	Management interface	10.3.1.2/24	-
		VLANIF 30	10.6.1.1/24	0000-5e00-0104
		VLANIF 40	10.7.2.2/24	-
		10GE1/0/29	10.7.1.2/24	-
		10GE1/0/30	10.9.1.1/24	-
	10GE1/0/31	10.9.2.1/24	-	
SwitchE		10GE1/0/1	10.8.1.2/24	-
		10GE1/0/2	10.9.2.2/24	-
SwitchF		10GE1/0/1	10.9.1.2/24	-
		10GE1/0/2	10.8.2.2/24	-
SeGWA		GigabitEthernet 3/0/0	10.10.0.1/24	-

Device Name	Interface	IP Address	Virtual MAC Address
	Uplink interface	Floating IP address: 10.6.1.3/24	-
	Downlink interface	Floating IP address: 10.5.1.3/24	-
SeGWB	GigabitEthernet 3/0/0	10.10.0.2/24	-
	Uplink interface	Floating IP address: 10.6.1.3/24	-
	Downlink interface	Floating IP address: 10.5.1.3/24	-
Network segment where servers are located	-	10.4.1.0/24	-

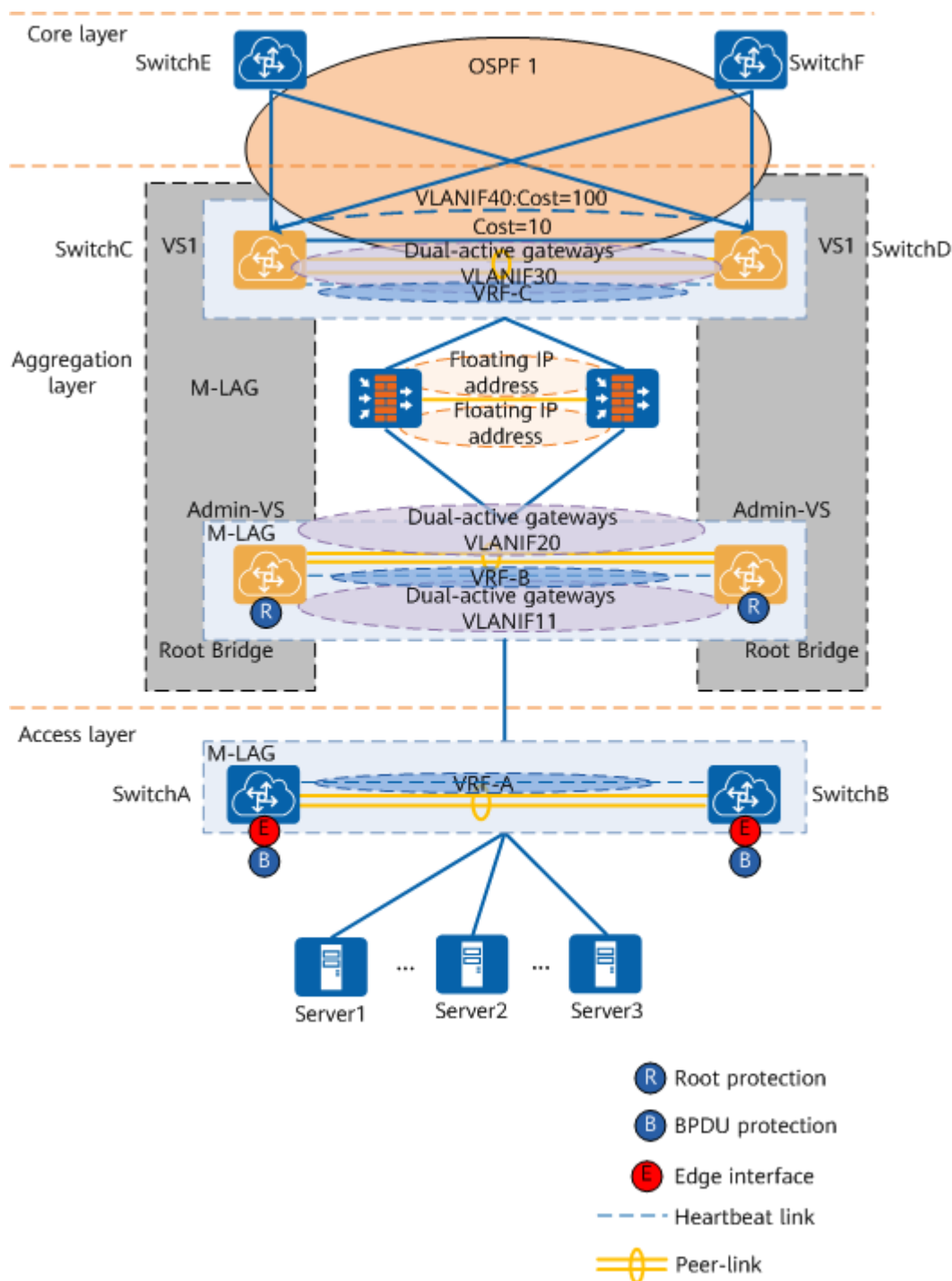
Requirement Analysis

The customer wants to build a stable large Layer 2 network where the dual-homing mode is used to ensure reliability and traffic is load balanced between links to improve link utilization. VSs are deployed at the aggregation layer to improve cabinet utilization. To ensure the server service security, the SeGWs are connected in bypass mode at the aggregation layer to provide security defense.

- Devices at the core and aggregation layers establish cross connections to load balance traffic based on ECMP (preferential forwarding of local traffic is used with M-LAG at the aggregation layer).
- VSs are assigned at the aggregation layer and Layer 3 firewalls are deployed to improve the cabinet use efficiency.
- Security gateways at the aggregation layer use the routing mode (static route), are enabled with the Hot Redundancy Protocol (HRP), and work in active/standby mode to enhance network robustness.
- M-LAG is deployed at the aggregation and access layers to form a loop-free topology.

Figure 1-17 shows the logical networking after M-LAG, VSs, and firewalls are deployed.

Figure 1-17 Networking for configuring M-LAG, VSs, and firewalls in bypass mode



Configuration Roadmap

The configuration roadmap is as follows:

1. Configure VSs on SwitchC and SwitchD.
2. Configure M-LAG between SwitchA and SwitchB, between Admin-VS (SwitchC) and Admin-VS (SwitchD), and between SwitchC-VS1 and SwitchD-VS1 to implement dual-homing access. When access and aggregation

switches work normally, links load balance traffic and a fault of any aggregation switch does not affect services. High service reliability is therefore ensured.

- Configure SwitchC and SwitchD as root bridges and enable root protection on downstream interfaces to ensure that the interfaces can forward traffic normally. Configure interfaces on SwitchA and SwitchB connected to user terminals as edge interfaces to accelerate route convergence and enable BPDU protection to enhance network stability.
 - Create VLANIF interfaces on Admin-VS (SwitchC), Admin-VS (SwitchD), SwitchC-VS1, and SwitchD-VS1 and configure IP addresses and MAC addresses for the VLANIF interfaces as the user-side gateway and next hop of the firewall.
3. Configure security gateways at the aggregation layer to use the routing mode, enable HRP, and configure them to work in active/standby mode to enhance network robustness.
 4. Enable OSPF on aggregation and core switches.

Procedure

Step 1 Configure VS assignment.

SwitchC is used as an example. The configuration of SwitchD is similar to the configuration of SwitchC, and is not mentioned here.

1. Create VS1 in port group mode and configure VS1 to use the default logical resource specifications. Allocate 10GE1/0/24 to 10GE1/0/47 and 10GE2/0/24 to 10GE2/0/47 to VS1.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchC
[*HUAWEI] commit
[~SwitchC] admin
[~SwitchC-admin] virtual-system vs1
[*SwitchC-admin-vs:vs1] port-mode group
[*SwitchC-admin-vs:vs1] assign interface 10GE 1/0/24
Warning: All configurations of the interfaces will be deleted. Interfaces 10GE1/0/24-47 of the same
group will be assigned. Continue?
[Y/N]: y
[*SwitchC-admin-vs:vs1] assign interface 10GE 2/0/24
Warning: All configurations of the interfaces will be deleted. Interfaces 10GE2/0/24-47 of the same
group will be assigned. Continue?
[Y/N]: y
[*SwitchC-admin-vs:vs1] quit
[*SwitchC-admin] commit
```

2. Verify the configuration.

Check VS1 information.

```
[~SwitchC-admin] display virtual-system name vs1 verbose
Name       : vs1
Status     : running
Description :
Create time : 2016-10-13 09:34:22
Port mode  : group
System MAC : 00e0-fc0c-0d05
Assigned slot(s)
pvmb       : 5
pvmb       : 6
CPU(s)
slot 5     : 0%
slot 6     : 0%
```

```
Memory(s)
slot 5    : 5%, 202632/3884636 (Used Kbytes/Max Kbytes)
slot 6    : 2%, 201272/8021592 (Used Kbytes/Max Kbytes)
Assigned interface(s)
10GE1/0/24, slot 1
10GE1/0/25, slot 1
10GE1/0/26, slot 1
10GE1/0/27, slot 1
10GE1/0/28, slot 1
10GE1/0/29, slot 1
10GE1/0/30, slot 1
10GE1/0/31, slot 1
10GE1/0/32, slot 1
10GE1/0/33, slot 1
10GE1/0/34, slot 1
10GE1/0/35, slot 1
10GE1/0/36, slot 1
10GE1/0/37, slot 1
10GE1/0/38, slot 1
10GE1/0/39, slot 1
10GE1/0/40, slot 1
10GE1/0/41, slot 1
10GE1/0/42, slot 1
10GE1/0/43, slot 1
10GE1/0/44, slot 1
10GE1/0/45, slot 1
10GE1/0/46, slot 1
10GE1/0/47, slot 1
10GE2/0/24, slot 2
10GE2/0/25, slot 2
10GE2/0/26, slot 2
10GE2/0/27, slot 2
10GE2/0/28, slot 2
10GE2/0/29, slot 2
10GE2/0/30, slot 2
10GE2/0/31, slot 2
10GE2/0/32, slot 2
10GE2/0/33, slot 2
10GE2/0/34, slot 2
10GE2/0/35, slot 2
10GE2/0/36, slot 2
10GE2/0/37, slot 2
10GE2/0/38, slot 2
10GE2/0/39, slot 2
10GE2/0/40, slot 2
10GE2/0/41, slot 2
10GE2/0/42, slot 2
10GE2/0/43, slot 2
10GE2/0/44, slot 2
10GE2/0/45, slot 2
10GE2/0/46, slot 2
10GE2/0/47, slot 2
Assigned resource(s)
u4route   : 60000(Max)
m4route   : 1000(Max)
u6route   : 16000(Max)
m6route   : 100(Max)
vlan      : 4063(Max)
vpn-instance : 4096(Max)
cpu        : 5(weight)
memory     : 100(ratio-threshold)
mpls       : enable
trill      : enable
mcast      : enable
```

Step 2 Configure M-LAG.

1. Configure V-STP, dual-active detection links, DFS groups, peer-links, and M-LAG member interfaces on SwitchA, SwitchB, Admin-VS (SwitchC), Admin-VS (SwitchD), SwitchC-VS1, and SwitchD-VS1.

The dual-active detection links are connected to management interfaces to implement interworking, DFS groups must be bound to IP addresses of management interfaces to ensure communication, and management interfaces are bound to VPN instances to implement isolation.

It is recommended that Eth-Trunk member interfaces of the peer-link be deployed on different cards to prevent the peer-link fault caused by a single-point failure.

Configure SwitchA.

Configure the Eth-Trunk on SwitchA connected to servers as the edge interface and enable BPDU protection on the Eth-Trunk. The Eth-Trunk on the access switch connected to Server 1 is used as an example.

The uplink interface of the server connected to the switch needs to be bound to an aggregated link, and the link aggregation modes on the server and switch must be consistent.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchA
[*HUAWEI] commit
[~SwitchA] stp mode rstp
[*SwitchA] stp v-stp enable
[*SwitchA] stp flush disable
[*SwitchA] ip vpn-instance VRF-A //Create VRF-A.
[*SwitchA-vpn-instance-VRF-A] ipv4-family
[*SwitchA-vpn-instance-VRF-A-af-ipv4] route-distinguisher 100:1
[*SwitchA-vpn-instance-VRF-A-af-ipv4] vpn-target 111:1 both
[*SwitchA-vpn-instance-VRF-A-af-ipv4] quit
[*SwitchA-vpn-instance-VRF-A] quit
[*SwitchA] interface meth 0/0/0
[*SwitchA-Meth0/0/0] ip binding vpn-instance VRF-A //Bind the management interface to VRF-A.
[*SwitchA-Meth0/0/0] ip address 10.1.1.1 24
[*SwitchA-Meth0/0/0] quit
[*SwitchA] dfs-group 1
[*SwitchA-dfs-group-1] source ip 10.1.1.1 vpn-instance VRF-A peer 10.1.1.2 //Configure the IPv4
address and VPN instance bound to the DFS group.
[*SwitchA-dfs-group-1] dual-active detection enhanced enable
[*SwitchA-dfs-group-1] priority 150
[*SwitchA-dfs-group-1] quit
[*SwitchA] interface eth-trunk 0
[*SwitchA-Eth-Trunk0] trunkport 10ge 1/0/4
[*SwitchA-Eth-Trunk0] trunkport 10ge 2/0/4 //Configure inter-card member interfaces of the Eth-
Trunk of the peer-link.
[*SwitchA-Eth-Trunk0] mode lacp-static
[*SwitchA-Eth-Trunk0] peer-link 1
[*SwitchA-Eth-Trunk0] port vlan exclude 1
[*SwitchA-Eth-Trunk0] quit
[*SwitchA] vlan batch 11
[*SwitchA] interface eth-trunk 10
[*SwitchA-Eth-Trunk10] mode lacp-dynamic
[*SwitchA-Eth-Trunk10] port link-type access
[*SwitchA-Eth-Trunk10] port default vlan 11
[*SwitchA-Eth-Trunk10] trunkport 10ge 1/0/1
[*SwitchA-Eth-Trunk10] dfs-group 1 m-lag 1
[*SwitchA-Eth-Trunk10] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[*SwitchA-Eth-Trunk10] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchA-Eth-Trunk10] quit
```

```
[*SwitchA] interface eth-trunk 20
[SwitchA-Eth-Trunk20] mode lacp-dynamic
[SwitchA-Eth-Trunk20] port link-type access
[SwitchA-Eth-Trunk20] port default vlan 11
[SwitchA-Eth-Trunk20] trunkport 10ge 1/0/2
[SwitchA-Eth-Trunk20] dfs-group 1 m-lag 2
[SwitchA-Eth-Trunk20] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[SwitchA-Eth-Trunk20] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[SwitchA-Eth-Trunk20] quit
[SwitchA] interface eth-trunk 30
[SwitchA-Eth-Trunk30] mode lacp-dynamic
[SwitchA-Eth-Trunk30] port link-type access
[SwitchA-Eth-Trunk30] port default vlan 11
[SwitchA-Eth-Trunk30] trunkport 10ge 1/0/3
[SwitchA-Eth-Trunk30] dfs-group 1 m-lag 3
[SwitchA-Eth-Trunk30] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[SwitchA-Eth-Trunk30] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[SwitchA-Eth-Trunk30] quit
[SwitchA] stp bpdu-protection //Enable BPDU protection on the edge interface.
[SwitchA] interface eth-trunk 40
[SwitchA-Eth-Trunk40] mode lacp-static
[SwitchA-Eth-Trunk40] port link-type trunk
[SwitchA-Eth-Trunk40] undo port trunk allow-pass vlan 1
[SwitchA-Eth-Trunk40] port trunk allow-pass vlan 11
[SwitchA-Eth-Trunk40] trunkport 10ge 1/0/6 to 1/0/7
[SwitchA-Eth-Trunk40] dfs-group 1 m-lag 4
[SwitchA-Eth-Trunk40] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[SwitchA-Eth-Trunk40] quit
[SwitchA] lacp m-lag priority 10
[SwitchA] lacp m-lag system-id 00e0-fc12-3456
[SwitchA] interface 10ge 1/0/9
[SwitchA-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[SwitchA-10GE1/0/9] quit
[SwitchA] commit
```

Configure SwitchB.

Configure the Eth-Trunk on SwitchB connected to servers as the edge interface and enable BPDU protection on the Eth-Trunk. The Eth-Trunk on the access switch connected to Server 1 is used as an example.

The uplink interface of the server connected to the switch needs to be bound to an aggregated link, and the link aggregation modes on the server and switch must be consistent.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchB
[*HUAWEI] commit
[~SwitchB] stp mode rstp
[SwitchB] stp v-stp enable
[SwitchB] stp flush disable
[SwitchB] ip vpn-instance VRF-A //Create VRF-A.
[SwitchB-vpn-instance-VRF-A] ipv4-family
[SwitchB-vpn-instance-VRF-A-af-ipv4] route-distinguisher 100:2
[SwitchB-vpn-instance-VRF-A-af-ipv4] vpn-target 111:1 both
[SwitchB-vpn-instance-VRF-A-af-ipv4] quit
[SwitchB-vpn-instance-VRF-A] quit
[SwitchB] interface meth 0/0/0
```

```
[*SwitchB-MEth0/0/0] ip binding vpn-instance VRF-A //Bind the management interface to VRF-A.
[*SwitchB-MEth0/0/0] ip address 10.1.1.2 24
[*SwitchB-MEth0/0/0] quit
[*SwitchB] dfs-group 1
[*SwitchB-dfs-group-1] source ip 10.1.1.2 vpn-instance VRF-A peer 10.1.1.1 //Configure the IPv4
address and VPN instance bound to the DFS group.
[*SwitchB-dfs-group-1] dual-active detection enhanced enable
[*SwitchB-dfs-group-1] priority 120
[*SwitchB-dfs-group-1] quit
[*SwitchB] interface eth-trunk 0
[*SwitchB-Eth-Trunk0] trunkport 10ge 1/0/4
[*SwitchB-Eth-Trunk0] trunkport 10ge 2/0/4
[*SwitchB-Eth-Trunk0] mode lacp-static
[*SwitchB-Eth-Trunk0] peer-link 1
[*SwitchB-Eth-Trunk0] port vlan exclude 1
[*SwitchB-Eth-Trunk0] quit
[*SwitchB] vlan batch 11
[*SwitchB] interface eth-trunk 10
[*SwitchB-Eth-Trunk10] mode lacp-dynamic
[*SwitchB-Eth-Trunk10] port link-type access
[*SwitchB-Eth-Trunk10] port default vlan 11
[*SwitchB-Eth-Trunk10] trunkport 10ge 1/0/1
[*SwitchB-Eth-Trunk10] dfs-group 1 m-lag 1
[*SwitchB-Eth-Trunk10] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[*SwitchB-Eth-Trunk10] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchB-Eth-Trunk10] quit
[*SwitchB] interface eth-trunk 20
[*SwitchB-Eth-Trunk20] mode lacp-dynamic
[*SwitchB-Eth-Trunk20] port link-type access
[*SwitchB-Eth-Trunk20] port default vlan 11
[*SwitchB-Eth-Trunk20] trunkport 10ge 1/0/2
[*SwitchB-Eth-Trunk20] dfs-group 1 m-lag 2
[*SwitchB-Eth-Trunk20] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[*SwitchB-Eth-Trunk20] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchB-Eth-Trunk20] quit
[*SwitchB] interface eth-trunk 30
[*SwitchB-Eth-Trunk30] mode lacp-dynamic
[*SwitchB-Eth-Trunk30] port link-type access
[*SwitchB-Eth-Trunk30] port default vlan 11
[*SwitchB-Eth-Trunk30] trunkport 10ge 1/0/3
[*SwitchB-Eth-Trunk30] dfs-group 1 m-lag 3
[*SwitchB-Eth-Trunk30] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[*SwitchB-Eth-Trunk30] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchB-Eth-Trunk30] quit
[*SwitchB] stp bpdu-protection //Enable BPDU protection on the edge interface.
[*SwitchB] interface eth-trunk 40
[*SwitchB-Eth-Trunk40] mode lacp-static
[*SwitchB-Eth-Trunk40] port link-type trunk
[*SwitchB-Eth-Trunk40] undo port trunk allow-pass vlan 1
[*SwitchB-Eth-Trunk40] port trunk allow-pass vlan 11
[*SwitchB-Eth-Trunk40] trunkport 10ge 1/0/6 to 1/0/7
[*SwitchB-Eth-Trunk40] dfs-group 1 m-lag 4
[*SwitchB-Eth-Trunk40] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchB-Eth-Trunk40] quit
[*SwitchB] lacp m-lag priority 10
[*SwitchB] lacp m-lag system-id 00e0-fc12-3456
[*SwitchB] interface 10ge 1/0/9
```

```
[*SwitchB-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchB-10GE1/0/9] quit
[*SwitchB] commit
```

Configure Admin-VS (SwitchC).

Configure Admin-VS (SwitchC) and Admin-VS (SwitchD) as the root bridges of the STP network, and enable root protection on Eth-Trunks of aggregation switches connected to access switches so that the Eth-Trunks can forward traffic normally.

```
<SwitchC> system-view
[~SwitchC] stp mode rstp
[*SwitchC] stp root primary //Configure the aggregation device as the root bridge of the STP
network.
[*SwitchC] stp bridge-address 00e0-fc39-1300 //Configure the MAC address of the root bridge
(MAC address of the master device).
[*SwitchC] stp v-stp enable
[*SwitchC] stp flush disable
[*SwitchC] ip vpn-instance VRF-B //Create VRF-B.
[*SwitchC-vpn-instance-VRF-B] ipv4-family
[*SwitchC-vpn-instance-VRF-B-af-ipv4] route-distinguisher 101:1
[*SwitchC-vpn-instance-VRF-B-af-ipv4] vpn-target 111:1 both
[*SwitchC-vpn-instance-VRF-B-af-ipv4] quit
[*SwitchC-vpn-instance-VRF-B] quit
[*SwitchC] interface meth 0/0/0
[*SwitchC-Meth0/0/0] ip binding vpn-instance VRF-B //Bind the management interface to VRF-B.
[*SwitchC-Meth0/0/0] ip address 10.2.1.1 24
[*SwitchC-Meth0/0/0] quit
[*SwitchC] dfs-group 1
[*SwitchC-dfs-group-1] source ip 10.2.1.1 vpn-instance VRF-B peer 10.2.1.2 //Configure the IPv4
address and VPN instance bound to the DFS group.
[*SwitchC-dfs-group-1] dual-active detection enhanced enable
[*SwitchC-dfs-group-1] priority 150
[*SwitchC-dfs-group-1] quit
[*SwitchC] interface eth-trunk 0
[*SwitchC-Eth-Trunk0] trunkport 10ge 1/0/3
[*SwitchC-Eth-Trunk0] trunkport 10ge 2/0/3 //Configure inter-card member interfaces of the Eth-
Trunk of the peer-link.
[*SwitchC-Eth-Trunk0] mode lacp-static
[*SwitchC-Eth-Trunk0] peer-link 1
[*SwitchC-Eth-Trunk0] port vlan exclude 1
[*SwitchC-Eth-Trunk0] quit
[*SwitchC] vlan batch 11 20
[*SwitchC] interface eth-trunk 30
[*SwitchC-Eth-Trunk30] mode lacp-static
[*SwitchC-Eth-Trunk30] port link-type trunk
[*SwitchC-Eth-Trunk30] undo port trunk allow-pass vlan 1
[*SwitchC-Eth-Trunk30] port trunk allow-pass vlan 11
[*SwitchC-Eth-Trunk30] trunkport 10ge 1/0/1 to 1/0/2
[*SwitchC-Eth-Trunk30] dfs-group 1 m-lag 1
[*SwitchC-Eth-Trunk30] stp root-protection //Enable root protection.
[*SwitchC-Eth-Trunk30] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchC-Eth-Trunk30] quit
[*SwitchC] interface eth-trunk 40
[*SwitchC-Eth-Trunk40] mode lacp-static
[*SwitchC-Eth-Trunk40] port link-type trunk
[*SwitchC-Eth-Trunk40] undo port trunk allow-pass vlan 1
[*SwitchC-Eth-Trunk40] port trunk allow-pass vlan 20
[*SwitchC-Eth-Trunk40] trunkport 10ge 1/0/5
[*SwitchC-Eth-Trunk40] dfs-group 1 m-lag 2
[*SwitchC-Eth-Trunk40] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
```



```
[*SwitchC-Eth-Trunk40] quit
[*SwitchC] interface eth-trunk 50
[*SwitchC-Eth-Trunk50] mode lacp-static
[*SwitchC-Eth-Trunk50] port link-type trunk
[*SwitchC-Eth-Trunk50] undo port trunk allow-pass vlan 1
[*SwitchC-Eth-Trunk50] port trunk allow-pass vlan 20
[*SwitchC-Eth-Trunk50] trunkport 10ge 1/0/6
[*SwitchC-Eth-Trunk50] dfs-group 1 m-lag 3
[*SwitchC-Eth-Trunk50] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchC-Eth-Trunk50] quit
[*SwitchC] lacp m-lag priority 10
[*SwitchC] lacp m-lag system-id 00e0-fc12-3451
[*SwitchC] interface 10ge 1/0/9
[*SwitchC-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchC-10GE1/0/9] quit
[*SwitchC] commit
[*SwitchC] quit
```

Configure SwitchC-VS1.

```
<SwitchC> switch virtual-system vs1
<SwitchC-vs1> system-view
[~SwitchC-vs1] stp mode rstp
[*SwitchC-vs1] stp v-stp enable
[*SwitchC-vs1] stp flush disable
[*SwitchC-vs1] ip vpn-instance VRF-C //Create VRF-C.
[*SwitchC-vs1-vpn-instance-VRF-C] ipv4-family
[*SwitchC-vs1-vpn-instance-VRF-C-af-ipv4] route-distinguisher 102:1
[*SwitchC-vs1-vpn-instance-VRF-C-af-ipv4] vpn-target 111:1 both
[*SwitchC-vs1-vpn-instance-VRF-C-af-ipv4] quit
[*SwitchC-vs1-vpn-instance-VRF-C] quit
[*SwitchC-vs1] interface meth 0/0/0
[*SwitchC-vs1-MEth0/0/0] ip binding vpn-instance VRF-C //Bind the management interface to
VRF-C.
[*SwitchC-vs1-MEth0/0/0] ip address 10.3.1.1 24
[*SwitchC-vs1-MEth0/0/0] quit
[*SwitchC-vs1] dfs-group 1
[*SwitchC-vs1-dfs-group-1] source ip 10.3.1.1 vpn-instance VRF-C peer 10.3.1.2 //Configure the
IPv4 address and VPN instance bound to the DFS group.
[*SwitchC-vs1-dfs-group-1] dual-active detection enhanced enable
[*SwitchC-vs1-dfs-group-1] priority 150
[*SwitchC-vs1-dfs-group-1] quit
[*SwitchC-vs1] interface eth-trunk 0
[*SwitchC-vs1-Eth-Trunk0] trunkport 10ge 1/0/32
[*SwitchC-vs1-Eth-Trunk0] trunkport 10ge 2/0/32
[*SwitchC-vs1-Eth-Trunk0] mode lacp-static
[*SwitchC-vs1-Eth-Trunk0] peer-link 1
[*SwitchC-vs1-Eth-Trunk0] port vlan exclude 1
[*SwitchC-vs1-Eth-Trunk0] quit
[*SwitchC-vs1] vlan batch 30
[*SwitchC-vs1] interface eth-trunk 60
[*SwitchC-vs1-Eth-Trunk60] mode lacp-static
[*SwitchC-vs1-Eth-Trunk60] port link-type trunk
[*SwitchC-vs1-Eth-Trunk60] undo port trunk allow-pass vlan 1
[*SwitchC-vs1-Eth-Trunk60] port trunk allow-pass vlan 30
[*SwitchC-vs1-Eth-Trunk60] trunkport 10ge 1/0/34
[*SwitchC-vs1-Eth-Trunk60] dfs-group 1 m-lag 2
[*SwitchC-vs1-Eth-Trunk60] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchC-vs1-Eth-Trunk60] quit
[*SwitchC-vs1] interface eth-trunk 70
[*SwitchC-vs1-Eth-Trunk70] mode lacp-static
[*SwitchC-vs1-Eth-Trunk70] port link-type trunk
[*SwitchC-vs1-Eth-Trunk70] undo port trunk allow-pass vlan 1
```

```
[*SwitchC-vs1-Eth-Trunk70] port trunk allow-pass vlan 30
[*SwitchC-vs1-Eth-Trunk70] trunkport 10ge 1/0/35
[*SwitchC-vs1-Eth-Trunk70] dfs-group 1 m-lag 3
[*SwitchC-vs1-Eth-Trunk70] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchC-vs1-Eth-Trunk70] quit
[*SwitchC-vs1] lacp m-lag priority 10
[*SwitchC-vs1] lacp m-lag system-id 00e0-fc12-3452
[*SwitchC-vs1] interface 10ge 1/0/39
[*SwitchC-vs1-10GE1/0/39] shutdown //Shut down the interface not in use. 10GE 1/0/39 is used as
an example.
[*SwitchC-vs1-10GE1/0/39] quit
[*SwitchC-vs1] commit
[~SwitchC-vs1] quit
```

Configure Admin-VS (SwitchD).

Configure Admin-VS (SwitchC) and Admin-VS (SwitchD) as the root bridges of the STP network, and enable root protection on Eth-Trunks of aggregation switches connected to access switches so that the Eth-Trunks can forward traffic normally.

```
<SwitchD> system-view
[~SwitchD] stp mode rstp
[*SwitchD] stp root primary //Configure the aggregation device as the root bridge of the STP
network.
[*SwitchD] stp bridge-address 00e0-fc39-1300 //Configure the MAC address of the root bridge
(MAC address of the master device).
[*SwitchD] stp v-stp enable
[*SwitchD] stp flush disable
[*SwitchD] ip vpn-instance VRF-B //Create VRF-B.
[*SwitchD-vpn-instance-VRF-B] ipv4-family
[*SwitchD-vpn-instance-VRF-B-af-ipv4] route-distinguisher 101:2
[*SwitchD-vpn-instance-VRF-B-af-ipv4] vpn-target 111:1 both
[*SwitchD-vpn-instance-VRF-B-af-ipv4] quit
[*SwitchD-vpn-instance-VRF-B] quit
[*SwitchD] interface meth 0/0/0
[*SwitchD-MEth0/0/0] ip binding vpn-instance VRF-B //Bind the management interface to VRF-B.
[*SwitchD-MEth0/0/0] ip address 10.2.1.2 24
[*SwitchD-MEth0/0/0] quit
[*SwitchD] dfs-group 1
[*SwitchD-dfs-group-1] source ip 10.2.1.2 vpn-instance VRF-B peer 10.2.1.1 //Configure the IPv4
address and VPN instance bound to the DFS group.
[*SwitchD-dfs-group-1] dual-active detection enhanced enable
[*SwitchD-dfs-group-1] priority 120
[*SwitchD-dfs-group-1] quit
[*SwitchD] interface eth-trunk 0
[*SwitchD-Eth-Trunk0] trunkport 10ge 1/0/3
[*SwitchD-Eth-Trunk0] trunkport 10ge 2/0/3 //Configure inter-card member interfaces of the Eth-
Trunk of the peer-link.
[*SwitchD-Eth-Trunk0] mode lacp-static
[*SwitchD-Eth-Trunk0] peer-link 1
[*SwitchD-Eth-Trunk0] port vlan exclude 1
[*SwitchD-Eth-Trunk0] quit
[*SwitchD] vlan batch 11 20
[*SwitchD] interface eth-trunk 30
[*SwitchD-Eth-Trunk30] mode lacp-static
[*SwitchD-Eth-Trunk30] port link-type trunk
[*SwitchD-Eth-Trunk30] undo port trunk allow-pass vlan 1
[*SwitchD-Eth-Trunk30] port trunk allow-pass vlan 11
[*SwitchD-Eth-Trunk30] trunkport 10ge 1/0/1 to 1/0/2
[*SwitchD-Eth-Trunk30] dfs-group 1 m-lag 1
[*SwitchD-Eth-Trunk30] stp root-protection //Enable root protection.
[*SwitchD-Eth-Trunk30] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
```

```
[*SwitchD-Eth-Trunk30] quit
[SwitchD] interface eth-trunk 40
[SwitchD-Eth-Trunk40] mode lacp-static
[SwitchD-Eth-Trunk40] port link-type trunk
[SwitchD-Eth-Trunk40] undo port trunk allow-pass vlan 1
[SwitchD-Eth-Trunk40] port trunk allow-pass vlan 20
[SwitchD-Eth-Trunk40] trunkport 10ge 1/0/5
[SwitchD-Eth-Trunk40] dfs-group 1 m-lag 2
[SwitchD-Eth-Trunk40] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[SwitchD-Eth-Trunk40] quit
[SwitchD] interface eth-trunk 50
[SwitchD-Eth-Trunk50] mode lacp-static
[SwitchD-Eth-Trunk50] port link-type trunk
[SwitchD-Eth-Trunk50] undo port trunk allow-pass vlan 1
[SwitchD-Eth-Trunk50] port trunk allow-pass vlan 20
[SwitchD-Eth-Trunk50] trunkport 10ge 1/0/6
[SwitchD-Eth-Trunk50] dfs-group 1 m-lag 3
[SwitchD-Eth-Trunk50] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[SwitchD-Eth-Trunk50] quit
[SwitchD] lacp m-lag priority 10
[SwitchD] lacp m-lag system-id 00e0-fc12-3451
[SwitchD] interface 10ge 1/0/9
[SwitchD-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[SwitchD-10GE1/0/9] quit
[SwitchD] commit
[~SwitchD] quit
```

Configure SwitchD-VS1.

```
<SwitchD> switch virtual-system vs1
<SwitchD-vs1> system-view
[~SwitchD-vs1] stp mode rstp
[*SwitchD-vs1] stp v-stp enable
[*SwitchD-vs1] stp flush disable
[*SwitchD-vs1] ip vpn-instance VRF-C //Create VRF-C.
[*SwitchD-vs1-vpn-instance-VRF-C] ipv4-family
[*SwitchD-vs1-vpn-instance-VRF-C-af-ipv4] route-distinguisher 102:2
[*SwitchD-vs1-vpn-instance-VRF-C-af-ipv4] vpn-target 111:1 both
[*SwitchD-vs1-vpn-instance-VRF-C-af-ipv4] quit
[*SwitchD-vs1-vpn-instance-VRF-C] quit
[*SwitchD-vs1] interface meth 0/0/0
[*SwitchD-vs1-MEth0/0/0] ip binding vpn-instance VRF-C //Bind the management interface to
VRF-C.
[*SwitchD-vs1-MEth0/0/0] ip address 10.3.1.2 24
[*SwitchD-vs1-MEth0/0/0] quit
[*SwitchD-vs1] dfs-group 1
[*SwitchD-vs1-dfs-group-1] source ip 10.3.1.2 vpn-instance VRF-C peer 10.3.1.1 //Configure the
IPv4 address and VPN instance bound to the DFS group.
[*SwitchD-vs1-dfs-group-1] dual-active detection enhanced enable
[*SwitchD-vs1-dfs-group-1] priority 120
[*SwitchD-vs1-dfs-group-1] quit
[*SwitchD-vs1] interface eth-trunk 0
[*SwitchD-vs1-Eth-Trunk0] trunkport 10ge 1/0/32
[*SwitchD-vs1-Eth-Trunk0] trunkport 10ge 2/0/32
[*SwitchD-vs1-Eth-Trunk0] mode lacp-static
[*SwitchD-vs1-Eth-Trunk0] peer-link 1
[*SwitchD-vs1-Eth-Trunk0] port vlan exclude 1
[*SwitchD-vs1-Eth-Trunk0] quit
[*SwitchD-vs1] vlan batch 30
[*SwitchD-vs1] interface eth-trunk 60
[*SwitchD-vs1-Eth-Trunk60] mode lacp-static
[*SwitchD-vs1-Eth-Trunk60] port link-type trunk
[*SwitchD-vs1-Eth-Trunk60] undo port trunk allow-pass vlan 1
```

```
[*SwitchD-vs1-Eth-Trunk60] port trunk allow-pass vlan 30
[*SwitchD-vs1-Eth-Trunk60] trunkport 10ge 1/0/34
[*SwitchD-vs1-Eth-Trunk60] dfs-group 1 m-lag 2
[*SwitchD-vs1-Eth-Trunk60] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchD-vs1-Eth-Trunk60] quit
[*SwitchD-vs1] interface eth-trunk 70
[*SwitchD-vs1-Eth-Trunk70] mode lacp-static
[*SwitchD-vs1-Eth-Trunk70] port link-type trunk
[*SwitchD-vs1-Eth-Trunk70] undo port trunk allow-pass vlan 1
[*SwitchD-vs1-Eth-Trunk70] port trunk allow-pass vlan 30
[*SwitchD-vs1-Eth-Trunk70] trunkport 10ge 1/0/35
[*SwitchD-vs1-Eth-Trunk70] dfs-group 1 m-lag 3
[*SwitchD-vs1-Eth-Trunk70] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchD-vs1-Eth-Trunk70] quit
[*SwitchD-vs1] commit
[*SwitchD-vs1] lacp m-lag priority 10
[*SwitchD-vs1] lacp m-lag system-id 00e0-fc12-3452
[*SwitchD-vs1] interface 10ge 1/0/39
[*SwitchD-vs1-10GE1/0/39] shutdown //Shut down the interface not in use. 10GE 1/0/39 is used
as an example.
[*SwitchD-vs1-10GE1/0/39] quit
[*SwitchD-vs1] commit
[~SwitchD-vs1] quit
```

2. Create VLANIF interfaces and configure IP addresses and MAC addresses for the VLANIF interfaces on Admin-VS (SwitchC), Admin-VS (SwitchD), SwitchC-VS1, and SwitchD-VS1, and create a dual-active gateway on VLANIF 11 as the user-side gateway, a dual-active gateway on VLANIF 20 as the downlink next hop of the firewall, and a dual active gateway on VLANIF 30 as the uplink next hop of the firewall. Configure static routes pointing to firewalls on Admin-VS (SwitchC) and Admin-VS (SwitchD), and configure static routes pointing to firewalls on SwitchC-VS1 and SwitchD-VS1.

Configure Admin-VS (SwitchC).

```
<SwitchC> system-view
[~SwitchC] interface vlanif 11
[*SwitchC-Vlanif11] ip address 10.4.1.1 24
[*SwitchC-Vlanif11] mac-address 0000-5e00-0102
[*SwitchC-Vlanif11] quit
[*SwitchC] interface vlanif 20
[*SwitchC-Vlanif20] ip address 10.5.1.1 24
[*SwitchC-Vlanif20] mac-address 0000-5e00-0103
[*SwitchC-Vlanif20] quit
[*SwitchC] ip route-static 0.0.0.0 0 10.5.1.3
[*SwitchC] commit
[~SwitchC] quit
```

Configure SwitchC-VS1.

```
<SwitchC> switch virtual-system vs1
<SwitchC-vs1> system-view
[~SwitchC-vs1] interface vlanif 30
[*SwitchC-vs1-Vlanif30] ip address 10.6.1.1 24
[*SwitchC-vs1-Vlanif30] mac-address 0000-5e00-0104
[*SwitchC-vs1-Vlanif30] quit
[*SwitchC-vs1] ip route-static 10.4.1.0 24 10.6.1.3
[*SwitchC-vs1] commit
[~SwitchC-vs1] quit
```

Configure Admin-VS (SwitchD).

```
<SwitchD> system-view
[~SwitchD] interface vlanif 11
[*SwitchD-Vlanif11] ip address 10.4.1.1 24
[*SwitchD-Vlanif11] mac-address 0000-5e00-0102
[*SwitchD-Vlanif11] quit
[*SwitchD] interface vlanif 20
[*SwitchD-Vlanif20] ip address 10.5.1.1 24
[*SwitchD-Vlanif20] mac-address 0000-5e00-0103
[*SwitchD-Vlanif20] quit
[*SwitchD] ip route-static 0.0.0.0 0 10.5.1.3
[*SwitchD] commit
[~SwitchD] quit
```

Configure SwitchD-VS1.

```
<SwitchD> switch virtual-system vs1
<SwitchD-vs1> system-view
[~SwitchD-vs1] interface vlanif 30
[*SwitchD-vs1-Vlanif30] ip address 10.6.1.1 24
[*SwitchD-vs1-Vlanif30] mac-address 0000-5e00-0104
[*SwitchD-vs1-Vlanif30] quit
[*SwitchD-vs1] ip route-static 10.4.1.0 24 10.6.1.3
[*SwitchD-vs1] commit
[~SwitchD-vs1] quit
```

Step 3 Configure SeGW A and SeGW B to work in routing mode and enable HRP.

1. Configure uplink and downlink interfaces of SeGW A and SeGW B.

Configure SeGW A.

```
<USG9000> system-view
[USG9000] sysname SeGWA
[SeGWA] interface eth-trunk 1
[SeGWA-Eth-Trunk1] mode lacp-static
[SeGWA-Eth-Trunk1] trunkport GigabitEthernet 1/0/0 to 1/0/1
[SeGWA-Eth-Trunk1] ip address 10.5.1.3 24 float master
[SeGWA-Eth-Trunk1] quit
[SeGWA] interface eth-trunk 2
[SeGWA-Eth-Trunk2] mode lacp-static
[SeGWA-Eth-Trunk2] trunkport GigabitEthernet 2/0/0 to 2/0/1
[SeGWA-Eth-Trunk2] ip address 10.6.1.3 24 float master
[SeGWA-Eth-Trunk2] quit
```

Configure SeGW B.

```
<USG9000> system-view
[USG9000] sysname SeGWB
[SeGWB] interface eth-trunk 1
[SeGWB-Eth-Trunk1] mode lacp-static
[SeGWB-Eth-Trunk1] trunkport GigabitEthernet 1/0/0 to 1/0/1
[SeGWB-Eth-Trunk1] ip address 10.5.1.3 24 float slave
[SeGWB-Eth-Trunk1] quit
[SeGWB] interface eth-trunk 2
[SeGWB-Eth-Trunk2] mode lacp-static
[SeGWB-Eth-Trunk2] trunkport GigabitEthernet 2/0/0 to 2/0/1
[SeGWB-Eth-Trunk2] ip address 10.6.1.3 24 float slave
[SeGWB-Eth-Trunk2] quit
```

2. Configure IP addresses for heartbeat interfaces of SeGW A and SeGW B.

Configure SeGW A.

```
[SeGWA] interface GigabitEthernet 3/0/0
[SeGWA-GigabitEthernet3/0/0] ip address 10.10.0.1 24
[SeGWA-GigabitEthernet3/0/0] quit
```

Configure SeGW B.

```
[SeGWB] interface GigabitEthernet 3/0/0
[SeGWB-GigabitEthernet3/0/0] ip address 10.10.0.2 24
[SeGWB-GigabitEthernet3/0/0] quit
```

3. Add uplink interfaces of SeGW A and SeGW B to the untrusted zone, downlink interfaces to the trusted zone, and heartbeat interfaces to the DMZ.

Configure SeGW A.

```
[SeGWA] firewall zone untrust
[SeGWA-zone-untrust] add interface eth-trunk 2
[SeGWA-zone-untrust] quit
[SeGWA] firewall zone trust
[SeGWA-zone-trust] add interface eth-trunk 1
[SeGWA-zone-trust] quit
[SeGWA] firewall zone dmz
[SeGWA-zone-dmz] add interface GigabitEthernet 3/0/0
[SeGWA-zone-dmz] quit
```

Configure SeGW B.

```
[SeGWB] firewall zone untrust
[SeGWB-zone-untrust] add interface eth-trunk 2
[SeGWB-zone-untrust] quit
[SeGWB] firewall zone trust
[SeGWB-zone-trust] add interface eth-trunk 1
[SeGWB-zone-trust] quit
[SeGWB] firewall zone dmz
[SeGWB-zone-dmz] add interface GigabitEthernet 3/0/0
[SeGWB-zone-dmz] quit
```

4. Specify the heartbeat interface and enable HRP.

Configure SeGW A.

```
[SeGWA] hrp interface GigabitEthernet 3/0/0 remote 10.10.0.2
[SeGWA] hrp enable
```

Configure SeGW B.

```
[SeGWB] hrp interface GigabitEthernet 3/0/0 remote 10.10.0.1
[SeGWB] hrp enable
```

5. Configure static routes, and specify next hops of uplink and downlink traffic of firewalls.

Configure SeGW A.

```
[SeGWA] ip route-static 0.0.0.0 0 10.6.1.111
[SeGWA] ip route-static 10.4.1.0 24 10.5.1.111
```

Configure SeGW B.

```
[SeGWB] ip route-static 0.0.0.0 0 10.6.1.111
[SeGWB] ip route-static 10.4.1.0 24 10.5.1.111
```

6. Configure security functions such as the security policy, IPS, and attack defense on SeGW A. The configuration of SeGW A is automatically backed up to SeGW B. For details, see the security gateway documentation.

Step 4 Enable OSPF on SwitchC-VS1, SwitchD-VS1, SwitchE, and SwitchF and use the main interface to establish the neighbor relationship.

Configure SwitchC-VS1.

```
<SwitchC> switch virtual-system vs1
<SwitchC-vs1> system-view
[~SwitchC-vs1] interface 10ge 1/0/29
[*SwitchC-vs1-10GE1/0/29] undo portswitch
[*SwitchC-vs1-10GE1/0/29] ip address 10.7.1.1 24
```

```
[*SwitchC-vs1-10GE1/0/29] ospf enable 1 area 0
[*SwitchC-vs1-10GE1/0/29] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchC-vs1-10GE1/0/29] ospf cost 10 //Set the OSPF cost of the interface to 10.
[*SwitchC-vs1-10GE1/0/29] quit
[*SwitchC-vs1] interface 10ge 1/0/30
[*SwitchC-vs1-10GE1/0/30] undo portswitch
[*SwitchC-vs1-10GE1/0/30] ip address 10.8.1.1 24
[*SwitchC-vs1-10GE1/0/30] ospf enable 1 area 0
[*SwitchC-vs1-10GE1/0/30] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchC-vs1-10GE1/0/30] quit
[*SwitchC-vs1] interface 10ge 1/0/31
[*SwitchC-vs1-10GE1/0/31] undo portswitch
[*SwitchC-vs1-10GE1/0/31] ip address 10.8.2.1 24
[*SwitchC-vs1-10GE1/0/31] ospf enable 1 area 0
[*SwitchC-vs1-10GE1/0/31] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchC-vs1-10GE1/0/31] quit
[*SwitchC-vs1] vlan batch 40
[*SwitchC-vs1] interface vlanif 40
[*SwitchC-vs1-Vlanif40] ip address 10.7.2.1 24
[*SwitchC-vs1-Vlanif40] ospf network-type p2p
[*SwitchC-vs1-Vlanif40] ospf cost 100
[*SwitchC-vs1-Vlanif40] quit
[*SwitchC-vs1] ospf 1
[*SwitchC-vs1-ospf-1] area 0
[*SwitchC-vs1-ospf-1-area-0.0.0.0] network 10.7.1.0 0.0.0.255
[*SwitchC-vs1-ospf-1-area-0.0.0.0] network 10.7.2.0 0.0.0.255
[*SwitchC-vs1-ospf-1-area-0.0.0.0] network 10.8.1.0 0.0.0.255
[*SwitchC-vs1-ospf-1-area-0.0.0.0] network 10.8.2.0 0.0.0.255
[*SwitchC-vs1-ospf-1-area-0.0.0.0] quit
[*SwitchC-vs1-ospf-1] import-route static
[*SwitchC-vs1-ospf-1] quit
[*SwitchC-vs1] commit
[~SwitchC-vs1] quit
```

Configure SwitchD-VS1.

```
<SwitchD> switch virtual-system vs1
<SwitchD-vs1> system-view
[~SwitchD-vs1] interface 10ge 1/0/29
[*SwitchD-vs1-10GE1/0/29] undo portswitch
[*SwitchD-vs1-10GE1/0/29] ip address 10.7.1.2 24
[*SwitchD-vs1-10GE1/0/29] ospf enable 1 area 0
[*SwitchD-vs1-10GE1/0/29] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchD-vs1-10GE1/0/29] ospf cost 10 //Set the OSPF cost of the interface to 10.
[*SwitchD-vs1-10GE1/0/29] quit
[*SwitchD-vs1] interface 10ge 1/0/30
[*SwitchD-vs1-10GE1/0/30] undo portswitch
[*SwitchD-vs1-10GE1/0/30] ip address 10.9.1.1 24
[*SwitchD-vs1-10GE1/0/30] ospf enable 1 area 0
[*SwitchD-vs1-10GE1/0/30] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchD-vs1-10GE1/0/30] quit
[*SwitchD-vs1] interface 10ge 1/0/31
[*SwitchD-vs1-10GE1/0/31] undo portswitch
[*SwitchD-vs1-10GE1/0/31] ip address 10.9.2.1 24
[*SwitchD-vs1-10GE1/0/31] ospf enable 1 area 0
[*SwitchD-vs1-10GE1/0/31] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchD-vs1-10GE1/0/31] quit
[*SwitchD-vs1] vlan batch 40
[*SwitchD-vs1] interface vlanif 40
[*SwitchD-vs1-Vlanif40] ip address 10.7.2.2 24
[*SwitchD-vs1-Vlanif40] ospf network-type p2p
[*SwitchD-vs1-Vlanif40] ospf cost 100
[*SwitchD-vs1-Vlanif40] quit
[*SwitchD-vs1] ospf 1
[*SwitchD-vs1-ospf-1] area 0
[*SwitchD-vs1-ospf-1-area-0.0.0.0] network 10.7.1.0 0.0.0.255
[*SwitchD-vs1-ospf-1-area-0.0.0.0] network 10.7.2.0 0.0.0.255
[*SwitchD-vs1-ospf-1-area-0.0.0.0] network 10.9.1.0 0.0.0.255
[*SwitchD-vs1-ospf-1-area-0.0.0.0] network 10.9.2.0 0.0.0.255
[*SwitchD-vs1-ospf-1-area-0.0.0.0] quit
```

```
[*SwitchD-vs1-ospf-1] import-route static
[*SwitchD-vs1-ospf-1] quit
[*SwitchD-vs1] commit
[~SwitchD-vs1] quit
```

Configure SwitchE.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchE
[*HUAWEI] commit
[~SwitchE] interface 10ge 1/0/1
[~SwitchE-10GE1/0/1] undo portswitch
[*SwitchE-10GE1/0/1] ip address 10.8.1.2 24
[*SwitchE-10GE1/0/1] ospf enable 1 area 0
[*SwitchE-10GE1/0/1] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchE-10GE1/0/1] quit
[*SwitchE] interface 10ge 1/0/2
[*SwitchE-10GE1/0/2] undo portswitch
[*SwitchE-10GE1/0/2] ip address 10.9.2.2 24
[*SwitchE-10GE1/0/2] ospf enable 1 area 0
[*SwitchE-10GE1/0/2] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchE-10GE1/0/2] quit
[~SwitchE] interface 10ge 1/0/9
[~SwitchE-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an example.
[*SwitchE-10GE1/0/9] quit
[*SwitchE] ospf 1
[*SwitchE-ospf-1] area 0
[*SwitchE-ospf-1-area-0.0.0.0] network 10.8.1.0 0.0.0.255
[*SwitchE-ospf-1-area-0.0.0.0] network 10.9.2.0 0.0.0.255
[*SwitchE-ospf-1-area-0.0.0.0] quit
[*SwitchE-ospf-1] quit
[*SwitchE] commit
```

Configure SwitchF.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchF
[*HUAWEI] commit
[~SwitchF] interface 10ge 1/0/1
[~SwitchF-10GE1/0/1] undo portswitch
[*SwitchF-10GE1/0/1] ip address 10.9.1.2 24
[*SwitchF-10GE1/0/1] ospf enable 1 area 0
[*SwitchF-10GE1/0/1] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchF-10GE1/0/1] quit
[*SwitchF] interface 10ge 1/0/2
[*SwitchF-10GE1/0/2] undo portswitch
[*SwitchF-10GE1/0/2] ip address 10.8.2.2 24
[*SwitchF-10GE1/0/2] ospf enable 1 area 0
[*SwitchF-10GE1/0/2] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchF-10GE1/0/2] quit
[~SwitchF] interface 10ge 1/0/9
[~SwitchF-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an example.
[*SwitchF-10GE1/0/9] quit
[*SwitchF] ospf 1
[*SwitchF-ospf-1] area 0
[*SwitchF-ospf-1-area-0.0.0.0] network 10.8.2.0 0.0.0.255
[*SwitchF-ospf-1-area-0.0.0.0] network 10.9.1.0 0.0.0.255
[*SwitchF-ospf-1-area-0.0.0.0] quit
[*SwitchF-ospf-1] quit
[*SwitchF] commit
```

Step 5 Verify the configuration.

- Run the **display dfs-group** command to check M-LAG information.
Check information about the M-LAG with DFS group ID 1. The M-LAG composed of SwitchA and SwitchB is used as an example. The configuration of the M-LAG composed of Admin-VS (SwitchC) and Admin-VS (SwitchD) and the configuration of the M-LAG composed of SwitchC-VS1 and SwitchD-VS1 are similar.


```
[~SwitchA] display dfs-group 1 m-lag
*          : Local node
Heart beat state : OK
Node 1 *
  Dfs-Group ID : 1
  Priority      : 150
  Address      : ip address 10.1.1.1 vpn-instance VRF-
A
  State        : Master
  Causation    : -
  System ID    : 00e0-fc95-7c01
  SysName     : SwitchA
  Version     : V100R006C00
  Device Type  : CE12800
Node 2
  Dfs-Group ID : 1
  Priority      : 120
  Address      : ip address 10.1.1.2 vpn-instance VRF-
A
  State        : Backup
  Causation    : -
  System ID    : 00e0-fc95-7c11
  SysName     : SwitchB
  Version     : V100R006C00
  Device Type  : CE12800
```

Check M-LAG information on SwitchA.

```
[~SwitchA] display dfs-group 1 node 1 m-lag brief
* - Local node

M-Lag ID   Interface   Port State   Status
  1   Eth-Trunk 10   Up          active(*)-active
  2   Eth-Trunk 20   Up          active(*)-active
  3   Eth-Trunk 30   Up          active(*)-active
  4   Eth-Trunk 40   Up          active(*)-active
```

In the preceding information, the value of **Heart beat state** is **OK**, indicating that the dual-active detection status is normal. SwitchA is used as Node 1, its priority is 150, and its status is **Master**. SwitchB is used as Node 2, its priority is 120, and its status is **Backup**. The value of **Causation** is -, the values of **Port State** of Node 1 and Node 2 are both **Up**, and the M-LAG status of both Node 1 and Node 2 is **active**, indicating that the M-LAG configuration is correct.

- Run the **display hrp state** command on SeGW A to check the HRP status. The following information indicates that the HRP is set up successfully.

```
HRP_M[SeGWA] display hrp state
Role: active, peer: standby
Running priority: 51008, peer: 51008
Core state: normal, peer: normal
Backup channel usage: 0%
Stable time: 0 days, 18 hours, 41 minutes
```

----End

Configuration Files

- SwitchA configuration file

```
#
sysname SwitchA
#
dfs-group 1
priority 150
source ip 10.1.1.1 vpn-instance VRF-A peer 10.1.1.2
dual-active detection enhanced enable
#
```

```

vlan batch 11
#
stp mode rstp
stp v-stp enable
stp bpdu-protection
stp flush disable
#
lacp m-lag system-id 00e0-fc12-3456
lacp m-lag priority 10
#
ip vpn-instance VRF-A
ipv4-family
 route-distinguisher 100:1
 vpn-target 111:1 export-extcommunity
 vpn-target 111:1 import-extcommunity
#
interface MEth0/0/0
 ip binding vpn-instance VRF-A
 ip address 10.1.1.1 255.255.255.0
#
interface Eth-Trunk0
 mode lacp-static
 peer-link 1
 port vlan exclude 1
#
interface Eth-Trunk10
 port default vlan 11
 stp edged-port enable
 mode lacp-dynamic
 dfs-group 1 m-lag 1
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk20
 port default vlan 11
 stp edged-port enable
 mode lacp-dynamic
 dfs-group 1 m-lag 2
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk30
 port default vlan 11
 stp edged-port enable
 mode lacp-dynamic
 dfs-group 1 m-lag 3
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk40
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 11
 mode lacp-static
 dfs-group 1 m-lag 4
 storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
 eth-trunk 10
#
interface 10GE1/0/2
 eth-trunk 20
#
interface 10GE1/0/3
 eth-trunk 30
#
interface 10GE1/0/4
 eth-trunk 0
#
interface 10GE1/0/6
 eth-trunk 40
#

```

```
interface 10GE1/0/7
 eth-trunk 40
#
interface 10GE1/0/9
 shutdown
#
interface 10GE2/0/4
 eth-trunk 0
#
return
```

- SwitchB configuration file

```
#
sysname SwitchB
#
dfs-group 1
 priority 120
 source ip 10.1.1.2 vpn-instance VRF-A peer 10.1.1.1
 dual-active detection enhanced enable
#
vlan batch 11
#
stp mode rstp
stp v-stp enable
stp bpdu-protection
stp flush disable
#
lACP m-lag system-id 00e0-fc12-3456
lACP m-lag priority 10
#
ip vpn-instance VRF-A
 ipv4-family
  route-distinguisher 100:2
  vpn-target 111:1 export-extcommunity
  vpn-target 111:1 import-extcommunity
#
interface MEth0/0/0
 ip binding vpn-instance VRF-A
 ip address 10.1.1.2 255.255.255.0
#
interface Eth-Trunk0
 mode lacp-static
 peer-link 1
 port vlan exclude 1
#
interface Eth-Trunk10
 port default vlan 11
 stp edged-port enable
 mode lacp-dynamic
 dfs-group 1 m-lag 1
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk20
 port default vlan 11
 stp edged-port enable
 mode lacp-dynamic
 dfs-group 1 m-lag 2
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk30
 port default vlan 11
 stp edged-port enable
 mode lacp-dynamic
 dfs-group 1 m-lag 3
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk40
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 11
```

```
mode lacp-static
dfs-group 1 m-lag 4
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 10
#
interface 10GE1/0/2
eth-trunk 20
#
interface 10GE1/0/3
eth-trunk 30
#
interface 10GE1/0/4
eth-trunk 0
#
interface 10GE1/0/6
eth-trunk 40
#
interface 10GE1/0/7
eth-trunk 40
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/4
eth-trunk 0
#
return
```

- Admin-VS (SwitchC) configuration file

```
#
sysname SwitchC
#
dfs-group 1
priority 150
source ip 10.2.1.1 vpn-instance VRF-B peer 10.2.1.2
dual-active detection enhanced enable
#
vlan batch 11 20
#
stp bridge-address 00e0-fc39-1300
stp mode rstp
stp v-stp enable
stp instance 0 root primary
stp flush disable
#
lacp m-lag system-id 00e0-fc12-3451
lacp m-lag priority 10
#
ip vpn-instance VRF-B
ipv4-family
route-distinguisher 101:1
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface Vlanif11
ip address 10.4.1.1 255.255.255.0
mac-address 0000-5e00-0102
#
interface Vlanif20
ip address 10.5.1.1 255.255.255.0
mac-address 0000-5e00-0103
#
interface MEth0/0/0
ip binding vpn-instance VRF-B
ip address 10.2.1.1 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
```

```
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk30
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk40
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 20
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk50
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 20
mode lacp-static
dfs-group 1 m-lag 3
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 30
#
interface 10GE1/0/2
eth-trunk 30
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
eth-trunk 40
#
interface 10GE1/0/6
eth-trunk 50
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
ip route-static 0.0.0.0 0.0.0.0 10.5.1.3
#
admin
virtual-system vs1
port-mode group
resource u4route upper-limit 60000
resource m4route upper-limit 1000
resource u6route upper-limit 16000
resource m6route upper-limit 100
resource vlan upper-limit 4063
resource mpls enable
resource trill enable
resource mcast enable
resource vpn-instance upper-limit 4096
resource cpu weight 5
resource memory ratio-threshold 100
assign interface 10GE1/0/24
assign interface 10GE1/0/25
assign interface 10GE1/0/26
assign interface 10GE1/0/27
```

```
assign interface 10GE1/0/28
assign interface 10GE1/0/29
assign interface 10GE1/0/30
assign interface 10GE1/0/31
assign interface 10GE1/0/32
assign interface 10GE1/0/33
assign interface 10GE1/0/34
assign interface 10GE1/0/35
assign interface 10GE1/0/36
assign interface 10GE1/0/37
assign interface 10GE1/0/38
assign interface 10GE1/0/39
assign interface 10GE1/0/40
assign interface 10GE1/0/41
assign interface 10GE1/0/42
assign interface 10GE1/0/43
assign interface 10GE1/0/44
assign interface 10GE1/0/45
assign interface 10GE1/0/46
assign interface 10GE1/0/47
assign interface 10GE2/0/24
assign interface 10GE2/0/25
assign interface 10GE2/0/26
assign interface 10GE2/0/27
assign interface 10GE2/0/28
assign interface 10GE2/0/29
assign interface 10GE2/0/30
assign interface 10GE2/0/31
assign interface 10GE2/0/32
assign interface 10GE2/0/33
assign interface 10GE2/0/34
assign interface 10GE2/0/35
assign interface 10GE2/0/36
assign interface 10GE2/0/37
assign interface 10GE2/0/38
assign interface 10GE2/0/39
assign interface 10GE2/0/40
assign interface 10GE2/0/41
assign interface 10GE2/0/42
assign interface 10GE2/0/43
assign interface 10GE2/0/44
assign interface 10GE2/0/45
assign interface 10GE2/0/46
assign interface 10GE2/0/47
#
return
```

- SwitchC-VS1 configuration file

```
#
sysname vs1
#
dfs-group 1
priority 150
source ip 10.3.1.1 vpn-instance VRF-C peer 10.3.1.2
dual-active detection enhanced enable
#
vlan batch 30 40
#
stp mode rstp
stp v-stp enable
stp flush disable
#
lACP m-lag system-id 00e0-fc12-3452
lACP m-lag priority 10
#
ip vpn-instance VRF-C
ipv4-family
route-distinguisher 102:1
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
```

```
#
interface Vlanif30
ip address 10.6.1.1 255.255.255.0
mac-address 0000-5e00-0104
#
interface Vlanif40
ip address 10.7.2.1 255.255.255.0
ospf cost 100
ospf network-type p2p
#
interface MEth0/0/0
ip binding vpn-instance VRF-B
ip address 10.3.1.1 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk60
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 30
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk70
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 30
mode lacp-static
dfs-group 1 m-lag 3
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/29
undo portswitch
ip address 10.7.1.1 255.255.255.0
ospf cost 10
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/30
undo portswitch
ip address 10.8.1.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/31
undo portswitch
ip address 10.8.2.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/32
eth-trunk 0
#
interface 10GE1/0/34
eth-trunk 60
#
interface 10GE1/0/35
eth-trunk 70
#
interface 10GE1/0/39
shutdown
#
interface 10GE2/0/32
eth-trunk 0
#
```

```
ip route-static 10.4.1.0 255.255.255.0 10.6.1.3
#
ospf 1
import-route static
area 0.0.0.0
network 10.7.1.0 0.0.0.255
network 10.7.2.0 0.0.0.255
network 10.8.1.0 0.0.0.255
network 10.8.2.0 0.0.0.255
#
return
```

- Admin-VS (SwitchD) configuration file

```
#
sysname SwitchD
#
dfs-group 1
priority 120
source ip 10.2.1.2 vpn-instance VRF-B peer 10.2.1.1
dual-active detection enhanced enable
#
vlan batch 11 20
#
stp bridge-address 00e0-fc39-1300
stp mode rstp
stp v-stp enable
stp instance 0 root primary
stp flush disable
#
lACP m-lag system-id 00e0-fc12-3451
lACP m-lag priority 10
#
ip vpn-instance VRF-B
ipv4-family
route-distinguisher 101:2
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface Vlanif11
ip address 10.4.1.1 255.255.255.0
mac-address 0000-5e00-0102
#
interface Vlanif20
ip address 10.5.1.1 255.255.255.0
mac-address 0000-5e00-0103
#
interface MEth0/0/0
ip binding vpn-instance VRF-B
ip address 10.2.1.2 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk30
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 100 mbps
#
interface Eth-Trunk40
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 20
mode lacp-static
dfs-group 1 m-lag 2
```



```
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk50
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 20
mode lacp-static
dfs-group 1 m-lag 3
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 30
#
interface 10GE1/0/2
eth-trunk 30
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
eth-trunk 40
#
interface 10GE1/0/6
eth-trunk 50
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
ip route-static 0.0.0.0 0.0.0.0 10.5.1.3
#
admin
virtual-system vs1
port-mode group
resource u4route upper-limit 60000
resource m4route upper-limit 1000
resource u6route upper-limit 16000
resource m6route upper-limit 100
resource vlan upper-limit 4063
resource mpls enable
resource trill enable
resource mcast enable
resource vpn-instance upper-limit 4096
resource cpu weight 5
resource memory ratio-threshold 100
assign interface 10GE1/0/24
assign interface 10GE1/0/25
assign interface 10GE1/0/26
assign interface 10GE1/0/27
assign interface 10GE1/0/28
assign interface 10GE1/0/29
assign interface 10GE1/0/30
assign interface 10GE1/0/31
assign interface 10GE1/0/32
assign interface 10GE1/0/33
assign interface 10GE1/0/34
assign interface 10GE1/0/35
assign interface 10GE1/0/36
assign interface 10GE1/0/37
assign interface 10GE1/0/38
assign interface 10GE1/0/39
assign interface 10GE1/0/40
assign interface 10GE1/0/41
assign interface 10GE1/0/42
assign interface 10GE1/0/43
assign interface 10GE1/0/44
assign interface 10GE1/0/45
```

```

assign interface 10GE1/0/46
assign interface 10GE1/0/47
assign interface 10GE2/0/24
assign interface 10GE2/0/25
assign interface 10GE2/0/26
assign interface 10GE2/0/27
assign interface 10GE2/0/28
assign interface 10GE2/0/29
assign interface 10GE2/0/30
assign interface 10GE2/0/31
assign interface 10GE2/0/32
assign interface 10GE2/0/33
assign interface 10GE2/0/34
assign interface 10GE2/0/35
assign interface 10GE2/0/36
assign interface 10GE2/0/37
assign interface 10GE2/0/38
assign interface 10GE2/0/39
assign interface 10GE2/0/40
assign interface 10GE2/0/41
assign interface 10GE2/0/42
assign interface 10GE2/0/43
assign interface 10GE2/0/44
assign interface 10GE2/0/45
assign interface 10GE2/0/46
assign interface 10GE2/0/47
#
return

```

- SwitchD-VS1 configuration file

```

#
sysname vs1
#
dfs-group 1
priority 120
source ip 10.3.1.2 vpn-instance VRF-C peer 10.3.1.1
dual-active detection enhanced enable
#
vlan batch 30 40
#
stp mode rstp
stp v-stp enable
stp flush disable
#
lACP m-lag system-id 00e0-fc12-3452
lACP m-lag priority 10
#
ip vpn-instance VRF-C
ipv4-family
route-distinguisher 102:2
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface Vlanif30
ip address 10.6.1.1 255.255.255.0
mac-address 0000-5e00-0104
#
interface Vlanif40
ip address 10.7.2.2 255.255.255.0
ospf cost 100
ospf network-type p2p
#
interface MEth0/0/0
ip binding vpn-instance VRF-B
ip address 10.3.1.2 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1

```

```
#
interface Eth-Trunk60
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 30
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk70
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 30
mode lacp-static
dfs-group 1 m-lag 3
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/29
undo portswitch
ip address 10.7.1.2 255.255.255.0
ospf cost 10
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/30
undo portswitch
ip address 10.9.1.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/31
undo portswitch
ip address 10.9.2.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/32
eth-trunk 0
#
interface 10GE1/0/34
eth-trunk 60
#
interface 10GE1/0/35
eth-trunk 70
#
interface 10GE1/0/39
shutdown
#
interface 10GE2/0/32
eth-trunk 0
#
ip route-static 10.4.1.0 255.255.255.0 10.6.1.3
#
ospf 1
import-route static
area 0.0.0.0
network 10.7.1.0 0.0.0.255
network 10.7.2.0 0.0.0.255
network 10.9.1.0 0.0.0.255
network 10.9.2.0 0.0.0.255
#
return
```

- SwitchE configuration file

```
#
sysname SwitchE
#
interface 10GE1/0/1
undo portswitch
ip address 10.8.1.2 255.255.255.0
```

```
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/2
undo portswitch
ip address 10.9.2.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/9
shutdown
#
ospf 1
area 0.0.0.0
network 10.8.1.0 0.0.0.255
network 10.9.2.0 0.0.0.255
#
return
```

- SwitchF configuration file

```
#
sysname SwitchF
#
interface 10GE1/0/1
undo portswitch
ip address 10.9.1.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/2
undo portswitch
ip address 10.8.2.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/9
shutdown
#
ospf 1
area 0.0.0.0
network 10.8.2.0 0.0.0.255
network 10.9.1.0 0.0.0.255
#
return
```

- SeGW A configuration file

```
#
sysname SeGWA
#
hrp enable
hrp interface GigabitEthernet 3/0/0 remote 10.10.0.2
#
interface Eth-Trunk1
mode lacp-static
trunkport GigabitEthernet 1/0/0 to 1/0/1
ip address 10.5.1.3 24 float master
#
interface Eth-Trunk2
mode lacp-static
trunkport GigabitEthernet 2/0/0 to 2/0/1
ip address 10.6.1.3 24 float master
#
interface GigabitEthernet3/0/0
ip address 10.10.0.1 24
#
firewall zone trust
set priority 85
add interface eth-trunk 1
#
firewall zone dmz
```

```
set priority 50
add interface GigabitEthernet 3/0/0
#
firewall zone untrust
set priority 5
add interface eth-trunk 2
#
ip route-static 0.0.0.0 0 10.6.1.111
ip route-static 10.4.1.0 24 10.5.1.111
#
return
```

- SeGW B configuration file

```
#
sysname SeGWB
#
hrp enable
hrp interface GigabitEthernet 3/0/0 remote 10.10.0.1
#
interface Eth-Trunk1
mode lacp-static
trunkport GigabitEthernet 1/0/0 to 1/0/1
ip address 10.5.1.3 24 float slave
#
interface Eth-Trunk2
mode lacp-static
trunkport GigabitEthernet 2/0/0 to 2/0/1
ip address 10.6.1.3 24 float slave
#
interface GigabitEthernet3/0/0
ip address 10.10.0.2 24
#
firewall zone trust
set priority 85
add interface eth-trunk 1
#
firewall zone dmz
set priority 50
add interface GigabitEthernet 3/0/0
#
firewall zone untrust
set priority 5
add interface eth-trunk 2
#
ip route-static 0.0.0.0 0 10.6.1.111
ip route-static 10.4.1.0 24 10.5.1.111
#
return
```

1.16 Configuring M-LAG and VRRP Across Data Centers

Applicable Products and Versions

This example applies to CE12800 series switches running V100R006C00 or later versions, CE8800, CE7800, CE6800, and CE5800 series switches running V200R002C50 or later versions, CE9800 series switches running V200R020C00 or later versions, CE12800E series switches running V200R005C00 or later versions, and CloudEngine 16800 series switches running V200R005C20 or later versions.

For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

Figure 1-18 shows the data center network across areas, and core, aggregation, and access switches are connected through 10GE interfaces. This example uses the CE12804.

Figure 1-18 Networking for configuring M-LAG and VRRP across data centers

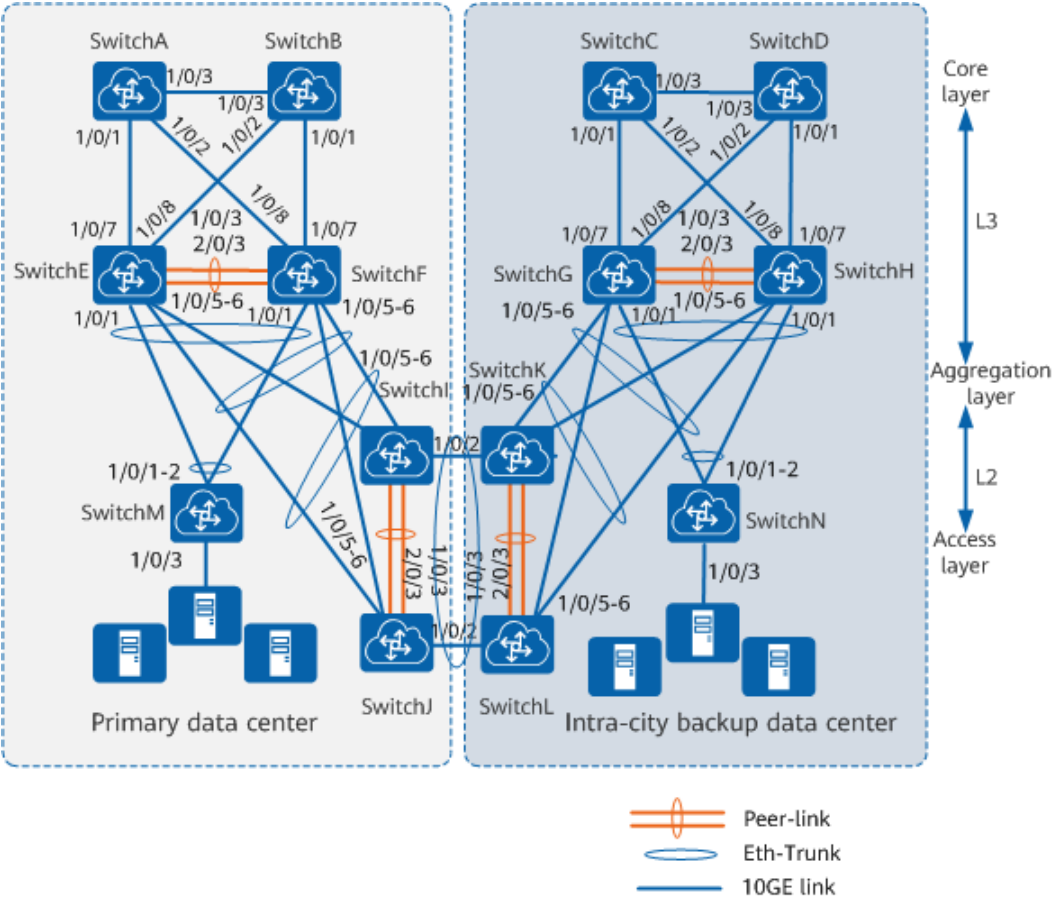


Table 1-16 Data preparation

Device Name	Interface Number	IP Address	Connected Device and Interface Number
SwitchA	10GE1/0/1	10.1.1.1/24	SwitchE: 10GE1/0/7
	10GE1/0/2	10.1.2.1/24	SwitchF: 10GE1/0/8
	10GE1/0/3	10.1.3.1/24	SwitchB: 10GE1/0/3
SwitchB	10GE1/0/1	10.1.4.1/24	SwitchF: 10GE1/0/7
	10GE1/0/2	10.1.5.1/24	SwitchE: 10GE1/0/8
	10GE1/0/3	10.1.3.2/24	SwitchA: 10GE1/0/3

Device Name	Interface Number	IP Address	Connected Device and Interface Number
SwitchC	10GE1/0/1	10.2.1.1/24	SwitchG: 10GE1/0/7
	10GE1/0/2	10.2.2.1/24	SwitchH: 10GE1/0/8
	10GE1/0/3	10.2.3.1/24	SwitchD: 10GE1/0/3
SwitchD	10GE1/0/1	10.2.4.1/24	SwitchH: 10GE1/0/7
	10GE1/0/2	10.2.5.1/24	SwitchG: 10GE1/0/8
	10GE1/0/3	10.2.3.2/24	SwitchC: 10GE1/0/3
SwitchE	Management interface	10.1.6.1/24	-
	VLANIF 11	10.1.7.1/24 Virtual IP address: 10.1.7.111	-
	Eth-Trunk 5: 10GE1/0/1	-	SwitchM: Eth-Trunk 15
	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	-	SwitchF: Eth-Trunk 0
		-	
	Eth-Trunk 10 • 10GE1/0/5 • 10GE1/0/6	-	SwitchI: 10GE1/0/5
		-	SwitchJ: 10GE1/0/6
	10GE1/0/7	10.1.1.2/24	SwitchA: 10GE1/0/1
	10GE1/0/8	10.1.5.2/24	SwitchB: 10GE1/0/2
SwitchF	Management interface	10.1.6.2/24	-
	VLANIF 11	10.1.7.1/24 Virtual IP address: 10.1.7.111	-
	Eth-Trunk 5: 10GE1/0/1	-	SwitchM: Eth-Trunk 15
	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	-	SwitchE: Eth-Trunk 0
		-	

Device Name	Interface Number	IP Address	Connected Device and Interface Number
	Eth-Trunk 10 • 10GE1/0/5 • 10GE1/0/6	-	SwitchJ: 10GE1/0/5
		-	SwitchI: 10GE1/0/6
	10GE1/0/7	10.1.4.2/24	SwitchB: 10GE1/0/1
	10GE1/0/8	10.1.2.2/24	SwitchA: 10GE1/0/2
SwitchG	Management interface	10.2.6.1/24	-
	VLANIF 11	10.1.7.3/24 Virtual IP address: 10.1.7.111	-
	Eth-Trunk 5: 10GE1/0/1	-	SwitchN: Eth-Trunk 30
	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	- -	SwitchH: Eth-Trunk 0
	Eth-Trunk 10 • 10GE1/0/5 • 10GE1/0/6	- -	SwitchK: 10GE1/0/5 SwitchL: 10GE1/0/6
	10GE1/0/7	10.2.1.2/24	SwitchC: 10GE1/0/1
	10GE1/0/8	10.2.5.2/24	SwitchD: 10GE1/0/2
SwitchH	Management interface	10.2.6.2/24	-
	VLANIF 11	10.1.7.3/24 Virtual IP address: 10.1.7.111	-
	Eth-Trunk 5: 10GE1/0/1	-	SwitchN: Eth-Trunk 30
	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	- -	SwitchG: Eth-Trunk 0
	Eth-Trunk 10 • 10GE1/0/5 • 10GE1/0/6	- -	SwitchL: 10GE1/0/5 SwitchK: 10GE1/0/6

Device Name	Interface Number	IP Address	Connected Device and Interface Number
	10GE1/0/7	10.2.4.2/24	SwitchD: 10GE1/0/1
	10GE1/0/8	10.2.2.2/24	SwitchC: 10GE1/0/2
SwitchI	Management interface	10.1.8.1/24	-
	Eth-Trunk 40 10GE1/0/2	-	SwitchK: 10GE1/0/2
	Eth-Trunk 0 10GE1/0/3 10GE2/0/3	-	SwitchJ: Eth-Trunk 0
		-	
	Eth-Trunk 35 10GE1/0/5 10GE1/0/6	-	SwitchE: 10GE1/0/5
		-	SwitchF: 10GE1/0/6
SwitchJ	Management interface	10.1.8.2/24	-
	Eth-Trunk 40 10GE1/0/2	-	SwitchL: 10GE1/0/2
	Eth-Trunk 0 10GE1/0/3 10GE2/0/3	-	SwitchI: Eth-Trunk 0
		-	
	Eth-Trunk 35 10GE1/0/5 10GE1/0/6	-	SwitchF: 10GE1/0/5
		-	SwitchE: 10GE1/0/6
SwitchK	Management interface	10.2.8.1/24	-
	Eth-Trunk 50 10GE1/0/2	-	SwitchI: 10GE1/0/2
	Eth-Trunk 0 10GE1/0/3 10GE2/0/3	-	SwitchL: Eth-Trunk 0
		-	
	Eth-Trunk 45 10GE1/0/5 10GE1/0/6	-	SwitchG: 10GE1/0/5
		-	SwitchH: 10GE1/0/6

Device Name	Interface Number	IP Address	Connected Device and Interface Number
SwitchL	Management interface	10.2.8.2/24	-
	Eth-Trunk 50 • 10GE1/0/2	-	SwitchJ: 10GE1/0/2
	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	-	SwitchK: Eth-Trunk 0
		-	
	Eth-Trunk 45 • 10GE1/0/5 • 10GE1/0/6	-	SwitchH: 10GE1/0/5
		-	SwitchG: 10GE1/0/6
SwitchM	Eth-Trunk 15 • 10GE1/0/1 • 10GE1/0/2	-	SwitchE: 10GE1/0/1 SwitchF: 10GE1/0/1
	10GE1/0/3	-	Server
SwitchN	Eth-Trunk 30 • 10GE1/0/1 • 10GE1/0/2	-	SwitchG: 10GE1/0/1 SwitchH: 10GE1/0/1
	10GE1/0/3	-	Server

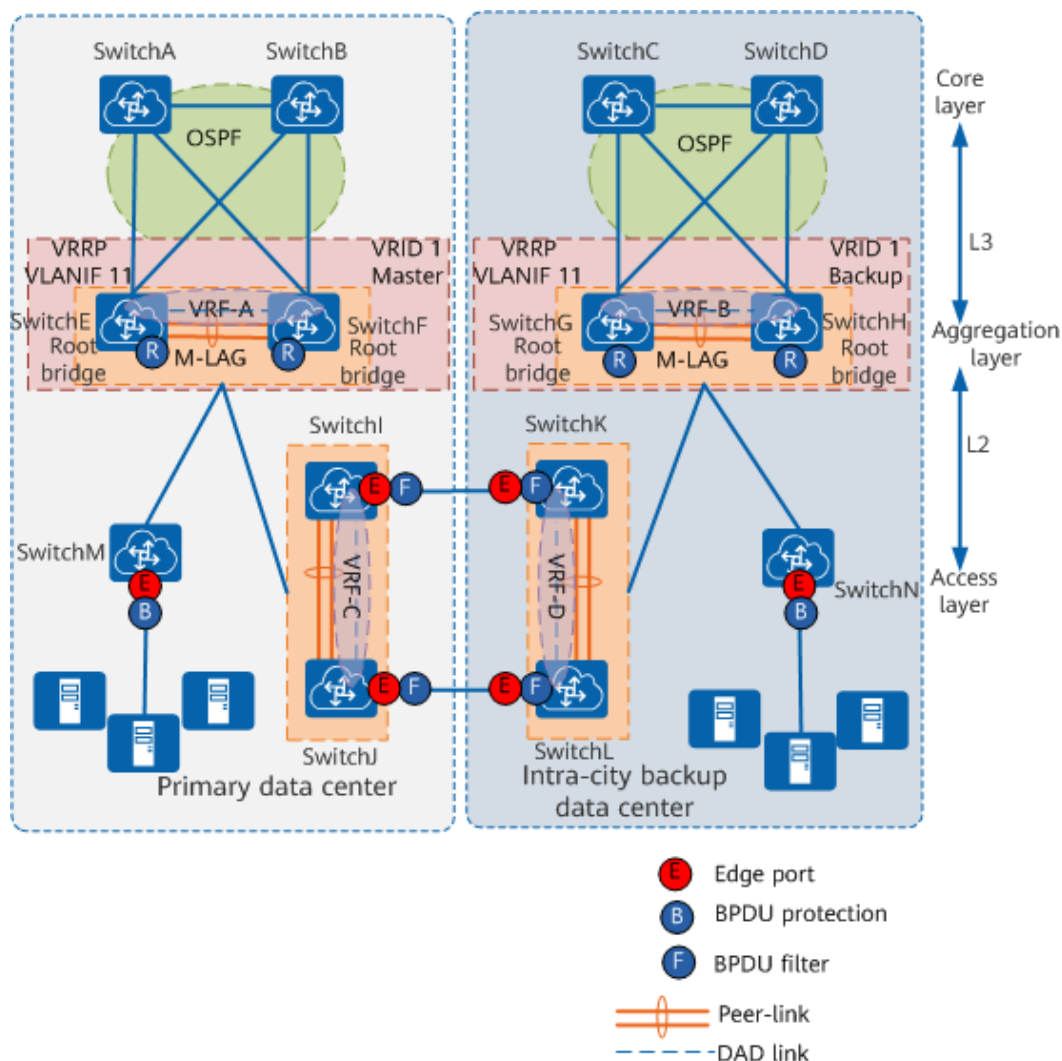
Requirement Analysis

To improve network reliability and serviceability, more and more users establish at least two data centers, and more data centers are deployed in different areas. The customer requires that the intra-city backup data center should ensure reliability. The aggregation and access layers use multi-level M-LAG interconnection to ensure reliability, improve the link usage, and extend the scale of dual-homing networking, meeting customer requirements. VRRP is deployed at the aggregation layer to provide dual-active gateways.

- Devices at the core and aggregation layers establish cross connections to ensure device-level reliability and load balance traffic based on ECMP (preferential forwarding of local traffic is used with M-LAG at the aggregation layer).
- The primary and backup data centers are connected through M-LAG and form a long-distance loop-free topology.

Figure 1-19 shows the logical networking after M-LAG and VRRP are deployed.

Figure 1-19 Networking for configuring M-LAG and VRRP across data centers



Configuration Roadmap

The configuration roadmap is as follows:

1. Configure M-LAG between SwitchE and SwitchF and between SwitchG and SwitchH to implement dual-homing access at the access layer. Configure M-LAG between SwitchI and SwitchJ and between SwitchK and SwitchL to implement multi-level LAG interconnection between aggregation layers of primary and backup data centers. When aggregation switches work normally, links load balance traffic and a fault of any aggregation switch does not affect services. High service reliability is therefore ensured.
 - Configure SwitchE, SwitchF, SwitchG, and SwitchH as root bridges and enable root protection on downstream interfaces to ensure that the interfaces can forward traffic normally. Configure the edge interface and BPD filter interface between the SwitchI and SwitchJ and between SwitchK and SwitchL so that STP calculation between primary and backup data centers is isolated. Configure interfaces on SwitchM and SwitchN connected to user terminals as edge interfaces to accelerate

route convergence and enable BPDU protection to enhance network stability.

- Create VLANIF interfaces on SwitchE, SwitchF, SwitchG, and SwitchH, assign IP addresses and MAC addresses to VLANIF interfaces, and create VRRP groups on VLANIF interfaces to implement active-active gateways. There are dual master devices in the primary data center and two backup devices in the backup data center.

NOTE

To enable two devices of the M-LAG to be simulated into one device to perform VRRP negotiation with the M-LAG of the remote data center, note the following points:

- In a data center, configure the same interface IP address and virtual MAC address when creating VRRP groups. However, the primary and backup data centers must use different interface IP addresses and virtual MAC addresses.
- The configured virtual MAC address must be different from the VRRP virtual MAC address, and the VRRP virtual MAC address depends on the VRID. For example, the VRID of the VRRP group composed of SwitchE and SwitchF is 1, so the MAC address of the VRRP group is 00-00-5E-00-01-01.

2. Enable OSPF on aggregation and core switches to ensure Layer 3 connectivity.

Procedure

Step 1 Configure M-LAG.

1. Configure V-STP, heartbeat links, DFS groups, peer-links, and M-LAG member interfaces on SwitchE, SwitchF, SwitchG, SwitchH, SwitchI, SwitchJ, SwitchK, and SwitchL.

Heartbeat links are connected to management interfaces to implement interworking, DFS groups must be bound to IP addresses of management interfaces to ensure communication, and management interfaces are bound to VRF instances to implement isolation.

It is recommended that Eth-Trunk member interfaces of the peer-link be deployed on different cards to prevent a peer-link fault caused by a single point of failure on a card.

Configure SwitchE.

Configure SwitchE as the root bridge of the STP network, and enable root protection on the Eth-Trunk of SwitchE connected to the access switch so that the Eth-Trunk can forward traffic normally.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchE
[*HUAWEI] commit
[~SwitchE] stp mode rstp
[*SwitchE] stp root primary //Configure the aggregation device as the root bridge of the STP
network.
[*SwitchE] stp bridge-address 200b-c739-1300 //Configure the bridge MAC address of the root
bridge (MAC address of the DFS master device).
[*SwitchE] stp v-stp enable
[*SwitchE] stp flush disable
[*SwitchE] ip vpn-instance VRF-A //Create VRF-A.
[*SwitchE-vpn-instance-VRF-A] ipv4-family
[*SwitchE-vpn-instance-VRF-A-af-ipv4] route-distinguisher 100:1
[*SwitchE-vpn-instance-VRF-A-af-ipv4] vpn-target 111:1 both
[*SwitchE-vpn-instance-VRF-A-af-ipv4] quit
```

```
[*SwitchE-vpn-instance-VRF-A] quit
[SwitchE] interface meth 0/0/0
[*SwitchE-Meth0/0/0] ip binding vpn-instance VRF-A //Bind the management interface to VRF-A.
[*SwitchE-Meth0/0/0] ip address 10.1.6.1 24
[*SwitchE-Meth0/0/0] quit
[SwitchE] dfs-group 1
[*SwitchE-dfs-group-1] source ip 10.1.6.1 vpn-instance VRF-A peer 10.1.6.2 //Bind an IPv4
address and a VPN instance to the DFS group.
[*SwitchE-dfs-group-1] dual-active detection enhanced enable
[*SwitchE-dfs-group-1] priority 150
[*SwitchE-dfs-group-1] quit
[SwitchE] interface eth-trunk 0
[*SwitchE-Eth-Trunk0] trunkport 10ge 1/0/3
[*SwitchE-Eth-Trunk0] trunkport 10ge 2/0/3 //Configure interfaces on different cards as member
interfaces of the Eth-Trunk that functions as the peer-link.
[*SwitchE-Eth-Trunk0] mode lacp-static
[*SwitchE-Eth-Trunk0] peer-link 1
[*SwitchE-Eth-Trunk0] port vlan exclude 1
[*SwitchE-Eth-Trunk0] quit
[SwitchE] vlan batch 11
[SwitchE] interface eth-trunk 5
[*SwitchE-Eth-Trunk5] mode lacp-static
[*SwitchE-Eth-Trunk5] port link-type trunk
[*SwitchE-Eth-Trunk5] undo port trunk allow-pass vlan 1
[*SwitchE-Eth-Trunk5] port trunk allow-pass vlan 11
[*SwitchE-Eth-Trunk5] trunkport 10ge 1/0/1
[*SwitchE-Eth-Trunk5] dfs-group 1 m-lag 1
[*SwitchE-Eth-Trunk5] stp root-protection //Enable root protection on the interface.
[*SwitchE-Eth-Trunk5] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchE-Eth-Trunk5] quit
[SwitchE] interface eth-trunk 10
[*SwitchE-Eth-Trunk10] mode lacp-static
[*SwitchE-Eth-Trunk10] port link-type trunk
[*SwitchE-Eth-Trunk10] undo port trunk allow-pass vlan 1
[*SwitchE-Eth-Trunk10] port trunk allow-pass vlan 11
[*SwitchE-Eth-Trunk10] trunkport 10ge 1/0/5 to 1/0/6
[*SwitchE-Eth-Trunk10] dfs-group 1 m-lag 2
[*SwitchE-Eth-Trunk10] stp root-protection //Enable root protection on the interface.
[*SwitchE-Eth-Trunk10] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchE-Eth-Trunk10] quit
[SwitchE] lacp m-lag priority 10
[SwitchE] lacp m-lag system-id 00e0-fc00-0000
[SwitchE] interface 10ge 1/0/9
[*SwitchE-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchE-10GE1/0/9] quit
[SwitchE] commit
```

Configure SwitchF.

Configure SwitchF as the root bridge of the STP network, and enable root protection on the Eth-Trunk of SwitchF connected to the access switch so that the Eth-Trunk can forward traffic normally.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchF
[*HUAWEI] commit
[~SwitchF] stp mode rstp
[SwitchF] stp root primary //Configure the aggregation device as the root bridge of the STP
network.
[SwitchF] stp bridge-address 200b-c739-1300 //Configure the bridge MAC address of the root
bridge (MAC address of the DFS master device).
[SwitchF] stp v-stp enable
```

```
[*SwitchF] stp flush disable
[SwitchF] ip vpn-instance VRF-A //Create VRF-A.
[SwitchF-vpn-instance-VRF-A] ipv4-family
[SwitchF-vpn-instance-VRF-A-af-ipv4] route-distinguisher 100:2
[SwitchF-vpn-instance-VRF-A-af-ipv4] vpn-target 111:1 both
[SwitchF-vpn-instance-VRF-A-af-ipv4] quit
[SwitchF-vpn-instance-VRF-A] quit
[SwitchF] interface meth 0/0/0
[SwitchF-Meth0/0/0] ip binding vpn-instance VRF-A //Bind the management interface to VRF-A.
[SwitchF-Meth0/0/0] ip address 10.1.6.2 24
[SwitchF-Meth0/0/0] quit
[SwitchF] dfs-group 1
[SwitchF-dfs-group-1] source ip 10.1.6.2 vpn-instance VRF-A peer 10.1.6.1 //Bind an IPv4
address and a VPN instance to the DFS group.
[SwitchF-dfs-group-1] dual-active detection enhanced enable
[SwitchF-dfs-group-1] priority 120
[SwitchF-dfs-group-1] quit
[SwitchF] interface eth-trunk 0
[SwitchF-Eth-Trunk0] trunkport 10ge 1/0/3
[SwitchF-Eth-Trunk0] trunkport 10ge 2/0/3
[SwitchF-Eth-Trunk0] mode lacp-static
[SwitchF-Eth-Trunk0] peer-link 1
[SwitchF-Eth-Trunk0] port vlan exclude 1
[SwitchF-Eth-Trunk0] quit
[SwitchF] vlan batch 11
[SwitchF] interface eth-trunk 5
[SwitchF-Eth-Trunk5] mode lacp-static
[SwitchF-Eth-Trunk5] port link-type trunk
[SwitchF-Eth-Trunk5] undo port trunk allow-pass vlan 1
[SwitchF-Eth-Trunk5] port trunk allow-pass vlan 11
[SwitchF-Eth-Trunk5] trunkport 10ge 1/0/1
[SwitchF-Eth-Trunk5] dfs-group 1 m-lag 1
[SwitchF-Eth-Trunk5] stp root-protection //Enable root protection on the interface.
[SwitchF-Eth-Trunk5] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[SwitchF-Eth-Trunk5] quit
[SwitchF] interface eth-trunk 10
[SwitchF-Eth-Trunk10] mode lacp-static
[SwitchF-Eth-Trunk10] port link-type trunk
[SwitchF-Eth-Trunk10] undo port trunk allow-pass vlan 1
[SwitchF-Eth-Trunk10] port trunk allow-pass vlan 11
[SwitchF-Eth-Trunk10] trunkport 10ge 1/0/5 to 1/0/6
[SwitchF-Eth-Trunk10] dfs-group 1 m-lag 2
[SwitchF-Eth-Trunk10] stp root-protection //Enable root protection on the interface.
[SwitchF-Eth-Trunk10] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[SwitchF-Eth-Trunk10] quit
[SwitchF] lacp m-lag priority 10
[SwitchF] lacp m-lag system-id 00e0-fc00-0000
[SwitchF] interface 10ge 1/0/9
[SwitchF-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[SwitchF-10GE1/0/9] quit
[SwitchF] commit
```

The configurations of SwitchG and SwitchH are similar to the configurations of SwitchE and SwitchF, and the configuration details are not mentioned here.

Configure SwitchI.

Configure the Eth-Trunk on SwitchI connected to the backup data center as the edge interface and BPDU filter interface so that STP calculation between primary and backup data centers is isolated.

```
<HUAWEI> system-view
[~HUAWEI] sysname Switch1
[*HUAWEI] commit
[~Switch1] stp mode rstp
[*Switch1] stp v-stp enable
[*Switch1] stp flush disable
[*Switch1] ip vpn-instance VRF-C //Create VRF-C.
[*Switch1-vpn-instance-VRF-C] ipv4-family
[*Switch1-vpn-instance-VRF-C-af-ipv4] route-distinguisher 102:1
[*Switch1-vpn-instance-VRF-C-af-ipv4] vpn-target 111:1 both
[*Switch1-vpn-instance-VRF-C-af-ipv4] quit
[*Switch1-vpn-instance-VRF-C] quit
[*Switch1] interface meth 0/0/0
[*Switch1-Meth0/0/0] ip binding vpn-instance VRF-C //Bind the management interface to VRF-C.
[*Switch1-Meth0/0/0] ip address 10.1.8.1 24
[*Switch1-Meth0/0/0] quit
[*Switch1] dfs-group 1
[*Switch1-dfs-group-1] source ip 10.1.8.1 vpn-instance VRF-C peer 10.1.8.2 //Bind an IPv4 address
and a VPN instance to the DFS group.
[*Switch1-dfs-group-1] dual-active detection enhanced enable
[*Switch1-dfs-group-1] priority 150
[*Switch1-dfs-group-1] quit
[*Switch1] interface eth-trunk 0
[*Switch1-Eth-Trunk0] trunkport 10ge 1/0/3
[*Switch1-Eth-Trunk0] trunkport 10ge 2/0/3
[*Switch1-Eth-Trunk0] mode lacp-static
[*Switch1-Eth-Trunk0] peer-link 1
[*Switch1-Eth-Trunk0] port vlan exclude 1
[*Switch1-Eth-Trunk0] quit
[*Switch1] vlan batch 11
[*Switch1] interface eth-trunk 35
[*Switch1-Eth-Trunk35] mode lacp-static
[*Switch1-Eth-Trunk35] port link-type trunk
[*Switch1-Eth-Trunk35] undo port trunk allow-pass vlan 1
[*Switch1-Eth-Trunk35] port trunk allow-pass vlan 11
[*Switch1-Eth-Trunk35] trunkport 10ge 1/0/5 to 1/0/6
[*Switch1-Eth-Trunk35] dfs-group 1 m-lag 1
[*Switch1-Eth-Trunk35] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*Switch1-Eth-Trunk35] quit
[*Switch1] interface eth-trunk 40
[*Switch1-Eth-Trunk40] mode lacp-static
[*Switch1-Eth-Trunk40] port link-type trunk
[*Switch1-Eth-Trunk40] undo port trunk allow-pass vlan 1
[*Switch1-Eth-Trunk40] port trunk allow-pass vlan 11
[*Switch1-Eth-Trunk40] trunkport 10ge 1/0/2
[*Switch1-Eth-Trunk40] dfs-group 1 m-lag 2
[*Switch1-Eth-Trunk40] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[*Switch1-Eth-Trunk40] stp bpdu-filter enable //Configure the Eth-Trunk as the BPDU filter
interface.
[*Switch1-Eth-Trunk40] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*Switch1-Eth-Trunk40] quit
[*Switch1] lacp m-lag priority 10
[*Switch1] lacp m-lag system-id 00e0-fc00-0002
[*Switch1] interface 10ge 1/0/9
[*Switch1-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*Switch1-10GE1/0/9] quit
[*Switch1] commit
```

Configure Switch1.

Configure the Eth-Trunk on SwitchJ connected to the backup data center as the edge interface and BPDU filter interface so that STP calculation between primary and backup data centers is isolated.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchJ
[*HUAWEI] commit
[~SwitchJ] stp mode rstp
[*SwitchJ] stp v-stp enable
[*SwitchJ] stp flush disable
[*SwitchJ] ip vpn-instance VRF-C //Create VRF-C.
[*SwitchJ-vpn-instance-VRF-C] ipv4-family
[*SwitchJ-vpn-instance-VRF-C-af-ipv4] route-distinguisher 102:2
[*SwitchJ-vpn-instance-VRF-C-af-ipv4] vpn-target 111:1 both
[*SwitchJ-vpn-instance-VRF-C-af-ipv4] quit
[*SwitchJ-vpn-instance-VRF-C] quit
[*SwitchJ] interface meth 0/0/0
[*SwitchJ-Meth0/0/0] ip binding vpn-instance VRF-C //Bind the management interface to VRF-C.
[*SwitchJ-Meth0/0/0] ip address 10.1.8.2 24
[*SwitchJ-Meth0/0/0] quit
[*SwitchJ] dfs-group 1
[*SwitchJ-dfs-group-1] source ip 10.1.8.2 vpn-instance VRF-C peer 10.1.8.1 //Bind an IPv4
address and a VPN instance to the DFS group.
[*SwitchJ-dfs-group-1] dual-active detection enhanced enable
[*SwitchJ-dfs-group-1] priority 120
[*SwitchJ-dfs-group-1] quit
[*SwitchJ] interface eth-trunk 0
[*SwitchJ-Eth-Trunk0] trunkport 10ge 1/0/3
[*SwitchJ-Eth-Trunk0] trunkport 10ge 2/0/3
[*SwitchJ-Eth-Trunk0] mode lacp-static
[*SwitchJ-Eth-Trunk0] peer-link 1
[*SwitchJ-Eth-Trunk0] port vlan exclude 1
[*SwitchJ-Eth-Trunk0] quit
[*SwitchJ] vlan batch 11
[*SwitchJ] interface eth-trunk 35
[*SwitchJ-Eth-Trunk35] mode lacp-static
[*SwitchJ-Eth-Trunk35] port link-type trunk
[*SwitchJ-Eth-Trunk35] undo port trunk allow-pass vlan 1
[*SwitchJ-Eth-Trunk35] port trunk allow-pass vlan 11
[*SwitchJ-Eth-Trunk35] trunkport 10ge 1/0/5 to 1/0/6
[*SwitchJ-Eth-Trunk35] dfs-group 1 m-lag 1
[*SwitchJ-Eth-Trunk35] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchJ-Eth-Trunk35] quit
[*SwitchJ] interface eth-trunk 40
[*SwitchJ-Eth-Trunk40] mode lacp-static
[*SwitchJ-Eth-Trunk40] port link-type trunk
[*SwitchJ-Eth-Trunk40] undo port trunk allow-pass vlan 1
[*SwitchJ-Eth-Trunk40] port trunk allow-pass vlan 11
[*SwitchJ-Eth-Trunk40] trunkport 10ge 1/0/2
[*SwitchJ-Eth-Trunk40] dfs-group 1 m-lag 2
[*SwitchJ-Eth-Trunk40] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[*SwitchJ-Eth-Trunk40] stp bpdu-filter enable //Configure the Eth-Trunk as the BPDU filter
interface.
[*SwitchJ-Eth-Trunk40] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface. (For CE8800, CE7800, CE6800 except the CE6870EI and
CE6875EI, and CE5800 series switches, this command can be run only in the GE interface view, 10GE
interface view, 25GE interface view, 40GE interface view, 100GE interface view, and port group view).
[*SwitchJ-Eth-Trunk40] quit
[*SwitchJ] lacp m-lag priority 10
[*SwitchJ] lacp m-lag system-id 00e0-fc00-0002
[*SwitchJ] interface 10ge 1/0/9
[*SwitchJ-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchJ-10GE1/0/9] quit
[*SwitchJ] commit
```


The configurations of SwitchK and SwitchL are similar to the configurations of SwitchI and SwitchJ, and the configuration details are not mentioned here.

2. Create VLANIF interfaces on SwitchE, SwitchF, SwitchG, and SwitchH and configure IP addresses for the VLANIF interfaces, and create a VRRP group as the user-side gateway on VLANIF 11. There are dual master devices in the primary data center and two backup devices in the backup data center.

Configure SwitchE.

```
[~SwitchE] interface vlanif 11
[*SwitchE-Vlanif11] ip address 10.1.7.1 24
[*SwitchE-Vlanif11] mac-address 0000-5e00-0102
[*SwitchE-Vlanif11] vrrp vrid 1 virtual-ip 10.1.7.111
[*SwitchE-Vlanif11] vrrp vrid 1 priority 120
[*SwitchE-Vlanif11] quit
[*SwitchE] commit
```

Configure SwitchF.

```
[~SwitchF] interface vlanif 11
[*SwitchF-Vlanif11] ip address 10.1.7.1 24
[*SwitchF-Vlanif11] mac-address 0000-5e00-0102
[*SwitchF-Vlanif11] vrrp vrid 1 virtual-ip 10.1.7.111
[*SwitchF-Vlanif11] vrrp vrid 1 priority 120
[*SwitchF-Vlanif11] quit
[*SwitchF] commit
```

Configure SwitchG.

```
[~SwitchG] interface vlanif 11
[*SwitchG-Vlanif11] ip address 10.1.7.3 24
[*SwitchG-Vlanif11] mac-address 0000-5e00-0103
[*SwitchG-Vlanif11] vrrp vrid 1 virtual-ip 10.1.7.111
[*SwitchG-Vlanif11] quit
[*SwitchG] commit
```

Configure SwitchH.

```
[~SwitchH] interface vlanif 11
[*SwitchH-Vlanif11] ip address 10.1.7.3 24
[*SwitchH-Vlanif11] mac-address 0000-5e00-0103
[*SwitchH-Vlanif11] vrrp vrid 1 virtual-ip 10.1.7.111
[*SwitchH-Vlanif11] quit
[*SwitchH] commit
```

3. Add uplink interfaces on SwitchM and SwitchN to an Eth-Trunk and configure interfaces connected to servers as edge interfaces and BPDU filter interfaces. The interface connected to Server 1 is used as an example.

Configure SwitchM.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchM
[HUAWEI] commit
[~SwitchM] stp mode rstp
[*SwitchM] vlan batch 11
[*SwitchM] interface eth-trunk 15
[*SwitchM-Eth-Trunk15] mode lacp-static
[*SwitchM-Eth-Trunk15] port link-type trunk
[*SwitchM-Eth-Trunk15] port trunk allow-pass vlan 11
[*SwitchM-Eth-Trunk15] trunkport 10ge 1/0/1 to 1/0/2
[*SwitchM-Eth-Trunk15] quit
[*SwitchM] interface 10ge 1/0/3
[*SwitchM-10GE1/0/3] port default vlan 11
[*SwitchM-10GE1/0/3] stp edged-port enable //Configure the interface as the edge interface.
[*SwitchM-10GE1/0/3] quit
[*SwitchM] interface 10ge 1/0/9
```

```
[*SwitchM-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchM-10GE1/0/9] quit
[*SwitchM] stp bpdu-protection //Enable BPDU protection on the edge interface.
[*SwitchM] commit
```

The configuration of SwitchN is similar to the configuration of SwitchM,
and the configuration details are not mentioned here.

Step 2 Enable OSPF on SwitchA, SwitchB, SwitchC, SwitchD, SwitchE, SwitchF, SwitchG,
and SwitchH.

Configure SwitchA.

```
[~SwitchA] interface 10ge 1/0/1
[*SwitchA-10GE1/0/1] undo portswitch
[*SwitchA-10GE1/0/1] ip address 10.1.1.1 24
[*SwitchA-10GE1/0/1] ospf enable 1 area 0
[*SwitchA-10GE1/0/1] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchA-10GE1/0/1] quit
[*SwitchA] interface 10ge 1/0/2
[*SwitchA-10GE1/0/2] undo portswitch
[*SwitchA-10GE1/0/2] ip address 10.1.2.1 24
[*SwitchA-10GE1/0/2] ospf enable 1 area 0
[*SwitchA-10GE1/0/2] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchA-10GE1/0/2] quit
[*SwitchA] interface 10ge 1/0/3
[*SwitchA-10GE1/0/3] undo portswitch
[*SwitchA-10GE1/0/3] ip address 10.1.3.1 24
[*SwitchA-10GE1/0/3] ospf enable 1 area 0
[*SwitchA-10GE1/0/3] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchA-10GE1/0/3] quit
[*SwitchA] interface 10ge 1/0/9
[*SwitchA-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an example.
[*SwitchA-10GE1/0/9] quit
[*SwitchA] ospf 1
[*SwitchA-ospf-1] area 0
[*SwitchA-ospf-1-area-0.0.0.0] quit
[*SwitchA-ospf-1] quit
[*SwitchA] commit
```

The configurations of SwitchB, SwitchC, and SwitchD are similar to the
configuration of SwitchA, and the configuration details are not mentioned here.

Configure SwitchE.

```
[~SwitchE] interface 10ge 1/0/7
[*SwitchE-10GE1/0/7] undo portswitch
[*SwitchE-10GE1/0/7] ip address 10.1.1.2 24
[*SwitchE-10GE1/0/7] ospf enable 1 area 0
[*SwitchE-10GE1/0/7] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchE-10GE1/0/7] quit
[*SwitchE] interface 10ge 1/0/8
[*SwitchE-10GE1/0/8] undo portswitch
[*SwitchE-10GE1/0/8] ip address 10.1.5.2 24
[*SwitchE-10GE1/0/8] ospf enable 1 area 0
[*SwitchE-10GE1/0/8] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchE-10GE1/0/8] quit
[*SwitchE] ospf 1
[*SwitchE-ospf-1] import-route direct //Configure the switch to import direct routes. You can configure
a routing policy to filter out unnecessary routes according to the actual networking.
[*SwitchE-ospf-1] area 0
[*SwitchE-ospf-1-area-0.0.0.0] quit
[*SwitchE-ospf-1] quit
[*SwitchE] commit
```

The configurations of SwitchF, SwitchG, and SwitchH are similar to the
configuration of SwitchE, and the configuration details are not mentioned here.

Step 3 Verify the configuration.

- Run the **display dfs-group** command to check M-LAG information.

Check information about the M-LAG with DFS group 1. (The M-LAG composed of SwitchE and SwitchF is used as an example. The M-LAG composed of SwitchG and SwitchH, the M-LAG composed of SwitchI and SwitchJ, and the M-LAG composed of SwitchK and SwitchL are similar.)

```
[~SwitchE] display dfs-group 1 m-lag
```

```
*          : Local node
Heart beat state : OK
Node 1 *
  Dfs-Group ID  : 1
  Priority      : 150
  Address       : ip address 10.1.6.1
  State        : Master
  Causation     : -
  System ID     : ac94-8400-df01
  SysName      : SwitchE
  Version      : V100R006C00
  Device Type   : CE12800
Node 2
  Dfs-Group ID  : 1
  Priority      : 120
  Address       : ip address 10.1.6.2
  State        : Backup
  Causation     : -
  System ID     : 0025-9e95-7c21
  SysName      : SwitchF
  Version      : V100R006C00
  Device Type   : CE12800
```

Check M-LAG information on SwitchE.

```
[~SwitchE] display dfs-group 1 node 1 m-lag brief
```

```
* - Local node

M-Lag ID   Interface   Port State   Status
  1   Eth-Trunk 5    Up    active(*)-active
  2   Eth-Trunk 10   Up    active(*)-active
```

In the preceding information, the value of **Heart beat state** is **OK**, indicating that the heartbeat status is normal. SwitchE is used as Node 1, its priority is 150, and its status is **Master**. SwitchF is used as Node 2, its priority is 120, and its status is **Backup**. The value of **Causation** is -, the values of **Port State** of Node 1 and Node 2 are both **Up**, and the M-LAG status of both Node 1 and Node 2 is **active**, indicating that the M-LAG configuration is correct.

- Run the **display vrrp** command on SwitchE, SwitchF, SwitchG and SwitchH. You can see that SwitchE and SwitchF are in Master state and SwitchG and SwitchH are in Backup state.

```
[~SwitchE] display vrrp verbose
```

```
Vlanif11 | Virtual Router 1
State      : Master
Virtual IP  : 10.1.7.111
Master IP   : 10.1.7.1
PriorityRun  : 120
PriorityConfig : 120
MasterPriority : 120
Preempt     : YES Delay Time : 0s Remain :
--
Hold Multiplier: 3
TimerRun     : 1s
TimerConfig  : 1s
Auth Type    : NONE
Virtual MAC   : 0000-5e00-0101
Check TTL    : YES
Config Type   : Normal
```

```
Create Time      : 2016-03-14 14:26:29
Last Change Time : 2016-03-14
14:26:35
[~SwitchF] display vrrp verbose
Vlanif11 | Virtual Router 1
State      : Master
Virtual IP  : 10.1.7.111
Master IP   : 10.1.7.1
PriorityRun  : 120
PriorityConfig : 120
MasterPriority : 120
Preempt     : YES Delay Time : 0s Remain :
--
Hold Multiplier: 3
TimerRun     : 1s
TimerConfig  : 1s
Auth Type    : NONE
Virtual MAC   : 0000-5e00-0101
Check TTL    : YES
Config Type  : Normal
Create Time  : 2016-03-14 14:26:29
Last Change Time : 2016-03-14
14:26:35
[~SwitchG] display vrrp verbose
Vlanif11 | Virtual Router 1
State      : Backup
Virtual IP  : 10.1.7.111
Master IP   : 10.1.7.3
PriorityRun  : 100
PriorityConfig : 100
MasterPriority : 120
Preempt     : YES Delay Time : 0s Remain : --
Hold Multiplier: 3
TimerRun     : 1s
TimerConfig  : 1s
Auth Type    : NONE
Virtual MAC   : 0000-5e00-0101
Check TTL    : YES
Config Type  : Normal
Create Time  : 2016-03-14 14:26:29
Last Change Time : 2016-03-14 14:26:35
[~SwitchH] display vrrp verbose
Vlanif11 | Virtual Router 1
State      : Backup
Virtual IP  : 10.1.7.111
Master IP   : 10.1.7.3
PriorityRun  : 100
PriorityConfig : 100
MasterPriority : 120
Preempt     : YES Delay Time : 0s Remain : --
Hold Multiplier: 3
TimerRun     : 1s
TimerConfig  : 1s
Auth Type    : NONE
Virtual MAC   : 0000-5e00-0101
Check TTL    : YES
Config Type  : Normal
Create Time  : 2016-03-14 14:26:29
Last Change Time : 2016-03-14 14:26:35
```

----End

Configuration Files

- SwitchA configuration file

```
#
sysname SwitchA
#
```

```
interface 10GE1/0/1
 undo portswitch
 ip address 10.1.1.1 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/2
 undo portswitch
 ip address 10.1.2.1 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/3
 undo portswitch
 ip address 10.1.3.1 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/9
 shutdown
#
ospf 1
 area 0.0.0.0
#
return
```

- SwitchB configuration file

```
#
sysname SwitchB
#
interface 10GE1/0/1
 undo portswitch
 ip address 10.1.4.1 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/2
 undo portswitch
 ip address 10.1.5.1 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/3
 undo portswitch
 ip address 10.1.3.2 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/9
 shutdown
#
ospf 1
 area 0.0.0.0
#
return
```

- SwitchC configuration file

```
#
sysname SwitchC
#
interface 10GE1/0/1
 undo portswitch
 ip address 10.2.1.1 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/2
 undo portswitch
 ip address 10.2.2.1 255.255.255.0
 ospf network-type p2p
```

```
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/3
undo portswitch
ip address 10.2.3.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/9
shutdown
#
ospf 1
area 0.0.0.0
#
return
```

- SwitchD configuration file

```
#
sysname SwitchD
#
interface 10GE1/0/1
undo portswitch
ip address 10.2.4.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/2
undo portswitch
ip address 10.2.5.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/3
undo portswitch
ip address 10.2.3.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/9
shutdown
#
ospf 1
area 0.0.0.0
#
return
```

- SwitchE configuration file

```
#
sysname SwitchE
#
dfs-group 1
priority 150
source ip 10.1.6.1 vpn-instance VRF-A peer 10.1.6.2
dual-active detection enhanced enable
#
vlan batch 11
#
stp bridge-address 200b-c739-1300
stp mode rstp
stp v-stp enable
stp instance 0 root primary
stp flush disable
#
lACP m-lag system-id 00e0-fc00-0000
lACP m-lag priority 10
#
ip vpn-instance VRF-A
ipv4-family
route-distinguisher 100:1
vpn-target 111:1 export-extcommunity
```

```
vpn-target 111:1 import-extcommunity
#
interface Vlanif11
ip address 10.1.7.1 255.255.255.0
vrrp vrid 1 virtual-ip 10.1.7.111
vrrp vrid 1 priority 120
mac-address 0000-5e00-0102
#
interface MEth0/0/0
ip binding vpn-instance VRF-A
ip address 10.1.6.1 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk5
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk10
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 5
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
eth-trunk 10
#
interface 10GE1/0/6
eth-trunk 10
#
interface 10GE1/0/7
undo portswitch
ip address 10.1.1.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/8
undo portswitch
ip address 10.1.5.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
ospf 1
import-route direct
area 0.0.0.0
```

```
#
return
• SwitchF configuration file
#
sysname SwitchF
#
dfs-group 1
priority 120
source ip 10.1.6.2 vpn-instance VRF-A peer 10.1.6.1
dual-active detection enhanced enable
#
vlan batch 11
#
stp bridge-address 200b-c739-1300
stp mode rstp
stp v-stp enable
stp instance 0 root primary
stp flush disable
#
lacp m-lag system-id 00e0-fc00-0000
lacp m-lag priority 10
#
ip vpn-instance VRF-A
ipv4-family
route-distinguisher 100:2
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface Vlanif11
ip address 10.1.7.1 255.255.255.0
vrrp vrid 1 virtual-ip 10.1.7.111
vrrp vrid 1 priority 120
mac-address 0000-5e00-0102
#
interface MEth0/0/0
ip binding vpn-instance VRF-A
ip address 10.1.6.2 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk5
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk10
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 5
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
eth-trunk 10
```



```
#
interface 10GE1/0/6
eth-trunk 10
#
interface 10GE1/0/7
undo portswitch
ip address 10.1.4.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/8
undo portswitch
ip address 10.1.2.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
ospf 1
import-route direct
area 0.0.0.0
#
return
```

- SwitchG configuration file

```
#
sysname SwitchG
#
dfs-group 1
priority 150
source ip 10.2.6.1 vpn-instance VRF-B peer 10.2.6.2
dual-active detection enhanced enable
#
vlan batch 11
#
stp bridge-address 200b-c739-1300
stp mode rstp
stp v-stp enable
stp instance 0 root primary
stp flush disable
#
lacp m-lag system-id 00e0-fc00-0001
lacp m-lag priority 10
#
ip vpn-instance VRF-B
ipv4-family
route-distinguisher 101:1
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface Vlanif11
ip address 10.1.7.3 255.255.255.0
vrrp vrid 1 virtual-ip 10.1.7.111
mac-address 0000-5e00-0103
#
interface MEth0/0/0
ip binding vpn-instance VRF-B
ip address 10.2.6.1 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk5
port link-type trunk
```

```
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk10
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 5
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
eth-trunk 10
#
interface 10GE1/0/6
eth-trunk 10
#
interface 10GE1/0/7
undo portswitch
ip address 10.2.1.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/8
undo portswitch
ip address 10.2.5.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
ospf 1
import-route direct
area 0.0.0.0
#
return
```

- SwitchH configuration file

```
#
sysname SwitchH
#
dfs-group 1
priority 120
source ip 10.2.6.2 vpn-instance VRF-B peer 10.2.6.1
dual-active detection enhanced enable
#
vlan batch 11
#
stp bridge-address 200b-c739-1300
stp mode rstp
stp v-stp enable
stp instance 0 root primary
stp flush disable
#
lacp m-lag system-id 00e0-fc00-0001
```

```
lacp m-lag priority 10
#
ip vpn-instance VRF-B
ipv4-family
 route-distinguisher 101:2
 vpn-target 111:1 export-extcommunity
 vpn-target 111:1 import-extcommunity
#
interface Vlanif11
 ip address 10.1.7.3 255.255.255.0
 vrrp vrid 1 virtual-ip 10.1.7.111
 mac-address 0000-5e00-0103
#
interface MEth0/0/0
 ip binding vpn-instance VRF-B
 ip address 10.2.6.2 255.255.255.0
#
interface Eth-Trunk0
 mode lacp-static
 peer-link 1
 port vlan exclude 1
#
interface Eth-Trunk5
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 11
 stp root-protection
 mode lacp-static
 dfs-group 1 m-lag 1
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk10
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 11
 stp root-protection
 mode lacp-static
 dfs-group 1 m-lag 2
 storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
 eth-trunk 5
#
interface 10GE1/0/3
 eth-trunk 0
#
interface 10GE1/0/5
 eth-trunk 10
#
interface 10GE1/0/6
 eth-trunk 10
#
interface 10GE1/0/7
 undo portswitch
 ip address 10.2.4.2 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/8
 undo portswitch
 ip address 10.2.2.2 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
#
interface 10GE1/0/9
 shutdown
#
interface 10GE2/0/3
 eth-trunk 0
```

```
#
ospf 1
import-route direct
area 0.0.0.0
#
return
```

- Switch1 configuration file

```
#
sysname Switch1
#
dfs-group 1
priority 150
source ip 10.1.8.1 vpn-instance VRF-C peer 10.1.8.2
dual-active detection enhanced enable
#
vlan batch 11
#
stp mode rstp
stp v-stp enable
stp flush disable
#
lACP m-lag system-id 00e0-fc00-0002
lACP m-lag priority 10
#
ip vpn-instance VRF-C
ipv4-family
route-distinguisher 102:1
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface MEth0/0/0
ip binding vpn-instance VRF-C
ip address 10.1.8.1 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk35
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk40
stp edged-port enable
stp bpdu-filter enable
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/2
eth-trunk 40
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
eth-trunk 35
#
interface 10GE1/0/6
eth-trunk 35
```

```
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
return
```

- SwitchJ configuration file

```
#
sysname SwitchJ
#
dfs-group 1
priority 120
source ip 10.1.8.2 vpn-instance VRF-C peer 10.1.8.1
dual-active detection enhanced enable
#
vlan batch 11
#
stp mode rstp
stp v-stp enable
stp flush disable
#
lACP m-lag system-id 00e0-fc00-0002
lACP m-lag priority 10
#
ip vpn-instance VRF-C
ipv4-family
route-distinguisher 102:2
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface MEth0/0/0
ip binding vpn-instance VRF-C
ip address 10.1.8.2 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk35
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk40
stp edged-port enable
stp bpdu-filter enable
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/2
eth-trunk 40
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
eth-trunk 35
#
```

```
interface 10GE1/0/6
eth-trunk 35
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
return
```

- SwitchK configuration file

```
#
sysname SwitchK
#
dfs-group 1
priority 150
source ip 10.2.8.1 vpn-instance VRF-D peer 10.2.8.2
dual-active detection enhanced enable
#
vlan batch 11
#
stp mode rstp
stp v-stp enable
stp flush disable
#
lacp m-lag system-id 00e0-fc00-0003
lacp m-lag priority 10
#
ip vpn-instance VRF-D
ipv4-family
route-distinguisher 103:1
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface MEth0/0/0
ip binding vpn-instance VRF-D
ip address 10.2.8.1 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk45
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk50
stp edged-port enable
stp bpdu-filter enable
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/2
eth-trunk 50
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
```

```
eth-trunk 45
#
interface 10GE1/0/6
eth-trunk 45
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
return
```

- SwitchL configuration file

```
#
sysname SwitchK
#
dfs-group 1
priority 120
source ip 10.2.8.2 vpn-instance VRF-D peer 10.2.8.1
dual-active detection enhanced enable
#
vlan batch 11
#
stp mode rstp
stp v-stp enable
stp flush disable
#
lacp m-lag system-id 00e0-fc00-0003
lacp m-lag priority 10
#
ip vpn-instance VRF-D
ipv4-family
route-distinguisher 103:2
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface MEth0/0/0
ip binding vpn-instance VRF-D
ip address 10.2.8.2 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk45
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk50
stp edged-port enable
stp bpdu-filter enable
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/2
eth-trunk 50
#
interface 10GE1/0/3
eth-trunk 0
```

```
#
interface 10GE1/0/5
eth-trunk 45
#
interface 10GE1/0/6
eth-trunk 45
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
return
```

- SwitchM configuration file

```
#
sysname SwitchM
#
vlan batch 11
#
stp mode rstp
stp bpdu-protection
#
interface Eth-Trunk15
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
mode lacp-static
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 15
#
interface 10GE1/0/2
eth-trunk 15
#
interface 10GE1/0/3
port default vlan 11
stp edged-port enable
#
interface 10GE1/0/9
shutdown
#
return
```

- SwitchN configuration file

```
#
sysname SwitchN
#
vlan batch 11
#
stp mode rstp
stp bpdu-protection
#
interface Eth-Trunk30
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
mode lacp-static
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 30
#
interface 10GE1/0/2
eth-trunk 30
#
interface 10GE1/0/3
port default vlan 11
```



```
stp edged-port enable
#
interface 10GE1/0/9
shutdown
#
return
```

1.17 Configuring M-LAG and VRRP/VRRP6 Across Data Centers

Applicable Products and Versions

This example applies to the CE12800 of V100R006C00 or later, the CE8800/CE7800/CE6800/CE5800 of V200R002C50 or later, the CE9800 of V200R020C00 or later, the CE12800E of V200R005C00 or later, the CloudEngine 16800 of V200R005C20 or later.

For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

[Figure 1-20](#) shows the data center network across areas, and core, aggregation, and access switches are connected through 10GE interfaces. This example uses the CE12804.

Figure 1-20 Networking for configuring M-LAG and VRRP/VRRP6 across data centers

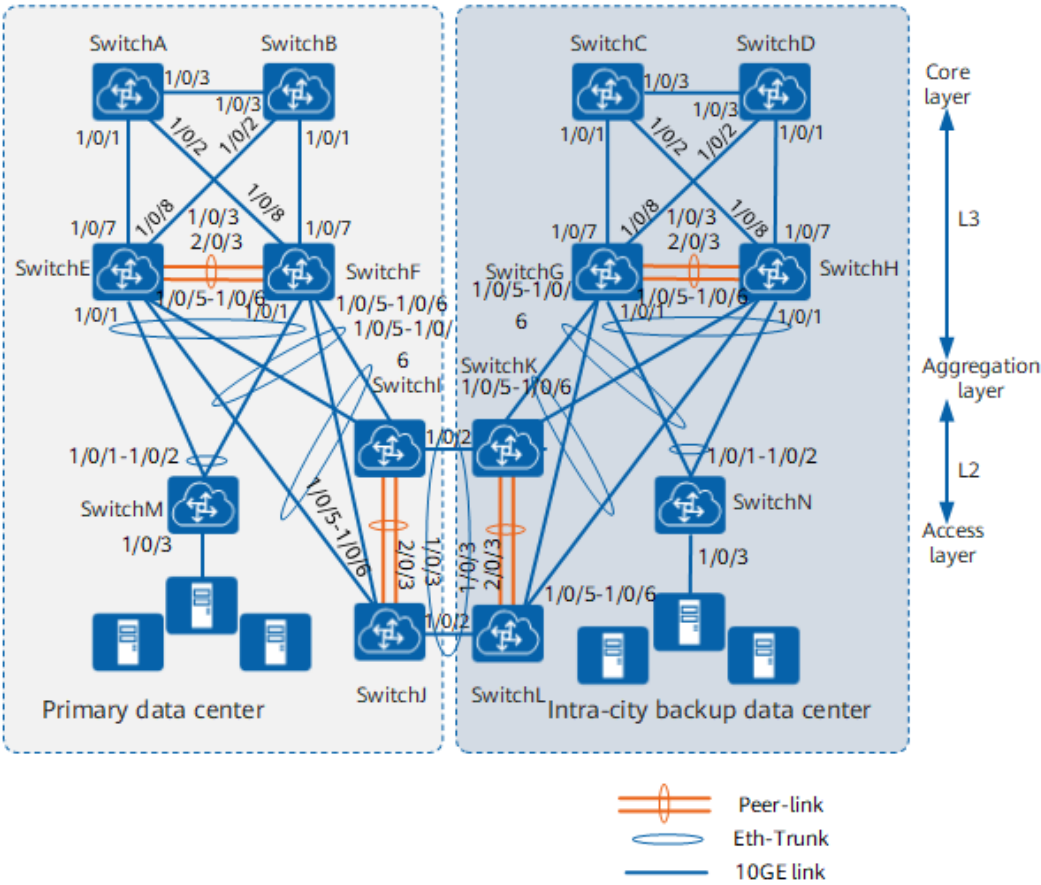


Table 1-17 Data preparation

Device Name	Interface Number	IP Address	Connected Device and Interface Number
SwitchA	10GE1/0/1	10.1.1.1/24 2001:db8:1::1/64	SwitchE: 10GE1/0/7
	10GE1/0/2	10.1.2.1/24 2001:db8:2::1/64	SwitchF: 10GE1/0/8
	10GE1/0/3	10.1.3.1/24 2001:db8:3::1/64	SwitchB: 10GE1/0/3
SwitchB	10GE1/0/1	10.1.4.1/24 2001:db8:4::1/64	SwitchF: 10GE1/0/7
	10GE1/0/2	10.1.5.1/24 2001:db8:1:5::1/64	SwitchE: 10GE1/0/8

Device Name	Interface Number	IP Address	Connected Device and Interface Number
	10GE1/0/3	10.1.3.2/24 2001:db8:1:3:2/64	SwitchA: 10GE1/0/3
SwitchC	10GE1/0/1	10.2.1.1/24 2001:db8:2:1::1/64	SwitchG: 10GE1/0/7
	10GE1/0/2	10.2.2.1/24 2001:db8:2:2::1/64	SwitchH: 10GE1/0/8
	10GE1/0/3	10.2.3.1/24 2001:db8:2:3::1/64	SwitchD: 10GE1/0/3
SwitchD	10GE1/0/1	10.2.4.1/24 2001:db8:2:4::1/64	SwitchH: 10GE1/0/7
	10GE1/0/2	10.2.5.1/24 2001:db8:2:5::1/64	SwitchG: 10GE1/0/8
	10GE1/0/3	10.2.3.2/24 2001:db8:2:3::2/64	SwitchC: 10GE1/0/3
SwitchE	Management interface	10.1.6.1/24	-
	VLANIF11	10.1.7.1/24 2001:db8:1:7::1/64 Virtual IP address: 10.1.7.111 2001:db8:1:7::111	-
	Eth-Trunk 5: 10GE1/0/1	-	SwitchM: Eth-Trunk 15
	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	- -	SwitchF: Eth-Trunk 0
	Eth-Trunk 10 • 10GE1/0/5 • 10GE1/0/6	- -	SwitchI: 10GE1/0/5 SwitchJ: 10GE1/0/6
	10GE1/0/7	10.1.1.2/24 2001:db8:1:1::2/64	SwitchA: 10GE1/0/1
	10GE1/0/8	10.1.5.2/24 2001:db8:1:5::2/64	SwitchB: 10GE1/0/2

Device Name	Interface Number	IP Address	Connected Device and Interface Number
SwitchF	Management interface	10.1.6.2/24	-
	VLANIF 11	10.1.7.1/24 2001:db8:1:7::1/64 Virtual IP address: 10.1.7.111 2001:db8:1:7::111	-
	Eth-Trunk 5: 10GE1/0/1	-	SwitchM: Eth-Trunk 15
	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	-	SwitchE: Eth-Trunk 0
		-	
	Eth-Trunk 10 • 10GE1/0/5 • 10GE1/0/6	-	SwitchJ: 10GE1/0/5
		-	SwitchI: 10GE1/0/6
	10GE1/0/7	10.1.4.2/24 2001:db8:1:4::2/64	SwitchB: 10GE1/0/1
	10GE1/0/8	10.1.2.2/24 2001:db8:1:2::2/64	SwitchA: 10GE1/0/2
SwitchG	Management interface	10.2.6.1/24	-
	VLANIF 11	10.1.7.3/24 2001:db8:1:7::3/64 Virtual IP address: 10.1.7.111 2001:db8:1:7::111	-
	Eth-Trunk 5: 10GE1/0/1	-	SwitchN: Eth-Trunk 30
	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	-	SwitchH: Eth-Trunk 0
		-	
	Eth-Trunk 10 • 10GE1/0/5 • 10GE1/0/6	-	SwitchK: 10GE1/0/5
		-	SwitchL: 10GE1/0/6

Device Name	Interface Number	IP Address	Connected Device and Interface Number
	10GE1/0/7	10.2.1.2/24 2001:db8:2:1::2/64	SwitchC: 10GE1/0/1
	10GE1/0/8	10.2.5.2/24 2001:db8:2:5::2/64	SwitchD: 10GE1/0/2
SwitchH	Management interface	10.2.6.2/24	-
	VLANIF11	10.1.7.3/24 2001:db8:1:7::3/64 Virtual IP address: 10.1.7.111 2001:db8:1:7::111	-
	Eth-Trunk 5: 10GE1/0/1	-	SwitchN: Eth-Trunk 30
	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	- -	SwitchG: Eth-Trunk 0
	Eth-Trunk 10 • 10GE1/0/5 • 10GE1/0/6	- -	SwitchL: 10GE1/0/5 SwitchK: 10GE1/0/6
	10GE1/0/7	10.2.4.2/24 2001:db8:2:4::2/64	SwitchD: 10GE1/0/1
	10GE1/0/8	10.2.2.2/24 2001:db8:2:2::2/64	SwitchC: 10GE1/0/2
	Management interface	10.1.8.1/24	-
	Eth-Trunk 40 • 10GE1/0/2	-	SwitchK: 10GE1/0/2
SwitchI	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	- -	SwitchJ: Eth-Trunk 0
	Eth-Trunk 35 • 10GE1/0/5 • 10GE1/0/6	- -	SwitchE: 10GE1/0/5 SwitchF: 10GE1/0/6

Device Name	Interface Number	IP Address	Connected Device and Interface Number
SwitchJ	Management interface	10.1.8.2/24	-
	Eth-Trunk 40 • 10GE1/0/2	-	SwitchL: 10GE1/0/2
	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	-	SwitchI: Eth-Trunk 0
		-	
	Eth-Trunk 35 • 10GE1/0/5 • 10GE1/0/6	-	SwitchF: 10GE1/0/5
		-	SwitchE: 10GE1/0/6
SwitchK	Management interface	10.2.8.1/24	-
	Eth-Trunk 50 • 10GE1/0/2	-	SwitchI: 10GE1/0/2
	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	-	SwitchL: Eth-Trunk 0
		-	
	Eth-Trunk 45 • 10GE1/0/5 • 10GE1/0/6	-	SwitchG: 10GE1/0/5
		-	SwitchH: 10GE1/0/6
SwitchL	Management interface	10.2.8.2/24	-
	Eth-Trunk 50 • 10GE1/0/2	-	SwitchJ: 10GE1/0/2
	Eth-Trunk 0 • 10GE1/0/3 • 10GE2/0/3	-	SwitchK: Eth-Trunk 0
		-	
	Eth-Trunk 45 • 10GE1/0/5 • 10GE1/0/6	-	SwitchH: 10GE1/0/5
		-	SwitchG: 10GE1/0/6
SwitchM	Eth-Trunk 15 • 10GE1/0/1 • 10GE1/0/2	-	SwitchE: 10GE1/0/1 SwitchF: 10GE1/0/1
	10GE1/0/3	-	Server

Device Name	Interface Number	IP Address	Connected Device and Interface Number
SwitchN	Eth-Trunk 30 10GE1/0/1 10GE1/0/2	-	SwitchG: 10GE1/0/1 SwitchH: 10GE1/0/1
	10GE1/0/3	-	Server

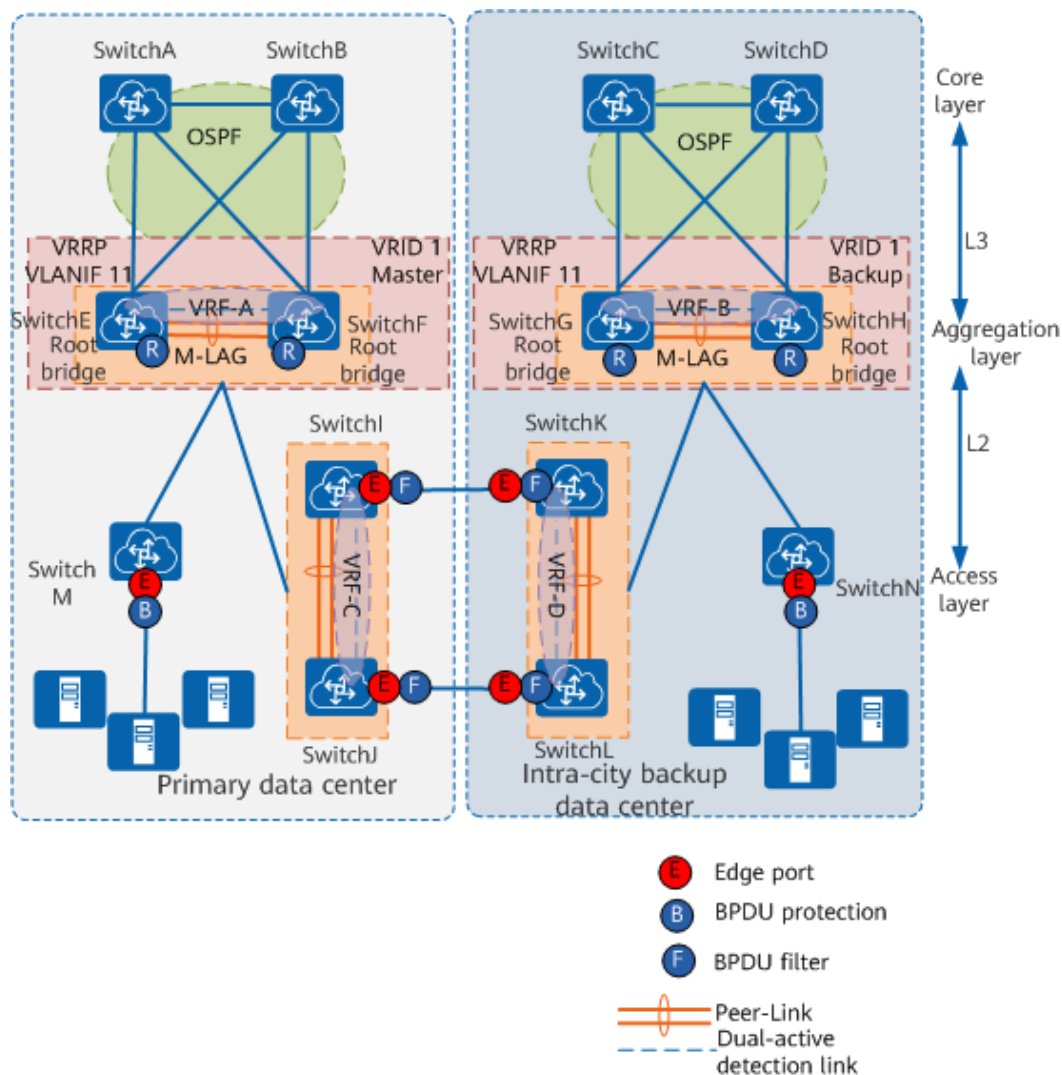
Requirement Analysis

To improve network reliability and serviceability, more and more users establish at least two data centers, and more data centers are deployed in different areas. The customer requires that the intra-city backup data center should ensure reliability. The aggregation and access layers use multi-level M-LAG interconnection to ensure reliability, improve the link usage, and extend the scale of dual-homing networking, meeting customer requirements. VRRP/VRRP6 is deployed at the aggregation layer to provide dual-active gateways.

- Devices at the core and aggregation layers establish cross connections to ensure device-level reliability and load balance traffic based on ECMP (preferential forwarding of local traffic is used with M-LAG at the aggregation layer).
- The primary and backup data centers are connected through M-LAG and form a long-distance loop-free topology.

Figure 1-21 shows the logical networking after M-LAG and VRRP/VRRP6 are deployed.

Figure 1-21 Networking for configuring M-LAG and VRRP/VRRP6 across data centers



Configuration Roadmap

The configuration roadmap is as follows:

1. Configure M-LAG between SwitchE and SwitchF and between SwitchG and SwitchH to implement dual-homing access at the access layer. Configure M-LAG between SwitchI and SwitchJ and between SwitchK and SwitchL to implement multi-level LAG interconnection between aggregation layers of primary and backup data centers. When aggregation switches work normally, links load balance traffic and a fault of any aggregation switch does not affect services. High service reliability is therefore ensured.
 - Configure SwitchE, SwitchF, SwitchG, and SwitchH as root bridges and enable root protection on downstream interfaces to ensure that the interfaces can forward traffic normally. Configure the edge interface and BPDU filter interface between the SwitchI and SwitchJ and between SwitchK and SwitchL so that STP calculation between primary and backup data centers is isolated. Configure interfaces on SwitchM and SwitchN connected to user terminals as edge interfaces to accelerate

route convergence and enable BPDU protection to enhance network stability.

- Create VLANIF interfaces on SwitchE, SwitchF, SwitchG, and SwitchH, assign IP addresses and MAC addresses to VLANIF interfaces, and create VRRP/VRRP6 groups on VLANIF interfaces to implement active-active gateways. There are dual master devices in the primary data center and two backup devices in the backup data center.

NOTE

To enable two devices of the M-LAG to be simulated into one device to perform VRRP/VRRP6 negotiation with the M-LAG of the remote data center, note the following points:

- In a data center, configure the same interface IP address and virtual MAC address when creating VRRP/VRRP6 groups. However, the primary and backup data centers must use different interface IP addresses and virtual MAC addresses.
- The configured virtual MAC address must be different from the VRRP/VRRP6 virtual MAC address, and the VRRP/VRRP6 virtual MAC address depends on the VRID. For example, the VRID of the VRRP/VRRP6 group composed of SwitchE and SwitchF is 1, so the MAC address of the VRRP/VRRP6 group is 00-00-5E-00-01-01.

2. Enable OSPF on aggregation and core switches to ensure Layer 3 connectivity.

Procedure

Step 1 Configure M-LAG.

1. Configure V-STP, heartbeat links, DFS groups, peer-links, and M-LAG member interfaces on SwitchE, SwitchF, SwitchG, SwitchH, SwitchI, SwitchJ, SwitchK, and SwitchL.

Heartbeat links are connected to management interfaces to implement interworking, DFS groups must be bound to IP addresses of management interfaces to ensure communication, and management interfaces are bound to VRF instances to implement isolation.

It is recommended that Eth-Trunk member interfaces of the peer-link be deployed on different cards to prevent the peer-link fault caused by a single-point failure.

Configure SwitchE.

Configure SwitchE as the root bridge of the STP network, and enable root protection on the Eth-Trunk of SwitchE connected to the access switch so that the Eth-Trunk can forward traffic normally.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchE
[*HUAWEI] commit
[~SwitchE] stp mode rstp
[*SwitchE] stp root primary //Configure the aggregation device as the root bridge of the STP
network.
[*SwitchE] stp bridge-address 200b-c739-1300 //Configure the MAC address of the root bridge
(MAC address of the master device).
[*SwitchE] stp v-stp enable
[*SwitchE] stp flush disable
[*SwitchE] ip vpn-instance VRF-A //Create VRF-A.
[*SwitchE-vpn-instance-VRF-A] ipv4-family
[*SwitchE-vpn-instance-VRF-A-af-ipv4] route-distinguisher 100:1
```

```
[*SwitchE-vpn-instance-VRF-A-af-ipv4] vpn-target 111:1 both
[*SwitchE-vpn-instance-VRF-A-af-ipv4] quit
[*SwitchE-vpn-instance-VRF-A] quit
[*SwitchE] interface meth 0/0/0
[*SwitchE-Meth0/0/0] ip binding vpn-instance VRF-A //Bind the management interface to VRF-A.
[*SwitchE-Meth0/0/0] ip address 10.1.6.1 24
[*SwitchE-Meth0/0/0] quit
[*SwitchE] dfs-group 1
[*SwitchE-dfs-group-1] source ip 10.1.6.1 vpn-instance VRF-A peer 10.1.6.2 //Configure the IPv4
address and VPN instance bound to the DFS group.
[*SwitchE-dfs-group-1] dual-active detection enhanced enable
[*SwitchE-dfs-group-1] priority 150
[*SwitchE-dfs-group-1] quit
[*SwitchE] interface eth-trunk 0
[*SwitchE-Eth-Trunk0] trunkport 10ge 1/0/3
[*SwitchE-Eth-Trunk0] trunkport 10ge 2/0/3 //Configure inter-card member interfaces of the Eth-
Trunk of the peer-link.
[*SwitchE-Eth-Trunk0] mode lacp-static
[*SwitchE-Eth-Trunk0] peer-link 1
[*SwitchE-Eth-Trunk0] port vlan exclude 1
[*SwitchE-Eth-Trunk0] quit
[*SwitchE] vlan batch 11
[*SwitchE] interface eth-trunk 5
[*SwitchE-Eth-Trunk5] mode lacp-static
[*SwitchE-Eth-Trunk5] port link-type trunk
[*SwitchE-Eth-Trunk5] undo port trunk allow-pass vlan 1
[*SwitchE-Eth-Trunk5] port trunk allow-pass vlan 11
[*SwitchE-Eth-Trunk5] trunkport 10ge 1/0/1
[*SwitchE-Eth-Trunk5] dfs-group 1 m-lag 1
[*SwitchE-Eth-Trunk5] stp root-protection //Enable root protection.
[*SwitchE-Eth-Trunk5] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface(The CE8800/CE7800/CE6800/CE5800 Series except CE6870EI
and CE6875EI only support GE interface view, 10GE interface view, 25GE interface view, 40GE
interface view, 100GE interface view and port group view).
[*SwitchE-Eth-Trunk5] quit
[*SwitchE] interface eth-trunk 10
[*SwitchE-Eth-Trunk10] mode lacp-static
[*SwitchE-Eth-Trunk10] port link-type trunk
[*SwitchE-Eth-Trunk10] undo port trunk allow-pass vlan 1
[*SwitchE-Eth-Trunk10] port trunk allow-pass vlan 11
[*SwitchE-Eth-Trunk10] trunkport 10ge 1/0/5 to 1/0/6
[*SwitchE-Eth-Trunk10] dfs-group 1 m-lag 2
[*SwitchE-Eth-Trunk10] stp root-protection //Enable root protection.
[*SwitchE-Eth-Trunk10] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface(The CE8800/CE7800/CE6800/CE5800 Series except CE6870EI
and CE6875EI only support GE interface view, 10GE interface view, 25GE interface view, 40GE
interface view, 100GE interface view and port group view).
[*SwitchE-Eth-Trunk10] quit
[*SwitchE] lacp m-lag priority 10
[*SwitchE] lacp m-lag system-id 00e0-fc00-0000
[*SwitchE] interface 10ge 1/0/9
[*SwitchE-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchE-10GE1/0/9] quit
[*SwitchE] commit
```

Configure SwitchF.

Configure SwitchF as the root bridge of the STP network, and enable root protection on the Eth-Trunk of SwitchF connected to the access switch so that the Eth-Trunk can forward traffic normally.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchF
[*HUAWEI] commit
[~SwitchF] stp mode rstp
[*SwitchF] stp root primary //Configure the aggregation device as the root bridge of the STP
network.
[*SwitchF] stp bridge-address 200b-c739-1300 //Configure the MAC address of the root bridge
```

```
(MAC address of the master device).
[SwitchF] stp v-stp enable
[SwitchF] stp flush disable
[SwitchF] ip vpn-instance VRF-A //Create VRF-A.
[SwitchF-vpn-instance-VRF-A] ipv4-family
[SwitchF-vpn-instance-VRF-A-af-ipv4] route-distinguisher 100:2
[SwitchF-vpn-instance-VRF-A-af-ipv4] vpn-target 111:1 both
[SwitchF-vpn-instance-VRF-A-af-ipv4] quit
[SwitchF-vpn-instance-VRF-A] quit
[SwitchF] interface meth 0/0/0
[SwitchF-Meth0/0/0] ip binding vpn-instance VRF-A //Bind the management interface to VRF-A.
[SwitchF-Meth0/0/0] ip address 10.1.6.2 24
[SwitchF-Meth0/0/0] quit
[SwitchF] dfs-group 1
[SwitchF-dfs-group-1] source ip 10.1.6.2 vpn-instance VRF-A peer 10.1.6.1 //Configure the IPv4
address and VPN instance bound to the DFS group.
[SwitchF-dfs-group-1] dual-active detection enhanced enable
[SwitchF-dfs-group-1] priority 120
[SwitchF-dfs-group-1] quit
[SwitchF] interface eth-trunk 0
[SwitchF-Eth-Trunk0] trunkport 10ge 1/0/3
[SwitchF-Eth-Trunk0] trunkport 10ge 2/0/3
[SwitchF-Eth-Trunk0] mode lacp-static
[SwitchF-Eth-Trunk0] peer-link 1
[SwitchF-Eth-Trunk0] port vlan exclude 1
[SwitchF-Eth-Trunk0] quit
[SwitchF] vlan batch 11
[SwitchF] interface eth-trunk 5
[SwitchF-Eth-Trunk5] mode lacp-static
[SwitchF-Eth-Trunk5] port link-type trunk
[SwitchF-Eth-Trunk5] undo port trunk allow-pass vlan 1
[SwitchF-Eth-Trunk5] port trunk allow-pass vlan 11
[SwitchF-Eth-Trunk5] trunkport 10ge 1/0/1
[SwitchF-Eth-Trunk5] dfs-group 1 m-lag 1
[SwitchF-Eth-Trunk5] stp root-protection //Enable root protection.
[SwitchF-Eth-Trunk5] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface(The CE8800/CE7800/CE6800/CE5800 Series except CE6870EI
and CE6875EI only support GE interface view, 10GE interface view, 25GE interface view, 40GE
interface view, 100GE interface view and port group view).
[SwitchF-Eth-Trunk5] quit
[SwitchF] interface eth-trunk 10
[SwitchF-Eth-Trunk10] mode lacp-static
[SwitchF-Eth-Trunk10] port link-type trunk
[SwitchF-Eth-Trunk10] undo port trunk allow-pass vlan 1
[SwitchF-Eth-Trunk10] port trunk allow-pass vlan 11
[SwitchF-Eth-Trunk10] trunkport 10ge 1/0/5 to 1/0/6
[SwitchF-Eth-Trunk10] dfs-group 1 m-lag 2
[SwitchF-Eth-Trunk10] stp root-protection //Enable root protection.
[SwitchF-Eth-Trunk10] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface(The CE8800/CE7800/CE6800/CE5800 Series except CE6870EI
and CE6875EI only support GE interface view, 10GE interface view, 25GE interface view, 40GE
interface view, 100GE interface view and port group view).
[SwitchF-Eth-Trunk10] quit
[SwitchF] lacp m-lag priority 10
[SwitchF] lacp m-lag system-id 00e0-fc00-0000
[SwitchF] interface 10ge 1/0/9
[SwitchF-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[SwitchF-10GE1/0/9] quit
[SwitchF] commit
```

The configurations of SwitchG and SwitchH are similar to the configurations of SwitchE and SwitchF, and the configuration details are not mentioned here.

Configure SwitchI.

Configure the Eth-Trunk on SwitchI connected to the backup data center as the edge interface and BPDU filter interface so that STP calculation between primary and backup data centers is isolated.

```
<HUAWEI> system-view
[~HUAWEI] sysname Switch1
[*HUAWEI] commit
[~Switch1] stp mode rstp
[*Switch1] stp v-stp enable
[*Switch1] stp flush disable
[*Switch1] ip vpn-instance VRF-C //Create VRF-C.
[*Switch1-vpn-instance-VRF-C] ipv4-family
[*Switch1-vpn-instance-VRF-C-af-ipv4] route-distinguisher 102:1
[*Switch1-vpn-instance-VRF-C-af-ipv4] vpn-target 111:1 both
[*Switch1-vpn-instance-VRF-C-af-ipv4] quit
[*Switch1-vpn-instance-VRF-C] quit
[*Switch1] interface meth 0/0/0
[*Switch1-Meth0/0/0] ip binding vpn-instance VRF-C //Bind the management interface to VRF-C.
[*Switch1-Meth0/0/0] ip address 10.1.8.1 24
[*Switch1-Meth0/0/0] quit
[*Switch1] dfs-group 1
[*Switch1-dfs-group-1] source ip 10.1.8.1 vpn-instance VRF-C peer 10.1.8.2 //Configure the IPv4
address and VPN instance bound to the DFS group.
[*Switch1-dfs-group-1] dual-active detection enhanced enable
[*Switch1-dfs-group-1] priority 150
[*Switch1-dfs-group-1] quit
[*Switch1] interface eth-trunk 0
[*Switch1-Eth-Trunk0] trunkport 10ge 1/0/3
[*Switch1-Eth-Trunk0] trunkport 10ge 2/0/3
[*Switch1-Eth-Trunk0] mode lacp-static
[*Switch1-Eth-Trunk0] peer-link 1
[*Switch1-Eth-Trunk0] port vlan exclude 1
[*Switch1-Eth-Trunk0] quit
[*Switch1] vlan batch 11
[*Switch1] interface eth-trunk 35
[*Switch1-Eth-Trunk35] mode lacp-static
[*Switch1-Eth-Trunk35] port link-type trunk
[*Switch1-Eth-Trunk35] undo port trunk allow-pass vlan 1
[*Switch1-Eth-Trunk35] port trunk allow-pass vlan 11
[*Switch1-Eth-Trunk35] trunkport 10ge 1/0/5 to 1/0/6
[*Switch1-Eth-Trunk35] dfs-group 1 m-lag 1
[*Switch1-Eth-Trunk35] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface(The CE8800/CE7800/CE6800/CE5800 Series except CE6870EI
and CE6875EI only support GE interface view, 10GE interface view, 25GE interface view, 40GE
interface view, 100GE interface view and port group view).
[*Switch1-Eth-Trunk35] quit
[*Switch1] interface eth-trunk 40
[*Switch1-Eth-Trunk40] mode lacp-static
[*Switch1-Eth-Trunk40] port link-type trunk
[*Switch1-Eth-Trunk40] undo port trunk allow-pass vlan 1
[*Switch1-Eth-Trunk40] port trunk allow-pass vlan 11
[*Switch1-Eth-Trunk40] trunkport 10ge 1/0/2
[*Switch1-Eth-Trunk40] dfs-group 1 m-lag 2
[*Switch1-Eth-Trunk40] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[*Switch1-Eth-Trunk40] stp bpdu-filter enable //Configure the Eth-Trunk as the BPDU filter
interface.
[*Switch1-Eth-Trunk40] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface(The CE8800/CE7800/CE6800/CE5800 Series except CE6870EI
and CE6875EI only support GE interface view, 10GE interface view, 25GE interface view, 40GE
interface view, 100GE interface view and port group view).
[*Switch1-Eth-Trunk40] quit
[*Switch1] lacp m-lag priority 10
[*Switch1] lacp m-lag system-id 00e0-fc00-0002
[*Switch1] interface 10ge 1/0/9
[*Switch1-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*Switch1-10GE1/0/9] quit
[*Switch1] commit
```

Configure Switch1.

Configure the Eth-Trunk on SwitchJ connected to the backup data center as the edge interface and BPDU filter interface so that STP calculation between primary and backup data centers is isolated.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchJ
[*HUAWEI] commit
[~SwitchJ] stp mode rstp
[*SwitchJ] stp v-stp enable
[*SwitchJ] stp flush disable
[*SwitchJ] ip vpn-instance VRF-C //Create VRF-C.
[*SwitchJ-vpn-instance-VRF-C] ipv4-family
[*SwitchJ-vpn-instance-VRF-C-af-ipv4] route-distinguisher 102:2
[*SwitchJ-vpn-instance-VRF-C-af-ipv4] vpn-target 111:1 both
[*SwitchJ-vpn-instance-VRF-C-af-ipv4] quit
[*SwitchJ-vpn-instance-VRF-C] quit
[*SwitchJ] interface meth 0/0/0
[*SwitchJ-Meth0/0/0] ip binding vpn-instance VRF-C //Bind the management interface to VRF-C.
[*SwitchJ-Meth0/0/0] ip address 10.1.8.2 24
[*SwitchJ-Meth0/0/0] quit
[*SwitchJ] dfs-group 1
[*SwitchJ-dfs-group-1] source ip 10.1.8.2 vpn-instance VRF-C peer 10.1.8.1 //Configure the IPv4
address and VPN instance bound to the DFS group.
[*SwitchJ-dfs-group-1] dual-active detection enhanced enable
[*SwitchJ-dfs-group-1] priority 120
[*SwitchJ-dfs-group-1] quit
[*SwitchJ] interface eth-trunk 0
[*SwitchJ-Eth-Trunk0] trunkport 10ge 1/0/3
[*SwitchJ-Eth-Trunk0] trunkport 10ge 2/0/3
[*SwitchJ-Eth-Trunk0] mode lacp-static
[*SwitchJ-Eth-Trunk0] peer-link 1
[*SwitchJ-Eth-Trunk0] port vlan exclude 1
[*SwitchJ-Eth-Trunk0] quit
[*SwitchJ] vlan batch 11
[*SwitchJ] interface eth-trunk 35
[*SwitchJ-Eth-Trunk35] mode lacp-static
[*SwitchJ-Eth-Trunk35] port link-type trunk
[*SwitchJ-Eth-Trunk35] undo port trunk allow-pass vlan 1
[*SwitchJ-Eth-Trunk35] port trunk allow-pass vlan 11
[*SwitchJ-Eth-Trunk35] trunkport 10ge 1/0/5 to 1/0/6
[*SwitchJ-Eth-Trunk35] dfs-group 1 m-lag 1
[*SwitchJ-Eth-Trunk35] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface(The CE8800/CE7800/CE6800/CE5800 Series except CE6870EI
and CE6875EI only support GE interface view, 10GE interface view, 25GE interface view, 40GE
interface view, 100GE interface view and port group view).
[*SwitchJ-Eth-Trunk35] quit
[*SwitchJ] interface eth-trunk 40
[*SwitchJ-Eth-Trunk40] mode lacp-static
[*SwitchJ-Eth-Trunk40] port link-type trunk
[*SwitchJ-Eth-Trunk40] undo port trunk allow-pass vlan 1
[*SwitchJ-Eth-Trunk40] port trunk allow-pass vlan 11
[*SwitchJ-Eth-Trunk40] trunkport 10ge 1/0/2
[*SwitchJ-Eth-Trunk40] dfs-group 1 m-lag 2
[*SwitchJ-Eth-Trunk40] stp edged-port enable //Configure the Eth-Trunk as the edge interface.
[*SwitchJ-Eth-Trunk40] stp bpdu-filter enable //Configure the Eth-Trunk as the BPDU filter
interface.
[*SwitchJ-Eth-Trunk40] storm suppression broadcast cir 10 mbps //Set the CIR of broadcast
packets to 10 Mbit/s on this interface(The CE8800/CE7800/CE6800/CE5800 Series except CE6870EI
and CE6875EI only support GE interface view, 10GE interface view, 25GE interface view, 40GE
interface view, 100GE interface view and port group view).
[*SwitchJ-Eth-Trunk40] quit
[*SwitchJ] lacp m-lag priority 10
[*SwitchJ] lacp m-lag system-id 00e0-fc00-0002
[*SwitchJ] interface 10ge 1/0/9
[*SwitchJ-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchJ-10GE1/0/9] quit
[*SwitchJ] commit
```

The configuration of SwitchK and SwitchL is similar to the configuration of SwitchI and SwitchJ, and the configuration details are not mentioned here.

2. Create VLANIF interfaces on SwitchE, SwitchF, SwitchG, and SwitchH and configure IP addresses for the VLANIF interfaces, and create a VRRP/VRRP6 group as the user-side gateway on VLANIF 11. There are dual master devices in the primary data center and two backup devices in the backup data center.

Configure SwitchE.

```
[~SwitchE] interface vlanif 11
[*SwitchE-Vlanif11] ip address 10.1.7.1 24
[*SwitchE-Vlanif11] mac-address 0000-5e00-0102
[*SwitchE-Vlanif11] vrrp vrid 1 virtual-ip 10.1.7.111
[*SwitchE-Vlanif11] vrrp vrid 1 priority 120
[*SwitchE-Vlanif11] ipv6 enable
[*SwitchE-Vlanif11] ipv6 address 2001:db8:1:7::1/64
[*SwitchE-Vlanif11] vrrp6 vrid 1 virtual-ip FE80:1:7::111 link-local
[*SwitchE-Vlanif11] vrrp6 vrid 1 virtual-ip 2001:db8:1:7::111
[*SwitchE-Vlanif11] vrrp6 vrid 1 priority 120
[*SwitchE-Vlanif11] ipv6 nd na glean
[*SwitchE-Vlanif11] quit
[*SwitchE] commit
```

Configure SwitchF.

```
[~SwitchF] interface vlanif 11
[*SwitchF-Vlanif11] ip address 10.1.7.1 24
[*SwitchF-Vlanif11] mac-address 0000-5e00-0102
[*SwitchF-Vlanif11] vrrp vrid 1 virtual-ip 10.1.7.111
[*SwitchF-Vlanif11] vrrp vrid 1 priority 120
[*SwitchF-Vlanif11] ipv6 enable
[*SwitchF-Vlanif11] ipv6 address 2001:db8:1:7::1/64
[*SwitchF-Vlanif11] vrrp6 vrid 1 virtual-ip FE80:1:7::111 link-local
[*SwitchF-Vlanif11] vrrp6 vrid 1 virtual-ip 2001:db8:1:7::111
[*SwitchF-Vlanif11] vrrp6 vrid 1 priority 120
[*SwitchF-Vlanif11] ipv6 nd na glean
[*SwitchF-Vlanif11] quit
[*SwitchF] commit
```

Configure SwitchG.

```
[~SwitchG] interface vlanif 11
[*SwitchG-Vlanif11] ip address 10.1.7.3 24
[*SwitchG-Vlanif11] mac-address 0000-5e00-0103
[*SwitchG-Vlanif11] vrrp vrid 1 virtual-ip 10.1.7.111
[*SwitchG-Vlanif11] ipv6 enable
[*SwitchG-Vlanif11] ipv6 address 2001:db8:1:7::3/64
[*SwitchG-Vlanif11] vrrp6 vrid 1 virtual-ip FE80:1:7::111 link-local
[*SwitchG-Vlanif11] vrrp6 vrid 1 virtual-ip 2001:db8:1:7::111
[*SwitchG-Vlanif11] ipv6 nd na glean
[*SwitchG-Vlanif11] quit
[*SwitchG] commit
```

Configure SwitchH.

```
[~SwitchH] interface vlanif 11
[*SwitchH-Vlanif11] ip address 10.1.7.3 24
[*SwitchH-Vlanif11] mac-address 0000-5e00-0103
[*SwitchH-Vlanif11] vrrp vrid 1 virtual-ip 10.1.7.111
[*SwitchH-Vlanif11] ipv6 enable
[*SwitchH-Vlanif11] ipv6 address 2001:db8:1:7::3/64
[*SwitchH-Vlanif11] vrrp6 vrid 1 virtual-ip FE80:1:7::111 link-local
[*SwitchH-Vlanif11] vrrp6 vrid 1 virtual-ip 2001:db8:1:7::111
[*SwitchH-Vlanif11] ipv6 nd na glean
[*SwitchH-Vlanif11] quit
[*SwitchH] commit
```

3. Add uplink interfaces on SwitchM and SwitchN to an Eth-Trunk and configure interfaces connected to servers as edge interfaces and BPDU filter interfaces. The interface connected to Server 1 is used as an example.

Configure SwitchM.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchM
[*HUAWEI] commit
[~SwitchM] stp mode rstp
[*SwitchM] vlan batch 11
[*SwitchM] interface eth-trunk 15
[*SwitchM-Eth-Trunk15] mode lacp-static
[*SwitchM-Eth-Trunk15] port link-type trunk
[*SwitchM-Eth-Trunk15] port trunk allow-pass vlan 11
[*SwitchM-Eth-Trunk15] trunkport 10ge 1/0/1 to 1/0/2
[*SwitchM-Eth-Trunk15] quit
[*SwitchM] interface 10ge 1/0/3
[*SwitchM-10GE1/0/3] port default vlan 11
[*SwitchM-10GE1/0/3] stp edged-port enable //Configure the interface as the edge interface.
[*SwitchM-10GE1/0/3] quit
[*SwitchM] interface 10ge 1/0/9
[*SwitchM-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an
example.
[*SwitchM-10GE1/0/9] quit
[*SwitchM] stp bpdu-protection //Enable BPDU protection.
[*SwitchM] commit
```

The configuration of SwitchN is similar to the configuration of SwitchM,
and the configuration details are not mentioned here.

- Step 2** Enable OSPF on SwitchA, SwitchB, SwitchC, SwitchD, SwitchE, SwitchF, SwitchG,
and SwitchH.

Configure SwitchA.

```
[~SwitchA] interface 10ge 1/0/1
[*SwitchA-10GE1/0/1] undo portswitch
[*SwitchA-10GE1/0/1] ip address 10.1.1.1 24
[*SwitchA-10GE1/0/1] ospf enable 1 area 0
[*SwitchA-10GE1/0/1] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchA-10GE1/0/1] quit
[*SwitchA] interface 10ge 1/0/2
[*SwitchA-10GE1/0/2] undo portswitch
[*SwitchA-10GE1/0/2] ip address 10.1.2.1 24
[*SwitchA-10GE1/0/2] ospf enable 1 area 0
[*SwitchA-10GE1/0/2] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchA-10GE1/0/2] quit
[*SwitchA] interface 10ge 1/0/3
[*SwitchA-10GE1/0/3] undo portswitch
[*SwitchA-10GE1/0/3] ip address 10.1.3.1 24
[*SwitchA-10GE1/0/3] ospf enable 1 area 0
[*SwitchA-10GE1/0/3] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchA-10GE1/0/3] quit
[*SwitchA] interface 10ge 1/0/9
[*SwitchA-10GE1/0/9] shutdown //Shut down the interface not in use. 10GE 1/0/9 is used as an example.
[*SwitchA-10GE1/0/9] quit
[*SwitchA] ospf 1
[*SwitchA-ospf-1] area 0
[*SwitchA-ospf-1-area-0.0.0.0] quit
[*SwitchA-ospf-1] quit
[*SwitchA] commit
[~SwitchA] ospfv3 1
[*SwitchA-ospfv3-1] router-id 10.1.1.1
[*SwitchA-ospfv3-1] area 0.0.0.1
[*SwitchA] interface 10ge 1/0/1
[*SwitchA-10GE1/0/1] undo portswitch
[*SwitchA-10GE1/0/1] ipv6 enable
[*SwitchA-10GE1/0/1] ipv6 address 2001:db8:1:1::1 64
```

```
[*SwitchA-10GE1/0/1] ospfv3 1 area 0.0.0.1
[*SwitchA-10GE1/0/1] quit
[*SwitchA] interface 10ge 1/0/2
[*SwitchA-10GE1/0/2] undo portswitch
[*SwitchA-10GE1/0/2] ipv6 enable
[*SwitchA-10GE1/0/2] ipv6 address 2001:db8:1:2::1 64
[*SwitchA-10GE1/0/2] ospfv3 1 area 0.0.0.1
[*SwitchA-10GE1/0/2] quit
[*SwitchA] interface 10ge 1/0/3
[*SwitchA-10GE1/0/3] undo portswitch
[*SwitchA-10GE1/0/3] ipv6 enable
[*SwitchA-10GE1/0/3] ipv6 address 2001:db8:1:3::1 64
[*SwitchA-10GE1/0/3] ospfv3 1 area 0.0.0.1
[*SwitchA-10GE1/0/3] quit
[*SwitchA] commit
```

The configurations of SwitchB, SwitchC, and SwitchD are similar to the configuration of SwitchA, and the configuration details are not mentioned here.

Configure SwitchE.

```
[~SwitchE] interface 10ge 1/0/7
[*SwitchE-10GE1/0/7] undo portswitch
[*SwitchE-10GE1/0/7] ip address 10.1.1.2 24
[*SwitchE-10GE1/0/7] ospf enable 1 area 0
[*SwitchE-10GE1/0/7] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchE-10GE1/0/7] quit
[*SwitchE] interface 10ge 1/0/8
[*SwitchE-10GE1/0/8] undo portswitch
[*SwitchE-10GE1/0/8] ip address 10.1.5.2 24
[*SwitchE-10GE1/0/8] ospf enable 1 area 0
[*SwitchE-10GE1/0/8] ospf network-type p2p //Set the network type of the interface to P2P.
[*SwitchE-10GE1/0/8] quit
[*SwitchE] ospf 1
[*SwitchE-ospf-1] import-route direct //Configure the switch to import direct routes. You can configure
a routing policy to filter unnecessary routes.
[*SwitchE-ospf-1] area 0
[*SwitchE-ospf-1-area-0.0.0.0] quit
[*SwitchE-ospf-1] quit
[*SwitchE] commit
[~SwitchE] ospfv3 1
[*SwitchE-ospfv3-1] router-id 10.1.1.2
[*SwitchE-ospfv3-1] area 0.0.0.1
[*SwitchE] interface 10ge 1/0/7
[*SwitchE-10GE1/0/7] undo portswitch
[*SwitchE-10GE1/0/7] ipv6 enable
[*SwitchE-10GE1/0/7] ipv6 address 2001:db8:1:1::2 64
[*SwitchE-10GE1/0/7] ospfv3 1 area 0.0.0.1
[*SwitchE-10GE1/0/7] quit
[*SwitchE] interface 10ge 1/0/8
[*SwitchE-10GE1/0/8] undo portswitch
[*SwitchE-10GE1/0/8] ipv6 enable
[*SwitchE-10GE1/0/8] ipv6 address 2001:db8:1:5::2 64
[*SwitchE-10GE1/0/8] ospfv3 1 area 0.0.0.1
[*SwitchE-10GE1/0/8] quit
[*SwitchE] commit
```

The configurations of SwitchF, SwitchG, and SwitchH are similar to the configuration of SwitchE, and the configuration details are not mentioned here.

Step 3 Verify the configuration.

- Run the **display dfs-group** command to check M-LAG information.
Check information about the M-LAG with DFS group 1. (The M-LAG composed of SwitchE and SwitchF is used as an example. The M-LAG composed of SwitchG and SwitchH, the M-LAG composed of SwitchI and SwitchJ, and the M-LAG composed of SwitchK and SwitchL are similar.)

```
[~SwitchE] display dfs-group 1 m-lag
*          : Local node
```



```
Heart beat state : OK
Node 1 *
  Dfs-Group ID : 1
  Priority      : 150
  Address       : ip address 10.1.6.1
  State         : Master
  Causation     : -
  System ID     : ac94-8400-df01
  SysName       : SwitchE
  Version       : V100R006C00
  Device Type   : CE12800
Node 2
  Dfs-Group ID : 1
  Priority      : 120
  Address       : ip address 10.1.6.2
  State         : Backup
  Causation     : -
  System ID     : 0025-9e95-7c21
  SysName       : switchF
  Version       : V100R006C00
  Device Type   : CE12800
```

Check M-LAG information on SwitchE.

```
[~SwitchE] display dfs-group 1 node 1 m-lag brief
* - Local node
```

M-Lag ID	Interface	Port State	Status
1	Eth-Trunk 5	Up	active(*)-active
2	Eth-Trunk 10	Up	active(*)-active

In the preceding information, the value of **Heart beat state** is **OK**, indicating that the heartbeat status is normal. SwitchE is used as Node 1, its priority is 150, and its status is **Master**. SwitchF is used as Node 2, its priority is 120, and its status is **Backup**. The value of **Causation** is -, the values of **Port State** of Node 1 and Node 2 are both **Up**, and the M-LAG status of both Node 1 and Node 2 is **active**, indicating that the MC-LAG configuration is correct.

- Run the **display vrrp** command on SwitchE, SwitchF, SwitchG and SwitchH. You can see that SwitchE and SwitchF are in Master state and SwitchG and SwitchH are in Backup state.

```
[~SwitchE] display vrrp verbose
Vlanif11 | Virtual Router 1
State      : Master
Virtual IP  : 10.1.7.111
Master IP   : 10.1.7.1
PriorityRun  : 120
PriorityConfig : 120
MasterPriority : 120
Preempt     : YES Delay Time : 0s Remain :
--
Hold Multiplier: 3
TimerRun     : 1s
TimerConfig  : 1s
Auth Type    : NONE
Virtual MAC   : 0000-5e00-0101
Check TTL    : YES
Config Type   : Normal
Create Time   : 2016-03-14 14:26:29
Last Change Time : 2016-03-14
14:26:35
[~SwitchF] display vrrp verbose
Vlanif11 | Virtual Router 1
State      : Master
Virtual IP  : 10.1.7.111
Master IP   : 10.1.7.1
PriorityRun  : 120
PriorityConfig : 120
MasterPriority : 120
```

```
Preempt      : YES  Delay Time : 0s  Remain :  
--  
Hold Multiplier: 3  
TimerRun     : 1s  
TimerConfig  : 1s  
Auth Type    : NONE  
Virtual MAC   : 0000-5e00-0101  
Check TTL    : YES  
Config Type   : Normal  
Create Time   : 2016-03-14 14:26:29  
Last Change Time : 2016-03-14  
14:26:35  
[~SwitchG] display vrrp verbose  
Vlanif11 | Virtual Router 1  
State        : Backup  
Virtual IP    : 10.1.7.111  
Master IP     : 10.1.7.3  
PriorityRun   : 100  
PriorityConfig : 100  
MasterPriority : 120  
Preempt      : YES  Delay Time : 0s  Remain : --  
Hold Multiplier: 3  
TimerRun     : 1s  
TimerConfig  : 1s  
Auth Type    : NONE  
Virtual MAC   : 0000-5e00-0101  
Check TTL    : YES  
Config Type   : Normal  
Create Time   : 2016-03-14 14:26:29  
Last Change Time : 2016-03-14 14:26:35  
[~SwitchH] display vrrp verbose  
Vlanif11 | Virtual Router 1  
State        : Backup  
Virtual IP    : 10.1.7.111  
Master IP     : 10.1.7.3  
PriorityRun   : 100  
PriorityConfig : 100  
MasterPriority : 120  
Preempt      : YES  Delay Time : 0s  Remain : --  
Hold Multiplier: 3  
TimerRun     : 1s  
TimerConfig  : 1s  
Auth Type    : NONE  
Virtual MAC   : 0000-5e00-0101  
Check TTL    : YES  
Config Type   : Normal  
Create Time   : 2016-03-14 14:26:29  
Last Change Time : 2016-03-14 14:26:35
```

- Run the **display vrrp6** command on SwitchE, SwitchF, SwitchG and SwitchH. You can see that SwitchE and SwitchF are in Master state and SwitchG and SwitchH are in Backup state.

```
[~SwitchE] display vrrp6 verbose  
Vlanif11 | Virtual Router 1  
State      : Master  
Virtual IP  :  
2001:db8:1:7::111  
Master IP   : 2001:db8:1:7::1  
PriorityRun  : 120  
PriorityConfig : 120  
MasterPriority : 120  
Preempt     : YES  Delay Time : 0s  Remain :  
--  
Hold Multiplier: 3  
TimerRun      : 1s  
TimerConfig   : 1s  
Auth Type     : NONE  
Virtual MAC    : 0000-5e00-0101  
Check TTL     : YES
```

```
Config Type : Normal
Create Time : 2022-05-14 14:26:29
Last Change Time : 2022-05-14 14:26:35
[~SwitchF] display vrrp6 verbose
Vlanif11 | Virtual Router 1
State : Master
Virtual IP :
2001:db8:1:7::111
Master IP : 2001:db8:1:7::1
PriorityRun : 120
PriorityConfig : 120
MasterPriority : 120
Preempt : YES Delay Time : 0s Remain :
--
Hold Multiplier: 3
TimerRun : 1s
TimerConfig : 1s
Auth Type : NONE
Virtual MAC : 0000-5e00-0101
Check TTL : YES
Config Type : Normal
Create Time : 2022-05-14 14:26:29
Last Change Time : 2022-05-14 14:26:35
[~SwitchG] display vrrp6 verbose
Vlanif11 | Virtual Router 1
State : Backup
Virtual IP : 2001:db8:1:7::111
Master IP : 2001:db8:1:7::3
PriorityRun : 100
PriorityConfig : 100
MasterPriority : 120
Preempt : YES Delay Time : 0s Remain : --
Hold Multiplier: 3
TimerRun : 1s
TimerConfig : 1s
Auth Type : NONE
Virtual MAC : 0000-5e00-0101
Check TTL : YES
Config Type : Normal
Create Time : 2022-05-14 14:26:29
Last Change Time : 2022-05-14 14:26:35
[~SwitchH] display vrrp6 verbose
Vlanif11 | Virtual Router 1
State : Backup
Virtual IP : 2001:db8:1:7::111
Master IP : 2001:db8:1:7::3
PriorityRun : 100
PriorityConfig : 100
MasterPriority : 120
Preempt : YES Delay Time : 0s Remain : --
Hold Multiplier: 3
TimerRun : 1s
TimerConfig : 1s
Auth Type : NONE
Virtual MAC : 0000-5e00-0101
Check TTL : YES
Config Type : Normal
Create Time : 2022-05-14 14:26:29
Last Change Time : 2022-05-14 14:26:35
```

----End

Configuration Files

- SwitchA configuration file

```
#
sysname SwitchA
#
```

```
interface 10GE1/0/1
 undo portswitch
 ip address 10.1.1.1 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
 ipv6 enable
 ipv6 address 2001:db8:1:1::1/64
 ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/2
 undo portswitch
 ip address 10.1.2.1 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
 ipv6 enable
 ipv6 address 2001:db8:1:2::1/64
 ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/3
 undo portswitch
 ip address 10.1.3.1 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
 ipv6 enable
 ipv6 address 2001:db8:1:3::1/64
 ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/9
 shutdown
#
ospf 1
 area 0.0.0.0
#
ospfv3 1
 area 0.0.0.1
#
return
```

- SwitchB configuration file

```
#
sysname SwitchB
#
interface 10GE1/0/1
 undo portswitch
 ip address 10.1.4.1 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
 ipv6 enable
 ipv6 address 2001:db8:1:4::1/64
 ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/2
 undo portswitch
 ip address 10.1.5.1 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
 ipv6 enable
 ipv6 address 2001:db8:1:5::1/64
 ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/3
 undo portswitch
 ip address 10.1.3.2 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
 ipv6 enable
 ipv6 address 2001:db8:1:3::2/64
 ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/9
```

```
shutdown
#
ospf 1
area 0.0.0.0
#
ospfv3 1
area 0.0.0.1
#
return
```

- SwitchC configuration file

```
#
sysname SwitchC
#
interface 10GE1/0/1
undo portswitch
ip address 10.2.1.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
ipv6 enable
ipv6 address 2001:db8:2:1::1/64
ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/2
undo portswitch
ip address 10.2.2.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
ipv6 enable
ipv6 address 2001:db8:2:2::1/64
ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/3
undo portswitch
ip address 10.2.3.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
ipv6 enable
ipv6 address 2001:db8:2:3::1/64
ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/9
shutdown
#
ospf 1
area 0.0.0.0
#
ospfv3 1
area 0.0.0.1
#
return
```

- SwitchD configuration file

```
#
sysname SwitchD
#
interface 10GE1/0/1
undo portswitch
ip address 10.2.4.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
ipv6 enable
ipv6 address 2001:db8:2:4::1/64
ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/2
undo portswitch
ip address 10.2.5.1 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
```

```
ipv6 enable
ipv6 address 2001:db8:2:5::1/64
ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/3
undo portswitch
ip address 10.2.3.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
ipv6 enable
ipv6 address 2001:db8:2:3::2/64
ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/9
shutdown
#
ospf 1
area 0.0.0.0
#
ospfv3 1
area 0.0.0.1
#
return
```

- SwitchE configuration file

```
#
sysname SwitchE
#
dfs-group 1
priority 150
source ip 10.1.6.1 vpn-instance VRF-A peer 10.1.6.2
dual-active detection enhanced enable
#
vlan batch 11
#
stp bridge-address 200b-c739-1300
stp mode rstp
stp v-stp enable
stp instance 0 root primary
stp flush disable
#
lacp m-lag system-id 00e0-fc00-0000
lacp m-lag priority 10
#
ip vpn-instance VRF-A
ipv4-family
route-distinguisher 100:1
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface Vlanif11
ipv6 enable
ip address 10.1.7.1 255.255.255.0
ipv6 address 2001:db8:1:7::1/64
vrrp vrid 1 virtual-ip 10.1.7.111
vrrp vrid 1 priority 120
vrrp6 vrid 1 virtual-ip FE80:1:7::111 link-local
vrrp6 vrid 1 virtual-ip 2001:db8:1:7::111
vrrp6 vrid 1 priority 120
mac-address 0000-5e00-0102
ipv6 nd na glean
#
interface MEth0/0/0
ip binding vpn-instance VRF-A
ip address 10.1.6.1 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
```

```
#
interface Eth-Trunk5
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk10
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 5
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
eth-trunk 10
#
interface 10GE1/0/6
eth-trunk 10
#
interface 10GE1/0/7
undo portswitch
ip address 10.1.1.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
ipv6 enable
ipv6 address 2001:db8:1:1::2/64
ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/8
undo portswitch
ip address 10.1.5.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
ipv6 enable
ipv6 address 2001:db8:1:5::2/64
ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
ospf 1
import-route direct
area 0.0.0.0
#
ospfv3 1
area 0.0.0.1
#
return
```

- SwitchF configuration file

```
#
sysname SwitchF
#
dfs-group 1
priority 120
```

```
source ip 10.1.6.2 vpn-instance VRF-A peer 10.1.6.1
dual-active detection enhanced enable
#
vlan batch 11
#
stp bridge-address 200b-c739-1300
stp mode rstp
stp v-stp enable
stp instance 0 root primary
stp flush disable
#
lACP m-lag system-id 00e0-fc00-0000
lACP m-lag priority 10
#
ip vpn-instance VRF-A
ipv4-family
route-distinguisher 100:2
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface Vlanif11
ipv6 enable
ip address 10.1.7.1 255.255.255.0
ipv6 address 2001:db8:1:7::1/64
vrrp vrid 1 virtual-ip 10.1.7.111
vrrp vrid 1 priority 120
vrrp6 vrid 1 virtual-ip FE80:1:7::111 link-local
vrrp6 vrid 1 virtual-ip 2001:db8:1:7::111
vrrp6 vrid 1 priority 120
mac-address 0000-5e00-0102
ipv6 nd na glean
#
interface MEth0/0/0
ip binding vpn-instance VRF-A
ip address 10.1.6.2 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk5
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk10
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 5
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
eth-trunk 10
#
interface 10GE1/0/6
```



```
eth-trunk 10
#
interface 10GE1/0/7
undo portswitch
ip address 10.1.4.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
ipv6 enable
ipv6 address 2001:db8:1:4::2/64
ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/8
undo portswitch
ip address 10.1.2.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
ipv6 enable
ipv6 address 2001:db8:1:2::2/64
ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
ospf 1
import-route direct
area 0.0.0.0
#
ospfv3 1
area 0.0.0.1
#
return
```

- SwitchG configuration file

```
#
sysname SwitchG
#
dfs-group 1
priority 150
source ip 10.2.6.1 vpn-instance VRF-B peer 10.2.6.2
dual-active detection enhanced enable
#
vlan batch 11
#
stp bridge-address 200b-c739-1300
stp mode rstp
stp v-stp enable
stp instance 0 root primary
stp flush disable
#
lACP m-lag system-id 00e0-fc00-0001
lACP m-lag priority 10
#
ip vpn-instance VRF-B
ipv4-family
route-distinguisher 101:1
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface Vlanif11
ipv6 enable
ip address 10.1.7.3 255.255.255.0
ipv6 address 2001:db8:1:7::3/64
vrrp vrid 1 virtual-ip 10.1.7.111
vrrp6 vrid virtual-ip FE80:1:7::111 link-local
vrrp6 vrid virtual-ip 2001:db8:1:7::111
mac-address 0000-5e00-0103
ipv6 nd na glean
```

```
#
interface MEth0/0/0
ip binding vpn-instance VRF-B
ip address 10.2.6.1 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk5
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk10
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 5
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
eth-trunk 10
#
interface 10GE1/0/6
eth-trunk 10
#
interface 10GE1/0/7
undo portswitch
ip address 10.2.1.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
ipv6 enable
ipv6 address 2001:db8:2:1::2/64
ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/8
undo portswitch
ip address 10.2.5.2 255.255.255.0
ospf network-type p2p
ospf enable 1 area 0.0.0.0
ipv6 enable
ipv6 address 2001:db8:2:5::2/64
ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
ospf 1
import-route direct
area 0.0.0.0
#
ospfv3 1
```

```
area 0.0.0.1
#
return
```

- SwitchH configuration file

```
#
sysname SwitchH
#
dfs-group 1
priority 120
source ip 10.2.6.2 vpn-instance VRF-B peer 10.2.6.1
dual-active detection enhanced enable
#
vlan batch 11
#
stp bridge-address 200b-c739-1300
stp mode rstp
stp v-stp enable
stp instance 0 root primary
stp flush disable
#
lACP m-lag system-id 00e0-fc00-0001
lACP m-lag priority 10
#
ip vpn-instance VRF-B
ipv4-family
route-distinguisher 101:2
vpn-target 111:1 export-extcommunity
vpn-target 111:1 import-extcommunity
#
interface Vlanif11
ipv6 enable
ip address 10.1.7.3 255.255.255.0
ipv6 address 2001:db8:1:7::3/64
vrrp vrid 1 virtual-ip 10.1.7.111
vrrp6 vrid 1 virtual-ip FE80:1:7::111 link-local
vrrp6 vrid 1 virtual-ip 2001:db8:1:7::111
mac-address 0000-5e00-0103
ipv6 nd na glean
#
interface MEth0/0/0
ip binding vpn-instance VRF-B
ip address 10.2.6.2 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk5
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk10
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 5
#
```

```
interface 10GE1/0/3
 eth-trunk 0
#
interface 10GE1/0/5
 eth-trunk 10
#
interface 10GE1/0/6
 eth-trunk 10
#
interface 10GE1/0/7
 undo portswitch
 ip address 10.2.4.2 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
 ipv6 enable
 ipv6 address 2001:db8:2:4::2/64
 ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/8
 undo portswitch
 ip address 10.2.2.2 255.255.255.0
 ospf network-type p2p
 ospf enable 1 area 0.0.0.0
 ipv6 enable
 ipv6 address 2001:db8:2:2::2/64
 ospfv3 1 area 0.0.0.1
#
interface 10GE1/0/9
 shutdown
#
interface 10GE2/0/3
 eth-trunk 0
#
ospf 1
 import-route direct
 area 0.0.0.0
#
ospfv3 1
 area 0.0.0.1
#
return
```

- SwitchI configuration file

```
#
sysname SwitchI
#
dfs-group 1
 priority 150
 source ip 10.1.8.1 vpn-instance VRF-C peer 10.1.8.2
 dual-active detection enhanced enable
#
vlan batch 11
#
stp mode rstp
 stp v-stp enable
 stp flush disable
#
lacp m-lag system-id 00e0-fc00-0002
 lacp m-lag priority 10
#
ip vpn-instance VRF-C
 ipv4-family
  route-distinguisher 102:1
  vpn-target 111:1 export-extcommunity
  vpn-target 111:1 import-extcommunity
#
interface MEth0/0/0
 ip binding vpn-instance VRF-C
 ip address 10.1.8.1 255.255.255.0
#
```

```
interface Eth-Trunk0
 mode lacp-static
 peer-link 1
 port vlan exclude 1
#
interface Eth-Trunk35
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 11
 mode lacp-static
 dfs-group 1 m-lag 1
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk40
 stp edged-port enable
 stp bpdu-filter enable
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 11
 stp root-protection
 mode lacp-static
 dfs-group 1 m-lag 2
 storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/2
 eth-trunk 40
#
interface 10GE1/0/3
 eth-trunk 0
#
interface 10GE1/0/5
 eth-trunk 35
#
interface 10GE1/0/6
 eth-trunk 35
#
interface 10GE1/0/9
 shutdown
#
interface 10GE2/0/3
 eth-trunk 0
#
return
```

- SwitchJ configuration file

```
#
sysname SwitchJ
#
dfs-group 1
 priority 120
 source ip 10.1.8.2 vpn-instance VRF-C peer 10.1.8.1
 dual-active detection enhanced enable
#
vlan batch 11
#
stp mode rstp
 stp v-stp enable
 stp flush disable
#
lacp m-lag system-id 00e0-fc00-0002
 lacp m-lag priority 10
#
ip vpn-instance VRF-C
 ipv4-family
  route-distinguisher 102:2
  vpn-target 111:1 export-extcommunity
  vpn-target 111:1 import-extcommunity
#
interface MEth0/0/0
 ip binding vpn-instance VRF-C
```

```
ip address 10.1.8.2 255.255.255.0
#
interface Eth-Trunk0
 mode lacp-static
 peer-link 1
 port vlan exclude 1
#
interface Eth-Trunk35
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 11
 mode lacp-static
 dfs-group 1 m-lag 1
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk40
 stp edged-port enable
 stp bpdu-filter enable
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 11
 stp root-protection
 mode lacp-static
 dfs-group 1 m-lag 2
 storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/2
 eth-trunk 40
#
interface 10GE1/0/3
 eth-trunk 0
#
interface 10GE1/0/5
 eth-trunk 35
#
interface 10GE1/0/6
 eth-trunk 35
#
interface 10GE1/0/9
 shutdown
#
interface 10GE2/0/3
 eth-trunk 0
#
return
```

- SwitchK configuration file

```
#
sysname SwitchK
#
dfs-group 1
 priority 150
 source ip 10.2.8.1 vpn-instance VRF-D peer 10.2.8.2
 dual-active detection enhanced enable
#
vlan batch 11
#
stp mode rstp
 stp v-stp enable
 stp flush disable
#
lacp m-lag system-id 00e0-fc00-0003
 lacp m-lag priority 10
#
ip vpn-instance VRF-D
 ipv4-family
  route-distinguisher 103:1
  vpn-target 111:1 export-extcommunity
  vpn-target 111:1 import-extcommunity
#
```

```
interface MEth0/0/0
 ip binding vpn-instance VRF-D
 ip address 10.2.8.1 255.255.255.0
#
interface Eth-Trunk0
 mode lacp-static
 peer-link 1
 port vlan exclude 1
#
interface Eth-Trunk45
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 11
 mode lacp-static
 dfs-group 1 m-lag 1
 storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk50
 stp edged-port enable
 stp bpdu-filter enable
 port link-type trunk
 undo port trunk allow-pass vlan 1
 port trunk allow-pass vlan 11
 stp root-protection
 mode lacp-static
 dfs-group 1 m-lag 2
 storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/2
 eth-trunk 50
#
interface 10GE1/0/3
 eth-trunk 0
#
interface 10GE1/0/5
 eth-trunk 45
#
interface 10GE1/0/6
 eth-trunk 45
#
interface 10GE1/0/9
 shutdown
#
interface 10GE2/0/3
 eth-trunk 0
#
return
```

- SwitchL configuration file

```
#
sysname SwitchK
#
dfs-group 1
 priority 120
 source ip 10.2.8.2 vpn-instance VRF-D peer 10.2.8.1
 dual-active detection enhanced enable
#
vlan batch 11
#
stp mode rstp
 stp v-stp enable
 stp flush disable
#
lacp m-lag system-id 00e0-fc00-0003
 lacp m-lag priority 10
#
ip vpn-instance VRF-D
 ipv4-family
  route-distinguisher 103:2
  vpn-target 111:1 export-extcommunity
```

```
vpn-target 111:1 import-extcommunity
#
interface MEth0/0/0
ip binding vpn-instance VRF-D
ip address 10.2.8.2 255.255.255.0
#
interface Eth-Trunk0
mode lacp-static
peer-link 1
port vlan exclude 1
#
interface Eth-Trunk45
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
mode lacp-static
dfs-group 1 m-lag 1
storm suppression broadcast cir 10 mbps
#
interface Eth-Trunk50
stp edged-port enable
stp bpdu-filter enable
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
stp root-protection
mode lacp-static
dfs-group 1 m-lag 2
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/2
eth-trunk 50
#
interface 10GE1/0/3
eth-trunk 0
#
interface 10GE1/0/5
eth-trunk 45
#
interface 10GE1/0/6
eth-trunk 45
#
interface 10GE1/0/9
shutdown
#
interface 10GE2/0/3
eth-trunk 0
#
return
```

- SwitchM configuration file

```
#
sysname SwitchM
#
vlan batch 11
#
stp mode rstp
stp bpdu-protection
#
interface Eth-Trunk15
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
mode lacp-static
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 15
#
interface 10GE1/0/2
```



```
eth-trunk 15
#
interface 10GE1/0/3
port default vlan 11
stp edged-port enable
#
interface 10GE1/0/9
shutdown
#
return
```

- SwitchN configuration file

```
#
sysname SwitchN
#
vlan batch 11
#
stp mode rstp
stp bpdu-protection
#
interface Eth-Trunk30
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 11
mode lacp-static
storm suppression broadcast cir 10 mbps
#
interface 10GE1/0/1
eth-trunk 30
#
interface 10GE1/0/2
eth-trunk 30
#
interface 10GE1/0/3
port default vlan 11
stp edged-port enable
#
interface 10GE1/0/9
shutdown
#
return
```

1.18 Example for Configuring Dynamic Routing Protocols (OSPF/OSPFv3/BGP/BGP4+) for Communication with an M-LAG

Applicable Products and Versions

This example applies to CE16800, CE6881, CE6820, CE6863, CE6863E, CE5881, CE6881K, CE6881E, and CE6863K series switches running V200R020C00 or later versions.

This example applies to CE12800, CE8850EI, CE8861EI, CE8868EI, CE6856HI, CE6857EI, CE6857E, CE6857S, CE6857F, CE6865EI, CE6865E, CE8850E-32CQ-EI, and CE6870EI series switches running V200R021C00 or later versions.

For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

In [Figure 1-22](#), SwitchB and SwitchC establish an M-LAG, and M-LAG member interfaces on SwitchB and SwitchC support dynamic routing protocols. You can configure dynamic routing protocols on SwitchA so that SwitchA can communicate with the M-LAG through Layer 3 routes.

Figure 1-22 Network diagram for configuring dynamic routing protocols for communication with an M-LAG

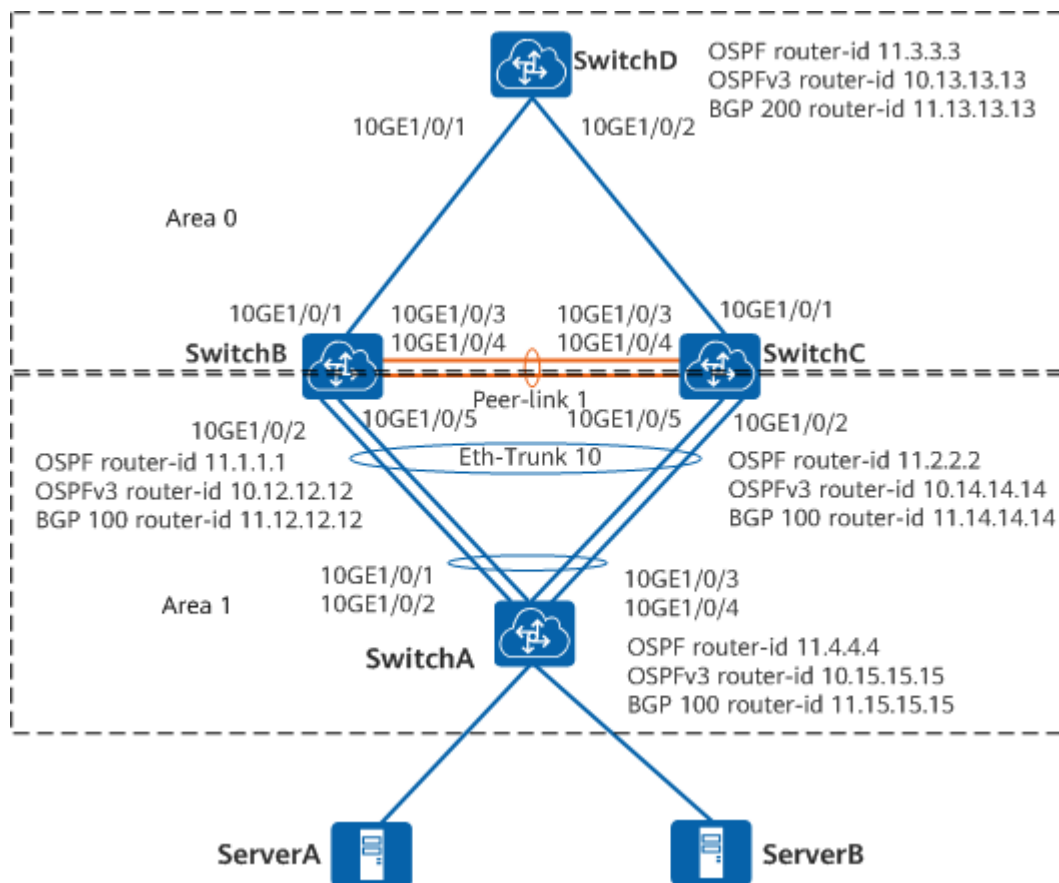


Table 1-18 Data plan

Device Name	Interface Number	VLAN and IP Address	Connected Device and Interface Number
SwitchA	Eth-Trunk 10 10GE1/0/1 10GE1/0/2 10GE1/0/3 10GE1/0/4	VLAN 100: IPv4 address 10.100.0.2/24 VLAN 101: IPv6 address 2001:DB8:101::2/64 VLAN 102: IPv4 address 10.102.0.2/24 VLAN 103: IPv6 address 2001:DB8:103::2/64	SwitchB: Eth-Trunk 10 10GE1/0/2 10GE1/0/3
			SwitchC: Eth-Trunk 10 10GE1/0/2 10GE1/0/3
SwitchB	Management interface	10.200.1.1/24	-
	10GE1/0/1	IPv4 address: 192.168.1.1/24 IPv6 address: 2001:DB8:1::1/64	SwitchD: 10GE1/0/1
	Eth-Trunk 1 10GE1/0/3 10GE1/0/4	-	SwitchC: Eth-Trunk 1 10GE1/0/3 10GE1/0/4
	Eth-Trunk 10 10GE1/0/2 10GE1/0/5	VLAN 100: IPv4 address 10.100.0.1/24 M-LAG IPv4 address 10.100.0.3/24 VLAN 101: IPv6 address 2001:DB8:101::1/64 M-LAG link-local address FE80:1000::1 VLAN 102: IPv4 address 10.102.0.1/24 M-LAG IPv4 address 10.102.0.3/24 VLAN 103: IPv6 address 2001:DB8:103::1/64 M-LAG IPv6 address 2001:DB8:103::3/64	SwitchA: Eth-Trunk 10 10GE1/0/1 10GE1/0/2
	Loopback 1	10.2.2.2/32	-
	Loopback 2	10.3.3.3/32	-

Device Name	Interface Number	VLAN and IP Address	Connected Device and Interface Number
SwitchC	Management interface	10.200.2.1/24	-
	10GE1/0/1	192.168.2.1/24 2001:DB8:2::1/64	SwitchD: 10GE1/0/2
	Eth-Trunk 1 • 10GE1/0/3 • 10GE1/0/4	-	SwitchB: Eth-Trunk 1 • 10GE1/0/3 • 10GE1/0/4
	Eth-Trunk 10 • 10GE1/0/2 • 10GE1/0/5	VLAN 100: IPv4 address 10.100.0.1/24 M-LAG IPv4 address 10.100.0.4/24 VLAN 101: IPv6 address 2001:DB8:101::1/64 M-LAG link-local address FE80:1000::2 VLAN 102: IPv4 address 10.102.0.1/24 M-LAG IPv4 address 10.102.0.4/24 VLAN 103: IPv6 address 2001:DB8:103::1/64 M-LAG IPv6 address 2001:DB8:103::4/64	SwitchA: Eth-Trunk 10 • 10GE1/0/3 • 10GE1/0/4
	Loopback 1	10.2.2.2/32	-
	Loopback 2	10.4.4.4/32	-
	Loopback 3	10.8.8.8/32	-
SwitchD	10GE1/0/1	192.168.1.2/24 2001:DB8:1::2/64	SwitchB: 10GE1/0/1
	10GE1/0/2	192.168.2.2/24 2001:DB8:2::2/64	SwitchC: 10GE1/0/1
	Loopback 1	10.1.1.1/32	-

Configuration Roadmap

The configuration roadmap is as follows:

1. Configure SwitchA, SwitchB, and SwitchC to establish an M-LAG.
 - Configure V-STP on SwitchB and SwitchC.
 - Configure a DFS group on SwitchB and SwitchC, and bind IP addresses of their management interfaces to the DFS group.
 - Configure a peer-link between SwitchB and SwitchC.
 - Configure M-LAG member interfaces on SwitchB and SwitchC and an Eth-Trunk interface on SwitchA.
 - Configure Monitor Link on SwitchB and SwitchC.

NOTE

Monitor Link associates uplink and downlink interfaces to prevent user-side traffic from being discarded due to an uplink fault.

2. Configure OSPF over M-LAG.
 - Configure OSPF on SwitchB, SwitchC, and SwitchD to implement Layer 3 connectivity.
 - Configure IPv4 addresses for OSPF over M-LAG on SwitchB and SwitchC.

NOTE

Configure different M-LAG IPv4 addresses on SwitchB and SwitchC. Otherwise, neighbor relationships of the routing protocol cannot be established.

- Configure an IP address for SwitchA to establish OSPF neighbor relationships with M-LAG member devices.
3. Configure OSPFv3 over M-LAG.
 - Configure OSPFv3 on SwitchB, SwitchC, and SwitchD to implement Layer 3 connectivity.
 - Configure link-local addresses for OSPFv3 over M-LAG on SwitchB and SwitchC.

NOTE

Configure different M-LAG link-local addresses on SwitchB and SwitchC. Otherwise, neighbor relationships of the routing protocol cannot be established.

- Configure an IPv6 address for SwitchA to establish OSPFv3 neighbor relationships with M-LAG member devices.
4. Configure BGP over M-LAG.
 - Configure BGP on SwitchB, SwitchC, and SwitchD to implement Layer 3 connectivity.
 - Configure IPv4 addresses for BGP over M-LAG on SwitchB and SwitchC.

NOTE

Configure different M-LAG IPv4 addresses on SwitchB and SwitchC. Otherwise, peer relationships of the routing protocol cannot be established.

- Configure SwitchA to establish BGP peer relationships with M-LAG member devices (SwitchB and SwitchC).

5. Configure BGP4+ over M-LAG.
 - Configure BGP4+ on SwitchB, SwitchC, and SwitchD to implement Layer 3 connectivity.
 - Configure IPv6 addresses for BGP4+ over M-LAG on SwitchB and SwitchC.

 NOTE

Configure different M-LAG IPv6 addresses on SwitchB and SwitchC. Otherwise, peer relationships of the routing protocol cannot be established.

- Configure SwitchA to establish BGP4+ peer relationships with M-LAG member devices (SwitchB and SwitchC) and import routes.

Procedure

Step 1 Configure SwitchA, SwitchB, and SwitchC to establish an M-LAG.

1. Configure V-STP on SwitchB and SwitchC.

Configure SwitchB. The configuration of SwitchC is similar to that of SwitchB.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchB
[*HUAWEI] commit
[~SwitchB] stp mode rstp
[SwitchB] stp v-stp enable
[*SwitchB] commit
```

2. Configure a DFS group on SwitchB and SwitchC, and bind IP addresses of their management interfaces to the DFS group.

Ensure that SwitchB and SwitchC can communicate at Layer 3 through their management interfaces.

Configure SwitchB. The configuration of SwitchC is similar to that of SwitchB.

```
[~SwitchB] dfs-group 1
[SwitchB-dfs-group-1] source ip 10.200.1.1 peer 10.200.2.1
[SwitchB-dfs-group-1] dual-active detection enhanced enable
[SwitchB-dfs-group-1] quit
[*SwitchB] commit
```

3. Configure a peer-link between SwitchB and SwitchC.

Configure SwitchB. The configuration of SwitchC is similar to that of SwitchB.

```
[~SwitchB] interface eth-trunk 1
[SwitchB-Eth-Trunk1] mode lacp-static
[SwitchB-Eth-Trunk1] trunkport 10ge 1/0/3
[SwitchB-Eth-Trunk1] trunkport 10ge 1/0/4
[SwitchB-Eth-Trunk1] peer-link 1
[SwitchB-Eth-Trunk1] quit
[*SwitchB] commit
```

4. Configure M-LAG member interfaces on SwitchB and SwitchC and an Eth-Trunk interface on SwitchA.

Configure SwitchB. The configuration of SwitchC is similar to that of SwitchB.

```
[~SwitchB] interface eth-trunk 10
[SwitchB-Eth-Trunk10] mode lacp-static
```

```
[*SwitchB-Eth-Trunk10] port link-type trunk
[*SwitchB-Eth-Trunk10] lacp mixed-rate link enable
[*SwitchB-Eth-Trunk10] trunkport 10ge 1/0/2
[*SwitchB-Eth-Trunk10] trunkport 10ge 1/0/5
[*SwitchB-Eth-Trunk10] dfs-group 1 m-lag 1
[*SwitchB-Eth-Trunk10] quit
[*SwitchB] commit
```

Configure SwitchA.

```
<HUAWEI> system-view
[~HUAWEI] sysname SwitchA
[*HUAWEI] commit
[~SwitchA] interface eth-trunk 10
[*SwitchA-Eth-Trunk10] mode lacp-static
[*SwitchA-Eth-Trunk10] port link-type trunk
[*SwitchA-Eth-Trunk10] trunkport 10ge 1/0/1 to 1/0/4
[*SwitchA-Eth-Trunk10] quit
[*SwitchA] commit
```

5. Configure Monitor Link on SwitchB and SwitchC.

Configure SwitchB. The configuration of SwitchC is similar to that of SwitchB.

```
[~SwitchB] monitor-link group 1
[*SwitchB-mtlk-group1] port 10ge 1/0/1 uplink
[*SwitchB-mtlk-group1] port eth-trunk 10 downlink 1
[*SwitchB-mtlk-group1] quit
[*SwitchB] commit
```

Step 2 Configure OSPF over M-LAG.

1. Configure OSPF on SwitchB, SwitchC, and SwitchD to implement Layer 3 connectivity.

Configure SwitchB. The configurations of SwitchC and SwitchD are similar to the configuration of SwitchB. When configuring OSPF, configure the switches to advertise the 32-bit IP addresses of loopback interfaces.

```
[~SwitchB] interface loopback 1
[*SwitchB-LoopBack1] ip address 10.2.2.2 32
[*SwitchB-LoopBack1] quit
[*SwitchB] interface loopback 2
[*SwitchB-LoopBack2] ip address 10.3.3.3 32
[*SwitchB-LoopBack2] quit
[*SwitchB] interface 10ge 1/0/1
[*SwitchB-10GE1/0/1] undo portswitch
[*SwitchB-10GE1/0/1] ip address 192.168.1.1 24
[*SwitchB-10GE1/0/1] quit
[*SwitchB] ospf 1 router-id 11.1.1.1
[*SwitchB-ospf-1] area 0
[*SwitchB-ospf-1-area-0.0.0.0] network 10.2.2.2 0.0.0.0
[*SwitchB-ospf-1-area-0.0.0.0] network 10.3.3.3 0.0.0.0
[*SwitchB-ospf-1-area-0.0.0.0] network 192.168.1.0 0.0.0.255
[*SwitchB-ospf-1-area-0.0.0.0] quit
[*SwitchB-ospf-1] quit
[*SwitchB] commit
```

After OSPF is configured successfully, SwitchB, SwitchC, and SwitchD can learn the IP addresses of each other's loopback interface through OSPF and successfully ping each other.

2. Configure IPv4 addresses for OSPF over M-LAG on SwitchB and SwitchC.

Configure SwitchB.

```
[~SwitchB] vlan 100
[*SwitchB-vlan100] quit
```

```
[*SwitchB] interface vlanif 100
[*SwitchB-Vlanif100] ip address 10.100.0.1 255.255.255.0
[*SwitchB-Vlanif100] m-lag ip address 10.100.0.3 255.255.255.0
[*SwitchB-Vlanif100] ospf source sub-address 10.100.0.3
[*SwitchB-Vlanif100] mac-address 0000-5e00-0101
[*SwitchB-Vlanif100] quit
[*SwitchB] commit
[~SwitchB] interface eth-trunk 10
[*SwitchB-Eth-Trunk10] port trunk allow-pass vlan 100
[*SwitchB-Eth-Trunk10] quit
[*SwitchB] commit
[~SwitchB] ospf
[~SwitchB-ospf-1] area 1
[~SwitchB-ospf-1-area-0.0.0.1] network 10.100.0.0 0.0.0.255
[*SwitchB-ospf-1-area-0.0.0.1] quit
[*SwitchB-ospf-1] quit
[*SwitchB] commit
```

Configure SwitchC.

```
[~SwitchC] vlan 100
[*SwitchC-vlan100] quit
[*SwitchC] interface vlanif 100
[*SwitchC-Vlanif100] ip address 10.100.0.1 255.255.255.0
[*SwitchC-Vlanif100] m-lag ip address 10.100.0.4 255.255.255.0
[*SwitchC-Vlanif100] ospf source sub-address 10.100.0.4
[*SwitchC-Vlanif100] mac-address 0000-5e00-0101
[*SwitchC-Vlanif100] quit
[*SwitchC] commit
[~SwitchC] interface eth-trunk 10
[*SwitchC-Eth-Trunk10] port trunk allow-pass vlan 100
[*SwitchC-Eth-Trunk10] quit
[~SwitchC] ospf
[~SwitchC-ospf-1] area 1
[~SwitchC-ospf-1-area-0.0.0.1] network 10.100.0.0 0.0.0.255
[*SwitchC-ospf-1-area-0.0.0.1] quit
[*SwitchC-ospf-1] quit
[*SwitchC] commit
```

3. Configure an IP address for SwitchA to establish OSPF neighbor relationships with M-LAG member devices.

Configure SwitchA.

```
[~SwitchA] vlan 100
[*SwitchA-vlan100] quit
[*SwitchA] interface vlanif 100
[*SwitchA-Vlanif100] ip address 10.100.0.2 255.255.255.0
[*SwitchA-Vlanif100] quit
[*SwitchA] commit
[~SwitchA] interface eth-trunk 10
[*SwitchA-Eth-Trunk10] port trunk allow-pass vlan 100
[*SwitchA-Eth-Trunk10] quit
[*SwitchA] ospf 1 router-id 11.4.4.4
[*SwitchA-ospf-1] area 1
[*SwitchA-ospf-1-area-0.0.0.1] network 10.100.0.0 0.0.0.255
[*SwitchA-ospf-1-area-0.0.0.1] quit
[*SwitchA-ospf-1] quit
[*SwitchA] commit
```

Step 3 Configure OSPFv3 over M-LAG.

1. Configure OSPFv3 on SwitchB, SwitchC, and SwitchD to implement Layer 3 connectivity.

Configure SwitchB. The configurations of SwitchC and SwitchD are similar to the configuration of SwitchB.

```
[~SwitchB] interface 10ge 1/0/1
[*SwitchB-10GE1/0/1] ipv6 enable
```



```
[*SwitchB-10GE1/0/1] ipv6 address 2001:db8:1::1/64
[*SwitchB-10GE1/0/1] quit
[*SwitchB] ospfv3 1
[*SwitchB-ospfv3-1] router-id 10.12.12.12
[*SwitchB-ospfv3-1] area 0
[*SwitchB-ospfv3-1-area-0.0.0.0] quit
[*SwitchB-ospfv3-1] quit
[*SwitchB] commit
[~SwitchB] interface 10ge 1/0/1
[*SwitchB-10GE1/0/1] ospfv3 1 area 0
[*SwitchB-10GE1/0/1] quit
[*SwitchB] commit
```

After OSPFv3 is configured successfully, SwitchB, SwitchC, and SwitchD can establish OSPFv3 neighbor relationships and the neighbor status is Full.

```
[~SwitchD] display ospfv3 peer
OSPFv3 Process (1)
Total number of peer(s): 2
Peer(s) in full state: 2
OSPFv3 Area (0.0.0.0)
Neighbor ID    Pri State          Dead Time  Interface    Instance ID
10.12.12.12    1 Full/Backup      00:00:37  10GE1/0/1    0
10.14.14.14    1 Full/DROther     00:00:38  10GE1/0/2    0
```

2. Configure link-local addresses for OSPFv3 over M-LAG on SwitchB and SwitchC.

Configure SwitchB.

```
[~SwitchB] vlan 101
[*SwitchB-vlan101] quit
[*SwitchB] interface vlanif 101
[*SwitchB-Vlanif101] ipv6 enable
[*SwitchB-Vlanif101] ipv6 address 2001:db8:101::1/64
[*SwitchB-Vlanif101] ospfv3 1 area 0.0.0.1
[*SwitchB-Vlanif101] mac-address 0000-5e00-0102
[*SwitchB-Vlanif101] m-lag ipv6 address FE80:1000::1 link-local
[*SwitchB-Vlanif101] quit
[*SwitchB] commit
[~SwitchB] interface eth-trunk 10
[*SwitchB-Eth-Trunk10] port trunk allow-pass vlan 101
[*SwitchB-Eth-Trunk10] quit
[*SwitchB] commit
```

Configure SwitchC.

```
[~SwitchC] vlan 101
[*SwitchC-vlan101] quit
[*SwitchC] interface vlanif 101
[*SwitchC-Vlanif101] ipv6 enable
[*SwitchC-Vlanif101] ipv6 address 2001:db8:101::1/64
[*SwitchC-Vlanif101] ospfv3 1 area 0.0.0.1
[*SwitchC-Vlanif101] mac-address 0000-5e00-0102
[*SwitchC-Vlanif101] m-lag ipv6 address FE80:1000::2 link-local
[*SwitchC-Vlanif101] quit
[*SwitchC] commit
[~SwitchC] interface eth-trunk 10
[*SwitchC-Eth-Trunk10] port trunk allow-pass vlan 101
[*SwitchC-Eth-Trunk10] quit
[*SwitchC] commit
```

3. Configure an IP address for SwitchA to establish OSPFv3 neighbor relationships with M-LAG member devices.

Configure SwitchA.

```
[~SwitchA] ospfv3 1
[*SwitchA-ospfv3-1] router-id 10.15.15.15
[*SwitchA-ospfv3-1] area 1
```

```
[*SwitchA-ospfv3-1-area-0.0.0.1] quit
[*SwitchA-ospfv3-1] quit
[*SwitchA] commit
[~SwitchA] vlan 101
[*SwitchA-vlan101] quit
[*SwitchA] interface vlanif 101
[*SwitchA-Vlanif101] ipv6 enable
[*SwitchA-Vlanif101] ipv6 address 2001:db8:101::2/64
[*SwitchA-Vlanif101] ospfv3 1 area 0.0.0.1
[*SwitchA-Vlanif101] quit
[*SwitchA] commit
[~SwitchA] interface eth-trunk 10
[*SwitchA-Eth-Trunk10] port trunk allow-pass vlan 101
[*SwitchA-Eth-Trunk10] quit
[*SwitchA] commit
```

Step 4 Configure BGP over M-LAG.

1. Configure BGP on SwitchB, SwitchC, and SwitchD to implement Layer 3 connectivity.

Configure a BGP peer for SwitchB.

```
[~SwitchB] bgp 100
[*SwitchB-bgp] router-id 11.12.12.12
[*SwitchB-bgp] peer 192.168.1.2 as-number 200
[*SwitchB-bgp] quit
[*SwitchB] commit
```

Configure a BGP peer for SwitchC.

```
[~SwitchC] bgp 100
[*SwitchC-bgp] router-id 11.14.14.14
[*SwitchC-bgp] peer 192.168.2.2 as-number 200
[*SwitchC-bgp] quit
[*SwitchC] commit
```

Configure BGP peers for SwitchD.

```
[~SwitchD] bgp 200
[*SwitchD-bgp] router-id 11.13.13.13
[*SwitchD-bgp] peer 192.168.1.1 as-number 100
[*SwitchD-bgp] peer 192.168.2.1 as-number 100
[*SwitchD-bgp] quit
[*SwitchD] commit
```

After BGP is configured successfully, SwitchB, SwitchC, and SwitchD can establish BGP peer relationships and the peer status is Established.

```
[~SwitchD] display bgp peer
Status codes: * - Dynamic
BGP local router ID : 11.13.13.13
Local AS number : 200
Total number of peers : 2
Peers in established state : 2
Total number of dynamic peers : 0
```

Peer	V	AS	MsgRcvd	MsgSent	OutQ	Up/Down	State	PrefRcv
192.168.1.1	4	100	3	3	0	00:00:45	Established	0
192.168.2.1	4	100	7	7	0	00:03:52	Established	0

2. Configure IPv4 addresses for BGP over M-LAG on SwitchB, SwitchC, and SwitchA.

Configure SwitchB.

```
[~SwitchB] vlan 102
[*SwitchB-vlan102] quit
[*SwitchB] interface vlanif 102
[*SwitchB-Vlanif102] ip address 10.102.0.1 255.255.255.0
```

```
[*SwitchB-Vlanif102] m-lag ip address 10.102.0.3 255.255.255.0
[*SwitchB-Vlanif102] mac-address 0000-5e00-0103
[*SwitchB-Vlanif102] quit
[*SwitchB] commit
[~SwitchB] interface eth-trunk 10
[*SwitchB-Eth-Trunk10] port trunk allow-pass vlan 102
[*SwitchB-Eth-Trunk10] quit
[*SwitchB] commit
```

Configure SwitchC.

```
[~SwitchC] vlan 102
[*SwitchC-vlan102] quit
[*SwitchC] interface vlanif 102
[*SwitchC-Vlanif102] ip address 10.102.0.1 255.255.255.0
[*SwitchC-Vlanif102] m-lag ip address 10.102.0.4 255.255.255.0
[*SwitchC-Vlanif102] mac-address 0000-5e00-0103
[*SwitchC-Vlanif102] quit
[*SwitchC] commit
[~SwitchC] interface eth-trunk 10
[*SwitchC-Eth-Trunk10] port trunk allow-pass vlan 102
[*SwitchC-Eth-Trunk10] quit
[*SwitchC] commit
```

Configure SwitchA.

```
[~SwitchA] vlan 102
[*SwitchA-vlan102] quit
[*SwitchA] interface vlanif 102
[*SwitchA-Vlanif102] ip address 10.102.0.2 255.255.255.0
[*SwitchA-Vlanif102] quit
[*SwitchA] commit
[~SwitchA] interface eth-trunk 10
[*SwitchA-Eth-Trunk10] port trunk allow-pass vlan 102
[*SwitchA-Eth-Trunk10] quit
[*SwitchA] commit
```

3. Configure SwitchA to establish BGP peer relationships with M-LAG member devices (SwitchB and SwitchC).

Configure SwitchA.

```
[~SwitchA] bgp 100
[*SwitchA-bgp] router-id 11.15.15.15
[*SwitchA-bgp] peer 10.102.0.3 as-number 100
[*SwitchA-bgp] peer 10.102.0.4 as-number 100
```

Configure SwitchB.

```
[~SwitchB] bgp 100
[~SwitchB-bgp] peer 10.102.0.2 as-number 100
[*SwitchB-bgp] peer 10.102.0.2 connect-interface Vlanif102 10.102.0.3
```

Configure SwitchC.

```
[~SwitchC] bgp 100
[~SwitchC-bgp] peer 10.102.0.2 as-number 100
[*SwitchC-bgp] peer 10.102.0.2 connect-interface Vlanif102 10.102.0.4
```

Step 5 Configure BGP4+ over M-LAG.

1. Configure BGP4+ on SwitchB, SwitchC, and SwitchD to implement Layer 3 connectivity.

Configure a BGP4+ peer for SwitchB and import routes.

```
[~SwitchB] bgp 100
[~SwitchB-bgp] peer 2001:db8:1::2 as-number 200
[*SwitchB-bgp] ipv6-family unicast
[*SwitchB-bgp-af-ipv6] peer 2001:db8:1::2 enable
```

```
Warning: This operation will reset the peer session. Continue? [Y/N]:y
[*SwitchB-bgp-af-ipv6] network 2001:db8:1::2 64
[*SwitchB-bgp-af-ipv6] quit
[*SwitchB-bgp] quit
[*SwitchB] commit
```

Configure a BGP4+ peer for SwitchC and import routes.

```
[~SwitchC] bgp 100
[~SwitchC-bgp] peer 2001:db8:2::2 as-number 200
[*SwitchC-bgp] ipv6-family unicast
[*SwitchC-bgp-af-ipv6] peer 2001:db8:2::2 enable
Warning: This operation will reset the peer session. Continue? [Y/N]:y
[*SwitchC-bgp-af-ipv6] network 2001:db8:2::2 64
[*SwitchC-bgp-af-ipv6] quit
[*SwitchC-bgp] quit
[*SwitchC] commit
```

Configure BGP4+ peers for SwitchD and import routes.

```
[~SwitchD] bgp 200
[~SwitchD-bgp] peer 2001:db8:1::1 as-number 100
[~SwitchD-bgp] peer 2001:db8:2::1 as-number 100
[*SwitchD-bgp] ipv6-family unicast
[*SwitchD-bgp-af-ipv6] peer 2001:db8:1::1 enable
Warning: This operation will reset the peer session. Continue? [Y/N]:y
[*SwitchD-bgp-af-ipv6] peer 2001:db8:2::1 enable
Warning: This operation will reset the peer session. Continue? [Y/N]:y
[*SwitchD-bgp-af-ipv6] network 2001:db8:1::1 64
[*SwitchD-bgp-af-ipv6] network 2001:db8:2::1 64
[*SwitchD-bgp-af-ipv6] quit
[*SwitchD-bgp] quit
[*SwitchD] commit
```

After BGP4+ is configured successfully, SwitchB, SwitchC, and SwitchD can establish BGP4+ peer relationships and the peer status is Established.

```
[~SwitchD] display bgp ipv6 peer
Status codes: * - Dynamic
BGP local router ID : 11.13.13.13
Local AS number : 200
Total number of peers : 2
Peers in established state
Total number of dynamic peers : 0
Peer V AS MsgRcvd MsgSent OutQ Up/Down State
PrefRcv
2001:db8:1::1 4 100 4 4 0 00:00:19 Established 1
2001:db8:2::1 4 100 4 4 0 00:00:18 Established 1
```

2. Configure IPv6 addresses for BGP4+ over M-LAG on SwitchB and SwitchC.

Configure SwitchB.

```
[~SwitchB] vlan 103
[*SwitchB-vlan103] quit
[*SwitchB] interface vlanif 103
[*SwitchB-Vlanif103] ipv6 enable
[*SwitchB-Vlanif103] ipv6 address 2001:db8:103::1/64
[*SwitchB-Vlanif103] mac-address 0000-5e00-0104
[*SwitchB-Vlanif103] m-lag ipv6 address 2001:db8:103::3
[*SwitchB-Vlanif103] quit
[*SwitchB] commit
[~SwitchB] interface eth-trunk 10
[*SwitchB-Eth-Trunk10] port trunk allow-pass vlan 103
[*SwitchB-Eth-Trunk10] quit
[*SwitchB] commit
```

Configure SwitchC.

```
[~SwitchC] vlan 103
[*SwitchC-vlan103] quit
```

```
[*SwitchC] interface vlanif 103
[*SwitchC-Vlanif103] ipv6 enable
[*SwitchC-Vlanif103] ipv6 address 2001:db8:103::1/64
[*SwitchC-Vlanif103] mac-address 0000-5e00-0104
[*SwitchC-Vlanif103] m-lag ipv6 address 2001:db8:103::4
[*SwitchC-Vlanif103] quit
[*SwitchC] commit
[~SwitchC] interface eth-trunk 10
[*SwitchC-Eth-Trunk10] port trunk allow-pass vlan 103
[*SwitchC-Eth-Trunk10] quit
[*SwitchC] commit
```

Configure SwitchA.

```
[~SwitchA] vlan 103
[*SwitchA-vlan103] quit
[*SwitchA] interface vlanif 103
[*SwitchA-Vlanif103] ipv6 enable
[*SwitchA-Vlanif103] ipv6 address 2001:db8:103::2/64
[*SwitchA-Vlanif103] quit
[*SwitchA] commit
[~SwitchA] interface eth-trunk 10
[*SwitchA-Eth-Trunk10] port trunk allow-pass vlan 103
[*SwitchA-Eth-Trunk10] quit
[*SwitchA] commit
```

3. Configure SwitchA to establish BGP4+ peer relationships with M-LAG member devices (SwitchB and SwitchC) and import routes.

Configure SwitchA.

```
[~SwitchA] bgp 100
[~SwitchA-bgp] peer 2001:db8:103::3 as-number 100
[*SwitchA-bgp] peer 2001:db8:103::4 as-number 100
[*SwitchA-bgp] ipv6-family unicast
[*SwitchA-bgp-af-ipv6] peer 2001:db8:103::3 enable
Warning: This operation will reset the peer session. Continue? [Y/N]:y
[*SwitchA-bgp-af-ipv6] peer 2001:db8:103::4 enable
Warning: This operation will reset the peer session. Continue? [Y/N]:y
[*SwitchA-bgp-af-ipv6] quit
[*SwitchA-bgp] quit
[*SwitchA] commit
```

Configure SwitchB.

```
[~SwitchB] bgp 100
[~SwitchB-bgp] peer 2001:db8:103::2 as-number 100
[*SwitchB-bgp] peer 2001:db8:103::2 connect-interface Vlanif 103 2001:db8:103::3
[*SwitchB-bgp] ipv6-family unicast
[*SwitchB-bgp-af-ipv6] peer 2001:db8:103::2 enable
Warning: This operation will reset the peer session. Continue? [Y/N]:y
[*SwitchB-bgp-af-ipv6] quit
[*SwitchB-bgp] quit
[*SwitchB] commit
```

Configure SwitchC.

```
[~SwitchC] bgp 100
[~SwitchC-bgp] peer 2001:db8:103::2 as-number 100
[*SwitchC-bgp] peer 2001:db8:103::2 connect-interface Vlanif 103 2001:db8:103::4
[*SwitchC-bgp] ipv6-family unicast
[*SwitchC-bgp-af-ipv6] peer 2001:db8:103::2 enable
Warning: This operation will reset the peer session. Continue? [Y/N]:y
[*SwitchC-bgp-af-ipv6] quit
[*SwitchC-bgp] quit
[*SwitchC] commit
```

Step 6 Verify the configuration.

- Run the **display dfs-group 1 m-lag** command to check M-LAG information.

Check information about the M-LAG with DFS group ID 1.

```
[~SwitchB] display dfs-group 1 m-lag
```

```
*          : Local node
Heart beat state : OK
Node 1 *
  Dfs-Group ID   : 1
  Priority        : 100
  Address         : ip address 10.200.1.1
  State          : Master
  Causation       : -
  System ID      : 0025-9e95-7c11
  SysName        : SwitchB
  Version        : V200R020C00
  Device Type    : CE16800
Node 2
  Dfs-Group ID   : 1
  Priority        : 100
  Address         : ip address 10.200.2.1
  State          : Backup
  Causation       : -
  System ID      : 0025-9e95-7c31
  SysName        : SwitchC
  Version        : V200R020C00
  Device Type    : CE16800
```

Check M-LAG information on SwitchB.

```
[~SwitchB] display dfs-group 1 node 1 m-lag brief
```

```
* - Local node
```

M-Lag ID	Interface	Port State	Status	Consistency-check
1	Eth-Trunk 10	Up	active(*)-active	--

Failed reason:

- 1 -- Relationship between vlan and port is inconsistent
- 2 -- STP configuration under the port is inconsistent
- 3 -- STP port priority configuration is inconsistent
- 4 -- LACP mode of M-LAG is inconsistent
- 5 -- M-LAG configuration is inconsistent
- 6 -- The number of M-LAG members is inconsistent

Check M-LAG information on SwitchC.

```
[~SwitchC] display dfs-group 1 node 2 m-lag brief
```

```
* - Local node
```

M-Lag ID	Interface	Port State	Status	Consistency-check
1	Eth-Trunk 10	Up	active(*)-active	--

Failed reason:

- 1 -- Relationship between vlan and port is inconsistent
- 2 -- STP configuration under the port is inconsistent
- 3 -- STP port priority configuration is inconsistent
- 4 -- LACP mode of M-LAG is inconsistent
- 5 -- M-LAG configuration is inconsistent
- 6 -- The number of M-LAG members is inconsistent

- Run the **display ospf peer brief** command on SwitchB, SwitchC, and SwitchA to check OSPF neighbor information.

Check OSPF neighbor information on SwitchB.

```
[~SwitchB] display ospf peer brief
```

```
OSPF Process 1 with Router ID 11.1.1.1
Peer Statistic Information
Total number of peer(s): 3
Peer(s) in full state: 3
```

Area Id	Interface	Neighbor id	State
0.0.0.0	10GE1/0/1	11.3.3.3	Full
0.0.0.0	Vlanif100	11.2.2.2	Full
0.0.0.1	Vlanif100	11.4.4.4	Full

Check OSPF neighbor information on SwitchC.

```
[~SwitchC] display ospf peer brief
OSPF Process 1 with Router ID 11.2.2.2
Peer Statistic Information
Total number of peer(s): 3
Peer(s) in full state: 3
-----
Area Id      Interface      Neighbor id    State
0.0.0.0      10GE1/0/1      11.3.3.3      Full
0.0.0.0      Vlanif100      11.1.1.1      Full
0.0.0.1      Vlanif100      11.4.4.4      Full
```

Check OSPF neighbor information on SwitchA.

```
[~SwitchA] display ospf peer brief
OSPF Process 1 with Router ID 11.4.4.4
Peer Statistic Information
Total number of peer(s): 2
Peer(s) in full state: 2
-----
Area Id      Interface      Neighbor id    State
0.0.0.1      Vlanif100      11.1.1.1      Full
0.0.0.1      Vlanif100      11.2.2.2      Full
```

- Run the **display ospfv3 peer** command on SwitchB, SwitchC, and SwitchA to check OSPFv3 neighbor information.

Check OSPFv3 neighbor information on SwitchB.

```
[~SwitchB] display ospfv3 peer
OSPFv3 Process (1)
Total number of peer(s): 3
Peer(s) in full state: 3
OSPFv3 Area (0.0.0.0)
Neighbor ID  Pri State      Dead Time  Interface      Instance ID
10.13.13.13  1 Full/Backup 00:00:37   10GE1/0/1      0
10.14.14.14  1 Full/DROther 00:00:38   Vlanif101      0

OSPFv3 Area (0.0.0.1)
Neighbor ID  Pri State      Dead Time  Interface      Instance ID
10.15.15.15  1 Full/DR     00:00:40   Vlanif101      0
```

Check OSPFv3 neighbor information on SwitchC.

```
[~SwitchC] display ospfv3 peer
OSPFv3 Process (1)
Total number of peer(s): 3
Peer(s) in full state: 3
OSPFv3 Area (0.0.0.0)
Neighbor ID  Pri State      Dead Time  Interface      Instance ID
10.13.13.13  1 Full/DR     00:00:37   10GE1/0/1      0
10.12.12.12  1 Full/DROther 00:00:38   Vlanif101      0

OSPFv3 Area (0.0.0.1)
Neighbor ID  Pri State      Dead Time  Interface      Instance ID
10.15.15.15  1 Full/DR     00:00:40   Vlanif101      0
```

Check OSPFv3 neighbor information on SwitchA.

```
[~SwitchA] display ospfv3 peer
OSPFv3 Process (1)
Total number of peer(s): 2
Peer(s) in full state: 2
OSPFv3 Area (0.0.0.1)
Neighbor ID  Pri State      Dead Time  Interface      Instance ID
10.14.14.14  1 Full/DROther 00:00:38   Vlanif101      0
10.12.12.12  1 Full/Backup  00:00:40   Vlanif101      0
```

- Run the **display bgp peer** command on SwitchB, SwitchC, and SwitchA to check BGP peer information.

Check BGP peer information on SwitchB.

```
[~SwitchB] display bgp peer
Status codes: * - Dynamic
```

```
BGP local router ID : 11.12.12.12
Local AS number : 100
Total number of peers : 2
Peers in established state : 2
Total number of dynamic peers : 0
```

Peer	V	AS	MsgRcvd	MsgSent	OutQ	Up/Down	State
10.102.0.2	4	100	30	30	0 00:24:37	Established	0
192.168.1.2	4	200	64	66	0 00:54:13	Established	0

Check BGP peer information on SwitchC.

```
[~SwitchC] display bgp peer
```

```
Status codes: * - Dynamic
BGP local router ID : 11.14.14.14
Local AS number : 100
Total number of peers : 59
Peers in established state : 2
Total number of dynamic peers : 0
```

Peer	V	AS	MsgRcvd	MsgSent	OutQ	Up/Down	State	PrefRcv
10.102.0.2	4	100	34	33	0 00:26:56	Established	0	
192.168.2.2	4	200	71	72	0 00:59:44	Established	0	

Check BGP peer information on SwitchA.

```
[~SwitchA] display bgp peer
```

```
2020-07-07 21:06:02.443
Status codes: * - Dynamic
BGP local router ID : 11.15.15.15
Local AS number : 100
Total number of peers : 2
Peers in established state : 2
Total number of dynamic peers : 0
```

Peer	V	AS	MsgRcvd	MsgSent	OutQ	Up/Down	State	PrefRcv
10.102.0.3	4	100	29	29	0 00:23:22	Established	0	
10.102.0.4	4	100	29	30	0 00:23:18	Established	0	

- Run the **display bgp ipv6 peer** command on SwitchB, SwitchC, and SwitchA to check BGP4+ peer information.

Check BGP4+ peer information on SwitchB.

```
[~SwitchB] display bgp ipv6 peer
```

```
Status codes: * - Dynamic
BGP local router ID : 11.12.12.12
Local AS number : 100
Total number of peers : 2
Peers in established state : 2
Total number of dynamic peers : 0
```

Peer	V	AS	MsgRcvd	MsgSent	OutQ	Up/Down	State
2001:DB8:1::2	4	200	61	61	0 00:50:07	Established	2
2001:DB8:103::2	4	100	5	6	0 00:02:18	Established	0

Check BGP4+ peer information on SwitchC.

```
[~SwitchC] display bgp ipv6 peer
```

```
Status codes: * - Dynamic
BGP local router ID : 11.14.14.14
Local AS number : 100
Total number of peers : 2
Peers in established state : 2
Total number of dynamic peers : 0
```

Peer	V	AS	MsgRcvd	MsgSent	OutQ	Up/Down	State	PrefRcv
2001:DB8:2::2	4	200	76	75	0 01:03:13	Established	2	
2001:DB8:103::2	4	100	21	22	0 00:15:55	Established	0	

Check BGP4+ peer information on SwitchA.


```
[~SwitchA] display bgp ipv6 peer
Status codes: * - Dynamic
BGP local router ID : 11.15.15.15
Local AS number : 100
Total number of peers : 2
Peers in established state : 2
Total number of dynamic peers : 0
```

Peer	V	AS	MsgRcvd	MsgSent	OutQ	Up/Down	State	PrefRcv
2001:DB8:103::3	4	100	7	7	0	00:03:02	Established	1
2001:DB8:103::4	4	100	7	6	0	00:02:33	Established	1

----End

Configuration Files

- SwitchA configuration file

```
#
sysname SwitchA
#
vlan batch 100 to 103
#
ospfv3 1
router-id 10.15.15.15
area 0.0.0.1
#
interface Vlanif100
ip address 10.100.0.2 255.255.255.0
#
interface Vlanif101
ipv6 enable
ipv6 address 2001:DB8:101::2/64
ospfv3 1 area 0.0.0.1
#
interface Vlanif102
ip address 10.102.0.2 255.255.255.0
#
interface Vlanif103
ipv6 enable
ipv6 address 2001:DB8:103::2/64
#
interface Eth-Trunk10
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 100 101 102 103
mode lacp-static
lacp mixed-rate link enable
#
interface 10GE1/0/1
eth-trunk 10
#
interface 10GE1/0/2
eth-trunk 10
#
interface 10GE1/0/3
eth-trunk 10
#
interface 10GE1/0/4
eth-trunk 10
#
bgp 100
router-id 11.15.15.15
peer 10.102.0.3 as-number 100
peer 10.102.0.4 as-number 100
peer 2001:DB8:103::3 as-number 100
peer 2001:DB8:103::4 as-number 100
#
ipv4-family unicast
```

```
peer 10.102.0.3 enable
peer 10.102.0.4 enable
#
ipv6-family unicast
peer 2001:DB8:103::3 enable
peer 2001:DB8:103::4 enable
#
ospf 1 router-id 11.4.4.4
area 0.0.0.1
network 10.100.0.0 0.0.0.255
#
return
```

- SwitchB configuration file

```
#
sysname Leaf
#
dfs-group 1
source ip 10.200.1.1 peer 10.200.2.1
dual-active detection enhanced enable
#
vlan batch 100 to 103
#
stp mode rstp
stp v-stp enable
#
ospfv3 1
router-id 10.12.12.12
area 0.0.0.0
area 0.0.0.1
#
interface Vlanif100
ip address 10.100.0.1 255.255.255.0
ospf source sub-address 10.100.0.3
mac-address 0000-5e00-0101
m-lag ip address 10.100.0.3 255.255.255.0
#
interface Vlanif101
ipv6 enable
ipv6 address 2001:DB8:101::1/64
ospfv3 1 area 0.0.0.1
mac-address 0000-5e00-0102
m-lag ipv6 address FE80:1000::1 link-local
#
interface Vlanif102
ip address 10.102.0.1 255.255.255.0
mac-address 0000-5e00-0103
m-lag ip address 10.102.0.3 255.255.255.0
#
interface Vlanif103
ipv6 enable
ipv6 address 2001:DB8:103::1/64
mac-address 0000-5e00-0104
m-lag ipv6 address 2001:DB8:103::3
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk10
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 100 101 102 103
mode lacp-static
lacp mixed-rate link enable
dfs-group 1 m-lag 1
#
interface 10GE1/0/1
undo portswitch
ipv6 enable
```

```
ip address 192.168.1.1 255.255.255.0
ipv6 address 2001:DB8:1::1/64
ospfv3 1 area 0.0.0.0
#
interface 10GE1/0/2
eth-trunk 10
#
interface 10GE1/0/5
eth-trunk 10
#
interface 10GE1/0/3
eth-trunk 1
#
interface 10GE1/0/4
eth-trunk 1
#
interface LoopBack1
ip address 10.2.2.2 255.255.255.255
#
interface LoopBack2
ip address 10.3.3.3 255.255.255.255
#
monitor-link group 1
port 10ge 1/0/1 uplink
port eth-trunk 10 downlink 1
#
bgp 100
router-id 11.12.12.12
peer 10.102.0.2 as-number 100
peer 10.102.0.2 connect-interface Vlanif102 10.102.0.3
peer 192.168.1.2 as-number 200
peer 2001:DB8:1::2 as-number 200
peer 2001:DB8:103::2 as-number 100
peer 2001:DB8:103::2 connect-interface Vlanif103 2001:DB8:103::3
#
ipv4-family unicast
peer 10.102.0.2 enable
peer 192.168.1.2 enable
#
ipv6-family unicast
network 2001:DB8:1:: 64
peer 2001:DB8:1::2 enable
peer 2001:DB8:103::2 enable
#
ospf 1 router-id 11.1.1.1
area 0.0.0.0
network 10.2.2.2 0.0.0.0
network 10.3.3.3 0.0.0.0
network 192.168.1.0 0.0.0.255
area 0.0.0.1
network 10.100.0.0 0.0.0.255
#
return
```

- SwitchC configuration file

```
#
sysname SwitchC
#
dfs-group 1
source ip 10.200.2.1 peer 10.200.1.1
dual-active detection enhanced enable
#
vlan batch 100 to 103
#
stp mode rstp
stp v-stp enable
#
ospfv3 1
router-id 10.14.14.14
area 0.0.0.0
```

```
area 0.0.0.1
#
interface Vlanif100
ip address 10.100.0.1 255.255.255.0
ospf source sub-address 10.100.0.4
mac-address 0000-5e00-0101
m-lag ip address 10.100.0.4 255.255.255.0
#
interface Vlanif101
ipv6 enable
ipv6 address 2001:db8:101::1/64
ospfv3 1 area 0.0.0.1
mac-address 0000-5e00-0102
m-lag ipv6 address FE80:1000::2 link-local
#
interface Vlanif102
ip address 10.102.0.1 255.255.255.0
mac-address 0000-5e00-0103
m-lag ip address 10.102.0.4 255.255.255.0
#
interface Vlanif103
ipv6 enable
ipv6 address 2001:DB8:103::1/64
mac-address 0000-5e00-0104
m-lag ipv6 address 2001:DB8:103::4
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk10
port link-type trunk
undo port trunk allow-pass vlan 1
port trunk allow-pass vlan 100 101 102 103
mode lacp-static
lacp mixed-rate link enable
dfs-group 1 m-lag 1
#
interface 10GE1/0/1
undo portswitch
ipv6 enable
ip address 192.168.2.1 255.255.255.0
ipv6 address 2001:db8:2::1/64
ospfv3 1 area 0.0.0.0
#
interface 10GE1/0/2
eth-trunk 10
#
interface 10GE1/0/5
eth-trunk 10
#
interface 10GE1/0/3
eth-trunk 1
#
interface 10GE1/0/4
eth-trunk 1
#
interface LoopBack1
ip address 10.2.2.2 255.255.255.255
#
interface LoopBack2
ip address 10.4.4.4 255.255.255.255
#
monitor-link group 1
port 10ge 1/0/1 uplink
port eth-trunk 10 downlink 1
#
bgp 100
router-id 11.14.14.14
```

```
peer 10.102.0.2 as-number 100
peer 10.102.0.2 connect-interface Vlanif102 10.102.0.4
peer 192.168.2.2 as-number 200
peer 2001:DB8:2::2 as-number 200
peer 2001:DB8:103::2 as-number 100
peer 2001:DB8:103::2 connect-interface Vlanif103 2001:DB8:103::4
#
ipv4-family unicast
peer 10.102.0.2 enable
peer 192.168.2.2 enable
#
ipv6-family unicast
network 2001:DB8:2:: 64
peer 2001:DB8:2::2 enable
peer 2001:DB8:103::2 enable
#
ospf 1 router-id 11.2.2.2
area 0.0.0.0
network 10.2.2.2 0.0.0.0
network 10.4.4.4 0.0.0.0
network 192.168.2.0 0.0.0.255
area 0.0.0.1
network 10.100.0.0 0.0.0.255
#
return
```

- SwitchD configuration file

```
#
sysname SwitchD
#
ospfv3 1
router-id 10.13.13.13
area 0.0.0.0
#
interface 10GE1/0/1
undo portswitch
ipv6 enable
ip address 192.168.1.2 255.255.255.0
ipv6 address 2001:db8:1::2/64
ospfv3 1 area 0.0.0.0
#
interface 10GE1/0/2
undo portswitch
ipv6 enable
ip address 192.168.2.2 255.255.255.0
ipv6 address 2001:db8:2::2/64
ospfv3 1 area 0.0.0.0
#
interface LoopBack1
ip address 10.1.1.1 255.255.255.255
#
bgp 200
router-id 11.13.13.13
peer 192.168.1.1 as-number 100
peer 192.168.2.1 as-number 100
peer 2001:db8:1::1 as-number 100
peer 2001:db8:2::1 as-number 100
#
ipv4-family unicast
peer 192.168.1.1 enable
peer 192.168.2.1 enable
#
ipv6-family unicast
network 2001:db8:1:: 64
network 2001:db8:2:: 64
peer 2001:db8:1::1 enable
peer 2001:db8:2::1 enable
#
ospf 1 router-id 11.3.3.3
area 0.0.0.0
```

```
network 10.1.1.1 0.0.0.0
network 192.168.1.0 0.0.0.255
network 192.168.2.0 0.0.0.255
#
return
```

1.19 Example for Configuring IPv4 Layer 3 Multicast Over VXLAN When the Multicast Source is Located on an External Network

Applicable Products and Versions

This example applies to CE12800, CE8800, CE7800, and CE6800 series switches running V200R005C10 or later versions. (A switch running V200R005C00 cannot function as a border leaf switch when configured with the IP multicast over VXLAN function.) Among these series, the CE12800E, CE6880EI, CE6881, CE5881, CE6881K, CE6820, CE6863, CE6863E, CE6863K, CE6881E, CE6850EI, CE6810EI, and CE6810LI do not support IPv4 Layer 3 multicast over VXLAN.

The CE8860EI, CE8850EI, CE7855EI, CE7850EI, CE6860EI, CE6856HI, CE6855HI, CE6851HI, CE6850HI, and CE6850U-HI cannot function as border leaf switches.

For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

On the IPv4 network shown in [Networking diagram for configuring IPv4 Layer 3 multicast over VXLAN when the multicast source is located on an external network](#), Borderleaf1, Borderleaf2, Leaf1, and Leaf2 function as distributed VXLAN gateways. In addition, the two border leaf switches and two leaf switches set up two M-LAG systems to ensure high reliability. The multicast source is located on the external network, and the multicast receiver is located on the VXLAN overlay network. The multicast source is directly connected to the CE. The CE is connected to the VXLAN network through the active-active gateways Borderleaf1 and Borderleaf2. The multicast receiver is connected to the VXLAN network through Layer 2 sub-interfaces on the active-active gateways Leaf1 and Leaf2, and belongs to BD 10. The multicast receiver orders a program without specifying the multicast source. The multicast group address is 225.1.1.1.

To meet these requirements, configure IPv4 Layer 3 multicast on the VXLAN network where distributed gateways are deployed.

Figure 1-23 Networking diagram for configuring IPv4 Layer 3 multicast over VXLAN when the multicast source is located on an external network

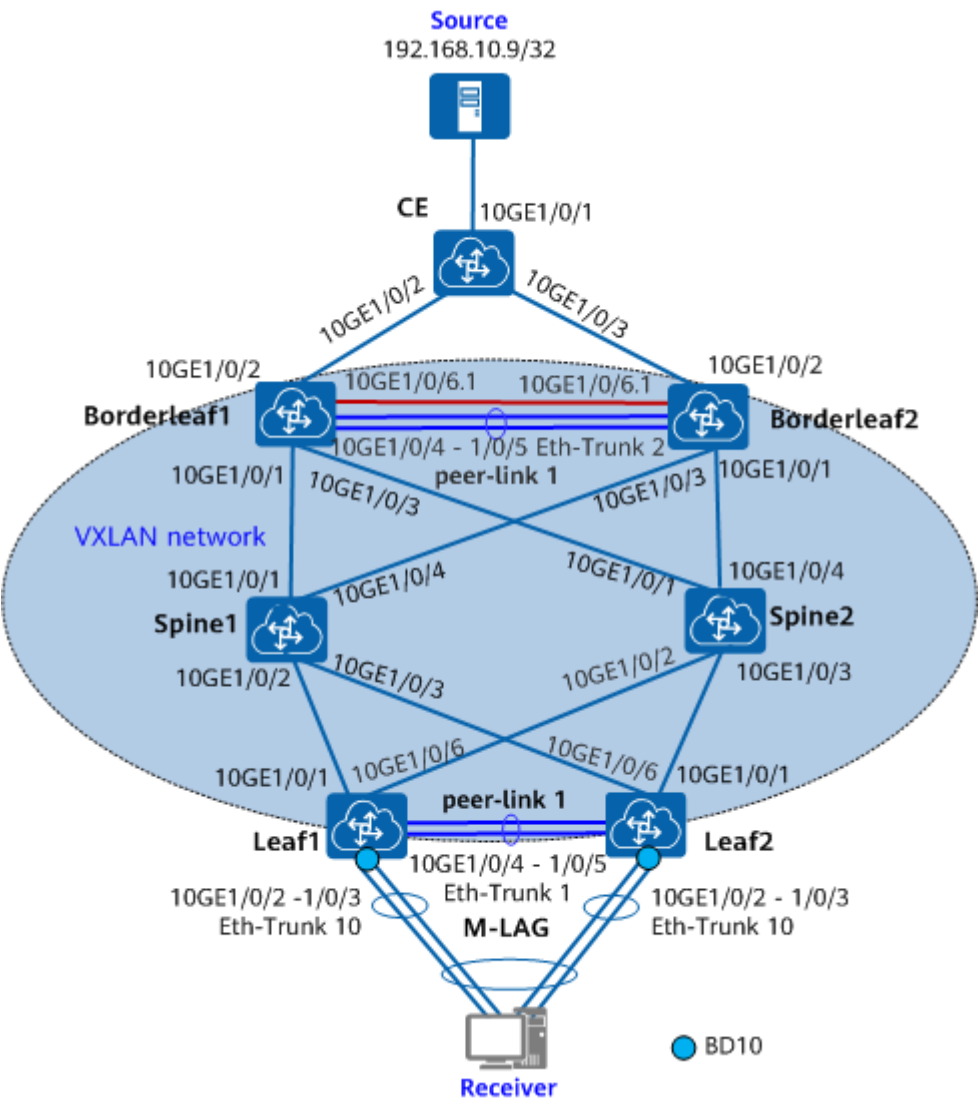


Table 1-19 Data plan

Device	Interface	IP Address	Device to Connect
CE	10GE1/0/1	192.168.10.2/24	Source
	10GE1/0/2	192.168.20.2/24	Borderleaf1
	10GE1/0/3	192.168.30.2/24	Borderleaf2
Borderleaf1	10GE1/0/1	10.1.1.1/24	Spine1
	10GE1/0/2	192.168.20.1/24	CE
	VLANIF 3000	10.10.21.1/24	Borderleaf2
	10GE1/0/3	10.1.11.1/24	Spine2

Device	Interface	IP Address	Device to Connect
	10GE1/0/6.1	10.6.1.1/24	Borderleaf2
	LoopBack1	1.1.1.1/32	NA
	LoopBack2	1.1.1.110/32	NA
	LoopBack3	1.1.1.10/32 (VPN RP)	NA
	MEth0/0/0	10.10.11.1/24	NA
Borderleaf2	10GE1/0/1	10.1.44.1/24	Spine2
	10GE1/0/2	192.168.30.1/24	CE
	VLANIF 3000	10.10.21.2/24	Borderleaf1
	10GE1/0/3	10.1.4.1/24	Spine1
	10GE1/0/6.1	10.6.1.2/24	Borderleaf1
	LoopBack1	2.2.2.2/32	NA
	LoopBack2	1.1.1.110/32	NA
	MEth0/0/0	10.10.11.2/24	NA
Spine1	10GE1/0/1	10.1.1.2/24	Borderleaf1
	10GE1/0/2	10.1.2.2/24	Leaf1
	10GE1/0/3	10.1.3.2/24	Leaf2
	10GE1/0/4	10.1.4.2/24	Borderleaf2
	LoopBack1	5.5.5.5/32 (public network RP)	NA
	LoopBack2	5.5.5.1/32	NA
Spine2	10GE1/0/1	10.1.11.2/24	Borderleaf1
	10GE1/0/2	10.1.22.2/24	Leaf1
	10GE1/0/3	10.1.33.2/24	Leaf2
	10GE1/0/4	10.1.44.2/24	Borderleaf2
	LoopBack1	5.5.5.5/32 (public network RP)	NA
	LoopBack2	5.5.5.2/32	NA
Leaf1	10GE1/0/1	10.1.2.1/24	Spine1
	VBDIF 10	192.168.10.1/24	Receiver
	VLANIF 4000	10.10.20.1/24	Leaf2

Device	Interface	IP Address	Device to Connect
	10GE1/0/6	10.1.22.1/24	Spine2
	LoopBack1	3.3.3.3/32	NA
	LoopBack2	2.2.2.210/32	NA
	MEth0/0/0	10.10.10.1/24	NA
Leaf2	10GE1/0/1	10.1.33.1/24	Spine1
	VBDIF 10	192.168.10.1/24	Receiver
	VLANIF 4000	10.10.20.2/24	Leaf1
	10GE1/0/6	10.1.3.1/24	Spine1
	LoopBack1	4.4.4.4/32	NA
	LoopBack2	2.2.2.210/32	NA
	MEth0/0/0	10.10.10.2/24	NA

Configuration Roadmap

NOTE

To address the preceding requirements, ensure that in addition to the peer-link, there is a direct Layer 3 link between Borderleaf1 and Borderleaf2 that function as the M-LAG master and backup devices. In addition, the PIM protocol must be enabled on the Layer 3 link, and the M-LAG master and backup devices must have the same multicast configuration.

The configuration roadmap is as follows:

1. Configure M-LAG between Borderleaf1 and Borderleaf2, and between Leaf1 and Leaf2.
2. Configure each leaf to set up VXLAN tunnels using BGP EVPN, configure the leaf switches to function as distributed gateways, and enable the hosts connected to these leaf nodes to communicate at Layer 3 through VPN routes.
3. Configure BUM multicast replication for the Layer 3 VNI of the L3VPN instance on each leaf.
4. Configure BGP MVPN peer relationships between Borderleaf1, Borderleaf2, Leaf1, and Leaf2.
5. Configure the VXLAN I-PMSI tunnel on each leaf.
6. On the interface of each leaf that is bound to the L3VPN instance, enable PIM-SM to create a multicast routing table for the VPN.
7. On each leaf, add uplink and downlink interfaces to a Monitor Link group.
8. Configure IGMP on each VBDIF interface and physical interface that connect multicast devices to user subnets.
9. Configure VPN Layer 3 multicast on Borderleaf1, Borderleaf2, and the CE.

Data Preparation

To complete the configuration, you need to prepare the following data according to [Data plan](#):

- VLAN ID of the receiver on the VXLAN overlay network (VLAN 10)
- BD ID (BD 10)
- VNI ID of the BD (VNI 10)
- L3VPN instance name (mcast1) and VNI ID of the L3VPN instance (VNI 5010)
- RDs of the EVPN instances (11:1 and 11:2) and RTs of the EVPN instances (12:1, 12:2, and 13:1)
- RD of the L3VPN instance (1:1), RT of the L3VPN instance (1:1), and RT used for interoperation between the L3VPN instance and the EVPN instance (13:1)
- Multicast group address for BUM multicast replication based on the Layer 3 VNI of the L3VPN instance (225.0.0.1)
- MVPN ID of each leaf (IP address of the Loopback2 interface)
- Local VPN instance ID of each leaf (1)
- Static RP in the public network instance (5.5.5.5) and that in the VPN instance (1.1.1.10)

Procedure

Step 1 Configure V-STP-based M-LAG on Borderleaf1 and Borderleaf2, and configure a direct Layer 3 link between Layer 3 sub-interfaces of the M-LAG master and backup devices.

Configure Borderleaf1. The configuration on Borderleaf2 is similar to the configuration on Borderleaf1. For details, see the configuration file.

```
<HUAWEI> system-view
[~HUAWEI] sysname Borderleaf1
[*HUAWEI] commit
[~Borderleaf1] interface meth 0/0/0
[~Borderleaf1-Meth0/0/0] ip address 10.10.11.1 24
[*Borderleaf1-Meth0/0/0] quit
[*Borderleaf1] stp mode rstp
[*Borderleaf1] stp v-stp enable
[*Borderleaf1] dfs-group 1
[*Borderleaf1-dfs-group-1] source ip 10.10.11.1
[*Borderleaf1-dfs-group-1] quit
[*Borderleaf1] interface eth-trunk 2
[*Borderleaf1-Eth-Trunk2] trunkport 10ge 1/0/4 to 1/0/5
[*Borderleaf1-Eth-Trunk2] mode lacp-static
[*Borderleaf1-Eth-Trunk2] peer-link 1
[*Borderleaf1-Eth-Trunk2] quit
[*Borderleaf1] interface 10ge 1/0/6
[*Borderleaf1-10GE1/0/6] undo portswitch
[*Borderleaf1-10GE1/0/6] quit
[*Borderleaf1] commit
[~Borderleaf1] interface 10ge 1/0/6.1
[*Borderleaf1-10GE1/0/6.1] ip address 10.6.1.1 24
[*Borderleaf1-10GE1/0/6.1] quit
[*Borderleaf1] commit
```

Step 2 Configure V-STP-based M-LAG on Leaf1 and Leaf2.

Configure Leaf1. The configuration on Leaf2 is similar to the configuration on Leaf1. For details, see the configuration file.

NOTE

If the link between Leaf1 and the VXLAN network fails, Leaf1 will discard all user traffic that reaches Leaf1, because no uplink interface is available. In this case, you can configure a Monitor Link group to associate the uplink and downlink interfaces of Leaf1. If the uplink interface of Leaf1 is Down, the downlink interface also goes Down. As a result, user-side traffic is not forwarded through Leaf1. This prevents traffic from being discarded.

```
[~Leaf1] interface meth 0/0/0
[~Leaf1-MEth0/0/0] ip address 10.10.10.1 32
[~Leaf1-MEth0/0/0] quit
[*Leaf1] stp mode rstp
[*Leaf1] stp v-stp enable
[*Leaf1] dfs-group 1
[*Leaf1-dfs-group-1] source ip 10.10.10.1
[*Leaf1-dfs-group-1] quit
[*Leaf1] interface eth-trunk 1
[*Leaf1-Eth-Trunk1] trunkport 10ge 1/0/4 to 1/0/5
[*Leaf1-Eth-Trunk1] mode lacp-static
[*Leaf1-Eth-Trunk1] peer-link 1
[*Leaf1-Eth-Trunk1] quit
[*Leaf1] interface eth-trunk 10
[*Leaf1-Eth-Trunk10] trunkport 10ge 1/0/2 to 1/0/3
[*Leaf1-Eth-Trunk10] mode lacp-static
[*Leaf1-Eth-Trunk10] dfs-group 1 m-lag 1
[*Leaf1-Eth-Trunk10] stp edged-port enable
[*Leaf1-Eth-Trunk10] quit
[*Leaf1] commit
```

Step 3 Assign IP addresses to the interfaces of the switches and configure a unicast routing protocol.

Configure IP addresses and masks for the interfaces according to [Data plan](#), and configure OSPF on each switch to ensure that the switches can communicate at the network layer. The configurations on the CE, Borderleaf2, Leaf1, Leaf2, Spine1, and Spine2 are similar to the configuration on Borderleaf1. For details, see the configuration file.

```
[~Borderleaf1] interface loopback 1
[*Borderleaf1-LoopBack1] ip address 1.1.1.1 32
[*Borderleaf1-LoopBack1] quit
[~Borderleaf1] interface loopback 2
[*Borderleaf1-LoopBack2] ip address 1.1.1.110 32
[*Borderleaf1-LoopBack2] quit
[*Borderleaf1] interface 10ge 1/0/1
[*Borderleaf1-10GE1/0/1] undo portswitch
[*Borderleaf1-10GE1/0/1] ip address 10.1.1.1 24
[*Borderleaf1-10GE1/0/1] quit
[*Borderleaf1] interface 10ge 1/0/3
[*Borderleaf1-10GE1/0/3] undo portswitch
[*Borderleaf1-10GE1/0/3] ip address 10.1.11.1 24
[*Borderleaf1-10GE1/0/3] quit
[*Borderleaf1] ospf
[*Borderleaf1-ospf-1] area 0
[*Borderleaf1-ospf-1-area-0.0.0.0] network 1.1.1.1 0.0.0.0
[*Borderleaf1-ospf-1-area-0.0.0.0] network 1.1.1.110 0.0.0.0
[*Borderleaf1-ospf-1-area-0.0.0.0] network 10.1.1.0 0.0.0.255
[*Borderleaf1-ospf-1-area-0.0.0.0] network 10.1.11.0 0.0.0.255
[*Borderleaf1-ospf-1-area-0.0.0.0] network 10.6.1.0 0.0.0.255
[*Borderleaf1-ospf-1-area-0.0.0.0] network 10.10.11.0 0.0.0.255
[*Borderleaf1-ospf-1-area-0.0.0.0] quit
[*Borderleaf1-ospf-1] quit
[*Borderleaf1] commit
```

After OSPF is configured successfully, the switches can learn the interface address of each other using OSPF and successfully ping each other. For example, if you

ping the Loopback1 interface address of Borderleaf2 on Borderleaf1, the output is as follows:

```
[~Borderleaf1] ping 2.2.2.2
PING 2.2.2.2: 56 data bytes, press CTRL_C to break
Reply from 2.2.2.2: bytes=56 Sequence=1 ttl=254 time=2 ms
Reply from 2.2.2.2: bytes=56 Sequence=2 ttl=254 time=1 ms
Reply from 2.2.2.2: bytes=56 Sequence=3 ttl=254 time=1 ms
Reply from 2.2.2.2: bytes=56 Sequence=4 ttl=254 time=1 ms
Reply from 2.2.2.2: bytes=56 Sequence=5 ttl=254 time=1 ms

--- 2.2.2.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 1/1/1 ms
```

Step 4 For CE12800 series switches, set the card interoperability mode to enhanced mode.

 NOTE

When the card interoperability mode is enhanced mode, only FD/FDA/FD1/FG/FG1/SD series interface cards and SFUF/SFUG can be installed on the switch. After completing the configuration, save the configuration and restart the switch for the configuration to take effect. You are advised to restart the switch after all the configurations are completed and saved. After the switch restarts, run the **save** command again to save the configuration.

Configure Borderleaf1. The configurations on the other switches are similar to the configuration on Borderleaf1. For details, see the configuration file.

```
[~Borderleaf1] set forward capability enhanced
Warning: Current configuration should be committed and saved, and it will take effect after reboot.
Continue? [Y/N]: y
[*Borderleaf1] commit
```

Step 5 Configure the VXLAN.

If the CE12800, CE16800 equipped with A series cards, CE6870EI, and CE6875EI are used as VXLAN gateways, configure the VXLAN tunnel mode and enable the VXLAN ACL extension function before configuring VXLAN. If other models are deployed as VXLAN gateways, skip this step. The configurations on Borderleaf2, Leaf1, and Leaf2 are similar to the configuration on Borderleaf1. For details, see the configuration file.

```
[~Borderleaf1] ip tunnel mode vxlan
[*Borderleaf1] assign forward nvo3 acl extend enable
Info: The configuration will take effect after system reboot.
[*Borderleaf1] commit
```

 NOTE

After configuring the VXLAN tunnel mode and enabling the VXLAN ACL extension function, you need to save the configuration and restart the device for the configuration to take effect. You are advised to restart the device after all the configurations are complete and saved.

Configure service access points on Leaf1 and Leaf2.

Configure Leaf1. The configuration on Leaf2 is similar to the configuration on Leaf1. For details, see the configuration file.

```
[~Leaf1] bridge-domain 10
[*Leaf1-bd10] quit
[*Leaf1] interface eth-trunk 10.1 mode l2
[*Leaf1-Eth-Trunk10.1] encapsulation dot1q vid 10
[*Leaf1-Eth-Trunk10.1] bridge-domain 10
[*Leaf1-Eth-Trunk10.1] quit
[*Leaf1] commit
```

Configure BGP EVPN on each leaf node as the VXLAN control plane protocol and configure BGP EVPN peers. The configurations on the other leaf nodes are similar to the configuration on Borderleaf1. For details, see the configuration file.

```
[~Borderleaf1] evpn-overlay enable
[*Borderleaf1] bgp 100
[*Borderleaf1] router-id 1.1.1.1
[*Borderleaf1-bgp] peer 2.2.2.2 as-number 100
[*Borderleaf1-bgp] peer 2.2.2.2 connect-interface LoopBack1
[*Borderleaf1-bgp] peer 3.3.3.3 as-number 100
[*Borderleaf1-bgp] peer 3.3.3.3 connect-interface LoopBack1
[*Borderleaf1-bgp] peer 4.4.4.4 as-number 100
[*Borderleaf1-bgp] peer 4.4.4.4 connect-interface LoopBack1
[*Borderleaf1-bgp] l2vpn-family evpn
[*Borderleaf1-bgp-af-evpn] peer 2.2.2.2 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Borderleaf1-bgp-af-evpn] peer 3.3.3.3 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Borderleaf1-bgp-af-evpn] peer 4.4.4.4 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Borderleaf1-bgp-af-evpn] quit
[*Borderleaf1-bgp] quit
[*Borderleaf1] commit
```

Configure an L3VPN instance on Borderleaf1 and Borderleaf2. The configuration on Borderleaf2 is similar to the configuration on Borderleaf1. For details, see the configuration file.

```
[~Borderleaf1] ip vpn-instance mcast1
[*Borderleaf1-vpn-instance-mcast1] vxlan vni 5010
[*Borderleaf1-vpn-instance-mcast1] ipv4-family
[*Borderleaf1-vpn-instance-mcast1-af-ipv4] route-distinguisher 1:1
[*Borderleaf1-vpn-instance-mcast1-af-ipv4] vpn-target 1:1
[*Borderleaf1-vpn-instance-mcast1-af-ipv4] vpn-target 13:1 evpn
[*Borderleaf1-vpn-instance-mcast1-af-ipv4] quit
[*Borderleaf1-vpn-instance-mcast1] quit
[*Borderleaf1] commit
```

Configure an L3VPN instance and an EVPN instance on Leaf1 and Leaf2. The configuration on Leaf2 is similar to the configuration on Leaf1. For details, see the configuration file.

```
[~Leaf1] ip vpn-instance mcast1
[*Leaf1-vpn-instance-mcast1] vxlan vni 5010
[*Leaf1-vpn-instance-mcast1] ipv4-family
[*Leaf1-vpn-instance-mcast1-af-ipv4] route-distinguisher 1:1
[*Leaf1-vpn-instance-mcast1-af-ipv4] vpn-target 1:1
[*Leaf1-vpn-instance-mcast1-af-ipv4] vpn-target 13:1 evpn
[*Leaf1-vpn-instance-mcast1-af-ipv4] quit
[*Leaf1-vpn-instance-mcast1] quit
[*Leaf1] bridge-domain 10
[*Leaf1-bd10] vxlan vni 10
[*Leaf1-bd10] evpn
[*Leaf1-bd10-evpn] route-distinguisher 11:1
[*Leaf1-bd10-evpn] vpn-target 12:1
[*Leaf1-bd10-evpn] vpn-target 13:1
[*Leaf1-bd10-evpn] quit
[*Leaf1-bd10] quit
[*Leaf1] commit
```

Configure ingress replication on each leaf node.

- Configure Borderleaf1.

```
[~Borderleaf1] interface nve 1
[*Borderleaf1-Nve1] source 1.1.1.110
[*Borderleaf1-Nve1] vni 5010 head-end peer-list protocol bgp
[*Borderleaf1-Nve1] mac-address 0000-5e00-0101
[*Borderleaf1-Nve1] quit
[*Borderleaf1] commit
```

- Configure Borderleaf2. Borderleaf1 and Borderleaf2 function as active-active gateways, so you must ensure that the IP addresses and MAC addresses of the NVE interfaces on the two devices are the same.

```
[~Borderleaf2] interface nve 1
[*Borderleaf2-Nve1] source 1.1.1.110
[*Borderleaf2-Nve1] vni 5010 head-end peer-list protocol bgp
[*Borderleaf2-Nve1] mac-address 0000-5e00-0101
[*Borderleaf2-Nve1] quit
[*Borderleaf2] commit
```

- Configure Leaf1.

```
[~Leaf1] interface nve 1
[*Leaf1-Nve1] source 2.2.2.210
[*Leaf1-Nve1] mac-address 0000-5e00-0102
[*Leaf1-Nve1] vni 5010 head-end peer-list protocol bgp
[*Leaf1-Nve1] quit
[*Leaf1] commit
```

- Configure Leaf2. Leaf1 and Leaf2 function as active-active gateways, so you must ensure that the IP addresses and MAC addresses of the NVE interfaces on the two devices are the same.

```
[~Leaf2] interface nve 1
[*Leaf2-Nve1] source 2.2.2.210
[*Leaf2-Nve1] mac-address 0000-5e00-0102
[*Leaf2-Nve1] vni 5010 head-end peer-list protocol bgp
[*Leaf2-Nve1] quit
[*Leaf2] commit
```

Configure Leaf1 and Leaf2 to function as Layer 3 VXLAN gateways.

- If a CE8860EI, CE8850EI, CE7855EI, CE7850EI, CE6860EI, CE6856HI, CE6855HI, CE6851HI, CE6850HI, or CE6850U-HI functions as a Layer 3 VXLAN gateway, configure a service loopback interface on the switch before configuring the switch as the Layer 3 VXLAN gateway. Skip this step if other CE switch models are configured as Layer 3 VXLAN gateways. The configuration on Leaf2 is similar to the configuration on Leaf1. For details, see the configuration file.

```
[~Leaf1] interface eth-trunk 11
[*Leaf1-Eth-Trunk11] service type tunnel
[*Leaf1-Eth-Trunk11] quit
[*Leaf1] interface 10ge 1/0/7
[*Leaf1-10GE1/0/7] eth-trunk 11
[*Leaf1-10GE1/0/7] quit
[*Leaf1] commit
```

NOTE

- A member interface must be an idle physical interface that does not carry any service. There is no requirement for the status of a physical interface.
 - Ensure that the bandwidth of the Eth-Trunk interface (multicast loopback interface) is at least twice the bandwidth required for transmitting traffic on the Layer 3 VXLAN gateway. For example, if traffic is sent from users to the gateway at a rate of 10 Gbit/s across the VXLAN network, add two 10GE interfaces to the Eth-Trunk as the service loopback interfaces.
 - In an M-LAG system comprised of the preceding devices, if these devices connect to the external network, the IPv4 Layer 3 multicast over VXLAN function cannot be configured together with the PIM protocol. That is, M-LAG member devices cannot function as the border leaf nodes when running IPv4 Layer 3 multicast over VXLAN.
- Configure the Layer 3 VXLAN gateway on Leaf1.

```
[~Leaf1] interface vbdif 10
[*Leaf1-Vbdif10] ip binding vpn-instance mcast1
[*Leaf1-Vbdif10] ip address 192.168.10.1 24
[*Leaf1-Vbdif10] mac-address 0000-5e00-0104
[*Leaf1-Vbdif10] vxlan anycast-gateway enable
```

```
[*Leaf1-Vbdif10] arp collect host enable
[*Leaf1-Vbdif10] quit
[*Leaf1] commit
```

- Configure the Layer 3 VXLAN gateway on Leaf2. Leaf1 and Leaf2 function as active-active gateways, so you must ensure that the IP addresses and MAC addresses of the VBDIF interfaces on the two devices are the same.

```
[~Leaf2] interface vbdif 10
[*Leaf2-Vbdif10] ip binding vpn-instance mcast1
[*Leaf2-Vbdif10] ip address 192.168.10.1 24
[*Leaf2-Vbdif10] mac-address 0000-5e00-0104
[*Leaf2-Vbdif10] vxlan anycast-gateway enable
[*Leaf2-Vbdif10] arp collect host enable
[*Leaf2-Vbdif10] quit
[*Leaf2] commit
```

Configure BGP on each leaf to advertise IRB routes to its peers. The configurations on the other leaf nodes are similar to the configuration on Borderleaf1. For details, see the configuration file.

```
[~Borderleaf1] bgp 100
[~Borderleaf1-bgp] l2vpn-family evpn
[~Borderleaf1-bgp-af-evpn] peer 2.2.2.2 advertise irb
[*Borderleaf1-bgp-af-evpn] peer 3.3.3.3 advertise irb
[*Borderleaf1-bgp-af-evpn] peer 4.4.4.4 advertise irb
[*Borderleaf1-bgp-af-evpn] quit
[*Borderleaf1-bgp] quit
[*Borderleaf1] commit
```

Configure BGP on Borderleaf1 to advertise IP prefix routes to its peers.

```
[~Borderleaf1] bgp 100
[~Borderleaf1-bgp] ipv4-family vpn-instance mcast1
[*Borderleaf1-bgp-mcast1] advertise l2vpn evpn
[*Borderleaf1-bgp-mcast1] network 1.1.1.10 32
[*Borderleaf1-bgp-mcast1] quit
[*Borderleaf1-bgp] quit
[*Borderleaf1] commit
```

Configure public network Layer 3 multicast on each leaf and spine switch. The configurations on Borderleaf1, Borderleaf2, Leaf1, Leaf2, and Spine2 are similar to the configuration on Spine1. For details, see the configuration file.

```
[~Spine1] multicast routing-enable
[*Spine1] interface loopback 1
[*Spine1-LoopBack1] pim sm
[*Spine1-LoopBack1] quit
[*Spine1] interface 10ge 1/0/1
[*Spine1-10GE1/0/1] pim sm
[*Spine1-10GE1/0/1] quit
[*Spine1] interface 10ge 1/0/2
[*Spine1-10GE1/0/2] pim sm
[*Spine1-10GE1/0/2] quit
[*Spine1] interface 10ge 1/0/3
[*Spine1-10GE1/0/3] pim sm
[*Spine1-10GE1/0/3] quit
[*Spine1] interface 10ge 1/0/4
[*Spine1-10GE1/0/4] pim sm
[*Spine1-10GE1/0/4] quit
[*Spine1] pim
[*Spine1-pim] static-rp 5.5.5.5
[*Spine1-pim] quit
[*Spine1] commit
```

Configure the Loopback1 interface addresses on Spine1 and Spine2 as the Anycast RP addresses, and configure Loopback2 interface addresses on Spine1 and Spine2 as the Anycast RP local addresses. The configuration on Spine2 is similar to the configuration on Spine1. For details, see the configuration file.

```
[~Spine1] pim
[~Spine1-pim] anycast-rp 5.5.5.5
[*Spine1-pim-anycast-rp-5.5.5.5] local-address 5.5.5.1
[*Spine1-pim-anycast-rp-5.5.5.5] commit
[~Spine1-pim-anycast-rp-5.5.5.5] peer 5.5.5.2
[*Spine1-pim-anycast-rp-5.5.5.5] quit
[*Spine1-pim] quit
[*Spine1] commit
```

Enable PIM for Layer 3 links on Borderleaf1 and Borderleaf2. The configuration on Borderleaf2 is similar to the configuration on Borderleaf1. For details, see the configuration file.

```
[~Borderleaf1] interface 10ge 1/0/6
[~Borderleaf1-10GE1/0/6] pim sm
[~Borderleaf1-10GE1/0/6] quit
[*Borderleaf1] commit
[~Borderleaf1] interface 10ge 1/0/6.1
[~Borderleaf1-10GE1/0/6.1] dot1q termination vid 1 //Set the VLAN ID for Dot1q termination on the
Layer 3 sub-interface based on the VLAN tag carried in packets.
[~Borderleaf1-10GE1/0/6.1] pim sm
[~Borderleaf1-10GE1/0/6.1] quit
[*Borderleaf1] commit
```

Configure BUM multicast replication for the Layer 3 VNI of the L3VPN instance on each leaf. The configurations on the other leaf nodes are similar to the configuration on Borderleaf1. For details, see the configuration file.

```
[~Borderleaf1] interface nve 1
[~Borderleaf1-Nve1] vni 5010 mcast-group 225.0.0.1
[*Borderleaf1-Nve1] quit
[*Borderleaf1] commit
```

Add VLANIF interfaces on Borderleaf1 and Borderleaf2 to multicast groups so that the M-LAG master and backup devices can synchronize VXLAN-encapsulated multicast packets through the peer-link. The configuration on Borderleaf2 is similar to the configuration on Borderleaf1. For details, see the configuration file.

```
[~Borderleaf1] vlan 3000
[*Borderleaf1-vlan3000] quit
[*Borderleaf1] interface vlanif 3000
[*Borderleaf1-Vlanif3000] ip address 10.10.21.1 24
[*Borderleaf1-Vlanif3000] vxlan multicast-group member enable
[*Borderleaf1-Vlanif3000] pim sm
[*Borderleaf1-Vlanif3000] quit
[*Borderleaf1] commit
```

Add VLANIF interfaces on Leaf1 and Leaf2 to multicast groups so that the M-LAG master and backup devices can synchronize VXLAN-encapsulated multicast packets through the peer-link. The configuration on Leaf2 is similar to the configuration on Leaf1. For details, see the configuration file.

```
[~Leaf1] vlan 4000
[*Leaf1-vlan4000] quit
[*Leaf1] interface vlanif 4000
[*Leaf1-Vlanif4000] ip address 10.10.20.1 24
[*Leaf1-Vlanif4000] vxlan multicast-group member enable
[*Leaf1-Vlanif4000] pim sm
[*Leaf1-Vlanif4000] quit
[*Leaf1] commit
```

After the preceding configurations are complete, VXLAN tunnels can be established between leaf nodes. You can run the **display vxlan tunnel** command to view VXLAN tunnel information. For example, in the command output on Borderleaf1, the **State** field is **up**, indicating that the VXLAN tunnel is reachable. If the value of the **Type** field is **static**, the destination IP address is statically

configured. If the value is **dynamic**, the destination IP address is dynamically learned through the protocol.

```
[~Borderleaf1] display vxlan tunnel
Number of vxlan tunnel : 2
Tunnel ID   Source       Destination    State Type    Uptime
-----
4026531841  1.1.1.110      225.0.0.1     up   static  04:44:52
4026531845  1.1.1.110      2.2.2.210     up   dynamic 04:40:52
```

Step 6 Configure BGP MVPN peers on each leaf.

Configure Borderleaf1. The configurations on the other leaf nodes are similar to the configuration on Borderleaf1. For details, see the configuration file.

```
[~Borderleaf1] bgp 100
[~Borderleaf1-bgp] ipv4-family mvpn
[*Borderleaf1-bgp-af-mvpn] peer 2.2.2.2 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Borderleaf1-bgp-af-mvpn] peer 3.3.3.3 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Borderleaf1-bgp-af-mvpn] peer 4.4.4.4 enable
Warning: This operation will reset the peer session. Continue? [Y/N]: y
[*Borderleaf1-bgp-af-mvpn] quit
[*Borderleaf1-bgp] quit
[*Borderleaf1] commit
```

Step 7 Configure the VXLAN I-PMSI tunnel on each leaf.

Configure Borderleaf1. The MVPN ID configured on the distributed VXLAN gateway must be the VTEP IP address of the current gateway.

```
[~Borderleaf1] multicast mvpn 1.1.1.110
[*Borderleaf1] ip vpn-instance mcast1
[*Borderleaf1-vpn-instance-mcast1] ipv4-family
[*Borderleaf1-vpn-instance-mcast1-af-ipv4] multicast routing-enable
[*Borderleaf1-vpn-instance-mcast1-af-ipv4] multicast mvpn route-import local-admin-id 1
[*Borderleaf1-vpn-instance-mcast1-af-ipv4] mvpn
[*Borderleaf1-vpn-instance-mcast1-af-ipv4-mvpn] c-multicast signaling bgp
[*Borderleaf1-vpn-instance-mcast1-af-ipv4-mvpn] spt-only mode
[*Borderleaf1-vpn-instance-mcast1-af-ipv4-mvpn] ipmsi-tunnel
[*Borderleaf1-vpn-instance-mcast1-af-ipv4-mvpn-ipmsi] vxlan static
[*Borderleaf1-vpn-instance-mcast1-af-ipv4-mvpn-ipmsi] quit
[*Borderleaf1-vpn-instance-mcast1-af-ipv4-mvpn] quit
[*Borderleaf1-vpn-instance-mcast1-af-ipv4] quit
[*Borderleaf1-vpn-instance-mcast1] quit
[*Borderleaf1] commit
```

Configure Borderleaf2. Borderleaf1 and Borderleaf2 are active-active gateways. Ensure that the two devices have the same MVPN ID, which is set to the VTEP IP address of the devices.

```
[~Borderleaf2] multicast mvpn 1.1.1.110
[*Borderleaf2] ip vpn-instance mcast1
[*Borderleaf2-vpn-instance-mcast1] ipv4-family
[*Borderleaf2-vpn-instance-mcast1-af-ipv4] multicast routing-enable
[*Borderleaf2-vpn-instance-mcast1-af-ipv4] multicast mvpn route-import local-admin-id 1
[*Borderleaf2-vpn-instance-mcast1-af-ipv4] mvpn
[*Borderleaf2-vpn-instance-mcast1-af-ipv4-mvpn] c-multicast signaling bgp
[*Borderleaf2-vpn-instance-mcast1-af-ipv4-mvpn] spt-only mode
[*Borderleaf2-vpn-instance-mcast1-af-ipv4-mvpn] ipmsi-tunnel
[*Borderleaf2-vpn-instance-mcast1-af-ipv4-mvpn-ipmsi] vxlan static
[*Borderleaf2-vpn-instance-mcast1-af-ipv4-mvpn-ipmsi] quit
[*Borderleaf2-vpn-instance-mcast1-af-ipv4-mvpn] quit
[*Borderleaf2-vpn-instance-mcast1-af-ipv4] quit
[*Borderleaf2-vpn-instance-mcast1] quit
[*Borderleaf2] commit
```

Configure Leaf1.

```
[~Leaf1] multicast mvpn 2.2.2.210
[*Leaf1] ip vpn-instance mcast1
[*Leaf1-vpn-instance-mcast1] ipv4-family
[*Leaf1-vpn-instance-mcast1-af-ipv4] multicast routing-enable
[*Leaf1-vpn-instance-mcast1-af-ipv4] multicast mvpn route-import local-admin-id 1
[*Leaf1-vpn-instance-mcast1-af-ipv4] mvpn
[*Leaf1-vpn-instance-mcast1-af-ipv4-mvpn] c-multicast signaling bgp
[*Leaf1-vpn-instance-mcast1-af-ipv4-mvpn] spt-only mode
[*Leaf1-vpn-instance-mcast1-af-ipv4-mvpn] ipmsi-tunnel
[*Leaf1-vpn-instance-mcast1-af-ipv4-mvpn-ipmsi] vxlan static
[*Leaf1-vpn-instance-mcast1-af-ipv4-mvpn-ipmsi] quit
[*Leaf1-vpn-instance-mcast1-af-ipv4-mvpn] quit
[*Leaf1-vpn-instance-mcast1-af-ipv4] quit
[*Leaf1-vpn-instance-mcast1] quit
[*Leaf1] commit
```

Configure Leaf2. Leaf1 and Leaf2 are active-active gateways. Ensure that the two devices have the same MVPN ID, which is set to the VTEP IP address of the devices.

```
[~Leaf2] multicast mvpn 2.2.2.210
[*Leaf2] ip vpn-instance mcast1
[*Leaf2-vpn-instance-mcast1] ipv4-family
[*Leaf2-vpn-instance-mcast1-af-ipv4] multicast routing-enable
[*Leaf2-vpn-instance-mcast1-af-ipv4] multicast mvpn route-import local-admin-id 1
[*Leaf2-vpn-instance-mcast1-af-ipv4] mvpn
[*Leaf2-vpn-instance-mcast1-af-ipv4-mvpn] c-multicast signaling bgp
[*Leaf2-vpn-instance-mcast1-af-ipv4-mvpn] spt-only mode
[*Leaf2-vpn-instance-mcast1-af-ipv4-mvpn] ipmsi-tunnel
[*Leaf2-vpn-instance-mcast1-af-ipv4-mvpn-ipmsi] vxlan static
[*Leaf2-vpn-instance-mcast1-af-ipv4-mvpn-ipmsi] quit
[*Leaf2-vpn-instance-mcast1-af-ipv4-mvpn] quit
[*Leaf2-vpn-instance-mcast1-af-ipv4] quit
[*Leaf2-vpn-instance-mcast1] quit
[*Leaf2] commit
```

Step 8 Configure PIM-SM and IGMP on the VBDIF interface.

Configure Leaf1.

```
[~Leaf1] interface vbdif 10
[~Leaf1-Vbdif10] pim sm
[*Leaf1-Vbdif10] igmp enable
[*Leaf1-Vbdif10] quit
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] interface vbdif 10
[~Leaf2-Vbdif10] pim sm
[*Leaf2-Vbdif10] igmp enable
[*Leaf2-Vbdif10] quit
[*Leaf2] commit
```

Step 9 Configure a VPN static RP on each leaf node.

Configure Borderleaf1.

```
[~Borderleaf1] interface loopback 3
[~Borderleaf1-LoopBack2] ip binding vpn-instance mcast1
[*Borderleaf1-LoopBack2] ip address 1.1.1.10 32
[*Borderleaf1-LoopBack2] pim sm
[*Borderleaf1-LoopBack2] quit
[*Borderleaf1] pim vpn-instance mcast1
[*Borderleaf1-pim-mcast1] static-rp 1.1.1.10
[*Borderleaf1-pim-mcast1] quit
[*Borderleaf1] commit
```

Configure Borderleaf2.

```
[*Borderleaf2] pim vpn-instance mcast1
[*Borderleaf2-pim-mcast1] static-rp 1.1.1.10
[*Borderleaf2-pim-mcast1] quit
[*Borderleaf2] commit
```

Configure Leaf1.

```
[*Leaf1] pim vpn-instance mcast1
[*Leaf1-pim-mcast1] static-rp 1.1.1.10
[*Leaf1-pim-mcast1] quit
[*Leaf1] commit
```

Configure Leaf2.

```
[*Leaf2] pim vpn-instance mcast1
[*Leaf2-pim-mcast1] static-rp 1.1.1.10
[*Leaf2-pim-mcast1] quit
[*Leaf2] commit
```

Step 10 On each leaf, add uplink and downlink interfaces to a Monitor Link group.

Configure Leaf1.

```
[~Leaf1] monitor-link group 1
[*Leaf1-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf1-mtlk-group1] port 10ge 1/0/6 uplink
[*Leaf1-mtlk-group1] port eth-trunk 10 downlink 1
[*Leaf1-mtlk-group1] quit
[*Leaf1] commit
```

Configure Leaf2.

```
[~Leaf2] monitor-link group 1
[*Leaf2-mtlk-group1] port 10ge 1/0/1 uplink
[*Leaf2-mtlk-group1] port 10ge 1/0/6 uplink
[*Leaf2-mtlk-group1] port eth-trunk 10 downlink 1
[*Leaf2-mtlk-group1] quit
[*Leaf2] commit
```

Step 11 Configure VPN Layer 3 multicast on Borderleaf1, Borderleaf2, and the CE.

Configure Borderleaf1. The configuration on Borderleaf2 is similar to the configuration on Borderleaf1. For details, see the configuration file.

```
[~Borderleaf1] interface 10ge 1/0/2
[*Borderleaf1-10GE1/0/2] undo portswitch
[*Borderleaf1-10GE1/0/2] ip binding vpn-instance mcast1
[*Borderleaf1-10GE1/0/2] ip address 192.168.20.1 24
[*Borderleaf1-10GE1/0/2] pim sm
[*Borderleaf1-10GE1/0/2] igmp enable
[*Borderleaf1-10GE1/0/2] quit
[*Borderleaf1] ip route-static vpn-instance mcast1 192.168.10.2 24 10ge 1/0/2
[*Borderleaf1] bgp 100
[*Borderleaf1-bgp] ipv4-family vpn-instance mcast1
[*Borderleaf1-bgp-mcast1] import-route static
[*Borderleaf1-bgp-mcast1] quit
[*Borderleaf1-bgp] quit
[*Borderleaf1] commit
```

Configure the CE.

```
[~CE] multicast routing-enable
[*CE] interface 10ge 1/0/1
[*CE-10GE1/0/1] pim sm
[*CE-10GE1/0/1] igmp enable
[*CE-10GE1/0/1] quit
[*CE] interface 10ge 1/0/2
[*CE-10GE1/0/2] pim sm
[*CE-10GE1/0/2] igmp enable
[*CE-10GE1/0/2] quit
```

```
[*CE] interface 10ge 1/0/3
[*CE-10GE1/0/2] pim sm
[*CE-10GE1/0/2] igmp enable
[*CE-10GE1/0/2] quit
[*CE] ip route-static 192.168.10.1 24 10ge 1/0/2
[*CE] commit
```

----End

Verifying the Configuration

Run the **display bgp mvpn all peer** command on each leaf node to query BGP MVPN peer information. For example, the information displayed on Borderleaf1 is as follows:

```
[~Borderleaf1] display bgp mvpn all peer
BGP local router ID      : 1.1.1.1
Local AS number         : 100
Total number of peers    : 3
Peers in established state : 3
```

Peer	V	AS	MsgRcvd	MsgSent	OutQ	Up/Down	State	PrefRcv
2.2.2.2	4	100	1860	1859	0	04:43:30	Established	1
3.3.3.3	4	100	3219	3221	0	04:43:39	Established	1
4.4.4.4	4	100	4559	4561	0	04:44:39	Established	1

After a receiver initiates a request to join a multicast group, run the **display pim vpn-instance mcast1 routing-table** command on each leaf to query the information about the PIM routing table of the VPN.

- The command output on Borderleaf1 shows that the multicast receiver joins the multicast group through BGP.

```
[~Borderleaf1] display pim vpn-instance mcast1 routing-table
VPN-Instance: mcast1
Total 1 (*, G) entry; 1 (S, G) entry

(*, 225.1.1.1)
RP: 1.1.1.10 (local)
Protocol: pim-sm, Flag: WC
UpTime: 04:43:57
Upstream interface: 10GE1/0/2
Upstream neighbor: 192.168.20.2
RPF prime neighbor: 192.168.20.2
Downstream interface(s) information:
Total number of downstreams: 1
1: pseudo
Protocol: BGP, UpTime: 04:43:57, Expires: -

(192.168.20.9, 225.1.1.1)
RP: 1.1.1.10 (local)
Protocol: pim-sm, Flag: SPT LOC ACT
UpTime: 04:43:57
Upstream interface: 10GE1/0/2
Upstream neighbor: 192.168.20.2
RPF prime neighbor: 192.168.20.2
Downstream interface(s) information:
Total number of downstreams: 1
1: pseudo
Protocol: pim-sm, UpTime: 04:43:57, Expires: -
```

- The command output on Leaf1 shows that the upstream interface in the VPN PIM routing entry for the receiver is Through-BGP and the downstream interface is VBDIF 10.

```
[~Leaf1] display pim vpn-instance mcast1 routing-table
VPN-Instance: mcast1
Total 1 (S, G) entry; 1 (S, G) entry
```

```
(* , 225.1.1.1)
RP: 1.1.1.10
Protocol: pim-sm, Flag: WC
UpTime: 04:44:18
Upstream interface: through-BGP
  Upstream neighbor: 1.1.1.1
  RPF prime neighbor: 1.1.1.1
Downstream interface(s) information:
Total number of downstreams: 1
  1: Vbdf10
    Protocol: igmp, UpTime: 04:44:18, Expires: -

(192.168.20.9, 225.1.1.1)
RP: 1.1.1.10
Protocol: pim-sm, Flag: SPT ACT
UpTime: 04:44:18
Upstream interface: through-BGP
  Upstream neighbor: 1.1.1.1
  RPF prime neighbor: 1.1.1.1
Downstream interface(s) information:
Total number of downstreams: 1
  1: Vbdf10
    Protocol: pim-sm, UpTime: 04:44:18, Expires: -
```

Configuration Files

- CE configuration file

```
#
sysname CE
#
multicast routing-enable
#
interface 10GE1/0/1
undo portswitch
ip address 192.168.10.2 255.255.255.0
pim sm
igmp enable
#
interface 10GE1/0/2
undo portswitch
ip address 192.168.20.2 255.255.255.0
pim sm
igmp enable
#
interface 10GE1/0/3
undo portswitch
ip address 192.168.30.2 255.255.255.0
pim sm
igmp enable
#
ip route-static 192.168.10.0 255.255.255.0 10ge 1/0/2
#
return
```

- Borderleaf1 configuration file

```
#
sysname Borderleaf1
#
set forward capability enhanced //This step is required only on the CE12800.
#
assign forward nvo3 acl extend enable //This step is required only on the CE12800, CE6870EI, and CE6875EI.
#
dfs-group 1
source ip 10.10.11.1
#
vlan batch 3000
#
stp mode rstp
```

```
stp v-stp enable
#
evpn-overlay enable
#
multicast routing-enable
#
multicast mvpn 1.1.1.110
#
ip vpn-instance mcast1
  ipv4-family
    route-distinguisher 1:1
    vpn-target 1:1 export-extcommunity
    vpn-target 13:1 export-extcommunity evpn
    vpn-target 1:1 import-extcommunity
    vpn-target 13:1 import-extcommunity evpn
    multicast routing-enable
    multicast mvpn route-import local-admin-id 1
  mvpn
    c-multicast signaling bgp
    spt-only mode
    ipmsi-tunnel
    vxlan static
  vxlan vni 5010
#
interface Vlanif3000
  ip address 10.10.21.1 255.255.255.0
  pim sm
  vxlan multicast-group member enable
#
interface MEth0/0/0
  ip address 10.10.11.1 255.255.255.0
#
interface Eth-Trunk2
  mode lacp-static
  peer-link 1
#
ip route-static vpn-instance mcast1 192.168.10.2 24 10GE1/0/2
#
interface 10GE1/0/1
  undo portswitch
  ip address 10.1.1.1 255.255.255.0
  pim sm
#
interface 10GE1/0/2
  undo portswitch
  ip binding vpn-instance mcast1
  ip address 192.168.20.1 24
  pim sm
  igmp enable
#
interface 10GE1/0/3
  undo portswitch
  ip address 10.1.11.1 255.255.255.0
  pim sm
#
interface 10GE1/0/4
  eth-trunk 2
#
interface 10GE1/0/5
  eth-trunk 2
#
interface 10GE1/0/6
  undo portswitch
  pim sm
#
interface 10GE1/0/6.1
  ip address 10.6.1.1 255.255.255.0
  dot1q termination vid 1
  pim sm
```

```
#
interface LoopBack1
 ip address 1.1.1.1 255.255.255.255
 pim sm
#
interface LoopBack2
 ip address 1.1.1.110 255.255.255.255
 pim sm
#
interface LoopBack3
 ip binding vpn-instance mcast1
 ip address 1.1.1.10 255.255.255.255
 pim sm
#
interface Nve1
 source 1.1.1.110
 vni 5010 head-end peer-list protocol bgp
 vni 5010 mcast-group 225.0.0.1
 mac-address 0000-5e00-0101
#
bgp 100
 router-id 1.1.1.1
 peer 2.2.2.2 as-number 100
 peer 2.2.2.2 connect-interface LoopBack1
 peer 3.3.3.3 as-number 100
 peer 3.3.3.3 connect-interface LoopBack1
 peer 4.4.4.4 as-number 100
 peer 4.4.4.4 connect-interface LoopBack1
#
ipv4-family unicast
 peer 2.2.2.2 enable
 peer 3.3.3.3 enable
 peer 4.4.4.4 enable
#
ipv4-family mvpn
 policy vpn-target
 peer 2.2.2.2 enable
 peer 3.3.3.3 enable
 peer 4.4.4.4 enable
#
ipv4-family vpn-instance mcast1
 network 1.1.1.10 255.255.255.255
 import-route static
 advertise l2vpn evpn
#
l2vpn-family evpn
 policy vpn-target
 peer 2.2.2.2 enable
 peer 2.2.2.2 advertise irb
 peer 3.3.3.3 enable
 peer 3.3.3.3 advertise irb
 peer 4.4.4.4 enable
 peer 4.4.4.4 advertise irb
#
ospf 1
 area 0.0.0.0
 network 1.1.1.1 0.0.0.0
 network 1.1.1.110 0.0.0.0
 network 10.1.1.0 0.0.0.255
 network 10.1.11.0 0.0.0.255
 network 10.6.1.0 0.0.0.255
 network 10.10.11.0 0.0.0.255
#
pim
 static-rp 5.5.5.5
#
pim vpn-instance mcast1
 static-rp 1.1.1.10
```

```
#
return
• Borderleaf2 configuration file
#
sysname Borderleaf2
#
set forward capability enhanced //This step is required only on the CE12800.
#
assign forward nvo3 acl extend enable //This step is required only on the CE12800, CE6870EI, and
CE6875EI.
#
dfs-group 1
source ip 10.10.11.2
#
vlan batch 3000
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
multicast routing-enable
#
multicast mvpn 1.1.1.110
#
ip vpn-instance mcast1
ipv4-family
route-distinguisher 1:1
vpn-target 1:1 export-extcommunity
vpn-target 13:1 export-extcommunity evpn
vpn-target 1:1 import-extcommunity
vpn-target 13:1 import-extcommunity evpn
multicast routing-enable
multicast mvpn route-import local-admin-id 1
mvpn
c-multicast signaling bgp
spt-only mode
ipmsi-tunnel
vxlan static
vxlan vni 5010
#
interface Vlanif3000
ip address 10.10.21.2 255.255.255.0
pim sm
vxlan multicast-group member enable
#
interface MEth0/0/0
ip address 10.10.11.2 255.255.255.0
#
interface Eth-Trunk2
mode lacp-static
peer-link 1
#
ip route-static vpn-instance mcast1 192.168.10.2 24 10GE1/0/2
#
interface 10GE1/0/1
undo portswitch
ip address 10.1.44.1 255.255.255.0
pim sm
#
interface 10GE1/0/2
undo portswitch
ip binding vpn-instance mcast1
ip address 192.168.30.1 24
pim sm
igmp enable
#
interface 10GE1/0/3
undo portswitch
```



```
ip address 10.1.4.1 255.255.255.0
pim sm
#
interface 10GE1/0/4
eth-trunk 2
#
interface 10GE1/0/5
eth-trunk 2
#
interface 10GE1/0/6
undo portswitch
pim sm
#
interface 10GE1/0/6.1
ip address 10.6.1.2 255.255.255.0
dot1q termination vid 1
pim sm
#
interface LoopBack1
ip address 2.2.2.2 255.255.255.255
pim sm
#
interface LoopBack2
ip address 1.1.1.110 255.255.255.255
pim sm
#
interface Nve1
source 1.1.1.110
vni 5010 head-end peer-list protocol bgp
vni 5010 mcast-group 225.0.0.1
mac-address 0000-5e00-0101
#
bgp 100
router-id 2.2.2.2
peer 1.1.1.1 as-number 100
peer 1.1.1.1 connect-interface LoopBack1
peer 3.3.3.3 as-number 100
peer 3.3.3.3 connect-interface LoopBack1
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack1
#
ipv4-family unicast
peer 1.1.1.1 enable
peer 3.3.3.3 enable
peer 4.4.4.4 enable
#
ipv4-family mvpn
policy vpn-target
peer 1.1.1.1 enable
peer 3.3.3.3 enable
peer 4.4.4.4 enable
#
ipv4-family vpn-instance mcast1
import-route static
#
l2vpn-family evpn
policy vpn-target
peer 1.1.1.1 enable
peer 1.1.1.1 advertise irb
peer 3.3.3.3 enable
peer 3.3.3.3 advertise irb
peer 4.4.4.4 enable
peer 4.4.4.4 advertise irb
#
ospf 1
area 0.0.0.0
network 2.2.2.2 0.0.0.0
network 1.1.1.110 0.0.0.0
network 10.1.4.0 0.0.0.255
```

```
network 10.1.44.0 0.0.0.255
network 10.6.1.0 0.0.0.255
network 10.10.11.0 0.0.0.255
#
pim
static-rp 5.5.5.5
#
pim vpn-instance mcast1
static-rp 1.1.1.10
#
return
```

- Leaf1 configuration file

```
#
sysname Leaf1
#
set forward capability enhanced //This step is required only on the CE12800.
#
assign forward nvo3 acl extend enable //This step is required only on the CE12800, CE6870EI, and
CE6875EI.
#
dfs-group 1
source ip 10.10.10.1
#
vlan batch 4000
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
multicast routing-enable
#
multicast mvpn 2.2.2.210
#
ip vpn-instance mcast1
ipv4-family
route-distinguisher 1:1
vpn-target 1:1 export-extcommunity
vpn-target 13:1 export-extcommunity evpn
vpn-target 1:1 import-extcommunity
vpn-target 13:1 import-extcommunity evpn
multicast routing-enable
multicast mvpn route-import local-admin-id 1
mvpn
c-multicast signaling bgp
spt-only mode
ipmsi-tunnel
vxlan static
vxlan vni 5010
#
bridge-domain 10
vxlan vni 10
evpn
route-distinguisher 11:1
vpn-target 12:1 export-extcommunity
vpn-target 13:1 export-extcommunity
vpn-target 12:1 import-extcommunity
vpn-target 13:1 import-extcommunity
#
interface Vbdif10
ip binding vpn-instance mcast1
ip address 192.168.10.1 255.255.255.0
mac-address 0000-5e00-0104
pim sm
igmp enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Vlanif4000
```

```

ip address 10.10.20.1 255.255.255.0
pim sm
vxlan multicast-group member enable
#
interface MEth0/0/0
ip address 10.10.10.1 255.255.255.0
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk11 //This step is required only on the CE8860EI, CE8850EI, CE7855EI, CE7850EI,
CE6860EI, CE6856HI, CE6855HI, CE6851HI, CE6850HI, and CE6850U-HI.
service type tunnel
#
interface Eth-Trunk10
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 1
#
interface Eth-Trunk10.1 mode l2
encapsulation dot1q vid 10
bridge-domain 10
#
interface 10GE1/0/1
undo portswitch
ip address 10.1.2.1 255.255.255.0
pim sm
#
interface 10GE1/0/2
eth-trunk 10
#
interface 10GE1/0/3
eth-trunk 10
#
interface 10GE1/0/4
eth-trunk 1
#
interface 10GE1/0/5
eth-trunk 1
#
interface 10GE1/0/6
undo portswitch
ip address 10.1.22.1 255.255.255.0
pim sm
#
interface 10GE1/0/7 //This step is required only on the CE8860EI, CE8850EI, CE7855EI, CE7850EI,
CE6860EI, CE6856HI, CE6855HI, CE6851HI, CE6850HI, and CE6850U-HI.
eth-trunk 11
#
interface LoopBack1
ip address 3.3.3.3 255.255.255.255
pim sm
#
interface LoopBack2
ip address 2.2.2.210 255.255.255.255
pim sm
#
interface Nve1
source 2.2.2.210
vni 5010 head-end peer-list protocol bgp
vni 5010 mcast-group 225.0.0.1
mac-address 0000-5e00-0102
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/6 uplink
port Eth-Trunk10 downlink 1
#

```

```
bgp 100
router-id 3.3.3.3
peer 1.1.1.1 as-number 100
peer 1.1.1.1 connect-interface LoopBack1
peer 2.2.2.2 as-number 100
peer 2.2.2.2 connect-interface LoopBack1
peer 4.4.4.4 as-number 100
peer 4.4.4.4 connect-interface LoopBack1
#
ipv4-family unicast
peer 1.1.1.1 enable
peer 2.2.2.2 enable
peer 4.4.4.4 enable
#
ipv4-family mvpn
policy vpn-target
peer 1.1.1.1 enable
peer 2.2.2.2 enable
peer 4.4.4.4 enable
#
l2vpn-family evpn
policy vpn-target
peer 1.1.1.1 enable
peer 1.1.1.1 advertise irb
peer 2.2.2.2 enable
peer 2.2.2.2 advertise irb
peer 4.4.4.4 enable
peer 4.4.4.4 advertise irb
#
ospf 1
area 0.0.0.0
network 3.3.3.3 0.0.0.0
network 2.2.2.210 0.0.0.0
network 10.1.2.0 0.0.0.255
network 10.1.22.0 0.0.0.255
network 10.10.10.0 0.0.0.255
#
pim
static-rp 5.5.5.5
#
pim vpn-instance mcast1
static-rp 1.1.1.10
#
return
```

- Leaf2 configuration file

```
#
sysname Leaf2
#
set forward capability enhanced //This step is required only on the CE12800.
#
assign forward nvo3 acl extend enable //This step is required only on the CE12800, CE6870EI, and
CE6875EI.
#
dfs-group 1
source ip 10.10.10.2
#
vlan batch 4000
#
stp mode rstp
stp v-stp enable
#
evpn-overlay enable
#
multicast routing-enable
#
multicast mvpn 2.2.2.210
#
ip vpn-instance mcast1
ipv4-family
```

```
route-distinguisher 1:1
vpn-target 1:1 export-extcommunity
vpn-target 13:1 export-extcommunity evpn
vpn-target 1:1 import-extcommunity
vpn-target 13:1 import-extcommunity evpn
multicast routing-enable
multicast mvpn route-import local-admin-id 1
mvpn
c-multicast signaling bgp
spt-only mode
ipmsi-tunnel
vxlan static
vxlan vni 5010
#
bridge-domain 10
vxlan vni 10
evpn
route-distinguisher 11:1
vpn-target 12:1 export-extcommunity
vpn-target 13:1 export-extcommunity
vpn-target 12:1 import-extcommunity
vpn-target 13:1 import-extcommunity
#
interface Vbdf10
ip binding vpn-instance mcast1
ip address 192.168.10.1 255.255.255.0
mac-address 0000-5e00-0104
pim sm
igmp enable
vxlan anycast-gateway enable
arp collect host enable
#
interface Vlanif4000
ip address 10.10.20.2 255.255.255.0
pim sm
vxlan multicast-group member enable
#
interface MEth0/0/0
ip address 10.10.10.2 255.255.255.0
#
interface Eth-Trunk1
mode lacp-static
peer-link 1
#
interface Eth-Trunk11 //This step is required only on the CE8860EI, CE8850EI, CE7855EI, CE7850EI,
CE6860EI, CE6856HI, CE6855HI, CE6851HI, CE6850HI, and CE6850U-HI.
service type tunnel
#
interface Eth-Trunk10
stp edged-port enable
mode lacp-static
dfs-group 1 m-lag 1
#
interface Eth-Trunk10.1 mode l2
encapsulation dot1q vid 10
bridge-domain 10
#
interface 10GE1/0/1
undo portswitch
ip address 10.1.33.1 255.255.255.0
pim sm
#
interface 10GE1/0/2
eth-trunk 10
#
interface 10GE1/0/3
eth-trunk 10
#
interface 10GE1/0/4
```

```
eth-trunk 1
#
interface 10GE1/0/5
eth-trunk 1
#
interface 10GE1/0/6
undo portswitch
ip address 10.1.3.1 255.255.255.0
pim sm
#
interface 10GE1/0/7 //This step is required only on the CE8860EI, CE8850EI, CE7855EI, CE7850EI,
CE6860EI, CE6856HI, CE6855HI, CE6851HI, CE6850HI, and CE6850U-HI.
eth-trunk 11
#
interface LoopBack1
ip address 4.4.4.4 255.255.255.255
pim sm
#
interface LoopBack2
ip address 2.2.2.210 255.255.255.255
pim sm
#
interface Nve1
source 2.2.2.210
vni 5010 head-end peer-list protocol bgp
vni 5010 mcast-group 225.0.0.1
mac-address 0000-5e00-0102
#
monitor-link group 1
port 10GE1/0/1 uplink
port 10GE1/0/6 uplink
port Eth-Trunk10 downlink 1
#
bgp 100
router-id 4.4.4.4
peer 1.1.1.1 as-number 100
peer 1.1.1.1 connect-interface LoopBack1
peer 2.2.2.2 as-number 100
peer 2.2.2.2 connect-interface LoopBack1
peer 3.3.3.3 as-number 100
peer 3.3.3.3 connect-interface LoopBack1
#
ipv4-family unicast
peer 1.1.1.1 enable
peer 2.2.2.2 enable
peer 3.3.3.3 enable
#
ipv4-family mvpn
policy vpn-target
peer 1.1.1.1 enable
peer 2.2.2.2 enable
peer 3.3.3.3 enable
#
l2vpn-family evpn
policy vpn-target
peer 1.1.1.1 enable
peer 1.1.1.1 advertise irb
peer 2.2.2.2 enable
peer 2.2.2.2 advertise irb
peer 3.3.3.3 enable
peer 3.3.3.3 advertise irb
#
ospf 1
area 0.0.0.0
network 4.4.4.4 0.0.0.0
network 2.2.2.210 0.0.0.0
network 10.1.3.0 0.0.0.255
network 10.1.33.0 0.0.0.255
network 10.10.10.0 0.0.0.255
```

```
#
pim
static-rp 5.5.5.5
#
pim vpn-instance mcast1
static-rp 1.1.1.10
#
return
```

- Spine1 configuration file

```
#
sysname Spine1
#
multicast routing-enable
#
interface 10GE1/0/1
undo portswitch
ip address 10.1.1.2 255.255.255.0
pim sm
#
interface 10GE1/0/2
undo portswitch
ip address 10.1.2.2 255.255.255.0
pim sm
#
interface 10GE1/0/3
undo portswitch
ip address 10.1.3.2 255.255.255.0
pim sm
#
interface 10GE1/0/4
undo portswitch
ip address 10.1.4.2 255.255.255.0
pim sm
#
interface LoopBack1
ip address 5.5.5.5 255.255.255.255
pim sm
#
interface LoopBack2
ip address 5.5.5.1 255.255.255.255
pim sm
#
ospf 1
area 0.0.0.0
network 5.5.5.5 0.0.0.0
network 5.5.5.1 0.0.0.0
network 10.1.1.0 0.0.0.255
network 10.1.2.0 0.0.0.255
network 10.1.3.0 0.0.0.255
network 10.1.4.0 0.0.0.255
#
pim
static-rp 5.5.5.5
anycast-rp 5.5.5.5
local-address 5.5.5.1
peer 5.5.5.2
#
return
```

- Spine2 configuration file

```
#
sysname Spine2
#
multicast routing-enable
#
interface 10GE1/0/1
undo portswitch
ip address 10.1.11.2 255.255.255.0
pim sm
```

```
#
interface 10GE1/0/2
undo portswitch
ip address 10.1.22.2 255.255.255.0
pim sm
#
interface 10GE1/0/3
undo portswitch
ip address 10.1.33.2 255.255.255.0
pim sm
#
interface 10GE1/0/4
undo portswitch
ip address 10.1.44.2 255.255.255.0
pim sm
#
interface LoopBack1
ip address 5.5.5.5 255.255.255.255
pim sm
#
interface LoopBack2
ip address 5.5.5.2 255.255.255.255
pim sm
#
ospf 1
area 0.0.0.0
network 5.5.5.5 0.0.0.0
network 5.5.5.2 0.0.0.0
network 10.1.11.0 0.0.0.255
network 10.1.22.0 0.0.0.255
network 10.1.33.0 0.0.0.255
network 10.1.44.0 0.0.0.255
#
pim
static-rp 5.5.5.5
anycast-rp 5.5.5.5
local-address 5.5.5.2
peer 5.5.5.1
#
return
```

1.20 Configuring an Intelligent Lossless Network

Recommended Models

Table 1-20 lists the recommended models for different application scenarios of an intelligent lossless network.

Table 1-20 Recommended models

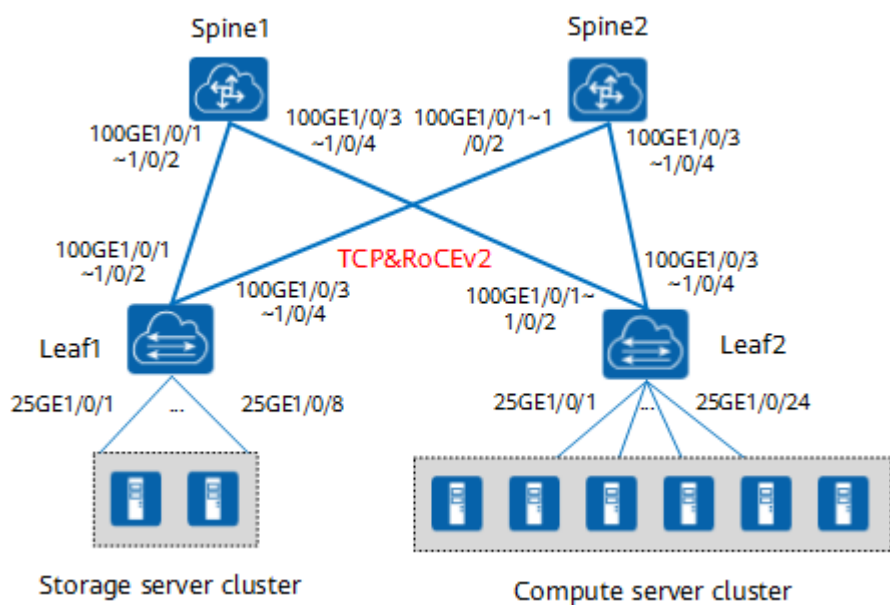
Application Scenario	Leaf	Spine
Storage	CE6865-48S8CQ-EI	CE8850-64CQ-EI/ CloudEngine 16800 (configured with CE- MPUE series MPUs)

Application Scenario	Leaf	Spine
AI GPU	CE8861-4C-EI, CE8850-64CQ-EI	CE8850-64CQ-EI/ CloudEngine 16800 (configured with CE- MPUE series MPUs)
HPC	CE8850-64CQ-EI	CE8850-64CQ-EI/ CloudEngine 16800 (configured with CE- MPUE series MPUs)
Remarks	For details about device selection and recommended networking, see the <i>CloudEngine 16800, 8800, and 6800 Series Switches AI Fabric Deployment Best Practices</i> .	

Networking Requirements

Figure 1-24 shows the networking of a RoCEv2 high-performance application, where both TCP and RoCEv2 traffic is transmitted and all servers support the RoCEv2 protocol and have the DCQCN function enabled. Compute and storage servers are deployed in the same PoD, and the ratio of compute nodes to storage nodes is 3:1. Leaf and spine switches are fully meshed through 100GE links. Servers are connected to leaf switches through 25GE links, and the oversubscription ratio is 1:1. In this example, a CE6865-48S8CQ-EI is used as a leaf switch, and a CloudEngine 16800 (configured with CE-MPUE series MPUs) is used as a spine switch.

Figure 1-24 Networking diagram for configuring an intelligent lossless network



Priority Planning

Based on the service traffic characteristics, priorities in this example are planned as follows:

- Set the priority of CNP traffic to 6, scheduling mode to PQ, and DSCP value to 25.
- Set the priority of RoCEv2 traffic to 4, scheduling mode to DRR, weight to 65%, and DSCP value to 24.
- Set the priority of TCP traffic to 1, scheduling mode to DRR, weight to 15%, and DSCP value to 7.
- Set the priority of O&M traffic to 0, scheduling mode to DRR, and weight to 5%.
- Reserve priorities 2, 3, and 5 for future use.

Configuration Roadmap

NOTE

- The parameter settings in this example are for reference only. You need to configure each device based on the traffic model in the actual networking. For more information, refer to the *CloudEngine 16800, 8800, and 6800 Series Switches AI Fabric Deployment Best Practices*.
- In this example, the IP addresses and routes for interconnection between spine switches, leaf switches, and servers have been configured, and there are reachable routes between servers.

The configuration roadmap is as follows:

- Configure leaf switches.
 - a. Configure PFC. Before configuring PFC, you need to configure priority mapping.
 - b. Configure PFC deadlock detection.
 - c. Configure the low-latency network function. After this function is configured, automatic buffer optimization and dynamic ECN threshold are enabled for lossless queues by default. You can optimize the two functions.
 - d. Enable the AI ECN function. Before enabling this function, you need to disable the dynamic ECN function. The AI ECN function is supported only in V200R019C10 and later versions. If a device does not support this function, you can configure the dynamic or static ECN function.
 - e. Configure the fast CNP function.
- Configure spine switches. If CE8850-64CQ-EI switches are used as the spine switches, the configuration of spine switches is similar to that of leaf switches. The CloudEngine 16800 (configured with CE-MPUE series MPUs) is used as an example.
 - a. Configure PFC. Before configuring PFC, you need to configure priority mapping.
 - b. Configure PFC deadlock detection.
 - c. Optimize the buffer space.

- d. Enable the AI ECN function. This function is supported only in V200R019C10 and later versions.
- Configure server NICs. (The detailed procedures are not provided.)
 - a. Configure NICs to work in RoCEv2 mode.
 - b. Configure the RoCEv2 link setup mode.
 - c. Configure NICs to trust DSCP values, and configure the DSCP values of RoCEv2 and CNP packets.
 - d. Enable PFC for the priority of RoCEv2 packets on NICs.
 - e. Enable DCQCN for the priority of RoCEv2 packets on NICs.

Procedure

The following describes the configurations on Leaf1. The configurations on Leaf2 are similar.

Step 1 Configure PFC.

1. Configure priority mapping and congestion scheduling.

In this example, the DSCP value of RoCEv2 packets is 24, and the DSCP value of CNP packets is 25. Configure a priority mapping profile in the DiffServ domain as follows to map the priority of RoCEv2 packets to priority 4 (queue 4) and the priority of CNP packets to priority 6 (queue 6), and map the DSCP value 7 to priority 1:

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf1
[HUAWEI] commit
[~Leaf1] diffserv domain ds1
[*Leaf1-dsdomain-ds1] ip-dscp-inbound 24 phb af4 green
[*Leaf1-dsdomain-ds1] ip-dscp-inbound 25 phb cs6 green
[*Leaf1-dsdomain-ds1] ip-dscp-inbound 7 phb af1 green
[*Leaf1-dsdomain-ds1] quit
[*Leaf1] port-group all_using
[*Leaf1-port-group-all_using] group-member 100ge 1/0/1 to 100ge 1/0/4
[*Leaf1-port-group-all_using] group-member 25ge 1/0/1 to 25ge 1/0/8
[*Leaf1-port-group-all_using] quit
[*Leaf1] commit
[~Leaf1] port-group all_using
[*Leaf1-port-group-all_using] trust dscp
[*Leaf1-port-group-all_using] trust upstream ds1
[*Leaf1-port-group-all_using] quit
[*Leaf1] commit
```

Configure the congestion scheduling mode for each queue. By default, queues on an interface use the PQ scheduling mode. Therefore, queue 6 can use the default scheduling mode to ensure preferential scheduling of CNP packets.

```
[~Leaf1] port-group all_using
[*Leaf1-port-group-all_using] qos drr 0 1 4
[*Leaf1-port-group-all_using] qos queue 0 drr weight 5
[*Leaf1-port-group-all_using] qos queue 1 drr weight 15
[*Leaf1-port-group-all_using] qos queue 4 drr weight 65
[*Leaf1-port-group-all_using] quit
[*Leaf1] commit
```

2. Configure PFC for the priority of RoCEv2 traffic.

Configure the queue with priority 4 to carry RoCEv2 traffic on the network. To implement this, enable PFC for priority 4 on each interface and enable PFC based on the priority mapped from the DSCP value.

```
[~Leaf1] dcb pfc
[~Leaf1-dcb-pfc-default] priority 4
[~Leaf1-dcb-pfc-default] quit
[*Leaf1] port-group all_using
[*Leaf1-port-all_using] dcb pfc enable mode manual
[*Leaf1-port-all_using] quit
[*Leaf1] dcb pfc dscp-mapping enable slot 1
[*Leaf1] commit
```

NOTE

After the preceding configurations are complete, RoCEv2 traffic is transmitted in the queue with priority 4, which is a lossless queue.

CE6865EI, CE6865E, CE8850E-32CQ-EI switches are used as leaf switches. On Leaf1, 12 ports are used; on Leaf2, 28 ports are used. You can change the dynamic threshold for triggering PFC frames to 5 to improve the performance of RoCEv2 services.

```
<Leaf1> system-view
[~Leaf1] port-group all_using
[*Leaf1-port-group-all_using] dcb pfc buffer 4 xoff dynamic 5
[*Leaf1-port-group-all_using] quit
[*Leaf1] commit
```

Step 2 Configure PFC deadlock detection.

Set the PFC deadlock detection interval and recovery time to 100 ms for lossless queues, and configure the switch to disable PFC when five PFC deadlocks occur within 20s.

```
[~Leaf1] dcb pfc
[*Leaf1-dcb-pfc-default] dcb pfc deadlock-detect interval 10
[*Leaf1-dcb-pfc-default] priority 4 deadlock-detect time 10
[*Leaf1-dcb-pfc-default] priority 4 deadlock-recovery time 10
[*Leaf1-dcb-pfc-default] priority 4 turn-off threshold 5
[*Leaf1-dcb-pfc-default] quit
[*Leaf1] commit
```

NOTE

After the configuration is complete, if you need to modify the PFC deadlock detection configuration, run the **shutdown** command to disable the PFC-enabled interface to prevent configuration failures caused by deadlock recovery on the switch.

Step 3 Configure the low-latency network function.

1. Configure the low-latency network function on Leaf1. This function takes effect after the switch restarts. After the configuration is successful, automatic buffer optimization and dynamic ECN threshold for lossless queues are enabled by default.

```
[~Leaf1] low-latency fabric
[*Leaf1-low-latency-fabric] quit
[*Leaf1] commit
[~Leaf1] quit
<Leaf1> save
Warning: The current configuration will be written to the device. Continue? [Y/N]: y
<Leaf1> reboot
Warning: The system will reboot. Continue? [Y/N]: y
```

2. Optimize the buffer space of lossless queues.

In this example, both TCP and RoCEv2 traffic is transmitted on the network. Therefore, you can manually reduce the threshold for the queue of TCP traffic. This ensures more shared buffer space of the chip available to lossless RoCEv2 traffic.

```
[~Leaf1] port-group all_using
[*Leaf1-port-group-all_using] qos buffer queue 1 shared-threshold dynamic 1
[*Leaf1-port-group-all_using] quit
[*Leaf1] commit
```

Step 4 Enable the AI ECN function.

```
[~Leaf1] low-latency fabric
[~Leaf1-low-latency-fabric] undo qos dynamic-ecn-threshold enable
[*Leaf1-low-latency-fabric] quit
[*Leaf1] commit
[~Leaf1] ai-service
[*Leaf1-ai-service] ai-ecn
[*Leaf1-ai-service-ai-ecn] ai-ecn enable
[*Leaf1-ai-service-ai-ecn] quit
[*Leaf1-ai-service] quit
[*Leaf1] commit
```

Step 5 Enable the fast CNP function.

```
[~Leaf1] low-latency fabric
[*Leaf1-low-latency-fabric] qos fast-cnp enable
[*Leaf1-low-latency-fabric] quit
[*Leaf1] commit
```

----End

The following describes the configurations on Spine1. The configurations on Spine2 are similar.

Step 1 Configure PFC.

1. Configure priority mapping and congestion scheduling.

In this example, the DSCP value of RoCEv2 packets is 24, and the DSCP value of CNP packets is 25. Configure a priority mapping profile in the DiffServ domain as follows to map the priority of RoCEv2 packets to priority 4 (queue 4) and the priority of CNP packets to priority 6 (queue 6), and map the DSCP value 7 to priority 1:

```
<HUAWEI> system-view
[~HUAWEI] sysname Spine1
[*HUAWEI] commit
[~Spine1] diffserv domain ds1
[*Spine1-dsdomain-ds1] ip-dscp-inbound 24 phb af4 green
[*Spine1-dsdomain-ds1] ip-dscp-inbound 25 phb cs6 green
[*Spine1-dsdomain-ds1] ip-dscp-inbound 7 phb af1 green
[*Spine1-dsdomain-ds1] quit
[*Spine1] port-group all_using
[*Spine1-port-group-all_using] group-member 100ge 1/0/1 to 100ge 1/0/2
[*Spine1-port-group-all_using] quit
[*Spine1] commit
[~Spine1] port-group all_using
[*Spine1-port-group-all_using] trust upstream ds1
[*Spine1-port-group-all_using] quit
[*Spine1] commit
```

Configure the congestion scheduling mode for each queue. By default, queues on an interface use the PQ scheduling mode. Therefore, queue 6 can use the default scheduling mode to ensure preferential scheduling of CNP packets.

```
[~Spine1] port-group all_using
[*Spine1-port-group-all_using] qos drr 0 1 4
[*Spine1-port-group-all_using] qos queue 0 drr weight 5
[*Spine1-port-group-all_using] qos queue 1 drr weight 15
[*Spine1-port-group-all_using] qos queue 4 drr weight 65
[*Spine1-port-group-all_using] quit
[*Spine1] commit
```

2. Configure PFC for the priority of RoCEv2 traffic.

Configure the queue with priority 4 to carry RoCEv2 traffic on the network. To implement this, enable PFC for priority 4 on each interface and enable PFC based on the priority mapped from the DSCP value.

```
[~Spine1] dcb pfc
[~Spine1-dcb-pfc-default] priority 4
[*Spine1-dcb-pfc-default] quit
[*Spine1] port-group all_using
[*Spine1-port-all_using] dcb pfc enable mode manual
[*Spine1-port-all_using] quit
[*Spine1] dcb pfc dscp-mapping enable slot 1
[*Spine1] commit
```

 NOTE

After the preceding configurations are complete, RoCEv2 traffic is transmitted in the queue with priority 4, which is a lossless queue.

CloudEngine 16800 switches are used as the spine switches. Set the XOFF parameter to 3000 cells and the headroom buffer to 2000 cells to improve the performance of RoCEv2 services.

```
<Spine1> system-view
[~Spine1] port-group all_using
[*Spine1-port-group-all_using] dcb pfc buffer 4 xoff static 3000 hdrm 2000
[*Spine1-port-group-all_using] quit
[*Spine1] commit
```

Step 2 Configure PFC deadlock detection.

Set the hardware-based PFC deadlock detection interval and recovery time of lossless queues to 1500 ms.

```
[~Spine1] dcb pfc
[*Spine1-dcb-pfc-default] dcb pfc deadlock-detect interval 100
[*Spine1-dcb-pfc-default] priority 4 deadlock-detect time 15
[*Spine1-dcb-pfc-default] priority 4 deadlock-recovery time 15
[*Spine1-dcb-pfc-default] quit
[*Spine1] commit
```

 NOTE

After the configuration is complete, if you need to modify the PFC deadlock detection configuration, run the **shutdown** command to disable the PFC-enabled interface to prevent configuration failures caused by deadlock recovery on the switch.

Step 3 Optimize the buffer space.

In this example, both TCP and RoCEv2 traffic is transmitted on the network. Therefore, you can manually reduce the threshold for the queue of TCP traffic and increase the threshold for the queue of RoCEv2 traffic. This ensures more shared buffer space of the chip available to lossless RoCEv2 traffic.

```
[~Spine1] port-group all_using
[*Spine1-port-group-all_using] qos buffer queue 1 shared-threshold dynamic 1
[*Spine1-port-group-all_using] qos buffer queue 4 shared-threshold dynamic 15
[*Spine1-port-group-all_using] quit
[*Spine1] commit
```

Step 4 Enable the AI ECN function.

```
[~Spine1] ai-service
[*Spine1-ai-service] ai-ecn
[*Spine1-ai-service-ai-ecn] ai-ecn enable
[*Spine1-ai-service-ai-ecn] quit
```

```
[*Spine1-ai-service] quit
[*Spine1] commit
```

----End

Verifying the Configuration

- Check the PFC threshold and headroom value.
[~Leaf1] **display dcb pfc buffer interface 100ge1/0/1**
Xon: PFC backpressure stop threshold
Xoff: PFC backpressure threshold
Hdrm: Headroom buffer threshold
Guaranteed: PFC guaranteed buffer threshold
The actual PFC backpressure stop threshold is the higher value between the value of xon and the difference between the value of xoff and the value of xon-offset.
C:cells B:bytes K:kilobytes M:megabytes D:dynamic alpha

Interface	Queue	Guaranteed	Xon	Xon-Offset	Xoff	Hdrm
100GE1/0/1	4	8(C)	200(C)	20(C)	5(D)	630(C)

- Check the numbers of PFC deadlocks and recovery times. If the values of **DeadlockNum** and **RecoveryNum** are 0, no deadlock is triggered.
[~Leaf1] **display dcb pfc interface 100ge 1/0/1**

Interface	Queue	Received(Frames)	ReceivedRate(pps)	DeadlockNum
		Transmitted(Frames)	TransmittedRate(pps)	RecoveryNum
100GE1/0/1	4	0	0	0
		0	0	0

- Check the enabling status of the AI ECN function and the calculated ECN threshold.
[~Leaf1] **display ai-ecn calculated state interface 100ge 1/0/1**
*: Indicates the queue where AI ECN takes effect.
AI-ECN State: enabled

Interface	Queue	Low-Threshold (Byte)	High-Threshold (Byte)	Probability (%)
100GE1/0/1	0	0	0	0
	1	0	0	0
	2	0	0	0
	3	0	0	0
	*4	4896	18874080	40
	5	0	0	0
	6	0	0	0
	7	0	0	0

1.21 Example for Configuring the INC Function for an Intelligent Lossless Network

Applicable Products and Versions

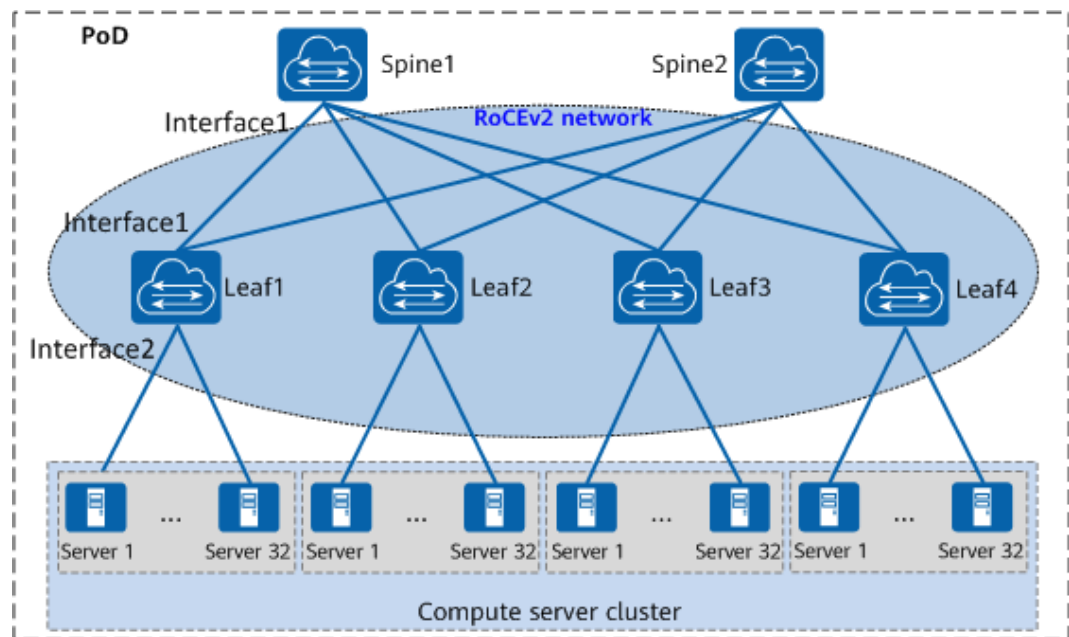
This example applies to CE8850-64CQ-EI switch running V200R020C10 or later versions.

For details about the mapping between software versions and switch models, see the [Hardware Center](#).

Networking Requirements

As shown in [Figure 1-25](#), an RoCEv2 HPC service is deployed in the compute server cluster in a PoD of a DC. The compute switches (Leaf1, Leaf2, Leaf3, and Leaf4) connect to the upstream spine switches and downstream compute servers. To ensure low latency, zero packet loss, and high throughput for this HPC service, you can configure PFC, AI ECN threshold of lossless queues, and INC on the switches.

Figure 1-25 Networking diagram for configuring the INC function for an intelligent lossless network



NOTE

- CE8850-64CQ-EI switches are deployed as spine and leaf nodes.
- In [Figure 1-25](#), Interface1 and Interface2 indicate 100GE 1/0/1 and 100GE 1/0/2, respectively.

Configuration Roadmap

NOTE

- The server NICs must support the RoCEv2 and DCQCN functions.
- The parameter settings in this example are for reference only. You need to configure each device based on the traffic model on your network.

The configuration roadmap is as follows:

1. Configure the DSCP priority-based PFC function and configure the queue that carries RoCEv2 traffic as a lossless queue.
2. Enable basic functions of the intelligent lossless network.
3. Configure buffer optimization of lossless queues.
4. Enable the AI ECN threshold function for lossless queues.

5. Configure fast ECN.
6. Configure the INC function.

Procedure

Step 1 Configure the DSCP priority-based PFC function.

Enable the PFC function for the queue with priority 3 on an interface of Leaf1 and Spine1 to carry RoCEv2 traffic. The configurations on Leaf2 to Leaf4 are similar to those on Leaf1, and the configurations on Spine2 are similar to those on Spine1. The detailed configurations are not described here.

```
<HUAWEI> system-view
[~HUAWEI] sysname Leaf1
[*HUAWEI] commit
[~Leaf1] dcb pfc
[*Leaf1-dcb-pfc-default] priority 3
[*Leaf1-dcb-pfc-default] quit
[*Leaf1] interface 100ge 1/0/1
[*Leaf1-100GE1/0/1] dcb pfc enable mode manual
[*Leaf1-100GE1/0/1] quit
[*Leaf1] interface 100ge 1/0/2
[*Leaf1-100GE1/0/2] dcb pfc enable mode manual
[*Leaf1-100GE1/0/2] quit
[*Leaf1] commit
<HUAWEI> system-view
[~HUAWEI] sysname Spine1
[*HUAWEI] commit
[~Spine1] dcb pfc
[*Spine1-dcb-pfc-default] priority 3
[*Spine1-dcb-pfc-default] quit
[*Spine1] interface 100ge 1/0/1
[*Spine1-100GE1/0/1] dcb pfc enable mode manual
[*Spine1-100GE1/0/1] quit
[*Spine1] commit
```

Enable Leaf1 and Spine1 to implement PFC based on the priority after DSCP mapping. The configurations on Leaf2 to Leaf4 are similar to those on Leaf1, and the configurations on Spine2 are similar to those on Spine1. The detailed configurations are not described here.

```
[~Leaf1] dcb pfc dscp-mapping enable slot 1
[*Leaf1] interface 100ge 1/0/2
[*Leaf1-100GE1/0/1] trust dscp
[*Leaf1-100GE1/0/1] quit
[*Leaf1] commit
[~Spine1] dcb pfc dscp-mapping enable slot 1
[*Spine1] commit
```

Step 2 Enable basic functions of the intelligent lossless network.

Configure the low-latency network function on Leaf1, which takes effect after the switch restarts. The switch then automatically enables basic functions of the intelligent lossless network. The configurations on Leaf2, Leaf3, Leaf4, Spine1, and Spine2 are similar and are not described here.

```
[~Leaf1] low-latency fabric
Info: Please save the configuration and reboot the system to enable the operation.
[*Leaf1-low-latency-fabric] quit
[*Leaf1] commit
[~Leaf1] quit
<Leaf1> save
Warning: The current configuration will be written to the device. Continue? [Y/N]: y
<Leaf1> reboot
Warning: The system will reboot. Continue? [Y/N]: y
```

Step 3 Configure buffer optimization of lossless queues.

NOTE

After the low-latency network function is configured, automatic buffer optimization of lossless queues is enabled by default. The buffer space automatically set by the device may not meet the requirement, for example, when the device throughput is high but the latency is high, or PFC is triggered frequently. In this case, you can disable automatic buffer optimization of lossless queues and manually configure buffer optimization for lossless queues.

Manually configure buffer optimization of lossless queues on Leaf1 to increase the headroom buffer of the corresponding queue to 330 cells, if packet loss occurs on an uplink interface of Leaf1. The configurations on Leaf2, Leaf3, Leaf4, Spine1, and Spine2 are similar to those on Leaf1 and are not described here.

```
<Leaf1> system-view
[~Leaf1] low-latency fabric
[~Leaf1-low-latency-fabric] undo qos buffer auto-configuration enable
Info: Please save the configuration and reboot the system to enable the operation.
[*Leaf1-low-latency-fabric] quit
<Leaf1> save
Warning: The current configuration will be written to the device. Continue? [Y/N]: y
<Leaf1> reboot
Warning: The system will reboot. Continue? [Y/N]: y
<Leaf1> system-view
[~Leaf1] interface 100ge 1/0/2
[*Leaf1-100GE1/0/2] dcb pfc buffer 3 xoff dynamic 4 hdrm 330
[*Leaf1-100GE1/0/2] quit
[*Leaf1] commit
```

Display the buffer optimization configuration of lossless queues on Leaf1.

```
[~Leaf1] display dcb pfc buffer
Xon:      PFC backpressure stop threshold
Xoff:     PFC backpressure threshold
Hdrm:     Headroom buffer threshold
Guaranteed: PFC guaranteed buffer threshold
The actual PFC backpressure stop threshold is the higher value between the value
of xon and the difference between the value of xoff and the value of xon-offset.
C:cells B:bytes K:kilobytes M:megabytes D:dynamic alpha
```

Interface	Queue	Guaranteed	Xon	Xon-Offset	Xoff	Hdrm
100GE1/0/2	3	10(C)	50(C)	20(C)	4(D)	330(C)

Step 4 Enable the AI ECN threshold function for lossless queues.

NOTE

The AI ECN threshold function takes effect only for the queue with the highest priority in the default PFC profile bound to an interface in Up state. You can adjust the priorities in the default PFC profile to adjust the queue for which AI ECN is enabled.

After the low-latency network function is enabled, the dynamic ECN threshold of all lossless queues is enabled on the device by default. The AI ECN threshold function and the dynamic ECN threshold function are mutually exclusive. You can run the **undo qos dynamic-ecn-threshold enable** command to disable the dynamic ECN threshold of all lossless queues.

Enable the AI ECN threshold function for lossless queues on Leaf1. The configurations on Leaf2, Leaf3, Leaf4, Spine1, and Spine2 are similar and are not described here.

```
[~Leaf1] low-latency fabric
[~Leaf1-low-latency-fabric] undo qos dynamic-ecn-threshold enable
[*Leaf1-low-latency-fabric] quit
[*Leaf1] commit
```

```
[~Leaf1] ai-service
[~Leaf1-ai-service] ai-ecn
[*Leaf1-ai-service-ai-ecn] ai-ecn enable
[*Leaf1-ai-service-ai-ecn] quit
[*Leaf1-ai-service] quit
[*Leaf1] commit
```

Step 5 Configure fast ECN. The fast ECN function is available when a switch starts. No command for enabling or disabling this function is provided.

Step 6 Configure the INC function.

Enable the INC function on Leaf1. The configurations on Leaf2 to Leaf4 are similar and are not described here.

```
[~Leaf1] interface eth-trunk 10
[*Leaf1-Eth-Trunk10] service type inc
[*Leaf1-Eth-Trunk10] quit
[*Leaf1] ai-service
[*Leaf1-ai-service] inc
[*Leaf1-ai-service-inc] inc-hierarchy hierarchy-1
[*Leaf1-ai-service-inc] source 10.10.10.10
[*Leaf1-ai-service-inc] inc enable
[*Leaf1-ai-service-inc] quit
[*Leaf1] commit
[~Leaf1] quit
<Leaf1> save
Warning: The current configuration will be written to the device. Continue? [Y/N]: y
<Leaf1> reboot
Warning: The system will reboot. Continue? [Y/N]: y
```

Enable the INC function on Spine1. The configurations on Spine2 are similar and are not described here.

```
[~Spine1] interface eth-trunk 10
[*Spine1-Eth-Trunk10] service type inc
[*Spine1-Eth-Trunk10] quit
[*Spine1] ai-service
[*Spine1-ai-service] inc
[*Spine1-ai-service-inc] inc-hierarchy hierarchy-2
[*Spine1-ai-service-inc] source 20.20.20.20
[*Spine1-ai-service-inc] inc enable
[*Spine1-ai-service-inc] quit
[*Spine1] commit
[~Spine1] quit
<Spine1> save
Warning: The current configuration will be written to the device. Continue? [Y/N]: y
<Spine1> reboot
Warning: The system will reboot. Continue? [Y/N]: y
```

----End

Verifying the Configuration

Display the configuration of the INC function.

```
[~Leaf1] display inc communicator
-----
CommID  Hierarchy  LocalRank  GlobalRank  Root
-----
1       1          2          4           20.20.20.20
-----
```